Self-Adaptive Kernels as Evolving Measures: Transport Adaptive Control

Mathematical Foundations of Self-Adaptive Kernels, version 3: a complete mathematical treatment

Delhi Technological University

October 2026

Version 3. This document supersedes version 2 and, before it, version 1, *Mathematical Foundations of Self-Adaptive Kernels Achieving Lyapunov Convergence*. Section 2 states what survived from version 1 and what was withdrawn; version 3 adds Lemma E.0 (explicit constants for Theorem E), Theorems C′, D′, I and J and Proposition K (Table 1); a second increment adds the sharp slow-regime rate (Lemma E.0′), uniform-in-time propagation of chaos in the leak-dominated regime (Theorem D″) and the stationary law of the dithered finite-size loop with its non-equilibrium current (Theorem J′, Proposition J′.2); a third increment adds the explicit stationary parameter spread of the dithered linearised loop (Corollary E.0′.3) and the slow-regime Lyapunov function for non-periodic persistently exciting regressors (Lemma E.0″) and uniform-in-time propagation of chaos in the learning regime for the averaged slow loop (Theorem D‴, with the finite-m weight-sensitivity law of Proposition D‴.5, the real loop on a warm start of Corollary D‴.7, the leak's prior as a design input of the window, Proposition D‴.9, and the barycentric OT leak's mass constraint, which removes the finite-size floor, Proposition D‴.12 — in the real loop too, where the collapsed subspace carries one K_0-atom orbit for every commensurate size, Corollary D‴.14 and Proposition D‴.15, and Theorem J holds uniformly in the number of kernels in the per-particle norm, Theorem D‴.18, while a soft (entropic) mass constraint equalises the masses only down to a critical excess, Proposition D‴.20; and the mass-constrained energy satisfies a Polyak–Łojasiewicz inequality whose constant does not depend on the number of kernels, which gives a measure-level basin with an exponential rate and, at the nearest-atom leak, explains the floor as a spurious minimum at every admissible count vector, Theorem D‴.21). This is research in progress and has not been peer reviewed. Code, tests, experiments and review records: https://github.com/Dutta-06/Lyapunov-Convergence-Kernels-Showcase.

Copyright © 2026 Odwitiyo Dutta. All rights reserved.

Abstract

This is a complete mathematical treatment of adaptive tracking control with Gaussian radial-basis-function (RBF) networks whose weights, centres and widths are adapted online. Part I is the projection-based adaptive RBF controller of version 1. An audit found its stability argument to be a correct but standard composition of known results, with two claims wrong as written (a non-convex "projection" for centre separation and an implicit-function-theorem argument that is not a proof); it contains no new theorem. We state it with explicit assumptions, a Hessian-free disturbance bound and an RMS bound, and list the errata to version 1. Part II develops transport adaptive control: the network is an empirical measure on a lifted parameter space, and the adaptive law is its transport. Theorems A–F, C′ and D′ are proved; A, B, C, C′, E and F are also checked by tests. Theorem E is self-contained: an explicit strict Lyapunov function (Lemma E.0) gives the radius, overshoot and rate of the label-free convergence under tangential excitation. Theorem C′ is the OT σ-modification toward a prior with a different number of atoms, through the barycentric projection of the optimal plan (Lloyd's algorithm in continuous time); Theorem D′ is propagation of chaos for the dithered closed loop, at the dimension-free rate $m^{-1/2}$. Proposition G is formal

and checked directly. Theorem H shows that the averaged mean-field loop with saturated features, σ-modification and dither is mean-field Langevin dynamics and converges globally and exponentially to the regularised optimum; this is proved for the averaged system, the reduction to it is formal, and the real loop is a conjecture. At finite gain the averaged loop is in general not a gradient flow of any free energy in the $W_{G^{-1}}$ metric: Proposition H.7 (frozen-measure part proved) exhibits a semi-gradient term, whose performative defect is non-reciprocal in the checked instances ($n = 1, 2$), and an orbit-area circulation kernel ($n \geq 2$). With a second-order averaging term they predict a closed-loop effect for which a targeted literature search found no precedent: traversing the reference orbit backwards changes the learned kernels by $O(1/k_e)$, although the occupation measure is unchanged. At $k_e = 40$ the prediction matches to 3% in size and 11% as a vector (cosine 0.994). Proposition H.8, proved assuming the chain rule for the perturbed flow, gives input-to-state stability of the averaged system. Theorem I characterises the non-reciprocity at order k_e^{-2}: for $n = 1$ the averaged loop is reciprocal exactly when the residual vanishes on the orbit, and no open set of measures is reciprocal for any n. Theorem J proves the deterministic finite-m case of the averaging conjecture: every hyperbolic critical point of the regularised loss has a unique nearby periodic orbit of the real closed loop with the same stability type, whose period mean is the first-order prediction up to $O(k_e^{-2})$. Proposition K shows that a collapsed configuration is a strict local minimum iff the leak gain exceeds a critical value σ_{crit}, and that in a window around σ_{crit} the direction in which the reference is traversed decides whether the kernels separate. A second increment adds the sharp slow-regime rate (Lemma E.0′), uniform-in-time propagation of chaos in the leak-dominated regime (Theorem D″) and the stationary law of the dithered finite-size loop with its non-equilibrium current (Theorem J′, Proposition J′.2). A third increment adds the explicit stationary parameter spread of the dithered linearised loop (Corollary E.0′.3) and the slow-regime Lyapunov function for non-periodic persistently exciting regressors (Lemma E.0″) and uniform-in-time propagation of chaos in the learning regime for the averaged slow loop (Theorem D‴, with the finite-m weight-sensitivity law of Proposition D‴.5, the real loop on a warm start of Corollary D‴.7, the leak's prior as a design input of the window, Proposition D‴.9, and the barycentric OT leak's mass constraint, which removes the finite-size floor, Proposition D‴.12 — in the real loop too, where the collapsed subspace carries one K_0-atom orbit for every commensurate size, Corollary D‴.14 and Proposition D‴.15, and Theorem J holds uniformly in the number of kernels in the per-particle norm, Theorem D‴.18, while a soft (entropic) mass constraint equalises the masses only down to a critical excess, Proposition D‴.20; and the mass-constrained energy satisfies a Polyak–Łojasiewicz inequality whose constant does not depend on the number of kernels, which gives a measure-level basin with an exponential rate and, at the nearest-atom leak, explains the floor as a spurious minimum at every admissible count vector, Theorem D‴.21).

Contents

1 Introduction **6**
- 1.1 The idea . . . 6
- 1.2 Why the classical theory needs it . . . 6
- 1.3 What the closed loop actually is . . . 6
- 1.4 Conventions and verification . . . 7
- 1.5 Contributions and their status . . . 7
- 1.6 Where the new mathematics lies . . . 20
- 1.7 Organisation . . . 21

2 Relation to version 1 **21**
- 2.1 What version 1 claimed . . . 21
- 2.2 The audit's verdict . . . 22
- 2.3 Errata to version 1 . . . 23
- 2.4 What survives and what is withdrawn . . . 25
- 2.5 How Part II answers the structural defects . . . 26

I The projection-based controller **28**

3 Setting and standing assumptions **28**

4 The algorithm **29**

5 Lemmas and the boundedness theorem **29**
- 5.1 Lemmas . . . 29
- 5.2 Theorem 1: uniform ultimate boundedness . . . 31

6 The no-bias claim, stated precisely **32**

7 Prior art **32**

8 What the experiments show **33**
- 8.1 Robustness mechanism versus geometry . . . 33
- 8.2 Kernel-regression baselines . . . 34
- 8.3 The bound against data . . . 35
- 8.4 Summary . . . 37

II Transport adaptive control **40**

9 Setting **40**

10 Theorem A: identification **42**

11 Theorem B: the optimal-assignment Lyapunov function **44**

12 Theorem C: optimal-transport σ-modification **48**

13 Theorem C′: the OT leak toward a prior with a different number of atoms **52**

14 Theorem D: the mean-field closed loop **57**

15 Theorem E: label-free local convergence under tangential excitation **60**

16 Theorem F: the transport cost of adaptation **76**

17 Theorem D′: propagation of chaos for the dithered closed loop on finite horizons **78**
Theorem D″: uniform-in-time propagation of chaos in the leak-dominated regime . . . 81

18 Theorem D‴: uniform-in-time propagation of chaos in the learning regime (the averaged slow loop) **84**

19 Proposition G: the slow geometry **157**

20 Theorem H: Langevin adaptive control and global convergence to the regularised optimum **161**

21 Finite gain: semi-gradient and non-reciprocal structure **167**
21.1 Frozen-measure structure of the exactly averaged loop 168
21.2 The oscillation correction and the orientation of the reference orbit 171
21.3 Reciprocity at order k_e^{-2}: a pairwise criterion and its characterisation 173
21.4 Input-to-state stability of $(\mathfrak{A})$ and the route to Conjecture H.6 176
21.5 Evidence from the real closed loop 177

22 Theorem J: the finite-m closed loop as a perturbed gradient flow **178**

23 Proposition K: collapsed configurations, the critical leak, and the orientation of the orbit as a design input **184**

24 Theorem J′: the stationary law of the dithered finite-m closed loop **188**

III Evidence and outlook **199**

25 Numerical evidence **199**
25.1 Theorem B: the label-free Lyapunov function 199
25.2 Theorem C: OT σ-modification 200
25.3 Theorem C′: the barycentric OT leak 201
25.4 Theorem D: convergence in m 202
25.5 Theorem D′: propagation of chaos for the dithered loop 202
25.6 Theorem D‴: uniform-in-time propagation of chaos in the learning regime 203
25.7 Proposition D‴.9: measure-valued priors and the design of the window 204
25.8 Proposition D‴.12: the OT leak's mass constraint against the weight fluctuation 205
25.9 Corollary D‴.14: the real loop with the OT leak 206
25.10 Corollary D‴.14 in two dimensions 206
25.11 Proposition D‴.15: the transverse spectrum of a lifted orbit 206
25.12 Remark D‴.16: dither and the mass constraint 207
25.13 Theorem D‴.18: Theorem J uniformly in m 207
25.14 Proposition D‴.20: the entropic OT leak 208
25.15 Theorem D‴.21: a measure-level basin for the OT leak 208
25.16 Theorem D‴.23: adaptive priors 209
25.17 Proposition D‴.24: the floor is the leak's bias; an adaptive prior without a mass constraint 210
25.18 Corollary D‴.26: perfect tracking with an adaptive prior in the real loop 212
25.19 Lemma D‴.26a: the exact slow subspace: Corollary D‴.26(c) and Proposition D‴.30(ii) proved for the linearised loop 213

25.20Proposition D‴.27: a projected adaptive prior 214
25.21Proposition D‴.28: the floor removal is a gauge property 214
25.22Proposition D‴.29: second-order plants: the sliding-variable reduction; a cogging-torque servo 215
25.23Proposition D‴.30: measurement noise at perfect tracking: coherent, high-passed, and the errors in variables 216
25.24Proposition D‴.31: the adaptive prior is a washout leak; the desired regressor makes the noise's own persistent force a least-squares force 217
25.25Theorem E: label-free convergence under excitation 218
25.26Lemma E.0 and Theorem E with explicit constants 218
25.27Theorem F: transport cost 221
25.28Proposition G: slow geometry 221
25.29Theorem H: Langevin adaptive control 222
25.30Remark H.0 and Proposition H.7: the exactly averaged closed loop 224
25.31Remark H.7a: the orientation of the reference orbit 225
25.32Theorem I: reciprocity at order k_e^{-2} 225
25.33Theorem J: periodic orbits of the finite-m closed loop 226
25.34Proposition K: collapsed configurations and the orientation of the orbit 227
25.35Theorem J′: the stationary law of the dithered single-kernel loop 228

26 Research program **228**

A Verification map **236**

1 Introduction

1.1 The idea

An adaptive approximator with nonlinearly parametrised features can be read as an *empirical measure* $\mu = \frac{1}{m}\sum_i \delta_{\vartheta_i}$ on a lifted parameter space. Here the features are Gaussian RBF kernels with adaptive weights, centres and widths. The adaptive law transports this measure, and the control output $\hat{f} = \int \Phi \, \mathrm{d}\mu$ is linear in μ. Stability, robustness modifications, excitation, the many-kernel limit and the slow evolution of kernel geometry all acquire formulations on Wasserstein space. So does the global optimality of kernel placement. These formulations do not depend on how the kernels are labelled.

1.2 Why the classical theory needs it

This work identifies two structural defects of the classical labelled analysis, that of version 1 of this manuscript and of the literature it builds on.

- The classical Lyapunov analyses depend on labels. A network that is a relabelling of the ideal has zero function error but a large parameter error.
- There is no object that describes where the kernels go, or whether their placement is good.

A third apparent defect was that constants grow with the number of kernels. It turned out to be an artefact of the normalisation. The mean-field normalisation removes it for labelled and label-free analyses alike (Remark A.1).

1.3 What the closed loop actually is

- At infinite feedback gain, the averaged closed loop is a Wasserstein gradient flow. With dither it is a mean-field Langevin dynamics (Theorem H).
- At finite gain it is in general *not* a gradient flow of any free energy in the $W_{G^{-1}}$ metric. This already holds for $n = 1$, so it is not a two-dimensional effect (Proposition H.7). The first correction has two parts at a frozen measure, and a third along the real loop.
  - The first part is a *semi-gradient* term. The adaptive law differentiates the loss on the closed-loop state distribution while ignoring that this distribution depends on the parameters, as in performative prediction. Like semi-gradient TD, the resulting update is not the gradient of any objective. Its performative defect is present for every n; for $n = 1$ it is non-reciprocal exactly when the residual is non-zero on the orbit (Theorem I(iii)), and it is non-reciprocal in the checked $n = 2$ instance.
  - The second part is an antisymmetric interaction between kernels ($n \geq 2$). Its kernel is an integral of $\sum_k \det[\nabla\tilde{\Phi}_k(\vartheta), \nabla\tilde{\Phi}_k(\vartheta')]$ over the region that the reference orbit encloses (Stokes).
  - Along the real loop, a third part comes from the parameters' oscillation within each period. It is a second-order averaging term (Remark H.7a).
- Together these predict a new closed-loop effect: traversing the reference orbit backwards changes the learned kernels by $O(1/k_e)$, although the occupation measure is unchanged. At $k_e = 40$ the prediction matches to 3% in size and 11% as a vector (cosine 0.994). The vector error halves with each doubling of k_e (`experiments/exp_orbit_orientation.py`).
- The gradient structure survives as an input-to-state Lyapunov function (Proposition H.8). This reduces the global conjecture for the real loop to an averaging estimate.

1.4 Conventions and verification

Status labels. Every result of Part II is labelled *proved*, *formal* or *conjectural*, and the label is given in a status line after the result.

- "Proved" means proved here. "Formal" means a derivation without error control.
- "Checked" means a test evaluates the statement, or every inequality in it, along simulated trajectories or on sampled states.
- The novelty column of Table 1 follows the verdicts in `LITERATURE_TRANSPORT.md`.

In Part I, every step of the stability argument is either proved or verified numerically by `tests/test_theorem_to_code.py`.

Verification and review.

- Every inequality the tests can evaluate is checked numerically along simulated trajectories, and every number in this document is produced by a script in the accompanying repository (the experiments are in `experiments/`); no table is edited by hand.
- The theory went through three rounds of automated multi-agent adversarial review, the manuscript through a fourth that compared it line by line with the theory, the results added in version 3 (Lemma E.0, Theorems C′, D′, I and J, Proposition K) through a fifth, and the second increment — the sharp slow-regime rate (Lemma E.0′), uniform-in-time propagation of chaos in the leak-dominated regime (Theorem D″) and the stationary law of the dithered finite-size loop with its non-equilibrium current (Theorem J′, Proposition J′.2) — through a sixth, and the third increment — the explicit stationary parameter spread of the dithered linearised loop (Corollary E.0′.3) and the slow-regime Lyapunov function for non-periodic persistently exciting regressors (Lemma E.0″) and uniform-in-time propagation of chaos in the learning regime for the averaged slow loop (Theorem D‴, with the finite-m weight-sensitivity law of Proposition D‴.5, the real loop on a warm start of Corollary D‴.7, the leak's prior as a design input of the window, Proposition D‴.9, and the barycentric OT leak's mass constraint, which removes the finite-size floor, Proposition D‴.12 — in the real loop too, where the collapsed subspace carries one K_0-atom orbit for every commensurate size, Corollary D‴.14 and Proposition D‴.15, and Theorem J holds uniformly in the number of kernels in the per-particle norm, Theorem D‴.18, while a soft (entropic) mass constraint equalises the masses only down to a critical excess, Proposition D‴.20; and the mass-constrained energy satisfies a Polyak–Łojasiewicz inequality whose constant does not depend on the number of kernels, which gives a measure-level basin with an exponential rate and, at the nearest-atom leak, explains the floor as a spurious minimum at every admissible count vector, Theorem D‴.21) — through a seventh (the first two) and an eighth to a sixteenth (the third). Every confirmed issue and its fix is logged in `review/REVIEW_LOG.md`, and the raw reports of rounds 1–3 and 5–16 are kept beside it.
- The claim-by-claim novelty audit is in `LITERATURE_TRANSPORT.md`.

Appendix A maps each checked statement to the test or experiment that checks it.

1.5 Contributions and their status

Table 1 lists the results of Part II with their status, in the conventions of Section 1.4, and with the novelty verdict of the literature audit.

Table 1: Contributions of Part II and their honest status. Status conventions: Section 1.4.

result	statement	status	novelty (audit)
A	The adaptive law is the particle form of the Wasserstein gradient flow of a μ-linear potential. Projected rates coincide only for a per-kernel projection ($m \geq 2$).	proved and checked	interpretive; ingredients known [14], [20], [51], [69]
B	$V_{\mathrm{OT}} = \frac{1}{2}\|e\|^2 + \frac{1}{2}W^2$ satisfies a Dini inequality with a two-regime matched remainder. It gives conditional RMS bounds along (L), and along (L_σ) with a leak term.	proved and checked, including a moving ideal, $\varepsilon \neq 0$, switches, an active projection, and the failure of the plain bound along (L_σ)	packaging new; ingredients known [84], [1], [68], [16]
C	OT σ-modification: exact dissipation for selection solutions; a gap-protected strict inequality with $\sqrt{C} \leq b + 2\eta$; a local exponential estimate, m-uniform for a single-atom prior and $O(m^{-1/2})$ otherwise; reshuffling by the labelled leak	proved and checked; the C(iv) instance satisfies every hypothesis	novel as a control construction; C(i) and the obstruction are known [80], [9]
C′	The OT leak toward a prior with any number of atoms, through the barycentric projection of the optimal plan: dissipation of $W(\mu, \nu)$ up to the within-cell variance of the plan; the fixed points of the pure leak are the equal-mass centroidal configurations of the prior (Lloyd's algorithm in continuous time, the G-gradient flow of the quantisation energy); the induced-coupling bound $\sqrt{C_{\mathrm{bar}}} \leq b + 2\eta$; a Lyapunov inequality and a local exponential estimate with the bias b_{bar} := distance of the ideal from the centroids of its prior cells ($\leq b$)	proved for vertex-selection solutions (existence assumed) and for generalised solutions (which exist); checked (dissipation at every step, convergence to a Lloyd fixed point, the coupling bound and plan stability, the Lyapunov inequality along the closed loop)	the law is the continuous-time Lloyd descent of coverage control [19] with equal-mass (capacity-constrained) OT plans [5], used as a σ-modification leak; new here are the induced-coupling bound, the Lyapunov inequality with the bias b_{bar} and the transfer reading; the quantisation side is Lloyd/CVT theory [57]; it reduces to Theorem C for $m_0 = m$
D	Local well-posedness of the mean-field loop; a Dobrushin estimate; a tube bound and global existence under a self-consistent growth condition; the mean-field limit on the existence interval	proved; consistency in m checked numerically (the constant is not checked)	technique anticipated [29, 72, 73]

continued on the next page

Table 1, continued

result	statement	status	novelty (audit)
E	Label-free local exponential convergence under tangential PE, in a uniform tube, at fixed m, with explicit radius, overshoot and rate (Lemma E.0); exponential merging of solutions. In the slow-adaptation regime a second strict Lyapunov function (Lemma E.0′: second-order slaving and a second-order averaging correction, with the peak factors of the regressor as data) has the averaged rate $\lambda_{\min}\langle\Psi^\top\Psi\rangle/k$ up to a factor $1-\delta$, and under parameter dither bounds the stationary parameter spread by $(1+O(\varepsilon))k\|\Sigma\|_F^2/(2(1-\delta)\lambda_{\min})$ (Corollary E.0′.3); for non-periodic PE regressors the same construction with an exponential average over a window gives the rate $(1-\delta-\delta_2)\lambda_T/k$, λ_T the weighted PE level of the window (Lemma E.0″)	proved, self-contained; linearisation checked; the strict Lyapunov inequalities checked pointwise (E.0′ by a certificate over all states) and the explicit envelope checked on the linearised and on the implemented loop; convergence observed. The rate of E.0 is conservative in a quantified way (Remark E.0a); the rate of E.0′ is within 12% of the exact rate above its threshold	partially anticipated [16], [34]; the strict Lyapunov function is of the Mazenc–Malisoff type
F	Along (L), the Wasserstein length of learning is $\le \lambda_G^{1/2} L_1 \int\|e\|$; leak versions	proved and checked, including the leak clause, a tight leak instance with $e\equiv 0$ and a tight velocity instance	elementary
G	Averaged joint descent on the occupation loss. The slow geometry follows the gradient flow of the profile loss in the metric induced on the slow manifold, which is plain variable projection only when that manifold is flat ($S\ll 1$).	formal; checked directly, including a counterexample to plain VarPro at $S=O(1)$	a fixed-m, closed-loop analogue of the two-timescale/VarPro training of [59], [6]
H	The averaged mean-field loop with saturated features, σ-modification and dither is mean-field Langevin dynamics. It converges globally and exponentially to the regularised optimum, with the log-Sobolev constant given in control quantities.	proved for ($\mathfrak{A}$); the reduction to ($\mathfrak{A}$) is formal (Remark H.0); real loop: Conjecture H.6	a bridge to [15], [64]; new in adaptive control
H.7	At a frozen measure, the exactly averaged loop is ($\mathfrak{A}$) plus an explicit $O(1/k_e)$ correction: a semi-gradient term, whose performative defect is present for every n, and an orbit-area circulation kernel ($n\ge 2$). Along the real loop, a second-order averaging term is added (Remark H.7a).	Frozen-measure part proved. Stokes identity and parity hold to machine precision. The expansion is checked through its $O(1/k_e)$ convergence. Non-reciprocity of the exact second variation is checked to 3% (asserted; 1–1.5% measured) and characterised by Theorem I. The full first-order theory predicts the orientation dependence of the learned kernels in the real closed loop to 3% in size and 11% as a vector (cosine 0.994, $k_e=40$).	no precedent found (targeted search in the third review round); ingredients known (higher-order averaging; performative prediction [71]; semi-gradient TD [88], [85])

continued on the next page

Table 1, continued

result	statement	status	novelty (audit)
I	Reciprocity at order k_e^{-2} is a pairwise property: the antisymmetric kernel $A_{\text{even}} + 2K$ is a coboundary iff it vanishes. For $n = 1$, $A_\mu \equiv 0$ iff the residual vanishes on the orbit, and the performative non-reciprocity is bounded by the residual. For $n \geq 2$ at perfect fit, $A_\mu \equiv 0$ iff the oriented orbit carries no circulation ($n = 2$: zero winding function) and the residual's Jacobian is symmetric along the orbit; $A_\mu = 2K$ when the residual vanishes near the orbit. A_{even} is the curl of the residual paired with the features plus the residual paired with the Lie bracket of the feature fields. No open set of measures is reciprocal, for any n: the averaged loop is nowhere a gradient flow at order k_e^{-2}.	proved; every formula checked against the generic second-variation code and, by the round-5 referee, against the exact second variation; the characterisations checked on perfect-fit, curl and zero-current instances	new; the analytic tools (injectivity of Gaussian convolution; Stokes with winding numbers) are classical
D′	Finite-horizon propagation of chaos for the dithered closed loop (L_τ): by synchronous coupling, $\mathbb{E}\sup_{t\leq T}\big(\|x^m - \bar{x}\| + \frac{1}{m}\sum_i \|\vartheta_i - \bar{\vartheta}_i\|\big) \leq TK_\Phi e^{\Lambda T} m^{-1/2}$, with Λ independent of m and τ, and a per-particle sup bound of the same order; the plant and the learned function converge at the rate $m^{-1/2}$, whose exponent is dimension-free, and only $W_1(\mu^m, \bar{\mu})$ carries $m^{-1/p}$; under a dominant leak (small-gain condition $\kappa_e \kappa_\vartheta > L_{b'} L_\vartheta$) the bound is uniform in time (Theorem D″), a leak-dominated regime in which the network does not learn (at $\tau = 0$ it collapses to one kernel; at $\tau > 0$ it stays in the dither cloud of the prior)	proved; checked: the coupled distance to a 4096-particle reference decays like $m^{-0.47\pm0.06}$ (24 seeds, $\tau = 0$; -0.46 ± 0.05 with dither) and like $(1/m - 1/M)^{0.94\pm0.08}$, the law of the shared-particle coupling	the technique is Sznitman's, and the dimension-free exponent for the state is its standard output; new in that the plant is in the loop, with constants in control quantities

continued on the next page

Table 1, continued

result	statement	status	novelty (audit)
D‴	Uniform-in-time propagation of chaos *in the learning regime* for the averaged slow loop ($\mathfrak{A}_m$) at $\tau = 0$ (Theorem J′'s leading order): under the synchronous coupling with the mean-field characteristic flow, the interaction term of the coupling identity is $-\|\hat{f}_\Theta - \hat{f}_{\bar\Theta}\|^2_{L^2(\rho)} + O(D^2)$ — flat convexity of the loss makes it contract whatever the gains — and only the residual-weighted feature curvature $R[\mu](\vartheta) = \langle r_\mu \nabla^2_\vartheta \tilde\Phi\rangle_\rho$ (Proposition K's R_2) can expand; if along the mean-field flow $\sigma_0 I - G^{1/2} R G^{1/2} \succeq \kappa I$ on a neighbourhood of the support from a time s_1 on (the strict second-order optimality condition of the fitted configuration, compatible with $\sigma_0 < \sigma_{\rm crit}$), then $\sup_s \mathbb{E}\frac{1}{m}\sum\|\vartheta_i - \bar\vartheta_i\|^2_{G^{-1}} \le 4e^{2\hat\Lambda_1}(L^G_\vartheta)^2 K^2_\Phi/(\kappa^2 m)$ plus a tail exponentially small in m (uniformly in time under exponential convergence of the flow, over horizons e^{cm} without it), no particle changes basin, and the learned function converges at $m^{-1/2}$ uniformly in time while the network learns; the window $\kappa_R[\mu_\infty] < \sigma_0 < \sigma_{\rm crit}$ is where learning and uniformity coexist; at finite m the stationary coupled distance follows the weight-sensitivity law $\mathbb{E}D_\infty = (1/m - 1/M)\sum_k \bar w_k \operatorname{tr}(\tilde S_k \tilde C \tilde S_k^\top)$, $S = \partial\Theta^*/\partial w$ the sensitivity of the stationary atoms to the cluster weights (Proposition D‴.5, the sharp form of the theorem's constant); the leak's prior is a design input of the window: with a measure-valued prior (nearest-atom leak) placed on the plant's structure the stationary residual vanishes and the margin is maximal, $\kappa = \sigma_0$, while the floors at parameter and function level rise (1.6–2.2× on the instance, with the amplitudes), and the sign of the residual at a cluster sets its admissible weights — overshoot splits, undershoot stabilises, so a shrunk prior enlarges the admissible set (Proposition D‴.9, Remark D‴.11);	proved for the averaged slow loop (synchronous coupling, a bootstrap on the mean square and the per-particle distance, McDiarmid for the reference's fluctuation); the hypothesis is a computable property of the mean-field flow; it fails for a connected cold start whose mass splits between basins (the support threads the ridges), where the coupling carries a floor; the real loop at finite k_e: settled on a warm start by Corollary D‴.7 through Theorem J (both loops converge to their periodic orbits, whose distance is Proposition D‴.5's up to $O(1/k_e)$), open for a cold start (Remark D‴.4); dither: the entropy method of [11] (imported); the theorem's regime $m \ge m_0$ is asymptotic ($m_0 \sim 10^{23}$ on the instance) and Proposition D‴.5 is formal; numerics: on a three-bump instance at $\sigma_0 = 0.15\sigma_{\rm crit}$ the window is realised ($\kappa_R[\mu^*](N_{0.1}) = 0.0051 < \sigma_0 = 0.0071 < \sigma_{\rm crit} = 0.047$) and (H) holds along a warm-start run; the coupled distance saturates (window maxima within 15% over 900 slow units) with slope -1.28 ± 0.30 in m, and the per-seed weight-sensitivity law predicts it within 25% in 9 of 12 cases with per-cluster cosines 0.92–1.00; the exact stationary distance of the replicated weighted configuration reproduces the measured one within 2.3% in all pairs, one $m = 16$ seed with inadmissible weights sitting on a saddle; a connected cold start grows and carries a floor of pairs in different basins whose fraction 0.20, 0.08, 0.04 ($m = 16, 64, 256$) decays like $m^{-0.64\pm0.08}$; the real loop at $k_e = 25$ follows its averaged slow dynamics on the same clouds to within 0.7%;	the two ingredients — the sign of the interaction from flat convexity, and the curvature at the actual rather than the worst-case residual — are new in propagation of chaos for adaptive control; the decomposition Gram + residual×curvature is the one of the interpolation literature ([56]); the window between Proposition K's collapsed instability and the fitted configuration's stability is new;

continued on the next page

Table 1, continued

result	statement	status	novelty (audit)
D‴, cont.	the barycentric OT leak of Theorem C′ toward the same prior constrains the cluster masses to the prior's — the nearest-atom leak conserves them, and the floor is the memory of the initial cloud that this conservation law carries — so on the sizes commensurate with the masses the finite-m stationary measure is the mean-field one exactly, both floors vanish and the fit loses the weight bias, while on other sizes at most $K_0 - 1$ split particles bridge the clusters, with floors $O(1/m^2)$ at function level and $O(1/m)$ in W^2 against the nearest-atom leak's $1/m$ and $m^{-1/2}$ (Proposition D‴.12); in the real loop the collapsed-by-class subspace is invariant and carries the weighted K_0-atom loop, so with the OT leak the lift of one K_0-atom periodic orbit is a periodic orbit of every commensurate size — the same orbit measure, learned function and tracking error exactly, the real loop's finite-size floor being zero — while with the nearest-atom leak it is the lift of the orbit of the realised weights, whose distance between sizes is Corollary D‴.7's floor (Corollary D‴.14); the Floquet spectrum of a lifted orbit is the K_0-atom orbit's together with each class's ghost spectrum of multiplicity $n_k - 1$, independent of m (Proposition D‴.15); with dither the nearest-atom leak equalises the masses only by Arrhenius crossings of the Voronoi ridges (barrier ≈ 0.05 on the instance: frozen for $350/\sigma_0$ at $\tau_0 = 0.005$), while under the OT leak a crossing particle stays assigned to its own atom and returns, so the counts are balanced 88–99.6% of the time at the same crossing barrier (Remark D‴.16); every trajectory of the OT dynamics converges to a Clarke-critical point of its energy (Kurdyka–Łojasiewicz for the definable, semiconcave-plus-smooth J_m^{OT}), so the stationary statement is reached from every initial condition in the basin (Proposition D‴.17)	for the OT leak the frozen-plan equivalence and the stationary statement are proved (total unimodularity, c-cyclical monotonicity, the Hessian of D‴.5), the quantisation bounds formal, and Theorem D‴'s coupling applies only to a balanced sample (probability $\approx 0.83/m$ under an i.i.d. start, and even there formal: Step 8 is written for i.i.d. particles) — for a generic sample the shared-particle coupling keeps the transit's mismatch for ever and a coupling theorem for the OT dynamics is open; numerics on the same clouds under the two leaks: cluster weights conserved exactly against equalised by $s \approx 100$ slow units ($\ln 2/\sigma_0 = 98$), function-level and W^2 distances at 10^{-4} and 10^{-3} of the nearest-atom leak's after $4.2/\sigma_0$ and still falling on the commensurate sizes while the synchronous D sits at 0.95 of its mismatch plateau, the quantisation laws $m \cdot W^2(\mu_m, \nu) \approx 20$ (slope -0.98 ± 0.03) and $m \cdot \mathrm{fit} \approx 0.55$ (squared slope -1.81 ± 0.05) on the incommensurate ones, the reference's fit error $0.012 \to 8 \cdot 10^{-5}$; the real loop at $k_e = 25$ on the same clouds: learned functions of $m = 15$, 60, 510 within squared distances $7.5 \cdot 10^{-9}$–$2.7 \cdot 10^{-8}$ of each other under the OT leak against 10^{-4}–10^{-3} under the nearest-atom leak (the lifted orbits' distance to 1.0003 at the one pair with admissible weights; three of the six nearest-atom systems sit at saddle weights and have split), rms tracking error 0.9–$7 \cdot 10^{-6}$ against $3 \cdot 10^{-4}$–$1.9 \cdot 10^{-3}$ (on a planar plant with four kernels: 10^{-7} against 10^{-2} and $4 \cdot 10^{-6}$–$1.6 \cdot 10^{-5}$ against 2–$7 \cdot 10^{-3}$); the lifted orbit's transverse spectrum equals the ghosts' and the collective block the K_0-atom loop's to 10^{-4} relative (invariance defects $\leq 10^{-7}$)	the finite-size floor as a conservation law of the leak, broken by the plan's mass constraint so that the particle system's stationary measure is exactly the mean-field one, is new; the lift of a weighted K_0-atom orbit as the exact periodic orbit of every commensurate size, with an m-independent Floquet spectrum (the orbit-level form of Proposition K's block structure), is new in adaptive control, where the size of the approximator is usually a nuisance parameter

continued on the next page

Table 1, continued

result	statement	status	novelty (audit)
D‴.18	Theorem J uniformly in m at a replicated configuration: for weights with margins (collective block and separation blocks $\geq \lambda_0$) the real loop's orbit near $\Theta^*_m(w)$ is the lift of the weighted K_0-atom orbit, unique in the per-particle ball of radius $\rho/2$ (inside the $\|\|\|\cdot\|\|\|$-ball of radius ρ where the period map contracts) and attracting every solution started in it at the rate $(1-\kappa)\min\{\lambda_{\mathrm{coll}}, \lambda_{\mathrm{sep}}\}/k_e$, with k_0, ρ, C independent of m; in the unnormalised norm of Theorem J the basin is m-independent too but shrinks per particle like $m^{-1/2}$ and the threshold k_0 grows like $\sqrt{m}$, and for the nearest-atom leak no mean-square (W_2-type) basin is possible — one particle in another cell changes the weights (Remark D‴.19)	proved as an adaptation of Theorem J's proof in the per-particle norm (three structural facts: the plant sees the class means up to second order in the separations; each particle's drift depends on the others only through the plant; the linearisation is block diagonal with an m-independent spectrum); numerics: the Floquet exponents within 0.56–3.22% of the averaged prediction and the ghosts' to $3 \cdot 10^{-4}$; on the matching prior fitted rates 0.95–1.05 of the prediction for $m \geq 15$ and constants 1.03–1.29, on a moving orbit 0.96–1.16 and 1.04–1.23, all from radii 31–261 times the proved one; across a cell boundary the nearest-atom leak ends at other weights (two of them inadmissible) while the OT leak restores the masses q	uniform-in-m averaging for an adaptive law with a mean-field interaction: the size of the approximator drops out of the stability constants — new in adaptive control, where Lyapunov and averaging constants usually grow with the number of parameters
D‴.20	The entropic OT leak (Sinkhorn plan, regularisation ε) is a smooth gradient flow of $m\bar{L} + (\sigma_0 m/2)\mathrm{OT}_\varepsilon$ that interpolates between the OT leak ($\varepsilon \to 0$) and the σ-modification toward the barycentre of the prior ($\varepsilon \to \infty$); at a collapsed configuration the separation block is $\sigma_0(I - (2/\varepsilon)\,\mathrm{Cov}^G_k) - R^G_k$ with Cov^G_k the conditional covariance of the atoms, so where $A_k := \sigma_0 I - R^G_k \succ 0$ an over-weighted class is a saddle exactly below the critical regularisation $\varepsilon_c = 2\sigma_0 \lambda_{\max}(A_k^{-1/2}\,\mathrm{Cov}^G_k\,A_k^{-1/2})$ (implicit, along the branch $\Theta^*_\varepsilon(w)$), and a split moves its excess toward an atom its conditional plan charges, which need not reduce it; at fixed ε the masses are equalised only down to a critical excess $\delta_c(\varepsilon) \approx w_k p_c(\varepsilon)$, and at large m the floor returns as Proposition D‴.5's law at the entropic stationary point	(i)–(iii) proved, the separation block checked entrywise against the finite-difference Hessian; (iv) proved under $A_k \succ 0$ (which fails at large excesses and small ε), the conditional plan charging both under-weighted atoms (Cov^G_k of rank two, the rank-one formula 1.22–1.31 times ε_c); (v) formal; numerics: $\varepsilon_c = 5.1$, 13.5, 23.1, 65.0 for excesses 0.02–0.2; at $\varepsilon = 1$ every size equalises; at $\varepsilon = 4$ every system ends, after up to 6000 slow units, at stable weights with residual excesses up to 0.017, of the order of $\delta_c(4) = 0.012$–0.016; at $\varepsilon = 16$ and 64 the stable systems keep their sampled weights, with floors 0.23–0.91 of the nearest-atom leak's	the critical temperature of deterministic annealing is the precedent ([77]), with the same roles (particles as codevectors, atoms as data), and the $\varepsilon \to 0/\infty$ interpolation is that of the Sinkhorn divergences ([21]); new are its role as the threshold of mass equalisation in an adaptive law, where the conditional covariance of a collapsed class is an effect of its excess, and the consequence that on the commensurate sizes only the hard constraint is floor-free for every m

continued on the next page

Table 1, continued

result	statement	status	novelty (audit)
D‴.21	A measure-level basin for the OT leak: at commensurate sizes two exact identities along the plan segments — the energy, and its gradient paired with the displacement toward the stationary atom of one's own class — give a Polyak–Łojasiewicz inequality $\|\nabla J\|^2 \geq \lambda \cdot m(\mathcal{J} - \mathcal{J}^*)$ with λ set by the residual-weighted curvature along the segments and independent of m (sharp, $2\sigma_0$, at a matching prior); the excess energy decays exponentially from an m-independent sublevel set (tiny on the instance: none of its configurations has a particle in another cell below $m \approx 2 \cdot 10^5$), and a posteriori wherever the plan-segment margin is positive, which covers configurations with a fixed fraction of the particles in other cells; under a curvature hypothesis at the stationary residual the replicated configuration is the unique global minimum among the empirical measures for every m, while the nearest-atom energy has a spurious minimum at every admissible count vector (the energy-landscape form of the floor)	(i), (ii), (iv), (v), (vi) proved, (iii) proved for vertex-selection solutions (their existence assumed); the mean-field form and the transfer to the real loop formal (Remark D‴.22); numerics: the identities to $6 \cdot 10^{-13}$ along 64 runs, the margin positive from $s \leq 300$, the observed rate at least λ throughout and λ within 1.5% of the observed $2\sigma_0$ at the end (matching prior); the nearest-atom runs at admissible counts stop at the energy of the replicated configuration of their sampled weights, at inadmissible ones (5 of 8 per prior at $m \leq 60$) they split below it	the Polyak–Łojasiewicz inequality ([74]), the generalised geodesics of AGS [2, §9.2] and PL inequalities through optimal maps for functionals that are not geodesically convex ([7]) are known; new is the PL inequality along the generalised geodesic based at the prior, with the loss linearised at the stationary residual and the counts making the constant independent of m
D‴.23	Adaptive priors: the OT leak toward a prior whose atoms move toward the class barycentres at a rate $\eta > 0$ is the gradient flow of $\bar{L} + (\sigma_0/2)W^2(\mu_\Theta, \nu_Z)$ in a weighted metric; its replicated stationary points carry no leak bias (the critical points of the unregularised K_0-atom loss), its non-replicated ones the K_0-cluster bias, and boundedness becomes a hypothesis (the leak no longer confines the parameters); at an exact representation it linearises, for every m, into the separations (rate σ_0) and 2×2 collective pairs whose slower rates satisfy $\eta h_j/(h_j + \sigma_0 + \eta) < \lambda_j^- < \min\{h_j, \eta\}$ ($h_j, \eta > 0$), h_j the measure-space excitation spectrum — a learned prior removes the fixed prior's bias $\sigma_0(H_L + \sigma_0 I)^{-1}x^0$ mode by mode, at the rate of the excitation; as $\eta \to \infty$ a capacity-constrained K_0-means regulariser; the transit drags the prior, so the adaptation should start from the fixed-prior stationary point	(i)–(iii) proved ((i) for bounded solutions), (iv) formal, (v) sketched (the radius stated, its constants not derived), (vi) the monotonicity of $\mathcal{J}$ proved and the drag numerical; numerics: the K_0-atom modes decay at λ_j^-; from separated starts the m-particle systems depart from the K_0-atom run at second order in the separations, consistent with the same expected size at $m = 15$ and 510 (five draws); started from the fixed-prior stationary point every size $m = 15$–510 follows the K_0-atom run (per-seed ratios 0.995–1.000 at $\eta = \sigma_0$, 0.968–1.001 at $\eta = \infty$), with fit errors $1.6 \cdot 10^{-4}$ and 3.4–$3.5 \cdot 10^{-5}$ against the fixed prior's $9.78 \cdot 10^{-3}$	proximal point / iterated Tikhonov regularisation (its per-mode filter factors), elastic averaging (its quadratic analysis) and soft weight-sharing are the precedents ([90]); new are their transfer to the mass-constrained particle system uniformly in m (H_L as the measure-space excitation spectrum, the separations decoupled at σ_0), and the transit's drag with its gate; the limits (the lost a-priori bound, the K_0-cluster bias) are the precedents' own, specific here in their particle form and in the run-off from an exact fit

continued on the next page

Table 1, continued

result	statement	status	novelty (audit)
D‴.24	The finite-size floor is the leak's bias at the sampled weights: under the nearest-atom leak toward a fixed matching prior the function-level weight-sensitivity is $F = \hat{J}\sigma_0(H_L + \sigma_0 I)^{-1}\partial_w x^0$, the weight-derivative of the bias, and the floor law decomposes on the excitation spectrum with the band-pass weight $h_j\sigma_0^2/(h_j+\sigma_0)^2 \le \sigma_0/4$, centred at $h = \sigma_0$ — at the samples where the fixed prior's replicated point is stable (D‴.5-admissible); the nearest-atom leak toward an adaptive prior is the OT leak with the masses minimised out (the cell masses), and its stationary replicated configurations are the critical points of the unregularised weighted loss, the exact representation of the sampled weights among them — no function-level floor at any representable sample, every separation relaxing at σ_0 (no splitting), perfect tracking an invariant set of the real loop — while the coupled distance of the amplitude–mass gauge grows (the sample's weight fluctuation moved from the bias into the parameters)	(i) F exact at q, the bias formula first order, the law (B) formal (the delta method, as D‴.9), (ii) proved inside the cells and across crossings for vertex-selection solutions, (iii) proved for the stationary points, (iv) the boundary case formal, (vi) the invariance proved; numerics: at the stable samples the exact bias at the sampled weights reproduces the observed floor (ratio 1.0000), while 5 of the 12 samples are fixed-prior saddles and split; the law is asymptotic (the exact floor's expectation 2.8, 1.5, 1.1 times it at $m = 15$, 60, 240); under the adaptive prior the function-level distance decreases without a floor (at $s = 6000$ at most $2.0 \cdot 10^{-6}$ at $\eta = \sigma_0$ over the representable samples) and the coupled distance is 1.7–4.8 times the fixed prior's, while W^2 is the cross-class transport of order $\|\delta w\|_1$ under both priors; the samples outside W have probability 0.36, 0.074, 0.0021 at $m = 15$, 60, 240; in the real loop the tracking error follows the averaged prediction (ratios 0.987–1.001)	the $1/m$ fluctuations of mean-field networks and their vanishing at an interpolating limit ([89], [91]), the delta method and the filter-factor bound of ridge regression are classical; new are the floor as the bias's weight-slope, mode by mode, a leak toward a learned prior as the way to restore the interpolation property under regularisation, and the design table of Remark D‴.25
D‴.26	Perfect tracking with an adaptive prior in the real loop: the lift of the exact representation of representable sampled weights, with the prior at its atoms and the plant on the reference, is a T_0-periodic solution whose Floquet exponents are σ_0/k_e exactly on the $(m - K_0)p$ separations — every separation stable, also where the fixed prior's replicated orbit is a saddle — and the weighted K_0-atom loop's on the collapsed subspace for every m; where $\partial_x \hat{f} = \partial_x f$ on the orbit (an exact representation) these are $(1/k_e)[\lambda_j^{\pm}(w) + c_j^{\pm}(w)/k_e^2 + O(k_e^{-4})]$ — in slow time the odd orders vanish (the variational equation's symmetry with the slow block at $-\varepsilon$, the time reversal of the reversed reference's: the slow generator's spectrum is even; the first-order correction is skew), and c_j, explicit, contains the error filter's attenuation (≤ 0), the slaving's part (≥ 0) and averaging terms, its sign not fixed; so perfect tracking is exponentially stable (locally, for each m) at the rate $(1/k_e)\min\{\sigma_0, \lambda_1^-(w) + c_1^-(w)/k_e^2 + \dots\}$	(a), (b) proved, exactly; (c) proved for $x_m \in C^4$ at simple eigenvalues, the stability where $H_L(w) \succ 0$ (Lemma D‴.26a: the slaving exact, the $O(k_e^{-4})$ remainder bounded, the parity an identity), c_j explicit; the basin uniform in m not proved; numerics: at $(6, 5, 4)/15$, $m = 15$ and 30, $k_e = 25$ and 50, $\eta = \sigma_0$ and ∞: the separations at σ_0/k_e to $\le 1.7 \cdot 10^{-6}$, the collective exponents the K_0-atom loop's to $\le 6.4 \cdot 10^{-4}$; by the variational equation to $k_e = 400$ on two references (one with a non-zero first-order correction): the first-order deviations falling by $\to 4$ per doubling (every exponent below its prediction on the instance; 13.6% in the slowest mode at $k_e = 25$), the second-order residuals by $\to 16$, $c_1/\lambda_1 = -100.6$ of which the filter's $-102.1 = -0.989\omega_1^2$; with the gain scaled by 30 two $c_j > 0$, those exponents above their prediction by $c_j/(\lambda_j k_e^2)$	Floquet theory, averaging and the Magnus expansion's commutator term, the averaging analysis of adaptive loops (the PE level), and leaks compatible with exact tracking (*e*-modification; a centre that follows the parameters) are classical, and (a)–(b) are Corollary D‴.14(ii) and Proposition D‴.15 at zero residual; new are the zero-residual lift at every representable sample with its exactly m-independent spectrum, and the explicit second-order coefficient of this loop (the even expansion is the classical consequence of time symmetry, [99])

continued on the next page

Table 1, continued

result	statement	status	novelty (audit)
D‴. 26a	The exact slow subspace of the perfect-tracking linearisation: Chang's periodic decoupling P (a contraction) makes the slaving of D‴.26 an exact block-triangularisation, so the slow Floquet multipliers are exactly the slow block's; its generator $\log\Phi(\varepsilon)/(\varepsilon T_0)$ has an expansion to $O(\varepsilon^4)$ with a bounded remainder (the Magnus series convergent for $\varepsilon T_0\|A\| < \pi$); the slow block at $-\varepsilon$, defined by the same contraction, is the time reversal of the reversed reference's at ε, so with the variational equation's symmetry the slow generator's whole spectrum is even, $\operatorname{eig}\bar{A}(-\varepsilon) = \operatorname{eig}\bar{A}(\varepsilon)$, and the rates are exactly even; and Chang's second transformation decouples the slow coordinates' drift from the fast ones exactly, their noise $\varepsilon b\nu$, $b = -(\dot{K}+aK)^\top - \varepsilon(\ddot{K}+KK^\top K + K\Lambda^\top + a\dot{K})^\top + O(\varepsilon^2)$, shared with the fast block — at leading order the high-passed $-\varepsilon(\dot{K}+aK)^\top\nu$ —, which gives D‴.30(ii)'s floor as a period average	proved (the constants not explicit; ε_0 also set by the spectral gap and $\bar{A}_0$'s margin): D‴.26(c) with a bounded $O(k_e^{-4})$ remainder, D‴.30(ii) for the linearised loop under white noise as a period average with the relative remainder $O(1/k_e)$ (at $\eta = 0$ with $K = \hat{J}$, $\Lambda = \sigma_0 I$); numerics ($\eta = 0$, σ_0, ∞; $k_e = 12.5$–1600): the invariance residual $\le 3.8\cdot10^{-15}$, $P - P_2 = O(\varepsilon^3)$ and $P - P_{(3)} = O(\varepsilon^4)$, the slow block's rates the full equation's to $1.0\cdot10^{-8}$, the rates at $\pm\varepsilon$ equal to $5.9\cdot10^{-12}$ of the largest ($1.1\cdot10^{-9}$ per mode) while the slowest mode's first-order deviation reaches 29.6%, a mismatch breaking it at $O(1/k_e)$; $b - b_0 - \varepsilon b_1 = O(\varepsilon^2)$; the weakest mode's covariance deviation the slow noise's attenuation 2.4–2.5 times the rate's correction, of opposite sign; the floor's period average at the formula to $3\cdot10^{-5}$ at $k_e = 1600$, pointwise ±1% (the jitter)	Chang's decoupling — with its time-varying and stochastic versions (Kokotović–Khalil–O'Reilly Ch. 4–5 and App. A, Haddad–Kokotović, Yang–Zhu; Berglund–Gentz's slow manifolds under noise) —, the Magnus series' convergence and the reversal argument (the adjoint relation of symmetric methods) are classical ([102], [99]); new is their application here: the exact slaving, the parity as an identity at negative ε, the slow noise identified exactly
D‴.27	A projected adaptive prior restores the leak's a-priori bound: with the prior's atoms started in and confined to a compact convex set $\mathcal{Z}$ (the projected flow; at $\eta = \infty$ the projected class barycentres), every particle's distance from $\mathcal{Z}$ is at most $B/\sigma_0 + (d(0) - B/\sigma_0)_+ e^{-\sigma_0 s}$, B the worst-case learning force, for every m, η and target; $\mathcal{J}$ is non-increasing (at $\eta = \infty$ in its selected form, along a Lloyd-consistent plan; the reduced energy need not decrease); inside $\mathcal{Z}$ nothing changes (Theorem D‴.23's bias removal and linearisation), and where the limit has an atom on the boundary — necessarily when no stationary point has its prior inside — a bounded bias remains; at $\eta = \infty$ on collapsed classes the leak is the gradient of the exterior penalty $(\sigma_0/2)\operatorname{dist}^2(\vartheta, \mathcal{Z})$, a switching σ-modification toward $\mathcal{Z}$	(i), (iii) proved ((iii)'s local convergence as Theorem D‴.23(v), sketched); (ii) the monotonicity proved, the convergence argued at $\eta = \infty$ and formal at finite η; (iv) the identification at $\eta = \infty$ proved, the boundary atoms and the bias outside numerical; numerics: beyond the saturation both η converge to the same bounded point (fit error 0.1415; the projected midpoint rule consistent, the trajectory error $6.3\cdot10^{-7}$–$3.8\cdot10^{-8}$ at $\mathrm{d}t = 0.25$–0.0625), particles within 0.92 of $\mathcal{Z}$ (bound ≤ 3686); the box's amplitude bound 0.75–0.99 of $a_{\max}$ trading the fit error 0.1676–0.1101 between the fixed prior's 0.1750 and the unprojected limit 0.1050, tighter boxes (which exclude the exact atoms) worse than the fixed prior; at $\eta = \infty$ the reduced energy rising by 26% along a Lloyd-consistent plan while the selected energy falls; on the four-bump target the box stops the width's run-off at its face; at the misplaced prior the box never approached	parameter projection, the exterior quadratic penalty and the switching σ-modification (which keeps the ideal properties inside its set) are classical, and at $\eta = \infty$ on collapsed classes the construction is the latter; new is its transfer to the learned centre of a mass-constrained K_0-atom prior at finite η, compatible with Theorem D‴.23's bias removal inside $\mathcal{Z}$

continued on the next page

Table 1, continued

result	statement	status	novelty (audit)
D‴.28	The adaptive prior's floor removal is a gauge property: the masses enter the network only through $w_k a(\alpha_k)$, so for any nondegenerate critical point $\vartheta^b(q)$ of the unregularised loss — exact or not — its gauge transform $G_w\vartheta^b(q)$ is a critical point at every sampled weight vector of W_g with the same function: at a strict local minimum no function-level floor at the samples of W_g with positive margins, whatever the approximation error; the fixed matching prior's floor is $\sigma_0\hat{J}(\tilde{A}+\sigma_0 I)^{-1}\tilde{g}$ with the full Hessian $\tilde{A}=H_L-\tilde{R}$; for a scalar amplitude per kernel the residual curvature has a zero amplitude row at a critical point and scales exactly by q_k/w_k under the gauge, so the adaptive prior's splitting margins are $\sigma_0-(q_k/w_k)\lambda_k^+$	(i), (iii) proved exactly, about the stationary points; (ii) as D‴.24(i); the dynamics' convergence, and with it the floor's vanishing, numerical; numerics: with a fourth bump that three kernels cannot represent (fit errors 0.4–2.5%), the critical point continued in w from $\vartheta^b(q)$ is the gauge transform to $2\cdot10^{-11}$ with $\hat{f}$ unchanged to $6\cdot10^{-13}$, the adaptive prior's sensitivity 10^{-11}–$3\cdot10^{-10}$ of the fixed prior's, the fixed prior's equal to the full-Hessian formula to $2\cdot10^{-9}$ (the Gauss–Newton form off by $7\cdot10^{-4}$–$5.6\cdot10^{-3}$), the margin formula exact to 10^{-14} and the dynamics' separation eigenvalues to $6\cdot10^{-12}$; $P(w^m\in W_g)=0.52$ at $m=15$; the dynamics: under the fixed prior the floor stays and the fit error above the approximant's; under the adaptive priors the floor falls by orders of magnitude and the fit error tends to the approximant's at the samples inside W_g; outside it the systems approach the box minimum on the branch (1.11–1.59 times the approximant's fit error)	the homogeneity of the conic view ([16], [23]) and the rescaling symmetries of networks ([98]) are classical; new are the gauge as the mechanism of the floor removal, robust to approximation error, and the exact margin formula for a scalar amplitude
D‴.29	Second-order plants: with the sliding variable $s=\dot{e}+\lambda e$ and $u=\ddot{x}_m-\lambda\dot{e}-k_s s-\hat{f}_\Theta(z)$, the error channel is $\dot{s}=-k_s s+r_\Theta(z)$, $\dot{e}=-\lambda e+s$ — Theorem J's with s in place of e — so as $k_s\to\infty$ the parameters follow the averaged slow loop along the regressor's orbit $z_m(t)$, every averaged-loop statement of Section 18 transfers with the occupation measure of z_m, and the position error is the residual filtered by $(\mathrm{d}/\mathrm{d}t+\lambda)^{-1}(\mathrm{d}/\mathrm{d}t+k_s)^{-1}$; kernels on the coordinate f depends on see the orbit as an interval, while isotropic phase-plane kernels on a circular orbit ($\omega_0=1$ in isotropic units) are over-parametrised (von Mises restrictions: in general position one zero Gauss–Newton eigenvalue per kernel; on other orbits weak excitation)	(i), (iv) exact; (ii), (iii) formal (first-order averaging with the fast pair (s,e); Theorem J's steps carry over for the smooth fixed-prior leak once the error-channel steps are redone; switching targets beyond the first-order theory); numerics: a position servo with a cogging torque whose four-kernel local minimum $\vartheta^b(q)$ (fit error 1.46%; the loss's infimum lies at the saturation, $\le 0.64\%$) is Proposition D‴.28's regime: from clouds around $\vartheta^b(q)$ the fixed prior's floor $1.85\cdot10^{-3}$–$4.49\cdot10^{-5}$ ($m=16$–128), the adaptive prior's 38–780 times lower, the OT leak at the approximant, σ-modification toward 0 biased (0.328), and no leak as good (0.0180–0.0146); from an engineer's model (fit error 0.200) the fixed designs stay biased (0.048–0.10) and the adaptive prior removes most of the bias, slowly (0.022–0.027 at $s=2000$; the K_0-atom flow 0.0155 at $s=10^5$); the real servo's position error at the exactly filtered prediction to 1.0001–1.0029 (the slaved one 0.931–0.996), no design drifting; noise-free, so the leaks' robustness is not exercised	the filtered error and Gaussian networks in adaptive control are classical (Sanner–Slotine), and so is the partial excitation of Gaussian networks along an orbit; new are the transfer of the transport leaks, their floors and the adaptive and projected priors to second-order plants, and the exact fibre of isotropic phase-plane kernels on a circular orbit

continued on the next page

Table 1, continued

result	statement	status	novelty (audit)
D‴.30	Measurement noise at perfect tracking: the noise is coherent — a replicated start stays replicated under every noise path, so the floor does not average out with m (invariance, not attraction: near the lift the separations grow above a noise level, which drifts along the trajectory with the developing errors-in-variables deviation); to first order the loop high-passes the measured error's noise, $((\mathrm{d}/\mathrm{d}t - a)/(\mathrm{d}/\mathrm{d}t + k_e))\nu$, and the function-level floor is $(N_0/2k_e)[\sum_j Q_{jj}(h_j+\eta)/(h_j+\eta+\sigma_0) + \operatorname{tr} H^2]$, $Q = \langle(\dot{\hat{J}} + a\hat{J})^\top(\dot{\hat{J}} + a\hat{J})\rangle$ — the prior's mobility enters the Tikhonov filter factor as $h_j + \eta$, and the adaptive prior's floor is $(N_0/2k_e)\operatorname{tr} Q$ whatever the spectrum; to second order the noise inside the regressor exerts a mean force (the errors in variables) whose stationary response is $(H + \sigma_0 I)^{-1}$ under the fixed prior and H^{-1} under the adaptive prior, for every $\eta > 0$	(i) exact as invariance, its threshold formal at the lift, the long-time threshold open; (ii) proved for the linearised loop under white noise as a period average (Lemma D‴.26a), (iii) formal, their closed forms exact for the averaged system; numerics: the floor per unit N_0 0.1597 (fixed) to 0.1757 ($\eta = \infty$), the weakest mode's variance $\times 24.1$ ($\times 28.0$ in the exact linearised loop at $k_e = 25$); the exact periodic covariance of the linearised loop at the prediction to 0.976–1.00001 ($k_e = 25$–800: the relative error $O(1/k_e)$, its first-order term from the slope a, the slope-free control numerically even in $1/k_e$); the errors-in-variables bias 491 (fixed) and 2397 (adaptive) per $(\mathbb{E}\nu^2)^2$ averaged, 504 and 2653 in the exact linearised loop, the parameters' fluctuation negligible; the real loop (12 common noise paths, two noise levels) within 1.75 standard errors of the prediction, the mean part 89% at std 0.01; the lift $m = 15$ identical to the K_0-atom run to rounding; the separations' threshold std 0.031 at the lift; on antithetic pairs the noise-induced separation rate $\lambda_{\mathrm{eff}} = 0.327$, 0.318, 0.294 at std 0.01, 0.03, 0.05 against the lift's 0.298 and the transient's 0.326–0.328, the small-noise limit 0.329 ± 0.003, the rate vanishing at std 0.0299 over 300 time units	measurement noise, the errors-in-variables bias, the diffusion approximation, the LMS misadjustment, leaky LMS, common noise and the lack-of-excitation drift are classical; new are the floor's interpolation in the prior's mobility ($h_j + \eta$ in the filter factor), the driving matrix Q of the sensitivity-filtered measurement, and their combination with the coherence of the K_0-atom identification and its noise threshold at the lift

continued on the next page

Table 1, continued

result	statement	status	novelty (audit)
D‴.31	The adaptive prior is a washout leak: the prior lags the class barycentre by its exponentially forgotten increments, so the leak acts in full on the separations and, for integral plans, on the barycentre through the high-pass $i\omega/(i\omega+\eta)$ — mode j's stiffness $h_j+\sigma_0 i\omega/(i\omega+\eta)$ — which is its bias removal and its lost robustness to persistent forces on the barycentres; along every affine time-invariant prior the stationary response R and the bias B toward a misplaced anchor satisfy $RH+B=I$ (per mode $h_j r_{jj}+b_{jj}=1$), so no prior gives both; a force's function-level response is at most the norm of its smallest equivalent target perturbation, under every leak with $S+S^\top\succeq 0$; and the desired regressor (the adaptive law's regressor at the reference) keeps the lift, the linearisation and the first-order noise response while turning the noise's own persistent force into the least-squares force of $g=k_e\mathbb{E}[e]$, its response at most $\frac{1}{4}\|\partial_x^2 f\|_\rho^2(\mathbb{E}\nu^2)^2$ uniformly in k_e, B and the spectrum — the parameters' fluctuation adding a force of the same order that is not the learning force of a mean tracking error, so no bound in the noise's moments applies	(i)–(iii) proved; (iv)(a)–(b) proved at the lift and for the linearised equations, (iv)(c) formal, its noise part's consequences exact for the averaged system, the parameters' part numerical; numerics: the adaptive prior's bias 2397, 0.471, 0.0526 per $(\mathbb{E}\nu^2)^2$ (noise part; MR, DR, DR*) at $k_e=25$, with the parameters' fluctuation 2398, 0.486, 0.0481 (DR +56% at $k_e=B=100$, where its noise part nearly cancels); MR over DR $3.0\cdot10^3$–$2.1\cdot10^8$ over $k_e=25$–200; R and B from the priors' dynamic blocks, $RH+B=I$ to $1.4\cdot10^{-14}$, also for non-commuting leaks, where $HR+B-I$ reaches 0.47–0.79; the real loop on common noise paths: $\mathbb{E}\|\hat{f}-f\|^2$ 8.7 times lower under DR (predicted 9.2), the MR − DR mean difference at its prediction to 0.5–5.3% (0.4–1.8 sampling errors)	washout filters, low-frequency learning, the desired regressor, the Tikhonov / Kalman split and the minimum-norm bound are classical; new are their transfer to the transport leaks (the washout reading of the adaptive prior, uniform in m) and the quantification; the parameters' own errors in variables is the classical second-order bias of a constant-gain stochastic approximation ([100]: Benveniste–Métivier–Priouret, Dieuleveut–Durmus–Bach, Box), new only the observation that the desired regressor does not remove it
J	At finite m, without dither, in the scaling $\sigma=\sigma_0/k_e$: every hyperbolic critical point of the m-particle regularised loss J_m has a unique nearby T_0-periodic orbit of the real closed loop, at distance $O(1/k_e)$, with the same stability type; the parameter Floquet exponents are $\frac{1}{k_e}\lambda(G_m^{1/2}\nabla^2 J_m G_m^{1/2})+O(k_e^{-2})$; the basin of a minimum is uniform in k_e; the period mean is $\Theta^*-J^{-1}(b_{\text{frozen}}+b_{\text{osc}})+O(k_e^{-2})$	proved (tube and slaving, periodic Riccati block-triangularisation, second-order averaging, Poincaré map); checked by Newton shooting for the exact orbit, its monodromy matrix and its period mean against the averaged Hessian and the first-order prediction, at $k_e=10$–80	the first step of Section 26, branch 1; the tools (two-time-scale averaging under attraction) are classical [48, Ch. 10–11], [78, Ch. 5], [81, Ch. 4]

continued on the next page

Table 1, continued

result	statement	status	novelty (audit)
K	At a collapsed critical point (all kernels equal) the Hessian of J_m splits by permutation symmetry into a separation block $\sigma_0 G^{-1} - R_2$, with R_2 the Hessian of the first-variation potential at the atom (independent of m), and a collective block; the collapsed configuration is a strict local minimum iff the atom minimises the single-kernel objective and $\sigma_0 > \sigma_{\mathrm{crit}} := \lambda_{\max}(G^{1/2} R_2 G^{1/2})$. The orientation-odd k_e^{-2} terms of the real loop (U_{odd}, b_{osc} and the third derivative of J_m against the first-order offset) shift the separation Floquet exponents of the collapsed orbit by $O(k_e^{-2})$, identically for the $(m-1)$-fold multiplet, and in a window of width $O(1/k_e)$ around σ_{crit} the direction in which the reference is traversed decides whether the kernels separate	proved ((iii)–(iv) given the second-order normal form, Section 23); checked by shooting on instance A: the orientation difference of the softest separation exponent scales like $k_e^{-2.18}$ and matches the prediction from the averaged data to 3% at $k_e = 80$; at a leak gain computed from that prediction, one traversal leaves the collapsed configuration stable and the other makes it unstable ($k_e = 10$ and 20)	new as a design statement; the splitting is the isotypic decomposition of the permutation action, and the stability criterion is the local Chizat–Bach condition at a Dirac mass
J′	At finite m with dither $\tau = \tau_0/k_e$ and leak $\sigma = \sigma_0/k_e$: the closed loop has an invariant law (unique under a smoothness hypothesis) whose parameter marginal is the Gibbs measure $\exp(-J_m/\tau_0)$ up to $O(1/k_e)$, with the explicit first-order correction $\varepsilon \int \varphi g \, \mathrm{d}\pi$ driven by the first-order drift b of Theorem J as a vector field on parameter space (g solves $\tau_0 \nabla \cdot (\pi G \nabla g) = \nabla \cdot (\pi b)$); the stationary tracking error is $(2/k_e^2)\pi(\bar{L}_m) + O(k_e^{-3})$; the stationary parameter process is not reversible, with a circulation $\varepsilon[\int \Theta \wedge j_1 + \pi\langle u \wedge \dot{u}\rangle] + O(\varepsilon^2)$, where $j_1 = \pi(b - \tau_0 G \nabla g)$ is the first-order stationary current, nonzero exactly when b is not a G-gradient (Proposition J′.2: the finite-m form of Theorem I's reciprocity defect plus a Hessian commutator from second-order averaging; never a gradient for $m \geq 3$)	proved (the Poisson-equation expansion of the invariant measure with the fast error and phase eliminated; three standard theorems imported and named); checked on a single-kernel instance: the first-order prediction removes the $O(1/k_e)$ discrepancy between the real loop's stationary means and the Gibbs measure, the orientation-odd part of the means (the dither version of Remark H.7a) matches its prediction, and the current machinery is checked on the first-order averaged SDE	the technique is the Pavliotis–Stuart [70] / Mattingly–Stuart–Tretyakov [60] expansion; new in its object (a hypoelliptic learning loop with a plant) and in the identification of the first-order non-equilibrium current with the reciprocity defect of Theorem I
H.8	$(\mathfrak{A})$ is input-to-state stable under drift perturbations, with gain $1/(\tau\rho_\tau)$	proved, assuming the chain rule for the perturbed flow	elementary given [15]

1.6 Where the new mathematics lies

The new mathematics this viewpoint requires lies in the open problems of Section 26.

1. Infinite-horizon averaging for McKean–Vlasov closed loops. Propositions H.7 and H.8 reduce it to controlling a near-identity averaging transformation along the moving measure. Theorem J settles the deterministic finite-m case.
2. Uniform-in-time propagation of chaos with a plant in the loop.
3. The theory of the non-reciprocal, semi-gradient mean-field dynamics that the closed loop actually produces. Theorem I characterises its non-reciprocity (completely for $n = 1$ and at

perfect fit; by an explicit bilinear form otherwise, and nowhere reciprocal on an open set); its long-time behaviour is open. Proposition K turns one of its effects into a design rule: near the critical leak, the orientation of the reference orbit decides whether a collapsed network separates.

4. Global convergence without dither.

1.7 Organisation

Section 2 relates this document to version 1: what version 1 claimed, the audit's verdict, the errata, what survives and what is withdrawn, and how Part II answers the structural defects of Section 1.2. Part I treats the projection-based controller: the setting and standing assumptions (Section 3), the algorithm (Section 4), the lemmas and the uniform-ultimate-boundedness theorem (Section 5), the "no bias" claim stated precisely (Section 6), prior art and the honest novelty statement (Section 7), and the experiments (Section 8). Part II develops transport adaptive control in Sections 9–23: the setting (Section 9), Theorems A–F together with Theorem C′ and Theorem D′ (Sections 10–17), Proposition G (Section 19), Theorem H (Section 20), the finite-gain structure of Propositions H.7 and H.8 and the reciprocity criterion of Theorem I (Section 21), Theorem J (Section 22) and Proposition K (Section 23). Part III collects the numerical evidence (Section 25) and the research program (Section 26). Appendix A is the verification map.

2 Relation to version 1

Version 1 of this document, *Mathematical Foundations of Self-Adaptive Kernels Achieving Lyapunov Convergence* (14 pages), presented the controller of Part I. An audit (`ASSESSMENT.md`) re-examined its theory, its code and its experiments; every numerical statement of the audit was regenerated by running the code, and the scripts and raw files are named next to each claim. A revised theory (`THEORY_REVISED.md`, Part I of this document) replaced its stability argument. This section records what version 1 claimed, the audit's verdict, the errata, what survives and what is withdrawn, and how Part II answers the structural defects of Section 1.2.

2.1 What version 1 claimed

Version 1 was subtitled *Uniform Ultimate Boundedness via Projection-Based Adaptive RBF Control Without Persistent Bias of Leaky Modification*. It treated the plant $\dot{x} = f(x,t) + u$ on a compact domain with the certainty-equivalence control $u = \dot{x}_m - k_e e - \hat{f}(x,t)$ and the network $\hat{f} = W^\top \phi(x, c, \sigma)$, adapted weights, centres and widths by projected gradient laws, and used the Lyapunov candidate $V = \frac{1}{2} e^\top e + \frac{1}{2}\operatorname{tr}(\tilde{W}^\top \Gamma_W^{-1} \tilde{W}) + \frac{1}{2}\tilde{c}^\top \Gamma_c^{-1} \tilde{c} + \frac{1}{2}\tilde{\sigma}^\top \Gamma_\sigma^{-1} \tilde{\sigma}$. Its claims were the following.

- *Central contribution* (abstract): "a rigorous proof that a fully shape-shifting kernel architecture — with online adaptation of weights, centers, and bandwidths — achieves Uniform Ultimate Boundedness (UUB) of the tracking error without the persistent steady-state bias introduced by classical leaky modification schemes (σ-modification, e-modification)".
- *Key mechanism* (abstract, §1.3): "the strict decoupling of offline Tikhonov regularization, which guarantees the well-posedness of the ideal parameter target, from the online smooth projection operator, which provides hard parameter boundedness", with an implicit-function-theorem (IFT) argument giving finite velocity bounds V_W, V_c, V_σ for the ideal parameters.
- *A "critical and non-trivial" result* (abstract, §7.2): the projection's hard cap on parameter energy "converts the standard boundary-only Lyapunov descent argument into a global

interior argument", proving convergence to the infimal invariant set without LaSalle's invariance principle or Barbalat's lemma, with the ultimate bound

$$\limsup_{t\to\infty}\|e(t)\| \le \sqrt{R_{\mathrm{UUB}}^2 + 2V_{\max,\Theta}}.$$

- Centre separation $\|c_i - c_j\| \ge \delta > 0$, "maintained via the projection set construction" (§1.2); and a corollary that a standard persistent-excitation condition on the regressor is insufficient to guarantee parameter convergence, so that alternative mechanisms, such as integral concurrent learning (ICL), are required (§8).
- §9 summarised five architectural choices and the failure mode each was said to close: $\sigma_{\min} > 0$ (Hessian blow-up near $\sigma \to 0$); offline Tikhonov regularisation (circular self-consistency of the projection geometry); a smooth convex projection operator (a cancellation gap when the projection is active); the projection's hard energy cap (the LaSalle and Barbalat requirements); and stability from the Lyapunov function and the projection alone (the circularity of ICL, which needs bounded states).
- §10 stated that "the theoretical core of this manuscript is mathematically complete and internally consistent", and that the remaining items were "completeness additions and empirical validation requirements . . . , not theoretical repairs": a constructive feasibility theorem; empirical validation (Lorenz-driven tracking against σ- and e-modification "to prove bias elimination", a Monte Carlo sweep "verifying R_{UUB} bound is not violated", an ablation of fixed against adaptive centres); and a novelty framing in which the architecture "eliminates this tradeoff entirely", the projection introduces "zero bias into the Lyapunov derivative", and the result is "not a marginal improvement but a structural result enabled by the strict decoupling of offline regularization from online projection".

2.2 The audit's verdict

The theory is a correct but standard composition of known results (Taylor-linearised inner-layer adaptation, boundary-layer projection, level-set UUB), presented with language that overstates its novelty and its conclusion. Two of its claims are wrong as written (non-convex "projection" for centre separation; an IFT argument that is not a proof) and one was not implemented (the Γ-weighted projection). The experiments contained a hand-chosen "theoretical" threshold, a confounded headline comparison against baselines that failed representationally, GP baselines handicapped by stale predictions and a mis-placed inducing grid, invalid significance tests on autocorrelated samples, and the public figure omitted the one benchmark ($n = 3$) where the method is weak and the one baseline (NTK) that beats it. After revision, the implemented algorithm is exactly the proved one and is machine-checked; the honest experiments show a real, consistent advantage of projection over σ- and e-modification on identical geometry (5/5 seeds in every cell) and a modest but consistent gain from adaptive geometry, while the a-priori bound is shown to be 915× to 10 954× loose.

Novelty. Table 2 compares the ingredients of version 1 with prior work. Net: no new theorem. What can honestly be claimed after revision (Section 7) is a machine-checked theorem-to-code map, an explicit Hessian-free disturbance bound, the RMS form of the bound, and a controlled mechanism-versus-geometry experiment.

Integrity findings. Table 3 lists the integrity findings of the audit, each verified by running code, with the evidence and the fix. File names refer to the accompanying repository.

Table 2: Novelty of the ingredients of version 1 (audit).

ingredient	prior source	status of the claim of version 1
Adaptive RBF control with projection-type robustification	Sanner & Slotine 1992 [79]; Ioannou & Sun 1996 [47]	known
Adapting centres/widths by first-order Taylor expansion with bounded remainder	Lewis, Yesildirek & Liu 1996 [54]; Polycarpou 1996 [75]; Fabri & Kadirkamanathan 1996 [28]; Ge & Wang 2002 [33]	known; version 1 cites none of these
Smooth projection with ε boundary layer and inequality (P)	Pomet & Praly 1992 [76]; Lavretsky & Gibson 2011 [52]; Lavretsky & Wise 2013 [53]	known; the showcase README called it "the core theoretical contribution"
Projection introduces no bias in $\dot{V}$ when $\theta^* \in \Omega$	Ioannou & Sun 1996 [47, §4.4]	known
Level-set ("global interior") UUB argument	Khalil [48, Thm. 4.18]; Lewis et al. 1999 [55]	known; called "critical and non-trivial"
Simultaneous adaptation of W, c, σ for Gaussian RBFs with all three projected	—	the only combinational novelty; minor

Recommendation. The audit recommended that version 1 not be submitted in its form, but re-framed as an empirical-plus-verified-implementation paper: the factorial result is a clean, reproducible finding, and the theory should be the revised one (Sections 3–6) with the prior work cited.

2.3 Errata to version 1

Table 4 lists the errata to version 1. Section numbers in the second column refer to version 1; "README" refers to the public showcase README that accompanied it.

Table 4: Errata to version 1.

Table 3: Integrity findings of the audit of version 1 and its code (each verified by running code).

#	finding	evidence	fix
I1	Implemented centre update $\neq$ proved update: a pairwise repulsion term broke the projection inequality the proof needs	`tests/test_theorem_to_code.py` (oracle plant): inequality violated at 6054 of 20000 samples with the legacy term, 0 of 20000 without; the term also failed its own purpose (min separation 0.07 vs $\delta = 0.5$)	removed from the algorithm (`enforce_separation=False` default); tracking unchanged on the headline run (mean steady-state error $0.0767 \to 0.0774$, max $0.46 \to 0.40$)
I2	Projection applied the isotropic formula to any Γ; the proof needs the Γ-weighted operator	`tests/test_controller_paths.py`	Γ-weighted projection implemented; fast/dense paths agree to 10^{-12}
I3	"R_{UUB} threshold" in the Monte Carlo figure was the constant 0.5	`tier1_monte_carlo.py` line 69: "A chosen arbitrary bound"	`exp_monte_carlo_honest.py`: only exactly checkable statements (200/200 hard-bound and ISS checks)
I4	"Theoretical R_{UUB}" in the k_e sweep used invented constants and an underived formula	`tier2_ruub_sweep.py`, former `estimate_D_max`	`exp_oracle_ruub.py`: all constants computable; a-priori bound $915\times$–$10\,954\times$ loose; RMS bound with measured disturbance holds 48/48
I5	Headline figure: fixed-grid baselines with a grid not covering the reference; their error (≈ 15) exceeded open loop, so the figure showed representational failure, not "bias"	`results/tier1_lorenz_tracking.png` vs ablation figure (projection W-only fails identically)	`exp_bias_factorial.py`: 3 mechanisms $\times$ 3 geometries, covering grid, 5 seeds
I6	Wilcoxon tests on $\sim$12 000 time samples with lag-1 autocorrelation 0.9999; the README's "$p > 0.3$ vs Exact GP" is contradicted by its own data ($p = 1.6 \times 10^{-11}$)	recomputed from `results/*.npy`	seed-level statistics only
I7	$n = 3$ benchmark and NTK baseline removed from the public figure (commits "exclude n=3 and NTK")	`results/n3/*.npy`, `results/*/ntk_*.npy`; `plot_comparison_hd_complete.py`	complete table regenerated: proposed ranks 3/5, 2/5, 3/5, 2/5
I8	GP baselines held a stale prediction for 50–100 steps; sparse GP inducing points on a diagonal line over $x \in [-3, 3]$ while the reference spans ± 20	`sim1`–`sim3`	`kernel_baselines_fair.py`: per-step prediction, 9×3 covering grid, seeds
I9	"$O(mn)$ per step" claim false with the $O(m^2 n)$ repulsion	`projection.py`	true after I1
I10	"Concurrent Learning" hook used the current residual only	former `controller.py`	removed
I11	Lyapunov monitor requires θ^*; never used by any experiment, so "Lyapunov descent verification" was never run	`integrator.py`	oracle module and tests perform it on a representable plant

#	section	statement in version 1	status
E1	1.2, 9	Centre separation $\|c_i - c_j\| \geq \delta$ "maintained via the projection set construction" with "convex projection boundaries"	**Wrong.** The separation set is non-convex; no projection onto it has property (P). The implemented repulsion term violated (P) in 30% of samples and did not keep the separation (min distance 0.07 vs $\delta = 0.5$). Removed from the algorithm; not needed by the proof.
E2	1.3	Offline Tikhonov regularisation of the ideal target: "This strict separation eliminates the circularity"	**Irrelevant to the proof.** Optimality of θ^* is never used (Section 3, remarks on the assumptions).
E3	1.3	IFT gives finite velocity bounds V_W, V_c, V_σ	**Not a proof.** The Hessian of a Tikhonov-regularised non-convex loss at a minimiser is PSD, not necessarily PD; the IFT hypothesis is asserted, not established. Replaced by Assumption A4.
E4	2.1	$D_{\max}$ from the Taylor remainder, "a finite geometric scalar"	**Valid but not explicit.** Replaced by the explicit, Hessian-free Lemma 2. The code's former `estimate_D_max` $= \varepsilon(1 + 2W_{\max}c_{\max})$ had no derivation and is deleted.
E5	4	$\operatorname{tr}(\tilde{W}^\top \Gamma_W^{-1} \tilde{W})$ with Γ_W "positive-definite learning rate matrices"	**Type error** for $\Gamma_W \in \mathbb{R}^{mn \times mn}$; the correct form is $\operatorname{vec}(\tilde{W})^\top \Gamma_W^{-1} \operatorname{vec}(\tilde{W})$. Harmless for scalar Γ.
E6	5	Projection inequality "holds as an absolute mathematical certainty — the cancellation gap does not exist"	**True only under A3** (θ^* in the inner set) and only for the Γ-weighted operator; the previous code applied the isotropic formula to any Γ. Fixed and tested.
E7	7.2	"A critical and non-trivial consequence of the projection operator"; its hard energy cap "converts what would be a boundary-only argument … into a global interior argument"	**Standard** level-set / comparison argument (Khalil [48, Thm. 4.18]). Correct, not new; also "strict monotone descent through every intermediate level set until the infimal set is penetrated" overstates: entry into every level above $c_{\inf}$ is finite-time, entry into $c_{\inf}$ itself only asymptotic.
E8	abstract, 7.3, 10.3	Projection gives UUB "without the persistent steady-state bias" (abstract); "no modification bias is introduced" (7.3); "zero bias into the Lyapunov derivative", "not a marginal improvement but a structural result" (10.3)	**Overclaimed.** The ultimate bound $\sqrt{R_{\text{UUB}}^2 + 2V_{\max,\Theta}}$ contains the parameter-energy term; see Section 6. The tighter, bias-free statement is the RMS bound (iii).
E9	8	"a standard Persistent Excitation (PE) condition on the regressor ϕ is insufficient to guarantee $\tilde{\Theta}(t) \to 0$"; convergence "requires alternative mechanisms", such as ICL	Parameter convergence is not treated; the remark is speculation and the code's former "Concurrent Learning" hook used only the current residual (not recorded data) and is removed.
E10	README	"100% of the independent simulation trials converged strictly below the calculated R_{UUB} threshold"	**False.** The threshold 0.5 was a hand-chosen constant (`tier1_monte_carlo.py`: "A chosen arbitrary bound"). Replaced by `exp_monte_carlo_honest.py` / `exp_oracle_ruub.py`.
E11	README	"$O(mn)$ time"	**Was false** with the $O(m^2 n)$ pairwise repulsion; true now that it is off by default.

2.4 What survives and what is withdrawn

What survives.

- *The corrected theorem.* Theorem 1 of Section 5 holds under the explicit Assumptions A1–A5, for an algorithm identical to the implemented one. Its pointwise ultimate bound (ii) is the conclusion of version 1; it is the textbook level-set argument, and its radius contains the parameter-energy term $V_{\max,\Theta}$. Its RMS bound (iii) does not contain that term and is the form of the result that can be confronted with data; its reference bound (iv) holds whatever the adaptation does. The ingredients are the projection property (P) (Lemma 1), the explicit Hessian-free disturbance bound (Lemma 2) and the Lyapunov derivative (Lemma 3), verified numerically along simulated trajectories. Nothing in the

stability theorem is new (Section 7). The "no bias" claim survives only in the precise form of Section 6: projection removes the *Lyapunov-derivative* bias of leakage, not the ultimate *tracking* bound's dependence on the parameter set.

- *The projection-versus-leakage finding* (Section 8). On identical geometry, projection beats both σ- and e-modification in every cell of a mechanism-by-geometry factorial, 5/5 seeds, with paired confidence intervals excluding zero. Within projection, adaptive centres improve on the fixed grid by 0.012 (5/5) and adaptive widths by a further 0.035 (5/5). So the method's real advantage is the robustness mechanism, with geometry adaptation a smaller second-order gain, close to the reverse of the story told by the original figure.
- *The comparison with kernel regression, with its limits* (Section 8). Against kernel-regression baselines the method is never the most accurate. Its edge is cost: it is cheaper per step than every GP baseline at every n and than NTK except at $n = 3$, and in the fair $n = 1$ rerun, with per-step prediction for every method, it has the cheapest step. Every kernel-regression baseline (GP and NTK, including the fair $n = 1$ rerun) is trained on noise-free samples of the true $f(x, t)$, which is equivalent to measuring the state derivative exactly, while the adaptive law uses only the tracking error.

What is withdrawn.

- Earlier versions of the public README stated that the method's steady-state error was "statistically indistinguishable (Wilcoxon signed-rank $p > 0.3$) from both the Exact GP and the Sparse GP", that "100% of the independent simulation trials converged strictly below the calculated R_{UUB} threshold", and announced "Curse of Dimensionality Solved". Those statements are withdrawn: the significance test was invalid, the threshold was an arbitrary constant, and the omitted $n = 3$ and NTK results contradict them.
- The "zenith" language is dropped: "absolute mathematical certainty", "curse of dimensionality solved", "structural result". None of it survives the data. Claims of a "structural" advantage over leaky modifications are withdrawn.
- The claim that centre separation is "maintained via the projection set construction" is withdrawn (erratum E1); no centre-separation constraint is needed by the proof.

2.5 How Part II answers the structural defects

Section 1.2 named two structural defects of the classical labelled analysis, that of version 1 and of the literature it builds on: its Lyapunov analyses depend on labels, and it has no object that describes where the kernels go or whether their placement is good. Part II addresses both by working with the measure $\mu = \frac{1}{m}\sum_i \delta_{\vartheta_i}$, which does not see the labels, and with its transport.

- *Labels.* The Lyapunov function of Theorem B, $V_{\mathrm{OT}} = \frac{1}{2}\|e\|^2 + \frac{1}{2}W(\mu_t, \mu_t^*)^2$, measures the parameter error by the optimal-assignment (Wasserstein) distance W, a minimum over all assignments of kernels to ideal atoms (Section 9). The labelled σ-modification is not a functional of μ; Theorem C replaces it by an optimal-transport σ-modification, Theorem C$'$ extends the OT leak to priors with a different number of atoms through the barycentric projection of the optimal plan, and Proposition C.5 shows that the labelled leak reshuffles kernels. Theorem E gives label-free local exponential convergence under tangential persistent excitation, with explicit radius, overshoot and rate (Lemma E.0).
- *Where the kernels go.* The measure μ_t is that object. By Theorem A the adaptive law is the particle form of the Wasserstein gradient flow of a μ-linear potential; Theorems D and D$'$ treat the mean-field (many-kernel) loop, without and with dither; Theorem F bounds the Wasserstein length of learning by the integrated tracking error; Proposition G describes the slow geometry (formal). Whether the placement is good is the subject of

Theorem H: with saturated features, σ-modification and dither, the averaged mean-field loop converges globally and exponentially to the regularised optimum. This is proved for $(\mathfrak{A})$; the reduction to $(\mathfrak{A})$ is formal, and the real loop is Conjecture H.6, which Propositions H.7 and H.8 reduce to an averaging estimate. Theorem J proves its deterministic finite-m case, Theorem I characterises the non-reciprocity of the averaged loop at finite gain, and Proposition K turns the orientation of the reference orbit into a design input for collapsed configurations.

- *Constants growing with m.* The third apparent defect is an artefact of the normalisation; the mean-field normalisation removes it for labelled and label-free analyses alike (Remark A.1).

The status of each of these results is that of Table 1.

Part I
The projection-based controller

3 Setting and standing assumptions

This part replaces the stability argument of version 1 with a statement whose assumptions are explicit, whose algorithm is exactly the one in `src/core/` of the accompanying repository, and whose every step is either proved here or verified numerically by `tests/test_theorem_to_code.py`. The errata to version 1 are listed in Section 2; Section 7 positions the result against prior work.

Notation. $\|\cdot\|$ is the Euclidean norm for vectors and the Frobenius norm for matrices; $\mathrm{vec}(\cdot)$ is the row-major vectorisation used throughout the code; there are m kernels and n states.

Plant, reference and error.

$$\dot{x} = f(x,t) + u, \qquad e = x - x_m(t), \qquad u = \dot{x}_m - k_e e - \hat{f}(x), \tag{3.1}$$

$$\hat{f}(x) = W^\top \varphi(x, c, \sigma), \qquad \varphi_i(x) = \exp\Big(-\frac{\|x - c_i\|^2}{2\sigma_i^2}\Big). \tag{3.2}$$

Assumption A1 (operation on a compact set)**.** There is a compact set $X \subset \mathbb{R}^n$ with $x(t) \in X$ for all $t \geq 0$, and $F_{\max} := \sup_{x \in X,\, t \geq 0} \|f(x,t)\| < \infty$. *(Version 1 assumes "a compact spatial domain X" in the same way. For globally bounded f the assumption is discharged by Theorem 1(iv); for the Lorenz plant it is a genuine assumption.)*

Assumption A2 (reference)**.** x_m is C^1 with x_m, $\dot{x}_m$ bounded, and $\dot{x}_m$ is available to the controller.

Assumption A3 (admissible ideal parameters)**.** There is a curve $\theta^*(t) = (W^*(t), c^*(t), \sigma^*(t))$ lying for all t in the *inner* projection sets

$$\|W^*\| \leq W_{\max} - \varepsilon, \qquad \|c^*\| \leq c_{\max} - \varepsilon, \qquad \sigma_{\min} + \varepsilon \leq \sigma_i^* \leq \sigma_{\max} - \varepsilon.$$

Assumption A4 (ideal-parameter regularity)**.** $\theta^*(\cdot)$ is C^1 with $\|\dot{W}^*\| \leq V_W$, $\|\dot{c}^*\| \leq V_c$, $\|\dot{\sigma}^*\| \leq V_\sigma$. (For a time-invariant plant one may take θ^* constant, $V_W = V_c = V_\sigma = 0$.)

Assumption A5 (gains and geometry)**.** $k_e > 0$; Γ_W, Γ_c symmetric positive definite on $\mathbb{R}^{mn}$ (a zero Γ freezes that group, see Remark 3); Γ_σ diagonal positive; $0 < \varepsilon < \min\big(W_{\max}, c_{\max}, (\sigma_{\max} - \sigma_{\min})/2\big)$; $\sigma_{\min} > 0$.

Remarks on the assumptions.

- **The proof never uses optimality of θ^*.** θ^* is an analytic device; any admissible C^1 curve works, and the approximation quality of the network enters the result only through the *actual* size of the disturbance $d(t)$ defined below, not through the bound. The Tikhonov-regularised definition of θ^* in version 1 and its implicit-function-theorem (IFT) argument for A4 are therefore unnecessary — and, as written, not a proof (erratum E3, Section 2).
- **No centre-separation constraint is needed.** The set $\{c : \|c_i - c_j\| \geq \delta \ \forall\, i \neq j\}$ is not convex, so no projection onto it has property (P) below; the claim of version 1 that separation is "maintained via the projection set construction" is withdrawn (erratum E1, Section 2). The bounded-Jacobian property that the proof needs follows from $\sigma \geq \sigma_{\min}$ alone.

4 The algorithm

The algorithm is identical to the one implemented in `src/core/controller.py`:

$$\begin{aligned} \dot{W} &= \mathrm{Proj}_W\big(W,\ \Gamma_W\,\varphi e^\top\big), \\ \dot{c} &= \mathrm{Proj}_c\big(c,\ \Gamma_c\,\varphi_c'^\top We\big), \\ \dot{\sigma} &= \mathrm{Proj}_\sigma\big(\sigma,\ \Gamma_\sigma\,\varphi_\sigma'^\top We\big), \end{aligned} \tag{4.1}$$

with $\varphi_c' = \partial\varphi/\partial\,\mathrm{vec}(c) \in \mathbb{R}^{m\times mn}$ (row i has the single non-zero block $\varphi_i(x-c_i)^\top/\sigma_i^2$) and $\varphi_\sigma' = \mathrm{diag}\big(\varphi_i\|x-c_i\|^2/\sigma_i^3\big)$.

Norm-ball projection (W and c), boundary layer of width ε.

$$\mathrm{Proj}(\theta,\Gamma y) = \begin{cases} \Gamma y - h(\theta)\,\Gamma\theta\,\dfrac{\theta^\top\Gamma y}{\theta^\top\Gamma\theta} & \text{if } h(\theta)>0 \text{ and } \theta^\top\Gamma y>0, \\ \Gamma y & \text{otherwise,} \end{cases} \tag{4.2}$$

$$h(\theta) = \begin{cases} 0 & \text{for } \|\theta\| \le \theta_{\max}-\varepsilon, \\ \big((\|\theta\|-\theta_{\max}+\varepsilon)/\varepsilon\big)^2 & \text{in between,} \\ 1 & \text{for } \|\theta\| \ge \theta_{\max}. \end{cases} \tag{4.3}$$

Box projection (σ, component-wise). The update component is multiplied by $\mathrm{clip}\big((\sigma_i-\sigma_{\min})/\varepsilon, 0, 1\big)$ when it points below the lower face and by $\mathrm{clip}\big((\sigma_{\max}-\sigma_i)/\varepsilon, 0, 1\big)$ when it points above the upper face.

Discrete time. The simulations integrate with fixed-step RK4 and re-clip the parameters to their sets after every stage (`src/core/integrator.py`). The theorem is a continuous-time statement; the clip is a safety net that is inactive if the step is small enough (property (P) already removes the outward component on the boundary).

5 Lemmas and the boundedness theorem

5.1 Lemmas

Lemma 1 (projection property (P))**.** *Under A3 and A5, for every θ in its set and every y,*

$$\tilde\theta^\top\Gamma^{-1}\,\mathrm{Proj}(\theta,\Gamma y) \le \tilde\theta^\top y, \qquad \tilde\theta := \theta-\theta^*. \tag{P}$$

Proof. If the projection is inactive the two sides are equal. Otherwise

$$\tilde\theta^\top\Gamma^{-1}\big(\mathrm{Proj}\ -\Gamma y\big) = -h\,\frac{\theta^\top\Gamma y}{\theta^\top\Gamma\theta}\,\tilde\theta^\top\theta,$$

with $h>0$ and $\theta^\top\Gamma y>0$, and $\tilde\theta^\top\theta = \|\theta\|^2 - \theta^{*\top}\theta \ge \|\theta\|\big(\|\theta\|-\|\theta^*\|\big) \ge 0$ because $h>0$ forces $\|\theta\| \ge \theta_{\max}-\varepsilon \ge \|\theta^*\|$. For the box, component i, with s_i the clip factor applied to that component: $\tilde\sigma_i(s_i-1)y_i \le 0$ because $s_i \le 1$, and on the lower face $y_i<0$ while $\tilde\sigma_i = \sigma_i-\sigma_i^* \le \sigma_{\min}+\varepsilon-\sigma_i^* \le 0$ (symmetrically on the upper face). □

Status. Proved.

Checked by: `tests/test_projection.py` (randomised; isotropic and non-isotropic Γ; vector, matrix and box), together with the counterexample showing that (P) fails when θ^* is outside the inner set.

Lemma 2 (lumped disturbance)**.** *Define*

$$d := f - \hat{f} + \tilde{W}^\top \varphi + W^\top \varphi_c' \,\tilde{c} + W^\top \varphi_\sigma' \,\tilde{\sigma}. \tag{5.1}$$

Then identically

$$\dot{e} = -k_e e - \tilde{W}^\top \varphi - W^\top \varphi_c' \,\tilde{c} - W^\top \varphi_\sigma' \,\tilde{\sigma} + d, \tag{5.2}$$

and under A1, A3,

$$\|d\| \le D_{\max} := F_{\max} + 3W_{\max}\sqrt{m} + W_{\max}\big(2c_{\max}J_c + \sqrt{m}\,(\sigma_{\max} - \sigma_{\min})J_\sigma\big), \tag{5.3}$$
$$J_c = \mathrm{e}^{-1/2}/\sigma_{\min}, \qquad J_\sigma = 2\mathrm{e}^{-1}/\sigma_{\min}.$$

Proof. The identity is the definition of d substituted into $\dot{e} = -k_e e + f - \hat{f}$. For the bound: $\|f\| \le F_{\max}$; $\|\hat{f}\| \le \|W\|\|\varphi\| \le W_{\max}\sqrt{m}$ $(0 < \varphi_i \le 1)$; $\|\tilde{W}\| \le 2W_{\max}$; $\|\tilde{c}\| \le 2c_{\max}$; $\|\tilde{\sigma}\| \le \sqrt{m}\,(\sigma_{\max} - \sigma_{\min})$; the rows of φ_c' have disjoint supports, so

$$\|\varphi_c'\|_2 = \max_i \frac{\varphi_i \|x - c_i\|}{\sigma_i^2} \le \sup_{r \ge 0} \frac{r\,\mathrm{e}^{-r^2/(2\sigma^2)}}{\sigma^2} = \frac{\mathrm{e}^{-1/2}}{\sigma} \le J_c,$$

and $\|\varphi_\sigma'\|_2 = \max_i \varphi_i \|x - c_i\|^2/\sigma_i^3 \le 2\mathrm{e}^{-1}/\sigma \le J_\sigma$. □

Status. Proved.

Checked by: `tests/test_rbf_kernel.py` checks the two suprema by brute force.

Remark 1 (local form)**.** Writing $\varphi^* = \varphi(x; c^*, \sigma^*)$ and expanding to first order in $(\tilde{c}, \tilde{\sigma})$ gives

$$d = \varepsilon + W^{*\top} O_{\mathrm{ho}} + \tilde{W}^\top \varphi_c' \,\tilde{c} + \tilde{W}^\top \varphi_\sigma' \,\tilde{\sigma}, \qquad \varepsilon := f - W^{*\top}\varphi^*,$$

with O_{ho} the Taylor remainder — the decomposition of version 1. Near convergence ($\tilde{\theta} \to 0$) this shows $\|d\| \lesssim \|\varepsilon\| + O(\|\tilde{\theta}\|^2)$, which is why the controller performs far better than $D_{\max}$ suggests; but it does not yield a smaller global constant than Lemma 2, which needs no Hessian and no Taylor expansion.

Lemma 3 (Lyapunov derivative)**.** *Let*

$$V = \tfrac{1}{2} e^\top e + \tfrac{1}{2}\,\mathrm{vec}(\tilde{W})^\top \Gamma_W^{-1}\,\mathrm{vec}(\tilde{W}) + \tfrac{1}{2}\tilde{c}^\top \Gamma_c^{-1}\tilde{c} + \tfrac{1}{2}\tilde{\sigma}^\top \Gamma_\sigma^{-1}\tilde{\sigma}.$$

Along trajectories of the algorithm,

$$\begin{aligned}
\dot{V} &= -k_e\|e\|^2 + e^\top d + \mathrm{cross} - \Omega^*, \\
\mathrm{cross} &= \langle \tilde{W},\, \Gamma_W^{-1}\dot{W} - \varphi e^\top \rangle + \tilde{c}^\top\big(\Gamma_c^{-1}\dot{c} - \varphi_c'^\top W e\big) + \tilde{\sigma}^\top\big(\Gamma_\sigma^{-1}\dot{\sigma} - \varphi_\sigma'^\top W e\big) \le 0, \\
\Omega^* &= \langle \tilde{W},\, \Gamma_W^{-1}\dot{W}^* \rangle + \tilde{c}^\top \Gamma_c^{-1}\dot{c}^* + \tilde{\sigma}^\top \Gamma_\sigma^{-1}\dot{\sigma}^*, \qquad |\Omega^*| \le C_\Omega, \\
C_\Omega &:= 2W_{\max}\|\Gamma_W^{-1}\|_2 V_W + 2c_{\max}\|\Gamma_c^{-1}\|_2 V_c + \sqrt{m}\,(\sigma_{\max} - \sigma_{\min})\|\Gamma_\sigma^{-1}\|_2 V_\sigma.
\end{aligned} \tag{5.4}$$

Proof. Differentiate V, substitute the identity of Lemma 2 for $\dot{e}$, and use $e^\top \tilde{W}^\top \varphi = \langle \tilde{W},\, \varphi e^\top \rangle$, $e^\top W^\top \varphi_c' \,\tilde{c} = \tilde{c}^\top \varphi_c'^\top W e$, $e^\top W^\top \varphi_\sigma' \,\tilde{\sigma} = \tilde{\sigma}^\top \varphi_\sigma'^\top W e$. That cross ≤ 0 is Lemma 1 applied to the three groups; the bound on Ω^* is Cauchy–Schwarz with A3, A4. □

Status. Proved.

Checked by: `tests/test_theorem_to_code.py`: the identity (to machine precision) and cross ≤ 0 (at every sample) are verified along simulated trajectories, on a plant that is exactly representable with known constant θ^*.

5.2 Theorem 1: uniform ultimate boundedness

Theorem 1 (uniform ultimate boundedness)**.** *Assume A1–A5, let*

$$R_{\mathrm{UUB}} := \frac{D_{\max}}{2k_e} + \sqrt{\frac{D_{\max}^2}{4k_e^2} + \frac{C_\Omega}{k_e}},$$

and let $V_{\max,\Theta}$ be the supremum over the projection sets of the parameter part of V.

(i) ***Descent outside a ball.*** *$\dot V \le -k_e\|e\|^2 + D_{\max}\|e\| + C_\Omega$, hence $\dot V < 0$ whenever $\|e\| > R_{\mathrm{UUB}}$.*

(ii) ***Pointwise ultimate bound.*** *Every solution satisfies $V(t) \le \max\{V(0), c_{\inf}\}$ with $c_{\inf} := \frac{1}{2}R_{\mathrm{UUB}}^2 + V_{\max,\Theta}$, every level $c > c_{\inf}$ is reached in finite time, and*

$$\limsup_{t\to\infty}\|e(t)\| \le \sqrt{R_{\mathrm{UUB}}^2 + 2V_{\max,\Theta}}.$$

(iii) ***RMS bound (the quantitatively meaningful one).*** *For every $T_1 < T_2$,*

$$\mathrm{RMS}_{[T_1,T_2]}(\|e\|) \le \frac{D_{\max}}{2k_e} + \sqrt{\frac{D_{\max}^2}{4k_e^2} + \frac{C_\Omega}{k_e} + \frac{V(T_1)}{k_e(T_2-T_1)}},$$

so $\limsup_{T_2\to\infty}\mathrm{RMS}_{[T_1,T_2]}(\|e\|) \le R_{\mathrm{UUB}}$ — without the parameter-energy term of (ii). The same inequality holds with $D_{\max}$ and C_Ω replaced by the trajectory-wise suprema of $\|d(t)\|$ and $|\Omega^*(t)|$ on $[T_1,T_2]$.

(iv) ***Reference bound (adaptation-free).*** *Under A1 and the hard parameter bounds alone, $\dot e = -k_e e + (f - \hat f)$ with $\|f - \hat f\| \le F_{\max} + W_{\max}\sqrt{m} =: \tilde F$, so*

$$\|e(t)\| \le \mathrm{e}^{-k_e t}\|e(0)\| + \tilde F/k_e \qquad \text{for all } t,$$

whatever the adaptation does.

Proof. (i) Lemma 3 with Cauchy–Schwarz on $e^\top d$.

(ii) If $V > c_{\inf}$ then, since the parameter part of V is at most $V_{\max,\Theta}$, $\frac{1}{2}\|e\|^2 > \frac{1}{2}R_{\mathrm{UUB}}^2$, so $\|e\| > R_{\mathrm{UUB}}$ and $\dot V < 0$; more precisely, with $\Delta := V - c_{\inf} > 0$ one has $\|e\|^2 \ge R_{\mathrm{UUB}}^2 + 2\Delta$ and therefore

$$\dot V \le -g\Big(\sqrt{R_{\mathrm{UUB}}^2 + 2\Delta}\Big) =: -\psi(\Delta), \qquad g(r) = k_e r^2 - D_{\max} r - C_\Omega,$$

with g increasing for $r \ge R_{\mathrm{UUB}}$ and ψ of class $\mathcal{K}$. The comparison lemma gives $V(t) \le \max\{V(0), c_{\inf}\}$, finite-time entry into every sublevel set $\{V \le c\}$ with $c > c_{\inf}$, and $\limsup V \le c_{\inf}$, i.e. the stated bound on $\|e\|$ ($\|e\|^2 \le 2V$).

(iii) Integrate (i):

$$\begin{aligned} k_e\int_{T_1}^{T_2}\|e\|^2\,\mathrm{d}t &\le V(T_1) - V(T_2) + D_{\max}\int_{T_1}^{T_2}\|e\|\,\mathrm{d}t + C_\Omega(T_2-T_1) \\ &\le V(T_1) + D_{\max}\sqrt{T_2-T_1}\,\sqrt{\int_{T_1}^{T_2}\|e\|^2\,\mathrm{d}t} + C_\Omega(T_2-T_1) \end{aligned}$$

by Cauchy–Schwarz; with $S := \mathrm{RMS}_{[T_1,T_2]}(\|e\|)$ this reads

$$k_e S^2 - D_{\max} S - \big(C_\Omega + V(T_1)/(T_2-T_1)\big) \le 0,$$

and S is below the positive root.

(iv) Variation of constants. □

Status. Proved.

What (ii)–(iv) say together. Part (ii) is the conclusion of version 1; it is the textbook level-set argument (Khalil [48, Thm. 4.18]; Lewis et al. [55]) and its radius contains the parameter-energy reservoir $V_{\max,\Theta}$ — a bias-like term — so it does **not** establish that projection yields a tighter tracking bound than σ-modification. Part (iii) is tighter and is the statement that can be checked without unknown constants: replace $D_{\max}$ by the measured $\sup\|d\|$, see `experiments/exp_oracle_ruub.py`, where it is satisfied in 48/48 runs (Section 8). Part (iv) shows that mere boundedness of e is already a consequence of projection plus a stable error channel, independent of the adaptation law; the content of adaptive control is in making the *actual* $d(t)$ small, which no a-priori constant captures. In the oracle experiments the a-priori R_{UUB} exceeds the observed RMS error by factors of 915 to 10954.

Remark 2 (no LaSalle / Barbalat)**.** Version 1's claim that the argument needs neither LaSalle's invariance principle nor Barbalat's lemma is correct, and standard: UUB statements never need them. Nothing here claims asymptotic convergence of e or of $\tilde{\theta}$.

Remark 3 (frozen groups)**.** If $\Gamma_c = 0$ (fixed centres), drop the $\tilde{c}$ term from V; the identity of Lemma 2 still holds with $-W^\top \varphi_c' \tilde{c}$ absorbed into d, with the same bound. Likewise for σ.

Remark 4 (what is not proved)**.** Parameter convergence (it requires persistent excitation of a nonlinear-in-parameter regressor — not addressed); exact tracking; any statement about the size of the ideal approximation error ε.

6 The no-bias claim, stated precisely

With σ-modification,

$$\dot{V} \le -k_e\|e\|^2 - \tfrac{1}{2}\sigma_{\mathrm{mod}}\|\tilde{\theta}\|_\Gamma^2 + \tfrac{1}{2}\sigma_{\mathrm{mod}}\|\theta^*\|_\Gamma^2 + D\|e\| + C_\Omega:$$

the leak adds a positive constant and removes the parameter-energy reservoir, which gives exponential convergence of V to a ball whose radius grows with $\sigma_{\mathrm{mod}}\|\theta^*\|^2$. With projection,

$$\dot{V} \le -k_e\|e\|^2 + D\|e\| + C_\Omega:$$

no positive constant is added, but nothing dissipates parameter energy, which is why the pointwise bound (ii) of Theorem 1 keeps $V_{\max,\Theta}$. Projection therefore removes the *Lyapunov-derivative* bias of leakage (true, and known since Pomet–Praly [76] and Ioannou–Sun [47]), not the dependence of the ultimate *tracking* bound on the parameter set (false as stated in Sections 7.3 and 10.3 of version 1). Empirically the two mechanisms are compared on identical geometry in `experiments/exp_bias_factorial.py` (Section 8).

7 Prior art

- Adaptive Gaussian-network control with robust modification: Sanner and Slotine [79].
- Adaptation of inner-layer parameters via first-order Taylor expansion of the feature map with bounded remainder — the exact device of Sections 4 and 5: Lewis, Yesildirek and Liu [54]; Polycarpou [75]; Ge and Wang [33]; dynamic-structure RBF networks with tuned centres and widths: Fabri and Kadirkamanathan [28].
- Smooth projection with a boundary layer and the inequality (P), including the "no bias when $\theta^* \in \Omega$" observation: Pomet and Praly [76]; Ioannou and Sun [47, §4.4]; Lavretsky and Gibson [52]; Lavretsky and Wise [53, Ch. 11].
- σ- and e-modification and their bias: Ioannou and Kokotovic [46]; Narendra and Annaswamy [63].

- UUB via level sets: Khalil [48, Thm. 4.18]; Lewis, Jagannathan and Yesildirek [55].
- Concurrent learning (recorded-data stacks, finite excitation): Chowdhary and Johnson [17, 18].

Table 5 sets the ingredients of version 1 against these sources.

Table 5: Novelty of the ingredients of the stability argument.

Ingredient	Prior source	Status of the claim of version 1
Adaptive RBF control with projection-type robustification	Sanner and Slotine [79]; Ioannou and Sun [47]	known
Adapting centres/widths by first-order Taylor expansion with bounded remainder	Lewis, Yesildirek and Liu [54]; Polycarpou [75]; Fabri and Kadirkamanathan [28]; Ge and Wang [33]	known; version 1 cites none of these
Smooth projection with ε boundary layer and inequality (P)	Pomet and Praly [76]; Lavretsky and Gibson [52]; Lavretsky and Wise [53]	known; the public showcase README calls it "the core theoretical contribution"
Projection introduces no bias in $\dot{V}$ when $\theta^* \in \Omega$	Ioannou and Sun [47, §4.4]	known
Level-set ("global interior") UUB argument	Khalil [48, Thm. 4.18]; Lewis et al. [55]	known; called "critical and non-trivial"
Simultaneous adaptation of W, c, σ for Gaussian RBFs with all three projected	—	the only combinational novelty; minor

Net: no new theorem.

What is new here. Nothing in the stability theorem: Theorem 1(i)–(ii) is the composition of known pieces (Taylor-linearised inner-layer adaptation, boundary-layer projection and level-set UUB), applied to Gaussian RBFs with simultaneously adaptive weights, centres and widths. The contribution of the accompanying repository, after the revision of version 1, is (a) a theorem-to-code map in which the implemented and the proved algorithm coincide and are machine-checked (Lemmas 1–3 verified numerically along trajectories), (b) the explicit Hessian-free disturbance bound of Lemma 2 and the RMS bound (iii) of Theorem 1, which is the form of the result that can actually be confronted with data, and (c) a controlled empirical separation of "projection vs leakage" from "adaptive vs fixed geometry" (Section 6; `results/factorial/`). Claims of a "structural" advantage over leaky modifications are withdrawn.

8 What the experiments show

Every numerical statement in this section was regenerated from the scripts of the accompanying repository; the result file and the generating script are named with each table. Statistics are computed over seeds, not over autocorrelated time samples.

8.1 Robustness mechanism versus geometry

The factorial experiment `experiments/exp_bias_factorial.py` separates the two factors that the original headline figure (`results/tier1_lorenz_tracking.png`) confounded: there, the fixed-grid baselines used a grid not covering the reference, and their error (≈ 15) exceeded open loop, so the figure showed representational failure, not "bias". Here three robustness mechanisms (projection, σ-modification, e-modification, the leaky ones implemented on the same architecture

in `src/baselines/leaky_geometry.py`) are crossed with three geometries (fixed grid with only W adapted; adaptive W, c; adaptive W, c, σ) on Lorenz tracking with a covering grid, $m = 27$, $k_e = 10$ and 5 seeds (Tables 6 and 7, Figure 1).

Table 6: Robustness mechanism × geometry (Lorenz tracking, $m = 27$, $k_e = 10$). Seeds: 5; metric = steady-state $\|e\|$ over the last 40% of a 30 s run; mean ± 95% CI over seeds; max SS = maximum over the window, mean over seeds; "div" = diverged runs. Source: `results/factorial/table.md`.

geometry	mechanism	mean SS $\|e\|$	median SS	max SS	max $\|W\|_F$	min sep	div
fixed (W)	projection	0.1584 ± 0.0072	0.1288	0.559	47.5	1.365	0
fixed (W)	σ-mod	0.7327 ± 0.0223	0.5647	2.720	38.5	1.469	0
fixed (W)	e-mod	0.5619 ± 0.0306	0.4264	2.192	26.7	1.469	0
adaptive (W, c)	projection	0.1468 ± 0.0038	0.1020	0.751	40.9	0.039	0
adaptive (W, c)	σ-mod	0.4004 ± 0.0050	0.3468	1.863	32.3	0.078	0
adaptive (W, c)	e-mod	0.2327 ± 0.0024	0.1993	0.718	28.1	0.099	0
adaptive (W, c, σ)	projection	0.1119 ± 0.0043	0.0852	0.512	38.5	0.172	0
adaptive (W, c, σ)	σ-mod	0.3862 ± 0.0057	0.3422	1.655	30.9	0.215	0
adaptive (W, c, σ)	e-mod	0.2385 ± 0.0029	0.2103	0.820	27.7	0.239	0

Table 7: Paired (same seed) differences of mean steady-state $\|e\|$ for the factorial experiment. Top: projection minus baseline. Bottom: geometry effect within the projection mechanism. Source: `results/factorial/table.md`.

geometry	baseline	mean diff	95% CI	projection better in
fixed (W)	σ-mod	−0.5743	±0.0232	5/5 seeds
fixed (W)	e-mod	−0.4035	±0.0310	5/5 seeds
adaptive (W, c)	σ-mod	−0.2536	±0.0021	5/5 seeds
adaptive (W, c)	e-mod	−0.0858	±0.0025	5/5 seeds
adaptive (W, c, σ)	σ-mod	−0.2743	±0.0043	5/5 seeds
adaptive (W, c, σ)	e-mod	−0.1266	±0.0031	5/5 seeds
comparison		mean diff	95% CI	adaptive better in
adaptive (W, c) − fixed (W)		−0.0116	±0.0056	5/5
adaptive (W, c, σ) − adaptive (W, c)		−0.0349	±0.0026	5/5

Projection beats both leaky baselines in every cell, in 5/5 seeds, with paired confidence intervals excluding zero. Within projection, adaptive centres improve on the fixed grid by 0.0116 ± 0.0056 (5/5 seeds) and adaptive widths by a further 0.0349 ± 0.0026 (5/5 seeds). So the method's real advantage is the robustness mechanism, with geometry adaptation a smaller second-order gain — close to the reverse of the story told by the original headline figure.

8.2 Kernel-regression baselines

Table 8 is the comparison against kernel-regression baselines from saved single runs, with nothing excluded (in particular the benchmark $n = 3$ and the NTK baseline are included). The proposed method is never first: it ranks 3/5, 2/5, 3/5 and 2/5 at $n = 1, 2, 3, 6$. NTK is first at $n = 2$ and $n = 6$ (mean steady-state $\|e\|$ 0.0015 and 0.0013, against 0.0372 and 0.0044 for the proposed method), and the sparse GP is first at $n = 3$ (3.2729), where the error of

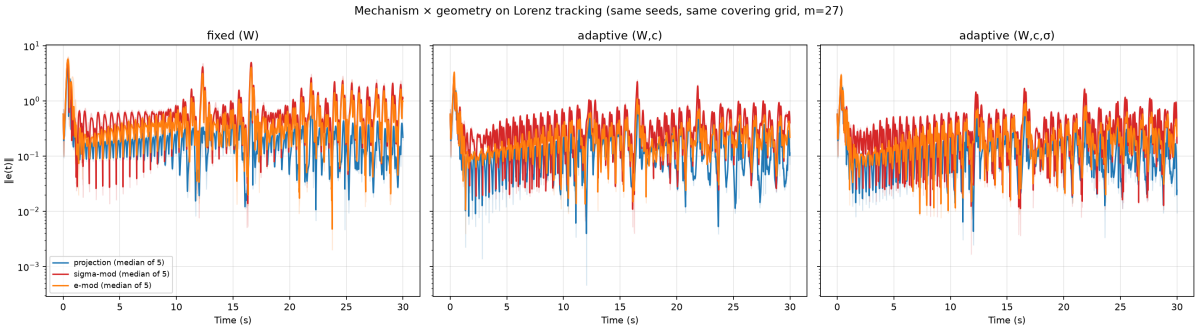


Figure 1: Mechanism × geometry on Lorenz tracking (same seeds, same covering grid, $m = 27$): $\|e(t)\|$ against time, logarithmic scale, median of the 5 seeds for projection, σ-modification and e-modification; shaded bands: minimum to maximum over the 5 seeds. Panels: fixed (W), adaptive (W, c), adaptive (W, c, σ). Generated by `experiments/exp_bias_factorial.py`.

the proposed method is 7.7190. Its genuine edge is cost against the GP baselines: the mean update time of the proposed method is 0.10, 0.22 and 0.60 ms per step at $n = 1, 2, 6$ and 4.11 ms at $n = 3$, against 52.92–1888.80 ms per update for the GP baselines, which update every 50–100 steps. NTK takes 1.59–13.22 ms per update; it updates every 50 steps at $n = 1$ and every step at $n = 2, 3, 6$ (`experiments/sim4_ntk_control.py`, `experiments/sim4_ntk_n2.py`, `experiments/sim4_ntk_n3.py`, `experiments/sim4_ntk_n6.py`), so it is cheaper than the proposed method only at $n = 3$ (2.35 against 4.11 ms per step). The comparison is not information-matched: every baseline, here and in the fair re-run below, is trained on noise-free samples of the true $f(x, t)$, which amounts to measuring the state derivative exactly, whereas the adaptive law uses only the tracking error e. With one run per cell no valid significance test is available.

In the original baseline scripts (`sim1`–`sim3`) the GP baselines held a stale prediction for 50–100 steps, and the inducing points of the sparse GP lay on a diagonal line over $x \in [-3, 3]$ while the reference spans ± 20. The fair re-run at $n = 1$ (`experiments/kernel_baselines_fair.py`, `experiments/run_fair_n1.py`, `experiments/plot_fair_n1.py`) uses per-step prediction for all methods, a covering inducing grid and 3 seeds (Table 9, Figure 2). The mean steady-state $\|e\|$ is 0.0591 for the exact GP, 0.1867 for the sparse GP, 0.1978 for the proposed method, 0.3087 for the deep GP and 0.4700 for NTK; the median online step is 6.309, 1.353, 0.060, 4.461 and 0.212 ms respectively. Once it is allowed to predict every step, the exact GP is 3–4× more accurate than the proposed method (mean steady-state $\|e\|$ 0.0591 against 0.1978), and more accurate in every seed (the proposed method has the lower mean in 0/3 seeds). The proposed method is on par with a fixed-grid sparse GP (the proposed method has the lower mean in 2/3 seeds) at 22.6× lower cost (median step 0.060 ms against 1.353 ms); its step is the cheapest of the five methods. This is the version of the comparison to quote.

8.3 The bound against data

Oracle bound study. The oracle experiment (`experiments/exp_oracle_ruub.py`) uses a plant that is exactly representable with constant θ^* ($m = 8$, $n = 2$), so every constant of Theorem 1 is computable; with θ^* constant, $C_\Omega = 0$ and $R_{\mathrm{UUB}} = D_{\max}/k_e$, reported as D_{theory}/k_e with $D_{\max}$ from Lemma 2. The "full-state bound" is $\sqrt{R_{\mathrm{UUB}}^2 + 2V_{\max,\Theta}}$ of Theorem 1(ii), and D_{meas} is the measured $\sup_{t\in[10,30]}\|d(t)\|$ over the evaluation window. Table 10 and Figure 3 cover $k_e \in \{2, 5, 10, 20\}$ and four learning-rate scales, 3 seeds each. The Lyapunov identity of Lemma 3 holds to 3.55×10^{-15} (maximal $|\dot{V} - (-k_e\|e\|^2 + e^\top d + \mathrm{cross})|$), property (P) is never violated, the hard bounds held, and the RMS bound of Theorem 1(iii) with the measured disturbance holds in 48/48 runs and tracks $1/k_e$. The a-priori R_{UUB} is 26.1–261.1 against observed RMS

Table 8: Higher-dimensional comparison, complete (nothing excluded). Single trajectory per cell; steady state = last 40% of 30 s. Step time = mean wall-clock time of the method's update call (GP: per update, every 50–100 steps; NTK: per update, every 50 steps at $n = 1$ and every step at $n = 2, 3, 6$; proposed: per step). No valid significance test is available (one run per cell). Generated by `experiments/plot_comparison_hd_complete.py`; source: `results/comparison_complete_table.md`.

benchmark	method	mean SS $\|e\|$	median SS	max SS	rank (mean SS)	mean update time (ms)
$n = 1$ (1-D plant, Lorenz ref.)	Proposed (Proj)	0.2342	0.1496	1.290	3/5	0.10
	Exact GP	0.2206	0.1415	1.016	1/5	273.78
	Sparse GP	0.2285	0.1914	0.874	2/5	52.92
	Deep GP	0.3963	0.2994	1.436	4/5	94.34
	NTK	0.4286	0.3318	1.324	5/5	13.22
$n = 2$ (Van der Pol)	Proposed (Proj)	0.0372	0.0137	0.310	2/5	0.22
	Exact GP	0.0555	0.0370	0.278	3/5	498.50
	Sparse GP	0.1771	0.0929	0.886	5/5	150.30
	Deep GP	0.1758	0.1169	1.097	4/5	415.37
	NTK	0.0015	0.0009	0.006	1/5	1.59
$n = 3$ (Lorenz plant)	Proposed (Proj)	7.7190	5.7197	30.476	3/5	4.11
	Exact GP	10.2774	8.8437	34.467	5/5	1888.80
	Sparse GP	3.2729	2.5099	19.176	1/5	264.64
	Deep GP	9.5344	8.3305	33.755	4/5	802.93
	NTK	5.7642	4.2944	18.240	2/5	2.35
$n = 6$ (coupled Duffing)	Proposed (Proj)	0.0044	0.0013	0.026	2/5	0.60
	Exact GP	0.0841	0.0695	0.279	3/5	668.01
	Sparse GP	0.1444	0.0872	0.670	5/5	523.83
	Deep GP	0.0953	0.0553	0.342	4/5	1607.48
	NTK	0.0013	0.0010	0.004	1/5	4.75

Table 9: Fair 1-D kernel comparison: per-step prediction for every method, covering inducing grid, seed-level statistics. Seeds: 3 (different reference and plant initial conditions). Steady state = last 40% of 30 s. Step time = median full online step (prediction + amortised training) after warm-up, same process type, 2 BLAS threads. Source: `results/fair_n1/table.md`.

method	mean SS $\|e\|$ (mean $\pm$ 95% CI over seeds)	max SS $\|e\|$ (mean)	median step (ms)	mean step (ms)	seeds where proposed has lower mean SS
Proposed (Proj)	0.1978 ± 0.0503	1.147	0.060	0.069	—
Exact GP	0.0591 ± 0.0061	0.780	6.309	13.936	0/3
Sparse GP (9×3 grid)	0.1867 ± 0.0210	0.699	1.353	2.732	2/3
Deep GP	0.3087 ± 0.1121	1.262	4.461	5.495	2/3
NTK	0.4700 ± 0.0250	1.438	0.212	0.503	3/3

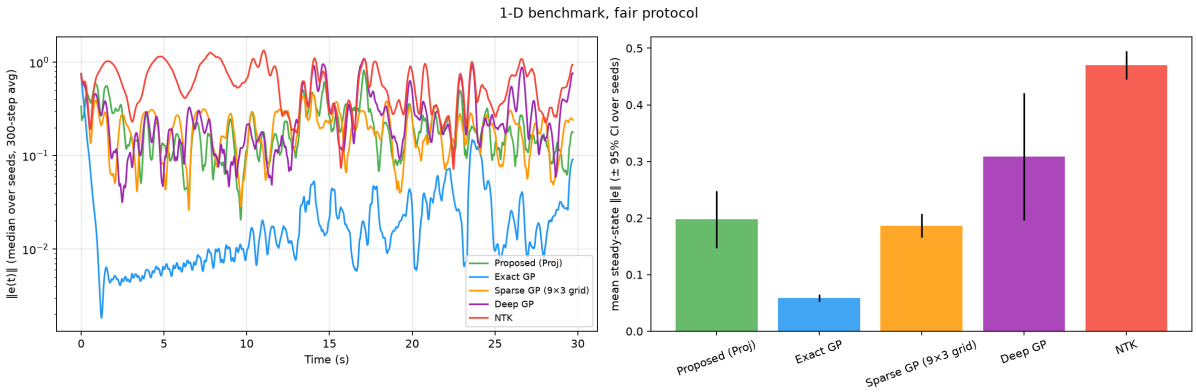


Figure 2: 1-D benchmark, fair protocol. Left: $\|e(t)\|$ (median over seeds, 300-step average), logarithmic scale, for the proposed method (Proj), the exact GP, the sparse GP (9×3 grid), the deep GP and NTK. Right: mean steady-state $\|e\|$ ($\pm$ 95% CI over seeds). Generated by `experiments/plot_fair_n1.py`; numbers in Table 9.

errors of 0.0087–0.0908; the ratio of the a-priori bound to the observed RMS error ranges from 915× to 10954×.

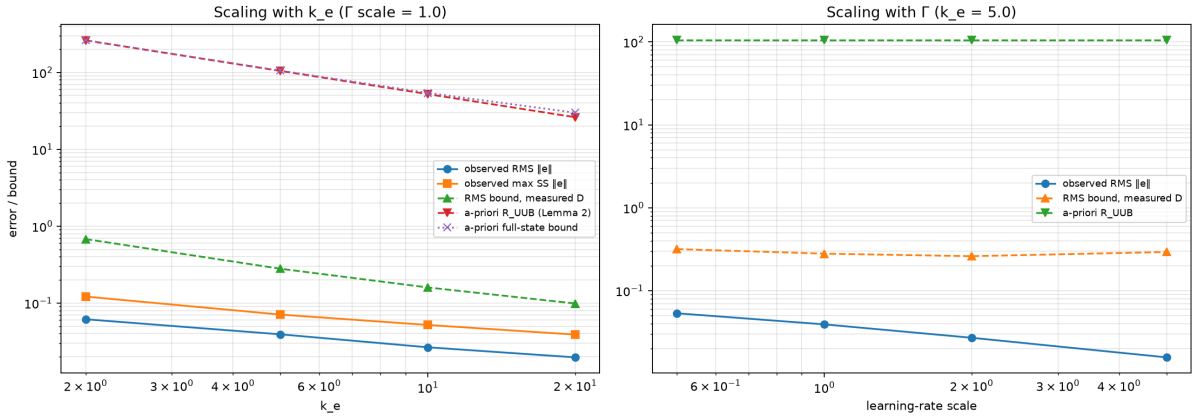


Figure 3: Oracle bound study, logarithmic axes. Left: scaling with k_e at learning-rate scale 1.0 — observed RMS $\|e\|$, observed maximal steady-state $\|e\|$, the RMS bound with the measured disturbance, the a-priori R_{UUB} (Lemma 2) and the a-priori full-state bound. Right: scaling with the learning-rate scale at $k_e = 5$ — observed RMS $\|e\|$, the RMS bound with the measured disturbance and the a-priori R_{UUB}. Generated by `experiments/exp_oracle_ruub.py`; numbers in Table 10.

Monte Carlo over initial conditions. On a plant that is not exactly representable only exactly checkable statements are tested (`experiments/exp_monte_carlo_honest.py`, Table 11): all 200 trajectories are finite, the hard parameter bounds held at every recorded step in 200/200 trials, and the a-posteriori ISS bound held in 200/200 trials (maximal ratio 0.979). No a-priori R_{UUB} threshold is drawn: $D_{\max}$ and C_Ω are not computable for this plant (the threshold 0.5 of the original figure was an arbitrary constant, not a calculated bound).

8.4 Summary

- On identical geometry, projection beats σ- and e-modification in every cell and in every seed (5/5), with paired confidence intervals excluding zero.

Table 10: Oracle R_{UUB} study (exactly representable plant, constant θ^*, $m = 8$, $n = 2$). Window $[10, 30]$ s; mean over 3 seeds. All bounds in the same units as $\|e\|$. "P viol" = samples with cross > 0 (must be 0); "inside" = hard bounds held. RMS bound (measured disturbance) satisfied in all 48 runs: True. Maximal identity error $|\dot{V} - (-k_e\|e\|^2 + e^\top d + \mathrm{cross})|$: 3.55×10^{-15}. Source: `results/oracle_ruub/table.md`.

k_e	Γ scale	RMS $\|e\|$	max SS $\|e\|$	RMS bound (D_{meas}, finite window)	D_{meas}/k_e	D_{theory}/k_e	full-state bound	ratio theory/RMS	P viol	inside
2	0.5	0.0908	0.1704	0.6791	0.5435	261.1	261.9	2875×	0	True
2	1	0.0616	0.1215	0.6801	0.6123	261.1	261.5	4239×	0	True
2	2	0.0383	0.0796	0.7139	0.6808	261.1	261.3	6824×	0	True
2	5	0.0238	0.0476	1.1766	1.1649	261.1	261.2	10954×	0	True
5	0.5	0.0533	0.1054	0.3170	0.1979	104.4	106.4	1958×	0	True
5	1	0.0392	0.0708	0.2793	0.2123	104.4	105.4	2664×	0	True
5	2	0.0270	0.0539	0.2601	0.2240	104.4	104.9	3866×	0	True
5	5	0.0157	0.0329	0.2925	0.2780	104.4	104.6	6662×	0	True
10	0.5	0.0398	0.0793	0.1970	0.0981	52.2	56.1	1313×	0	True
10	1	0.0266	0.0519	0.1596	0.1003	52.2	54.2	1965×	0	True
10	2	0.0194	0.0369	0.1395	0.1057	52.2	53.2	2688×	0	True
10	5	0.0117	0.0240	0.1335	0.1188	52.2	52.6	4466×	0	True
20	0.5	0.0285	0.0522	0.1290	0.0501	26.1	33.2	915×	0	True
20	1	0.0197	0.0389	0.0987	0.0494	26.1	29.9	1328×	0	True
20	2	0.0132	0.0260	0.0799	0.0503	26.1	28.0	1971×	0	True
20	5	0.0087	0.0168	0.0676	0.0534	26.1	26.9	3011×	0	True

Table 11: Monte Carlo over random initial conditions (Van der Pol, $m = 16$, $k_e = 5.0$, 200 trials). Source: `results/monte_carlo_honest/summary.md`.

check	result
all trajectories finite	200/200
hard parameter bounds held at every recorded step	200/200
a-posteriori ISS bound $\|e(t)\| \le \mathrm{e}^{-k_e(t-T_1)}\|e(T_1)\| + \sup\|f - \hat{f}\|/k_e$ held	200/200 (max ratio 0.979)
steady-state max $\|e\|$ (last 25%): median / 95th pct / max	0.0568 / 0.0938 / 0.1075
steady-state mean $\|e\|$: median / 95th pct / max	0.0302 / 0.0484 / 0.0593
$\sup\|f - \hat{f}\|$ in steady state / k_e: median / max	0.0627 / 0.1475
transient max $\|e\|$: max over trials	4.903

- Adaptive geometry is a smaller but consistent gain: 0.0116 for adaptive centres and a further 0.0349 for adaptive widths in mean steady-state $\|e\|$, 5/5 seeds each.
- Against kernel-regression baselines the proposed method is never the most accurate. Its edge is cost against the GP baselines (NTK is cheaper only at $n = 3$), and in the fair re-run at $n = 1$ it has the cheapest step of the five methods.
- In the fair re-run at $n = 1$ the exact GP is 3–4$\times$ more accurate than the proposed method.
- In the oracle study the a-priori bound exceeds the observed RMS error by factors of 915 to 10954, while the RMS bound with the measured disturbance holds in 48/48 runs.

Part II
Transport adaptive control

9 Setting

Plant, reference, control, error.

$$\dot{x} = f(x,t) + u, \qquad e = x - x_m(t), \qquad u = \dot{x}_m - k_e e - \hat{f}(x).$$

Lifted parameter. $\vartheta = (a, c, s) \in \mathbb{R}^n \times \mathbb{R}^n \times \mathbb{R}_{>0}$, and $p := 2n + 1$. The feature is $\Phi(x;\vartheta) := a\,\varphi(x;c,s) \in \mathbb{R}^n$, where $\varphi(x;c,s) = \exp\bigl(-\|x-c\|^2/(2s^2)\bigr)$.

Network as a measure.

$$\hat{f}(x) = \frac{1}{m}\sum_{i=1}^{m} \Phi(x;\vartheta_i) = \int \Phi(x;\vartheta)\,\mathrm{d}\mu(\vartheta), \qquad \mu := \frac{1}{m}\sum_{i=1}^{m} \delta_{\vartheta_i}.$$

The map $\mu \mapsto \hat{f}$ is linear.

Metric. $G = \mathrm{diag}(g_a I_n, g_c I_n, g_s) \succ 0$ is the gain matrix. Distances are measured in $\|v\|_{G^{-1}}^2 = v^\top G^{-1} v$, and $\lambda_G := \lambda_{\max}(G)$.

Admissible box and projection.

- **The box.** $\Theta_K := \{\|a\| \le a_{\max}\} \times \{\|c\| \le c_{\max}\} \times [s_{\min}, s_{\max}]$.
- **Layer and inner set.** ε_K is the width of the boundary layer. Shrinking each bound of Θ_K by ε_K gives the inner set Θ_K^ε, with interior $\operatorname{int}\Theta_K^\varepsilon$.
- **The operator.** $\mathrm{Proj}_K(\vartheta, y)$ is the per-particle, factor-wise projection (implemented in `src/core/projection.py` in the accompanying repository). It uses Pomet–Praly norm-ball operators [76] for a and c, with a quadratic ramp $h \in [0,1]$ (defined in Section 4), and a box operator for s, with a linear ramp.
- **Regularity.** The operator is locally Lipschitz in (ϑ, y). The norm-ball factors are C^1 at the inner edge of their layers. The box factor, and every factor at the outer edge, is only Lipschitz. The proofs use only Lipschitz continuity and ramp values in $[0,1]$.

Property (P). For $\vartheta \in \Theta_K$, $\vartheta' \in \Theta_K^\varepsilon$ and all y,

$$\langle \vartheta - \vartheta',\, G^{-1}\,\mathrm{Proj}_K(\vartheta, Gy)\rangle \le \langle \vartheta - \vartheta',\, y\rangle. \tag{P}$$

On $\partial\Theta_K$ the projected velocity has no outward component, so Θ_K is forward invariant (Nagumo; Pomet–Praly 1992 [76]).

Lemma P.1 (Proj_K is non-expansive)**.** *For every $\vartheta = (a,c,s) \in \Theta_K$ and $y = (y_a, y_c, y_s)$,*

$$\|\mathrm{Proj}_K(\vartheta, Gy)\|_{G^{-1}}^2 = \|y\|_G^2 - \sum_{b\in\{a,c,s\}} h_b(2-h_b)\,\frac{\bigl[(n_b^\top G_b y_b)_+\bigr]^2}{n_b^\top G_b n_b} \;\le\; \|y\|_G^2.$$

The terms are as follows.

- *For the balls, $n_a = a$ and $n_c = c$, with the quadratic ramps $h_a = h(\|a\|; a_{\max}, \varepsilon_K)$ and $h_c = h(\|c\|; c_{\max}, \varepsilon_K)$.*

- *For $s < s_{\min} + \varepsilon_K$, $n_s = -1$ and $h_s = 1 - \mathrm{clip}\big((s - s_{\min})/\varepsilon_K, 0, 1\big)$.*
- *For $s > s_{\max} - \varepsilon_K$, $n_s = +1$ and $h_s = 1 - \mathrm{clip}\big((s_{\max} - s)/\varepsilon_K, 0, 1\big)$.*
- *Otherwise $h_s = 0$.*

Proof. G is block-diagonal, so both squared norms split over the blocks. On an active ball block, $\mathrm{Proj}_b = G_b q_b$ with $q_b = y_b - h_b n_b\,(n_b^\top G_b y_b)/(n_b^\top G_b n_b)$. Expanding $q_b^\top G_b q_b$ gives the subtracted term. On an inactive block ($h_b = 0$ or $n_b^\top G_b y_b \le 0$), $\mathrm{Proj}_b = G_b y_b$. The s-block is the one-dimensional case. Finally $h_b \in [0,1]$, so $h_b(2 - h_b) \in [0,1]$. □

Checked by: `tests/test_transport.py`: `test_lemma_P1_projection_is_nonexpansive_and_tight_velocity_bound`.

Potential. $g_t(\vartheta) := e(t)^\top \Phi(x(t);\vartheta)$, and $F_t(\mu) := -\int g_t \,\mathrm{d}\mu = -e(t)^\top \hat{f}(x(t))$.

Adaptive law (the algorithm).

$$\dot{\vartheta}_i = \mathrm{Proj}_K\big(\vartheta_i,\, G\nabla_\vartheta g_t(\vartheta_i)\big), \qquad i = 1, \dots, m. \tag{L}$$

Ideal measure. For $j = 1, \dots, m$, $\vartheta_j^*(t) \in \Theta_K^\varepsilon$ is C^1 in t, and $\mu_t^* := \frac{1}{m}\sum_j \delta_{\vartheta_j^*(t)}$. Write $\varepsilon(x,t) := f(x,t) - \int \Phi(x;\cdot)\,\mathrm{d}\mu_t^*$, with $\|\varepsilon\| \le \varepsilon_{\max}$ on the operating set X, and set

$$V_*(t) := \Big(\frac{1}{m}\sum_j \|\dot{\vartheta}_j^*\|_{G^{-1}}^2\Big)^{1/2}, \qquad V_* := \sup_t V_*(t).$$

The proofs never use optimality of μ^*. It is an analytic device, and any admissible C^1 curve may be used.

Optimal-transport distance. For uniform atomic $\mu = \frac{1}{m}\sum \delta_{\vartheta_i}$ and $\nu = \frac{1}{m}\sum \delta_{\vartheta'_j}$,

$$W(\mu,\nu)^2 := \min_{\pi \in S_m} \frac{1}{m}\sum_i \|\vartheta_i - \vartheta'_{\pi(i)}\|_{G^{-1}}^2.$$

This is the Kantorovich 2-Wasserstein distance with ground cost $\|\cdot\|_{G^{-1}}^2$. For equal masses the optimum is attained at a permutation (Birkhoff). It is computed by linear assignment in $O(m^3)$. For general measures on Θ_K, W denotes the same Kantorovich distance.

Label-free Lyapunov function.

$$V_{\mathrm{OT}}(t) := \tfrac{1}{2}\|e(t)\|^2 + \tfrac{1}{2}W(\mu_t, \mu_t^*)^2 = \min_\pi V_\pi(t),$$
$$V_\pi := \tfrac{1}{2}\|e\|^2 + \frac{1}{2m}\sum_i \|\vartheta_i - \vartheta^*_{\pi(i)}\|_{G^{-1}}^2.$$

Standing hypotheses (S).

(S0) $k_e > 0$; $G = \mathrm{diag}(g_a I_n, g_c I_n, g_s) \succ 0$; $s_{\min} > 0$; and $0 < \varepsilon_K < \min\{a_{\max}, c_{\max}, (s_{\max} - s_{\min})/2\}$ (`src/core/projection.py` enforces the last condition).

(S1) f is continuous in (x,t) and locally Lipschitz in x, uniformly in t on bounded time intervals.

(S2) $x_m \in C^1$, and x_m and $\dot{x}_m$ are bounded.

(S3) $\vartheta_i(0) \in \Theta_K$.

(S4) The ϑ_j^* are C^1 with values in Θ_K^ε, and $V_* < \infty$.

(S5) $x(t) \in X$ on the interval considered. For bounded f, take $X := \bigcup_t \{x : \|x - x_m(t)\| \le r(F)\}$ with $F = \sup\|f\|$ (Theorem D(b)). Then $\varepsilon_{\max}$ is the supremum over this X, and it depends on $\|e(0)\|$.

Under (S0)–(S3) the right-hand side of (L) is continuous in t and locally Lipschitz in (x, ϑ). Solutions are classical (C^1) and unique on their maximal interval. Under (S4) as well, each V_π is C^1.

Notation.

- Times are written $t_0 \le t_1$ (or $T_1 < T_2$), and r is the integration variable in time.
- $\mathrm{RMS}_{[T_1,T_2]}\|e\| := \Big((T_2 - T_1)^{-1} \int_{T_1}^{T_2} \|e\|^2 \,\mathrm{d}r\Big)^{1/2}$.

Constants of the Gaussian. Closed forms (`gaussian_constants` in `src/core/transport.py` of the accompanying repository). With $u := \|x - c\|^2/s^2$ and $\varphi = e^{-u/2}$, on $s \ge s_{\min}$:

$$\|\nabla_c \varphi\| \le J_c := e^{-1/2}/s_{\min}, \qquad |\partial_s \varphi| \le J_s := 2e^{-1}/s_{\min}, \qquad J := (J_c^2 + J_s^2)^{1/2},$$
$$\|\nabla^2_{(c,s)} \varphi\| \le H_\varphi := \big(\max(h_{cc}, h_{ss}) + h_{cs}\big)/s_{\min}^2,$$
$$h_{cc} = 1, \qquad h_{ss} = 2e^{-1/2}, \qquad h_{cs} = \sqrt{u}\, e^{-u/2}|u - 2| \ \text{ evaluated at } u = (5 - \sqrt{17})/2.$$

J and H_φ are upper bounds. The true supremum of $\|\nabla_{(c,s)}\varphi\|$ is about $0.92/s_{\min}$, and that of $\|\nabla^2_{(c,s)}\varphi\|$ is about $1.57/s_{\min}^2$. The Hessian of the lifted potential is the block matrix

$$\begin{pmatrix} 0 & u\,\nabla\varphi^\top \\ \nabla\varphi\, u^\top & (u^\top a)\,\nabla^2\varphi \end{pmatrix}.$$

Hence

$$\sup_{x \in \mathbb{R}^n,\ \vartheta \in \Theta_K,\ \|u\|=1} \|\nabla^2_\vartheta \big(u^\top \Phi(x;\vartheta)\big)\| \le H_K := J + a_{\max} H_\varphi, \qquad K := \tfrac{1}{2} H_K. \tag{1}$$

A brute-force numerical check over the full lifted Hessian gives sup $\approx 0.76\, H_K$.

10 Theorem A: identification

Theorem A. *Assume* (S0).

(i) Assume scalar classical gains $\Gamma_W = \gamma_W I$, $\Gamma_c = \gamma_c I$ and $\Gamma_\sigma = \gamma_\sigma I$, and use the dictionary $W_i = a_i/m$, $\gamma_W = g_a/m$, $\gamma_c = m g_c$, $\gamma_\sigma = m g_s$.

- *The* raw (*unprojected*) *rates of* (L) *and of the classical law of Section 4 coincide.*
- *The* projected *rates coincide when the classical joint projections are replaced by per-kernel ones. The classical law projects W onto one Frobenius ball and* vec(c) *onto one Euclidean ball. The per-kernel version projects:*
  - *W_i onto $\|W_i\| \le a_{\max}/m$, with layer ε_K/m;*
  - *c_i onto $\|c_i\| \le c_{\max}$, with layer ε_K;*
  - *σ_i onto $[s_{\min}, s_{\max}]$, with layer ε_K.*
- *For $m \ge 2$, no joint ball, of any radius $W_{\max}$ and layer ε_W, reproduces the per-kernel field. For $m = 1$ the two coincide, with $W_{\max} = a_{\max}$ and layer ε_K.*

(ii) Fix a closed-loop trajectory $(x(\cdot), e(\cdot))$. Then:

- *The field* $v_t(\vartheta) := \mathrm{Proj}_K(\vartheta, G\nabla g_t(\vartheta))$ *is Lipschitz in* ϑ *on* Θ_K*, and* Θ_K *is forward invariant.*
- *For every* $\mu_0 \in \mathcal{P}(\Theta_K)$*,* $\mu_t := (X_t)_{\#}\mu_0$ *is a distributional solution of* $\partial_t\mu + \nabla\cdot(\mu v_t) = 0$*.*
- *On* $\operatorname{int}\Theta_K^\varepsilon$*,* $v_t = -G\nabla(\delta F_t/\delta\mu)$*. So at frozen time,* μ_t *is the Wasserstein gradient flow (metric* G^{-1}*) of the linear functional* F_t*, and* (L) *is its particle form.*
- F_t *is generated by the closed loop, so the coupled system is the one studied in Theorem D.*

(iii) ***Dictionary****, all in the* G^{-1} *metric:*

classical device	measure-level object
projection onto Θ_K	A velocity-level (Pomet–Praly) regularisation of the support constraint $\mu_t \in \mathcal{P}(\Theta_K)$. For $\varepsilon_K > 0$ it is not the gradient flow of any functional of μ. Formally, as $\varepsilon_K \to 0$, the field tends to the G^{-1}-projection of $G\nabla g_t$ onto the tangent cone of Θ_K, which is the velocity of the $W_{G^{-1}}$-gradient flow of $F_t + \iota_{\mathcal{P}(\Theta_K)}$ ([51, Prop. 3.1] for the exact W_2 projection).
σ-modification in the G^{-1} metric (leak outside the gain, $\dot\vartheta = G\nabla g - \sigma(\vartheta - \vartheta^0)$, as in ($\mathrm{L}_\sigma$) and Theorem H)	the potential energy $(\sigma/2)\int\|\vartheta - \vartheta^0\|^2_{G^{-1}}\,\mathrm{d}\mu$, which is geodesically σ-convex
classical σ-modification $\Gamma(y - \sigma\theta)$ (leak inside the gain; `src/baselines/sigma_mod.py` implements its W-block, with c and σ frozen)	For the full-vector leak, under (i), the object is $(\sigma/2)\int(\vartheta - \vartheta^0)^\top D(\vartheta - \vartheta^0)\,\mathrm{d}\mu$ with $D = \mathrm{diag}(I_n/m, mI_n, m)$. In the G^{-1} metric it is λ-convex with $\lambda = \sigma\cdot\min(g_a/m, mg_c, mg_s)$, which depends on m. For the W-block alone it is $(\sigma/2)\int(a - a^0)^\top(I_n/m)(a - a^0)\,\mathrm{d}\mu$.
σ-modification toward a labelled (ϑ_i^0)	not a functional of μ; it is a functional on the labelled lift $\mathbb{R}^{mp}$
OT σ-modification (Theorem C)	$(\sigma/2)W(\mu, \mu^0)^2$. For atomic μ with a unique optimal assignment, its W-gradient at atom i is $\sigma(\vartheta_i - \vartheta^0_{\tau(i)})$. At ties, ($\mathrm{L}_\sigma$) uses a measurable selection from the superdifferential $\{\sigma(\vartheta_i - \vartheta^0_{\tau(i)}) : \tau \text{ optimal}\}$.
e-modification	the same functionals with coefficient $\sigma\|e\|$
parameter dither (Theorem H)	the entropy $\tau\int\mu\log\mu$

Status. Proved and checked. Novelty (literature audit): interpretive; ingredients known [14], [20], [51], [69].

Checked by: `tests/test_transport.py`: `test_identification_with_classical_law` (raw rates in (i)); `test_identification_projected_rates_per_kernel` (projected rates and the only-if construction in (i)).

Proof. **(i) Gain map.** We have $\nabla_a g = \varphi e$, $\nabla_c g = (a^\top e)\nabla_c\varphi$ and $\partial_s g = (a^\top e)\partial_s\varphi$. Matching these with the classical rates gives the gain map.

Scaling identity. $(1/m)\,\mathrm{Proj}_{r,\varepsilon}(a, y) = \mathrm{Proj}_{r/m,\varepsilon/m}(a/m, y/m)$. Three facts give it:

- $h_{r/m,\varepsilon/m}(\|a\|/m) = h_{r,\varepsilon}(\|a\|)$;
- the direction $a(a^\top y)/\|a\|^2$ is homogeneous of degree 0 in a and of degree 1 in y;
- the activation $a^\top y > 0$ is invariant under scaling.

If the layer is not rescaled, the ramps differ.

Only if $(m \geq 2)$. Let $r_{\text{in}} := (a_{\max} - \varepsilon_K)/m$, and let the joint ball have radius $W_{\max}$, layer ε_W and $\rho := W_{\max} - \varepsilon_W$. In both cases below the driving term points outward.

- If $\rho < \sqrt{2}\, r_{\text{in}}$, take two rows of equal norm r with $\sqrt{2}\, r \in \big(\rho, \min(\sqrt{2}\, r_{\text{in}}, W_{\max})\big]$, parallel to e. The per-kernel field is inactive and the joint field is active.
- If $\rho \geq \sqrt{2}\, r_{\text{in}}$, take one row with norm in $\big(r_{\text{in}}, \min(a_{\max}/m, \rho)\big)$ and the others zero. The per-kernel field is active and the joint field is inactive.

The same argument applies to $\mathrm{vec}(c)$. For $m = 1$ the two operators coincide.

The test `test_identification_projected_rates_per_kernel` checks three things: the identity with active projections; that the layer must be rescaled; and the only-if construction against `ProjectionOperator.project_matrix` on five joint balls. Four are active against an inactive per-kernel field, and one is inactive against an active per-kernel field.

(ii) Proj_K is Lipschitz in ϑ. The ramp vanishes for $\|\vartheta\| \leq \theta_{\max} - \varepsilon_K$, so the direction $\vartheta/\|\vartheta\|$ is evaluated only away from 0. Also, ∇g_t is C^1 on $s \geq s_{\min}$.

- **Invariance.** Forward invariance follows from Nagumo's theorem, since $n_b^\top \mathrm{Proj}_b \leq 0$ on the boundary, with the outward normals n_b of Lemma P.1.
- **Continuity equation.** The push-forward of μ_0 by a Lipschitz flow solves the continuity equation (Golse [37]; Ambrosio–Gigli–Savaré [2, §8.1]).
- **Gradient form.** On $\operatorname{int} \Theta_K^\varepsilon$ the projection is inactive, so $v_t = G\nabla g_t = -G\nabla(\delta F_t/\delta\mu)$.

(iii) Potential-energy rows. These follow from the computation in the metric G^{-1}: for any V, $\nabla_{G^{-1}} V = G\nabla V$. For example, $V = (\sigma/2)\|\vartheta - \vartheta^0\|^2_{G^{-1}}$ gives $G\nabla V = \sigma(\vartheta - \vartheta^0)$.

Classical-leak row. Applying the gain map of (i) to the classical leak $\Gamma(y - \sigma\theta)$ gives the velocity $-\sigma GD(\vartheta - \vartheta^0)$.

Projection row. The G^{-1}-rescaled correction field has an asymmetric Jacobian. Take the example $m = 1$, $n = 1$, $G = I$, $x = 0.7$, $e = 1$, $\vartheta = (0.93, 0.2, 1.5)$, $a_{\max} = 1$, $\varepsilon_K = 0.15$ (so $h(0.93) = 0.284$) and $c_{\max} = 2$, with a in the layer. A numerical evaluation gives:

- the c-derivative of the a-component is -0.060;
- the a-derivative of the c-component is 0.

So the field is not a gradient. □

Remark A.1 (normalisation)**.** The algorithm's raw rates are unchanged. Two things do change: the admissible set becomes a product of per-kernel balls, and $a_{\max}$, not $\|W\|_F$, is held fixed in m.

- ***m*-free constants.** In this normalisation the structural constants k_e, K, L_0, L_1, λ_G and Δ are free of m for fixed G and $a_{\max}$. So are the constants of Theorems B, C(i)–(iii), D and F.
- ***m*-dependent data.** $\varepsilon_{\max}$ and V_* are data of the chosen m-atom ideal. The local constants of C(iv) and of Theorem E depend on m.
- **Classical convention.** Fixed classical γ's correspond to $g_a = m\gamma_W \propto m$ and $g_c, g_s \propto 1/m$. Weights speed up, centres and widths freeze, $\lambda_G \propto m$ and $\Delta \propto \sqrt{m}$.

11 Theorem B: the optimal-assignment Lyapunov function

Lemma B.1 (Dini derivative of a minimum)**.** *Suppose each V_π is C^1 and $V_{\mathrm{OT}} = \min_\pi V_\pi$. Then V_{OT} is locally Lipschitz, and $D^+ V_{\mathrm{OT}}(t) \leq \dot{V}_\pi(t)$ for every π active at t.*

Proof. For $h > 0$, $V_{\mathrm{OT}}(t + h) \leq V_\pi(t + h)$, and $V_{\mathrm{OT}}(t) = V_\pi(t)$. □

Lemma B.2 (matched remainder in two regimes; gradient bound)**.** *Let* $L_0 := 2a_{\max}$ *and* $L_1 := (1 + a_{\max}^2 J^2)^{1/2}$*. For* $\vartheta, \vartheta' \in \Theta_K$*:*

(a) the remainder $R := g_t(\vartheta') - g_t(\vartheta) - \langle \nabla g_t(\vartheta),\, \vartheta' - \vartheta \rangle$ *satisfies*

$$|R| \le \|e(t)\| \cdot \min\bigl\{ K\|\vartheta - \vartheta'\|^2,\ L_0 + L_1\|\vartheta - \vartheta'\| \bigr\};$$

(b) $\|\nabla_\vartheta g_t(\vartheta)\| \le L_1 \|e(t)\|$.

Proof. (b) We have $\nabla_a g = \varphi e$ and $\nabla_{(c,s)} g = (a^\top e)\nabla_{(c,s)}\varphi$, with $\varphi \le 1$ and $\|\nabla_{(c,s)}\varphi\| \le J$ on $s \ge s_{\min}$. So $\|\nabla g\|^2 \le \|e\|^2(1 + a_{\max}^2 J^2)$.

(a) If $e(t) = 0$, both sides vanish. Otherwise set $h := \vartheta' - \vartheta$.

- *Quadratic branch.* Θ_K is convex, so the Lagrange remainder of $\tau \mapsto g_t(\vartheta + \tau h)$ gives $R = \frac{1}{2} h^\top \nabla^2 g_t(\vartheta + \tau_0 h) h$ for some $\tau_0 \in (0,1)$. Since $g_t = \|e\| \cdot (u^\top \Phi)$ with $u = e/\|e\|$, the Hessian bound (1) of Section 9 gives this branch.
- *Linear branch.* It follows from $|g_t| \le \|e\| a_{\max}$ and (b). □

Theorem B. *Under (S), along every solution of (L), for every* t*, with* $W := W(\mu_t, \mu_t^*)$*:*

$$D^+ V_{\mathrm{OT}} \le -k_e\|e\|^2 + \|e\|\Bigl(\varepsilon_{\max} + \min\bigl\{K\lambda_G W^2,\ L_0 + L_1 \lambda_G^{1/2} W\bigr\}\Bigr) + V_*(t)\, W. \tag{B}$$

Status. Proved and checked, including a moving ideal, $\varepsilon \neq 0$, switches, an active projection, and the failure of the plain bound along (L_σ). Novelty (literature audit): packaging new; ingredients known [84], [1], [68], [16].

Checked by: `tests/test_transport.py`: `test_theorem_B_inequality_along_trajectories`, `test_theorem_B_moving_ideal_nonrepresentable_switches_and_active_projection`, `test_dini_derivative_of_V_OT_bounded_by_active_branch`, `test_permuted_ideal_labelled_vs_OT`; `experiments/exp_transport_permutation.py`.

Proof. Fix t and an optimal π, and write $\tilde\vartheta_i := \vartheta_i - \vartheta^*_{\pi(i)}$. Then

$$\dot V_\pi = e^\top \dot e + \frac{1}{m}\sum_i \langle \tilde\vartheta_i,\, G^{-1}\dot\vartheta_i\rangle - \frac{1}{m}\sum_i \langle \tilde\vartheta_i,\, G^{-1}\dot\vartheta^*_{\pi(i)}\rangle.$$

(a) $e^\top \dot e = -k_e\|e\|^2 + e^\top \varepsilon + \frac{1}{m}\sum_i \bigl[g_t(\vartheta^*_{\pi(i)}) - g_t(\vartheta_i)\bigr]$.

(b) By (P) with $\vartheta' = \vartheta^*_{\pi(i)}$, $\frac{1}{m}\sum_i \langle \tilde\vartheta_i,\, G^{-1}\dot\vartheta_i\rangle \le \frac{1}{m}\sum_i \langle \tilde\vartheta_i,\, \nabla g_t(\vartheta_i)\rangle$.

(c) Since π is a bijection and is optimal,

$$-\frac{1}{m}\sum_i \langle \tilde\vartheta_i,\, G^{-1}\dot\vartheta^*_{\pi(i)}\rangle \le \Bigl(\frac{1}{m}\sum_i \|\tilde\vartheta_i\|^2_{G^{-1}}\Bigr)^{1/2}\Bigl(\frac{1}{m}\sum_i \|\dot\vartheta^*_{\pi(i)}\|^2_{G^{-1}}\Bigr)^{1/2} = W \cdot V_*(t).$$

Summing gives

$$\dot V_\pi \le -k_e\|e\|^2 + \|e\|\varepsilon_{\max} + \frac{1}{m}\sum_i R_i + W\, V_*(t),$$

where each R_i is a matched remainder. Average Lemma B.2(a), using three facts:

- $\frac{1}{m}\sum_i \min\{A_i, B_i\} \le \min\{\mathrm{avg}\, A, \mathrm{avg}\, B\}$;
- $\frac{1}{m}\sum_i \|\tilde\vartheta_i\|^2 \le \lambda_G W^2$;
- $\frac{1}{m}\sum_i \|\tilde\vartheta_i\| \le \lambda_G^{1/2} W$.

Lemma B.1 then completes the proof. □

Corollary B.3 (ultimate bounds)**.** *Let* $\Delta := \operatorname{diam}_{G^{-1}}(\Theta_K)$*, so* $W \le \Delta$*. Define*

$$D^{\mathrm{OT}} := \varepsilon_{\max} + \min\{K\lambda_G\Delta^2,\ L_0 + L_1\lambda_G^{1/2}\Delta\}, \qquad C^{\mathrm{OT}} := V_*\Delta,$$
$$R^{\mathrm{OT}} := \frac{D^{\mathrm{OT}}}{2k_e} + \Big(\frac{(D^{\mathrm{OT}})^2}{4k_e^2} + \frac{C^{\mathrm{OT}}}{k_e}\Big)^{1/2}.$$

Then:

- $D^+V_{\mathrm{OT}} < 0$ *whenever* $\|e\| > R^{\mathrm{OT}}$*.*
- *Under (S5) on* $[0,\infty)$*,* $\limsup\|e\| \le \big((R^{\mathrm{OT}})^2 + \Delta^2\big)^{1/2}$*. This uses* $\sup \frac{1}{2}W^2 \le \Delta^2/2$*.*
- $\mathrm{RMS}_{[T_1,T_2]}\|e\| \le \dfrac{D^{\mathrm{OT}}}{2k_e} + \Big(\dfrac{(D^{\mathrm{OT}})^2}{4k_e^2} + \dfrac{C^{\mathrm{OT}}}{k_e} + \dfrac{V_{\mathrm{OT}}(T_1)}{k_e(T_2-T_1)}\Big)^{1/2}$.

Proof. As in Theorem 1 (Section 5). The Dini comparison lemma, applied with the continuous, V-independent right-hand side, gives the integrated form. □

Remark B.4 (what OT buys and what it does not)**.**

- **Globally, the OT constant is not smaller than the classical one.** Take the test box: $a_{\max} = 8$, $c_{\max} = 4$, $s_{\min} = 0.3$, $s_{\max} = 3$, $G = \mathrm{diag}(2, 1, 0.5)$ and $F_{\max} := a_{\max}$. Compare constants on the same per-kernel admissible set.

constant	value
classical Lemma 2 (Section 5), $F_{\max} + 3a_{\max} + 2a_{\max}c_{\max}J_c + a_{\max}(s_{\max} - s_{\min})J_s$	214, for every m
OT linear branch, crude form $L_0 + L_1\lambda_G^{1/2}\Delta$	533
OT linear branch, block split through W	382
OT linear branch, bounded per particle by $\|e\|\big(2a_{\max} + \|\tilde{a}_i\| + a_{\max}(J_c\|\tilde{c}_i\| + J_s\|\tilde{s}_i\|)\big)$	214

  Bounded per particle, the OT constant equals the classical one. An earlier comparison figure of "101–138" used a joint centre ball of radius $c_{\max}$. That radius is $\sqrt{m}$ times smaller than the radius $\sqrt{m}\,c_{\max}$ of the ball that circumscribes the per-kernel product of Theorem A.

- **Locally, the excess disturbance is quadratic in the matched error** (Corollary B.5).
- **Labels.** A labelled theory can make the same local statement only for a labelling fixed in advance, and only while the optimal assignment does not switch. (B) holds uniformly, with the minimum over π taken at each t.
- m**-uniformity** is a property of the normalisation (Remark A.1), not of OT.

Corollary B.5 (conditional bound)**.** *Along a solution of (L), suppose* $\sup_{[T_1,T_2]} W(\mu_t, \mu_t^*) \le \bar{\eta}$*. This quantity is available in oracle simulations, since it requires* μ_t^**, or it can be controlled a priori by Theorem E. With* $\bar{D} := \varepsilon_{\max} + \min\{K\lambda_G\bar{\eta}^2,\ L_0 + L_1\lambda_G^{1/2}\bar{\eta}\}$*,*

$$\mathrm{RMS}_{[T_1,T_2]}(\|e\|) \le \frac{\bar{D}}{2k_e} + \Big(\frac{\bar{D}^2}{4k_e^2} + \frac{V_*\bar{\eta}}{k_e} + \frac{V_{\mathrm{OT}}(T_1)}{k_e(T_2-T_1)}\Big)^{1/2}. \tag{11.1}$$

As $\bar{\eta} \to 0$ *and* $T_2 \to \infty$*, this tends to* $\varepsilon_{\max}/k_e$*, the value of the frozen-parameter error channel.*

Variant for (L_σ). *Take a solution of* (L_σ) *in the sense of Section 12 with* $\sup_{[T_1,T_2]} W(\mu_t, \mu_t^*) \le \bar{\eta}$ *and* $4\bar{\eta}(b + \bar{\eta}) < \Delta_{\mathrm{gap}}$*. For example,* $\bar{\eta}$ *can come from Theorem C(iv). Then, with* $\bar{D} := \varepsilon_{\max} + K\lambda_G\bar{\eta}^2$*,*

$$\mathrm{RMS}_{[T_1,T_2]}(\|e\|) \le \frac{\bar{D}}{2k_e} + \Big(\frac{\bar{D}^2}{4k_e^2} + \frac{V_*\bar{\eta} + \sigma_m b^2/4}{k_e} + \frac{V_{\mathrm{OT}}(T_1)}{k_e(T_2-T_1)}\Big)^{1/2}. \tag{11.2}$$

Proof of the variant. Apply Theorem C(iii) with $C = b(t)^2 \le b^2$, and use $-\sigma_m\eta^2 + \sigma_m\eta b \le \sigma_m b^2/4$. Without the gap condition, replace b^2 by $(b + 2\bar\eta)^2$ (from $\sqrt{C} \le b + 2\eta$ in C(ii)), or by $b^2 + 4\bar\eta R$, with $R := \Delta$ as in Theorem C. □

As $\bar\eta \to 0$ and $T_2 \to \infty$ the variant tends formally to $\varepsilon_{\max}/(2k_e) + \big(\varepsilon_{\max}^2/(4k_e^2) + \sigma_m b^2/(4k_e)\big)^{1/2}$. Along ($\mathrm{L}_\sigma$) with $b > 0$, the measured $\bar\eta$ stays of order b (about $0.9b$ in the test instance), so the bound is first order in b. The plain bound of Corollary B.5 is false along (L_σ). In the test instance, every hypothesis of C(iv) is asserted and $\sup W \le \bar\eta$ is measured. There, the bound of Corollary B.5 is exceeded by more than 3×, while the variant holds (`tests/test_transport.py::test_corollary_B5_fails_along_lsigma_and_its_leak_variant_holds`). Independent numerical checks found violations of 3.8–8.3×.

Status. The bound along (L) is not checked in the repository; its failure along (L_σ) and the (L_σ) variant are checked (Appendix A).

Corollary B.6 (approximation error as quantisation error)**.** *Suppose $f = \int \Phi \,\mathrm{d}\mu^*_\infty + \varepsilon_\infty$ for a probability measure μ^*_∞ on Θ^ε_K, with $\|\varepsilon_\infty\| \le \bar\varepsilon_\infty$. Then for every m-atom μ^*_m supported in Θ^ε_K,*

$$\varepsilon_{\max}(m) \le \bar\varepsilon_\infty + L_1 \lambda_G^{1/2}\, W(\mu^*_m, \mu^*_\infty).$$

Status. Not checked in the repository (Appendix A).

Proof. For any coupling γ,

$$\begin{aligned}\Big\| \int \Phi \,\mathrm{d}(\mu^*_m - \mu^*_\infty) \Big\| &\le \int \|\Phi(\vartheta) - \Phi(\vartheta')\| \,\mathrm{d}\gamma \le L_1 \lambda_G^{1/2} \int \|\vartheta - \vartheta'\|_{G^{-1}} \,\mathrm{d}\gamma \\ &\le L_1 \lambda_G^{1/2} \Big(\int \|\vartheta - \vartheta'\|^2_{G^{-1}} \,\mathrm{d}\gamma \Big)^{1/2}.\end{aligned}$$

Minimise over γ. The block constant $\big(g_a + a_{\max}^2(J_c^2 g_c + J_s^2 g_s)\big)^{1/2}$ may replace $L_1\lambda_G^{1/2}$. □

Rates.

(a) **First order.** Let μ^*_∞ be compactly supported and $p \ge 3$. Then there are equal-weight μ^*_m, with atoms (cell barycentres) in Θ^ε_K, such that $W(\mu^*_m, \mu^*_\infty) = O(m^{-1/p})$. This follows from an equal-mass dyadic decomposition (Chevallier 2018 [13]). For $p > 4$, i.i.d. atoms also achieve the rate in expectation (Fournier–Guillin 2015 [30]). If μ^*_∞ has a non-trivial absolutely continuous part, the order cannot be improved, even with free weights (Zador; Graf–Luschgy [38, Thm. 6.2 with Lemma 3.4]).

(b) **Second order.** Suppose μ^*_m is a centroid quantiser for an optimal plan; a best equal-weight approximation is one. Then the first-order term cancels, and by the Hessian bound (1) of Section 9 and convexity of Θ^ε_K,

$$\varepsilon_{\max}(m) \le \bar\varepsilon_\infty + \min\big\{ K\lambda_G W^2,\ L_1 \lambda_G^{1/2} W \big\} = \bar\varepsilon_\infty + O(m^{-2/p}).$$

(c) **Free of p.** I.i.d. atoms drawn from μ^*_∞ give, with positive probability, an m-atom ideal with

$$\varepsilon_{\max}(m) \le \bar\varepsilon_\infty + O\Big(a_{\max} \big(n \log(m \cdot \operatorname{diam} X)/m \big)^{1/2} + a_{\max}/(s_{\min}\sqrt{m}) \Big).$$

This is Maurey's empirical method; the constants are not optimised.

The exponent $1/p$ is therefore a property of controlling $\varepsilon_{\max}$ through W, as Theorem D(e) requires. It is not a property of the approximation error itself.

Remark B.7 (continuum form, via couplings)**.** Let $\mu_0 \in \mathcal{P}(\Theta_K)$ and $\mu_t = (X_t)_{\#}\mu_0$, and let μ_t^* be an absolutely continuous ideal curve in $(\mathcal{P}(\Theta_K^\varepsilon), W)$ with velocity field w_t. Set $V_* := \operatorname{ess\,sup}_t \|w_t\|_{L^2(\mu_t^*;G^{-1})}$. Its smallest admissible value is the metric derivative $|\dot\mu_t^*|_W$ (AGS [2, Thm. 8.3.1], after the isometry $\vartheta \mapsto G^{-1/2}\vartheta$). Then (B) holds for a.e. t.

Status. Not checked in the repository (Appendix A).

Sketch.

1. Push an optimal plan γ_t forward by $(X_{t,t+h}, Y_{t,t+h})$. Here Y is the flow of w, or the superposition measure of AGS [2, Thm. 8.2.1] when w is not Lipschitz.
2. This couples (μ_{t+h}, μ_{t+h}^*), so $D^+\big(\frac12 W^2\big) \le \int \langle \vartheta - \vartheta',\, v_t(\vartheta) - w_t(\vartheta')\rangle_{G^{-1}}\, \mathrm{d}\gamma_t$.
3. Integrate (P) and Lemma B.2 against γ_t. □

12 Theorem C: optimal-transport σ-modification

Let $\mu^0 = \frac1m \sum_k \delta_{\vartheta_k^0}$ be a prior configuration with $\vartheta_k^0 \in \Theta_K^\varepsilon$. The *OT σ-modified law* is

$$\begin{aligned} \dot\vartheta_i &= \mathrm{Proj}_K\big(\vartheta_i,\ G\nabla g_t(\vartheta_i) - \sigma_m(\vartheta_i - \vartheta^0_{\tau(i)})\big), \\ \tau(t) &\in \operatorname*{argmin}_{\rho}\ \frac1m \sum_i \|\vartheta_i - \vartheta^0_{\rho(i)}\|_{G^{-1}}^2. \end{aligned} \tag{L_σ}$$

Solution concept. The right-hand side of (L_σ) is discontinuous at assignment ties.

- **Selection solutions.** A solution is an absolutely continuous (x, ϑ) such that, for a.e. t, $\dot\vartheta_i = \mathrm{Proj}_K\big(\vartheta_i,\ G\nabla g_t(\vartheta_i) - \sigma_m(\vartheta_i - \vartheta^0_{\tau(t)(i)})\big)$ for some $\tau(t)$ optimal at $\vartheta(t)$. This is what `src/core/transport.py` in the accompanying repository integrates.
- **Filippov solutions.** These exist by Filippov's theorem, since the right-hand side is measurable and locally bounded. But they include sliding motions on repelling ties, along which (i) below fails. For example, take ideal = prior = $\{a = 0,\ a = 1\}$, both particles at $a = \frac12$ and $e \equiv 0$: this is a stationary Filippov solution.
- **What holds for which class.**
  - Parts (ii)–(iv) hold for every Filippov solution, with C taken as the maximum over optimal τ.
  - Part (i) and the exact decay hold for selection solutions.
  - For Filippov solutions, (i) weakens as stated below.
- **Regularity and existence.** V_{OT} and Z are absolutely continuous along solutions, and every inequality holds for a.e. t. Existence of selection solutions is proved for the pure leak (Lemma C.3). In general it is assumed: two-way ties are never attracting, by monotonicity of Proj_K, but multi-way ties, the degenerate ramp $h = 1$ and accumulating switch times are not handled.

Notation.

- $c_\rho(\vartheta) := \frac1m \sum_i \|\vartheta_i - \vartheta^0_{\rho(i)}\|_{G^{-1}}^2$.
- $\eta := W(\mu_t, \mu_t^*)$, and $R := \Delta = \mathrm{diam}_{G^{-1}}(\Theta_K)$.
- $b(t) := W(\mu_t^*, \mu^0)$, and $b := \sup_t b(t)$.

- $\Delta_{\mathrm{gap}}(t) := \min\{c_\rho(\vartheta^*(t)) - b(t)^2 : c_\rho(\vartheta^*(t)) > b(t)^2\}$, with $\min\emptyset := +\infty$; and $\Delta_{\mathrm{gap}} := \inf_t \Delta_{\mathrm{gap}}(t)$.
- $A_{\tau\tau'} := \frac{1}{m}\sum_i \|\vartheta^0_{\tau(i)} - \vartheta^0_{\tau'(i)}\|^2_{G^{-1}}$.

Theorem C. *Under* (S)*, along every solution of* (L_σ) *in the sense above:*

(i) Dissipation toward the prior (selection solutions). *Let* $Z := \frac{1}{2}W(\mu_t,\mu^0)^2$ *and* $\Gamma := \big(\frac{1}{m}\sum_i \|G\nabla g_t(\vartheta_i)\|^2_{G^{-1}}\big)^{1/2} \le \lambda_G^{1/2} L_1\|e\|$*. Then:*

- $D^+Z \le -2\sigma_m Z + (2Z)^{1/2}\Gamma(t)$*;*
- *hence* $W(\mu_t,\mu^0) \le \max\{W(\mu_0,\mu^0),\ \sup\Gamma/\sigma_m\}$*;*
- *for the pure-leak flow* ($\Gamma \equiv 0$)*, the decay* $W(\mu_t,\mu^0) = e^{-\sigma_m t}W(\mu_0,\mu^0)$ *is exact for every* $G \succ 0$ *(Lemma C.3).*

For a Filippov solution with weights θ_k *on tied optimal assignments* τ_k*, the bound weakens to*

$$D^+Z \le -2\sigma_m Z + \frac{\sigma_m}{2}\min_\tau \sum_k \theta_k A_{\tau\tau_k} + (2Z)^{1/2}\Gamma,$$

with the minimum over assignments τ *optimal at* t*.*

(ii) Induced coupling. *Let* π^* *be optimal from* μ *to* μ^**, and let* τ *be the optimal assignment from* μ *to* μ^0 *that the law uses at* t *(for Filippov solutions, any optimal* τ*). Set* $C := \frac{1}{m}\sum_i \|\vartheta^*_{\pi^*(i)} - \vartheta^0_{\tau(i)}\|^2_{G^{-1}}$*. Then*

$$C \le b(t)^2 + 4\eta R \qquad \text{and} \qquad \sqrt{C} \le b(t) + 2\eta,$$

and $C = b(t)^2$ *whenever* $4\eta(b(t)+\eta) < \Delta_{\mathrm{gap}}$*.*

(iii) Lyapunov inequality. *With* C *as in* (ii) *(for Filippov solutions, its maximum over optimal* τ*),*

$$D^+V_{\mathrm{OT}} \le -k_e\|e\|^2 + \|e\|\big(\varepsilon_{\max} + K\lambda_G\eta^2\big) + V_*\eta - \sigma_m\eta^2 + \sigma_m\eta C^{1/2}.$$

Under $4\eta(b+\eta) < \Delta_{\mathrm{gap}}$ *this becomes*

$$D^+V_{\mathrm{OT}} \le -k_e\|e\|^2 + \|e\|\big(\varepsilon_{\max} + K\lambda_G\eta^2\big) + V_*\eta - \frac{\sigma_m}{2}\eta^2 + \frac{\sigma_m}{2}b^2.$$

(iv) Local exponential estimate. *Suppose* $4\bar\eta(b+\bar\eta) < \Delta_{\mathrm{gap}}$ *and* $8K^2\lambda_G^2\bar\eta^2 \le k_e\sigma_m$*. Set* $\alpha := \min(k_e, \sigma_m/2)$ *and* $\beta := \varepsilon^2_{\max}/k_e + (\sigma_m/2)b^2 + 2V_*^2/\sigma_m$*, and suppose* $\max\{V_{\mathrm{OT}}(0), \beta/\alpha\} < \bar\eta^2/2$*. Then:*

- $\{V_{\mathrm{OT}} \le c\}$ *is forward invariant for every* $c \in \big[\max\{V_{\mathrm{OT}}(0), \beta/\alpha\},\ \bar\eta^2/2\big)$*;*
- $V_{\mathrm{OT}}(t) \le e^{-\alpha t}V_{\mathrm{OT}}(0) + \beta/\alpha$*;*
- *on this set* $\|e\| < \bar\eta$*, so* (S5) *holds with* $X := \bigcup_t\{x : \|x - x_m(t)\| \le \bar\eta\}$*.*

(v) The labelled leak. *Define* $b_{\mathrm{lab}}(\pi)(t) := \big(\frac{1}{m}\sum_i \|\vartheta^*_{\pi(i)}(t) - \vartheta^0_i\|^2_{G^{-1}}\big)^{1/2}$ *and* $b_{\mathrm{lab}}(\pi) := \sup_t b_{\mathrm{lab}}(\pi)(t)$*.*

- *For any fixed* π*, the labelled* σ*-modification (leak toward* ϑ^0_i*) satisfies the second display of* (iii) *at every state, with no gap condition. Here* b *is replaced by* $b_{\mathrm{lab}}(\pi)$*,* V_{OT} *by* V_π*, and* η *by the* π*-labelled error.*
- *Choosing* π *to minimise* $b_{\mathrm{lab}}(\pi)$ *gives the bias* $\min_\pi \sup_t b_{\mathrm{lab}}(\pi)(t) \ge b$*. Equality holds when one assignment is optimal for* (μ^*_t, μ^0) *at every* t*, for example a static ideal, or whenever* $\Delta_{\mathrm{gap}} > 0$*.*

- *The labelled leak does* ***not*** *satisfy* (iii) *for* V_{OT} (*Proposition C.5*).

Status. Proved and checked; the instance used to check (iv) satisfies every hypothesis of (iv). Not checked in the repository: the Filippov form of (i) and the inequality of (v). Novelty (literature audit): novel as a control construction; C(i) and the obstruction are known [80], [9].

Checked by: `tests/test_transport.py`:
(i) and Lemma C.3 (unprojected pure leak): `test_theorem_C_i_exact_dissipation_pure_leak`;
(ii): `test_lemma_C_ii_induced_coupling_bound`;
(iv): `test_theorem_C_iv_exponential_estimate`.
(v) and Proposition C.5, in `experiments/`: `exp_ot_sigma_mod.py` and `exp_sigma_mod_transient.py`.

Scope.

- **Finite gap.** If $\Delta_{\mathrm{gap}} < \infty$, that is, not all ideal–prior assignments tie, then $\Delta_{\mathrm{gap}} \le 2R^2/m$ always. Indeed, if every transposition of an optimal ρ_0 were cost-neutral, the matrix $S(i,j) := B(i,j) - (B(i,i) + B(j,j))/2$ built from $B(i,j) := \langle \vartheta_i^*, \vartheta^0_{\rho_0(j)} \rangle_{G^{-1}}$ would be antisymmetric, and every assignment would tie.
  - Hence (iv) needs $\bar\eta < \sqrt{\Delta_{\mathrm{gap}}}/2 \le R/\sqrt{2m}$.
  - $\beta/\alpha < \bar\eta^2/2$ forces $b < \bar\eta/\sqrt{2}$.
  - So (iv) is a near-prior local result of size $O(m^{-1/2})$ in W, and $O(1/m)$ in V_{OT}.
- **Infinite gap.** If $\Delta_{\mathrm{gap}} = +\infty$, for example for a single-atom prior (the classical σ-modification), the gap condition is void. Then (iv) is m-uniform; $\bar\eta$ is then fixed by $8K^2\lambda_G^2\bar\eta^2 \le k_e\sigma_m$ alone.
- **Switching.** $\Delta_{\mathrm{gap}} > 0$ forces the set of optimal ideal–prior assignments to be constant in t. If that set switches along a moving ideal, then $\Delta_{\mathrm{gap}} = 0$, and the second clause of (ii), the second display of (iii) and all of (iv) are vacuous.

Proof. (i) For a selection solution, at a.e. t the law uses an optimal $\tau(t)$, and $Z \le Z_\tau := \frac{1}{2}c_\tau$ with equality at t. By Lemma B.1 it suffices to bound $\dot Z_\tau$. Apply (P) with $\vartheta' = \vartheta^0_{\tau(i)}$ and Cauchy–Schwarz in the G^{-1} inner product: $\dot Z_\tau \le (2Z)^{1/2}\Gamma - 2\sigma_m Z$. Write $y := (2Z)^{1/2}$.

- Where $Z > 0$, $D^+ y \le -\sigma_m y + \Gamma$.
- At $Z = 0$, $D^+ y \le \big(\frac{1}{m}\sum \|\dot\vartheta_i\|^2_{G^{-1}}\big)^{1/2} \le \Gamma$, by Lemma P.1 (the leak term vanishes there).
- $\Gamma \le \lambda_G^{1/2} L_1 \|e\|$ by Lemma B.2(b).

The pure-leak case is Lemma C.3, and the projection is inactive there. In the layer, $a^\top(a^0 - a) \le \|a\|(\|a^0\| - \|a\|) \le 0$, and similarly for c. For s, the leak points away from the nearby face.

For Filippov velocities, the leak part averaged over the tie is

$$-\sigma_m \frac{1}{m}\sum_i \sum_k \theta_k \langle \vartheta_i - \vartheta^0_{\tau(i)},\, \vartheta_i - \vartheta^0_{\tau_k(i)} \rangle_{G^{-1}}.$$

By polarisation, and since $c_\tau = c_{\tau_k} = 2Z$ for tied optimal assignments, this equals $-2\sigma_m Z + (\sigma_m/2)\sum_k \theta_k A_{\tau\tau_k}$. Property (P) holds termwise for the convex combination. The bound is tight at the tie equilibrium of the solution-concept example ($0 = -\sigma_m/4 + \sigma_m/4$).

(ii) Relabel so that $\frac{1}{m}\sum \|\vartheta_i - \vartheta_i^*\|^2_{G^{-1}} = \eta^2$. For any ρ:

- $|c_\rho(\vartheta) - c_\rho(\vartheta^*)| \le 2\eta R$;
- $|\sqrt{c_\rho(\vartheta)} - \sqrt{c_\rho(\vartheta^*)}| \le \eta$, by the reverse triangle inequality for the $(1/m)$-weighted G^{-1} norm.

Let ρ_0 be optimal for $(\vartheta^*(t), \vartheta^0)$. Then

$$C = c_\tau(\vartheta^*) \le c_\tau(\vartheta) + 2\eta R \le c_{\rho_0}(\vartheta) + 2\eta R \le b(t)^2 + 4\eta R,$$
$$\sqrt{C} \le \sqrt{c_\tau(\vartheta)} + \eta \le \sqrt{c_{\rho_0}(\vartheta)} + \eta \le b(t) + 2\eta.$$

So $C - b(t)^2 \le 4\eta(b(t) + \eta) < \Delta_{\mathrm{gap}} \le \Delta_{\mathrm{gap}}(t)$. No assignment cost at time t lies strictly between $b(t)^2$ and $b(t)^2 + \Delta_{\mathrm{gap}}(t)$, so $C = b(t)^2$.

(iii) Apply Theorem B with $y = \nabla g_t - \sigma_m G^{-1}(\vartheta_i - \vartheta^0_{\tau(i)})$ in (P). With $\tilde{\vartheta}_i := \vartheta_i - \vartheta^*_{\pi^*(i)}$ as in the proof of Theorem B, the extra term is

$$-\sigma_m \frac{1}{m} \sum \langle \tilde{\vartheta}_i,\, \tilde{\vartheta}_i + \vartheta^*_{\pi^*(i)} - \vartheta^0_{\tau(i)} \rangle_{G^{-1}} \le -\sigma_m \eta^2 + \sigma_m \eta C^{1/2}.$$

Under the gap condition, (ii) gives $C^{1/2} = b(t) \le b$, and $\sigma_m \eta b \le (\sigma_m/2)(\eta^2 + b^2)$.

(iv) Apply Young's inequality three times:

- $\|e\|\varepsilon_{\max} \le (k_e/4)\|e\|^2 + \varepsilon^2_{\max}/k_e$;
- $\|e\| K \lambda_G \eta^2 \le (k_e/4)\|e\|^2 + (\sigma_m/8)\eta^2$ on $\{\eta \le \bar{\eta}\}$;
- $V_* \eta \le (\sigma_m/8)\eta^2 + 2V_*^2/\sigma_m$.

This gives $D^+ V_{\mathrm{OT}} \le -\alpha V_{\mathrm{OT}} + \beta$. Each set $\{V_{\mathrm{OT}} \le c\}$ with $\beta/\alpha < c < \bar{\eta}^2/2$ lies inside $\{\eta \le \bar{\eta}\}$ and has a negative boundary derivative. Intersect over $c > \beta/\alpha$ for the endpoint, and apply the comparison lemma.

(v) Steps (a)–(c) of Theorem B hold for any fixed π, so use the classical completion of squares for V_π.

- *Equality case.* If $\Delta_{\mathrm{gap}} > 0$, the costs minus $b(t)^2$ take values in $\{0\} \cup [\Delta_{\mathrm{gap}}, \infty)$ and are continuous in t, so the optimal set is constant.
- *Strict case.* If the optimal ideal–prior assignment changes along the ideal, the inequality can be strict. For example, $a^*_1(t) = t$ and $a^*_2(t) = 1 - t$ on $[0, 1]$, with prior $a \in \{0, 1\}$, give $b = 0.5$ against 1.0.
- *Not always strict.* For $a^*(t) = (u(t),\, 1 - u(t))$ with $u(t) = -0.025 + 0.575 \sin t$, the optimal assignment alternates between the identity and the swap, yet $\min_\pi \sup_t b_{\mathrm{lab}}(\pi)(t) = \sup_t b_{\mathrm{lab}}(\mathrm{id})(t) = 0.6 = b$ (computed numerically).

The negative statement is Proposition C.5. □

Lemma C.3 (no switching along the pure leak; exact decay for any $G \succ 0$)**.** *Let τ be optimal for $(\mu_{\mathrm{init}}, \mu^0)$ and $\dot{\vartheta}_i = -\sigma_m(\vartheta_i - \vartheta^0_{\tau(i)})$. Then τ stays optimal and $W(\mu_t, \mu^0) = e^{-\sigma_m t} W(\mu_{\mathrm{init}}, \mu^0)$.*

Proof. Put $d_i := \vartheta_i(0) - \vartheta^0_{\tau(i)}$, so $\vartheta_i(t) - \vartheta^0_{\tau(i)} = \lambda d_i$ with $\lambda = e^{-\sigma_m t}$. For any ρ, with $q_i := \vartheta^0_{\tau(i)} - \vartheta^0_{\rho(i)}$,

$$c_\rho(t) - c_\tau(t) = \frac{2\lambda}{m} \sum \langle d_i,\, q_i \rangle_{G^{-1}} + \frac{1}{m} \sum \|q_i\|^2_{G^{-1}}.$$

This is affine in λ. It is non-negative at $\lambda = 1$ by optimality and at $\lambda = 0$ trivially, hence on all of $[0, 1]$.

At an exact tie at $t = 0$, the chosen assignment becomes strictly optimal immediately against every ρ with $A_\rho := \frac{1}{m} \sum \|q_i\|^2 > 0$. Assignments with $A_\rho = 0$ induce the same velocity. So no selection solution of the pure leak switches. Filippov sliding on the tie is a different solution, excluded by the solution concept. □

Checked by: `test_theorem_C_i_exact_dissipation_pure_leak` in `tests/test_transport.py`: zero switches, and exact decay to integrator accuracy.

Proposition C.5 (the labelled leak reshuffles)**.** *There are configurations with* $b = b_{\mathrm{lab}}(\mathrm{id}) = 0$ *and* $e = 0$ *at which the labelled leak gives* $D^+V_{\mathrm{OT}} > 0$*.*

Example. Take $n = 1$, $G = I$ and $a_{\max} - \varepsilon_K \geq 1$. Let ideal = prior = $\{a = 0,\ a = 1\}$, with the other coordinates equal and in the inner set, and let the network weights be $(1 - \delta, \delta)$ with $0 < \delta < \frac{1}{2}$.

- The optimal assignment to the ideal is the swap.
- The labelled leak moves each particle toward the *other* ideal atom.
- So $D^+V_{\mathrm{OT}} = \sigma_m(\delta - \delta^2) > 0$. □

From a relabelled start, the labelled leak moves every kernel in a non-trivial cycle of the relabelling to its prior position. In the reshuffling experiment (`results/ot_sigma_mod/transient.md` in the accompanying repository):

- the steady states coincide;
- the labelled leak's particle path is 5.8× longer (2.26 against 0.39);
- the early RMS$\|e\|$ is only about 4% larger (0.0771 against 0.0743 on $[0, 5]$).

Theorem F does not price this motion. With a leak, Theorem F bounds transport from above by tracking error plus a leak term, and the leak can pay for all of it. With $\delta = 0$ above, $e \equiv 0$ while every particle crosses the domain.

Checked by: the reshuffling transient, `experiments/exp_sigma_mod_transient.py`. The value $D^+V_{\mathrm{OT}} = \sigma_m(\delta - \delta^2)$ is not checked in the repository.

Remark C.6 (the obstruction is known)**.**

- **Non-convexity.** For some ν, for example a two-atom ν in dimension ≥ 2, $\frac{1}{2}W^2(\cdot, \nu)$ is not λ-convex along geodesics for any λ (Ambrosio–Gigli–Savaré [2, Example 9.1.5]; Santambrogio [80, §4.5]). It is 1-semiconcave in every dimension [2, Thm. 7.3.2]. For $\nu = \delta_{\vartheta^0}$ it is a potential energy, and 1-convex.
- **Generalised geodesics.** $\frac{1}{2}W^2(\cdot, \nu)$ is 1-convex along generalised geodesics with base ν ([2, Lemma 9.2.1]; Carlen–Craig [9]).
- **Label-free leaks.**
  - A label-free leak that is λ-convex ($\lambda > 0$) on the lift $\mathbb{R}^{mp}$ has a unique, permutation-invariant minimiser, at which all particles coincide. So it cannot vanish at a μ^0 with distinct atoms.
  - 0-convex leaks that vanish at μ^0 exist, for example toward $\operatorname{conv}\operatorname{supp}\mu^0$. They do not single out μ^0.

So the trade-off reflects this geometry. On one side is a global inequality for a labelled V_π, which is large at relabelled starts (hence the reshuffling of Proposition C.5). On the other is a local, gap-protected inequality for the label-free V_{OT}.

13 Theorem C′: the OT leak toward a prior with a different number of atoms

Theorem C needs a prior with as many atoms as the network. Here the prior is any $\nu \in \mathcal{P}_2(\Theta_K^\varepsilon)$, for example the empirical measure of a network of a different size (the transfer problem of Section 26, branch 8), and the leak points at the barycentre of the prior mass that the optimal plan assigns to each particle. The dissipation toward the prior survives up to the quantisation

error, the fixed points of the pure leak are the equal-mass centroidal configurations of ν (the leak is Lloyd's algorithm in continuous time), and the Lyapunov bias is measured by the distance of the ideal from the *centroids* of its prior cells, not by $W(\mu^*, \nu)$.

The law. Let $\nu \in \mathcal{P}_2(\Theta_K^\varepsilon)$, for instance $\nu = \frac{1}{m_0}\sum_{j=1}^{m_0} \delta_{\vartheta_j^0}$ with any $m_0 \geq 1$. For $\mu = \frac{1}{m}\sum \delta_{\vartheta_i}$, let γ be a vertex optimal plan from μ to ν for the cost $\|\cdot\|_{G^{-1}}^2$ (for atomic ν a vertex of the transportation polytope; for general ν any optimal plan), and let

$$\bar{T}_i := m \int \vartheta' \, \mathrm{d}\gamma(\vartheta_i, \vartheta')$$

(the barycentric projection of γ at ϑ_i; for atomic ν, $\bar{T}_i = m\sum_j \gamma_{ij}\vartheta_j^0$),

$$V_{\mathrm{bar}}^2 := \frac{1}{m}\sum_i \|\vartheta_i - \bar{T}_i\|_{G^{-1}}^2, \qquad \mathrm{Var}_\gamma := \int \|\vartheta' - \bar{T}(\vartheta)\|_{G^{-1}}^2 \, \mathrm{d}\gamma(\vartheta, \vartheta'),$$
$$\text{so that} \qquad W(\mu,\nu)^2 = V_{\mathrm{bar}}^2 + \mathrm{Var}_\gamma.$$

The **barycentric OT leak** is

$$\dot{\vartheta}_i = \mathrm{Proj}_K\big(\vartheta_i,\ G\nabla g_t(\vartheta_i) - \sigma_m(\vartheta_i - \bar{T}_i)\big). \tag{$\mathrm{L}_{\sigma,\mathrm{bar}}$}$$

For atomic ν the plan is a vertex of the transportation polytope (row sums $1/m$, column sums $1/m_0$), computed by linear programming (`transport.optimal_plan`, `transport.barycentric_projection`); for $m_0 = m$ it is a permutation and ($\mathrm{L}_{\sigma,\mathrm{bar}}$) is ($\mathrm{L}_\sigma$). The right-hand side is discontinuous where the optimal plan is not unique, and here, unlike Section 12 (Lemma C.3), the plan does switch along solutions (numerics below), so two solution concepts are used. *Vertex-selection solutions* are absolutely continuous $\vartheta(t)$ with $\dot{\vartheta}_i = \mathrm{Proj}_K\big(\vartheta_i,\ G\nabla g_t - \sigma_m(\vartheta_i - \bar{T}_i^{\gamma(t)})\big)$ for a.e. t and $\gamma(t)$ a vertex optimal plan at $\vartheta(t)$; the time-stepped implementation produces these (switches at step times), and their existence for all time is assumed. *Generalised solutions* allow $\gamma(t)$ to be any measurable selection of optimal plans. The optimal face $F(\vartheta)$ of the transportation LP is a compact convex polytope, upper semicontinuous in ϑ (finitely many vertices, continuous costs), and $\gamma \mapsto \bar{T}^\gamma$ is linear, so the set of generalised velocities is compact, convex and upper semicontinuous in ϑ: it contains the Filippov regularisation of the vertex law, and generalised solutions exist on $[0,\infty)$ by the standard theorem for such differential inclusions (the projection is handled as in Section 12). Every statement below holds for both concepts, with $\bar{T}$ and V_{bar} computed from the plan the solution uses. Since Θ_K^ε is convex, $\bar{T}_i \in \Theta_K^\varepsilon$, so property (P) applies with $\vartheta' = \bar{T}_i$.

Notation. $\eta := W(\mu_t, \mu_t^*)$, $b := \sup_t W(\mu_t^*, \nu)$, and

$$b_{\mathrm{bar}}(t)^2 := \frac{1}{m}\sum_j \|\vartheta_j^*(t) - \bar{T}_j^*(t)\|_{G^{-1}}^2 = W(\mu_t^*,\nu)^2 - \mathrm{Var}_{\gamma^*(t)}, \qquad b_{\mathrm{bar}} := \sup_t b_{\mathrm{bar}}(t),$$

the *centroid defect of the ideal*, computed with an optimal plan $\gamma^*(t)$ from μ_t^* to ν. $\Delta_{\mathrm{gap}}(t)$ is the gap between the optimal and the next vertex cost of the transportation polytope at μ_t^* ($\min\emptyset := +\infty$), and $\Delta_{\mathrm{gap}} := \inf_t \Delta_{\mathrm{gap}}(t)$.

Theorem C′. *Under* (S)*, along every vertex-selection or generalised solution of* ($\mathrm{L}_{\sigma,\mathrm{bar}}$)*:*

(i) Dissipation toward the centroidal configurations. *With* $Z := \frac{1}{2}W(\mu_t,\nu)^2$ *and* Γ *as in Theorem C(i),*

$$D^+ Z \leq -\sigma_m V_{\mathrm{bar}}^2 + V_{\mathrm{bar}}\Gamma \leq -2\sigma_m\big(Z - \tfrac{1}{2}\mathrm{Var}_\gamma\big) + (2Z)^{1/2}\Gamma \qquad \textit{for a.e. } t.$$

For the pure leak ($\Gamma \equiv 0$), $W(\mu_t, \nu)$ is non-increasing, strictly decreasing wherever the plan in use is not centroidal ($V_{\mathrm{bar}} > 0$), and at every ω-limit point ϑ_∞ of a solution on the compact Θ_K some optimal plan γ_∞ at ϑ_∞ is centroidal: $\vartheta_{\infty,i} = \bar{T}_i^{\gamma_\infty}$ for all i — a vertex plan for a vertex-selection solution. Such configurations are the fixed points of the equal-mass Lloyd iteration $\vartheta \leftarrow \bar{T}^{\gamma}(\vartheta)$ with an optimal (vertex) plan; where the optimal plan is unique they are the critical points of the quantisation energy $\vartheta \mapsto \frac{1}{2}W^2\big(\frac{1}{m}\sum \delta_{\vartheta_i}, \nu\big)$, which is differentiable there, and the pure leak is its G-gradient flow with the factor m: $\dot{\vartheta}_i = -\sigma_m m\, G\, \partial_{\vartheta_i}\big[\frac{1}{2}W^2\big]$. They include degenerate configurations with coincident particles. For $m_0 = m$, $\mathrm{Var}_\gamma = 0$ and (i) *is Theorem C(i).*

(ii) Induced coupling. *Let π^* be optimal from μ to μ^*, and let γ be the plan the law uses at t. Put $C_{\mathrm{bar}} := \frac{1}{m}\sum_i \|\vartheta^*_{\pi^*(i)} - \bar{T}_i\|^2_{G^{-1}}$. Then*

$$\sqrt{C_{\mathrm{bar}}} \le b(t) + 2\eta \quad \textit{always},$$

and, for atomic ν, $C_{\mathrm{bar}} = b_{\mathrm{bar}}(t)^2$ whenever $4\eta(b(t)+\eta) < \Delta_{\mathrm{gap}}(t)$ and $\gamma^*(t)$ is unique.

(iii) Lyapunov inequality. *With C_{bar} as in* (ii),

$$D^+ V_{\mathrm{OT}} \le -k_e\|e\|^2 + \|e\|\big(\varepsilon_{\max} + K\lambda_G\eta^2\big) + V_*\eta - \sigma_m\eta^2 + \sigma_m\eta\sqrt{C_{\mathrm{bar}}},$$

and under the gap condition of (ii) *the leak terms are at most $-(\sigma_m/2)\eta^2 + (\sigma_m/2)b^2_{\mathrm{bar}}$.*

(iv) Local exponential estimate. *For atomic ν, Theorem C(iv) holds verbatim with b replaced by b_{bar}: under $4\bar{\eta}(b+\bar{\eta}) < \Delta_{\mathrm{gap}}$, uniqueness of $\gamma^*(t)$, and $8K^2\lambda_G^2\bar{\eta}^2 \le k_e\sigma_m$, with $\alpha := \min(k_e, \sigma_m/2)$ and $\beta := \varepsilon^2_{\max}/k_e + (\sigma_m/2)b^2_{\mathrm{bar}} + 2V_*^2/\sigma_m$, if $\max\{V_{\mathrm{OT}}(0), \beta/\alpha\} < \bar{\eta}^2/2$ then $V_{\mathrm{OT}}(t) \le e^{-\alpha t}V_{\mathrm{OT}}(0) + \beta/\alpha$ and $\{V_{\mathrm{OT}} \le c\}$ is forward invariant for $c \in \big[\max\{V_{\mathrm{OT}}(0), \beta/\alpha\},\ \bar{\eta}^2/2\big)$.*

(v) What the bias measures. *$b_{\mathrm{bar}} \le b$, with $b_{\mathrm{bar}} = 0$ exactly when the ideal is itself an equal-mass centroidal configuration of ν. So a prior with many atoms whose centroidal m-quantisation coincides with the ideal anchors the network without bias, although $W(\mu^*, \nu) > 0$; a prior with the right number of atoms but the wrong positions has $b_{\mathrm{bar}} = b$.*

Status. Proved and checked: dissipation at every step and convergence of the pure leak to an equal-mass Lloyd fixed point, the coupling bound and plan stability, the Lyapunov inequality along the closed loop (`tests/test_barycentric.py`, `exp_barycentric_leak.py`). Novelty (literature audit): new as a control construction; the quantisation side is Lloyd/CVT theory; it reduces to Theorem C for $m_0 = m$.

Scope of (ii)–(iv). The gap condition $4\eta(b+\eta) < \Delta_{\mathrm{gap}}(t)$ asks the network to be closer to the ideal than the optimal vertex of the LP is to the next one. For a prior with many atoms, Δ_{gap} is at most the cost of exchanging two atoms between two cells, which is $O(1/m_0)$, so the condition holds only in a small neighbourhood of the ideal; in the closed-loop experiment below $4\eta(0)(b+\eta(0))$ exceeds that bound by factors 83–175, and (iv) applies to no tested instance (the table reports the exchange bound on Δ_{gap}, $\eta(0)$ and $4\eta(0)(b+\eta(0))$ for each run). What the experiment confirms is the fixed-point structure, (i) and (v): the loop settles at the centroidal configuration of its prior, whose distance from the ideal is b_{bar}.

Proof. (i) At a.e. t the law uses an optimal plan γ, and for $h > 0$ the same plan couples μ_{t+h} and ν, so $W(\mu_{t+h}, \nu)^2 \le \sum_{ij}\gamma_{ij}\|\vartheta_i(t+h) - \vartheta^0_j\|^2_{G^{-1}}$ (for general ν, $\int\|\vartheta_i(t+h) - \vartheta'\|^2\,\mathrm{d}\gamma$). Differentiating the right-hand side at $h = 0$,

$$D^+ Z \le \sum_{ij}\gamma_{ij}\langle\vartheta_i - \vartheta^0_j,\, G^{-1}\dot{\vartheta}_i\rangle = \frac{1}{m}\sum_i\langle\vartheta_i - \bar{T}_i,\, G^{-1}\dot{\vartheta}_i\rangle,$$

because $\sum_j \gamma_{ij}(\vartheta_i-\vartheta_j^0)=\frac{1}{m}(\vartheta_i-\bar T_i)$. Apply (P) with $\vartheta'=\bar T_i\in\Theta_K^\varepsilon$ and $y=\nabla g_t-\sigma_m G^{-1}(\vartheta_i-\bar T_i)$: $\langle\vartheta_i-\bar T_i,\,G^{-1}\dot\vartheta_i\rangle\le\langle\vartheta_i-\bar T_i,\,\nabla g_t(\vartheta_i)\rangle-\sigma_m\|\vartheta_i-\bar T_i\|^2_{G^{-1}}$. Average and use Cauchy–Schwarz for the first term: $\frac{1}{m}\sum\langle\vartheta_i-\bar T_i,\,\nabla g_t\rangle\le V_{\mathrm{bar}}\Gamma\le(2Z)^{1/2}\Gamma$. The Pythagorean identity $W^2=V_{\mathrm{bar}}^2+\mathrm{Var}_\gamma$ follows from $\sum_j\gamma_{ij}(\vartheta_j^0-\bar T_i)=0$: for each i, $\sum_j\gamma_{ij}\|\vartheta_i-\vartheta_j^0\|^2=\sum_j\gamma_{ij}\big(\|\vartheta_i-\bar T_i\|^2+\|\bar T_i-\vartheta_j^0\|^2\big)$. For the pure leak, Z is non-increasing and $D^+Z\le-\sigma_m V_{\mathrm{bar}}(t)^2$, with V_{bar} computed from the plan in use (the coupling inequality above holds for every optimal plan, vertex or not), so Z converges and $\int_0^\infty V_{\mathrm{bar}}^2\,\mathrm{d}t\le Z(0)/\sigma_m<\infty$. V_{bar} itself is neither monotone nor continuous — it jumps at plan switches, upward as well as downward (numerics below) — so the ω-limit statement rests on the stability of the LP instead. Let ϑ_∞ be an ω-limit point and F_∞ the set of optimal plans at ϑ_∞ (for a vertex-selection solution, its finitely many vertices). For a fixed plan γ the barycentres $\bar T_i^\gamma$ do not depend on ϑ, so $V_{\mathrm{bar}}^\gamma(\vartheta):=\big(\frac{1}{m}\sum\|\vartheta_i-\bar T_i^\gamma\|^2_{G^{-1}}\big)^{1/2}$ is 1-Lipschitz in ϑ for the norm $\big(\frac{1}{m}\sum\|\cdot\|^2_{G^{-1}}\big)^{1/2}$; and the optimal face is upper semicontinuous — the plan costs are continuous in ϑ and the polytope has finitely many vertices — so every plan optimal at a ϑ close enough to ϑ_∞ lies in F_∞. Suppose no plan of F_∞ is centroidal; then $\delta:=\min_{\gamma\in F_\infty}V_{\mathrm{bar}}^\gamma(\vartheta_\infty)>0$ (a minimum over a compact set), and for $\|\vartheta-\vartheta_\infty\|<\delta/2$, $V_{\mathrm{bar}}(\vartheta)\ge\delta/2$ whatever optimal plan is in use. The speed is bounded by $\sigma_m\cdot\mathrm{diam}_{G^{-1}}(\Theta_K)$ (Lemma P.1), so the solution, which enters the ball of radius $\delta/4$ around ϑ_∞ at arbitrarily large times, either stays in the ball of radius $\delta/2$ from some time on or spends a time $\ge\delta/(4\sigma_m\,\mathrm{diam})$ in it on each of infinitely many visits; either way $\int V_{\mathrm{bar}}^2\,\mathrm{d}t=\infty$, a contradiction. Hence some plan of F_∞ — a vertex for vertex-selection solutions — is centroidal at ϑ_∞. (A two-way tie between vertex plans γ and ρ is not a sliding mode of the vertex law unless $\bar T^\gamma=\bar T^\rho$: the rate of change of $c_\rho-c_\gamma$ when following γ, plus that of $c_\gamma-c_\rho$ when following ρ, equals $(2\sigma_m/m)\sum_i\|\bar T_i^\gamma-\bar T_i^\rho\|^2_{G^{-1}}\ge0$, so at least one of the two plans stays optimal when followed. Ties between three or more plans are not analysed.) The gradient identity is the envelope theorem for the transportation LP at a configuration with a unique optimal plan: $\partial_{\vartheta_i}\big[\frac12\sum\gamma_{ij}\|\vartheta_i-\vartheta_j^0\|^2_{G^{-1}}\big]=\sum_j\gamma_{ij}G^{-1}(\vartheta_i-\vartheta_j^0)=\frac{1}{m}G^{-1}(\vartheta_i-\bar T_i)$, so $-\sigma_m(\vartheta_i-\bar T_i)=-\sigma_m m\,G\,\partial_{\vartheta_i}\big[\frac12W^2\big]$. (Where the plan is not unique, $\frac12W^2(\cdot,\nu)$ is not differentiable; the generalised velocities are then $-\sigma_m m\,G\,\xi$ with ξ ranging over the barycentric projections of the optimal plans — the elements of the (sub)differential of $\frac12W^2(\cdot,\nu)$ at an atomic μ, cf. Ambrosio–Gigli–Savaré [2, §10.4].)

(ii) Relabel so that $\frac{1}{m}\sum\|\vartheta_i-\vartheta_i^*\|^2_{G^{-1}}=\eta^2$. For any plan ρ with row sums $1/m$, the plan cost $c_\rho(\vartheta):=\sum\rho_{ij}\|\vartheta_i-\vartheta_j^0\|^2_{G^{-1}}$ is the squared norm of $(\vartheta_i-\vartheta_j^0)$ in $L^2(\rho)$, so $|\sqrt{c_\rho(\vartheta)}-\sqrt{c_\rho(\vartheta^*)}|\le\eta$ by the triangle inequality (the labels of ϑ and ϑ^* are matched). The barycentric part is dominated by the plan cost: $C_{\mathrm{bar}}=\frac{1}{m}\sum\|\vartheta_i^*-\bar T_i\|^2\le c_\gamma(\vartheta^*)$ because $\sum_j\gamma_{ij}\|\vartheta_i^*-\vartheta_j^0\|^2=\frac{1}{m}\|\vartheta_i^*-\bar T_i\|^2+\sum_j\gamma_{ij}\|\bar T_i-\vartheta_j^0\|^2$. Let γ^* be optimal for (ϑ^*,ν). Then

$$\sqrt{C_{\mathrm{bar}}}\le\sqrt{c_\gamma(\vartheta^*)}\le\sqrt{c_\gamma(\vartheta)}+\eta\le\sqrt{c_{\gamma^*}(\vartheta)}+\eta\le\sqrt{c_{\gamma^*}(\vartheta^*)}+2\eta=b(t)+2\eta.$$

For the equality case: $c_\gamma(\vartheta^*)-b(t)^2\le(b+2\eta)^2-b^2=4\eta(b+\eta)<\Delta_{\mathrm{gap}}(t)$; the vertex γ (a vertex by the choice of the law) has cost at ϑ^* strictly below the next vertex cost, so it is optimal at ϑ^*; if $\gamma^*(t)$ is unique, $\gamma=\gamma^*(t)$ and $C_{\mathrm{bar}}=\frac{1}{m}\sum\|\vartheta_i^*-\bar T_i^*\|^2=b_{\mathrm{bar}}(t)^2$. [For a generalised solution whose plan at t is not a vertex, the same chain applies to each vertex of the optimal face at ϑ, so every such vertex is optimal at ϑ^*, and uniqueness of $\gamma^*(t)$ again gives $\gamma=\gamma^*(t)$.]

(iii) Apply Theorem B with $y=\nabla g_t-\sigma_m G^{-1}(\vartheta_i-\bar T_i)$ in (P). The extra term is

$$-\sigma_m\frac{1}{m}\sum\langle\tilde\vartheta_i,\,\tilde\vartheta_i+\vartheta^*_{\pi^*(i)}-\bar T_i\rangle_{G^{-1}}\le-\sigma_m\eta^2+\sigma_m\eta\sqrt{C_{\mathrm{bar}}}.$$

Under the gap condition $\sqrt{C_{\mathrm{bar}}}=b_{\mathrm{bar}}(t)\le b_{\mathrm{bar}}$ and $\sigma_m\eta b_{\mathrm{bar}}\le(\sigma_m/2)(\eta^2+b_{\mathrm{bar}}^2)$.

(iv) is the proof of Theorem C(iv) with b_{bar} in place of b.

(v) $b_{\mathrm{bar}}(t)^2=W(\mu_t^*,\nu)^2-\mathrm{Var}_{\gamma^*(t)}\le W(\mu_t^*,\nu)^2$, with equality iff $\mathrm{Var}_{\gamma^*}=0$, that is, iff every cell of γ^* consists of a single atom of ν (coincident atoms counted once) — the case $m_0=m$,

but also $m_0 \mid m$ and priors with repeated atoms — and $b_{\mathrm{bar}} = 0$ iff $\vartheta_j^* = \bar{T}_j^*$ for all j (at a tie, with the plan $\gamma^*(t)$ fixed by the selection). □

Remark C′.1 (reading)**.** The barycentric leak replaces "pull each kernel to its prior kernel" by "pull each kernel to the centre of mass of the prior it is responsible for". Three consequences.

- *Transfer across sizes.* A network of m kernels can be anchored to a learned measure of $m_0 \neq m$ kernels. The bias in the Lyapunov inequality is b_{bar}, the distance of the ideal from the centroids of its cells, not the quantisation error $W(\mu^*, \nu)$ itself.
- *Quantisation on the fly.* With $e \equiv 0$ the law is Lloyd's algorithm for the equal-mass quantisation of ν in the G^{-1} metric, in continuous time — the continuous-time Lloyd descent of coverage control [19], with the equal-mass constraint of capacity-constrained quantisation [5]; so the controller's leak and the quantisation theory of Corollary B.6 (Section 26, branch 9) are the same object. Like Lloyd's algorithm it converges to a critical point of the quantisation energy, not necessarily the global minimiser.
- *Theorem F.* The leak clauses (i′)–(iii′) of Theorem F hold with $W(\mu_r, \nu)$ replaced by $V_{\mathrm{bar}}(r) \leq W(\mu_r, \nu)$, since $\|\dot{\vartheta}_i\|_{G^{-1}} \leq \|\nabla g_t\|_G + \sigma_m \|\vartheta_i - \bar{T}_i\|_{G^{-1}}$ by Lemma P.1 and Minkowski.

Checked (`tests/test_barycentric.py`). The plan reduces to the assignment for $m_0 = m$ (10^{-10}) and the Pythagorean split $W^2 = V_{\mathrm{bar}}^2 + \mathrm{Var}_\gamma$ holds (10^{-10}); along the pure leak W^2 is non-increasing at every step, the Dini bound $D^+ Z \leq -\sigma_m V_{\mathrm{bar}}^2$ holds at every state (within 5% of the first-order Euler quotient), the flow reaches $V_{\mathrm{bar}}^2 < 10^{-8}$, and the end point is a fixed point of the equal-mass Lloyd iteration with the same W^2; $\sqrt{C_{\mathrm{bar}}} \leq b + 2\eta$ on 90 random configurations, with $C_{\mathrm{bar}} = b_{\mathrm{bar}}^2$ at small η; the Lyapunov inequality (iii) holds at every recorded state of a closed loop with the barycentric leak toward a 24-atom prior. The numbers are in `results/barycentric_leak/table.md` (`exp_barycentric_leak.py`).

Numerics (`results/barycentric_leak/table.md`).

- *Pure leak toward a 48-atom prior* ($m = 6$, lifted dimension 3, $G = \mathrm{diag}(2, 1, 0.5)$, $\sigma_m = 1$, three random starts). $W^2(\mu_t, \nu)$ is non-increasing at every one of the 3000 steps; the flow stops where the centroid defect is at the rounding level ($V_{\mathrm{bar}}^2 \leq 2.2 \cdot 10^{-25}$) and $W^2 = \mathrm{Var}_\gamma$ (1.164, 1.213, 1.203 from 2.55, 2.30, 2.58), after 2–6 plan switches, at which V_{bar} jumps (upward as well as downward). The end points are fixed points of the equal-mass Lloyd iteration; the best of 20 Lloyd runs from random starts reaches 1.171, below two of the three end points and above the third: like Lloyd's algorithm, the leak finds a local minimiser of the quantisation energy, not always the global one. With $m_0 = m$ the same code reproduces the exact decay of Theorem C(i) to $5 \cdot 10^{-16}$.
- *Closed loop* (representable plant with an $m^* = 4$ ideal, $m = 4$, $k_e = 4$, $\sigma_m = 0.5$, three seeds). Three leaks of the same gain: (A) the OT leak toward the ideal itself, (B) the barycentric leak toward a 32-atom cloud of jittered copies of the ideal atoms, (C) the labelled leak toward a random 4-subset of the cloud. The network converges, in each case, to the configuration its leak anchors: (A) to the ideal ($W(\mu_T, \mu^*)$ and RMS$\|e\|$ below the printed precision); (B) to the equal-mass centroids of the cloud, at $W(\mu_T, \mu^*) = 0.167$, 0.097, 0.157, which equal the centroid defects b_{bar} of the ideal (0.168, 0.097, 0.159) although $W(\mu^*, \nu) = 0.45$, 0.33, 0.40; (C) to the subset, at $W(\mu_T, \mu^*) = 0.61$, 1.02, 0.35 against $W(\mu^*, \mathrm{subset}) = 0.65$, 1.03, 0.40. The steady tracking RMS is 0.002–0.007 for (B) and 0.03–0.04 for (C). So the loop settles at the centroidal configuration of its prior, at distance b_{bar} from the ideal, as the fixed-point statements (i) and (v) describe; the Lyapunov bound (iv) does not apply to these runs (scope paragraph: the exchange bound on Δ_{gap} and $4\eta(0)(b + \eta(0))$ are in the table). The comparison with (C) shows the cost of a prior of the

right size at the wrong positions, not an advantage of barycentric over labelled leaks as such: a labelled leak toward the cloud's own equal-mass m-quantiser would have the same bias b_{bar} by (v).

14 Theorem D: the mean-field closed loop

The state is $(x, \mu) \in \mathbb{R}^n \times \mathcal{P}(\Theta_K)$, with

$$\begin{gathered}\dot{x} = f(x,t) + \dot{x}_m - k_e(x - x_m) - \int \Phi(x;\vartheta)\,\mathrm{d}\mu_t, \qquad \partial_t \mu_t + \nabla \cdot (\mu_t v_t) = 0, \\ v_t = \mathrm{Proj}_K\big(\vartheta,\, G\nabla_\vartheta\big[(x - x_m)^\top \Phi(x;\vartheta)\big]\big).\end{gathered} \tag{14.1}$$

Solutions are Lagrangian: $\mu_t = (\Xi_t)_{\#}\mu_0$ for the flow Ξ_t of v_t. This is law (L). The OT leak is multivalued at ties and is excluded here.

Theorem D. *Assume* (S0)–(S2).

(a) Local well-posedness. *For every $(x_0, \mu_0) \in \mathbb{R}^n \times \mathcal{P}(\Theta_K)$ there is a unique Lagrangian solution. It continues for as long as x stays bounded.*

(b) Adaptation-free tube (m-uniform, μ^*-free). *For every $\mu \in \mathcal{P}(\Theta_K)$, $\|\int \Phi(x;\cdot)\,\mathrm{d}\mu\| \le a_{\max}$. For $F \ge 0$ let*

$$r(F) := \|e(0)\| + \frac{F + a_{\max}}{k_e}, \qquad S(F) := \sup\big\{\|f(x,t)\| : t \ge 0,\ \|x - x_m(t)\| \le r(F)\big\}.$$

Suppose some $F \ge 0$ satisfies $S(F) \le F$. Then every solution, for every m and every μ_0:

- *exists for all $t \ge 0$;*
- *satisfies $\|x(t) - x_m(t)\| \le r(F)$;*
- *satisfies $\|e(t)\| \le e^{-k_e t}\|e(0)\| + (1 - e^{-k_e t})(F + a_{\max})/k_e$.*

Examples:

- *For bounded f, $F = \sup\|f\|$ works for every $e(0)$.*
- *If $\|f(x,t)\| \le \alpha + \beta\|x\|$ with $\beta < k_e$, then*

$$F = \frac{\alpha + \beta\big(\sup\|x_m\| + \|e(0)\| + a_{\max}/k_e\big)}{1 - \beta/k_e}$$

  works.
- *The condition can fail for superlinear f. For the Van der Pol plant of the numerics, $S(F) - F \ge 51.6$ for every F.*

(c) Dobrushin estimate. *Let two solutions of the same loop (same f, x_m, k_e, G, Θ_K and ε_K) remain in a compact X on $[0,T]$. Then*

$$\begin{gathered}\|x_t - x'_t\| + W(\mu_t, \mu'_t) \le e^{\Lambda t}\big(\|x_0 - x'_0\| + W(\mu_0, \mu'_0)\big), \\ \Lambda := \max(L_\vartheta + L_2,\, L_x + M_x),\end{gathered} \tag{14.2}$$

and Λ is independent of m. The constants are:

- *L_ϑ: a Lipschitz constant of $\vartheta \mapsto v$ in the G^{-1} norm, uniform over $x \in X$ and $t \le T$. It is proportional to $\bar{E} := \sup_{X\times[0,T]}\|x - x_m(t)\|$, and it is $O(\bar{E}/\varepsilon_K)$ because of the ramp ($h' \le 2/\varepsilon_K$).*

- *L_x: a Lipschitz constant, Euclidean to G^{-1}, of $x \mapsto v(\vartheta; x, x - x_m(t))$, uniform over $\vartheta \in \Theta_K$, $t \le T$ and $x \in X$.*
- *M_x: $M_x := \sup_{t \le T} \operatorname{Lip}_X f(\cdot, t) + k_e + a_{\max} J_c$. Here $a_{\max} J_c$ bounds $\|\nabla_x \int \Phi(x; \cdot)\, \mathrm{d}\mu\|$ over $x \in \mathbb{R}^n$ and $\mu \in \mathcal{P}(\Theta_K)$.*
- *L_2: $L_2 := \lambda_G^{1/2} L_1$, with L_1 from Lemma B.2, is a W-Lipschitz constant of $\mu \mapsto \int \Phi(x; \cdot)\, \mathrm{d}\mu$. For any coupling, $\|\int \Phi(x; \cdot)\, \mathrm{d}(\mu - \mu')\| \le L_2 \big(\int \|\vartheta - \vartheta'\|_{G^{-1}}^2 \, \mathrm{d}\gamma\big)^{1/2}$.*

(d) Mean-field limit. *Let (x, μ) be the solution from (x_0, μ_0), and let T be smaller than its existence time. Put $X_1 := \{y : \operatorname{dist}(y, x([0,T])) \le 1\}$, and let Λ_1 be the Λ of (c) computed on $X_1 \times [0,T]$. Suppose $x_0^m \to x_0$ and $\mu_0^m \to \mu_0$ in W, and set $\delta_m := \|x_0^m - x_0\| + W(\mu_0^m, \mu_0)$. Then, for all large m:*

- *the m-particle loop exists on $[0,T]$ and stays in X_1;*
- *$\sup_{t \le T} \big(\|x_t^m - x_t\| + W(\mu_t^m, \mu_t)\big) \le e^{\Lambda_1 T} \delta_m \to 0$.*

Under the hypothesis of (b), for example for bounded f, the limit is global and T is arbitrary.

(e) Passage of (B) to the limit. *Work in the setting of (d), with $X := X_1$. Let $\mu_t^{*m} = \frac{1}{m} \sum_j \delta_{\vartheta_j^{*m}(t)}$ satisfy* (S4), *and let $\varepsilon_{\max}^m \ge \sup_{X \times [0,T]} \|f - \int \Phi \, \mathrm{d}\mu_t^{*m}\|$. Assume $\mu_t^{*m} \to \mu_t^*$ uniformly in W on $[0,T]$, $\sup_m \varepsilon_{\max}^m \le \varepsilon_{\max}$ and $\sup_m V_*^m \le V_*$. Only tails matter, so $\sup_m$ may be replaced by $\limsup_m$. Then:*

- *μ_t^* is V_*-Lipschitz in W, with values in $\mathcal{P}(\Theta_K^\varepsilon)$;*
- *the limit satisfies* (B) *in integrated form, and in Dini form for every $t \in [0,T)$, with constants $(k_e, \varepsilon_{\max}, K, L_0, L_1, \lambda_G, V_*)$.*

V_* depends on the approximating ideals; the intrinsic version is Remark B.7.

Status. Proved; consistency in m checked numerically (the constant is not checked). Not checked in the repository: (b), the constant of (c), and the one-sided constant of Remark D.1. Novelty (literature audit): technique anticipated [29, 72, 73].

Proof. (a) Write the loop as an ODE for $(x, \Xi) \in \mathbb{R}^n \times C(\Theta_K; \mathbb{R}^p)$, with the sup norm (and G^{-1} on $\mathbb{R}^p$):

$$\dot{x} = f(x,t) + \dot{x}_m - k_e(x - x_m) - \int \Phi(x; \Xi(\vartheta))\, \mathrm{d}\mu_0(\vartheta), \qquad \partial_t \Xi(\vartheta) = v(\Xi(\vartheta); x, t), \qquad \Xi_0 = \mathrm{id},$$

so that $\mu_t = \Xi_{t\#}\mu_0$. Extend v Lipschitz to a neighbourhood of Θ_K inside $\{s > s_{\min}/2\}$ (McShane). On $\{\|x\| \le R\}$, with the constants of (c), the right-hand side is Lipschitz:

- $\|v(\Xi(\cdot); x, t) - v(\Xi'(\cdot); x', t)\|_\infty \le L_\vartheta \|\Xi - \Xi'\|_\infty + L_x \|x - x'\|$;
- the x-field changes by at most $M_x \|x - x'\| + L_2 \|\Xi - \Xi'\|_\infty$, because $W(\Xi_\# \mu_0, \Xi'_\# \mu_0) \le \|\Xi - \Xi'\|_{L^2(\mu_0; G^{-1})}$.

Picard–Lindelöf gives a unique local solution. Each characteristic stays in Θ_K (Nagumo, as in A(ii)), so the solution continues while x is bounded. Golse [37, Thms. 1.3.2 and 1.4.3], and [29, 72, 73], are structural analogues (pairwise C^1 kernels, W_1), not the proof.

(b) While $\|e\| \le r(F)$, $\dot{e} = -k_e e + (f - \hat{f})$ with $\|f - \hat{f}\| \le F + a_{\max}$. Variation of constants gives

$$\|e(t)\| \le e^{-k_e t} \|e(0)\| + (1 - e^{-k_e t})(F + a_{\max})/k_e < r(F)$$

for every finite t, since $a_{\max} > 0$. So x never reaches the boundary of the tube (first exit), stays bounded, and (a) continues the solution.

(c) Let γ_0 be an optimal plan for (μ_0, μ_0'), let Ξ_t and Ξ_t' be the two flows, and set $D(t)^2 := \int \|\Xi_t(\vartheta) - \Xi_t'(\vartheta')\|_{G^{-1}}^2 \,\mathrm{d}\gamma_0$. Then $(\Xi_t, \Xi_t')_\# \gamma_0$ couples μ_t and μ_t', so $W(\mu_t, \mu_t') \le D(t)$. Split

$$v_t(\Xi) - v_t'(\Xi') = \big[v_t(\Xi) - v_t(\Xi')\big] + \big[v_t(\Xi') - v_t'(\Xi')\big]$$

and apply Cauchy–Schwarz:

$$\frac{\mathrm{d}}{\mathrm{d}t} D^2 \le 2L_\vartheta D^2 + 2L_x \|x - x'\| D, \qquad D^+ \|x - x'\| \le M_x \|x - x'\| + L_2 D.$$

Apply Grönwall to $D_\delta + \|x - x'\|$, with $D_\delta := (D^2 + \delta^2)^{1/2}$, and let $\delta \downarrow 0$.

(d) Let $\tau_m \le T$ be the first time x^m leaves X_1, with $\tau_m = T$ if it never leaves. On $[0, \tau_m]$ both solutions lie in X_1, so (c) gives $\|x_t^m - x_t\| \le e^{\Lambda_1 T} \delta_m < 1$ once m is large. Then $x^m(\tau_m)$ is interior to X_1, so $\tau_m = T$. By (a), the m-solution continues while it is bounded.

(e) Theorem B for each m gives $V^m(t_2) - V^m(t_1) \le \int R^m \le \int \bar{R}^m$. Here $\bar{R}^m$ is R^m with $\varepsilon_{\max}^m$ and $V_*^m(t)$ replaced by $\varepsilon_{\max}$ and V_*. By (d) and the triangle inequality for W, $e^m \to e$ and $W(\mu_t^m, \mu_t^{*m}) \to W(\mu_t, \mu_t^*)$ uniformly. So $V^m \to V$ and $\bar{R}^m \to \bar{R}$ uniformly, and $V(t_2) - V(t_1) \le \int \bar{R}$. $\bar{R}$ is continuous, which gives the Dini form. The identity coupling gives $W(\mu_t^{*m}, \mu_s^{*m}) \le V_* |t - s|$. This passes to the uniform limit, and Θ_K^ε is closed. □

Remark D.1 (a one-sided constant free of ε_K)**.** The coupling step uses v only through $\langle \Xi - \Xi', G^{-1}(v(\Xi) - v(\Xi')) \rangle$. So the one-sided constant

$$L_\vartheta^+ := \sup \frac{\langle \vartheta - \vartheta', G^{-1}(v(\vartheta) - v(\vartheta')) \rangle}{\|\vartheta - \vartheta'\|_{G^{-1}}^2}$$

may replace L_ϑ. Let $N(\theta, y) := h(\|\theta\|)(\hat{\theta}^\top y)_+ \hat{\theta}$ be the norm-ball correction, with $r = \|\theta\|$, $c = \hat{\theta}^\top \hat{\theta}'$ and $\alpha = (\hat{\theta}^\top y)_+$. Then

$$\langle \theta - \theta', N(\theta, y) - N(\theta', y) \rangle \ge -\frac{\|y\| \|\theta - \theta'\|^2}{2(\theta_{\max} - \varepsilon_K)}.$$

This follows from three facts:

- h is monotone;
- $|\alpha - \alpha'| \le \|y\| \sqrt{2(1 - c)}$;
- $\|\theta - \theta'\|^2 \ge 2\sqrt{r r'}\, |r - r'| \sqrt{2(1 - c)}$.

The box correction is monotone. Hence L_ϑ^+ stays bounded as $\varepsilon_K \to 0$, and so does $\Lambda := \max(L_\vartheta^+ + L_2,\, L_x + M_x)$. On the experiment's box, a numerical measurement gives $L_\vartheta^+ \approx 88$–98 for ε_K from 0.2 to 0.005. Over the same range the two-sided constant grows like $30/\varepsilon_K$. This measurement is not repeated in the repository's tests.

Numerics (`results/mean_field_m`). Halving $W(\mu_0^m, \mu_0^{2m})$ halves $W(\mu_T^m, \mu_T^{2m})$, with a nearly constant factor (1.64–1.69 over the four refinements). $\sup_t \|e^m - e^{2m}\|$ falls from 4.6×10^{-3} (16 against 32 particles) to below 10^{-4} (128 against 256).

This is a consistency check only.

- On the experiment's box the guaranteed factor $e^{\Lambda T}$ is at least e^{496}, from $L_2 = 33.08$ alone over $T = 15$.
- The condition of (b) fails for this Van der Pol plant, so the run is covered by the localised (d) through its observed boundedness, not by (b).

Checked by: `experiments/exp_mean_field_m.py` (consistency in m; not the constant $e^{\Lambda T}$).

15 Theorem E: label-free local convergence under tangential excitation

Assumptions.

- (S0) holds, in particular $k_e > 0$.
- **Uniform tube.** There is $\rho > 0$ such that $\varepsilon(x,t) = 0$ whenever $\|x - x_m(t)\| < \rho$, for all $t \geq 0$.
- **Interior ideal.** μ^* is constant, with $\vartheta_j^* \in \operatorname{int}\Theta_K^\varepsilon$. Put $\delta := \min_j \operatorname{dist}(\vartheta_j^*, \partial\Theta_K^\varepsilon) > 0$.
- $x_m \in C^1$, and x_m and $\dot{x}_m$ are bounded.
- Define $\Psi(t) := \big[\nabla_\vartheta \Phi(x_m(t);\vartheta_1^*), \dots, \nabla_\vartheta \Phi(x_m(t);\vartheta_m^*)\big] \in \mathbb{R}^{n\times mp}$.
- **(PE)** There are $T, \alpha > 0$ with $\int_t^{t+T} \Psi^\top \Psi \,\mathrm{d}r \succeq \alpha I_{mp}$ for all $t \geq 0$.

(PE) forces $a_j^* \neq 0$ and pairwise distinct pairs (c_j^*, s_j^*).

- If $a_j^* = 0$, the (c,s)-columns of block j vanish.
- If $(c_j^*, s_j^*) = (c_k^*, s_k^*)$, the kernel of $\int \Psi^\top \Psi \,\mathrm{d}r$ contains the vector with v in the a-slot of block j, $-v$ in the a-slot of block k, and zeros elsewhere.

In particular the ϑ_j^* are pairwise distinct, and

$$d^2 := \min_{\pi\neq\pi'} \frac{1}{m}\sum_i \|\vartheta_{\pi(i)}^* - \vartheta_{\pi'(i)}^*\|_{G^{-1}}^2 = \frac{2}{m}\cdot \min_{i\neq j} \|\vartheta_i^* - \vartheta_j^*\|_{G^{-1}}^2 > 0, \tag{15.1}$$

with $d^2 := +\infty$ for $m = 1$. The closed form holds because a non-identity permutation moves at least two indices, and the transposition of the G^{-1}-closest pair attains the bound.

Lemma E.0 (explicit strict Lyapunov function for the skew-symmetric mixed class)**.** *Let $\Psi : [0,\infty) \to \mathbb{R}^{n\times q}$ be C^1 with $\|\Psi(t)\| \leq \Psi_M$ and $\|\dot\Psi(t)\| \leq \Psi_D$ for all t, and persistently exciting: there are $T, \alpha > 0$ with $\int_t^{t+T} \Psi(r)^\top \Psi(r)\,\mathrm{d}r \succeq \alpha I_q$ for all $t \geq 0$. Let $k > 0$ and consider*

$$\dot e = -ke - \Psi(t)\zeta, \qquad \dot\zeta = \Psi(t)^\top e, \qquad w := (e,\zeta) \in \mathbb{R}^{n+q}. \tag{LTV}$$

Put $M := \Psi^\top\Psi$ and

$$Q(t) := \int_t^{t+T} (t+T-r)M(r)\,\mathrm{d}r, \qquad Q_M := \frac{T^2\Psi_M^2}{2}, \qquad c_1 := T^2\Psi_M^3 + Tk\Psi_M + T\Psi_D,$$
$$V(t,e,\zeta) := \tfrac12\|e\|^2 + \tfrac12\|\zeta\|^2 - \varepsilon\,\zeta^\top Q(t)\zeta + \varepsilon T\, e^\top\Psi(t)\zeta.$$

(i) Every admissible ε. *For every $\varepsilon > 0$ with $\varepsilon(Q_M + T\Psi_M/2) < \frac12$, V is C^1 and, along* (LTV),

$$p_{\mathrm{lo}}\|w\|^2 \leq V \leq p_{\mathrm{hi}}\|w\|^2, \qquad p_{\mathrm{lo/hi}} := \tfrac12 \mp \varepsilon(Q_M + T\Psi_M/2),$$
$$\dot V \leq -(k - \varepsilon T\Psi_M^2)\|e\|^2 - \varepsilon\alpha\|\zeta\|^2 + \varepsilon c_1\|e\|\|\zeta\|.$$

Hence $\dot V \leq -\gamma(\varepsilon)\|w\|^2$ with $\gamma(\varepsilon) := \lambda_{\min}\left[\begin{smallmatrix} a & -c/2 \\ -c/2 & b\end{smallmatrix}\right]$, $a := k - \varepsilon T\Psi_M^2$, $b := \varepsilon\alpha$, $c := \varepsilon c_1$, whenever this is positive, and then $\|w(t)\| \leq (p_{\mathrm{hi}}/p_{\mathrm{lo}})^{1/2} e^{-(\gamma/(2p_{\mathrm{hi}}))(t-t_0)}\|w(t_0)\|$ for all $t \geq t_0 \geq 0$.

(ii) Closed form. *With $\varepsilon := \min\{1/(4(Q_M + T\Psi_M/2)),\ k/(4T\Psi_M^2),\ k\alpha/c_1^2\}$, one has $p_{\mathrm{lo}} \geq \frac14$, $p_{\mathrm{hi}} \leq \frac34$ and $\dot V \leq -\gamma\|w\|^2$ with $\gamma := \min\{k/4, \varepsilon\alpha/2\}$. Hence*

$$\|w(t)\| \leq \sqrt{3}\cdot e^{-(2\gamma/3)(t-t_0)}\cdot\|w(t_0)\| \qquad \text{for all } t \geq t_0 \geq 0.$$

(iii) Dependence on Ψ_D and on time rescaling. *The rate tends to* 0 *as* $\Psi_D \to \infty$*, and it must:* $\Psi = \sin t^2$ *is bounded and PE with* $\Psi_D = \infty$*, and* (LTV) *is then not uniformly exponentially stable (part (f) of the proof). Under the time rescaling* $\Psi_\omega(t) := \Psi(\omega t)$*, with* $\Psi_D > 0$*, the data transform as* $(\alpha, T, \Psi_M, \Psi_D) \mapsto (\alpha/\omega, T/\omega, \Psi_M, \omega\Psi_D)$*, so* $c_1 = T\Psi_D \cdot (1 + O(1/\omega))$ *and the closed-form rate is* $k\alpha^2/(3\omega^2 T^2 \Psi_D^2) \cdot (1 + O(1/\omega))$ *as* $\omega \to \infty$*. This is the* ω^{-2} *law of Remark E.1.*

Proof. (a) The Gramian weight. $0 \preceq Q(t) \preceq \int_t^{t+T} (t + T - r)\Psi_M^2 \,\mathrm{d}r \cdot I = Q_M I$. Since M is continuous, Q is C^1 with $\dot{Q}(t) = \int_t^{t+T} M(r)\,\mathrm{d}r - TM(t)$: the upper limit contributes $(t + T - (t + T))M(t + T) = 0$, the lower limit contributes $-TM(t)$, and differentiating the weight under the integral gives $\int_t^{t+T} M$.
(b) Bounds on V. $|\zeta^\top Q\zeta| \le Q_M\|\zeta\|^2$ and $|e^\top\Psi\zeta| \le \Psi_M\|e\|\|\zeta\| \le (\Psi_M/2)(\|e\|^2 + \|\zeta\|^2)$.
(c) Three derivatives along (LTV).

$$\begin{aligned}
\frac{\mathrm{d}}{\mathrm{d}t}\tfrac{1}{2}(\|e\|^2 + \|\zeta\|^2) &= -k\|e\|^2 - e^\top\Psi\zeta + \zeta^\top\Psi^\top e = -k\|e\|^2,\\
\frac{\mathrm{d}}{\mathrm{d}t}(-\zeta^\top Q\zeta) &= -\zeta^\top\dot{Q}\zeta - 2\zeta^\top Q\dot{\zeta} = -\zeta^\top\Big(\int_t^{t+T} M\Big)\zeta + T\|\Psi\zeta\|^2 - 2\zeta^\top Q\Psi^\top e\\
&\le -\alpha\|\zeta\|^2 + T\|\Psi\zeta\|^2 - 2\zeta^\top Q\Psi^\top e,\\
\frac{\mathrm{d}}{\mathrm{d}t}(e^\top\Psi\zeta) &= \dot{e}^\top\Psi\zeta + e^\top\dot{\Psi}\zeta + e^\top\Psi\dot{\zeta} = -k\,e^\top\Psi\zeta - \|\Psi\zeta\|^2 + e^\top\dot{\Psi}\zeta + \|\Psi^\top e\|^2.
\end{aligned}$$

Adding them with weights 1, ε and εT, the $\|\Psi\zeta\|^2$ terms cancel:

$$\dot{V} \le -k\|e\|^2 - \varepsilon\alpha\|\zeta\|^2 - 2\varepsilon\zeta^\top Q\Psi^\top e - \varepsilon Tk\,e^\top\Psi\zeta + \varepsilon T\,e^\top\dot{\Psi}\zeta + \varepsilon T\|\Psi^\top e\|^2.$$

The last four terms are at most $2\varepsilon Q_M\Psi_M\|e\|\|\zeta\|$, $\varepsilon Tk\Psi_M\|e\|\|\zeta\|$, $\varepsilon T\Psi_D\|e\|\|\zeta\|$ and $\varepsilon T\Psi_M^2\|e\|^2$. This is (i); the quadratic form $-a\|e\|^2 - b\|\zeta\|^2 + c\|e\|\|\zeta\|$ is at most

$$-\lambda_{\min}\begin{bmatrix} a & -c/2 \\ -c/2 & b \end{bmatrix} \cdot \|w\|^2,$$

and $V \le p_{\mathrm{hi}}\|w\|^2$, $\|w\|^2 \le V/p_{\mathrm{lo}}$ give the estimate.
(d) The closed form. Young's inequality gives $\varepsilon c_1\|e\|\|\zeta\| \le (\varepsilon\alpha/2)\|\zeta\|^2 + (\varepsilon c_1^2/(2\alpha))\|e\|^2$. With $\varepsilon \le k\alpha/c_1^2$ the second coefficient is at most $k/2$, and with $\varepsilon \le k/(4T\Psi_M^2)$ the coefficient of $\|e\|^2$ in (i) is at least $k - k/4$. So $\dot{V} \le -(k/4)\|e\|^2 - (\varepsilon\alpha/2)\|\zeta\|^2 \le -\gamma\|w\|^2$. Since $V \le (3/4)\|w\|^2$, $\dot{V} \le -(4\gamma/3)V$, and $\|w\|^2 \le 4V$ gives

$$\|w(t)\|^2 \le 4V(t) \le 4e^{-(4\gamma/3)(t-t_0)}V(t_0) \le 3e^{-(4\gamma/3)(t-t_0)}\|w(t_0)\|^2.$$

(e) Rescaling. Ψ_ω has sup norm Ψ_M, derivative bound $\omega\Psi_D$, and

$$\int_t^{t+T/\omega} \Psi_\omega^\top\Psi_\omega \,\mathrm{d}r = \frac{1}{\omega}\int_{\omega t}^{\omega t+T} \Psi^\top\Psi \,\mathrm{d}r \succeq \frac{\alpha}{\omega}I.$$

Substitute into c_1 and the closed form.
(f) $\Psi = \sin t^2$ *is not uniformly exponentially stabilising* ($n = q = 1$, $k > 0$). Let $F(s) := \int_s^\infty \sin r^2\,\mathrm{d}r$, so that $\sin s^2 = -F'(s)$ and $|F(s)| \le C/s$ (Fresnel). From $w(t_0) = (0, \zeta_0)$, $e(t) = -\int_{t_0}^t e^{-k(t-s)}\sin(s^2)\zeta(s)\,\mathrm{d}s$; integrating by parts and using $|\zeta| \le \|w(t_0)\|$ (the energy $\frac{1}{2}\|w\|^2$ is non-increasing) and $|\dot{\zeta}| = |\Psi e| \le |e|$,

$$|e(t)| \le \frac{C}{t_0}(2 + 1)\|w(t_0)\| + \frac{C}{t_0}\int_{t_0}^t e^{-k(t-s)}|e(s)|\,\mathrm{d}s,$$

so $\sup_{t\ge t_0}|e| \le 3C\|w(t_0)\|/\big(t_0(1-C/(kt_0))\big)$ for $t_0 > C/k$. Hence

$$\tfrac{1}{2}\|w(t_0+\Delta)\|^2 \ge \tfrac{1}{2}\|w(t_0)\|^2 - k\Delta\Big(\frac{3C}{t_0}\Big)^2\|w(t_0)\|^2(1+o(1)) :$$

the decay over any fixed Δ tends to 0 as $t_0 \to \infty$, which excludes a uniform exponential rate. The three ratios quoted in step 3 of the proof of Theorem E (0.9715, 0.9942, 0.9980 for $t_0 = 5$, 20, 50 and $\Delta = 200$) match $1 - \frac{1}{8}\big(1/t_0 - 1/(t_0+200)\big)$ to three digits. □

Checked by: `tests/test_excitation.py`: the two bounds on V and $\dot V \le -\gamma\|w\|^2$ at random (t, w) for four regressors of different shapes and gains, with the closed-form ε and with the ε maximising the rate (`excitation.optimise_eps`; every admissible ε gives a valid Lyapunov function, so the maximum over the grid is a valid bound for the given PE data; the data Ψ_M, Ψ_D, α are themselves grid maxima and a refinement-checked quadrature—the PE level over the 20π window is computed with 800 Gauss–Legendre nodes, and `excitation.pe_level` raises if a 2× refinement disagrees—so the ordering against the exact rate is numerical); the envelope along trajectories; the ordering explicit rate ≤ exact Floquet rate; the ω^{-2} law; the slow-adaptation limit (a no-growth check of the ratio, not a saturation check); and the explicit nonlinear envelope of Theorem E on the implemented closed loop. The round-5 referee verified the pointwise form $\lambda_{\max}(N(t)) \le -\gamma$ of $\dot V \le -\gamma\|w\|^2$ over all w, with an independent quadrature, for ten regressor families ($q > n$, $q < n$, k from 10^{-3} to 10^4, Ψ_D/Ψ_M up to 100, non-periodic PE, near-degenerate PE) and eight choices of ε, without finding a violation.

Remark E.0a (how conservative the explicit rate is)**.** The construction is of the Mazenc–Malisoff type, a backward–forward integral of the PE Gramian as a cross term; its content here is the explicit, self-contained statement for the mixed class (LTV), which removes the dependence of Theorem E on [62]. The price is conservativeness, which the table of `exp_explicit_rate.py` → `results/explicit_rate/table.md` quantifies against the exact Floquet rate of the linearised loop.

- *Slow adaptation* ($T\Psi_M^2 \ll k$). To leading order in the gains, $c_1 \approx T(k\Psi_M + \Psi_D)$ and $\alpha = T\lambda_{\min}\langle\Psi^\top\Psi\rangle$ for a window equal to the period, so the closed-form bound $\frac{2}{3}\gamma_{\mathrm{cf}}$ of (ii) is $k\alpha^2/(3c_1^2)$, and the rate with the optimal ε of the same quadratic form is $k\alpha^2/c_1^2$ (maximise $\varepsilon\alpha - \varepsilon^2 c_1^2/(4k)$ over ε). The exact rate is $\lambda_{\min}\langle\Psi^\top\Psi\rangle/k$ when k is also large compared with the bandwidth of the reference (Remark E.1). Hence, in that doubly asymptotic regime,

$$\frac{\text{exact}}{\text{explicit}} \approx 3\kappa_\Psi\Big(1+\frac{\Psi_D}{k\Psi_M}\Big)^2 \text{ for (ii)}, \qquad \approx \kappa_\Psi\Big(1+\frac{\Psi_D}{k\Psi_M}\Big)^2 \text{ for the optimised } \varepsilon,$$

  with $\kappa_\Psi := \Psi_M^2/\lambda_{\min}\langle\Psi^\top\Psi\rangle$ the condition number of the averaged Gramian. The exact rate pays for ill-conditioning once, through $\lambda_{\min}$; the bound pays twice, because the Young step that absorbs the cross term $\varepsilon c_1\|e\|\|\zeta\|$ compares $c_1 \propto \Psi_M$ with $\alpha \propto \lambda_{\min}$ (Lemma E.0′ below removes this by slaving the error to the slow manifold before the Young step). On the ideal of the Theorem E tests ($m = 3$, $n = 2$, $k_e = 6$), driven by a two-tone reference of period 20π, with $G = g\cdot\mathrm{diag}(2,2,1)$: $\kappa_\Psi = 1.27\cdot10^5$, $(1+\Psi_D/(k_e\Psi_M))^2 = 3.84$, so the two formula values are $1.47\cdot10^6$ and $4.9\cdot10^5$. The measured ratio with the optimised ε falls from $6.56\cdot10^{10}$ at $g = 30$ to $4.22\cdot10^5$ at $g = 0.01$ ($5.63\cdot10^5$ with the closed-form ε, which the table evaluates with the exact 2×2 form and the exact p_{hi} rather than the bound (ii)). Two corrections account for the difference from $4.9\cdot10^5$: on this instance the exact rate is 0.66 of $\lambda_{\min}\langle\Psi^\top\Psi\rangle/k$ at every g ($k_e = 6$ is not large against the reference's bandwidth; the two last columns of the table), and the $T^2\Psi_M^3$ term is still 13% of c_1 at $g = 0.01$ ($(c_1/c_1^\infty)^2 = 1.31$); their product, $0.66\times1.31 = 0.86$, brings $4.9\cdot10^5$ to $4.2\cdot10^5$. The ratio is still decreasing at $g = 0.01$ and tends to about $0.66\cdot\kappa_\Psi(1+\Psi_D/(k_e\Psi_M))^2 \approx 3.2\cdot10^5$ as $g \to 0$. In the mean-field normalisation $\Psi_G \propto m^{-1/2}$, so large networks sit in this regime at fixed geometry.

- *Fast adaptation* ($T\Psi_M^2 \gtrsim k$). The ratio grows further with the excitation strength, through the $T^2\Psi_M^3$ term of c_1 and the Q_M term of ε.
- *Remark E.1's instance* ($m = n = 1$, $G = I$, $k_e = 3$, $x_m = 1.5\sin\omega t$). The ratio exact/explicit decreases from 954 at $\omega = 1$ to 17.6 at $\omega = 64$. Over $\omega \geq 4$ the log-log slopes are -1.97 (exact) and -1.56 (explicit); over the last octave, $\omega = 32 \to 64$, the explicit slope is -1.92, approaching the asymptotic ω^{-2} of (iii).
- *The window.* Any window length T is admissible with its own $\alpha(T)$. On the test ideal the best window is half a period.
- *Answered below.* Lemma E.0′ is a strict Lyapunov function whose rate is $\lambda_{\min}\langle\Psi^\top\Psi\rangle/k$ up to a factor $1-\delta$, above a threshold in k set by the peak factors of the regressor: in the slow regime it pays for the ill-conditioning once.

Lemma E.0′ (the slow-adaptation regime: a strict Lyapunov function whose rate is the averaged rate)**.** *Let Ψ be T-periodic and C^2, $M := \Psi^\top\Psi$, $\bar M := \frac{1}{T}\int_0^T M\,\mathrm{d}t \succ 0$, $\|v\|_{\bar M} := (v^\top \bar M v)^{1/2}$, $\lambda_{\min} := \lambda_{\min}(\bar M)$, and*

$$\Psi_M := \sup\|\Psi\|, \qquad \Psi_D := \sup\|\dot\Psi\|, \qquad \Psi_{DD} := \sup\|\ddot\Psi\|,$$
$$\rho := \sup_{t,\,v\neq 0}\frac{\|\Psi(t)v\|^2}{\|v\|_{\bar M}^2}, \qquad \rho_D := \sup\frac{\|\dot\Psi(t)v\|^2}{\|v\|_{\bar M}^2}, \qquad \rho_{DD} := \sup\frac{\|\ddot\Psi(t)v\|^2}{\|v\|_{\bar M}^2},$$

the peak factors*: the largest ratio, over directions, between the instantaneous excitation of a direction (by Ψ, $\dot\Psi$, $\ddot\Psi$) and its average excitation; $\rho \geq 1$, and all three are finite because $\bar M \succ 0$. Write $\varepsilon := 1/k$ and let*

$$P_1(t) := -\frac{2}{T}\int_t^{t+T}(t+T-r)\big(M(r)-\bar M\big)\,\mathrm{d}r, \qquad F := P_1M + MP_1,$$
$$S := \text{the zero-mean antiderivative of } F, \qquad P_2 := -(M-\bar M) + S,$$
$$e_2(\zeta,t) := -\varepsilon\Psi\zeta + \varepsilon^2\dot\Psi\zeta - \varepsilon^3\Psi M\zeta, \qquad w := e - e_2(\zeta,t),$$
$$V(e,\zeta,t) := \tfrac12\zeta^\top\big(I + \varepsilon P_1(t) + \varepsilon^2 P_2(t)\big)\zeta + \frac{\beta}{2}\|w\|^2,$$

where P_1 is the zero-mean antiderivative of $2(M-\bar M)$ and e_2 is the slow manifold of the error, expanded through order ε^3. F has zero mean: $\langle P_1M + MP_1\rangle = \langle P_1\rangle\bar M + \bar M\langle P_1\rangle + \frac12\big\langle\frac{\mathrm{d}}{\mathrm{d}t}(P_1^2)\big\rangle = 0$, so S exists and the second-order averaged correction $N_2 := \frac12\langle F\rangle$ vanishes identically. Put

$$\pi_1 := \tfrac{T}{2}\Psi_M(\sqrt\rho+1), \qquad \pi_2 := \Psi_M(\sqrt\rho+1) + \tfrac{T^2}{8}\Psi_M^3(2\sqrt\rho+1),$$
$$p_1 := \tfrac{T}{2}\Psi_M^2, \qquad p_2 := \Psi_M^2 + \tfrac{T^2}{4}\Psi_M^4,$$
$$c_1 := \sqrt\rho + \varepsilon\Psi_M\pi_1 + \varepsilon^2\Psi_M\pi_2,$$
$$c_2 := \varepsilon^2\sqrt{\rho_{DD}} + \varepsilon^3\big(3\Psi_D\Psi_M\sqrt\rho + 2\Psi_M^2\sqrt{\rho_D}\big) + \varepsilon^4\big(2\Psi_M^4\sqrt\rho + \Psi_D\Psi_M\sqrt{\rho_D}\big)$$
$$\quad + \varepsilon^5\big(\Psi_M^4\sqrt{\rho_D} + \Psi_D\Psi_M^3\sqrt\rho\big) + \varepsilon^6\Psi_M^6\sqrt\rho,$$
$$d_w := \varepsilon\Psi_M^2 + \varepsilon^2\Psi_D\Psi_M + \varepsilon^3\Psi_M^4,$$
$$r_\zeta := \varepsilon^3\big(\pi_2\Psi_M\sqrt\rho + \pi_1\Psi_M\sqrt{\rho_D} + \Psi_M^2\rho\big) + \varepsilon^4\big(\pi_2\Psi_M\sqrt{\rho_D} + \pi_1\Psi_M^3\sqrt\rho\big) + \varepsilon^5\pi_2\Psi_M^3\sqrt\rho,$$
$$\beta := 2c_1^2/\delta, \qquad c_\zeta := 1 - \delta - r_\zeta/\varepsilon.$$

Fix $\delta \in (0,1)$. If

$$\text{(A) } c_1c_2 \leq \delta/2, \qquad \text{(B) } d_w \leq k/4, \qquad \text{(C) } c_\zeta > 0, \qquad \text{(D) } \varepsilon p_1 + \varepsilon^2 p_2 < 1,$$

then along (LTV)*, for all (e,ζ,t),*

$$\dot V \leq -\varepsilon c_\zeta\,\zeta^\top\bar M\zeta - \frac{\beta k}{4}\|w\|^2 \leq -\gamma V, \qquad \gamma := \min\Big\{\frac{2c_\zeta\lambda_{\min}}{k(1+\varepsilon p_1+\varepsilon^2 p_2)},\ \frac{k}{2}\Big\},$$

and $\|(e,\zeta)(t)\| \le Ce^{-(\gamma/2)(t-t_0)}\|(e,\zeta)(t_0)\|$ *with*

$$C := \big(1+\varepsilon\Psi_M+\varepsilon^2\Psi_D+\varepsilon^3\Psi_M^3\big)^2\cdot\Big(\frac{\max\{1+\varepsilon p_1+\varepsilon^2 p_2,\ \beta\}}{\min\{1-\varepsilon p_1-\varepsilon^2 p_2,\ \beta\}}\Big)^{1/2}.$$

In the slow-adaptation limit ($k\to\infty$ at fixed Ψ) the conditions hold for every $\delta\in(0,1)$ once $k\ge k_0(\delta)$, with $k_0\approx(4\rho\rho_{DD}/\delta^2)^{1/4}$ when (A) binds (small δ, or $T\Psi_M^2\lesssim 1$; otherwise (C) sets k_0 through r_ζ, as on the sinusoidal regressor below), and the rate is $\gamma/2=(1-\delta)(1-\varepsilon p_1+O(k^{-2}))\lambda_{\min}/k$: the averaged rate $\lambda_{\min}\langle\Psi^\top\Psi\rangle/k$ of Remark E.1 up to the factor $1-\delta$ and the first-order deficit $\varepsilon p_1=T\Psi_M^2/(2k)$ of the quadratic form. The bound pays for the ill-conditioning of $\bar M$ once, through $\lambda_{\min}$, as the exact rate does; the peak factors enter only the threshold k_0 and the overshoot $C=O((\rho/\delta)^{1/2})$, not the rate.

Status. Proved (below); checked pointwise by a certificate—the supremum of $\dot V/V$ over all states and over a time grid is a generalised eigenvalue problem—on the sinusoidal regressor and on the slow-adaptation ideal of the Theorem E tests, where the guaranteed rate is within 12% ($\delta=0.1$) and 26% ($\delta=0.2$) of the exact rate at $k=400$ while Lemma E.0's is five orders of magnitude below it (`tests/test_excitation.py`, `exp_explicit_rate.py`, Table 12). The round-6 referee re-derived every step and certified the inequality independently on five regressor families without a failure (`review/REVIEW_round6_E0prime.md`).

Proof. Write M, Ψ, … for their values at t. Two facts about zero-mean antiderivatives are used throughout: if f is T-periodic with zero mean and g is its zero-mean antiderivative, then $\|g\|_\infty\le(T/4)\|f\|_\infty$—for the maximiser t^* of g, $g(t^*)=\frac1T\int_0^T(g(t^*)-g(s))\,\mathrm{d}s\le\frac1T\int_0^T\operatorname{dist}(t^*,s)\|f\|_\infty\,\mathrm{d}s=(T/4)\|f\|_\infty$ with dist the distance on the circle, and the same for $-g$—and the same bound holds for every bilinear coordinate $u^\top gv$, hence $\|g(t)v\|\le(T/4)\sup_r\|f(r)v\|$. As $\|M-\bar M\|\le\Psi_M^2$ (two positive semidefinite matrices of norm $\le\Psi_M^2$), this gives $\|P_1\|\le T\Psi_M^2/2=p_1$ and $\|P_1v\|\le(T/2)\sup\|(M-\bar M)v\|$, and $\|S\|\le(T/4)\|F\|_\infty$, $\|Sv\|\le(T/4)\sup\|Fv\|$.
(a) The fast variable. From $\dot e=-ke-\Psi\zeta$ and $ke_2=-\Psi\zeta+\varepsilon\dot\Psi\zeta-\varepsilon^2\Psi M\zeta$,

$$\dot e=-kw-\varepsilon\dot\Psi\zeta+\varepsilon^2\Psi M\zeta.$$

Differentiating e_2 along the flow, with $\dot\zeta=\Psi^\top e=\Psi^\top w-\varepsilon M\zeta+\varepsilon^2\Psi^\top\dot\Psi\zeta-\varepsilon^3M^2\zeta$ and $\dot M=\dot\Psi^\top\Psi+\Psi^\top\dot\Psi$, the terms of order ε and ε^2 in ζ cancel and

$$\begin{aligned}\dot w=&-kw+\varepsilon\Psi\Psi^\top w-\varepsilon^2\dot\Psi\Psi^\top w+\varepsilon^3\Psi M\Psi^\top w-\varepsilon^2\ddot\Psi\zeta+\varepsilon^3\big(2\dot\Psi M+\Psi\dot\Psi^\top\Psi+2\Psi\Psi^\top\dot\Psi\big)\zeta\\&-\varepsilon^4\big(2\Psi M^2+\dot\Psi\Psi^\top\dot\Psi\big)\zeta+\varepsilon^5\big(\Psi M\Psi^\top\dot\Psi+\dot\Psi M^2\big)\zeta-\varepsilon^6\Psi M^3\zeta.\end{aligned}$$

(b) The slow part of $\dot V$. With $H:=I+\varepsilon P_1+\varepsilon^2P_2$ (symmetric), $\frac{\mathrm{d}}{\mathrm{d}t}[\frac12\zeta^\top H\zeta]=\zeta^\top H\dot\zeta+\frac12\zeta^\top\dot H\zeta$ and $\dot H=2\varepsilon(M-\bar M)+\varepsilon^2(F-\dot M)$. Insert $\dot\zeta$ and collect by powers of ε. At order ε: $-\zeta^\top M\zeta+\zeta^\top(M-\bar M)\zeta=-\zeta^\top\bar M\zeta$. At order ε^2:

$$\begin{aligned}-\zeta^\top P_1M\zeta+\zeta^\top\Psi^\top\dot\Psi\zeta+\tfrac12\zeta^\top(F-\dot M)\zeta=&-\tfrac12\zeta^\top(P_1M+MP_1)\zeta+\tfrac12\zeta^\top\dot M\zeta\\&+\tfrac12\zeta^\top(P_1M+MP_1-\dot M)\zeta=0,\end{aligned}$$

using $\zeta^\top\Psi^\top\dot\Psi\zeta=\frac12\zeta^\top\dot M\zeta$: the choice of P_2 removes the whole second-order term, and this is where $N_2=0$ enters (otherwise $-\zeta^\top N_2\zeta$ would remain). The remaining terms are

$$R_\zeta:=\varepsilon^3\big(-\zeta^\top P_2M\zeta+\zeta^\top P_1\Psi^\top\dot\Psi\zeta-\zeta^\top M^2\zeta\big)+\varepsilon^4\big(\zeta^\top P_2\Psi^\top\dot\Psi\zeta-\zeta^\top P_1M^2\zeta\big)-\varepsilon^5\zeta^\top P_2M^2\zeta,$$

plus the coupling $\zeta^\top H\Psi^\top w$, so $\frac{\mathrm{d}}{\mathrm{d}t}[\frac12\zeta^\top H\zeta]=-\varepsilon\zeta^\top\bar M\zeta+R_\zeta+\zeta^\top H\Psi^\top w$.

(c) Per-direction bounds. For every v and t:

$$\|\Psi v\| \le \sqrt{\rho}\|v\|_{\bar M}, \qquad \|\dot\Psi v\| \le \sqrt{\rho_D}\|v\|_{\bar M}, \qquad \|\ddot\Psi v\| \le \sqrt{\rho_{DD}}\|v\|_{\bar M};$$
$$\|Mv\| \le \Psi_M\|\Psi v\| \le \Psi_M\sqrt{\rho}\|v\|_{\bar M}; \qquad \|\bar M v\| \le \lambda_{\max}(\bar M)^{1/2}\|v\|_{\bar M} \le \Psi_M\|v\|_{\bar M};$$
$$\|\Psi^\top\dot\Psi v\| \le \Psi_M\sqrt{\rho_D}\|v\|_{\bar M}; \qquad \|P_1 v\| \le \tfrac{T}{2}\sup_r\|(M(r)-\bar M)v\| \le \pi_1\|v\|_{\bar M};$$
$$\|Fv\| \le \|P_1 Mv\| + \|MP_1 v\| \le \tfrac{T}{2}\Psi_M^3\sqrt{\rho}\|v\|_{\bar M} + \Psi_M^2\pi_1\|v\|_{\bar M} = T\Psi_M^3(\sqrt{\rho}+\tfrac12)\|v\|_{\bar M},$$

so $\|Sv\| \le (T/4)\sup\|Fv\| = (T^2/8)\Psi_M^3(2\sqrt{\rho}+1)\|v\|_{\bar M}$ and $\|P_2 v\| \le \|(M-\bar M)v\| + \|Sv\| \le \pi_2\|v\|_{\bar M}$; also

$$\|P_2\| \le \Psi_M^2 + \tfrac{T}{4}\|F\|_\infty \le \Psi_M^2 + \tfrac{T}{4}\cdot 2\cdot\tfrac{T\Psi_M^2}{2}\Psi_M^2 = p_2.$$

Hence $|R_\zeta| \le r_\zeta\|\zeta\|_{\bar M}^2$, $|\zeta^\top H\Psi^\top w| = |\langle \Psi H\zeta, w\rangle| \le c_1\|\zeta\|_{\bar M}\|w\|$, and from (a), $\beta w^\top\dot w \le \beta\big[-(k-d_w)\|w\|^2 + c_2\|\zeta\|_{\bar M}\|w\|\big]$: each ζ-term of $\dot w$ is bounded through $\|\Psi\zeta\|$, $\|\dot\Psi\zeta\|$, $\|\ddot\Psi\zeta\|$, $\|M\zeta\|$ or $\|M^2\zeta\|$ (the terms of c_2, in the order of (a); the last one is $\|\Psi M^3\zeta\| \le \Psi_M\cdot\Psi_M^4\|M\zeta\| \le \Psi_M^6\sqrt{\rho}\|\zeta\|_{\bar M}$), the w-terms through operator norms (d_w).

(d) Assembly. $\dot V \le -\varepsilon\|\zeta\|_{\bar M}^2 + r_\zeta\|\zeta\|_{\bar M}^2 + (c_1+\beta c_2)\|\zeta\|_{\bar M}\|w\| - \beta(k-d_w)\|w\|^2$. By Young, $(c_1+\beta c_2)\|\zeta\|_{\bar M}\|w\| \le \delta\varepsilon\|\zeta\|_{\bar M}^2 + (c_1+\beta c_2)^2\|w\|^2/(4\delta\varepsilon)$, and $(c_1+\beta c_2)^2 \le 2c_1^2 + 2\beta^2 c_2^2$. With $\beta = 2c_1^2/\delta$ the term $2c_1^2/(4\delta\varepsilon)$ equals $\beta k/4$, and (A) gives $2\beta^2c_2^2/(4\delta\varepsilon) \le \beta k/4$, so with (B) the coefficient of $\|w\|^2$ is at most $-\beta(k-d_w-k/2) \le -\beta k/4$; the coefficient of $\|\zeta\|_{\bar M}^2$ is $-\varepsilon(1-\delta-r_\zeta/\varepsilon) = -\varepsilon c_\zeta$. This is the first inequality. Since $\frac12(1-\varepsilon p_1-\varepsilon^2 p_2)\|\zeta\|^2 + \frac{\beta}{2}\|w\|^2 \le V \le \frac12(1+\varepsilon p_1+\varepsilon^2 p_2)\|\zeta\|^2 + \frac{\beta}{2}\|w\|^2$ by (D), and $\zeta^\top\bar M\zeta \ge \lambda_{\min}\|\zeta\|^2$, $\dot V \le -\gamma V$ follows, and the envelope with C from $\|e\| \le \|w\| + (\varepsilon\Psi_M + \varepsilon^2\Psi_D + \varepsilon^3\Psi_M^3)\|\zeta\|$ and its converse. □

Remark E.0′.1 (reading)**.**

- *What is sharp and what is not.* The exact rate of (LTV) is $\lambda_{\min}/k\cdot(1+O(k^{-2}))$: the first-order averaging of $\dot\zeta = -(1/k)M\zeta$ after the slaving gives $\lambda_{\min}/k$, and the second-order correction to the symmetric part of the averaged generator is $N_2 = 0$ (Table 12 measures $1-\text{exact}/(\lambda_{\min}/k) \approx 29/k^2$ on the ideal; on $\Psi = (\sin t, \cos t)$ the deviation is of order k^{-4}). Lemma E.0′ recovers this rate up to the factor $c_\zeta/(1+\varepsilon p_1+\varepsilon^2 p_2) \to 1-\delta$, and δ can be taken $\to 0$ as $k\to\infty$ (condition (A) allows $\delta \gtrsim 2\sqrt{\rho\rho_{DD}}/k^2$). The overshoot constant $C = O((\rho/\delta)^{1/2})$ is an artefact of the quadratic form, not of the dynamics.
- *Two regimes, two lemmas.* Lemma E.0 needs only $(\alpha, T, \Psi_M, \Psi_D)$ and gives the ω^{-2} law of the fast regime; its rate is quadratic in $\lambda_{\min}$. Lemma E.0′ needs k above the threshold $k_0 \approx (4\rho\rho_{DD}/\delta^2)^{1/4}$, set by the peak factors of Ψ and its two derivatives, and gives the sharp rate. On the regressors of Theorem E the derivative peak factors are large—the directions that are poorly excited on average are excited in short bursts, so ρ_{DD}/ρ is of the order of the fourth power of the effective bandwidth—and k_0 is in the hundreds there, while on $\Psi = (\sin t, \cos t)$ ($\rho = \rho_D = \rho_{DD} = 2$) it is about 10, set by (C) rather than (A). A third-order slaving would replace ρ_{DD} by the peak factor of $\dddot\Psi$ at one order higher in ε and lower k_0 further; this is not pursued.
- *Hypotheses.* Only the periodicity of Ψ is used (through the zero-mean antiderivatives; a non-periodic PE regressor would need a windowed $\bar M(t)$ and its derivative). Theorem E's x_m is periodic, so Ψ_G is. Theorem E's steps 4–5 go through with this V in place of Lemma E.0's (Corollary E.0′.2 below): the nonlinear rate becomes $\gamma/4 \approx (1-\delta)\lambda_{\min}/(2k)$, while the explicit ball radius carries the anisotropy $p'_{\text{lo}}/p'_{\text{hi}} \approx 1/\beta$ of the quadratic form and can be smaller than Lemma E.0's.

Numerics (`exp_explicit_rate.py` → `results/explicit_rate/table.md`, table 4; reproduced in Table 12). The slow-adaptation ideal of Remark E.0a (table 2 of that file) at $g = 0.01$

($T = 20\pi$): $\lambda_{\min}(\bar{M}) = 2.16 \cdot 10^{-7}$, $\kappa(\bar{M}) = 7.8 \cdot 10^3$, $\Psi_M = 0.168$, $\Psi_D = 0.957$, $\Psi_{DD} = 8.39$, $T\Psi_M^2 = 1.78$; peak factors $\rho = 69.9$, $\rho_D = 3.7 \cdot 10^3$, $\rho_{DD} = 3.7 \cdot 10^5$ (the poorly excited directions are excited in bursts: $\rho_D/\rho \approx 53$ and $\rho_{DD}/\rho_D \approx 99$ are effective squared bandwidths); $\|N_2\| = 3 \cdot 10^{-21}$. The threshold of (A)–(D) is $k_0 = 226$ for $\delta = 0.2$ and 321 for $\delta = 0.1$, set by (A): $c_1 c_2 = 2.07$, 0.51, 0.13, 0.056, 0.032, 0.014 at $k = 50$, 100, 200, 300, 400, 600, the k^{-2} law of $\varepsilon^2\sqrt{\rho\rho_{DD}}$.

Table 12: Lemma E.0′ in the slow regime on the slow-adaptation ideal of Remark E.0a at $g = 0.01$ (`results/explicit_rate/table.md`, table 4).

k	exact Floquet rate	exact / ($\lambda_{\min}/k$)	guaranteed rate $\gamma/2$, $\delta = 0.2$ ($\delta = 0.1$)	exact / guaranteed	exact / Lemma E.0 (optimised ε)	$\sup \dot{V}/V$ on the grid, $-\gamma$ ($\delta = 0.1$)
300	$7.21 \cdot 10^{-10}$	0.9997	$5.70 \cdot 10^{-10}$ (below threshold)	1.26 (—)	$1.33 \cdot 10^5$	—
400	$5.41 \cdot 10^{-10}$	0.9998	$4.30 \cdot 10^{-10}$ ($4.84 \cdot 10^{-10}$)	1.26 (1.12)	$1.32 \cdot 10^5$	$-1.079 \cdot 10^{-9}$, $-9.68 \cdot 10^{-10}$
600	$3.61 \cdot 10^{-10}$	0.9999	$2.88 \cdot 10^{-10}$ ($3.24 \cdot 10^{-10}$)	1.25 (1.12)	$1.30 \cdot 10^5$	$-7.20 \cdot 10^{-10}$, $-6.47 \cdot 10^{-10}$

Above the threshold the guaranteed rate is within 26% ($\delta = 0.2$) and 12% ($\delta = 0.1$) of the exact rate—the factor $1/(1-\delta)$ of the lemma, with $c_\zeta = 0.80$ and 0.90—while Lemma E.0's optimised rate is $1.3 \cdot 10^5$ times smaller; the certificate $\sup \dot{V}/V \le -\gamma$ holds on the 128-point grid in every row where the conditions hold, with $\sup \dot{V}/V = -1.078 \cdot 10^{-9} \approx -2\lambda_{\min}/k$ at $k = 400$ (attained in the slow direction at $w = 0$). The exact rate deviates from $\lambda_{\min}/k$ by 1.15%, 0.29%, 0.07% at $k = 50$, 100, 200: the k^{-2} law that $N_2 = 0$ predicts. The overshoot constant is $C = \sqrt{\beta} \cdot (1 + O(1/k)) = 27$ ($\delta = 0.2$) and 38 ($\delta = 0.1$). Below k_0 the lemma makes no claim; Lemma E.0 does (Remark E.0a; table 2 of that file).

Checked by: `tests/test_excitation.py`: on $\Psi = (\sin t, \cos t)$ the computed $\lambda_{\min}$, ρ, ρ_D, ρ_{DD} equal 1/2, 2, 2, 2, $N_2 = 0$ to 10^{-12}, the threshold of (A)–(D) is below 12 for $\delta = 0.2$, the certificate $\sup \dot{V}/V \le -\gamma$ holds at $k = 20$, 50, 200, and the guaranteed rate is below the exact Floquet rate and within a factor 2 of it, the ratio decreasing towards $1/(1-\delta)$ (`test_lemma_E0prime_sinusoidal_regressor_certificate_and_sharpness`); on the slow-adaptation ideal of the Theorem E tests at $g = 0.01$ and $k = 400$, $\delta = 0.2$: $\kappa(\bar{M}) > 10^3$, the conditions hold, the certificate holds on a 128-point grid, the guaranteed rate is within a factor 1.5 of the exact rate and exceeds Lemma E.0's optimised rate by more than 10^4 (`test_lemma_E0prime_pays_for_ill_conditioning_once_on_the_theorem_E_ideal`).

Corollary E.0′.2 (Theorem E in the slow regime)**.** *Under the hypotheses of Theorem E, suppose the regressor Ψ_G of its step 3 satisfies the conditions (A)–(D) of Lemma E.0′ at $k = k_e$ for some δ, with constants $(\beta, \gamma, p_1, p_2)$ and $L_M := \varepsilon\Psi_{G,M} + \varepsilon^2\Psi_{G,D} + \varepsilon^3\Psi_{G,M}^3$ ($\varepsilon = 1/k_e$), and that $\beta L_M^2 < 1 - \varepsilon p_1 - \varepsilon^2 p_2$. Put*

$$p'_{\mathrm{lo}} := \tfrac{1}{2}\min\{\beta/2,\ 1 - \varepsilon p_1 - \varepsilon^2 p_2 - \beta L_M^2\},$$
$$p'_{\mathrm{hi}} := \tfrac{1}{2}\max\{\beta(1+L_M),\ 1 + \varepsilon p_1 + \varepsilon^2 p_2 + \beta L_M(1+L_M)\},$$
$$c'_0 := \min\Big\{\Big[\frac{\gamma p'_{\mathrm{lo}}}{4(p'_{\mathrm{hi}})^{1/2}\tilde{C}_R}\Big]^2,\ \frac{p'_{\mathrm{lo}} r^2}{\|S^{-1}\|^2}\Big\}, \qquad r'_0 := \Big(\frac{c'_0}{p'_{\mathrm{hi}}}\Big)^{1/2}, \qquad \lambda' := \frac{\gamma}{4}, \qquad C' := \Big(\frac{2p'_{\mathrm{hi}}}{p'_{\mathrm{lo}}}\Big)^{1/2},$$

with r, $\tilde{C}_R$, S as in Theorem E. Then the conclusions of Theorem E hold with (c_0, λ, C) replaced by $((r'_0)^2/2, \lambda', C')$. In the slow regime $\lambda' = (1-\delta)(1 - \varepsilon p_1 + O(k_e^{-2}))\lambda_{\min}(\bar{M}_G)/(2k_e)$, $\bar{M}_G := \langle \Psi_G^\top \Psi_G \rangle$: half the exact rate of the linearised loop, up to the factor $1-\delta$ and the first-order deficit εp_1.

Proof. Write $V := \frac{1}{2} w^\top \mathbf{H}(t) w$ for Lemma E.0′'s V in the coordinates $w = (e, \zeta)$ of step 3:

$$\mathbf{H} = \begin{bmatrix} \beta I & -\beta L \\ -\beta L^\top & H_\zeta + \beta L^\top L \end{bmatrix}$$

with $e_2 = L(t)\zeta$, $\|L\| \le L_M$ and $H_\zeta := I + \varepsilon P_1 + \varepsilon^2 P_2$. Then $w^\top \mathbf{H} w = \beta\|e - L\zeta\|^2 + \zeta^\top H_\zeta \zeta$, and $\|e - L\zeta\|^2 \le (1 + L_M)(\|e\|^2 + L_M\|\zeta\|^2)$, $\|e - L\zeta\|^2 \ge \frac{1}{2}\|e\|^2 - L_M^2\|\zeta\|^2$ give $2p'_{\mathrm{lo}}\|w\|^2 \le w^\top \mathbf{H} w \le 2p'_{\mathrm{hi}}\|w\|^2$. Along the nonlinear loop $\dot w = [A_0 + B(t)]w + \tilde R(t, w)$ with $\|\tilde R\| \le \tilde C_R\|w\|^2$ on $\{\|w\| < r/\|S^{-1}\|\}$ (step 4), so with Lemma E.0′,

$$\dot V \le -\gamma V + \langle \mathbf{H} w, \tilde R\rangle \le -\gamma V + \|\mathbf{H}^{1/2} w\|\,\|\mathbf{H}^{1/2}\|\,\tilde C_R\|w\|^2 \le -\gamma V + \frac{2(p'_{\mathrm{hi}})^{1/2}\tilde C_R}{p'_{\mathrm{lo}}} V^{3/2},$$

using $\|\mathbf{H}^{1/2} w\| = (2V)^{1/2}$, $\|\mathbf{H}^{1/2}\| = (2p'_{\mathrm{hi}})^{1/2}$ and $\|w\|^2 \le V/p'_{\mathrm{lo}}$. Hence $\dot V \le -(\gamma/2)V$ wherever $V^{1/2} \le \gamma p'_{\mathrm{lo}}/(4(p'_{\mathrm{hi}})^{1/2}\tilde C_R)$, and the sublevel set $\{V \le c'_0\}$—inside the ball where $\tilde R$ is bounded, by the second term of c'_0—is forward invariant with $V(t) \le e^{-(\gamma/2)(t-t_0)}V(t_0)$ on it. It contains $\{\|w\| \le r'_0\}$ because $V \le p'_{\mathrm{hi}}\|w\|^2$, and $\|w(t)\| \le (p'_{\mathrm{hi}}/p'_{\mathrm{lo}})^{1/2} e^{-(\gamma/4)(t-t_0)}\|w(t_0)\|$ follows. Step 5 of Theorem E is unchanged: $\|w\|^2 = 2V_\pi$, so $V_{\mathrm{OT}}(t_0) \le (r'_0)^2/2$ puts the state in the ball, $\|e\| + W \le \sqrt{2}\|w\|$ gives C', and uniqueness and the limits follow as there. The asymptotics of λ' are those of γ in Lemma E.0′. □

Numerics (`exp_explicit_rate.py` → `results/explicit_rate/table.md`, table 5; reproduced in Table 13). Remark E.1's single-kernel instance ($m = n = 1$, $\vartheta^* = (1.5, 0.3, 0.8)$, $G = 0.3 \cdot I$, $x_m = 1.5 \sin t$) at $k_e = 40$: $\lambda_{\min}(\bar M_G) = 5.41 \cdot 10^{-2}$, $\kappa(\bar M_G) = 6.7$, peak factors $\rho = 4.8$, $\rho_D = 30.3$, $\rho_{DD} = 461$, threshold $k_0 = 24.4$ ($\delta = 0.2$) and 33.5 ($\delta = 0.1$); $L_J = 4.62$ estimated from samples; exact linear rate $1.35 \cdot 10^{-3}$.

Table 13: Theorem E in the slow regime on Remark E.1's single-kernel instance at $k_e = 40$ (`results/explicit_rate/table.md`, table 5).

Lyapunov function	explicit ball radius r_0 (w-coordinates)	nonlinear rate λ	overshoot C	exact linear rate / λ
Lemma E.0, optimised ε (Theorem E)	$6.15 \cdot 10^{-6}$	$2.63 \cdot 10^{-5}$	1.43	51
Lemma E.0′, $\delta = 0.2$ (this corollary)	$1.53 \cdot 10^{-6}$	$4.53 \cdot 10^{-4}$	11.8	3.0
Lemma E.0′, $\delta = 0.1$ (this corollary)	$8.32 \cdot 10^{-7}$	$5.16 \cdot 10^{-4}$	17.1	2.6

On this well-conditioned regressor ($\kappa \approx 7$) Lemma E.0 is only an order of magnitude conservative, and the corollary trades the certified ball for the rate: λ' is within a factor 3 of the exact linear rate (the factor 2 of the Lyapunov argument times $1/(1-\delta)$), while r'_0 is four times smaller than Theorem E's r_0, carrying the anisotropy $p'_{\mathrm{lo}}/p'_{\mathrm{hi}} \approx 1/\beta$ of the quadratic form. On ill-conditioned regressors the rate improves by the factor of Remark E.0a (Table 12); the radius comparison was not computed there.

Checked by: `tests/test_excitation.py`: on the single-kernel instance at $k_e = 40$, $\delta = 0.2$, from a start with $\|w(0)\| \le r'_0$, the implemented closed loop satisfies $\|w(t)\| \le C' e^{-\lambda' t}\|w(0)\|$ at every sample over 1.5 e-folds of the explicit envelope ($\mathrm{d}t = 0.005$: the fixed-step loop has a discretisation floor $\approx 6 \cdot 10^{-6}$ at $\mathrm{d}t = 0.02$, above the explicit ball), with λ' more than ten times Theorem E's rate from Lemma E.0 and within a factor 4 of the exact linear rate (`test_corollary_E0prime_2_envelope_on_the_implemented_loop`).

Corollary E.0′.3 (the stationary parameter spread of the dithered linearised loop)**.** *Under the hypotheses of Lemma E.0′ (conditions (A)–(D) at (k, δ)), consider* (LTV) *with parameter dither,*

$$\dot e = -ke - \Psi(t)\zeta, \qquad \mathrm{d}\zeta = \Psi(t)^\top e\,\mathrm{d}t + \Sigma\,\mathrm{d}B_t, \tag{LTV$_\tau$}$$

with $\Sigma \in \mathbb{R}^{q\times q'}$ constant and B a standard Brownian motion. Let V be the strict Lyapunov function of Lemma E.0′ and $\mathbf{H}(t)$ its quadratic form ($V = \frac{1}{2}x^\top \mathbf{H}(t)x$, $x = (e,\zeta)$), $L(t) := -\varepsilon\Psi + \varepsilon^2\dot\Psi - \varepsilon^3\Psi M$ the slaving map ($e_2 = L(t)\zeta$), $L_M := \varepsilon\Psi_M + \varepsilon^2\Psi_D + \varepsilon^3\Psi_M^3 \geq \sup\|L\|$, and

$$\nu(t) := \tfrac{1}{2}\operatorname{tr}\big(\Sigma^\top \mathbf{H}_{\zeta\zeta}(t)\Sigma\big) = \tfrac{1}{2}\operatorname{tr}\big(\Sigma^\top(I + \varepsilon P_1 + \varepsilon^2 P_2 + \beta L^\top L)\Sigma\big),$$
$$\bar\nu := \sup_t \nu(t) \leq \tfrac{1}{2}\|\Sigma\|_F^2\big(1 + \varepsilon p_1 + \varepsilon^2 p_2 + \beta L_M^2\big).$$

Then, for every initial law with finite second moment,

$$\limsup_{t\to\infty} \mathbb{E}V(t,x_t) \leq \frac{\bar\nu}{\gamma}, \qquad \limsup_{t\to\infty} \mathbb{E}\|\zeta_t\|^2 \leq \frac{2\bar\nu}{\gamma(1-\varepsilon p_1 - \varepsilon^2 p_2)}, \qquad \limsup_{t\to\infty} \mathbb{E}\|w_t\|^2 \leq \frac{2\bar\nu}{\gamma\beta},$$

with γ the rate of Lemma E.0′; in the slow regime $\gamma = 2c_\zeta\lambda_{\min}(\bar M)/(k(1+\varepsilon p_1+\varepsilon^2 p_2))$, so that

$$\limsup \mathbb{E}\|\zeta_t\|^2 \leq \frac{k\|\Sigma\|_F^2\,(1+\varepsilon p_1+\varepsilon^2 p_2)(1+\varepsilon p_1+\varepsilon^2 p_2+\beta L_M^2)}{2c_\zeta(1-\varepsilon p_1-\varepsilon^2 p_2)\lambda_{\min}(\bar M)} = (1+O(\varepsilon))\,\frac{k\|\Sigma\|_F^2}{2(1-\delta)\lambda_{\min}(\bar M)}.$$

For comparison, the averaged Langevin dynamics $\dot\zeta = -\frac{1}{k}\bar M\zeta + \Sigma\dot B$ has the exact stationary covariance $\bar C = kX$ with $\bar M X + X\bar M = \Sigma\Sigma^\top$, so $\mathbb{E}\|\zeta\|^2 = k\operatorname{tr}X$, equal to $\frac{k}{2}\operatorname{tr}(\bar M^{-1}\Sigma\Sigma^\top)$ when $\Sigma\Sigma^\top$ commutes with $\bar M$: the bound is within $(1+O(\varepsilon))/(1-\delta)$ of it when $\bar M$ is isotropic, and pays the condition number of $\bar M$ in the trace otherwise (it replaces every $1/\lambda_i$ by $1/\lambda_{\min}$), as a scalar Lyapunov function must.

Status. Proved (Itô's formula and the pointwise inequality of Lemma E.0′). Checked (`exp_explicit_rate.py`, table 6; `tests/test_excitation.py`) against the exact periodic stationary covariance of (LTV$_\tau$), computed from the periodic Lyapunov equation.

Proof. $V(t,x) = \frac{1}{2}x^\top\mathbf{H}(t)x$ is C^1 in t and quadratic in x, so by Itô's formula along (LTV$_\tau$), with $\tilde\Sigma := [0;\Sigma] \in \mathbb{R}^{(n+q)\times q'}$,

$$\begin{aligned}\mathrm{d}V &= \big[\partial_t V + \nabla_x V\cdot A(t)x\big]\,\mathrm{d}t + \nabla_x V\cdot\tilde\Sigma\,\mathrm{d}B_t + \tfrac{1}{2}\operatorname{tr}\big(\tilde\Sigma^\top\mathbf{H}(t)\tilde\Sigma\big)\,\mathrm{d}t\\ &\leq \big[-\gamma V + \nu(t)\big]\,\mathrm{d}t + \nabla_x V\cdot\tilde\Sigma\,\mathrm{d}B_t,\end{aligned}$$

the inequality being Lemma E.0′'s $\dot V \leq -\gamma V$ pointwise in (t,x), and $\tilde\Sigma^\top\mathbf{H}\tilde\Sigma = \Sigma^\top\mathbf{H}_{\zeta\zeta}\Sigma$ with $\mathbf{H}_{\zeta\zeta} = I+\varepsilon P_1+\varepsilon^2P_2+\beta L^\top L$ from the block form of $\mathbf{H}$ (Corollary E.0′.2). The stochastic integral is a martingale because its integrand is linear in x and the linear SDE has bounded second moments (Gaussian with a bounded covariance), so $\frac{\mathrm{d}}{\mathrm{d}t}\mathbb{E}V \leq -\gamma\mathbb{E}V + \bar\nu$ and $\mathbb{E}V(t) \leq e^{-\gamma t}\mathbb{E}V(0) + \bar\nu/\gamma$. The bounds on $\mathbb{E}\|\zeta\|^2$ and $\mathbb{E}\|w\|^2$ follow from $V \geq \frac{1}{2}(1-\varepsilon p_1-\varepsilon^2 p_2)\|\zeta\|^2$ and $V \geq \frac{\beta}{2}\|w\|^2$; the estimate of $\bar\nu$ uses $\|P_1\| \leq p_1$, $\|P_2\| \leq p_2$, $\|L\| \leq L_M$. The averaged covariance is the stationary solution of the Lyapunov equation of the time-invariant averaged system. □

Numerics (`exp_explicit_rate.py` → `results/explicit_rate/table.md`, table 6; reproduced in Table 14). $\Sigma = (2\cdot10^{-3})^{1/2}I$; the exact stationary covariance from the periodic Lyapunov equation (RK4 with $k\cdot\mathrm{d}t \leq 1/4$; the period mean and the maximum of $\mathbb{E}\|\zeta\|^2$ coincide to four digits), the averaged value $k\cdot\operatorname{tr}X$, and the bound:

Table 14: Corollary E.0′.3: the explicit bound on the stationary parameter spread of the dithered (LTV) against the exact periodic stationary covariance and the averaged value (`results/explicit_rate/table.md`, table 6).

regressor ($\kappa(\bar{M})$)	k	δ	exact $\mathbb{E}\|\zeta\|^2$	averaged $k \cdot \operatorname{tr} X$	bound	bound / exact	$1/(1-\delta)$
Theorem E ideal of table 4 ($7.8 \cdot 10^3$)	300	0.2	$1.509 \cdot 10^6$	$1.509 \cdot 10^6$	$2.64 \cdot 10^7$	17.5	1.25
	400	0.2 / 0.1	$2.012 \cdot 10^6$	$2.012 \cdot 10^6$	$3.51 \cdot 10^7$ / $3.11 \cdot 10^7$	17.4 / 15.5	1.25 / 1.11
	600	0.2 / 0.1	$3.018 \cdot 10^6$	$3.018 \cdot 10^6$	$5.23 \cdot 10^7$ / $4.65 \cdot 10^7$	17.3 / 15.4	1.25 / 1.11
$\Psi = (\sin t, \cos t)$ (1)	20	0.2 / 0.1	$8.025 \cdot 10^{-2}$	$8.000 \cdot 10^{-2}$	0.216 / 0.202	2.69 / 2.52	1.25 / 1.11
	50	0.2 / 0.1	$2.001 \cdot 10^{-1}$	$2.000 \cdot 10^{-1}$	0.315 / 0.282	1.58 / 1.41	1.25 / 1.11
	200	0.2 / 0.1	$8.000 \cdot 10^{-1}$	$8.000 \cdot 10^{-1}$	1.051 / 0.935	1.31 / 1.17	1.25 / 1.11

The exact stationary spread is the averaged Langevin value up to $3 \cdot 10^{-4}$ (ideal) and $1.25/k^2$ (sinusoid): on these instances, where Σ is isotropic so that $\Sigma\Sigma^\top$ commutes with $\bar{M}$, the averaging prediction of the dithered loop's stationary covariance is exact to $O(k^{-2})$. When $\Sigma\Sigma^\top$ does not commute with $\bar{M}$ the deviation is $O(1/k)$ (referee round 7 measured $\pm 0.25/k$ on two-dimensional sinusoids with anisotropic Σ, the bound still holding with ratios 2.7–5.1). On the isotropic sinusoid the bound exceeds the exact value by a factor that falls toward $1/(1-\delta)$ (2.7, 1.6, 1.3 at $k = 20$, 50, 200 for $\delta = 0.2$; the excess is the $O(1/k)$ of the factors $(1 \pm \varepsilon p_1) \dots$ and βL_M^2). On the ideal, where one direction is excited $7.8 \cdot 10^3$ times less than the others, the bound is 15–18 times the exact value: it charges every one of the $q = 15$ directions the price of the worst, $(q/\lambda_{\min})/\sum_i (1/\lambda_i) \approx q = 15$ (up to $1/(1-\delta)$), the trace-level price of a scalar Lyapunov function. The exact $\mathbb{E}\|e\|^2$ is $\|\Sigma\|_F^2/(2k) = \operatorname{tr}(\bar{M}X)/k$ to five digits on both instances (the error rides on its slow manifold $e \approx \Psi\zeta/k$, and $\operatorname{tr}(\bar{M}X) = \frac{1}{2}\|\Sigma\|_F^2$ by the trace of the Lyapunov equation), while the bound on $\mathbb{E}\|w\|^2$ is coarse (factor $1/\beta$): on the sinusoid the exact stationary $\mathbb{E}\|w\|^2 = \mathbb{E}\|e - L\zeta\|^2$, computed from the periodic covariance and the slaving map $L = -\varepsilon\Psi + \varepsilon^2\dot{\Psi} - \varepsilon^3\Psi M$, is more than three orders of magnitude below the bound (asserted in the test below).

Checked by: `tests/test_excitation.py`: on $\Psi = (\sin t, \cos t)$ with $\Sigma = (2 \cdot 10^{-3})^{1/2} I$, the explicit bound dominates the exact period mean and the exact maximum of $\mathbb{E}\|\zeta\|^2$ at $k = 20$, 50, 200 for $\delta = 0.1$ and 0.2, its ratio to the exact value decreases toward $1/(1-\delta)$ (within 6% at $k = 200$), the grid value of $\bar{\nu}$ lies below its explicit bound, and the exact value agrees with the averaged $k \cdot \operatorname{tr} X$ within $2/k^2$ (`test_corollary_E0prime_3_dither_covariance_bound_against_the_periodic_lyapunov_solution`).

Lemma E.0″ (the slow regime for non-periodic persistently exciting regressors)**.** *Let $\Psi\colon [0,\infty) \to \mathbb{R}^{n\times q}$ be C^2 with $\|\Psi\| \le \Psi_M$, $\|\dot{\Psi}\| \le \Psi_D$, $\|\ddot{\Psi}\| \le \Psi_{DD}$, $M := \Psi^\top\Psi$, and fix a window $T > 0$. For a bounded continuous X let*

$$\langle X\rangle_T(t) := \frac{1}{T}\int_0^\infty e^{-s/T} X(t+s)\,\mathrm{d}s$$

(the forward exponential average; $T\frac{\mathrm{d}}{\mathrm{d}t}\langle X\rangle_T = \langle X\rangle_T - X$),

and $\bar{M} := \langle M\rangle_T$. Assume the exponentially weighted PE condition

$$\bar{M}(t) \succeq \lambda I_q \quad \text{for all } t \ge 0, \quad \text{with } \lambda > 0, \tag{PE$_T$}$$

which is equivalent to classical PE ($\int_t^{t+T} M \succeq \alpha I$ for all t gives (PE$_T$) *with $\lambda \ge e^{-1}\alpha/T$, and* (PE$_T$) *gives PE over windows $T' > T\log(\Lambda/\lambda)$ with level $T(\lambda - e^{-T'/T}\Lambda)$; proof below). Let $\Lambda := \sup_t \lambda_{\max}(\bar{M}(t)) \le \Psi_M^2$. Fix symmetric matrices $\hat{M}$, $\hat{F}$ (any; the long-time means of $\bar{M}$*

and $\bar F$ minimise the time-averaged squares of $\|\bar M-\hat M\|$, $\|\bar F-\hat F\|$, not necessarily the suprema below) and define

$$P_1:=-2T(\bar M-\hat M)\ \ (so\ \dot P_1=2(M-\bar M)),\qquad F:=P_1M+MP_1,$$
$$\bar F:=\langle F\rangle_T,\qquad S:=-T(\bar F-\hat F)\ \ (\dot S=F-\bar F),$$
$$P_2:=-(M-\bar M)+S,\qquad \bar N_2:=\tfrac12(\bar F-\dot{\bar M})=\tfrac12\big(\bar F-(\bar M-M)/T\big),$$

and e_2, w, V exactly as in Lemma E.0′ ($V=\frac12\zeta^\top(I+\varepsilon P_1+\varepsilon^2P_2)\zeta+\frac{\beta}{2}\|w\|^2$, $\varepsilon=1/k$). The constants are those of Lemma E.0′ with the period mean replaced by the exponential average, all of them computable from the regressor: the time-local peak factors

$$\rho:=\sup_{t,v}\frac{\|\Psi(t)v\|^2}{v^\top\bar M(t)v},\qquad \rho_D:=\sup\frac{\|\dot\Psi(t)v\|^2}{v^\top\bar M(t)v},\qquad \rho_{DD}:=\sup\frac{\|\ddot\Psi(t)v\|^2}{v^\top\bar M(t)v}$$
$$(all\ \le\Psi_\cdot^2/\lambda),$$

the operator and per-direction norms

$$p_1:=\sup_t\|P_1(t)\|=2T\sup\|\bar M-\hat M\|,\qquad p_2:=\sup_t\|P_2(t)\|\le\Psi_M^2+T\sup\|\bar F-\hat F\|,$$
$$\pi_1:=\sup_{t,v}\frac{\|P_1(t)v\|}{\|v\|_{\bar M(t)}}\le 2T\big(\sqrt{\Lambda}+\|\hat M\|/\sqrt{\lambda}\big),$$
$$\pi_2:=\sup_{t,v}\frac{\|P_2(t)v\|}{\|v\|_{\bar M(t)}}\le\Psi_M\sqrt{\rho}+\sqrt{\Lambda}+T\big(2p_1\Psi_M^2+\|\hat F\|\big)/\sqrt{\lambda},$$
$$n_2:=\sup_t\|\bar N_2(t)\|\le\tfrac12\big(\sup\|F\|+2\Psi_M^2/T\big)\le p_1\Psi_M^2+\Psi_M^2/T,$$

and, with these π_1, π_2, p_1, p_2, the quantities c_1, c_2, d_w, r_ζ and $\beta:=2c_1^2/\delta$ of Lemma E.0′ (same formulas), together with

$$\delta_2:=\varepsilon n_2/\lambda,\qquad c_\zeta:=1-\delta-\delta_2-r_\zeta/\varepsilon.$$

Fix $\delta\in(0,1)$. If (A) $c_1c_2\le\delta/2$, (B) $d_w\le k/4$, (C) $c_\zeta>0$ *and* (D) $\varepsilon p_1+\varepsilon^2p_2<1$, *then along* (LTV), *for all (e,ζ,t),*

$$\dot V\le-\varepsilon c_\zeta\,\zeta^\top\bar M(t)\zeta-\frac{\beta k}{4}\|w\|^2\le-\gamma V,\qquad \gamma:=\min\Big\{\frac{2c_\zeta\lambda}{k(1+\varepsilon p_1+\varepsilon^2p_2)},\ \frac{k}{2}\Big\},$$

and $\|(e(t),\zeta(t))\|\le Ce^{-\gamma(t-t_0)/2}\|(e(t_0),\zeta(t_0))\|$ with the C of Lemma E.0′. In the slow regime the guaranteed rate is

$$\gamma/2=(1-\delta-\delta_2)\big(1-\varepsilon p_1+O(k^{-2})\big)\,\lambda/k,\qquad \lambda=\inf_t\lambda_{\min}\big(\langle M\rangle_T(t)\big):$$

the exponentially weighted PE level of the window divided by k, up to the factor $1-\delta-\delta_2$, where $\delta_2=O(1/k)$ is the price of the residual second-order term (zero for a periodic Ψ with $\hat M=\langle M\rangle$ in the limit $T\to\infty$, where Lemma E.0′ applies directly).

Status. Proved (the proof of Lemma E.0′ with the changes listed below). Checked (`exp_pe_general.py`, `tests/test_excitation.py`) on a quasi-periodic regressor and on a regressor driven by a Lorenz orbit, where no periodic construction applies: the conditions hold from $k=20$ on for the quasi-periodic regressor and from $k=100$ on for the chaotic one (the derivative peak factors of a chaotic orbit are large), the certificate $\sup\dot V/V\le-\gamma$ holds on the time grid wherever they hold, and the guaranteed rate is within a factor 1.3–2.5 of the exact decay rate on the quasi-periodic regressor ($k=50$–200) and, with the window $T=20$, within a factor 3.5–5 on the chaotic one (shorter windows pay more: 7.5–11 at $T=10$ and 14–138 at $T=5$), where the worst exponential window is three times less exciting than the long-time average. Referee round 7 (`review/REVIEW_round7_E0p3_E0dp.md`) confirmed the proof and the code and found no counterexample on an attack set of barely exciting, bursty, near-resonant and chaotic regressors.

Proof. Follow the proof of Lemma E.0′ with $\bar{M}$, P_1, S, P_2 as defined here. (a) is unchanged (it does not involve the averages). In (b), $\dot{H} = 2\varepsilon(M-\bar{M})+\varepsilon^2(F-\bar{F}-\dot{M}+\dot{\bar{M}})$ exactly, by $\dot{P}_1 = 2(M-\bar{M})$ and $\dot{P}_2 = -\dot{M} + \dot{\bar{M}} + \dot{S} = -\dot{M} + \dot{\bar{M}} + F - \bar{F}$; the order-$\varepsilon$ terms give $-\zeta^\top \bar{M}\zeta$ as before, and the order-ε^2 terms give $-\zeta^\top P_1 M\zeta + \frac{1}{2}\zeta^\top \dot{M}\zeta + \frac{1}{2}\zeta^\top(F - \bar{F} - \dot{M} + \dot{\bar{M}})\zeta = \frac{1}{2}\zeta^\top(\dot{\bar{M}} - \bar{F})\zeta = -\zeta^\top \bar{N}_2\zeta$, since $\zeta^\top P_1 M\zeta = \frac{1}{2}\zeta^\top F\zeta$: the periodic cancellation $N_2 = 0$ is replaced by the residual $-\varepsilon^2\zeta^\top\bar{N}_2\zeta$, and R_ζ has the same form, so $\frac{\mathrm{d}}{\mathrm{d}t}[\frac{1}{2}\zeta^\top H\zeta] = -\varepsilon\zeta^\top(\bar{M} + \varepsilon\bar{N}_2)\zeta + R_\zeta + \zeta^\top H\Psi^\top w$. In (c), every per-direction bound is taken in the time-local norm $\|v\|_{\bar{M}(t)}$: $\|\Psi(t)v\| \le \sqrt{\rho}\|v\|_{\bar{M}(t)}$ and likewise for $\dot{\Psi}$, $\ddot{\Psi}$ by the definition of the peak factors; $\|M(t)v\| \le \Psi_M\sqrt{\rho}\|v\|_{\bar{M}(t)}$; $\|\bar{M}(t)v\| \le \sqrt{\Lambda}\|v\|_{\bar{M}(t)}$; $\|P_1(t)v\| = 2T\|(\bar{M}(t) - \hat{M})v\| \le 2T(\sqrt{\Lambda} + \|\hat{M}\|/\sqrt{\lambda})\|v\|_{\bar{M}(t)}$, using $\sqrt{\lambda}\|v\| \le \|v\|_{\bar{M}(t)}$ (this is where the exponential average pays off: $P_1(t)$ is a function of $\bar{M}(t)$ alone, with no integral over other times); for $\bar{F}(t)v$ and $S(t)v$, which do involve other times $r \ge t$, $\|F(r)v\| \le \|P_1(r)\|\|M(r)v\| + \|M(r)\|\|P_1(r)v\| \le 2p_1\Psi_M^2\|v\| \le (2p_1\Psi_M^2/\sqrt{\lambda})\|v\|_{\bar{M}(t)}$, an average of such bounds bounds $\|\bar{F}(t)v\|$, and $\|S(t)v\| \le T(\|\bar{F}(t)v\| + \|\hat{F}v\|)$, whence the bound on π_2; the stated bounds on p_2 and n_2 are $\|P_2\| \le \|M - \bar{M}\| + \|S\|$ and $\|\bar{N}_2\| \le \frac{1}{2}(\|\bar{F}\| + \|\dot{\bar{M}}\|)$ with $\|\dot{\bar{M}}\| = \|\bar{M} - M\|/T \le 2\Psi_M^2/T$ and $\|\bar{F}\| \le \sup\|F\| \le 2p_1\Psi_M^2$. The bounds $|R_\zeta| \le r_\zeta\|\zeta\|_{\bar{M}}^2$, $|\zeta^\top H\Psi^\top w| \le c_1\|\zeta\|_{\bar{M}}\|w\|$ and $\beta w^\top\dot{w} \le \beta[-(k-d_w)\|w\|^2 + c_2\|\zeta\|_{\bar{M}}\|w\|]$ follow with the same formulas (the suprema over v at the current time t of the norms that appear). In (d), the extra term is bounded by $\varepsilon^2|\zeta^\top\bar{N}_2\zeta| \le \varepsilon^2 n_2\|\zeta\|^2 \le (\varepsilon^2 n_2/\lambda)\zeta^\top\bar{M}\zeta = \varepsilon\delta_2\zeta^\top\bar{M}\zeta$, which turns $1-\delta-r_\zeta/\varepsilon$ into $c_\zeta = 1-\delta-\delta_2-r_\zeta/\varepsilon$; the rest of the assembly, the sandwich $\frac{1}{2}(1-\varepsilon p_1-\varepsilon^2 p_2)\|\zeta\|^2 + \frac{\beta}{2}\|w\|^2 \le V \le \frac{1}{2}(1+\varepsilon p_1+\varepsilon^2 p_2)\|\zeta\|^2 + \frac{\beta}{2}\|w\|^2$ and $\zeta^\top\bar{M}(t)\zeta \ge \lambda\|\zeta\|^2$ give $\dot{V} \le -\gamma V$. The equivalence of (PE_T) with classical PE: if $\int_t^{t+T} M \succeq \alpha I$ then $\bar{M}(t) \succeq \frac{1}{T}\int_0^T e^{-s/T}M(t+s)\,\mathrm{d}s \succeq \frac{e^{-1}}{T}\int_t^{t+T} M \succeq (e^{-1}\alpha/T)I$; conversely ($\mathrm{PE}_T$) gives, for every $T' > 0$,

$$\int_t^{t+T'} M \succeq \int_t^{t+T'} Me^{-(r-t)/T}\,\mathrm{d}r \succeq T\big(\bar{M}(t) - e^{-T'/T}\bar{M}(t+T')\big) \succeq T(\lambda - e^{-T'/T}\Lambda)I,$$

positive for $T' > T\log(\Lambda/\lambda)$ (the identity $\int_t^{t+T'} e^{-(r-t)/T}M(r)\,\mathrm{d}r = T[\bar{M}(t) - e^{-T'/T}\bar{M}(t+T')]$ follows from the definition of the exponential average). □

Remark E.0″.1 (reading)**.**

- *What changes and what does not.* The construction, the conditions and the rate formula are those of Lemma E.0′; the period mean becomes the exponential average over a window, the peak factors become time-local, and one new term appears: the residual $\bar{N}_2$, whose price $\delta_2 = \varepsilon n_2/\lambda$ vanishes like $1/k$. For a periodic regressor the exponential average with a finite window keeps residual oscillations of relative size $O(\text{period}/T)$, so Lemma E.0′ is sharper there; Lemma E.0″ is the version for references that are not periodic — quasi-periodic (several incommensurate frequencies) or chaotic (a reference along a strange attractor, as in the Lorenz-tracking experiments of Part I), where only PE is available.
- *The window is a design parameter.* The lemma holds for every $T > 0$; the guaranteed rate is $(1-\delta-\delta_2)\lambda_T/k$ with $\lambda_T = \inf\lambda_{\min}(\langle M\rangle_T)$, which increases toward the long-time excitation level as T grows, against $p_1 \propto T$, $n_2 \propto T$ and the threshold in k, which grow with T. The numerics show the optimum at a window of a few characteristic times of the regressor.
- *Exact rate.* For a non-periodic regressor the exact rate is the top Lyapunov exponent of (LTV), which the certificate does not need: the lemma is a pointwise inequality in (t, e, ζ), checked on a grid in t by the same generalised eigenvalue problem as Lemma E.0′.

Numerics (`exp_pe_general.py` → `results/explicit_rate/general_pe.md`). Two regressors that are not periodic, on $[0, 300]$ (grid step 0.005; the exponential averages formed from

an extended grid; the certificate $\sup \dot{V}/V$ over 400 times and all states; the exact rate the top Lyapunov exponent of (LTV) over 200 time units), with the measured constants (the grid suprema of $\|P_1\|$, $\|P_2\|$, the per-direction norms and $\|\bar{N}_2\|$, computable constants of the regressor like the peak factors).

Quasi-periodic $\Psi = (\sin t, \cos\sqrt{2}\,t)$ ($q = 2$; the long-time mean of M is $\frac{1}{2}I$, and the exact decay rate is $\lambda_\infty/k = 0.5/k$ to within 1%, 0.6% at $k = 20$ and 0.8% at $k = 400$): the window average has $\lambda_T = 0.297$, 0.391, 0.444 for $T = 2\pi$, 4π, 8π ($\rho = 4.0$, $\rho_D \approx 6.1$–6.4, $\rho_{DD} \approx 10.0$, $p_1 \approx 2.9$, $p_2 = 8.0$–9.2, $\pi_1 = 4.3$–4.7, $\pi_2 = 11.6$–13.0, $\|\bar{N}_2\| = 0.71$, 0.35, 0.17); the conditions hold from $k = 20$ on for both δ, the certificate holds at every row, and (Table 15)

Table 15: Lemma E.0″ on the quasi-periodic regressor $\Psi = (\sin t, \cos\sqrt{2}\,t)$: the guaranteed rate against the exact decay rate for two windows (`results/explicit_rate/general_pe.md`).

T	k	δ_2	guaranteed $\gamma/2$ ($\delta = 0.2$ / 0.1)	exact	exact/guaranteed	λ_T/k
4π	20	0.045	0.0100 / 0.0117	0.0249	2.49 / 2.13	0.0195
4π	50	0.018	0.00557 / 0.00631	0.00995	1.78 / 1.58	0.00781
4π	200	0.004	0.00153 / 0.00172	0.00249	1.63 / 1.45	0.00195
8π	20	0.020	0.0118 / 0.0137	0.0249	2.10 / 1.81	0.0222
8π	200	0.002	0.00174 / 0.00196	0.00249	1.43 / 1.27	0.00222

($T = 2\pi$: 3.69 → 2.16 for $\delta = 0.2$ and 3.10 → 1.92 for $\delta = 0.1$ from $k = 20$ to 200.) The ratio exact/guaranteed is the product of the window loss λ_∞/λ_T (1.69, 1.28, 1.13 for the three windows), the factor $1/(1 - \delta - \delta_2)$ and the $O(1/k)$ factors; the explicit closed-form constants satisfy the conditions from $k = 50$ on and give rates below the measured-constant ones by 22–79% at $k = 50$ (the longer the window, the larger the deficit), 5–19% at $k = 100$, at most 5% at $k = 200$ and about 1% at $k = 400$. All constants are suprema over $[0, 300]$ (the exponential averages are formed on an extended grid): on the quasi-periodic regressor they are identical to four digits on $[0, 10^4]$ (closed form); on the Lorenz regressor below, the suprema over $[0, 1500]$ are larger by up to 14% (p_1 5.31 → 5.67, p_2 19.4 → 20.7, π_2 49.4 → 51.9, $\|\bar{N}_2\|$ 0.42 → 0.48 at $T = 20$), which lowers the guaranteed rate at $k = 100$ by 3% and keeps the certificate (referee round 7): the constants of a chaotic regressor are those of the stretch of the attractor that the run visits.

Chaotic $\Psi = (x_1/20, (x_3 - 25)/20)$ *along a Lorenz orbit* ($\sigma = 10$, $r = 28$, $b = 8/3$, traversed five times slower than standard Lorenz time, $\dot{\Psi}$ by the chain rule through the vector field; $q = 2$): the exact decay rates are 0.00759, 0.00304, 0.00152, 0.00076, 0.00038 at $k = 20$, 50, 100, 200, 400 (an effective excitation level 0.15), while the worst exponential window has $\lambda_T = 0.015$, 0.024, 0.051 for $T = 5$, 10, 20 ($\Lambda = 0.55$, 0.36, 0.24; $\rho \approx 20$–26, $\rho_D \approx 110$–160, $\rho_{DD} \approx 2000$–3100; $\|\bar{N}_2\| = 1.05$, 0.34, 0.42: the residual does not follow a monotone rule in T). The conditions hold from $k = 100$ on (condition (A) fails below: the derivative peak factors of a chaotic orbit are large), the certificate holds at every admissible row, and at $T = 20$ the guaranteed rate is 0.00030 / 0.00035 ($k = 100$), 0.00018 / 0.00021 ($k = 200$) against the exact 0.00152 and 0.00076: exact/guaranteed = 5.0 / 4.3 and 4.2 / 3.7, of which the window loss $\lambda_\infty/\lambda_T \approx 3$ is the dominant factor. For a chaotic reference the uniform-in-time construction pays the least exciting stretch of the attractor; a longer window ($T = 20$ against the slowed Lorenz loop time ≈ 3.75) recovers part of it ($\lambda_T = 0.015 \to 0.051$ from $T = 5$ to 20) at the price of a larger p_2 and π_2 (9.9 → 19.4 and 13.7 → 49.4). An exploratory regressor with components $\tanh(x_i/8)$ traversed at standard Lorenz speed (not in the table) had ρ_{DD} of order 10^6–10^7 and a threshold beyond $k = 400$: the slow regime requires the reference to be slow relative to the loop, and the time scale of a chaotic reference is the designer's choice.

Checked by: `tests/test_excitation.py`: on $\Psi = (\sin t, \cos\sqrt{2}\,t)$ with $T = 4\pi$, the differential identities $T\dot{\bar{M}} = \bar{M} - M$ and $\dot{P}_1 = 2(M - \bar{M})$ on the grid (10^{-4}), the conditions at $k = 20$, 50, 200 for $\delta = 0.1$, 0.2 with $\delta_2 \le 5\%$, the certificate $\sup \dot{V}/V \le -\gamma$ on 400 times, and the guaranteed rate below the exact

decay rate (top Lyapunov exponent over 200 time units) within a factor 2.6, the ratio improving with k (`test_lemma_E0dprime_quasi_periodic_regressor_certificate_and_rate`).

Theorem E (with explicit constants)**.** *Let $\Psi_G := m^{-1/2}\Psi(I_m \otimes G^{1/2}) \in \mathbb{R}^{n\times mp}$, and let $(\alpha_G, T, \Psi_{G,M}, \Psi_{G,D})$ be PE data for it: $\int_t^{t+T} \Psi_G^\top \Psi_G \,\mathrm{d}r \succeq \alpha_G I$, $\|\Psi_G\| \le \Psi_{G,M}$, $\|\dot\Psi_G\| \le \Psi_{G,D}$. One may take $\alpha_G := \alpha\lambda_{\min}(G)/m$, $\Psi_{G,M} := m^{-1/2}\lambda_G^{1/2}\sup\|\Psi\|$ and $\Psi_{G,D} := m^{-1/2}\lambda_G^{1/2}\sup\|\dot\Psi\|$; $\dot\Psi$ is bounded because $x_m \in C^1$ with bounded derivative and the Gaussian's mixed derivatives are bounded for $s \ge s_{\min}$. Let $(\varepsilon, \gamma, p_{\mathrm{lo}}, p_{\mathrm{hi}}, g_V)$ be the constants of Lemma E.0 for the data $(k_e, \alpha_G, T, \Psi_{G,M}, \Psi_{G,D})$, with any admissible ε (for example the closed form of E.0(ii)), where $g_V := 1 + \varepsilon(2Q_{G,M} + T\Psi_{G,M})$ bounds $\|\partial V/\partial w\|/\|w\|$ (the Hessian of V is I plus a matrix of norm at most $\varepsilon T\Psi_{G,M} + 2\varepsilon Q_{G,M}$) and $Q_{G,M} := T^2\Psi_{G,M}^2/2$. Let $r := \min(\rho, \delta)$, let L_J be a Lipschitz constant, uniform in t, of the Jacobian of the closed-loop field F of step 2 on the ball D, and let $S := \mathrm{diag}\big(I_n,\, m^{-1/2}(I_m \otimes G^{-1/2})\big)$ be the similarity to the coordinates $w := (e, \zeta)$ of step 3. Put*

$$\tilde C_R := \|S\|\,\frac{L_J}{2}\,\|S^{-1}\|^2, \qquad r_w := \min\Big\{\frac{r}{\|S^{-1}\|},\ \frac{\gamma}{2g_V\tilde C_R}\Big\}, \qquad r_0 := r_w\Big(\frac{p_{\mathrm{lo}}}{p_{\mathrm{hi}}}\Big)^{1/2},$$

$$\lambda := \frac{\gamma}{4p_{\mathrm{hi}}}, \qquad c_0 := \frac{r_0^2}{2}, \qquad C := \Big(\frac{2p_{\mathrm{hi}}}{p_{\mathrm{lo}}}\Big)^{1/2} \quad (\le \sqrt{6} \text{ for the closed-form } \varepsilon).$$

If $V_{\mathrm{OT}}(t_0) \le c_0$, then:

- *the optimal assignment π at t_0 is unique, and it is the unique optimal assignment for all large t;*
- *$\|e(t)\| + W(\mu_t, \mu^*) \le Ce^{-\lambda(t-t_0)}\big(\|e(t_0)\| + W(\mu_{t_0}, \mu^*)\big)$ for $t \ge t_0$;*
- *$\vartheta_i(t) \to \vartheta^*_{\pi(i)}$. The labelled error*

$$E_{\mathrm{lab}}(t) := \Big(\frac{1}{m}\sum_i \|\vartheta_i(t) - \vartheta_i^*\|_{G^{-1}}^2\Big)^{1/2} \quad \text{tends to} \quad \Big(\frac{1}{m}\sum_i \|\vartheta^*_{\pi(i)} - \vartheta_i^*\|_{G^{-1}}^2\Big)^{1/2},$$

  which is non-zero for $\pi \ne \mathrm{id}$.

Every constant is an explicit function of $(k_e, \alpha_G, T, \Psi_{G,M}, \Psi_{G,D}, m, G, r, L_J)$, hence uniform over the class of regressors with given PE data $(T, \alpha, \sup\|\Psi\|, \sup\|\dot\Psi\|)$. With the closed-form ε, $\lambda \ge \frac{1}{3}\cdot\min\{k_e/4,\, \varepsilon\alpha_G/2\}$. The constants are conservative, in a quantified way (Remark E.0a); their role is to make the statement self-contained. L_J is defined as a supremum; a closed-form bound follows from the third-derivative suprema of the Gaussian and is not computed here (`excitation. jacobian_ lipschitz_ estimate` estimates it from samples for the tests). The previous version proved this theorem with existence constants, conditional on Morgan–Narendra [62, Thm. 1]; Lemma E.0 removes that citation.

Status. Proved, self-contained (Lemma E.0 replaces the citation of Morgan–Narendra [62]); linearisation checked; the strict Lyapunov inequalities checked pointwise (E.0′ by a certificate over all states) and the explicit envelope checked on the linearised and on the implemented loop; convergence observed. The rate of E.0 is conservative in a quantified way (Remark E.0a); the rate of E.0′ is within 12% of the exact rate above its threshold. Novelty (literature audit): partially anticipated [16], [34]; the strict Lyapunov function is of the Mazenc–Malisoff type.

Proof.

1. *Symmetry.* Relabelling the network, $(e, \vartheta) \mapsto (e, \vartheta\circ\sigma)$, maps solutions to solutions. It fixes e, is an isometry of the matched metric, and conjugates the dynamics near $(0, \vartheta^*\circ\pi)$ to those near $(0, \vartheta^*)$. So it suffices to treat $\pi = \mathrm{id}$.

2. *Domain and linearisation.* Let $r := \min(\rho, \delta)$ and $D := \{z = (e, \tilde{\vartheta}) \in \mathbb{R}^{n+mp} : \|z\|_2 < r\}$.

   - For $z \in D$ we have $\|e\| < \rho$, so $f = \hat{f}^*$.
   - Also $\|\tilde{\vartheta}_i\| < \delta$ for every i, so $\vartheta_i \in \operatorname{int} \Theta_K^\varepsilon$ and $\mathrm{Proj}_K = \mathrm{id}$.

   On $[0, \infty) \times D$ the closed loop is

$$F(t, z) = \Big(-k_e e + \hat{f}^*(x) - \hat{f}(x),\ \big(G \nabla_\vartheta \Phi(x; \vartheta_i)^\top e\big)_i\Big), \qquad x = x_m(t) + e.$$

   - It is C^1 in (t, z), since $x_m \in C^1$, and C^∞ in z.
   - Its derivatives through order two are bounded uniformly in t, because the Gaussian's derivatives through order three are bounded uniformly in x for $s \geq s_{\min}$.
   - Its Jacobian at 0 is

$$A(t) = \begin{pmatrix} -k_e I & -\frac{1}{m}\Psi \\ (I_m \otimes G)\Psi^\top & 0 \end{pmatrix},$$

     and the remainder is $O(\|z\|^2)$.
   - The Jacobian and the identity $QA + A^\top Q = \mathrm{diag}(-k_e I, 0)$, with $Q = \mathrm{diag}\big(I/2,\ (I \otimes G^{-1})/(2m)\big)$, are checked numerically by `test_theorem_E_linearisation_and_lyapunov_identity`.

3. *LTV stability, with explicit constants.* Put $w := (e, \zeta)$ with $\zeta := m^{-1/2}(I_m \otimes G^{-1/2})\tilde{\vartheta}$, that is $w = Sz$, and $\Psi_G := m^{-1/2}\Psi(I \otimes G^{1/2})$.

   - The system becomes $\dot{e} = -k_e e - \Psi_G \zeta$ and $\dot{\zeta} = \Psi_G^\top e$: this is (LTV) of Lemma E.0 with $k = k_e$ and $\Psi = \Psi_G$, that is, $\dot{w} = [A_0 + B(t)]w$ with $A_0 = \mathrm{diag}(-k_e I, 0)$ and B skew-symmetric, bounded, with bounded derivative.
   - The excitation transfers: $\int \Psi_G^\top \Psi_G \, \mathrm{d}r \succeq (\alpha \lambda_{\min}(G)/m) I = \alpha_G I$, and $\|\Psi_G\| \leq \Psi_{G,M}$, $\|\dot{\Psi}_G\| \leq \Psi_{G,D}$.
   - Lemma E.0 gives the strict Lyapunov function $V(t, w)$ with $p_{\mathrm{lo}}\|w\|^2 \leq V \leq p_{\mathrm{hi}}\|w\|^2$ and $\dot{V} \leq -\gamma\|w\|^2$ along the linear system, hence

$$\|w(t)\| \leq \Big(\frac{p_{\mathrm{hi}}}{p_{\mathrm{lo}}}\Big)^{1/2} e^{-(\gamma/(2p_{\mathrm{hi}}))(t - t_0)} \|w(t_0)\| :$$

     uniform exponential stability with explicit overshoot and rate.
   - The w-coordinates are the coordinates of the matched Lyapunov function: $\|\zeta\|^2 = \frac{1}{m}\sum_i \|\tilde{\vartheta}_i\|^2_{G^{-1}} = W_\pi^2$, so $\|w\|^2 = \|e\|^2 + W_\pi^2 = 2V_\pi$.
   - The bound on $\dot{\Psi}$ cannot be dropped. $\Psi = \sin t^2$ is bounded and PE, but for $\dot{e} = -e - \Psi\zeta$, $\dot{\zeta} = \Psi e$, from $w(t_0) = (e, \zeta) = (0, 1)$, a numerical computation gives the ratio $\|w(t_0 + 200)\|/\|w(t_0)\|$ as 0.9715, 0.9942 and 0.9980 for $t_0 = 5$, 20 and 50, and part (f) of the proof of Lemma E.0 shows that the decay over a fixed horizon tends to 0 as $t_0 \to \infty$. So that system is not UES. Consistently, the rate of Lemma E.0 tends to 0 as $\Psi_D \to \infty$.

4. *Nonlinear system, with an explicit ball.* On the closure of D (to which the Lipschitz bound extends) the field is $F(t, z) = A(t)z + R(t, z)$ with $\|R(t, z)\| \leq (L_J/2)\|z\|^2$ (Taylor with the Lipschitz Jacobian). In w-coordinates, $\dot{w} = [A_0 + B(t)]w + \tilde{R}(t, w)$ with $\tilde{R}(t, w) := S\,R(t, S^{-1}w)$, so $\|\tilde{R}(t, w)\| \leq \tilde{C}_R\|w\|^2$ on $\{\|w\| < r/\|S^{-1}\|\} \subset S(D)$. Along the nonlinear system, since $\|\partial V/\partial w\| \leq g_V\|w\|$,

$$\dot{V} \leq -\gamma\|w\|^2 + g_V \tilde{C}_R \|w\|^3 \leq -\frac{\gamma}{2}\|w\|^2 \qquad \text{whenever } \|w\| \leq r_w.$$

The sublevel set $\{V \le p_{\mathrm{lo}} r_w^2\}$ is contained in $\{\|w\| \le r_w\}$, so it is forward invariant, and it contains the ball $\{\|w\| \le r_0\}$, since $V \le p_{\mathrm{hi}}\|w\|^2$. On it $\dot V \le -(\gamma/(2p_{\mathrm{hi}}))V$, so for $\|w(t_0)\| \le r_0$,

$$\|w(t)\| \le \Big(\frac{p_{\mathrm{hi}}}{p_{\mathrm{lo}}}\Big)^{1/2} e^{-\lambda(t-t_0)}\|w(t_0)\|, \qquad \lambda = \frac{\gamma}{4p_{\mathrm{hi}}}.$$

These trajectories stay in D, where the closed loop equals F. (Khalil [48, Thm. 4.13] gives the same conclusion with existence constants; the explicit form is what Lemma E.0 adds.)

5. *Transfer to* $(\|e\|, W)$. Let π be optimal at t_0. Then $\|w(t_0)\|^2 = 2V_\pi(t_0) = 2V_{\mathrm{OT}}(t_0) \le 2c_0 = r_0^2$, so step 4 applies. Moreover $\|e\| + W \le \|e\| + W_\pi = \|e\| + \|\zeta\| \le \sqrt{2}\|w\|$ at every t, and $\|w(t_0)\| \le \|e(t_0)\| + \|\zeta(t_0)\| = \|e(t_0)\| + W(t_0)$. This gives the estimate with $C = (2p_{\mathrm{hi}}/p_{\mathrm{lo}})^{1/2}$.

   Uniqueness. Suppose $V_\pi, V_{\pi'} \le c_0$ with $\pi \ne \pi'$. Then the state lies in the invariant balls of step 4 around both $(0, \vartheta^* \circ \pi)$ and $(0, \vartheta^* \circ \pi')$. The solution would converge to two distinct equilibria, which is impossible.

   Limits. Hence $z(t) \to 0$, that is, $\vartheta_i(t) \to \vartheta^*_{\pi(i)}$. For $\pi' \ne \pi$,

$$V_{\pi'}(t) \to \frac{1}{2m}\sum_i \|\vartheta^*_{\pi(i)} - \vartheta^*_{\pi'(i)}\|^2_{G^{-1}} \ge \frac{d^2}{2} > 0,$$

   while $V_\pi(t) \to 0$. So π is the unique optimal assignment for all large t, and E_{lab} converges as stated. □

Checked by: `tests/test_transport.py`, `tests/test_excitation.py` and `experiments/exp_pe_convergence.py`.

- Linearisation: the Jacobian of the implemented loop equals $A(t)$, and $QA + A^\top Q = \mathrm{diag}(-k_e I, 0)$ (`test_theorem_E_linearisation_and_lyapunov_identity`).
- Explicit constants: on the implemented loop, from a start inside the explicit ball, $\|w(t)\| \le (p_{\mathrm{hi}}/p_{\mathrm{lo}})^{1/2} e^{-\lambda t}\|w(0)\|$ at every sample (L_J estimated from samples) (`test_theorem_E_explicit_nonlinear_envelope_on_the_implemented_loop`).
- Convergence: the matched error decays and the labelled error does not (qualitative); sliding-window excitation and fitted rates (`test_theorem_E_matched_error_converges_labelled_does_not`, `exp_pe_convergence.py`).
- Not checked in the repository: the sharpness of the explicit rate beyond the tables of Remark E.0a; Corollary E.2 and Remark E.3.

Remark E.1 (the rate)**.**

- **What converges.** "The parameters converge" is false for the labelled error and true for the matched one.
- **The rate is not set by the PE level.** $\lambda > 0$ exists for each admissible Ψ, but it depends on $\Psi(\cdot)$, not only on its PE level. No positive function of $\alpha\lambda_{\min}(G)/m$, k_e, T and $\sup\|\Psi_G\|$ alone bounds it below.
  - For $m = n = 1$, $\vartheta^* = (1.5, 0.3, 0.8)$, $G = I$, $k_e = 3$ and $x_m = 1.5\sin\omega t$ ($T = 2\pi$), these four quantities do not depend on ω. The exact Floquet rate, computed numerically, falls from 4.7×10^{-2} ($\omega = 1$) to 2.9×10^{-5} ($\omega = 64$), like ω^{-2}.
  - A lower bound must also involve $\sup\|\dot\Psi_G\|$ (step 3). Lemma E.0 gives one; it reproduces the ω^{-2} law (Remark E.0a).
  - Exact Floquet rates, computed numerically, are also not monotone in G or in k_e, and they decrease for large G. When k_e is large relative to the reference bandwidth, they approach $\lambda_{\min}\langle\Psi_G^\top\Psi_G\rangle/k_e$.
- **The numbers** (`results/pe_convergence`; $m = 3$, $n = 2$, six runs):

quantity	range
min over sliding 60 s windows of $\lambda_{\min}\big(\int_t^{t+60} \Psi^\top \Psi \,\mathrm{d}r\big)$	5.9×10^{-4} to 2.7×10^{-2}
fitted decay rate of the matched error, second half of the run	2.3×10^{-3} to $4.3 \times 10^{-3}\,\mathrm{s}^{-1}$
rank correlation between the two	0.60

The convergence test is therefore qualitative. The machine-checked part of the proof is the linearisation.

Corollary E.2 (uniform-in-time exponential merging under PE)**.** *Take two solutions of the same closed loop with $V_{\mathrm{OT}}(0) \le c_0$ and $V'_{\mathrm{OT}}(0) \le c_0$. Then*

$$\|x_t - x'_t\| + W(\mu_t, \mu'_t) \le Ce^{-\lambda t} R_0, \qquad R_0 := \|e(0)\| + \|e'(0)\| + W(\mu_0, \mu^*) + W(\mu'_0, \mu^*). \tag{15.2}$$

Let $\delta_0 := \|x_0 - x'_0\| + W(\mu_0, \mu'_0)$. Combine this with the coupling estimate of Theorem D(c) on $[0, T_0]$, with $T_0 := (\Lambda + \lambda)^{-1} \log(CR_0/\delta_0)$.

- *Both solutions stay in D (proof step 4), so $\|x - x_m(t)\| < r \le \rho$.*
- *The segment between x_t and x'_t lies in $B(x_m(t), \rho)$, where $f = \hat{f}^*$, and $\mathrm{Proj}_K = \mathrm{id}$.*
- *Hence Theorem D(c) holds with $M_x := k_e + 2a_{\max} J_c$, with L_x and L_ϑ computed for $\bar{E} = r$ and an inactive projection, and with $L_2 = \lambda_G^{1/2} L_1$. This Λ depends neither on T_0 nor on f outside the tube.*

This gives uniform-in-time Hölder continuity in the basin:

$$\sup_{t \ge 0} \big(\|x_t - x'_t\| + W(\mu_t, \mu'_t) \big) \le (CR_0)^{\Lambda/(\Lambda+\lambda)} \, \delta_0^{\lambda/(\Lambda+\lambda)}. \tag{15.3}$$

Status. Proved, through Theorem E (self-contained since Lemma E.0). Not checked in the repository.

Proof. $\|x_t - x'_t\| = \|e - e'\| \le \|e\| + \|e'\|$, and $W(\mu_t, \mu'_t) \le W(\mu_t, \mu^*) + W(\mu'_t, \mu^*)$. Bound each bracket by Theorem E. On $[0, T_0]$ use $e^{\Lambda T_0} \delta_0$; on $[T_0, \infty)$ use $CR_0 e^{-\lambda T_0}$. Note that $\delta_0 \le R_0 \le CR_0$. □

Remark E.3 (the collapse point)**.** Take the G^{-1}-closest pair (j, k). Set $e = 0$ and $\vartheta_j = \vartheta_k = (\vartheta_j^* + \vartheta_k^*)/2$, with all other particles ideal.

- At this point $V_{\mathrm{OT}} = d^2/8$, and there are two optimal assignments.
- By exchange symmetry, $\vartheta_j \equiv \vartheta_k$ for all t, so $W(\mu_t, \mu^*) \ge d/2$ forever.

The uniqueness argument of step 5 shows $c_0 < d^2/8$ automatically, and $d^2/8$ is sharp along this direction.

16 Theorem F: the transport cost of adaptation

Theorem F. *Assume* (S0)–(S3)*, and take $t_0 \le t_1$ in the maximal interval of existence (all $t \ge 0$ under the hypothesis of Theorem D(b)). μ^*,* (S4) *and* (S5) *are not needed. Along* (L)*:*

(i) $\big(\frac{1}{m} \sum_i \|\dot{\vartheta}_i\|^2_{G^{-1}}\big)^{1/2} \le \lambda_G^{1/2} L_1 \|e(t)\|$*;*

(ii) $W(\mu_{t_1}, \mu_{t_0}) \le \lambda_G^{1/2} L_1 \int_{t_0}^{t_1} \|e\| \,\mathrm{d}r$*. Hence the W-length of $r \mapsto \mu_r$ on $[t_0, t_1]$ is at most $\lambda_G^{1/2} L_1 \int_{t_0}^{t_1} \|e\| \,\mathrm{d}r$;*

(iii) $W(\mu_{T_2}, \mu_{T_1}) \le \lambda_G^{1/2} L_1 (T_2 - T_1) \cdot \mathrm{RMS}_{[T_1, T_2]} \|e\|$.

Along (L_σ) *with the OT leak, for a.e. t:*

(i′) $\big(\frac{1}{m}\sum_i \|\dot\vartheta_i\|^2_{G^{-1}}\big)^{1/2} \le \lambda_G^{1/2} L_1 \|e\| + \sigma_m W(\mu_t, \mu^0)$;

(ii′) $W(\mu_{t_1}, \mu_{t_0}) \le \int_{t_0}^{t_1} \big(\lambda_G^{1/2} L_1 \|e\| + \sigma_m W(\mu_r, \mu^0)\big)\,\mathrm{d}r$;

(iii′) $W(\mu_{T_2}, \mu_{T_1}) \le \lambda_G^{1/2} L_1 (T_2 - T_1) \cdot \mathrm{RMS}_{[T_1,T_2]} \|e\| + \sigma_m \int_{T_1}^{T_2} W(\mu_r, \mu^0)\,\mathrm{d}r$.

For the labelled leak, replace $W(\mu_r, \mu^0)$ by $d_{\mathrm{lab}}(r) := \big(\frac{1}{m}\sum_i \|\vartheta_i(r) - \vartheta_i^0\|^2_{G^{-1}}\big)^{1/2} \ge W(\mu_r, \mu^0)$. The term $W(\mu_r, \mu^0)$ does not suffice there.

All bounds also hold with $L_G := \big(g_a + a_{\max}^2 (g_c J_c^2 + g_s J_s^2)\big)^{1/2} \le \lambda_G^{1/2} L_1$ in place of $\lambda_G^{1/2} L_1$. Since $\max_i \|\dot\vartheta_i\|_{G^{-1}} \le L_G \|e\|$ along (L)*, also $W_\infty(\mu_{t_1}, \mu_{t_0}) \le L_G \int_{t_0}^{t_1} \|e\|\,\mathrm{d}r$.*

Status. Proved and checked, including the leak clause, a tight leak instance with $e \equiv 0$ and a tight velocity instance. Novelty (literature audit): elementary.

Proof. (i) By Lemma P.1 and Lemma B.2(b),

$$\|\dot\vartheta_i\|_{G^{-1}} \le \|\nabla g_t(\vartheta_i)\|_G \le \lambda_G^{1/2} \|\nabla g_t(\vartheta_i)\| \le \lambda_G^{1/2} L_1 \|e(t)\|.$$

Square and average over i.

(ii) Use the identity coupling $(\vartheta_i(t_0), \vartheta_i(t_1))$ and Minkowski's inequality in $L^2\big(\frac{1}{m}\sum_i; G^{-1}\big)$. For general measures, use the Lagrangian coupling $(X_{t_0}, X_{t_1})_\# \mu_0$, or Ambrosio–Gigli–Savaré [2, Thm. 8.3.1] after the isometry $\vartheta \mapsto G^{-1/2}\vartheta$. Summing over a partition gives the length statement.

(iii) Cauchy–Schwarz.

(i′) Apply Lemma P.1 with $y = \nabla g_t - \sigma_m G^{-1}(\vartheta_i - \vartheta^0_{\tau(i)})$. Then Minkowski's inequality gives $\|y\|_G \le \|\nabla g_t\|_G + \sigma_m \|\vartheta_i - \vartheta^0_{\tau(i)}\|_{G^{-1}}$. Average over i with τ optimal. At ties, Filippov velocities are convex combinations of fields with τ optimal at the current point. Each such field satisfies (i′), and the norm is convex.

(ii′) and (iii′) follow as (ii) and (iii).

For L_G: $\|\nabla g\|_G^2 = g_a \|\varphi e\|^2 + (a^\top e)^2 \big(g_c \|\nabla_c \varphi\|^2 + g_s (\partial_s \varphi)^2\big) \le \|e\|^2 L_G^2$. □

Reading. Along (L), moving the kernel measure by W costs at least $W/(\lambda_G^{1/2} L_1)$ of integrated tracking error. This is an exchange rate between learning and tracking.

Along a leaky law the proved statement is

$$\int_{t_0}^{t_1} \big(\lambda_G^{1/2} L_1 \|e\| + \sigma_m d_r\big)\,\mathrm{d}r \ge W(\mu_{t_1}, \mu_{t_0}),$$

with $d_r = W(\mu_r, \mu^0)$ for (L_σ) and $d_r = d_{\mathrm{lab}}(r)$ for the labelled leak. This gives no lower bound on $\int \|e\|$, because the leak can pay for the whole motion. Under (L_σ) there are runs with $e \equiv 0$ and $W(\mu_{10}, \mu_0) \approx 1.05$ (found numerically).

Numerical checks. The following are checked at every sample (with the particle path length $\ge$ the Wasserstein length):

- (i) and Lemma P.1, including a tight instance with ratio ≥ 0.99;
- (ii) along (L);
- (ii′) along (L_σ), on a grid of pairs, including an instance with $e \equiv 0$ where it is tight (ratio ≥ 0.99) and the (L) reading fails.

Checked by: `tests/test_transport.py`:
`test_lemma_P1_projection_is_nonexpansive_and_tight_velocity_bound` ((i) and Lemma P.1);
`test_theorem_F_wasserstein_length_bounded_by_integrated_error` ((ii) along (L));
`test_theorem_F_leak_clause_along_ot_sigma_mod` and
`test_theorem_F_leak_clause_is_tight_when_the_leak_pays_for_all_motion` ((ii′) along (L_σ)).

17 Theorem D′: propagation of chaos for the dithered closed loop on finite horizons

Theorem D covers the deterministic projected law (L). The route to Conjecture H.6 (Section 21) lists the mean-field limit of the dithered law (L_τ) as one of its three missing pieces. This section proves it on finite horizons, with an explicit $m^{-1/2}$ rate that does not depend on the temperature, by a synchronous coupling. The uniform-in-time statement is Section 26, branch 3.

Setting. The design of Theorem H: saturated features $\tilde\Phi$, $\|\tilde\Phi\| \le K_\Phi$, with all ϑ-derivatives bounded uniformly in x, and the m-particle closed loop

$$\begin{aligned} \dot x &= f(x) + \dot x_m - k_e e - \frac{1}{m}\sum_j \tilde\Phi(x;\vartheta_j), \qquad e = x - x_m(t), \\ \mathrm{d}\vartheta_i &= b(x,e,\vartheta_i)\,\mathrm{d}t + \sqrt{2\tau}\,G^{1/2}\,\mathrm{d}B_i, \qquad i = 1,\dots,m, \\ b(x,e,\vartheta) &:= G\nabla_\vartheta\big[e^\top\tilde\Phi(x;\vartheta)\big] - \sigma(\vartheta-\vartheta^0), \end{aligned} \tag{L_τ}$$

with independent Brownian motions B_i, i.i.d. initial particles $\vartheta_i(0) \sim \bar\mu_0 \in \mathcal{P}_2(\mathbb{R}^p)$ independent of the B_i, and a deterministic $x(0) = \bar x_0$. f is C^1 and $x_m \in C^1$ with $\dot x_m$ bounded; no autonomy or periodicity is needed. The **McKean–Vlasov closed loop** is

$$\begin{aligned} \dot{\bar x} &= f(\bar x) + \dot x_m - k_e \bar e - \int \tilde\Phi(\bar x;\vartheta)\,\mathrm{d}\bar\mu_t(\vartheta), \qquad \bar e = \bar x - x_m(t), \\ \mathrm{d}\bar\vartheta &= b(\bar x_t, \bar e_t, \bar\vartheta)\,\mathrm{d}t + \sqrt{2\tau}\,G^{1/2}\,\mathrm{d}B, \qquad \bar\mu_t := \mathrm{Law}(\bar\vartheta_t), \end{aligned} \tag{MV}$$

from the same $\bar x_0$ and $\bar\vartheta(0) \sim \bar\mu_0$. Here $\bar x_t$ is deterministic, and the particle $\bar\vartheta$ interacts with its own law only through $\bar x_t$.

Standing bound. Fix $T > 0$ and suppose both x^m (for every m and every realisation) and $\bar x$ remain in a compact set X on $[0,T]$. Since $\|\frac{1}{m}\sum\tilde\Phi\| \le K_\Phi$ and $\|\int\tilde\Phi\,\mathrm{d}\bar\mu\| \le K_\Phi$ whatever the parameters, the tube bound of Theorem D(b) (with $a_{\max}$ replaced by K_Φ) gives such an X for every T when its self-consistent growth condition holds, for example for bounded f; otherwise the statement is local in time, as in Theorem D(d). Put $E_X := \sup_{t\le T,\,x\in X}\|x - x_m(t)\|$ and let

$$\begin{gathered} L_f := \mathrm{Lip}_X f, \qquad L_x := \sup\|\partial_x\tilde\Phi\|, \qquad L_\vartheta := \sup\|\nabla_\vartheta\tilde\Phi\|, \\ L_b := \lambda_G\big(L_\vartheta + E_X\sup\|\partial_x\nabla_\vartheta\tilde\Phi\| + E_X\sup\|\nabla_\vartheta^2\tilde\Phi\|\big) + \sigma, \\ \Lambda := \max\{L_f + L_x + 2L_b,\ L_\vartheta + L_b\}, \end{gathered}$$

where L_b is a Lipschitz constant of $(x,e,\vartheta) \mapsto b$ on $X \times \{\|e\| \le E_X\} \times \mathbb{R}^p$ for the norm $\|x\| + \|e\| + \|\vartheta\|$. All of these are finite by saturation and independent of m and τ; the suprema of the $\tilde\Phi$-derivatives are over all $x \in \mathbb{R}^n$ (the saturated features and their derivatives are globally bounded), so the segments between x and $\bar x$ used below may leave X. The constants grow with the output dimension n ($K_\Phi = a_{\max}\sqrt{n}$), not with the lifted dimension p beyond that.

Theorem D′. *Under the standing bound:*

(a) Well-posedness. (MV) *has a unique solution* $(\bar x, \mathrm{Law}(\bar\vartheta))$ *on* $[0,T]$*, and* $\bar x$ *is deterministic.*

(b) Synchronous coupling. *Let* $\bar\vartheta_i$*,* $i = 1,\dots,m$*, be the solutions of* (MV)*'s particle equation driven by the same* B_i *and started at the same* $\vartheta_i(0)$ *as the* m*-particle system. Then, pathwise,*

$$\begin{aligned} \|x_t - \bar x_t\| + \frac{1}{m}\sum_i\|\vartheta_i(t) - \bar\vartheta_i(t)\| &\le e^{\Lambda t}\int_0^t \zeta_s\,\mathrm{d}s, \\ \zeta_s &:= \Big\|\frac{1}{m}\sum_j\tilde\Phi(\bar x_s;\bar\vartheta_j(s)) - \int\tilde\Phi(\bar x_s;\cdot)\,\mathrm{d}\bar\mu_s\Big\|, \end{aligned}$$

and since the $\bar\vartheta_j(s)$ are i.i.d. with law $\bar\mu_s$ and $\bar x_s$ is deterministic, $\mathbb{E}\zeta_s^2 \le K_\Phi^2/m$. Hence

$$\mathbb{E}\sup_{t\le T}\Big(\|x_t-\bar x_t\|+\frac{1}{m}\sum_i\|\vartheta_i(t)-\bar\vartheta_i(t)\|\Big)\le TK_\Phi e^{\Lambda T}m^{-1/2},$$
$$\mathbb{E}\sup_{t\le T}\Big(\|x_t-\bar x_t\|+\frac{1}{m}\sum_i\|\vartheta_i(t)-\bar\vartheta_i(t)\|\Big)^2\le T^2K_\Phi^2e^{2\Lambda T}m^{-1},$$

with constants independent of $\tau\ge 0$ and of $\bar\mu_0$. The exponent $-1/2$ does not depend on the dimension; the constants do, through K_Φ and the Lipschitz bounds of $\tilde\Phi$ (standing bound).

(c) Consequences.

- *Propagation of chaos: for each fixed k, the law of $(\vartheta_1,\dots,\vartheta_k)$ on $[0,T]$ converges to the k-fold product of the path law of $\bar\vartheta$, with the per-particle bound $\mathbb{E}\sup_{t\le T}\|\vartheta_i(t)-\bar\vartheta_i(t)\|\le 2(e^{L_bT}-1)TK_\Phi e^{\Lambda T}m^{-1/2}$ for every i. (Exchangeability alone gives only the fixed-t bound $\mathbb{E}\|\vartheta_i(t)-\bar\vartheta_i(t)\|=\mathbb{E}\bar\Delta(t)\le TK_\Phi e^{\Lambda T}m^{-1/2}$, since $\sup_t$ of a mean is not the mean of the $\sup_t$; the per-particle sup bound is the pathwise Grönwall estimate of the proof.)*
- *The learned function converges at the rate $m^{-1/2}$: at each fixed t,*

$$\begin{aligned}\mathbb{E}\Big\|\hat f^m(x_t)-\int\tilde\Phi(\bar x_t;\cdot)\,\mathrm{d}\bar\mu_t\Big\|&\le(L_x+L_\vartheta)TK_\Phi e^{\Lambda T}m^{-1/2}+\mathbb{E}\zeta_t\\&\le\big((L_x+L_\vartheta)Te^{\Lambda T}+1\big)K_\Phi m^{-1/2}\end{aligned}$$

  (the supremum over t would need a chaining argument for ζ and is not pursued).
- *The parameter measure converges in W_1 (Euclidean cost on $\mathbb{R}^p$; the G^{-1} metric used elsewhere changes it by a factor $\lambda_G^{\pm1/2}$) at the i.i.d. rate:*

$$\mathbb{E}W_1(\mu_t^m,\bar\mu_t)\le TK_\Phi e^{\Lambda T}m^{-1/2}+\mathbb{E}W_1(\bar\mu_t^{\mathrm{emp},m},\bar\mu_t),$$

  where $\bar\mu_t^{\mathrm{emp},m}$ is the empirical measure of m i.i.d. samples of the time-t marginal $\bar\mu_t$; the last term is $O(m^{-1/p})$ for $p>2$ (Fournier–Guillin [30]; applicable because $\bar\mu_t$ has a bounded second moment on $[0,T]$, the drift being linearly bounded). Only this term carries the dimension.

(d) $\tau=0$. *The statement holds for the deterministic saturated law with i.i.d. initial particles, where it gives, for this law (not the projected law of Theorem D(d)), the explicit $m^{-1/2}$ rate in place of a rate $\delta_m\to 0$ for the state and the paired particles.*

Status. Proved; consistency of the $m^{-1/2}$ rate checked by coupling m-particle systems to a 4096-particle reference (`exp_chaos.py`). Novelty (literature audit): the technique is Sznitman's; new in that the plant is in the loop and the rate for the state is dimension-free.

Proof. (a) Let X' be a compact neighbourhood of X on which f is Lipschitz. For a given curve $\bar x\in C([0,T_1];X')$, the particle equation of (MV) is an SDE with drift Lipschitz in ϑ (bounded $\nabla^2_\vartheta\tilde\Phi$ and the linear leak; the derivative bounds of $\tilde\Phi$ are global in x) and constant diffusion; it has a unique strong solution, whose law $\bar\mu_t[\bar x]$ depends Lipschitz-continuously on $\bar x$: by synchronous coupling of two solutions driven by $\bar x$ and $\bar x'$, $\frac{\mathrm{d}}{\mathrm{d}t}\|\bar\vartheta-\bar\vartheta'\|\le L_b\big(2\|\bar x-\bar x'\|+\|\bar\vartheta-\bar\vartheta'\|\big)$, so $W_1(\bar\mu_t[\bar x],\bar\mu_t[\bar x'])\le 2L_b\int_0^t e^{L_b(t-s)}\|\bar x_s-\bar x'_s\|\,\mathrm{d}s$. The plant input $\dot x_m-k_e\bar e-\int\tilde\Phi\,\mathrm{d}\bar\mu$ is bounded by $\sup\|\dot x_m\|+k_eE_{X'}+K_\Phi$ whatever the measure, so for T_1 small the plant solutions from $\bar x_0$ stay in X', and the map taking $\bar x$ to the solution of the plant ODE with input $\int\tilde\Phi(\cdot\,;\vartheta)\,\mathrm{d}\bar\mu_t[\bar x]$ sends $C([0,T_1];X')$ into itself and is a contraction there, since $\mu\mapsto\int\tilde\Phi(x;\cdot)\,\mathrm{d}\mu$ is W_1-Lipschitz with constant L_ϑ and f is Lipschitz on X'. This gives a unique solution on $[0,T_1]$ in the class of

curves with values in X'; the standing bound keeps it in X, so the local solutions concatenate to $[0,T]$, and uniqueness holds among solutions with $\bar{x} \in C([0,T];X)$. This is the Picard argument of Theorem D(a) with the Dobrushin estimate replaced by the coupling estimate. (Theorem D(b)'s tube bound, used in the standing bound for a law it does not formally cover, rests only on $\|\hat{f}\| \le a_{\max}$ — here K_Φ — and on the pathwise error ODE, so it applies.)

(b) Set $\Delta_i := \vartheta_i - \bar{\vartheta}_i$ and $\delta := x - \bar{x}$. The noise cancels in $\mathrm{d}\Delta_i$, so Δ_i is absolutely continuous with $\frac{\mathrm{d}}{\mathrm{d}t}\|\Delta_i\| \le \|b(x,e,\vartheta_i) - b(\bar{x},\bar{e},\bar{\vartheta}_i)\| \le L_b\big(\|x-\bar{x}\| + \|e-\bar{e}\| + \|\Delta_i\|\big) = L_b\big(2\|\delta\| + \|\Delta_i\|\big)$, using $e - \bar{e} = \delta$ (same x_m). For δ,

$$\dot{\delta} = f(x) - f(\bar{x}) - k_e\delta - \Big[\frac{1}{m}\sum_j \tilde{\Phi}(x;\vartheta_j) - \int \tilde{\Phi}(\bar{x};\cdot)\,\mathrm{d}\bar{\mu}_t\Big],$$

and the bracket splits as

$$\frac{1}{m}\sum_j\big[\tilde{\Phi}(x;\vartheta_j) - \tilde{\Phi}(\bar{x};\vartheta_j)\big] + \frac{1}{m}\sum_j\big[\tilde{\Phi}(\bar{x};\vartheta_j) - \tilde{\Phi}(\bar{x};\bar{\vartheta}_j)\big] + \Big[\frac{1}{m}\sum_j \tilde{\Phi}(\bar{x};\bar{\vartheta}_j) - \int \tilde{\Phi}(\bar{x};\cdot)\,\mathrm{d}\bar{\mu}_t\Big],$$

of norms at most $L_x\|\delta\|$, $L_\vartheta \frac{1}{m}\sum\|\Delta_j\|$ and ζ_t. Dropping the dissipative $-k_e\delta$, $\frac{\mathrm{d}}{\mathrm{d}t}\|\delta\| \le (L_f + L_x)\|\delta\| + L_\vartheta\bar{\Delta} + \zeta_t$ with $\bar{\Delta} := \frac{1}{m}\sum\|\Delta_j\|$, and $\frac{\mathrm{d}}{\mathrm{d}t}\bar{\Delta} \le L_b\big(2\|\delta\| + \bar{\Delta}\big)$. For $D := \|\delta\| + \bar{\Delta}$ this gives $\dot{D} \le \Lambda D + \zeta_t$ with $D(0) = 0$, and Grönwall gives the pathwise bound, whose right-hand side is non-decreasing in t. For the expectation, $\bar{x}_s$ is deterministic and the $\tilde{\Phi}(\bar{x}_s;\bar{\vartheta}_j(s))$ are i.i.d. bounded by K_Φ, so

$$\mathbb{E}\zeta_s^2 = \frac{1}{m}\mathbb{E}\big\|\tilde{\Phi}(\bar{x}_s;\bar{\vartheta}_1(s)) - \mathbb{E}\tilde{\Phi}(\bar{x}_s;\bar{\vartheta}_1(s))\big\|^2 \le K_\Phi^2/m,$$

and $\mathbb{E}\int_0^T \zeta_s\,\mathrm{d}s \le TK_\Phi m^{-1/2}$; for the second moment, $\mathbb{E}\big(\int_0^T \zeta\big)^2 \le T\int_0^T \mathbb{E}\zeta^2 \le T^2K_\Phi^2/m$.

(c) For the first item, $\frac{\mathrm{d}}{\mathrm{d}t}\|\Delta_i\| \le L_b\big(2\|\delta\| + \|\Delta_i\|\big)$ with $\|\delta(s)\| \le D(s) \le e^{\Lambda s}\int_0^s \zeta$ and $\Delta_i(0) = 0$ give, by Grönwall and $\Lambda \ge L_b$,

$$\|\Delta_i(t)\| \le 2L_b\int_0^t e^{L_b(t-s)}e^{\Lambda s}\Big(\int_0^s \zeta\Big)\,\mathrm{d}s \le 2(e^{L_bt} - 1)e^{\Lambda t}\int_0^t \zeta,$$

a pathwise bound whose right-hand side is non-decreasing in t; take $\sup_t$ and expectations. Convergence of the k-particle law follows from the pathwise coupling bound. For the second, $\|\hat{f}^m(x_t) - \int \tilde{\Phi}(\bar{x}_t;\cdot)\,\mathrm{d}\bar{\mu}_t\|$ is bounded by the same three terms as the bracket in (b), the first two by $(L_x + L_\vartheta)D(t)$ and the third by ζ_t. For the third, couple μ_t^m with the empirical measure of the $\bar{\vartheta}_j(t)$ through the identity pairing (cost $\frac{1}{m}\sum\|\Delta_j\|$) and use the triangle inequality.

(d) Nothing in the argument uses $\tau > 0$: the noise cancels in the coupling and only the i.i.d. sampling of the initial particles enters through ζ. □

Remark D′.1 (what carries the dimension)**.** The plant and the learned function see the particle measure only through $\int \tilde{\Phi}\,\mathrm{d}\mu$, the mean of a bounded function. Their fluctuation is $K_\Phi/\sqrt{m}$: the exponent does not see the dimension p of the lifted parameter, and the constant $K_\Phi = a_{\max}\sqrt{n}$ grows with the output dimension only. The curse of dimensionality of Corollary B.6 (rates $m^{-1/p}$) concerns the Wasserstein distance between the parameter measures, which the closed loop never evaluates. Theorem F, by contrast, bounds the transport of μ by the tracking error and is dimension-free as well.

Remark D′.2 (what is still missing for Conjecture H.6)**.** The constant $e^{\Lambda T}$ grows with the horizon; uniformity in time would follow from a contraction of the coupled system, which the plant coupling and the non-reciprocal averaged interaction (Proposition H.7) obstruct for the general law. For the averaged system, Remark H.5 points to the uniform-in-time result of [11]; with the plant in the loop the question is Section 26, branch 3.

Numerics (`exp_chaos.py` → `results/chaos/table.md`). An $M = 4096$ particle system stands in for (MV); for $m = 16, 32, \dots, 1024$ the m-particle system is driven by the first m initial particles and Brownian motions of the reference — the coupling of (b) — with $n = 1$, $k_e = 10$, $\sigma = 0.02$, $T = 20$, at $\tau = 0$ and $\tau = 2{\cdot}10^{-3}$, and $\sup_t D_m$, $D_m := \|x^m - x^M\| + \frac{1}{m}\sum_{i\le m}\|\vartheta_i^m - \vartheta_i^M\|$, is averaged over 24 seeds. Because the m-system shares its particles with the reference, the sampling fluctuation that drives D_m has variance proportional to $1/m - 1/M$ (the shared particles contribute $(1/m - 1/M)\sum_{i\le m}$, the others $(1/M)\sum_{i>m}$), so the predicted law for the mean is a constant times $(1/m - 1/M)^{1/2}$, and $m^{-1/2}$ for $m \ll M$.

	$\tau = 0$	$\tau = 2 \cdot 10^{-3}$
mean $\sup_t D_m$, $m = 16 \to 1024$	$1.49 \cdot 10^{-1} \to 2.17 \cdot 10^{-2}$	$1.52 \cdot 10^{-1} \to 2.32 \cdot 10^{-2}$
log-log slope of the mean against m over $m \le 256$ (bootstrap s.e. over seeds)	-0.47 ± 0.06	-0.46 ± 0.05
slope of the median	-0.46	-0.45
slope of the mean against $(1/m - 1/M)^{1/2}$, all m (predicted 1)	0.94 ± 0.08	0.92 ± 0.08
mean / $(1/m - 1/M)^{1/2}$, $m = 32 \dots 1024$	0.70–0.80	0.67–0.86

The rate is the $m^{-1/2}$ of (b) within one standard error, the same at both temperatures (the noise cancels in the coupling, (d)), and the two-scale law in $(1/m - 1/M)^{1/2}$ holds up to $m = M/4$. Two features of the data are worth recording. First, single seeds differ by more than an order of magnitude at fixed m (the table lists the minimum and maximum over seeds): $\sup_t D_m$ is governed by the projection of the initial sampling fluctuation on a few slowly amplified directions of the closed loop, so the expectation in (b) needs tens of seeds — a first version of this experiment with three seeds gave slopes -0.25 and -0.21 and was withdrawn (`review/REVIEW_LOG.md`, round 5, D2). Second, the measured ratio $\sup_t D_m/(1/m - 1/M)^{1/2} \approx 0.7$ is astronomically smaller than the constant $TK_\Phi e^{\Lambda T}$ of (b): on this horizon the theorem is a rate statement, not a quantitative bound.

Checked by: `exp_chaos.py`: the experiment only; there is no test for Theorem D′ (its hypotheses are structural and the coupling is the experiment itself).

Theorem D″: uniform-in-time propagation of chaos in the leak-dominated regime

Theorem D″ (uniform-in-time propagation of chaos in the leak-dominated regime)**.** *In the setting and with the constants of Theorem D′, assume the standing bound for all $t \ge 0$ and let $L_2 := \sup\|\nabla_\vartheta^2\tilde\Phi\|$, $L_{x\vartheta} := \sup\|\partial_x\nabla_\vartheta\tilde\Phi\|$ (over all x and ϑ; for $n > 1$ the operator norms are contracted with unit vectors as in (1) of Section 9), and*

$$\kappa_e := k_e - L_f - L_x, \qquad \kappa_\vartheta := \sigma - \lambda_G E_X L_2,$$
$$L_{b'} := \lambda_G(L_\vartheta + E_X L_{x\vartheta}), \qquad \det A := \kappa_e\kappa_\vartheta - L_{b'}L_\vartheta.$$

If $\kappa_e > 0$ and $\det A > 0$ (then $\kappa_\vartheta > 0$ as well) — the leak dominates the loop gain of the adaptive law — then, under the synchronous coupling of Theorem D′(b), for every $t \ge 0$ and every $\tau \ge 0$,

$$\mathbb{E}\|x_t - \bar x_t\| \le K_\Phi\kappa_\vartheta m^{-1/2}/\det A,$$
$$\mathbb{E}\frac{1}{m}\sum_i\|\vartheta_i(t) - \bar\vartheta_i(t)\| = \mathbb{E}\|\vartheta_1(t) - \bar\vartheta_1(t)\| \le K_\Phi L_{b'} m^{-1/2}/\det A,$$

with constants independent of t: propagation of chaos is uniform in time, at the rate $m^{-1/2}$. Moreover $\sup_t \mathbb{E}W_1(\mu_t^m, \bar\mu_t) \le K_\Phi L_{b'} m^{-1/2}/\det A + \sup_t \mathbb{E}W_1(\bar\mu_t^{\mathrm{emp},m}, \bar\mu_t)$, and the second term is finite: $\sup_t \int\|\vartheta - \vartheta^0\|^2\,\mathrm{d}\bar\mu_t \le \max\{\int\|\vartheta - \vartheta^0\|^2\,\mathrm{d}\bar\mu_0,\ ((\lambda_G E_X L_\vartheta)^2/\sigma + 2\tau\,\mathrm{tr}\,G)/\sigma\}$.

Status. Proved; checked numerically on one slow-adaptation instance (`exp_chaos_uniform.py`), where it also shows its limit. The contraction condition $\kappa_\vartheta > 0$ makes the particle map a contraction uniformly in t, so in the certified regime the particles of a single system stay together: at $\tau = 0$ they synchronise exponentially and the network collapses to one kernel; at $\tau > 0$ they form the dither cloud of the prior, $\mathcal{N}(\vartheta^0, (\tau/\sigma)G)$ to leading order, and the network's output stays at the level of its prior (Remark D″.1). In both cases the leak dominates the learning: the certified regime is the one in which the network does not learn. What the theorem adds to Theorem D′ is the removal of the horizon from the bound in that regime, with explicit constants (non-trivially at $\tau > 0$, where the coupled distance is of order $m^{-1/2}$ and not zero), and a precise statement of where the contraction method stops. Referee round 6 (`review/REVIEW_round6_Dpp.md`) confirmed the proof, corrected a double-counted factor 2 in the interconnection gain $L_{b'}$ (sharp constants below), and found that the run first reported as "learning regime" was a slow collapse to the dead kernel of an odd plant; the instance was changed, and the new long-horizon run in the learning regime shows the coupled distance growing in time — the uniform statement is supported only where the theorem proves it.

Proof. Keep the contracting terms that Theorem D′'s proof dropped. For $\delta := x - \bar{x}$, the term $-k_e\delta$ gives $D^+\|\delta\| \le -(k_e - L_f - L_x)\|\delta\| + L_\vartheta\bar{\Delta} + \zeta_t$ with the same decomposition of the bracket as in D′(b). For $\Delta_i := \vartheta_i - \bar{\vartheta}_i$, split $b(x,e,\vartheta_i) - b(\bar{x},\bar{e},\bar{\vartheta}_i) = [b(x,e,\vartheta_i) - b(\bar{x},\bar{e},\vartheta_i)] + [b(\bar{x},\bar{e},\vartheta_i) - b(\bar{x},\bar{e},\bar{\vartheta}_i)]$. Since $e - \bar{e} = \delta$, the first bracket is $G\nabla_\vartheta\tilde{\Phi}(x;\vartheta_i)^\top(e-\bar{e}) + G[\nabla_\vartheta\tilde{\Phi}(x;\vartheta_i) - \nabla_\vartheta\tilde{\Phi}(\bar{x};\vartheta_i)]^\top\bar{e}$, of norm at most $\lambda_G L_\vartheta\|\delta\| + \lambda_G L_{x\vartheta}\|\delta\|E_X = L_{b'}\|\delta\|$ on the tube. The second bracket is $G[\nabla_\vartheta\tilde{\Phi}(\bar{x};\vartheta_i) - \nabla_\vartheta\tilde{\Phi}(\bar{x};\bar{\vartheta}_i)]^\top\bar{e} - \sigma\Delta_i$, and $\langle\Delta_i, \cdot\rangle \le \lambda_G\|\bar{e}\|L_2\|\Delta_i\|^2 - \sigma\|\Delta_i\|^2 \le -\kappa_\vartheta\|\Delta_i\|^2$, so $D^+\|\Delta_i\| \le -\kappa_\vartheta\|\Delta_i\| + L_{b'}\|\delta\|$, and the same for the average $\bar{\Delta}$. The pair $u := (\|\delta\|, \bar{\Delta})$ thus satisfies $D^+u \le Au + (\zeta_t, 0)^\top$ componentwise, $u(0) = 0$, with the Metzler matrix $A := \left[\begin{smallmatrix} -\kappa_e & L_\vartheta \\ L_{b'} & -\kappa_\vartheta \end{smallmatrix}\right]$ (non-negative off-diagonal entries), whose flow e^{As} is entrywise non-negative. The comparison principle for cooperative systems gives $u(t) \le \int_0^t e^{A(t-s)}(\zeta_s, 0)^\top \mathrm{d}s$ entrywise. A is Hurwitz exactly when its trace is negative and its determinant positive; with $\kappa_e > 0$ and $\det A = \kappa_e\kappa_\vartheta - L_{b'}L_\vartheta > 0$ one has $\kappa_\vartheta > L_{b'}L_\vartheta/\kappa_e \ge 0$, so both hold, and then $\int_0^\infty e^{As}\,\mathrm{d}s = -A^{-1} = (\det A)^{-1}\left[\begin{smallmatrix} \kappa_\vartheta & L_\vartheta \\ L_{b'} & \kappa_e \end{smallmatrix}\right]$, entrywise non-negative. Taking expectations, with $\mathbb{E}\zeta_s \le (\mathbb{E}\zeta_s^2)^{1/2} \le K_\Phi m^{-1/2}$ for every s (the $\bar{\vartheta}_j(s)$ are i.i.d. with law $\bar{\mu}_s$ and $\bar{x}_s$ is deterministic, as in D′(b)), $\mathbb{E}u(t) \le K_\Phi m^{-1/2}(-A^{-1})e_1 = K_\Phi m^{-1/2}(\kappa_\vartheta, L_{b'})^\top/\det A$ for every t. The per-particle statement is exchangeability at fixed t (valid here, unlike for $\sup_t$: Theorem D′(c)); the W_1 statement is the triangle inequality as in D′(c). For the second moment of $\bar{\mu}_t$: $z(t) := \mathbb{E}\|\bar{\vartheta}_t - \vartheta^0\|^2$ satisfies, by Itô's formula with $\|G\nabla_\vartheta(\bar{e}^\top\tilde{\Phi})\| \le \lambda_G E_X L_\vartheta =: \beta$, $\dot{z} \le -2\sigma z + 2\beta\sqrt{z} + 2\tau\,\mathrm{tr}\,G \le -\sigma z + \beta^2/\sigma + 2\tau\,\mathrm{tr}\,G$, whence $z(t) \le \max\{z(0), (\beta^2/\sigma + 2\tau\,\mathrm{tr}\,G)/\sigma\}$. The noise cancels in Δ_i, so the coupling estimate does not depend on τ. □

Remark D″.1 (reading)**.**

- *The certified regime is the leak-dominated one.* The condition $\kappa_\vartheta = \sigma - \lambda_G E_X L_2 > 0$ says that $\vartheta \mapsto b(\bar{x}, \bar{e}, \vartheta)$ is a contraction uniformly in t. At $\tau = 0$ the same estimate applied to two particles of the *same* system gives $D^+\|\vartheta_i - \vartheta_j\| \le -\kappa_\vartheta\|\vartheta_i - \vartheta_j\|$: all particles synchronise exponentially, the network collapses to a single kernel, and the m-system and the M-system — which share their first m particles — become identical, so the coupled distance is not just bounded but zero (the numerics: particle spread and coupled distance at the rounding level). At $\tau > 0$ the two Brownian increments do not cancel in $\vartheta_i - \vartheta_j$; the particles form a cloud around the common mean motion whose second moment is bounded as in the proof, $\mathcal{N}(\vartheta^0, (\tau/\sigma)G)$ to leading order when the leak dominates (measured: rms spread within 3% of $(\tau\,\mathrm{tr}\,G/\sigma)^{1/2}$ in the referee's runs), the coupled distance is of order $m^{-1/2}$ and the bound is not trivial — but the network's output stays at the level of its prior: $\sup|\hat{f}|$ is a few percent of $\sup|f|$ in the certified runs. Either way, propagation of chaos is uniform in time because the leak has stopped the learning. This is the price of the contraction method, not of the closed loop: the learning regime, in which kernels separate

and the network fits the plant (σ below the critical leak of Proposition K at the collapsed critical point, the leak below the separating force of the residual), is where Theorems H, I and K live, where the averaged interaction is non-reciprocal, and where uniformity in time remains the open part of Section 26, branch 3.

- *What the conditions say in the scalings of the program.* On the a-priori tube of Theorem D(b) with $e(0) = 0$, $k_e E_X = F + K_\Phi$, so the certified leak is $\sigma_{\mathrm{cert}} := \lambda_G E_X L_2 + L_{b'} L_\vartheta / \kappa_e = [\lambda_G (F + K_\Phi) L_2 + \lambda_G L_\vartheta^2]/k_e + O(k_e^{-2})$: every fixed leak gain $\sigma > 0$ certifies uniform-in-time propagation of chaos for k_e large, provided the tracking error starts at zero (or $\|e(0)\| = O(1/k_e)$; otherwise $E_X \geq \|e(0)\|$ and the condition does not improve with k_e) — and stops the learning. In the scaling $\sigma = \sigma_0/k_e$ of Theorem J the condition reads $\sigma_0 > \lambda_G (F + K_\Phi) L_2 + \lambda_G L_\vartheta^2$, which implies that J_m is strongly convex in the G^{-1} metric (its Hessian is bounded below by $\sigma_0 - \lambda_G \sup_\Theta \|r_\Theta\| L_2 \geq \sigma_0 - \lambda_G (F + K_\Phi) L_2$ there) with a unique, collapsed minimiser.

- *Why the constants are conservative.* L_ϑ, L_2, $L_{x\vartheta}$ are suprema over the parameter box and the tube; for sharp kernels (small $s_{\min}$) they are large, so σ_{cert} can exceed the adaptation gains by an order of magnitude; the a-priori E_X is itself several times the measured $\sup\|\bar{e}\|$ (0.23 against 0.07 in the numerics below). The gap to the learning regime is not a matter of constants, however: in the numerics the coupled distance grows over the horizon once the kernels separate.

- *Technique.* The comparison system is the standard contraction argument for McKean–Vlasov systems with a confining potential, with the plant as a second contracting channel (rate κ_e) and the two interconnection gains L_ϑ (particles $\to$ plant) and $L_{b'}$ (plant $\to$ particles); $\det A > 0$ is the small-gain condition of that interconnection.

Numerics (`exp_chaos_uniform.py` $\to$ `results/chaos/uniform_table.md`; reproduced in Table 16). A slow-adaptation instance ($n = 1$, $k_e = 25$, $G = 0.05 \cdot \mathrm{diag}(4, 1, 1)$, saturated features with $a_{\max} = 4$, $s \geq 0.3$, prior $\vartheta^0 = 0$), a two-bump plant that is not odd, $f(x) = 1.6e^{-(x-1)^2/0.4} - 1.2e^{-(x+1)^2/0.4} + 0.1x$, on $x_m = 1.5 \sin t + 0.5 \sin 2t$ (so that the dead kernel is not a critical point; referee round 6), $m = 16$, 64, 256 coupled to an $M = 2048$ reference. The constants are exact: the Lipschitz constants of the saturated features in closed form, $L_x = L_\vartheta = a_{\max} e^{-1/2}/s_{\min} = 8.087$ and $L_2 = L_{x\vartheta} = a_{\max}/s_{\min}^2 = 44.4$ (a numerical search over $1.2 \cdot 10^5$ parameter samples with Nelder–Mead polish reproduces them to 10^{-8}), $L_f = 2.29$, and the a-priori tube radius of Theorem D(b) with $e(0) = 0$, $E_X = (F + K_\Phi)/k_e = 0.228$ ($F = 1.70$; the measured $\sup\|\bar{e}\|$ is 0.070): $\kappa_e = 14.6$, $L_{b'} = 3.64$, $\sigma_{\mathrm{cert}} = 4.04$. Proposition K gives the critical leak of the collapsed critical point (the best single kernel, $a = 0.136$, $c = 0.068$, $s = 1.15$): $\sigma_K = \sigma_{0,\mathrm{crit}}/k_e = 0.00144$ — the certified leak is 2800 times the leak below which the kernels separate. Four runs:

Table 16: Theorem D″ on the slow-adaptation instance: four runs at different leaks (`results/chaos/uniform_table.md`).

σ, τ	regime	T, seeds	reference network at T: rms spread in (a, c, s) (from 1.53), fit error $\|f - \hat{f}\|/\|f\|$ (from 1.00)	max D_m, first $\to$ last window, $m = 16$ and 256	slope against m (per-seed, $\pm$ s.e.)	bound $\sup_t \mathbb{E}D_m$, $m = 16$ / 256
4.85 ($1.2\sigma_{\text{cert}}$), 0.02	certified, with dither	80 s, 8	0.033 (the dither cloud: $(\tau \operatorname{tr} G/\sigma)^{1/2} = 0.035$), 1.000	$6.7 \cdot 10^{-4} \to 7.9 \cdot 10^{-4}$ and $1.6 \cdot 10^{-4} \to 1.9 \cdot 10^{-4}$	-0.51 ± 0.02	0.55 / 0.14
4.85, 0	certified, no dither	80 s, 8	$1.7 \cdot 10^{-14}$, 1.000	$2.5 \cdot 10^{-25} \to 0$ and $1.2 \cdot 10^{-26} \to 0$	—	0.55 / 0.14
0.505 ($\sigma_{\text{cert}}/8$), 0	leak-dominated, below the threshold	80 s, 8	$3.2 \cdot 10^{-14}$, 1.000	$7.2 \cdot 10^{-5} \to 4.3 \cdot 10^{-11}$ and $3.1 \cdot 10^{-5} \to 1.8 \cdot 10^{-11}$	-0.33 ± 0.20	—
0.00048 ($\sigma_K/3$), 0	learning regime	1000 s, 6	1.53, 0.632	$6.1 \cdot 10^{-2} \to 2.0 \cdot 10^{-1}$ and $1.3 \cdot 10^{-2} \to 3.9 \cdot 10^{-2}$	-0.62 ± 0.16	—

(Windows $(10, 20]$, $(20, 40]$, $(40, 80]$ for the 80 s runs and $(100, 200]$, $(200, 500]$, $(500, 1000]$ for the long run; the table reports $\mathbb{E} \max_t D_m$ over a window, which exceeds the theorem's $\sup_t \mathbb{E}D_m$.) At the certified leak with dither the particles stay in the cloud of the prior, the network's output stays at the level of the empty network (fit error 1.000), and the coupled distance is of order $m^{-1/2}$ (slope -0.51 ± 0.02), bounded in time (the last window, twice as long, has maxima 15–18% above the first) and seven hundred times below the bound: the theorem's content at $\tau > 0$. Without dither the reference collapses (spread 10^{-14}) and the coupled distance is zero to rounding; at $\sigma_{\text{cert}}/8$ the particles still synchronise and the distance decays with them. In the learning regime ($\sigma_K/3$, the collapsed configuration unstable) the network stays spread, fits the plant (fit error $1.00 \to 0.63$ at $T = 1000$ s with these slow gains) and the coupled distance follows the $m^{-1/2}$ law at each time but *grows* over the horizon — by a factor 3 between the windows $(100, 200]$ and $(500, 1000]$ for every m: on this horizon propagation of chaos in the learning regime is the finite-horizon statement of Theorem D′, not a uniform one. Whether the growth saturates on longer horizons (the leak's own time scale is $1/\sigma = 2000$ s) is open; the earlier claim that the distance stays bounded in a "spread" regime (version 4, first draft) rested on a run that was in fact a slow collapse, and is withdrawn (referee round 6, DPP-3).

Checked by: `exp_chaos_uniform.py`: the experiment only.

18 Theorem D‴: uniform-in-time propagation of chaos in the learning regime (the averaged slow loop)

Theorem D″ ends where the leak stops dominating the loop gain: in the learning regime its small-gain argument has nothing to say, and the long-horizon run of `exp_chaos_uniform.py` showed the coupled distance growing over the horizon. This section isolates the structure that the small-gain argument throws away. Under the synchronous coupling the interaction between the particles enters through the fitted function $\hat{f}_\Theta$, which is linear in the empirical measure; the loss $\bar{L}$ is a convex quadratic of it (flat convexity, Theorem H(i)); and the interaction term of the coupling identity is therefore the negative of a squared $L^2(\rho)$-norm at linear order — it contracts, whatever the gains. What remains and can expand is the self term, whose curvature is the Hessian of the first-variation potential, $\sigma_0 G^{-1} - R[\mu](\vartheta)$ with R the *residual-weighted*

curvature of the features — the same object as Proposition K's R_2 at a collapsed configuration. Proposition K says that the collapsed configuration is unstable when this curvature exceeds the leak *with the collapsed residual*; at a fitted configuration the residual is the approximation error, and the second-order optimality condition of the limit measure says that the curvature is *below* the leak at every atom. The theorem below turns this into a uniform-in-time statement for the averaged slow loop at $\tau = 0$, with explicit constants, in the regime in which the network learns.

Setting. The averaged slow dynamics of Section 24 at $\tau_0 = 0$ (Theorem J′'s leading order $\bar{L}_0$ without dither; slow time $s = t/k_e$, leak $\sigma_0 = k_e\sigma$, $f = f(x)$ continuous, ρ the occupation measure of the T_0-periodic reference, $F_\rho := \sup_{\operatorname{supp}\rho}\|f\|$, saturated features, $K_\Phi := \sup\|\tilde{\Phi}\|$), written as the G-preconditioned gradient flow of $J_m(\Theta) = m\bar{L}(\mu_\Theta) + \frac{\sigma_0}{2}\sum_i\|\vartheta_i - \vartheta^0\|^2_{G^{-1}}$:

$$\frac{\mathrm{d}\vartheta_i}{\mathrm{d}s} = -G\nabla_{\vartheta_i}J_m(\Theta) = -G\nabla_\vartheta\mathcal{V}[\mu_\Theta](\vartheta_i), \qquad i = 1..m, \tag{$\mathfrak{A}_m$}$$
$$\mathcal{V}[\mu](\vartheta) := V_L[\mu](\vartheta) + \frac{\sigma_0}{2}\|\vartheta - \vartheta^0\|^2_{G^{-1}},$$

with $V_L[\mu](\vartheta) = -\langle r_\mu^\top\tilde{\Phi}(\cdot\,;\vartheta)\rangle_\rho$ and $r_\mu = f - \hat{f}[\mu]$, $\hat{f}[\mu] := \int\tilde{\Phi}(\cdot\,;\vartheta')\,\mathrm{d}\mu(\vartheta')$, as in Section 20. Its mean-field equation is the Wasserstein gradient flow ($\mathfrak{A}$) of Section 20 at $\tau = 0$ in slow time, $\partial_s\mu_s = \nabla\cdot(\mu_s G\nabla\mathcal{V}[\mu_s])$, solved by the characteristic flow

$$\frac{\mathrm{d}X_s(\vartheta)}{\mathrm{d}s} = -G\nabla\mathcal{V}[\mu_s](X_s(\vartheta)), \qquad X_0 = \mathrm{id}, \qquad \mu_s = (X_s)_\#\mu_0 \tag{$\mathfrak{A}_0$}$$

(a McKean–Vlasov equation without noise whose vector field has linear growth through the leak, is Lipschitz in ϑ and, through r_μ, Lipschitz in μ for W_1; existence, uniqueness and the representation by characteristics follow from Dobrushin's argument [25] — the synchronous-coupling proof of Theorem D′(b) with the plant removed — and the velocity is bounded on the invariant ball of Step 0). The constants are taken in the G^{-1} metric on parameters ("whitened"):

$$L^G_\vartheta := \sup_{x,\vartheta}\|\nabla_\vartheta\tilde{\Phi}(x;\vartheta)G^{1/2}\|, \qquad L^G_2 := \sup_{x,\vartheta,\|u\|=1}\|G^{1/2}\nabla^2_\vartheta\big(u^\top\tilde{\Phi}(x;\vartheta)\big)G^{1/2}\|,$$

so that $\|\tilde{\Phi}(x;\vartheta) - \tilde{\Phi}(x;\vartheta')\| \le L^G_\vartheta\|\vartheta - \vartheta'\|_{G^{-1}}$, $\|\nabla_\vartheta\tilde{\Phi}(x;\vartheta)v\| \le L^G_\vartheta\|v\|_{G^{-1}}$ and $\|(\nabla_\vartheta\tilde{\Phi}(x;\vartheta) - \nabla_\vartheta\tilde{\Phi}(x;\vartheta'))v\| \le L^G_2\|\vartheta - \vartheta'\|_{G^{-1}}\|v\|_{G^{-1}}$ ($L^G_\vartheta \le \lambda_G^{1/2}L_\vartheta$ and $L^G_2 \le \lambda_G L_2$ in the constants of Section 17; `chaos_learning.whitened_lipschitz_constants` computes the suprema by search, with closed-form bounds). The *residual-weighted curvature* of μ at ϑ is the symmetric matrix

$$R[\mu](\vartheta) := \Big\langle\sum_k r^k_\mu(x)\nabla^2_\vartheta\tilde{\Phi}^k(x;\vartheta)\Big\rangle_\rho = -\nabla^2_\vartheta V_L[\mu](\vartheta), \qquad R^G := G^{1/2}RG^{1/2},$$
$$\kappa_R[\mu](A) := \sup_{\vartheta\in A}\lambda_{\max}\big(R^G[\mu](\vartheta)\big),$$

so that $G^{1/2}\nabla^2_\vartheta\mathcal{V}[\mu](\vartheta)G^{1/2} = \sigma_0 I - R^G[\mu](\vartheta)$; for $\mu = \delta_{\bar\vartheta}$ and $\vartheta = \bar\vartheta$ it is Proposition K's R_2 and $\sigma_{\mathrm{crit}}(\bar\vartheta) = \kappa_R[\delta_{\bar\vartheta}](\{\bar\vartheta\})$. It satisfies $\|R^G[\mu](\vartheta)\| \le L^G_2\langle\|r_\mu\|\rangle_\rho \le L^G_2(F_\rho + K_\Phi)$ and $\|R^G[\mu](\vartheta) - R^G[\nu](\vartheta)\| \le L^G_2\|\hat{f}[\mu] - \hat{f}[\nu]\|_{L^1(\rho)}$. $N_\varrho(A) := \{\vartheta : \mathrm{dist}_{G^{-1}}(\vartheta, A) \le \varrho\}$.
The coupling. $\vartheta_i(0)$, $i = 1..m$, i.i.d. with law μ_0 of compact support, $R_0 := \sup_{\operatorname{supp}\mu_0}\|\vartheta - \vartheta^0\|_{G^{-1}}$; $\Theta(s)$ solves ($\mathfrak{A}_m$) and the reference particles are $\bar\vartheta_i(s) := X_s(\vartheta_i(0))$, i.i.d. with law μ_s and $\bar\vartheta_i(s) \in \operatorname{supp}\mu_s$. Let $\delta_i := \vartheta_i - \bar\vartheta_i$,

$$D(s) := \frac{1}{m}\sum_i\|\delta_i(s)\|^2_{G^{-1}}, \qquad Y(s) := \|\hat{f}_{\bar\Theta(s)} - \hat{f}[\mu_s]\|_{L^2(\rho)},$$

Y being the sampling fluctuation of the reference's fitted function ($\mathbb{E}Y(s)^2 = \frac{1}{m}\langle\mathrm{Var}_{\mu_s}\tilde{\Phi}(x;\cdot)\rangle_\rho \le K^2_\Phi/m$). A priori (Step 0 below) every particle of either system stays in the ball $\|\vartheta - \vartheta^0\|_{G^{-1}} \le R_\Theta := \max\{R_0,\ L^G_\vartheta(F_\rho + K_\Phi)/\sigma_0\}$, so $D \le D_{\max} := 4R^2_\Theta$, and the characteristics move at speed at most $V_{\max} := L^G_\vartheta(F_\rho + K_\Phi) + \sigma_0 R_\Theta$.

Theorem D‴ (uniform-in-time propagation of chaos in the learning regime, $\tau = 0$)**.** *Assume*

(H) (non-degenerate stability along the mean-field flow) *there are* $s_1 \geq 0$, $\kappa > 0$ *and* $\varrho' > 0$ *such that*

$$\kappa_R[\mu_s]\big(N_{\varrho'}(\operatorname{supp}\mu_s)\big) \leq \sigma_0 - \kappa \qquad \text{for every } s \geq s_1,$$

and put $a(s) := \sigma_0 - \kappa_R[\mu_s](N_{\varrho'}(\operatorname{supp}\mu_s))$ *(so* $a \geq \kappa$ *on* $[s_1,\infty)$ *and* $a \geq \sigma_0 - L_2^G(F_\rho + K_\Phi)$ *everywhere) and* $\Lambda_1 := \int_0^{s_1}(\kappa - a(s))_+\,\mathrm{d}s < \infty$. *Define*

$$\varrho := \min\big\{(2\kappa)^{1/2}/L_2^G,\ \varrho'\kappa e^{-\Lambda_1}/(4(L_\vartheta^G)^2)\big\},$$
$$\theta := \min\big\{\kappa/(8L_2^G),\ \varrho\kappa e^{-\Lambda_1}/(4L_\vartheta^G),\ \varrho'\kappa e^{-\Lambda_1}/(4L_\vartheta^G)\big\},$$
$$\Delta := \theta/(4L_\vartheta^G V_{\max}), \qquad C_1 := 4e^{2\Lambda_1}(L_\vartheta^G)^2K_\Phi^2/\kappa^2, \qquad m_0 := 16K_\Phi^2/\theta^2.$$

Then for every $m \geq m_0$*:*

(a) (exponentially long horizons) *for every* $S > 0$ *and every* $s \in [0,S]$,

$$\mathbb{E}D(s) \leq C_1/m + 4R_\Theta^2(S/\Delta + 2)\exp\big(-m\theta^2/(32K_\Phi^2)\big);$$

in particular $\sup_{s\leq S}\mathbb{E}D(s) \leq (C_1 + 12R_\Theta^2)/m$ *for all* $S \leq \Delta e^{m\theta^2/(64K_\Phi^2)}$ *and* $m \geq m_0$ *with* $e^{m\theta^2/(64K_\Phi^2)} \geq m$.

(b) (uniform in time) *if moreover* (H′) *the characteristic flow converges exponentially on the support of* μ_0, $\sup_{\vartheta\in\operatorname{supp}\mu_0}\|X_s(\vartheta) - X_\infty(\vartheta)\|_{G^{-1}} \leq C_2e^{-\kappa_2 s}$ *for all* $s \geq 0$, *then for every* $s \geq 0$

$$\mathbb{E}D(s) \leq C_1/m + 4R_\Theta^2(R_{\mathrm{eff}}/\Delta + 3)\exp\big(-m\theta^2/(32K_\Phi^2)\big), \qquad R_{\mathrm{eff}} := \kappa_2^{-1}\big(\log(8L_\vartheta^GC_2/\theta)\big)_+.$$

(c) (the learned function and the particles) *in either case, with the same right-hand side* $B(s)$ *for* $\mathbb{E}D(s)$,

$$\mathbb{E}\|\hat f_{\Theta(s)} - \hat f[\mu_s]\|_{L^2(\rho)}^2 \leq 2(L_\vartheta^G)^2B(s) + 2K_\Phi^2/m,$$
$$\mathbb{E}W_2(\mu_{\Theta(s)},\mu_s) \leq B(s)^{1/2} + \mathbb{E}W_2(\mu_{\bar\Theta(s)},\mu_s),$$

and on the event $\mathcal{G}_S := \{\sup_{s\leq S}Y(s) \leq \theta\}$ *(case* (a)*;* $\mathcal{G}_\infty$ *in case* (b)*), whose complement is the tail above, every particle stays within* $3\varrho'/4$ *of its reference up to time* S*:* $\max_i\sup_{s\leq S}\|\delta_i(s)\|_{G^{-1}} \leq 3\varrho'/4$ *— no particle changes basin.*

The constants depend on $(\kappa,\varrho',\Lambda_1,L_\vartheta^G,L_2^G,K_\Phi,F_\rho,\sigma_0,R_0)$ *and not on* m, s *or* S.

Status. Proved (synchronous coupling with the three-term identity of Step 1, a bootstrap on the mean square and the per-particle distance, and McDiarmid's inequality [61] for the reference's sampling fluctuation; the finite-horizon part of the transient is Theorem D′'s mechanism and appears in Λ_1). Hypothesis (H) is a property of the deterministic mean-field flow and can be computed; it is the strict second-order optimality condition of the limit configuration with a margin and a neighbourhood (Remark D‴.1), and it is compatible with the learning regime $\sigma_0 < \sigma_{\mathrm{crit}}$ of Proposition K. It requires the support of μ_s to stay away from the ridges of $\mathcal{V}[\mu_s]$ for $s \geq s_1$: for an initial cloud whose support is a union of pieces each flowing into one basin (a warm start, or a cluster per bump) this holds; for a connected cold start whose mass splits between several basins it fails for every s (Remark D‴.2), and the numerics below show what the coupling does there. Numerics: `exp_chaos_learning.py` → `results/chaos/learning_table.md` (below). The theorem is about the averaged slow dynamics; its transfer to the real loop at finite k_e is open (Remark D‴.4). Its constants are worst-case and its regime is asymptotic: on the instance below m_0 is of order 10^{23} (Remark D‴.6), and the finite-m behaviour is described sharply by Proposition D‴.5. Referee round 8 (`review/REVIEW_round8_Dppp.md`) re-derived every constant of Steps 0–9, found no counterexample to (2.1), (3.1), (4.1) on an adversarial search, and corrected the wording of Remarks

D‴.1–D‴.2, the good event per horizon and the well-posedness sentence (all applied). Referee round 9 (`review/REVIEW_round9_D3b.md`) reviewed Proposition D‴.5, Corollary D‴.7 and the numerics: all hold with fixes, the main one being that the finite-m replicated configuration is a minimiser only for admissible weights (Proposition D‴.5(iv)) — one of the four $m = 16$ warm seeds sits on a saddle, which the numerics paragraph now says. Novelty (literature audit): the technique (synchronous coupling + bootstrap + bounded differences) is standard; the sign structure of the interaction and the local, residual-weighted hypothesis — and with them a uniform-in-time statement in a regime where the network learns — are new in propagation of chaos for adaptive control; the theorem's regime is asymptotic in m and the finite-m law is a delta-method statement, new in its object.

Literature audit (`LITERATURE_TRANSPORT.md`, D‴).

- *Prior.* Uniform-in-time propagation of chaos by synchronous coupling under *global* convexity of the confinement and convexity of the interaction is Malrieu's [58]; non-convex confinements with *small* Lipschitz interactions are handled by reflection coupling [26]; flat-convex energies with dither by the entropy method [11]. The Hessian decomposition "Gram + residual-weighted feature curvature" is the standard tool of the interpolation / PL-condition literature [56] and of lazy-training analyses, where the second term vanishes at interpolation.

- *Distinction.* (i) The interaction of the fitted network is neither small nor convex in the particle coordinates; its sign comes from the flat convexity of $\bar{L}$ ($\hat{f}$ linear in the measure), which makes $-\|\hat{f}_\Theta - \hat{f}_{\bar{\Theta}}\|^2$ the leading interaction term under the coupling — the finite-m, two-measure version of [56]'s decomposition. (ii) The convexity needed is local: $\sigma_0 I - G^{1/2} R[\mu_s] G^{1/2} \succeq \kappa I$ on a neighbourhood of the support of the mean-field flow, i.e. the strict second-order optimality condition of the fitted configuration, evaluated at the actual residual — not a global convexity (which would force the collapsed configuration to be stable and the network not to learn, Theorem D″). (iii) The window $\kappa_R[\mu_\infty] < \sigma_0 < \sigma_{\mathrm{crit}}$ between the collapsed instability (Proposition K) and the fitted stability is a design statement about the leak that has no counterpart in the cited work. (iv) The limits are stated with the result: a connected initial support that splits between basins violates the hypothesis for all times (the characteristic flow is a homeomorphism), and the coupling then has a floor of pairs in different basins; the real loop at finite k_e is not covered.

- *Proposition D‴.5 (the weight-sensitivity law).* At finite m the stationary coupled distance is the multinomial fluctuation of the cluster weights propagated through the implicit function $w \mapsto \Theta^*(w)$: a delta-method statement (the Gaussian limit of a smooth function of a multinomial), standard in statistics; what is specific is its object (the shared-particle coupling of a particle approximation to a stationary K-atom measure, with the covariance $(1/m - 1/M)(\operatorname{diag} \bar{w} - \bar{w}\bar{w}^\top)$) and its role as the sharp form of the theorem's constant (stiffness^{-1} $\times$ feature correlations against $\kappa^{-1} L^G_\vartheta K_\Phi$).

- *Proposition D‴.9 (measure-valued priors).* Leaks toward a single prior (σ-modification, Ioannou–Kokotović) are the standard of robust adaptive control; regularisation toward the nearest of several centres is standard in machine learning (mixture priors, prototype regularisers) and this document's own Theorem C′ leaks toward a prior *measure* through an optimal plan. The nearest-atom leak is the simplest measure-valued prior; what is new is its role in the window of Theorem D‴ — the prior's placement sets the stationary residual, hence the lower end of the window and the relaxation rate — and the finding that removing the bias raises the coupling's floors (on the instance the weight sensitivity rises with the amplitudes): a bias–variance trade-off stated for the particle system. Verdict: standard object, new design statement; the function-level sensitivity law is the delta method once more.

- *Proposition D‴.12 (the mass constraint).* Capacity-constrained and equal-mass quantisation through optimal plans, and semi-discrete optimal transport with its Laguerre cells, are classical ([5, 3]); the integrality of the transportation polytope and the forest structure of its vertices are textbook ([42]); c-cyclical monotonicity is the standard optimality criterion ([86]). What is new is the observation that the finite-m floor of the mean-field limit of an adaptive law is carried by a conservation law of its leak (the cluster masses under the nearest-atom leak), and that a design of the leak — the mass constraint of the OT plan — breaks it: on the commensurate sizes the stationary measure of the particle system coincides with the mean-field one, and the finite-m fluctuation of the plan is counted (at most $K_0 - 1$ split particles) rather than estimated. Propagation of chaos with an exactly m-independent stationary law is not in the literature we know of; the mean-field form of the barycentric leak for an atomic prior (the Laguerre-cell leak) was Section 26's open item. Numerics: `exp_ot_leak.py`. Verdict: textbook ingredients, new mechanism and new object (a conservation law of the leak as the source of the finite-size floor, broken by the plan's mass constraint).

- *Proposition D‴.17 (convergence of the OT dynamics).* The Kurdyka–Łojasiewicz route to the convergence of (sub)gradient trajectories of definable functions is standard ([8, 50]); its application to an adaptive law whose energy is a semiconcave transport cost plus a smooth loss is a routine verification (the saturated Gaussian features are exponential-analytic, the plan is a minimum over finitely many vertices). Verdict: imported tool, routine verification; the content is that the mass-constrained law converges without a rate, which separates the qualitative question (settled) from Theorem D‴'s quantitative one (open for the OT dynamics).

- *Proposition D‴.20 (the entropic OT leak).* Entropic OT, Sinkhorn's algorithm and barycentric mappings of entropic plans are standard ([21]), and so is the interpolation of the entropic cost between optimal transport and a kernel (MMD-type) discrepancy as ε runs from 0 to ∞ (Sinkhorn divergences, [21]); size-constrained soft assignment by Sinkhorn (equipartition) is related prior art for a soft mass constraint. A critical temperature equal to twice the largest eigenvalue of a conditional covariance is deterministic annealing's ([77]), with the same roles — the particles as the codevectors, the prior's atoms as the data; here the curvature of the fit enters, and the threshold is the largest generalised eigenvalue of the conditional covariance relative to $I - R^G/\sigma_0$. What is new is the soft mass constraint as an adaptive law's leak — a smooth gradient flow between the OT leak and the σ-modification toward the prior's barycentre — in which the conditional covariance of a collapsed class is an effect of its mass excess, and its consequence for propagation of chaos: the masses equalise only down to a critical excess, so the finite-size floor returns at large m for every positive regularisation. Verdict: standard ingredients, new object and new consequence (on the commensurate sizes only the hard constraint is floor-free for every m).

- *Theorem D‴.21 (a measure-level basin for the OT leak).* The Polyak–Łojasiewicz inequality and its linear convergence are classical ([74]); in Wasserstein space Łojasiewicz inequalities are known for λ-convex and displacement-convex functionals, and PL inequalities through optimal maps for some that are not geodesically convex — barycentre functionals, the distance of uniform point clouds to a density ([7]); the leak alone, $\frac{1}{2}W^2(\cdot,\nu)$, satisfies one with constant 2 (its metric slope is $W(\cdot,\nu)$; AGS [2, §10.4]). The theorem's plan segments are the generalised geodesic from μ^* to μ_Θ based at the prior (AGS [2, §9.2]), along which $\frac{1}{2}W^2(\cdot,\nu)$ is 1-convex, and turning convexity along such a curve into a slope bound is AGS's device. The adaptive law's energy is semiconvex along generalised geodesics only with a negative constant (the loss's curvature at the actual residual), so a global argument is unavailable. What is new is the PL inequality along that geodesic for this energy — the loss linearised exactly in the fitted function with the curvature taken at the

stationary residual, where the fixed counts of the mass constraint make the geodesic a particle interpolation and the fitted function's increment the average of the per-particle increments: its constant is set by the residual-weighted curvature along the segments and does not depend on m (sharp, $2\sigma_0$, at a matching prior); an m-independent energy basin, and uniqueness of the global minimum among empirical measures under a curvature hypothesis at the stationary residual; and the converse for the nearest-atom leak, whose energy has a spurious minimum at every admissible count vector — the energy-landscape form of the finite-size floor. Numerics: `exp_ot_basin.py`. Verdict: classical inequality and geometry (the generalised geodesic), new mechanism for a new energy (the counts as the source of uniformity in m).

Proof. Step 0 (a priori bounds). For either system, $\frac{\mathrm{d}}{\mathrm{d}s}\|\vartheta-\vartheta^0\|_{G^{-1}}^2 = 2\langle\vartheta-\vartheta^0,\, \langle\nabla_\vartheta(r^\top\tilde\Phi(\cdot\,;\vartheta))\rangle_\rho\rangle - 2\sigma_0\|\vartheta-\vartheta^0\|_{G^{-1}}^2$ with $|\langle\vartheta-\vartheta^0,\, \langle\nabla_\vartheta(r^\top\tilde\Phi)\rangle_\rho\rangle| = |\langle r^\top\nabla_\vartheta\tilde\Phi(\vartheta-\vartheta^0)\rangle_\rho| \le \langle\|r\|\rangle_\rho L_\vartheta^G\|\vartheta-\vartheta^0\|_{G^{-1}} \le (F_\rho+K_\Phi)L_\vartheta^G\|\vartheta-\vartheta^0\|_{G^{-1}}$, so $D^+\|\vartheta-\vartheta^0\|_{G^{-1}} \le L_\vartheta^G(F_\rho+K_\Phi)-\sigma_0\|\vartheta-\vartheta^0\|_{G^{-1}}$ and $\|\vartheta(s)-\vartheta^0\|_{G^{-1}} \le R_\Theta$; hence $\|\delta_i\|_{G^{-1}} \le 2R_\Theta$, $D \le D_{\max}$, and $\|\,\mathrm{d}\vartheta/\,\mathrm{d}s\|_{G^{-1}} = \|G^{1/2}\langle\nabla_\vartheta(r^\top\tilde\Phi)\rangle_\rho - \sigma_0 G^{-1/2}(\vartheta-\vartheta^0)\| \le V_{\max}$. Two Lipschitz facts: for Θ, $\bar\Theta$ with rows ϑ_j, $\bar\vartheta_j$ and $A_j(x) := \int_0^1 \nabla_\vartheta\tilde\Phi(x;\bar\vartheta_j+t\delta_j)\,\mathrm{d}t$,

$$\hat f_\Theta - \hat f_{\bar\Theta} = \frac{1}{m}\sum_j A_j\delta_j =: h, \qquad \|h\|_{L^2(\rho)} \le L_\vartheta^G\frac{1}{m}\sum_j\|\delta_j\|_{G^{-1}} \le L_\vartheta^G D^{1/2}, \tag{0.1}$$

$$\begin{aligned} &\|R^G[\mu_{\bar\Theta}](\vartheta) - R^G[\mu_s](\vartheta)\| \le L_2^G\|r_{\bar\Theta} - r_{\mu_s}\|_{L^1(\rho)} \le L_2^G Y,\\ &\|(\nabla_\vartheta\tilde\Phi(x;\vartheta_i) - A_i(x))v\| \le \frac{L_2^G}{2}\|\delta_i\|_{G^{-1}}\|v\|_{G^{-1}}. \end{aligned} \tag{0.2}$$

Step 1 (the identity). Since $\mathrm{d}\delta_i/\,\mathrm{d}s = -G[\nabla\mathcal{V}[\mu_\Theta](\vartheta_i) - \nabla\mathcal{V}[\mu_s](\bar\vartheta_i)]$ and $\langle\delta,\, Gv\rangle_{G^{-1}} = \langle\delta,\, v\rangle$,

$$\begin{aligned} \tfrac12\frac{\mathrm{d}D}{\mathrm{d}s} &= -\frac{1}{m}\sum_i\langle\delta_i,\, \nabla\mathcal{V}[\mu_\Theta](\vartheta_i) - \nabla\mathcal{V}[\mu_s](\bar\vartheta_i)\rangle = T_A + T_B + T_C,\\ T_A &:= -\frac{1}{m}\sum_i\langle\delta_i,\, \nabla\mathcal{V}[\mu_{\bar\Theta}](\vartheta_i) - \nabla\mathcal{V}[\mu_{\bar\Theta}](\bar\vartheta_i)\rangle,\\ T_B &:= -\frac{1}{m}\sum_i\langle\delta_i,\, \nabla V_L[\mu_\Theta](\vartheta_i) - \nabla V_L[\mu_{\bar\Theta}](\vartheta_i)\rangle,\\ T_C &:= -\frac{1}{m}\sum_i\langle\delta_i,\, \nabla V_L[\mu_{\bar\Theta}](\bar\vartheta_i) - \nabla V_L[\mu_s](\bar\vartheta_i)\rangle \end{aligned}$$

(the leak is linear and sits in T_A; the three brackets telescope to $\nabla\mathcal{V}[\mu_\Theta](\vartheta_i) - \nabla\mathcal{V}[\mu_s](\bar\vartheta_i)$). The self term T_A compares the *reference's* potential at the two points; the interaction term T_B compares the two empirical measures at the m-system's point; T_C is the reference's sampling error. `chaos_learning.AveragedSlowLoop.decomposition` evaluates the three terms and checks the identity to rounding.

Step 2 (the self term is local convexity). $\nabla\mathcal{V}[\mu_{\bar\Theta}](\vartheta_i) - \nabla\mathcal{V}[\mu_{\bar\Theta}](\bar\vartheta_i) = \int_0^1\nabla^2\mathcal{V}[\mu_{\bar\Theta}](\bar\vartheta_i+t\delta_i)\,\mathrm{d}t\cdot\delta_i = \sigma_0 G^{-1}\delta_i - \bar R_i\delta_i$ with $\bar R_i := \int_0^1 R[\mu_{\bar\Theta}](\bar\vartheta_i+t\delta_i)\,\mathrm{d}t$, so $T_A = -\sigma_0 D + \frac{1}{m}\sum_i\langle\delta_i,\, \bar R_i\delta_i\rangle$ and $\langle\delta,\, R\delta\rangle \le \lambda_{\max}(R^G)\|\delta\|_{G^{-1}}^2$. If $\|\delta_i\|_{G^{-1}} \le \varrho'$ the segment lies in $N_{\varrho'}(\operatorname{supp}\mu_s)$ ($\bar\vartheta_i \in \operatorname{supp}\mu_s$), and by (0.2) $\lambda_{\max}(R^G[\mu_{\bar\Theta}]) \le \kappa_R[\mu_s](N_{\varrho'}(\operatorname{supp}\mu_s)) + L_2^G Y$ there. Hence, as long as $\max_i\|\delta_i\|_{G^{-1}} \le \varrho'$,

$$T_A \le -\big(a(s) - L_2^G Y(s)\big)D. \tag{2.1}$$

Step 3 (the interaction term is a negative square at linear order).

$$\nabla V_L[\mu_\Theta](\vartheta_i) - \nabla V_L[\mu_{\bar\Theta}](\vartheta_i) = \langle\nabla_\vartheta\tilde\Phi(\cdot\,;\vartheta_i)^\top(\hat f_\Theta - \hat f_{\bar\Theta})\rangle_\rho = \langle\nabla_\vartheta\tilde\Phi(\cdot\,;\vartheta_i)^\top h\rangle_\rho,$$

so, with the $L^2(\rho;\mathbb{R}^n)$ inner product $\langle\langle\cdot,\cdot\rangle\rangle$,

$$\begin{aligned} T_B &= -\Big\langle\Big\langle \frac{1}{m}\sum_i \nabla_\vartheta\tilde\Phi(\cdot\,;\vartheta_i)\delta_i,\, h\Big\rangle\Big\rangle = -\|h\|^2 - \Big\langle\Big\langle \frac{1}{m}\sum_i (\nabla_\vartheta\tilde\Phi(\cdot\,;\vartheta_i) - A_i)\delta_i,\, h\Big\rangle\Big\rangle \\ &\le -\|h\|^2 + \|h\|\frac{L_2^G}{2}D \le \frac{(L_2^G)^2D^2}{16} \end{aligned} \tag{3.1}$$

by (0.2) and $\max_x(-x^2+cx) = c^2/4$. The leading part $-\|h\|^2 = -\|\hat f_\Theta - \hat f_{\bar\Theta}\|^2_{L^2(\rho)}$ is the flat convexity of $\bar L$: the interaction contracts, and only the Taylor remainder of the features, quadratic in D, is charged. This is the step that Theorem D″'s small-gain bound replaces by $+L_\vartheta^2 D$-type terms.

Step 4 (the sampling term). $\nabla V_L[\mu_{\bar\Theta}](\bar\vartheta_i) - \nabla V_L[\mu_s](\bar\vartheta_i) = \langle \nabla_\vartheta\tilde\Phi(\cdot\,;\bar\vartheta_i)^\top(\hat f_{\bar\Theta} - \hat f[\mu_s])\rangle_\rho$, so $|\langle\delta_i,\,\cdot\rangle| \le L_\vartheta^G\|\delta_i\|_{G^{-1}}Y$ and

$$T_C \le L_\vartheta^G D^{1/2} Y. \tag{4.1}$$

Step 5 (the differential inequalities). While $\max_i\|\delta_i\|_{G^{-1}} \le \varrho'$,

$$\tfrac12\frac{\mathrm{d}D}{\mathrm{d}s} \le -(a - L_2^G Y)D + \frac{(L_2^G)^2D^2}{16} + L_\vartheta^G D^{1/2}Y, \tag{5.1}$$

and per particle, from the same three brackets with $|T_{B,i}| \le \|\delta_i\|L_\vartheta^G\|h\| \le \|\delta_i\|(L_\vartheta^G)^2D^{1/2}$ by (0.1) and $|T_{C,i}| \le \|\delta_i\|L_\vartheta^G Y$,

$$D^+\|\delta_i\|_{G^{-1}} \le -(a - L_2^G Y)\|\delta_i\|_{G^{-1}} + (L_\vartheta^G)^2D^{1/2} + L_\vartheta^G Y. \tag{5.2}$$

Step 6 (the bootstrap on the good event). Fix a horizon $S \le \infty$, let $\mathcal{G}_S := \{\sup_{s\le S} Y(s) \le \theta\}$ and $T^* := \inf\{s \le S : D(s) \ge \varrho^2 \text{ or } \max_i\|\delta_i(s)\|_{G^{-1}} \ge \varrho'\}$ ($T^* > 0$ since $\delta(0) = 0$; everything below is on $[0,S]$). On $\mathcal{G}_S$ and $[0,T^*)$: $L_2^G Y \le \kappa/8$, $(L_2^G)^2D/16 \le (L_2^G)^2\varrho^2/16 \le \kappa/8$, and $L_\vartheta^G D^{1/2}Y \le (\kappa/8)D + (2(L_\vartheta^G)^2/\kappa)Y^2$ (Young), so (5.1) gives $\mathrm{d}D/\,\mathrm{d}s \le -2(a - 3\kappa/8)D + (4(L_\vartheta^G)^2/\kappa)Y^2$. Since $a - 3\kappa/8 \ge \kappa/2 - (\kappa - a)_+$, $\int_r^s(a-3\kappa/8) \ge (\kappa/2)(s-r) - \Lambda_1$ for $r \le s$, and Grönwall gives

$$\begin{aligned} D(s) \le Z(s) &:= 4e^{2\Lambda_1}(L_\vartheta^G)^2\kappa^{-1}\int_0^s e^{-\kappa(s-r)}Y(r)^2\,\mathrm{d}r \\ &\le 4e^{2\Lambda_1}(L_\vartheta^G)^2\kappa^{-2}\theta^2 \le \varrho^2/4 \qquad \text{on } \mathcal{G}_S\cap[0,T^*) \end{aligned} \tag{6.1}$$

by the second constraint on θ. Per particle, (5.2) with $D^{1/2} \le \varrho/2$ and $Y \le \theta$, and $\int_r^s(a-\kappa/8) \ge (\kappa/2)(s-r) - \Lambda_1$, gives $\|\delta_i(s)\|_{G^{-1}} \le 2e^{\Lambda_1}\kappa^{-1}[(L_\vartheta^G)^2\varrho/2 + L_\vartheta^G\theta] \le \varrho'/4 + \varrho'/2 = 3\varrho'/4$ by the second constraint on ϱ and the third on θ. Both thresholds are therefore strict on $[0,T^*)$, so $T^* = S$ on $\mathcal{G}_S$ by continuity: $D \le Z$ and $\max_i\|\delta_i\| \le 3\varrho'/4$ on $[0,S]$ on $\mathcal{G}_S$. Z does not involve the m-system.

Step 7 (expectation). For $s \le S$,

$$\begin{aligned} \mathbb{E}D(s) \le \mathbb{E}[Z(s)] + D_{\max}\mathbb{P}(\mathcal{G}_S^c) &\le 4e^{2\Lambda_1}(L_\vartheta^G)^2\kappa^{-1}\int_0^s e^{-\kappa(s-r)}\mathbb{E}Y(r)^2\,\mathrm{d}r + D_{\max}\mathbb{P}(\mathcal{G}_S^c) \\ &\le C_1/m + 4R_\Theta^2\mathbb{P}(\mathcal{G}_S^c), \end{aligned}$$

using $\mathbb{E}Y(r)^2 \le K_\Phi^2/m$.

Step 8 (the good event). $Y(s)$ is a function of the i.i.d. sample $(\vartheta_j(0))_{j\le m}$: $Y(s) = \|\frac1m\sum_j[\tilde\Phi(\cdot\,;X_s(\vartheta_j(0))) - \int\tilde\Phi(\cdot\,;X_s(\vartheta))\,\mathrm{d}\mu_0]\|_{L^2(\rho)}$. Changing one $\vartheta_j(0)$ changes it by at most $2K_\Phi/m$ (bounded differences), and $\mathbb{E}Y(s) \le (\mathbb{E}Y^2)^{1/2} \le K_\Phi/m^{1/2}$; McDiarmid's inequality [61]

gives $\mathbb{P}(Y(s) \ge K_\Phi/m^{1/2} + t) \le \exp(-mt^2/(2K_\Phi^2))$, so for $m \ge m_0 = 16K_\Phi^2/\theta^2$ ($K_\Phi/m^{1/2} \le \theta/4$, $t = \theta/4$)

$$\mathbb{P}(Y(s) \ge \theta/2) \le \exp\bigl(-m\theta^2/(32K_\Phi^2)\bigr) \qquad \text{for every fixed } s. \tag{8.1}$$

In time, $|Y(s) - Y(s')| \le 2L_\vartheta^G V_{\max}|s - s'|$ (each $X_s(\vartheta)$ moves at speed $\le V_{\max}$ and $\tilde\Phi$ is L_ϑ^G-Lipschitz, for the sample and for the mean), so on a grid of step $\Delta = \theta/(4L_\vartheta^G V_{\max})$, $Y \le \theta$ on $[s_k, s_k + \Delta]$ whenever $Y(s_k) \le \theta/2$. (a) The grid $\{k\Delta : k\Delta \le S\} \cup \{S\}$ has at most $S/\Delta + 2$ points, and a union bound with (8.1) gives $\mathbb{P}(\mathcal{G}_S^c) = \mathbb{P}(\sup_{s \le S} Y > \theta) \le (S/\Delta + 2)\exp(-m\theta^2/(32K_\Phi^2))$; with Step 7 this is (a). For $S \le \Delta e^{m\theta^2/(64K_\Phi^2)}$, $(S/\Delta + 2)e^{-m\theta^2/(32K_\Phi^2)} \le 3e^{-m\theta^2/(64K_\Phi^2)} \le 3/m$ when $e^{m\theta^2/(64K_\Phi^2)} \ge m$. (b) Under (H′), for $s \ge R_{\text{eff}}$ every $X_s(\vartheta)$ is within $\theta/(8L_\vartheta^G)$ of $X_\infty(\vartheta)$ on $\operatorname{supp}\mu_0$, so $Y(s) \le Y(\infty) + \theta/4$ with $Y(\infty) := \|\frac{1}{m}\sum_j[\tilde\Phi(\cdot\,; X_\infty(\vartheta_j(0))) - \int \tilde\Phi(\cdot\,; X_\infty)\,\mathrm{d}\mu_0]\|$, another bounded-differences statistic with $\mathbb{E}Y(\infty) \le K_\Phi/m^{1/2}$; the grid on $[0, R_{\text{eff}}]$ plus the point at infinity has at most $R_{\text{eff}}/\Delta + 3$ points, and $\mathbb{P}(\mathcal{G}_\infty^c) = \mathbb{P}(\sup_{s \ge 0} Y > \theta) \le (R_{\text{eff}}/\Delta + 3)\exp(-m\theta^2/(32K_\Phi^2))$.

Step 9 (the learned function and the measures). $\|\hat f_\Theta - \hat f[\mu_s]\| \le \|h\| + Y \le L_\vartheta^G D^{1/2} + Y$ by (0.1), so $\mathbb{E}\|\cdot\|^2 \le 2(L_\vartheta^G)^2\mathbb{E}D + 2K_\Phi^2/m$. The coupling $(\vartheta_i, \bar\vartheta_i)$ is a coupling of μ_Θ and $\mu_{\bar\Theta}$, so $W_2^2(\mu_\Theta, \mu_{\bar\Theta}) \le D$ (G^{-1} metric), and the triangle inequality gives (c). The per-particle statement is Step 6. □

Remark D‴.1 (what the hypothesis says, and the window)**.** (H) at $s = \infty$ reads: on a neighbourhood of every atom of the limit measure, $G^{1/2}\nabla^2\mathcal{V}[\mu_\infty]G^{1/2} = \sigma_0 I - R^G[\mu_\infty] \succeq \kappa I$ — each atom is a non-degenerate local minimiser of the first-variation potential $\mathcal{V}[\mu_\infty]$, with margin κ: the strict form of the second-order optimality condition for measures (the local Chizat–Bach condition, as in Proposition K(ii)). Its non-strict form, $\sigma_0 G^{-1} - R[\mu_\infty](\vartheta^*) \succeq 0$ at every atom ϑ^* of positive mass, is necessary for an atomic stationary measure to be a local minimiser of $\mathcal{G}$ in the Wasserstein sense: splitting an atom of mass w along a direction of negative curvature of $\mathcal{V}[\mu_\infty]$ lowers $\mathcal{G}$ at second order (Proposition K(i)–(ii) with the atom's own residual). Along the flow, (H) asks for this with the *current* residual r_{μ_s} from the time s_1 on. It does not mention the collapsed configuration: $\sigma_{\text{crit}} = \kappa_R[\delta_{\bar\vartheta}](\{\bar\vartheta\})$ is the same curvature *with the collapsed residual*, which is the residual of the best single kernel and is large when the plant needs several. The learning regime $\sigma_0 < \sigma_{\text{crit}}$ (the kernels separate) and the uniform regime (H) ($\kappa_R[\mu_\infty] \le \sigma_0 - \kappa$ at the fitted configuration) therefore coexist exactly when the fitted configuration is non-degenerately stable while the collapsed one is not — the window

$$\kappa_R[\mu_\infty]\bigl(N_{\varrho'}(\operatorname{supp}\mu_\infty)\bigr) < \sigma_0 < \sigma_{\text{crit}}(\bar\vartheta)$$

of leaks in which the network learns *and* propagation of chaos is uniform in time. Theorem D″'s condition, by contrast, asks the leak to dominate the worst-case residual curvature $\lambda_G L_2(F_\rho + K_\Phi)$ *and* the interaction's Lipschitz constant L_ϑ^2-type terms, which forces $\sigma_0 \ge \sigma_{\text{crit}}$. The two ingredients that remove the horizon here are (i) the sign of the interaction, which is the flat convexity of $\bar L$ (Step 3), and (ii) the residual-weighted curvature evaluated at the actual residual rather than at the worst case (Step 2). Both are properties of the lifted linear parametrisation $\hat f = \int \tilde\Phi\,\mathrm{d}\mu$, not of the particular features.

Remark D‴.2 (the transient, the support, and the floor of the coupling)**.** Two things limit the theorem, and both are visible in the numerics. (i) The constant carries $e^{2\Lambda_1}$, the finite-horizon amplification of Theorem D′ over the time the flow needs to enter the non-degenerate regime: for a warm start $\Lambda_1 = 0$ and $C_1 = 4(L_\vartheta^G)^2K_\Phi^2/\kappa^2$; for a cold start the expanding phase ($a(s) < \kappa$ while the residual is large) lasts a time of order $1/\sigma_0$ and Λ_1 is of order $\sigma_0^{-1} \cdot \sup(\kappa_R - \sigma_0)$ — the bound is uniform but quantitatively void, as D′'s $e^{\Lambda T}$ is. (ii) (H) is a statement about the ϱ'-neighbourhood of the *support* of μ_s. The characteristic flow is a homeomorphism, so a connected support stays connected: if the mass of a connected cold start splits between several basins of $\mathcal{V}[\mu_\infty]$, the support of μ_s threads through the ridges between them for every s, and

the filament's points converge to the saddles of $\mathcal{V}[\mu_\infty]$ on those ridges, where $\mathcal{V}$ has a direction of negative curvature and $\kappa_R > \sigma_0$ — (H) fails for every s_1 (asymptotically in s; pointwise the filament may well lie in a locally convex region for a while, referee round 8, D3-11), however little mass the filament carries. The theorem then says nothing, and the coupling can indeed carry a floor: a particle whose reference lies near a separatrix can be pushed to the other side by the $O(m^{-1/2})$ difference between the two drifts, after which the pair sits in different basins for ever, at a distance of the order of the basin separation. A fraction p_m of such pairs contributes $p_m \cdot O(R_\Theta^2)$ to D, so D follows the law of p_m (the μ_0-mass of a boundary layer of width $O(m^{-1/2})$ around the separatrices' preimage, hence $p_m = O(m^{-1/2})$ when μ_0 has a bounded density there), not $1/m$; the learned function is less sensitive, $\|\hat{f}_\Theta - \hat{f}_{\bar{\Theta}}\| \le L_\vartheta^G \frac{1}{m} \sum \|\delta_i\|_{G^{-1}} \le L_\vartheta^G (p_m \cdot 2R_\Theta + D_{\text{good}}^{1/2})$, which is still $O(m^{-1/2})$. The measure-weighted form of (H) — a bad set of small μ_s-mass, with the per-particle bootstrap run on the good particles only and the interaction split as in Step 3 — is the natural extension and is listed in Section 26; what the present theorem shows is where the clean statement holds (supports inside basins) and why the cold start of Theorem D″'s experiment was outside it on two counts (the transient and the connected support). (iii) The floor has a second source at small m, on a warm start: when the realised cluster weights fall outside the admissible set of Proposition D‴.5(iv) the replicated configuration is a saddle of J_m, the m-system lingers on it and then splits a cluster, so that pairs end in different basins although the mean-field flow satisfies (H); its probability is exponentially small in m but appreciable at $m = 16$ on the instance below.

Remark D‴.3 (dither)**.** At $\tau_0 > 0$ the synchronous coupling cancels the noise and the identity of Step 1 holds pathwise with the same three terms, but the reference law has full support and the particles visit the ridges of $\mathcal{V}$ with the Gibbs weight of the ridge, so (H) fails in the sense of (ii) above and the coupling carries a floor of that order. Uniform-in-time propagation of chaos at $\tau_0 > 0$ is the province of the entropy method: with dither, $(\mathfrak{A}_m)$ is the m-particle mean-field Langevin dynamics of the free energy F_τ of Theorem H in the G^{-1} metric (Corollary J′.3 is its stationary statement), and [11] proves uniform-in-time propagation of chaos for mean-field Langevin dynamics under flat convexity, regularity of the first variation and a uniform log-Sobolev inequality for the proximal Gibbs measures — the three hypotheses that Theorem H(i)–(ii) verifies for F_τ, with the constant ρ_τ of H(ii). That result is imported here, not re-derived; its constants are those of Theorem H and are exponentially small in $1/\tau_0$, where the constants of Theorem D‴ are those of the leak and the residual curvature. The leak-dominated regime is covered for the real loop and every $\tau \ge 0$ by Theorem D″.

Remark D‴.4 (the real loop)**.** Theorem D‴ is a statement about the averaged slow dynamics $(\mathfrak{A}_m)$. Theorems J and J′ identify $(\mathfrak{A}_m)$ as the leading order of the real loop's slow dynamics, with the corrections $\varepsilon \cdot b$ (Theorem J′'s b, not a gradient: Proposition J′.2) and the fast error eliminated at $O(\varepsilon^2)$; the transfer of a uniform-in-time statement through these corrections — a near-identity change of variables that is uniform in time — is the open item of Section 26, branch 1; on a warm start it is settled by Corollary D‴.7 below through Theorem J. What the theorem settles is the mechanism: in the learning regime the uniformity rests on the fitted configuration's non-degenerate stability and on the flat convexity of the loss, not on a dominant leak; the real-loop run on a warm start (numerics, table 5) tracks the averaged slow dynamics over the horizon run, at the same k_e at which Theorem D″'s cold start grew.

Proposition D‴.5 (the stationary coupled distance at finite m: the weight-sensitivity law)**.** *Let $\mu^* = \sum_{k=1}^K \bar{w}_k \delta_{\vartheta_k^*}$ be a K-atom stationary measure of $(\mathfrak{A}_0)$ (each ϑ_k^* a critical point of $\mathcal{V}[\mu^*]$), and for weights $w \in (0,\infty)^K$ let $\mathcal{G}_w(\Theta) := \frac{1}{2}\langle \|f - \sum_k w_k \tilde{\Phi}(\cdot\,;\vartheta_k)\|^2 \rangle_\rho + \frac{\sigma_0}{2} \sum_k w_k \|\vartheta_k - \vartheta^0\|_{G^{-1}}^2$ be the mean-field energy of $\mu_w := \sum_k w_k \delta_{\vartheta_k}$, so that $\nabla_{\vartheta_k} \mathcal{G}_w(\Theta) = w_k \nabla_\vartheta \mathcal{V}[\mu_w](\vartheta_k)$. Assume the K-atom Hessian $\nabla_\Theta^2 \mathcal{G}_{\bar{w}}(\Theta^*)$ is positive definite. Then:*

(i) (the weighted stationary point and its replication) *there is a smooth map $w \mapsto \Theta^*(w)$ on a neighbourhood of $\bar{w}$ with $\Theta^*(\bar{w}) = \Theta^*$ and $\nabla_\vartheta \mathcal{V}[\mu_{\Theta^*(w)}](\vartheta_k^*(w)) = 0$ for every k*

(implicit function theorem); $S(w) := \partial\Theta^*/\partial w$ ($S_k := \partial\vartheta_k^*/\partial w \in \mathbb{R}^{p\times K}$) is given by $S = -[\nabla^2_\Theta \mathcal{G}_w]^{-1}\partial_w\nabla_\Theta\mathcal{G}_w$ at $(\Theta^*(w), w)$. For m particles with n_k of them in cluster k ($w_k^m := n_k/m$), the configuration $\Theta_m^*(w^m)$ with n_k copies of $\vartheta_k^*(w^m)$ is a critical point of J_m ($\mu_\Theta = \mu_{w^m}$ there). Its whitened Hessian splits (Proposition K(i), applied cluster by cluster) into the separation blocks $\sigma_0 I - R^G[\mu_{w^m}](\vartheta_k^*(w^m))$, each with multiplicity $n_k - 1$, and the collective block $\operatorname{diag}(w^m)^{-1/2}\cdot G_K^{1/2}\nabla^2_\Theta\mathcal{G}_{w^m}G_K^{1/2}\cdot\operatorname{diag}(w^m)^{-1/2}$ *(the K-atom Hessian, whitened and weighted). Hence $\Theta_m^*(w^m)$ is a strict local minimiser of J_m exactly when w^m lies in the* admissible set

$$W_{\mathrm{adm}} := \big\{w : \sigma_0 G^{-1} - R[\mu_{\Theta^*(w)}](\vartheta_k^*(w)) \succ 0 \text{ for every } k \text{ and } \nabla^2\mathcal{G}_w(\Theta^*(w)) \succ 0\big\},$$

a neighbourhood of $\bar w$ (both conditions are open and hold at $\bar w$ by (H) and the assumption), and a saddle when the splitting condition fails at some cluster: a cluster can then split — by Proposition K's local separation inside its basin (sub-clusters along the unstable direction, the cluster weights unchanged) or by particles migrating to other basins; the numerics show both (this section, referee rounds 9 and 10).

(ii) (the stationary distance) *Under the shared-particle coupling of Theorem D‴ with i.i.d. cluster labels of law $\bar w$ (the m particles are the first m of the M reference particles), suppose $w^m, w^M \in W_{\mathrm{adm}}$ and that the m-system converges to $\Theta_m^*(w^m)$ and the reference to $\Theta_M^*(w^M)$. Then the coupling distance converges to*

$$\begin{aligned} D_\infty &= \sum_k w_k^m \|\vartheta_k^*(w^m) - \vartheta_k^*(w^M)\|^2_{G^{-1}} \\ &= \sum_k w_k^m \|S_k(w^M)(w^m - w^M)\|^2_{G^{-1}} + O(\|w^m - w^M\|^3), \end{aligned}$$

and, with $\tilde S_k := G^{-1/2}S_k(\bar w)$ (replacing $S(w^M)$ by $S(\bar w)$ costs a relative $O(\|w^M - \bar w\|)$) and $\mathbb{E}[(w^m - w^M)(w^m - w^M)^\top] = (1/m - 1/M)\tilde C$, $\tilde C := \operatorname{diag}(\bar w) - \bar w\bar w^\top$ (multinomial counts of the shared labels), conditionally on the admissibility of both weights,

$$\mathbb{E}[D_\infty \mid w^m, w^M \in W_{\mathrm{adm}}] = (1/m - 1/M)\cdot\sum_k \bar w_k \operatorname{tr}\big(\tilde S_k\tilde C\tilde S_k^\top\big) + O(m^{-3/2}) + O(m^{-1}M^{-1/2}) :$$

the $1/m$ law with an explicit constant, and the $(1/m - 1/M)$ law of the shared-particle coupling observed in Theorem D′'s numerics.

(iii) (the fluctuations) *As $M \to \infty$ ($w^M \to \bar w$) and then $m \to \infty$, $\sqrt{m}(w^m - \bar w) \to \mathcal{N}(0, \tilde C)$ and $\sqrt{m}\cdot(\vartheta_i - \bar\vartheta_i) \to S_{k(i)}\xi$ with $\xi \sim \mathcal{N}(0, \tilde C)$ for every particle i of cluster $k(i)$: the fluctuations of the coupling are Gaussian, carried entirely by the cluster weights, and all particles of a cluster are displaced by the same vector.*

(iv) (admissibility) *Let $\delta_w := \sup\{\delta : \|w - \bar w\|_\infty \le \delta \Rightarrow w \in W_{\mathrm{adm}}\} > 0$. By Hoeffding's inequality for each multinomial coordinate, $\mathbb{P}(w^m \notin W_{\mathrm{adm}}) \le 2K\exp(-2m\delta_w^2)$: the finite-$m$ configuration is a strict local minimiser with probability exponentially close to one, the exceptional event being the one on which the realised weights make the replicated configuration a saddle: a cluster of the m-system then splits, locally or by migration to other basins — in the latter case with Remark D‴.2's floor on a warm start.*

Status. Formal: (i) and the algebra of (ii)–(iv) are exact (the block splitting is Proposition K(i) cluster by cluster, confirmed on J_{10} to $8\cdot10^{-12}$ in referee round 9; implicit function theorem; the multinomial covariance of shared labels; Hoeffding), the convergence of both systems to the stated configurations is assumed (it is what Theorem D‴ proves for $m \ge m_0$ and what the numerics show for $m = 16$–256 at admissible weights; at inadmissible weights the m-system sits on the saddle for a long time and then leaves it, numerics below). Checked: `weighted_stationary_atoms`, `weight_sensitivity`, `predicted_`

`stationary_D` in `chaos_learning.py`; on the instance below the replicated weighted stationary point is a critical point of J_{16} to 10^{-10}, the linearisation $\vartheta^*(\bar w+\mathrm{d}w)-\vartheta^* = S\cdot \mathrm{d}w$ has a second-order remainder, and the per-cluster displacements of the coupled run point along $S_k(w^m - w^M)$ (test and table below). The proposition is the sharp form of Theorem D‴'s constant: $S = -H_w^{-1}\partial_w\nabla\mathcal{G}_w$ has the structure (stiffness)$^{-1}$ $\times$ (feature correlations), of which $C_1 = 4e^{2\Lambda_1}(L_\vartheta^G)^2K_\Phi^2/\kappa^2$ is the worst-case envelope (κ^{-1} for the stiffness, $L_\vartheta^G K_\Phi$ for the correlations).

Remark D‴.6 (the regime of the theorem, and what the numerics test)**.** The sample size $m_0 = 16K_\Phi^2/\theta^2$ from which the theorem's tail is in force is astronomically large on realistic instances (10^{23} below): θ carries $\varrho'\kappa^2e^{-2\Lambda_1}/(L_\vartheta^G)^3$ from the per-particle bootstrap of Step 6, because the interaction's effect on one particle is bounded by $(L_\vartheta^G)^2D^{1/2}$ and divided by the stiffness κ. This is not an artefact of the proof's constants only: Proposition D‴.5 says the particles of a cluster are displaced by $S_k(w^m - w^M)$, of order $\|S\|m^{-1/2}$ with $\|S\| \sim$ (stiffness)$^{-1}$, which exceeds any fixed ϱ' for m below $(\|S\|/\varrho')^2$ — on the instance below the $m = 16$ particles sit 0.4–1.4 whitened units from their reference copies, outside the neighbourhood $N_{\varrho'}$ in which $\mathcal{V}[\mu_{\bar\Theta}]$ is convex, at the replicated stationary configuration $\Theta_{16}^*(w^m)$ of their own J_{16} — a strict local minimiser when $w^m \in W_{\mathrm{adm}}$ and a saddle otherwise (Proposition D‴.5(i), (iv); one of the four $m = 16$ seeds below). Hypothesis (H) is nevertheless the right object: it is the condition under which the reference configuration is stable to splitting, and the m-system's configuration is the same configuration with perturbed weights. What the numerics test is therefore (a) hypothesis (H) at the stationary measure and along the flow, (b) the mechanism — the coupled distance saturates (no growth) and follows the $1/m$ law — and (c) the sharp constant of Proposition D‴.5; the theorem's own constant C_1 is reported for the record.

Corollary D‴.7 (the real loop on a warm start: uniform in time through Theorem J)**.** *Let $\mu^* = \sum_k \bar w_k\delta_{\vartheta_k^*}$ be a K-atom stationary measure as in Proposition D‴.5, satisfying the splitting condition $\sigma_0 G^{-1} - R[\mu^*](\vartheta_k^*) \succ 0$ at every atom and the non-degeneracy of the K-atom Hessian, and let $w \mapsto \Theta^*(w)$ be the weighted stationary point of D‴.5(i). Fix $m \le M$ and cluster weights $w^m, w^M \in W_{\mathrm{adm}}$ (Proposition D‴.5(i)), and let $\Theta_m^* :=$ the configuration with $n_k = mw_k^m$ copies of $\vartheta_k^*(w^m)$ (likewise Θ_M^*). Both are non-degenerate local minimisers of J_m and J_M (Proposition D‴.5(i)), hence hyperbolic critical points, and Theorem J applies to each loop: there are k_0, ρ_0, C (depending on Θ_m^* and Θ_M^*, hence on m, M and the weights) such that for $k_e \ge k_0$ the real closed loop* (CL) *with m kernels started at $\|\Theta(0) - \Theta_m^*\|_{G^{-1}} \le \rho_0$, $\|e(0)\| \le R$ converges to its T_0-periodic orbit Θ_p^m, and likewise the M-kernel loop to Θ_p^M. Under the shared-particle coupling (the m kernels are the first m of the M, the plants are driven by the same reference) the coupled distance $D(t) := \frac{1}{m}\sum_i\|\vartheta_i(t) - \bar\vartheta_i(t)\|_{G^{-1}}^2$ satisfies, for all $t \ge 0$,*

$$D(t)^{1/2} \le D_{\mathrm{orb}}^{1/2} + Ce^{-\mu t}\big(\|\Theta(0) - \Theta_p^m(0)\|_{G^{-1}} + \|x(0) - x_p^m(0)\|\big) + Ce^{-\mu t}\big(\|\bar\Theta(0) - \Theta_p^M(0)\|_{G^{-1}} + \|\bar x(0) - x_p^M(0)\|\big),$$

with $\mu = (\lambda_*/k_e)(1-\kappa_J)$ the rate of Theorem J(c) (κ_J its small parameter; the smaller of the two loops' rates) and

$$\begin{aligned} D_{\mathrm{orb}} &:= \sup_t \frac{1}{m}\sum_i \|\Theta_p^m(t)_i - \Theta_p^M(t)_i\|_{G^{-1}}^2 \\ &= \sum_k w_k^m\|\vartheta_k^*(w^m) - \vartheta_k^*(w^M)\|_{G^{-1}}^2 + O(1/k_e) \\ &= \sum_k w_k^m\|S_k(w^m - w^M)\|_{G^{-1}}^2 + O(\|w^m - w^M\|^3) + O(1/k_e)\,: \end{aligned}$$

the coupled distance of the real loop is bounded uniformly in time, converges to the distance between the two periodic orbits, and that distance is Proposition D‴.5's stationary distance up

to the $O(1/k_e)$ offsets of Theorem J(a),(d) (which differ between the two loops by $O(k_e^{-1}\|w^m - w^M\|) + O(k_e^{-2})$, the first-order offset of a replicated configuration being a smooth function of the K-atom data $(\Theta^*(w), w)$ by permutation equivariance). In particular, for cluster labels i.i.d. of law $\bar{w}$ and conditionally on both weights being admissible and both starts lying in their basins, $\mathbb{E}D_{\mathrm{orb}} = (1/m - 1/M)\sum_k \bar{w}_k \operatorname{tr}(\tilde{S}_k \tilde{C} \tilde{S}_k^\top) + O(m^{-3/2}) + O(1/k_e)$.

Status. Proved as stated (Theorem J(a),(c),(d) applied to each loop, the triangle inequality in the G^{-1} metric, and Proposition D‴.5(ii) for the distance between the critical points), under Theorem J's hypotheses for both configurations: the constants k_0, ρ_0, C, μ are those of Theorem J at Θ_m^* and Θ_M^*, and the corollary does not assert their uniformity in m or M (Remark D‴.8). It is the statement of Remark D‴.4 for a warm start: the real loop's coupled distance saturates, at a value set by the cluster weights. Consistent with the instance of this section's numerics (table 5: the real loop at $k_e = 25$ from the jittered mixture over 80 slow units, a sixth of the relaxation time $1/\kappa$, follows its averaged slow dynamics on the same initial clouds to within a few percent at every window end; the saturation at the orbit distance lies beyond the horizon run and is not checked).

Proof. Theorem J(c) for the m-kernel loop gives $\|\Theta(t) - \Theta_p^m(t)\|_{G^{-1}} \le Ce^{-\mu t}(\|\Theta(0) - \Theta_p^m(0)\|_{G^{-1}} + \|x(0) - x_p^m(0)\|)$ and the same for the M-kernel loop; the first m rows of the M-loop are the reference particles $\bar{\vartheta}_i$. For the $(m \times p)$-row norm $\frac{1}{m}\sum_i \|\cdot\|_{G^{-1}}^2 \le \|\cdot\|_{G^{-1}}^2/m \le \|\cdot\|_{G^{-1}}^2$ (the full configuration norm dominates the mean square over the first m rows), so by the triangle inequality $D(t)^{1/2} \le [\frac{1}{m}\sum_i \|\Theta_p^m(t)_i - \Theta_p^M(t)_i\|^2]^{1/2} +$ (the two transients). Theorem J(a) places each orbit within C/k_e of its critical point, uniformly in t, so the first term is $\sum_k w_k^m \|\vartheta_k^*(w^m) - \vartheta_k^*(w^M)\|_{G^{-1}}^2$ up to $O(1/k_e)$ (the two offsets are $O(1/k_e)$ each, with $O(k_e^{-2})$ remainders; at a replicated configuration the first-order offset is the same vector for all particles of a cluster and depends smoothly on the K-atom data $(\Theta^*(w), w)$, by the permutation equivariance of (CL) and the block splitting of Proposition D‴.5(i), so the two loops' offsets differ by $O(k_e^{-1}\|w^m - w^M\|) + O(k_e^{-2})$ — the two loops live on $\mathbb{R}^{mp}$ and $\mathbb{R}^{Mp}$, and it is this reduction to the K-atom data that makes them comparable), and Proposition D‴.5(ii) expands the distance between the critical points in $w^m - w^M$. The expectation statement is D‴.5(ii)'s. □

Remark D‴.8 (what the corollary does and does not say)**.** It transfers the uniform-in-time statement to the real loop in the regime where Theorem J applies to both loops — a warm start in the basins of two hyperbolic minimisers at a gain k_e above both thresholds. Theorem J's constants at a replicated configuration are controlled by the K-atom data (the Hessian of J_m there is the collective block $\nabla^2 \mathcal{G}_{w^m} \oplus$ the separation blocks $\sigma_0 G^{-1} - R_k$ of Proposition K(i), both independent of m beyond the weights), so the threshold and the basin are expected to be uniform in m for configurations with the same weights; the corollary does not prove this, and the cold start — where the coupled distance of the real loop grew in Theorem D″'s experiment — is outside it on the same two counts as in Remark D‴.2.

Proposition D‴.9 (the leak toward a measure-valued prior, and the design of the window)**.** *Let $\nu = \sum_{j=1}^{K_0} q_j \delta_{\vartheta_j^0}$ be a prior measure with distinct atoms and let the leak be the G-gradient of the distance-to-support potential*

$$V_\nu(\vartheta) := \tfrac{1}{2}\min_j \|\vartheta - \vartheta_j^0\|_{G^{-1}}^2, \qquad -G\nabla V_\nu(\vartheta) = -(\vartheta - \vartheta_{j(\vartheta)}^0), \qquad j(\vartheta) := \operatorname*{argmin}_j \|\vartheta - \vartheta_j^0\|_{G^{-1}}$$

(the nearest-atom leak*: inside each Voronoi cell of ν it is the σ-modification toward that cell's atom; the barycentric OT leak of Theorem C′ is its mass-constrained relative). Write $(\mathfrak{A}_m^\nu)$ and $(\mathfrak{A}_0^\nu)$ for $(\mathfrak{A}_m)$ and $(\mathfrak{A}_0)$ with $\sigma_0(\vartheta - \vartheta^0)$ replaced by $\sigma_0(\vartheta - \vartheta_{j(\vartheta)}^0)$, and $\mathcal{V}^\nu[\mu] := V_L[\mu] + \sigma_0 V_\nu$ for the first-variation potential.*

(i) (the theorems carry over) *Assume, in addition to the hypotheses of Theorem D‴ for $(\mathfrak{A}_0^\nu)$, that the assignment is locally constant along the flow: for every $s \ge 0$ the neighbourhood*

$N_{\varrho'}(\operatorname{supp}\mu_s)$ *meets no Voronoi boundary of* ν *(in particular when the gap between the distances to the two nearest atoms exceeds* $2\varrho'$ *on* $\operatorname{supp}\mu_s$*, the* G^{-1}*-distance to the boundary being at least half the gap). Then Theorem D‴, Proposition D‴.5 and Corollary D‴.7 hold for* $(\mathfrak{A}^\nu_m)$ *with the same constants,* R_Θ *replaced by* $\max\{R_0, L^G_\vartheta(F_\rho + K_\Phi)/\sigma_0\} + \max_j \|\vartheta^0_j - \vartheta^0_1\|_{G^{-1}}$ *and* ϑ^0 *by* $\vartheta^0_{j(\cdot)}$ *in the a priori bounds, and Theorem J's basin radius* ρ_0 *at the replicated configuration taken below* $\min_k \operatorname{dist}_{G^{-1}}(\vartheta^*_k, \partial\operatorname{cell}_k)$*. The masses* q_j *enter the nearest-atom leak only through* $\operatorname{supp}\nu$*; they matter in (ii), and they are what the mass constraint of the OT leak (Theorem C′, Remark D‴.10) acts on.*

(ii) (the stationary residual and the window) *If* ν *is placed on an exact representation of the plant on the orbit —* $K_0 = K$ *atoms* ϑ^0_k *with* $\sum_k q_k \tilde\Phi(\cdot\,;\vartheta^0_k) = f$ *on* $\operatorname{supp}\rho$ *— then* $\mu^* := \sum_k q_k \delta_{\vartheta^0_k}$ *is a stationary measure of* $(\mathfrak{A}^\nu_0)$ *with* $r_{\mu^*} = 0$*,* $R[\mu^*] \equiv 0$*, and the window of Remark D‴.1 has its maximal margin:* $\kappa_R[\mu^*](A) = 0$ *for every set* A *(*$R[\mu^*] \equiv 0$*) and* $\kappa = \sigma_0$*, the only restriction on* ϱ' *being (i)'s Voronoi hypothesis, against* $\kappa = \sigma_0 - \kappa_R[\mu^*_\sigma](N_{\varrho'})$ *for the* σ*-modification's stationary measure* μ^*_σ*, whose residual is the leak's bias. For a prior* ν_δ *displaced from the exact representation the residual of the stationary measure is* $O(\delta)$ *and* $\kappa_R[\mu^*_{\nu_\delta}] = O(\delta)$*: the margin decreases with the misplacement, and the window closes when the displacement makes the residual-weighted curvature at an atom reach* σ_0*.*

(iii) (the floors) *Under the shared-particle coupling the stationary distances are the weight-sensitivity laws of Proposition D‴.5 at parameter level,* $\mathbb{E}[D_\infty \mid w^m, w^M \in W_{\mathrm{adm}}] = (1/m - 1/M)\sum_k \bar w_k \operatorname{tr}(\tilde S_k \tilde C \tilde S_k^\top) + O(m^{-3/2}) + O(m^{-1}M^{-1/2})$*, and at function level,*

$$\mathbb{E}\|\hat f_{\Theta^*(w^m)} - \hat f_{\Theta^*(w^M)}\|^2_{L^2(\rho)} = (1/m - 1/M)\cdot \operatorname{tr}(F\tilde C F^\top)/N + O(m^{-3/2}),$$
$$F_{\cdot j} := \partial \hat f_w/\partial w_j = \tilde\Phi(\cdot\,;\vartheta^*_j) + \sum_k w_k \nabla_\vartheta \tilde\Phi(\cdot\,;\vartheta^*_k) S_{k\cdot j},$$

with S *the sensitivity of the weighted stationary point of* $(\mathfrak{A}^\nu_0)$ *to the weights. Neither floor vanishes for the matching prior: away from* $\bar w$ *the weighted stationary point does not stay on the zero-residual manifold (the leak pulls the amplitudes back to the prior while the fit asks for* $a_k \propto 1/w_k$*); on the instance both sensitivities rise with the amplitudes of the stationary atoms along the amplitude-scaled priors — a numerical trend, not a law (at a fixed amplitude the width changes the sensitivity, Table 17).*

Status. (i) proved (the proofs of Theorem D‴, Proposition D‴.5 and Corollary D‴.7 use the leak only through its curvature $\sigma_0 G^{-1}$ and its a priori bound — the latter goes through with $\operatorname{dist}(\cdot, \operatorname{supp}\nu)$ in place of $\|\cdot - \vartheta^0\|$, the upper Dini derivative of the minimum of quadratics along the flow being bounded by the active branch's. The local constancy of the assignment makes every segment $[\bar\vartheta_i, \vartheta_i]$ of Step 2 lie in one cell, and it is needed for every $s \geq 0$, not only from s_1 on, because (2.1) and (5.2) are used on $[0, s_1)$ as well: a pair straddling a boundary adds $(\sigma_0/m)\langle \delta_i,\, \vartheta^0_{j(\vartheta_i)} - \vartheta^0_{j(\bar\vartheta_i)}\rangle_{G^{-1}} \geq 0$ to $\frac{1}{2}\,\mathrm{d}D/\,\mathrm{d}s$ (the sum of the two cell inequalities; V_ν is semiconcave and its leak expands across the boundaries), a forcing of size up to $\sigma_0 \operatorname{diam}(\operatorname{supp}\nu)\sqrt{D}$ that none of the theorem's constants absorbs — referee round 10 found `decomposition`'s same-cell convention and the true $\frac{1}{2}\,\mathrm{d}D/\,\mathrm{d}s$ to differ by exactly this term on a straddling pair. The same hypothesis gives the well-posedness of $(\mathfrak{A}^\nu_0)$, whose vector field is discontinuous on the boundaries, and Theorem J applies to J^ν_m, which is C^2 away from the Voronoi boundaries, on a basin that meets none.) (ii) proved ($r_{\mu^*} = 0$ and $\nabla V_\nu(\vartheta^0_k) = 0$ give stationarity; $R[\mu^*] = \langle r_{\mu^*} \nabla^2 \tilde\Phi\rangle_\rho = 0$; the $O(\delta)$ statements are the smooth dependence of the weighted stationary point on the prior, by the implicit function theorem at μ^*, whose K-atom Hessian Gram $+ \sigma_0 \operatorname{diag}(q) G^{-1}$ is positive definite — on the instance its whitened, weighted eigenvalues are 0.0074–0.20, all at or above σ_0, referee round 10). (iii) formal as Proposition D‴.5 (the function-level law is the delta method applied to $w \mapsto \hat f_{\Theta^*(w)}$); the sign of the comparison with the σ-modification is a numerical finding (`exp_prior_design.py` → `results/chaos/prior_design.md`, below), not a theorem. Checked: `tests/test_prior_design.py` (the single-atom prior reproduces $(\mathfrak{A}_m)$; the gradient of J^ν_m against finite differences; the coupling identity

inside the cells; the matching prior's exact representation stationary with zero residual and zero curvature, margin σ_0 against the σ-modification's; the parameter- and function-level sensitivities, the latter against finite differences of the fitted function of the weighted stationary point).

Remark D‴.10 (design)**.** The leak's prior is a design input of the window. The σ-modification's single ϑ^0 biases every kernel toward one point and leaves a residual at the fitted configuration — the residual-weighted curvature that bounds the window from below and slows the relaxation ($\kappa = 0.00196$ on the instance of this section, the theorem's relaxation time $1/\kappa = 511$ slow units). A prior measure whose atoms sit on the plant's structure removes the bias and gives the maximal margin $\kappa = \sigma_0$ ($1/\kappa = 141$ slow units, 3.6 times shorter; the observed e-folding times of the coupled distance are 3.3 times shorter in the median, numerics below), at the price of a larger floor at both levels (the stationary atoms have the full amplitudes, to which the weight sensitivity is proportional): the classical trade-off between bias and variance, here between the window's margin and the coupling's floor (Proposition D‴.24 resolves it at the function level: the floor is the weight-slope of the bias, and an adaptive prior removes both, at the price of the collective rate, of the leak's a-priori bound — restored by projecting the prior, Proposition D‴.27 — and of a larger parameter-level floor). A misplaced prior does worse than no structured prior (centres off by 0.6, amplitudes $\times 1.2$ or widths $\times 0.7$ close the window, which the σ-modification keeps open with margin 0.0020), and a shrunk prior does better than the matching one (amplitudes $\times 0.5$–0.7: margin above σ_0 at the uniform weights, both floors lower, all four tested unbalanced weights admissible — with splitting margins above σ_0 at one of them and slightly below at the other three; numerics below and Remark D‴.11). The barycentric OT leak of Theorem C′ adds a mass constraint to the same geometry — each prior atom must receive its mass — and so acts on the cluster weights themselves, the source of the floor; its interplay with the weight fluctuation (an equal-mass Lloyd dynamics against the fit) is the natural next step (Section 26, branch 8).

Numerics (`exp_prior_design.py` → `results/chaos/prior_design.md`). The three-bump instance of Theorem D‴'s numerics at the same leak $\sigma_0 = 0.15\sigma_{\mathrm{crit}} = 0.00707$, with the nearest-atom leak toward a prior of three atoms placed at the exact representation $(3A_k, C_k, S_k)$ and misplaced from it, against the σ-modification ($\vartheta^0 = 0$); $\varrho' = 0.1$; the floors predicted for $m = 64$ against $M = 2048$.

The design table (table 1). Stationary three-atom measure μ_ν^* under each prior — fit error, κ_R at the atoms and over $N_{0.1}$, margin $\kappa = \sigma_0 - \kappa_R(N_{0.1})$, whitened $\|S_k\|$, predicted $\mathbb{E}D_\infty$ and $\mathbb{E}\|\hat{f}_{64} - \hat{f}_M\|^2$ (Table 17):

Table 17: Proposition D‴.9 on the three-bump instance: the stationary three-atom measure under each prior — fit error, residual curvature at the atoms and over $N_{0.1}$, margin, whitened weight sensitivities, and the predicted parameter- and function-level floors for $m = 64$ against $M = 2048$ (`results/chaos/prior_design.md`, table 1).

prior	fit error	κ_R at the atoms (max)	κ_R over $N_{0.1}$	margin κ	$\|S_k\|$	$\mathbb{E}D_\infty$ ($m = 64$), $\times 10^{-1}$	$\mathbb{E}\|\hat{f}_{64} - \hat{f}_M\|^2$, $\times 10^{-4}$
σ-modification, $\vartheta^0 = 0$	0.2439	+0.00411	+0.00512	+0.00196	8.38, 4.66, 8.32	2.010	4.287
matching prior (exact representation)	0.0000	+0.00000	+0.00000	+0.00707	13.01, 6.18, 8.91	3.227	9.334
centres shifted by +0.15 / +0.3 / +0.45 / +0.6	0.0098 / 0.0200 / 0.0311 / 0.0436	+0.00140 / +0.00298 / +0.00480 / +0.00698	+0.00157 / +0.00333 / +0.00534 / +0.00770	+0.00550 / +0.00374 / +0.00173 / −0.00063	12.9–13.0, 6.18, 8.92	3.21–3.22	9.40–9.51
amplitudes ×0.5 / ×0.7 / ×1.2	0.0921 / 0.0624 / 0.0861	−0.00039 / −0.00028 / +0.03684	−0.00033 / −0.00023 / +0.03875	+0.00740 / +0.00731 / −0.03167	9.76, 5.01, 7.47 / 10.59, 5.40, 7.92 / 17.80, 6.96, 10.11	1.991 / 2.303 / 5.242	7.106 / 7.657 / 18.02
widths ×0.7 / ×1.5 / ×2	0.2195 / 0.0581 / 0.1190	+0.03086 / +0.00113 / +0.00294	+0.03316 / +0.00126 / +0.00306	−0.02609 / +0.00581 / +0.00401	13.00, 2.82, 9.77 / 11.13, 7.19, 8.50 / 10.39, 7.27, 8.71	3.062 / 2.932 / 3.008	17.16 / 7.459 / 6.148

The matching prior makes the exact representation stationary with zero residual (fit error $1.2 \cdot 10^{-16}$), $\kappa_R = 0$ and the maximal margin $\kappa = \sigma_0 = 0.00707$, 3.6 times the σ-modification's 0.00196 (the theorem's relaxation time $1/\kappa$: 141 against 511 slow units); the σ-modification's bias (amplitudes 2.45, −2.29, 2.05 against $3A_k = 3.0, -2.1, 2.4$; widths 0.83, 0.70, 0.88 against 0.5, 0.4, 0.5) is removed. The floors move the other way: $\|S_k\| = 13.01, 6.18, 8.91$ against 8.38, 4.66, 8.32, $\mathbb{E}D_\infty$ 1.61 times and the function-level floor 2.18 times the σ-modification's. Misplacing the prior brings the residual curvature back and closes the window at centres shifted by +0.6 (margin −0.00063), at amplitudes ×1.2 (−0.03167) and at widths ×0.7 (−0.02609), while shrinking the amplitudes (×0.5, ×0.7) gives margins above σ_0 (+0.00740, +0.00731: κ_R is negative at every atom) with floors below the matching prior's ($\mathbb{E}D_\infty$ 0.199, 0.230; function level $7.1 \cdot 10^{-4}$, $7.7 \cdot 10^{-4}$) at fit errors 0.09, 0.06. The cell margins (the gap between the two nearest prior atoms along the configuration) are 1.2–6.3, above $2\varrho' = 0.2$ everywhere.

Admissible weights (table 2). The splitting margin of the replicated weighted stationary configuration $\Theta^*(w)$ under the three priors (Table 18):

Table 18: Proposition D‴.9 on the three-bump instance: the splitting margin $\sigma_0 - \max_k \kappa_R$ of the replicated weighted stationary configuration $\Theta^*(w)$ at four weight vectors under the σ-modification, the matching prior and the shrunk prior (`results/chaos/prior_design.md`, table 2).

weights w	σ-modification	matching prior	amplitudes ×0.7
$(7, 5, 4)/16$	+0.00257	−0.00544	+0.00733
$(9, 4, 3)/16$	−0.00219	−0.00579	+0.00570
$(10, 3, 3)/16$	−0.00377	−0.00415	+0.00536
$(0.5, 0.3, 0.2)$	−0.00062	−0.00653	+0.00663

The matching prior's admissible set is the smallest: any excess weight overshoots the plant (κ_R at the over-weighted atom $+0.0112$ to $+0.0136$), so even $(7,5,4)/16$ is a saddle, and the set is smallest in the direction of the largest bump (an excess ε on the first atom alone, $w = \bar{w} + \varepsilon(1.5e_1 - 0.5)$, has margins $+0.00279$, -0.00031, -0.00392 at $\varepsilon = 0.02$, 0.04, 0.08, while $\varepsilon = 0.08$ on the second or third atom alone keeps $+0.00208$, $+0.00316$); the σ-modification undershoots ($\hat{f}/f = 0.77$–0.78 at the first atom) and tolerates $(7,5,4)/16$ but not $(9,4,3)/16$ (Theorem D‴'s inadmissible seed); the shrunk prior keeps all four admissible, with margins $+0.00536$ to $+0.00733$ — above σ_0 at $(7,5,4)/16$ and slightly below it at the other three, where its over-weighted first atom overshoots the plant at the centre by 1–3% ($\hat{f}/f = 1.020$, 1.028, 1.007; κ_R there $+0.0014$, $+0.0017$, $+0.0004$; Remark D‴.11). At the realised weights of the matching-prior coupled runs below, two of the nine (seed, m) pairs are saddles ($m = 16$, seed 0 with $w = (6,5,5)/16$, margin -0.00052; $m = 64$, seed 1 with $w = (26,17,21)/64$, -0.00296), the other seven admissible ($+0.00056$ to $+0.00609$).

Coupled runs with the matching prior (table 3). $m = 16$, 64, 256 against $M = 2048$, warm start the mixture of Gaussians of width 0.02 around the exact representation, 300 slow units (2.1 relaxation times), three seeds. Along the run $\kappa_R[\mu^*]$ over the spheres is 0 ($r^* = 0$) and with the reference's own residual at most $0.00674 < \sigma_0$ (per-seed maxima 0.00634, 0.00674, 0.00155, each at the first record; the seed-mean curve peaks at 0.00488); the reference's fit error falls from 0.029 to 0.008 and its rms drift from $3.6\cdot10^{-3}$ to $3.0\cdot10^{-5}$ (seed means; per seed the fit error 0.041, 0.036, $0.011 \to 0.012$, 0.009, 0.003). Window maxima of D_m over $(0, 37.5]$, $(37.5, 75]$, $(75, 150]$, $(150, 300]$: $0.114 \to 0.193 \to 0.239 \to 0.250 \pm 0.017$ ($m = 16$), $0.0954 \to 0.157 \to 0.190 \to 0.197 \pm 0.12$ (64), $0.00933 \to 0.0140 \to 0.0166 \to 0.0173 \pm 0.010$ (256): last-to-previous 1.05, 1.04, 1.04 — saturated within 2.1 of the theorem's relaxation times $1/\kappa$; the observed late-time e-folding times of D_m (from the geometric decay of its increments over $(150, 300]$) are 48–84 slow units, median 59, against 107–306 (median 195) for the σ-modification's warm runs of Theorem D‴'s numerics — a factor 3.3 in the medians, against 3.6 in $1/\kappa$; slope -1.23 ± 0.40 in m (the function-level distance -0.63 ± 0.21). Per seed, the linearised law from the realised weights predicts the measured D_m at $s = 300$ with ratios 0.81–1.11 at parameter level (nine pairs) and 0.61–1.69 at function level, and the exact form of Proposition D‴.5(ii), $\sum_k w_k^m \|\vartheta_k^*(w^m) - \vartheta_k^*(w^M)\|^2_{G^{-1}}$, with ratios 0.987–0.998 in all nine pairs: the linearised law's deviations are its second-order remainder (its two lowest ratios, 0.81 and 0.82, sit at $|w^m - w^M|_\infty = 0.084$ and 0.033), not the saddles' doing. The two inadmissible pairs (ratios 0.91 and 0.81; exact 0.995 and 0.994) sit on their replicated saddles at $s = 300$; continued autonomously to $s = 6000$ and 9000 (referee round 10, `review/REVIEW_round10_D3c.md`), both m-systems leave the saddle by Proposition K's local split — the over-weighted cluster separates into two sub-clusters along the centre direction inside its own cell, the cluster weights unchanged — and the coupled distance stays on its plateau (-1.3% at $m = 64$), unlike the σ-modification's seed of round 9, which migrated between basins.

The function-level law for the σ-modification (table 4). Against the warm runs of Theorem D‴'s numerics ($M = 2048$, last window): the predicted root-mean-square $(1/m - 1/M)^{1/2}$ $(\operatorname{tr} F\tilde{C}F^\top/N)^{1/2}$ is 0.0419, 0.0207, 0.00984 against the measured means 0.0307 ± 0.0097, 0.0154 ± 0.0033, 0.00467 ± 0.0012 (ratios 0.73, 0.74, 0.48; a mean of norms over four seeds lies below the root-mean-square).

Reading. The prior is a design input on three counts. Its placement sets the stationary residual, hence the window's lower end and the relaxation rate (the matching prior: margin σ_0, the theorem's relaxation time 3.6 times shorter, the observed one 3.3 times); the amplitudes it induces set the floors (which rise with them along the amplitude-scaled priors, so removing the bias raises the floors by 1.6–2.2 times); and the sign of the residual it leaves at each cluster sets the admissible set of weights (overshoot splits, undershoot stabilises). A shrunk prior — amplitudes 0.5–0.7 of the plant's — wins on all three at a fit error of 0.06–0.09; a misplaced one (centres off by more than a bump width, amplitudes too large, widths too narrow) closes the window.

Remark D‴.11 (undershoot stabilises, overshoot splits)**.** For a Gaussian kernel, $\partial^2_{cc}\tilde{\Phi}(x;\vartheta) = -(a/s^2)(1-(x-c)^2/s^2)\varphi$ is negative at the centre, so the (c,c) entry of $R[\mu](\vartheta_k) = \langle r_\mu \nabla^2\tilde{\Phi}\rangle_\rho$ has the sign of $-a_k \cdot r_\mu$ near the centre: a cluster whose kernel overshoots the plant there (r of the sign opposite to a_k) gets $R \succ 0$ in the centre direction — the separation block $\sigma_0 G^{-1} - R$ loses margin, and the cluster splits once κ_R exceeds σ_0 — while an undershooting cluster gets $R \prec 0$ and a splitting margin above σ_0. For the matching prior this is exactly what Table 18 shows: at every tested weight the over-weighted atom overshoots at its centre ($\hat{f}/f = 1.07$–1.10), its R_{cc} is positive and the top eigenvector of R^G is the centre direction, $(0.01, 1.00, -0.02)$ in (a,c,s), while the under-weighted atoms have $R_{cc} < 0$; the shrunk prior undershoots at the uniform weights and at $(7,5,4)/16$, and at the three more unbalanced weights its over-weighted atom overshoots slightly ($\hat{f}/f = 1.007$–1.028), taking the margin just below σ_0. The two other instabilities of this section and Section 23 have a different anatomy (referee round 10): the σ-modification's inadmissible seed (weight $9/16$ on the first bump) still undershoots at its centre ($R_{cc} < 0$), its unstable direction being a mixed amplitude–width direction — the broad kernel (width 0.88 against the bump's 0.5) overshoots in its tails, through the cross-curvature $\langle r\varphi(x-c)^2/s^3\rangle_\rho$; and Proposition K's collapsed kernel owes its $R_{cc} = +0.055$ to the overshoot over the negative bump and to the undershoot of the outer bumps beyond its inflection points, where $1-(x-c)^2/s^2 < 0$, not to the region between the bumps. It is an observation about the Gaussian kernel's second derivative, not a theorem: the width and amplitude directions carry other signs, and the residual's profile matters; the splitting margin itself is the quantity to compute.

Proposition D‴.12 (the mass constraint removes the weight-fluctuation floor)**.** *Let $\nu = \sum_{j=1}^{K_0} q_j \delta_{\vartheta_j^0}$ be a prior measure with distinct atoms and masses $q_j > 0$, $\Delta_\nu := \min_{j\neq k}\|\vartheta_j^0 - \vartheta_k^0\|_{G^{-1}}$, and let the leak be the barycentric OT leak of Theorem C′ toward ν for the cost $\|\cdot\|^2_{G^{-1}}$: with $\gamma = \gamma(\Theta)$ a vertex optimal plan from μ_Θ to ν (row sums $1/m$, column sums q_j) and $\bar{T}_i := m\sum_j \gamma_{ij}\vartheta_j^0$ its barycentric projection,*

$$\frac{\mathrm{d}\vartheta_i}{\mathrm{d}s} = G\langle \nabla_\vartheta (r_\Theta \tilde{\Phi}(\cdot\,;\vartheta_i))\rangle_\rho - \sigma_0(\vartheta_i - \bar{T}_i), \tag{$\mathfrak{A}_m^{\mathrm{OT}}$}$$

the G-gradient flow of $J_m^{\mathrm{OT}}(\Theta) := m\bar{L}(\mu_\Theta) + (\sigma_0 m/2)W^2_{G^{-1}}(\mu_\Theta,\nu)$ wherever the optimal plan is unique (Danskin's theorem, as in Theorem C′(i)), with the vertex-selection solutions of Section 13 where it is not. Its mean-field form is the Laguerre-cell leak*: for μ giving no mass to their boundaries, the optimal plan from μ to the atomic ν is induced by the map $T_\mu(\vartheta) = \vartheta_j^0$ on the Laguerre cell*

$$L_j[\mu] := \big\{\vartheta : \|\vartheta - \vartheta_j^0\|^2_{G^{-1}} - \psi_j \le \|\vartheta - \vartheta_k^0\|^2_{G^{-1}} - \psi_k \text{ for all } k\big\},$$

the dual weights ψ fixed by $\mu(L_j[\mu]) = q_j$ (semi-discrete optimal transport [3]; for a disconnected support the boundaries in the gaps are not unique, but T_μ is unique on $\operatorname{supp}\mu$*), and ($\mathfrak{A}_0^{\mathrm{OT}}$) is ($\mathfrak{A}_0$) with the leak $-\sigma_0(\vartheta - T_{\mu_s}(\vartheta))$: the nearest-atom leak of Proposition D″.9 with the Voronoi cells replaced by cells of prescribed masses. Call m* commensurate *with ν if $mq_j \in \mathbb{N}$ for every j.*

(i) (conservation against constraint) *Along ($\mathfrak{A}_m^\nu$) on the good event $\mathcal{G}_S$ of Theorem D‴ (Proposition D‴.9(i)) no particle changes cell, so the cluster masses $w_k(\Theta_s) := (1/m)\#\{i : j(\vartheta_i(s)) = k\}$ are conserved: $w^m(s) = w^m(0)$, the multinomial sample of the initial cloud, and the stationary floor of Proposition D‴.5 is the memory of the initial condition that this conservation law carries. Along ($\mathfrak{A}_m^{\mathrm{OT}}$) the masses assigned to the atoms are constrained instead: $\sum_i \gamma_{ij}(s) = q_j$ for every s.*

(ii) (the frozen plan) *If m is commensurate, every vertex of the transportation polytope is integral — $\gamma_{ij} \in \{0, 1/m\}$: each particle is assigned to one atom $\pi(i)$ and atom j receives exactly mq_j particles — so $\bar{T}_i = \vartheta^0_{\pi(i)}$ and ($\mathfrak{A}_m^{\mathrm{OT}}$) is ($\mathfrak{A}_m^\nu$) with the Voronoi assignment $j(\cdot)$ replaced by the plan's assignment π, locally constant in Θ wherever the optimal plan is*

unique; there J_m^{OT} is C^2 with the leak's Hessian $\sigma_0 G^{-1} \otimes I$. Theorem D‴'s synchronous coupling applies to $(\mathfrak{A}_m^{\mathrm{OT}})$ only under the hypothesis of Proposition D‴.9(i) read with the plan — for every $s \geq 0$ the plan assigns each particle to the atom of its reference's Laguerre cell $L_\cdot[\mu_s]$, and $N_{\varrho'}(\operatorname{supp}\mu_s)$ meets no Laguerre boundary — and at $s = 0$ this requires the sample to be balanced, *$w^m(0) = q$ (the mean-field flow of the mixture start has masses q from the start): an event of probability of order $m^{-(K_0-1)/2}$ under an i.i.d. start — 0.053, 0.014, 0.0034, 0.0016 at $m = 15$, 60, 240, 510 on the instance, $\approx 0.83/m$. On that event Theorem D‴, Proposition D‴.5 and Corollary D‴.7 go through with the same constants except for Step 8, whose McDiarmid bound for the reference's sampling term is written for i.i.d. particles and would have to be redone for the stratified sample (mq_k particles i.i.d. within each class). For a generic sample the plan reassigns at least $\frac{1}{2}\|w^m(0) - q\|_1$ of the mass at $s = 0$ and the shared-particle coupling carries that mismatch for ever ((v)); what is proved about the OT dynamics in general is the local equivalence above and the stationary statement (iii).*

(iii) (the stationary measure does not depend on m) *Let $\Theta^*(q) = (\vartheta_1^*, \dots, \vartheta_{K_0}^*)$ be the weighted stationary point of $\mathcal{G}_q$ with the prior atoms (Proposition D‴.5 with $w = q$, each class k leaking toward ϑ_k^0 as in Proposition D‴.9) and $\varepsilon := \max_k \|\vartheta_k^* - \vartheta_k^0\|_{G^{-1}}$. If $\varepsilon < \Delta_\nu/2$ then, for every commensurate m, the replicated configuration $\Theta_m^*(q)$ (mq_k particles at ϑ_k^*) has the diagonal plan (class $k \to$ atom k) as its unique optimal plan, is a critical point of J_m^{OT}, and is a strict local minimiser iff $q \in W_{\mathrm{adm}}$ (its Hessian is Proposition D‴.5's, with separation blocks $\sigma_0 I - R_k^G$ of multiplicities $mq_k - 1$). The stationary measure $\mu^*(q) := \sum_k q_k \delta_{\vartheta_k^*}$ and the learned function $\hat{f}^*(q) = \sum_k q_k \tilde{\Phi}(\cdot\,; \vartheta_k^*)$ are therefore the same for every commensurate m, and the stationary floors vanish:*

$$W_{G^{-1}}(\mu_m^*, \mu_M^*) = 0, \qquad \|\hat{f}_m^* - \hat{f}_M^*\|_{L^2(\rho)} = 0 \qquad \text{for all commensurate } m, M,$$

against $\mathbb{E}[D_\infty \mid \mathrm{adm}] = (1/m - 1/M)\sum_k \bar{w}_k \operatorname{tr}(\tilde{S}_k \tilde{C} \tilde{S}_k^\top)$ and $(1/m - 1/M)\operatorname{tr}(F\tilde{C}F^\top)/N$ under the nearest-atom leak (Propositions D‴.5 and D‴.9(iii)), whose stationary measures are moreover at squared W_2-distance $\frac{1}{2}\|w^m - w^M\|_1 \cdot O(\Delta_\nu^2) = O(\Delta_\nu^2 m^{-1/2})$ from each other (W_2 itself $O(\Delta_\nu m^{-1/4})$) — the cross-cluster transport of the weight mismatch, which the shared-particle distance D does not see. The price is the bias of the masses: the stationary residual $r^*(q) = f - \sum_k q_k \tilde{\Phi}(\cdot\,; \vartheta_k^*)$ vanishes iff ν carries an exact representation of the plant with its masses, *$\sum_k q_k \tilde{\Phi}(\cdot\,; \vartheta_k^0) = f$ on $\operatorname{supp}\rho$; for masses q_δ displaced from those of the exact representation the residual is $O(\delta)$ and the window's margin decreases with δ (Proposition D‴.9(ii) with the masses as the parameter), and the designed masses must themselves be admissible, $q \in W_{\mathrm{adm}}$: past the admissible set the over-weighted equal-mass cluster splits (Remark D‴.11).*

(iv) (incommensurate m: the quantisation remainder) *For any m and any configuration a vertex optimal plan splits at most $K_0 - 1$ particles between two or more atoms (a vertex of the transportation polytope is supported on a forest of the bipartite graph with $m + K_0$ nodes, so on at most $m + K_0 - 1$ edges, every particle carrying at least one). At a stationary configuration whose unsplit particles are collapsed by class (the minimisers of Proposition D‴.5's type) a split particle sits where the fit force balances the pull toward its barycentric target $\bar{T}_i \in \operatorname{conv}\{\vartheta_j^0 : \gamma_{ij} > 0\}$ — a* bridge particle *between the clusters — so the stationary measure differs from $\mu^*(q)$ by at most $(K_0 - 1)/m$ of mass at the bridge positions and by the $O(K_0/m)$ shift of the class weights:*

$$W_{G^{-1}}^2(\mu_m^*, \mu^*(q)) \leq 4(K_0 - 1)R_\Theta^2/m + O(K_0^2\|S\|^2/m^2),$$
$$\|\hat{f}_m^* - \hat{f}^*(q)\|_{L^2(\rho)} \leq 2(K_0 - 1)K_\Phi/m + O(K_0(K_\Phi + L_\vartheta^G\|S\|)/m),$$

R_Θ the a priori radius of Step 0 (with the hull of $\operatorname{supp}\nu$). The orders are the content, not the constants — on the instance $R_\Theta = 2865$ makes the first bound $4 \cdot 10^6$ against a

measured 1.0–1.4 at $m = 16$, and the second 1.00 against 0.018: the function-level floor is $O(1/m^2)$ in square, one order in m below the $(1/m - 1/M)$ law, and the W^2 floor is $O(1/m)$ against the nearest-atom leak's $O(m^{-1/2})$ of (iii) — half an order gained in W^2, a full order at function level — with deterministic constants set by the plan's split pattern (measured $m \cdot W^2(\mu_m^*, \nu) \approx 20 = 0.5\Delta_\nu^2$ and $m \cdot \|\hat{f}_m^* - f\|/\|f\| \approx 0.55$).

(v) (the transit) *From a cloud with cluster masses $w^m \neq q$ the plan assigns at least $\frac{1}{2}\|w^m - q\|_1$ of the mass (up to the split particles) of the over-weighted clusters to under-weighted atoms, and these particles migrate between the clusters against the fit force at the leak's rate — for the pure leak a migrating particle crosses the Voronoi midpoint at $s = \ln 2/\sigma_0$. During the transit the synchronous coupling of Theorem D‴ sees the mass migrating in one system and not in the other at a distance of order Δ_ν, $D = O(\Delta_\nu^2\|w^m - w^M\|_1)$ — it is not a coupling the OT leak respects — $W^2_{G^{-1}}(\mu_m, \mu_M)$ is of the same order $O(\Delta_\nu^2 m^{-1/2})$, the sampling term of the cluster masses, and the function-level distance is $O(K_\Phi\|w^m - w^M\|_1) = O(m^{-1/2})$. After the transit the two measure-level distances vanish on the commensurate sizes ((iii)), while the synchronous D converges from below to the plateau $\sum_{k\neq k'} p_{kk'}\Delta_{kk'}^2$ of the mass reassigned in one system and not in the other (0.95–0.96 of it at $s = 600$ on the instance): the memory of the initial masses, erased from the stationary measure, is kept by the shared-particle coupling. A mean-field flow whose clusters start with masses $w \neq q$ has the same transit — the Laguerre boundaries cut the over-weighted clusters and the mass $\frac{1}{2}\|w - q\|_1$ migrates — but the mixture start of the numerics has mean-field masses q, so there the M-references transit only as finite samples and are poor stand-ins for the mean-field flow during the transit.*

Status. (i) proved on $\mathcal{G}_S$ (Theorem D‴'s per-particle bound keeps every particle within $3\varrho'/4$ of its reference, whose cell has margin ϱ'). (ii) the local equivalence proved (the transportation constraints are the incidence matrix of a bipartite graph, totally unimodular, so with integral right-hand sides every vertex is integral — Hoffman–Kruskal [42]; a unique optimal vertex of a linear program stays optimal under small perturbations of its cost; the Laguerre form of the mean-field leak is semi-discrete optimal transport [3]); the coupling statement formal — the proofs of Theorem D‴ and Proposition D‴.5 use the leak only through its Hessian and its a priori bound, as in Proposition D‴.9(i), the constancy of the plan being needed for every $s \geq 0$ (referee round 10), which at $s = 0$ is the balanced-sample event of probability $\approx 0.83/m$, and on it Step 8 would have to be redone for the stratified sample (referee round 11). (iii) proved (c-cyclical monotonicity [86]: for $\varepsilon < \Delta_\nu/2$ every off-diagonal pair costs at least $\frac{1}{2}(\Delta_\nu - \varepsilon)^2 > \frac{1}{2}\varepsilon^2 \geq$ every diagonal pair, so along any cycle the diagonal plan is strictly cheaper; the diagonal plan is therefore optimal, and every other plan, which differs from it by a nonnegative combination of cycle flows, costs strictly more; the Hessian is Proposition D‴.5(i)'s block splitting with the plan frozen; the bias statements are Proposition D‴.9(ii)'s implicit-function argument with the masses as the parameter). (iv) the count proved (the vertices of the transportation polytope are the plans supported on spanning forests, Klee–Witzgall [42]); the two bounds formal (they assume the stationary configuration's structure). (v) formal; the transit and the three distances are numerical (`exp_ot_leak.py` → `results/chaos/ot_leak.md`, below). Open: a coupling theorem for the OT dynamics from an i.i.d. start — Theorem D‴'s synchronous coupling needs a balanced sample and otherwise carries the transit's mismatch for ever (the plan reassigns $O(\sqrt{m})$ particles at $s = 0$, which the shared-particle pairing never re-matches); the natural candidate follows the plan, pairing a particle with its assigned atom's mean-field mass — so what is proved about the OT dynamics is the local equivalence (ii) and the stationary statement (iii). Checked: `tests/test_ot_leak.py` (the plan is a vertex with the prescribed masses, integral when m is commensurate and with at most $K_0 - 1$ split rows otherwise, and the single-atom prior gives the σ-modification; the leak is the G-gradient flow of J_m^{OT} against finite differences, for commensurate and incommensurate m; the replicated weighted configuration is stationary for three commensurate sizes with the diagonal plan and the same learned function — for the matching masses and for mismatched ones — with $W^2 = 0$ between the sizes and the splitting margin of Proposition D‴.5 (σ_0 at the matching masses);

the conservation of the cluster masses under the nearest-atom leak against their equalisation under the OT leak on a short run, with the mass in transit equal to $\frac{1}{2}\|w(0)-q\|_1$ and the fit error lower at the end; the forest bound on random clouds; the exact W^2 between empirical measures by replication against the transportation LP).

Remark D‴.13 (conservation, constraint, and what the mean-field limit forgets)**.** Theorem D‴ bounds the coupled distance uniformly in time, and Proposition D‴.5 says what the bound's constant is made of: the m-system's memory of its initial cluster masses, which the nearest-atom leak conserves. The OT leak replaces the conservation law by a constraint and the memory is erased from the stationary measure — the finite-m stationary measure *is* the mean-field one on the commensurate sizes, and within $(K_0-1)/m$ of mass of it otherwise — though not from the shared-particle coupling, whose synchronous distance keeps the plateau of the reassigned particles ((v)). This is a statement about the design of the leak, not about m: the same clouds under the nearest-atom leak keep their floors. Three consequences. (a) The masses join the atoms as design inputs of the leak's prior — both are read from the plant's structure (Proposition D‴.9(ii)) — and the admissible set W_{adm} of Proposition D‴.5, a probability statement about the realised weights under the nearest-atom leak, becomes a condition on the design, $q \in W_{\mathrm{adm}}$, under the OT leak. (b) The mean-field form of Theorem C′'s leak for an atomic prior is the Laguerre-cell leak; its finite-m fluctuation is not a random displacement of the cells but the split particles — at most K_0-1 — and the migrating ones during the transit. (c) A soft mass constraint (an entropic or penalised plan) would interpolate between the two leaks, with a floor set by the stiffness of the penalty; open (Section 26, branch 8), as is the transfer of a learned measure to a plant whose representation weights are unknown, where q itself would have to be learned.

Numerics (`exp_ot_leak.py` → `results/chaos/ot_leak.md`). The instance of Theorem D‴'s numerics at $\sigma_0 = 0.15\sigma_{\mathrm{crit}} = 0.00707$ with the matching prior of Proposition D‴.9's numerics — three atoms at the exact representation, masses $q = (1/3, 1/3, 1/3)$, the plant's representation weights, $\Delta_\nu = 6.26$ — under the nearest-atom leak and under the OT leak (plan recomputed every 0.2 slow units) on the same warm clouds (the jittered mixture of width 0.02), 600 slow units $= 4.2/\sigma_0$, four seeds; commensurate sizes $m = 15$, 60, 240 against $M = 510$ and incommensurate sizes $m = 16$, 64, 256 against $M = 512$.

Conservation against constraint (table 1). Under the nearest-atom leak no cluster weight changes along any run ($\max_s |w(s)-w(0)|_\infty = 0$ in all 32 systems). Under the OT leak the mass in transit at $s=0$ is $\frac{1}{2}\|w(0)-q\|_1 = 0.117$, 0.058, 0.027, 0.024 (seed means for $m = 15$, 60, 240, 510; per system 1–3, 1–5, 2–13 and 3–21 particles), it reaches zero, and the weights by nearest atom equal q exactly from $s_{\mathrm{eq}} = 93$–164 slow units on (median 104, against the pure leak's midpoint crossing $\ln 2/\sigma_0 = 98$); no plan is split. The fit error at the end falls from 0.067, 0.029, 0.017, 0.012 under the nearest-atom leak — the bias of the realised weights — to $2.8\cdot10^{-4}$, $1.2\cdot10^{-4}$, $1.2\cdot10^{-4}$, $0.8\cdot10^{-4}$ (seed means); the migrated particles are still 0.06–0.16 whitened units from their new atoms at $s = 600$ — the pure leak's remaining approach $\Delta_{kk'}e^{-\sigma_0\cdot600} = 0.09$–$0.10$ — and the fit force cancels their first-order effect on $\hat f$, so the OT finals are not yet stationary (maximal whitened drift 5–$8\cdot10^{-4}$, against $\le 10^{-3}$ and mostly $\le 10^{-4}$ under the nearest-atom leak).

The three distances on the commensurate sizes (table 2, last window $(450, 600]$*).* Function level $\|\hat f_m - \hat f_M\|^2_{L^2(\rho)}$: nearest-atom $2.5\cdot10^{-3}$, $5.2\cdot10^{-4}$, $5.9\cdot10^{-5}$ (stationary, last/previous window 0.97–1.00; the law $(1/m-1/M)\,\mathrm{tr}(F\tilde C F^\top)/N$ predicts $4.0\cdot10^{-3}$, $9.1\cdot10^{-4}$, $1.4\cdot10^{-4}$) against OT $4.9\cdot10^{-7}$, $4.9\cdot10^{-8}$, $7.5\cdot10^{-9}$ — ratios $2\cdot10^{-4}$, $9\cdot10^{-5}$, $1.3\cdot10^{-4}$ — still falling by a factor 0.02–0.035 per 150 slow units (a rate of 3.0–$3.8\sigma_0$: the displaced particles' first-order effect cancels, the class means agreeing to first order) toward the zero of (iii). $W^2_{G^{-1}}(\mu_m, \mu_M)$: nearest-atom 5.1, 2.7, 0.81 — the cross-cluster transport of the weight mismatch: the exact $K\times K$ transport between the weighted stationary measures at the realised weights reproduces it to 0.997–0.999 at the seven admissible pairs (0.85–0.99 at the five pairs with saddle weights,

where the m-system is not at the replicated configuration), the formula $\frac{1}{2}\|w^m - w^M\|_1\Delta_\nu^2 = 4.9$, 2.5, 0.70 to -7% to $+28\%$; slope -0.67 ± 0.13 in m, the sampling term of the cluster masses — against OT $7.9 \cdot 10^{-3}$, $2.3 \cdot 10^{-3}$, $6.9 \cdot 10^{-4}$ (ratios $1.5 \cdot 10^{-3}$, $0.9 \cdot 10^{-3}$, $0.9 \cdot 10^{-3}$), falling by 0.12 per 150 slow units ($e^{-2\sigma_0 \cdot 150} = 0.12$, the square of the spread's contraction at the margin σ_0). The synchronous D is the one distance the OT leak does not improve: 5.8, 3.3, 0.94 against 0.87, 0.25, 0.017, still rising (last/previous 1.08–1.10) toward the plateau $\sum_{\text{mismatched}} \Delta_{kk'}^2/m$ of the pairs reassigned in one system and not in the other (0.95–0.96 of it at $s = 600$), as (v) says: the coupling keeps the memory that the measure loses.

The incommensurate sizes (table 3). Every plan has one or two split particles ($\leq K_0 - 1 = 2$); the weights by nearest atom differ from q by the split particle's share, 0.042, 0.010, 0.0026, $0.0013 = (2/3)/m$; the unsplit particles in transit at $s = 0$ (0–2 at $m = 16$, 1–4 at 64, 2–11 at 256, 1–19 at 512) all arrive by the end. At the end the split particles sit between the clusters, at 0.28–0.58 Δ_ν from the nearest atom and within 0.03–0.6 (G^{-1} metric) of their barycentric target $\bar{T}_i$ — the bridge particles of (iv). Against the $M = 512$ reference, which carries its own two bridge particles: function level OT $3.4 \cdot 10^{-4}$, $3.1 \cdot 10^{-5}$, $1.1 \cdot 10^{-6}$ (the $m = 16$ value still relaxing, 0.82 per window — continued to $s = 2400$ by referee round 11 it settles at $2.8 \cdot 10^{-4}$, and $m = 64$ at $2.9 \cdot 10^{-5}$, with slope -2.00) against nearest-atom $2.5 \cdot 10^{-3}$, $5.0 \cdot 10^{-4}$, $4.4 \cdot 10^{-5}$, ratios 0.11–0.14, 0.063, 0.025; W^2 OT 1.16, 0.29, 0.041 (stationary) against nearest-atom 4.8, 2.7, 0.69, ratios 0.24, 0.10, 0.059. The quantisation laws of (iv) are read against the mean-field stationary measure $\mu^*(q)$, here ν itself: $W^2(\mu_m, \nu) = 1.21$, 0.33, 0.079, 0.038 for $m = 16$, 64, 256, 512, i.e. $m \cdot W^2 = 19.4$, 21.0, 20.1, $19.5 \approx 0.5\Delta_\nu^2$ with slope -0.98 ± 0.03, and the fit error 0.028, 0.0093, 0.0023, 0.0011, i.e. $m \cdot \text{fit} = 0.45$–$0.60$ with squared slope -1.81 ± 0.05 — the orders $1/m$ and $1/m^2$ of (iv), with constants set by the plan's split pattern (seeds with the same pattern reach identical plateaus, referee round 11) far below the worst case $(K_0 - 1)\Delta_\nu^2 = 78$; against the nearest-atom leak's fit errors 0.061, 0.035, 0.013, 0.012.

The masses as a design input (table 4). For $q_\delta = (1/3 + \delta, 1/3 - \delta/2, 1/3 - \delta/2)$ the weighted stationary point has fit error 0.0069, 0.013, 0.019, 0.029, 0.049 at $\delta = 0.01$, 0.02, 0.03, 0.05, 0.1 — the bias of the mass constraint, 0.69δ to 0.49δ, decreasing with δ — κ_R at the over-weighted atom $+0.0023$, $+0.0043$, $+0.0060$, $+0.0086$, $+0.0119$, margin κ $+0.0047$, $+0.0026$, $+0.0008$, -0.0019, -0.0054 and splitting margin $+0.0048$, $+0.0028$, $+0.0011$, -0.0015, -0.0049: the designed masses leave the admissible set between $\delta = 0.03$ and 0.05 — the asymmetry of the matching prior's admissible set in Proposition D‴.9's numerics (an excess of 0.04 on the largest bump is inadmissible) — while the diagonal plan stays the unique optimal plan at every row ($\varepsilon \leq 1.04 < \Delta_\nu/2 = 3.13$).

Reading. The two leaks start from the same clouds and end in different worlds. The nearest-atom leak keeps the clusters' sampled masses for ever, and its stationary distances are what Propositions D‴.5 and D‴.9 say: a function-level floor of order $1/m$, and a W^2 floor of order $m^{-1/2}$ set by the cross-cluster transport of the weight mismatch. The OT leak moves one to 21 particles across in about $\ln 2/\sigma_0$ slow units and then settles on the mean-field stationary measure: on the commensurate sizes every distance but the synchronous one decays to zero (three to four orders of magnitude below the nearest-atom leak's floors after $4.2/\sigma_0$), and the fit loses the weight bias; on the incommensurate sizes the floors are the one or two bridge particles — $O(1/m^2)$ at function level, $O(1/m)$ in W^2 — whose position between the clusters is the price of a mass constraint the size cannot meet. The masses are a design input with a bias and an admissibility constraint of their own.

Corollary D‴.14 (the real loop with the OT leak: one orbit for every commensurate size)**.** *Consider the real closed loop* (CL) *of Theorem J with the leak $\sigma(\vartheta_i - T_i)$, $\sigma = \sigma_0/k_e$, where T_i is the nearest-atom target $\vartheta^0_{j(\vartheta_i)}$ (Proposition D″.9) or the barycentric target $\bar{T}_i$ of the vertex plan (Proposition D‴.12). For a weight vector w with $mw \in \mathbb{N}^{K_0}$ and a class map $\pi_0 : \{1, \dots, m\} \to \{1, \dots, K_0\}$ with $|\pi_0^{-1}(k)| = mw_k$, let $\mathcal{C}_m(\pi_0) := \{\Theta : \vartheta_i = \vartheta_{\pi_0(i)}\}$ be the*

collapsed subspace, *on which* $\hat f_\Theta = \sum_k w_k \tilde\Phi(\cdot\,;\vartheta_k)$, *and let the* weighted K_0-atom loop (CL_w) *be* (CL) *with* $\hat f = \sum_k w_k \tilde\Phi(\cdot\,;\vartheta_k)$, *the* K_0 *parameters* ϑ_k *and the leaks* $\sigma(\vartheta_k - \vartheta_k^0)$.

(i) (invariance) $\mathcal{C}_m(\pi_0)$ *is invariant under* (CL) *with the nearest-atom leak as long as every class atom* ϑ_k *stays in the Voronoi cell of* ϑ_k^0, *and under* (CL) *with the OT leak as long as* $w = q$ *and* $\max_k \|\vartheta_k - \vartheta_k^0\|_{G^{-1}} < \Delta_\nu/2$; *on it* (CL) *is* ($\mathrm{CL}_w$). *With the OT leak and* $w \neq q$ *the subspace is not invariant: the plan assigns at least* $\frac{1}{2}\|w - q\|_1$ *of the mass of the over-weighted classes to other atoms, and those particles leave their class.*

(ii) (one orbit for every commensurate size) *Let* $q \in W_{\mathrm{adm}}$ *and let* Γ_q *be the* T_0*-periodic orbit of* (CL_q) *near* $\Theta^*(q)$ *given by Theorem J, whose proof applies to* (CL_w) *verbatim (its averaged drift is* $-(1/k_e)(\mathrm{diag}(w)^{-1} \otimes G)\nabla \mathcal{G}_w$, *the gradient of* $\mathcal{G}_w$ *in the metric* $\mathrm{diag}(w) \otimes G^{-1}$, *which is all its Steps 4–6 use), and assume* Γ_q *stays inside the cells,* $\max_k \|\vartheta_k^\Gamma(t) - \vartheta_k^0\|_{G^{-1}} < \Delta_\nu/2$ *— the cell condition of (i), which* W_{adm} *does not contain. For every commensurate* m *the lift of* Γ_q *to* $\mathcal{C}_m(\pi_0)$ *is a* T_0*-periodic orbit of* (CL) *with the OT leak. It is exponentially stable iff* Γ_q *is and the ghost monodromies of the classes contract (Proposition D‴.15 below) — at the matching masses, where the residual vanishes, the ghosts' exponent is exactly* $\sigma_0/k_e > 0$ *— and it is Theorem J's orbit* Γ_m *at* $\Theta_m^*(q)$, *with the basin (c) of that theorem, whenever it lies in Theorem J's uniqueness ball* $\|\Theta - \Theta_m^*(q)\|_{G_m^{-1}} \le \rho_0(m)$*: with* $\|\Theta\|_w^2 := \sum_k q_k \|\vartheta_k\|_{G^{-1}}^2$ *the lift's* G_m^{-1}*-distance to* $\Theta_m^*(q)$ *is* $\sqrt{m}$ *times the weighted distance, so the condition is* $\sqrt{m} \cdot \sup_t \|\Theta_p^q(t) - \Theta^*(q)\|_w \le \rho_0(m)$, *satisfied for* $k_e \ge \max\{k_0(m), \sqrt{m}\, C_q/\rho_0(m)\}$ *by Theorem J(a)'s* $\sup_t \|\Theta_p^q(t) - \Theta^*(q)\|_w \le C_q/k_e$. *Along the lift the orbit measure* $(1/m)\sum_i \delta_{\vartheta_i(t)} = \sum_k q_k \delta_{\vartheta_k(t)}$, *the learned function* $\sum_k q_k \tilde\Phi(\cdot\,;\vartheta_k(t))$, *the plant's orbit* $x_p(t)$ *and the tracking error* $e(t)$ *are the same for every commensurate* m *at every* t *— exactly, not up to* $O(1/k_e)$*: the finite-size floor of the real loop (Corollary D‴.7) is zero.*

(iii) (Corollary D‴.7 sharpened) *With the nearest-atom or the* σ*-modification leak and both realised weight vectors admissible,* $w^m, w^M \in W_{\mathrm{adm}}$ *(Corollary D‴.7's hypothesis: the replicated configurations are hyperbolic minima), the same argument lifts* Γ_{w^m}, *the orbit of* (CL_{w^m}) *with the realised weights: Theorem J's orbit near* $\Theta_m^*(w^m)$ *is the lift of* Γ_{w^m} *under the same proviso, so the real loop's coupled distance between two sizes is exactly the distance between two weighted* K_0*-atom orbits,*

$$D_{\mathrm{orb}}(t) = \sum_k w_k^m \|\vartheta_k^\Gamma(t; w^m) - \vartheta_k^\Gamma(t; w^M)\|_{G^{-1}}^2,$$

Corollary D‴.7's $O(1/k_e)$ *terms being the* k_e*-dependence of* $w \mapsto \Gamma_w$ *and nothing else; and the tracking error along the orbit is that of* (CL_{w^m}) *— the weight bias of the fit is a tracking error, which the mass constraint removes for every size at once. When* $w^m \notin W_{\mathrm{adm}}$ *the lift is a saddle orbit (a ghost exponent is negative, Proposition D‴.15) and the* m*-system leaves it by splitting its over-weighted class, as three of the six nearest-atom systems of the numerics below do.*

Status. (i) proved (the particle law depends on i only through ϑ_i and its target; on $\mathcal{C}_m(\pi_0)$ the targets agree within a class — by the cell condition for the nearest-atom leak, by the lemma of Proposition D‴.12(iii) for the OT leak — so the vector field is tangent to $\mathcal{C}_m(\pi_0)$ and coincides there with (CL_w)'s, and solutions are unique away from the cell boundaries and the plan switches). (ii), (iii) proved from (i), Proposition D‴.15 and Theorem J under the stated proviso and hypotheses (the cell condition along Γ_q; admissible weights in (iii)); the m-uniformity of Theorem J's thresholds $k_0(m)$, $\rho_0(m)$ at the replicated configuration is the open point of Corollary D‴.7 — its spectral part is settled by Proposition D‴.15 (the Floquet spectrum of the lift does not depend on m), the basin radius in the mean norm remains formal; without the proviso the lift is still a periodic orbit of (CL) for every commensurate m, with the stability Proposition D‴.15 gives it, and only its identification with Theorem J's orbit is conditional.

Checked: `tests/test_ot_leak.py` (the collapsed 15-system and the 3-atom loop integrate to the same configuration and plant state to 10^{-12} over 300 steps of the real loop with the OT leak at masses q, the atoms moving; the collapsed 15-system with weights $(6,5,4)/15$ stays collapsed under the nearest-atom leak, while under the OT leak one of its particles is assigned away from its class; two commensurate lifts, $m = 15$ and 30, keep $W^2 = 0$ and the same learned function along the real loop; the lifts of two weight vectors under the nearest-atom leak are different measures).

Numerics (`exp_ot_real_loop.py` → `results/chaos/ot_real_loop.md`). The real loop at $k_e = 25$ ($\sigma = \sigma_0/k_e$; RK4 plant with the control and the reference evaluated at the stages and the parameters frozen over the step, Euler parameters, $\mathrm{d}t = 10^{-3}$ — an earlier zero-order hold of the control had set a tracking floor of $4 \cdot 10^{-5}$, referee round 12) for 100 time units — 16 periods, 4 slow units — from the clouds at the end of Proposition D‴.12's averaged runs (contracted where the realised weights are admissible) (seeds 0 and 1; $m = 15$, 60 and $M = 510$; the OT runs' clouds carry the masses q, the nearest-atom runs' their conserved sampled weights) and from the collapsed lifts (the exact representation for the OT leak — the 3-atom loop — and the replicated $\Theta^*(w^m)$ for the nearest-atom leak); statistics over the last five periods on the reference orbit. Under the OT leak the learned functions of the three sizes agree to $\|\hat{f}_m - \hat{f}_M\|^2_{L^2(\mathrm{orbit})} = 7.5 \cdot 10^{-9}$–$2.7 \cdot 10^{-8}$ and with the 3-atom lift to $5.2 \cdot 10^{-10}$–$3.6 \cdot 10^{-8}$ — the clouds' within-class spread (0.013–0.033, the migrated particles still approaching their atoms) at second order — with fit errors along the orbit $3.5 \cdot 10^{-5}$–$2.9 \cdot 10^{-4}$ and rms tracking errors $0.9 \cdot 10^{-6}$–$7.3 \cdot 10^{-6}$ (the fit error over k_e; the collapsed lift tracks to $8 \cdot 10^{-12}$, the scheme's floor), and $W^2(\mu_m, \mu_M) = 3.3 \cdot 10^{-4}$–$1.4 \cdot 10^{-3}$ at the end. Under the nearest-atom leak they disagree by $1.1 \cdot 10^{-4}$–$2.5 \cdot 10^{-3}$, with fit errors 0.012–0.075 and rms tracking errors $3.2 \cdot 10^{-4}$–$1.9 \cdot 10^{-3}$ — the weight bias as a tracking error, 130 to 550 times the OT leak's per system — and $W^2(\mu_m, \mu_M) = 1.6$–5.5, the cross-cluster transport of the weight mismatch. Three of the six nearest-atom systems have inadmissible realised weights (seed 0, $m = 15$: $(6,5,4)/15$, splitting margin -0.0035; seed 1, $m = 15$: $(6,3,6)/15$, -0.0011; seed 1, $m = 60$: $(24,15,21)/60$, -0.0022): their lifts are saddle orbits and the m-systems are not on them — the over-weighted class has split (within-class spread 0.04–0.11 against 0.0006–0.0013 for the admissible systems), the distance to the lift, $4 \cdot 10^{-8}$–$3 \cdot 10^{-6}$, being the split's second-order effect on $\hat{f}$, and the lift-to-lift ratio at those pairs 0.960, 0.998, 0.961. At the one pair whose weights are both admissible (seed 0, $m = 60$ against $M = 510$) the disagreement is the distance between the lifted orbits to a ratio 1.0003, and the admissible m-systems sit within $2.5 \cdot 10^{-11}$ of their lifts: (iii) as stated, with its admissibility hypothesis visibly necessary.

Numerics in two dimensions (`exp_ot_2d.py` → `results/chaos/ot_2d.md`). The mechanism is dimension-free, and the real loop shows it on a planar plant: $\dot{x} = f(x) + u$ on $\mathbb{R}^2$ with f the sum of four Gaussian bumps with vector amplitudes, the Lissajous reference $x_m(t) = (2\sin t, 1.5\sin 2t)$, a saturated network with $p = 5$ parameters per kernel, $G = \mathrm{diag}(1, 1, \frac{1}{4}, \frac{1}{4}, \frac{1}{4})$, the matching prior (four atoms at the exact representation, masses $\frac{1}{4}$), $k_e = 25$ and $\sigma_0 = 0.05$ (the leak is a design input; $\ln 2/\sigma_0 = 14$ slow units), 2500 time units $= 100$ slow units $= 5/\sigma_0$ from the same jittered warm clouds with random class labels, sizes $m = 8$, 32 against $M = 128$, two seeds, the plant integrated as above. Under the nearest-atom leak the sampled cluster weights — $(3,2,2,1)/8$, $(7,4,12,9)/32$, … — never change, the fit errors along the orbit are 0.11–0.39, the sizes disagree by $\|\hat{f}_m - \hat{f}_M\|^2_{L^2(\mathrm{orbit})} = 5.5 \cdot 10^{-3}$–$2.3 \cdot 10^{-2}$ and the rms tracking error is $1.9 \cdot 10^{-3}$–$6.7 \cdot 10^{-3}$. Under the OT leak the weights reach $(\frac{1}{4}, \frac{1}{4}, \frac{1}{4}, \frac{1}{4})$ at 13–14 slow units in all six systems ($\ln 2/\sigma_0 = 14$), the fit errors fall to $2.1 \cdot 10^{-4}$–$8.7 \cdot 10^{-4}$ (the within-class spread 0.012–0.018 still contracting), the sizes agree to $5.0 \cdot 10^{-8}$–$1.6 \cdot 10^{-7}$ at the stopping time $5/\sigma_0$ — five orders of magnitude below the nearest-atom leak, and still falling by about 3.5 times per 10 slow units ($1.8 \cdot 10^{-12}$ at $10/\sigma_0$ in referee round 13's continued run) — and the rms tracking error is $3.9 \cdot 10^{-6}$–$1.6 \cdot 10^{-5}$, 290 to 480 times smaller per system: the weight bias is a tracking

error in any dimension, and the mass constraint removes it for every size at once.

Proposition D‴.15 (the transverse spectrum of a lifted orbit)**.** *Let $\Gamma = (x_p, \Theta_p)$ be a T_0-periodic orbit of* (CL) *— with any leak whose targets are locally constant near Γ and equal within each class — lying in the collapsed subspace $\mathcal{C}_m(\pi_0)$ with classes of sizes n_k, and let $\Gamma_K = (x_p, (\vartheta_{p,k})_k)$ be the K_0-atom orbit it lifts. For each class k let*

$$S_k := \Big\{\delta \in \mathbb{R}^{n+mp} : \delta x = 0,\ \delta_i = 0 \text{ for } i \notin \pi_0^{-1}(k),\ \sum_{i\in\pi_0^{-1}(k)} \delta_i = 0\Big\} \qquad \text{(dimension } (n_k-1)p\text{)},$$

and let M_k be the monodromy over one period of the ghost equation $\dot\delta = A_k(t)\delta$, $A_k(t) := \partial_\vartheta\big[G\nabla_\vartheta\big(e_p(t)^\top\tilde\Phi(x_p(t);\vartheta)\big)\big]\big|_{\vartheta=\vartheta_{p,k}(t)} - \sigma I$ *— the linearisation of one particle of class k driven by the orbit's plant state, with weight zero in $\hat f$. Then the monodromy matrix $\mathbb{M}$ of Γ leaves every S_k invariant, with $\mathbb{M}|_{S_k} = M_k \otimes I_{n_k-1}$, and on the complementary invariant subspace — the plant coordinates and the class means — it is the monodromy $\mathbb{M}_K$ of Γ_K as an orbit of the weighted K_0-atom loop* (CL_w)*. Hence the Floquet spectrum of Γ is that of Γ_K together with the spectrum of each M_k with multiplicity $n_k - 1$, independent of m; Γ is exponentially stable iff Γ_K is and every M_k is contracting; and when Γ_K is Theorem J's orbit of* (CL_w) *at $\Theta^*(w)$, the exponents of M_k are $(1/k_e)\lambda_j(\sigma_0 I - R_k^G) + O(k_e^{-2})$, $j = 1..p$, with $R_k^G = G^{1/2}R[\mu_{\Theta^*(w)}](\vartheta_k^*)G^{1/2}$ — Proposition K's separation condition and the splitting blocks of Proposition D‴.5(i), read along the orbit.*

Proof. Write the particle law as $\dot\vartheta_i = F(x, e, \vartheta_i) - \sigma(\vartheta_i - T_i)$, $F(x, e, \vartheta) := G\nabla_\vartheta(e^\top\tilde\Phi(x;\vartheta))$, and note that the plant's equation depends on Θ only through $\hat f_\Theta(x) = (1/m)\sum_i \tilde\Phi(x;\vartheta_i)$. Along Γ the linearised equations are $\delta\dot x = (\partial_x f - k_e I - \partial_x\hat f)\delta x - (1/m)\sum_i \partial_\vartheta\tilde\Phi(x_p;\vartheta_i)\delta_i$ and $\dot\delta_i = (\partial_x F + \partial_e F)\delta x + (\partial_\vartheta F - \sigma I)|_{\vartheta_i}\delta_i$ ($e = x - x_m$ gives $\delta e = \delta x$). For $\delta \in S_k$, $\delta x = 0$ and all ϑ_i with $i \in \pi_0^{-1}(k)$ equal $\vartheta_{p,k}(t)$, so $\sum_i \partial_\vartheta\tilde\Phi(x_p;\vartheta_i)\delta_i = \partial_\vartheta\tilde\Phi(x_p;\vartheta_{p,k})\sum_{i\in\pi_0^{-1}(k)}\delta_i = 0$: the plant is not perturbed to first order, each δ_i obeys $\dot\delta_i = A_k(t)\delta_i$ with the same A_k, and $\sum_{i\in\pi_0^{-1}(k)}\dot\delta_i = A_k\sum\delta_i = 0$. So S_k is invariant and the linearised flow acts on it as the ghost equation on every component, whence $\mathbb{M}|_{S_k} = M_k \otimes I_{n_k-1}$. The class-constant perturbations $\{\delta_i = u_{\pi_0(i)}\}$ together with δx form the complementary subspace; it is invariant because particles of one class have identical right-hand sides when their perturbations coincide, and there the variation of $\hat f$ is $\sum_k (n_k/m)\partial_\vartheta\tilde\Phi(x_p;\vartheta_{p,k})u_k = \sum_k w_k\partial_\vartheta\tilde\Phi u_k$ — the linearisation of (CL_w) along Γ_K. The spectral statements follow. For the last one, along Theorem J's orbit of (CL_w) the tracking error is $e_p(t) = r_{\Theta^*(w)}(x_m(t))/k_e + O(k_e^{-2})$ and $\vartheta_{p,k}(t) = \vartheta_k^*(w) + O(1/k_e)$ (Theorem J(a) with the error equation), so the ghost matrix is $A_k(t) = (1/k_e)\big[G\partial_\vartheta^2\big(r_{\Theta^*(w)}(x_m(t))\tilde\Phi(x_m(t);\vartheta_k^*)\big) - \sigma_0 I\big] + O(k_e^{-2})$: a T_0-periodic matrix of size $O(1/k_e)$ whose period mean is $(1/k_e)(GR_k - \sigma_0 I) + O(k_e^{-2})$, with $R_k = R[\mu_{\Theta^*(w)}](\vartheta_k^*)$ the residual-weighted curvature of Theorem D‴. First-order averaging of this linear periodic equation gives $M_k = \exp(T_0(1/k_e)(GR_k - \sigma_0 I)) + O(k_e^{-2})$, and $GR_k - \sigma_0 I$ is similar to $G^{1/2}R_kG^{1/2} - \sigma_0 I = -(\sigma_0 I - R_k^G)$: the exponents are $(1/k_e)\lambda_j(\sigma_0 I - R_k^G) + O(k_e^{-2})$, with no condition on m. □

Status. Proved (linear algebra along the orbit; first-order averaging of the ghost equation for the last claim, which needs only that Γ_K is Theorem J's orbit of (CL_w) — referee round 12 confirmed the finite-difference ghost monodromy against the matrix ODE to $3\cdot10^{-9}$ and against $\exp(T_0\bar A_k)$ to $7\cdot10^{-9}$, the prediction's remaining error being R at $\Theta^*(w)$ against R along $\Theta_p(t)$, falling 6–12 times from $k_e = 25$ to 50). The m-uniformity of the Floquet spectrum settles the spectral part of Corollary D‴.7's open point: the stability type and the rates of Theorem J(b)–(c) at a replicated configuration do not depend on m. Checked: `tests/test_ot_leak.py` (the lift of the 3-atom orbit to $m = 6$ at the matching masses is a fixed point of the Poincaré map to 10^{-10}; each class's separation subspace is invariant under its monodromy to 10^{-6} and carries the ghost's exponents, $\sigma_0/k_e = 2.83\cdot10^{-4}$, to 10^{-6}; the complement carries the 3-atom loop's exponents to 10^{-6}).

Numerics (`exp_lifted_orbit.py` → `results/chaos/lifted_orbit.md`). The real loop at $k_e = 25$, $\sigma = \sigma_0/k_e$; monodromy matrices by central differences of the Poincaré map (DOP853, rtol 10^{-11}). (A) The OT leak at the matching masses: the 3-atom orbit is the exact representation with zero tracking error, and its lifts to $m = 6$, 9, 15 are fixed points of the Poincaré map to 10^{-15}; every class's separation subspace is invariant to $2 \cdot 10^{-8}$ and carries the exponent $\sigma_0/k_e = 2.830 \cdot 10^{-4}$ with multiplicity $n_k - 1$ — the ghosts' exponent to $4 \cdot 10^{-5}$ relative — and the collective exponents $2.935 \cdot 10^{-4}$, $3.558 \cdot 10^{-4}$, $3.999 \cdot 10^{-4}$, $8.73 \cdot 10^{-4}$, $1.109 \cdot 10^{-3}$, $1.573 \cdot 10^{-3}$, $3.704 \cdot 10^{-3}$, $5.523 \cdot 10^{-3}$, $7.934 \cdot 10^{-3}$ are the same for $m = 3$, 6, 9 and 15 to 10^{-4} relative, within 0.6–3.2% of Theorem J(b)'s $\lambda_j(H_q)/k_e = 2.952 \cdot 10^{-4}, \ldots, 8.091 \cdot 10^{-3}$, H_q the whitened, weighted collective block of Proposition D‴.5(i). (B) The σ-modification at $w = (7, 5, 4)/16$ (admissible, splitting margin +0.00257) and $(9, 4, 3)/16$ (a saddle, −0.00219): the weighted 3-atom orbits by shooting (residuals 10^{-14}, period-start offsets from $\Theta^*(w)$ of $1.1 \cdot 10^{-2}$ and $1.6 \cdot 10^{-2}$ — Theorem J(a)'s $O(1/k_e)$), and their lifts to $m = 16$ fixed points to $5 \cdot 10^{-14}$. The separation exponents of each class equal the ghost's to $2 \cdot 10^{-5}$ relative (invariance defects $\leq 6 \cdot 10^{-8}$) and the prediction $(\sigma_0 - \lambda_j(R_k^G))/k_e$ within 0.2–4.1% (class 0 at $(7, 5, 4)/16$: $9.989 \cdot 10^{-5}$, $4.553 \cdot 10^{-4}$, $1.267 \cdot 10^{-3}$ against $1.029 \cdot 10^{-4}$, $4.572 \cdot 10^{-4}$, $1.275 \cdot 10^{-3}$); at $(9, 4, 3)/16$ the over-weighted class carries the negative exponent $-9.13 \cdot 10^{-5}$ (predicted $-8.77 \cdot 10^{-5}$) with multiplicity 8 — the saddle orbit of Theorem D‴'s inadmissible seed, Proposition K's split at the orbit level — while the collective exponents $(2.131 \cdot 10^{-4}, \ldots, 9.985 \cdot 10^{-3})$ are the weighted 3-atom orbit's to $2 \cdot 10^{-5}$ relative.

Remark D‴.16 (dither and the mass constraint)**.** With dither (Theorem J′: $\tau = \tau_0/k_e$, $\sigma = \sigma_0/k_e$) the leading-order stationary law of the slow variables is the Gibbs measure $\pi \propto \exp(-J_m/\tau_0)$, and the cluster masses are no longer conserved under the nearest-atom leak: a particle crosses a Voronoi ridge at the Arrhenius rate $\exp(-\Delta V/\tau_0)$, ΔV the single-particle barrier out of the configuration — at least the leak's part $\sigma_0 \Delta_\nu^2/8 = (\sigma_0/2)(\Delta_\nu/2)^2$ at the lowest ridge point, plus the fit's variation — and the counts sample the Gibbs law over the count vectors n: in the Laplace approximation multinomial$(n) \cdot \exp(-m\mathcal{G}^*_{n/m}/\tau_0)$ times the determinants of the separation and collective blocks of Proposition D‴.5(i), for $n/m \in W_{\mathrm{adm}}$ (at a saddle weight the split configuration lies lower, and its soft and unstable separation modes carry entropy that the quadratic approximation misses). Dither thus does on the Arrhenius time scale, and only in distribution, what the OT leak does deterministically in $\ln 2/\sigma_0$ slow units at $\tau_0 = 0$. Under the OT leak the Gibbs measure is $\pi^{\mathrm{OT}} \propto \exp(-J_m^{\mathrm{OT}}/\tau_0)$ (J_m^{OT} is locally Lipschitz, piecewise smooth, with Gaussian tails), the plan's masses are q by construction, and a particle that crosses a Voronoi ridge stays assigned to its own atom: with integral plans an exchange of the assignments of particles $a \in$ class j and $b \in$ class k lowers the cost only once $\langle \vartheta_a - \vartheta_b, \vartheta_k^0 - \vartheta_j^0 \rangle_{G^{-1}} > 0$, that is, once they have passed each other. So the counts by nearest atom leave q at the nearest-atom leak's ridge barrier, but only for a reversible excursion — beyond the ridge the leak toward the particle's own atom keeps pulling it back — of stationary probability of order $\exp(-\Delta V_{\mathrm{ridge}}/\tau_0)$; exchanges of assignments (whose barrier is at most the lowest tie point of two particles) leave the masses at q as well. The mass constraint does not raise the barrier of a crossing; it makes the crossing reversible.

Status. Formal (the Gibbs law of a locally Lipschitz potential with Gaussian tails; the Arrhenius estimate; the Laplace asymptotics).

Numerics (`exp_dither_ot.py` → `results/chaos/dither_ot.md`): the dithered averaged slow loop — Langevin in slow time (Euler–Maruyama, $\mathrm{d}s = 0.05$), with the Gibbs law $\exp(-J_m/\tau_0)$ — for $m = 15$ particles started at the counts $(6, 5, 4)$ under the matching prior at $\sigma_0 = 0.15\sigma_{\mathrm{crit}}$, at $\tau_0 = 0.005$, 0.01, 0.02 over 50 000, 200 000 and 50 000 slow units, two seeds. On the instance the single-particle barriers out of the balanced configuration — the lowest points of the Voronoi ridges with the fourteen other particles relaxed — are 0.040–0.074 (the leak's part 0.035), and the lowest exchange barrier between two classes is 0.083 (referee round 13).

Under the nearest-atom leak the record-level label changes (every 10 slow units; about half of them undone at the next change, recrossings) occur at $1.0 \cdot 10^{-5}$, $5.1 \cdot 10^{-4}$, $5.6 \cdot 10^{-3}$ per slow unit — an Arrhenius barrier of 0.048 (0.043–0.052 at 95%) from $\tau_0 = 0.01$ and 0.02, among the ridge minima, while $\tau_0 = 0.005$ has one event in 100 000 slow units — so at $\tau_0 = 0.005$ the counts $(6, 5, 4)$ stand for $350/\sigma_0$, and at $\tau_0 = 0.01$ and 0.02 the counts sample a distribution whose weight on $(5, 5, 5)$ is 0.54 and 0.25, against 0.58 and 0.33 for the energy–multinomial law $\mathrm{multinomial}(n) \cdot \exp(-m\mathcal{G}_n^*/\tau_0)$ — a comparison that cannot discriminate at $\tau_0 = 0.01$ (an integrated autocorrelation time of 7 400–23 000 slow units leaves 9–27 effective samples per seed, a standard error of about 0.1) and puts the law 2.1 standard errors above the data at $\tau_0 = 0.02$; the law under-weights the saddle count vectors $(6, 4, 5)$ and $(6, 5, 4)$ by 3.2 and 2.7 times (total-variation distance 0.20, against 0.07 between the two seeds): their soft and unstable separation modes carry the excess, the very factors the law omits. Under the OT leak the counts by nearest atom are balanced in every record after the initial transit at $\tau_0 = 0.005$ (no excursion in 50 000 slow units; 99.6% of the whole run), and in 99.4% and 88% of the records after it at $\tau_0 = 0.01$ and 0.02; the excursions are returns of the crossing particle — all 121 at $\tau_0 = 0.01$, 298 of 336 at 0.02, the other 38 exchanges — the label changes are about as frequent as the nearest-atom leak's (ratios 1.18–1.30), and the off-balance fraction ($6.3 \cdot 10^{-3}$, 0.117) has an effective barrier of 0.058, inside the ridge range.

Proposition D‴.17 (convergence of the OT dynamics)**.** *Let J_m^{OT} be the energy of Proposition D‴.12 for the saturated Gaussian features, with ρ the finite quadrature of the code or the period average of an analytic reference, on the a priori ball of Step 0.*

(i) (a Lyapunov function with a semiconcave part) $J_m^{\mathrm{OT}} = m\bar{L}(\mu_\Theta) + (\sigma_0 m/2) \min_\gamma \sum_{ij} \gamma_{ij} \|\vartheta_i - \vartheta_j^0\|^2_{G^{-1}}$ *is the minimum over the finitely many vertices γ of the transportation polytope of the smooth energies J_m^γ, so it is locally Lipschitz, the sum of a smooth function and a semiconcave one (a quadratic in Θ minus a maximum of linear functions), and its Clarke subdifferential at Θ is the convex hull of the gradients $\nabla J_m^\gamma(\Theta)$ of the active vertices. The vertex-selection solutions of $(\mathfrak{A}_m^{\mathrm{OT}})$ solve the differential inclusion $\mathrm{d}\Theta/\,\mathrm{d}s \in -G_m \partial J_m^{\mathrm{OT}}(\Theta)$; along them J_m^{OT} is non-increasing, with $\mathrm{d}J_m^{\mathrm{OT}}/\,\mathrm{d}s = -\|\nabla J_m^\gamma(\Theta)\|^2_{G_m}$ wherever the plan γ is unique, and at a tie the directional derivative along either active plan's drift is at most $-\|\nabla J_m^\gamma\|^2_{G_m}$.*

(ii) (convergence) J_m^{OT} *is definable in the o-minimal structure $\mathbb{R}_{\mathrm{an,exp}}$ on the a priori ball — finitely many compositions of polynomials, exponentials and the saturations* tanh *and* sig*, a finite quadrature sum (or an analytic function of Θ for the period average of an analytic reference), and a minimum over finitely many such functions; for the finite quadrature it is definable in $\mathbb{R}_{\mathrm{exp}}$ on all of $\mathbb{R}^{mp}$, and for the period average of an analytic reference in $\mathbb{R}_{\mathrm{an}}$ on the ball — so it satisfies the Kurdyka–Łojasiewicz inequality in the nonsmooth sense [8]. The change of variables $Y = G_m^{-1/2}\Theta$ maps $\mathrm{d}\Theta/\,\mathrm{d}s \in -G_m \partial J$ into $\mathrm{d}Y/\,\mathrm{d}s \in -\partial(J \circ G_m^{1/2})(Y)$, the form in which the theorems of [8] are stated; definable locally Lipschitz functions obey the chain rule along absolutely continuous curves, so along a subgradient trajectory $\mathrm{d}J/\,\mathrm{d}s = -\|\,\mathrm{d}\Theta/\,\mathrm{d}s\|^2_{G_m^{-1}}$ [8], and with the KL inequality every bounded subgradient trajectory has finite length and converges to a Clarke-critical point. Every vertex-selection and every generalised solution of $(\mathfrak{A}_m^{\mathrm{OT}})$ is such a trajectory, bounded by Step 0's a priori estimate with $\mathrm{conv}\,\mathrm{supp}\,\nu$; hence it converges, and so do the solutions of $(\mathfrak{A}_m^\nu)$ (a minimum over the Voronoi assignments) and of $(\mathfrak{A}_m)$ (smooth: Łojasiewicz's gradient inequality for real-analytic functions, in Kurdyka's form for definable ones [50]).*

(iii) (the limit) *At a Clarke-critical point whose optimal plan is unique, the plan is a vertex and the point is a critical point of the frozen energy J_m^γ; for commensurate m the plan is integral (an assignment π) and the limit is a critical point of the frozen energy J_m^π, every particle leaking toward its assigned atom — $\Theta_m^*(q)$, the replication of another critical point*

of $\mathcal{G}_q$, or a split configuration. (iii) does not classify these limits: on the instance $\mathcal{G}_q$ has a second local minimum (fit error 0.37), whose replication to $m = 15$ has the diagonal plan as its unique optimal plan and is a saddle of J_{15}^{OT} — one of its separation blocks is indefinite although $q \in W_{\mathrm{adm}}$, the admissible set being defined at the branch $\Theta^*(q)$ (referee round 13). A trajectory that enters the basin of $\Theta_m^*(q)$ converges to it: the stationary statement of Proposition D‴.12(iii) is reached dynamically from every initial condition in that basin, with the limit's masses the plan's masses q.

Status. (i) proved (Danskin's theorem and the structure of a minimum of smooth functions; the energy decrease is checked along a run with reassigned particles, and the two-sided decrease at a constructed tie, in `tests/test_ot_leak.py`). (ii) proved given the imported Kurdyka–Łojasiewicz theorem for Clarke subgradient trajectories of definable functions [8] and the definability of restricted analytic and exponential functions [50]; the KL exponent is not quantified, so the statement carries no rate — the uniform-in-m rate is Theorem D‴'s question, which this does not answer. (iii) proved. Under the pure leak an integral plan (commensurate m) never switches once optimal: $\Theta(s) = (1 - e^{-\sigma_0 s})T + e^{-\sigma_0 s}\Theta(0)$, and the cost difference between two plans is affine in Θ (the quadratic terms cancel), so it is the convex combination of its value at $\Theta(0)$, nonnegative, and at the targets T, positive because an integral plan costs nothing there and every other plan costs something for distinct atoms. A split plan can switch even under the pure leak — its barycentric targets are not atoms, and it need not be optimal at them (referee round 13: switches in 216 of 400 random incommensurate instances, none in 200 commensurate ones). With the fit, cold starts on the instance switch the plan in 2 of 6 runs at $m = 15$ and in all 6 at $m = 16$ at the experiments' σ_0, and in none at ten times that leak — the test's run, which is therefore a smooth gradient flow; the nonsmooth part of (i) is tested at the static tie.

Theorem D‴.18 (Theorem J uniformly in m at a replicated configuration)**.** *Let* (CL) *be the real closed loop of Theorem J with a leak whose targets are class-constant and locally constant near the configurations considered (the σ-modification; the nearest-atom leak with every class atom inside its Voronoi cell; the OT leak at $w = q$ with $\varepsilon < \Delta_\nu/2$, Proposition D‴.12(iii)), and let $\mathcal{W}$ be a compact set of weight vectors with $\min_k w_k \geq w_{\min} > 0$ such that for every $w \in \mathcal{W}$ the weighted stationary point $\Theta^*(w)$ of $\mathcal{G}_w$ is a non-degenerate minimum with margins*

$$\lambda_{\mathrm{coll}}(w) := \lambda_{\min}(H_w) \geq \lambda_0 > 0, \qquad \lambda_{\mathrm{sep}}(w) := \min_k \lambda_{\min}(\sigma_0 I - R_k^G(w)) \geq \lambda_0 > 0,$$

H_w the whitened, weighted collective block and $\sigma_0 I - R_k^G(w)$ the separation blocks of Proposition D‴.5(i) (so $\mathcal{W} \subset W_{\mathrm{adm}}$). For a configuration $\Theta = (\vartheta_1, \dots, \vartheta_m)$ with a class map π_0 whose class sizes are $n_k = mw_k$, $w \in \mathcal{W}$, write $u_i := \vartheta_i - \vartheta^*_{\pi_0(i)}(w)$, $\bar{u}_k$ for the class means of u, $u_i^\perp := u_i - \bar{u}_{\pi_0(i)}$, and

$$|||u||| := \max\Big\{ \Big(\sum_k w_k \|\bar{u}_k\|_{G^{-1}}^2 \Big)^{1/2},\ \max_i \|u_i^\perp\|_{G^{-1}} \Big\},$$

$$\textit{so that} \qquad \max_i \|u_i\|_{G^{-1}} \leq (1 + w_{\min}^{-1/2})|||u||| \qquad \textit{and} \qquad |||u||| \leq 2 \max_i \|u_i\|_{G^{-1}}$$

*(the second constant is sharp up to the factor $1 - 1/\max_k n_k$). There are k_0, ρ, C, κ depending on the plant, the reference, the features, σ_0, G, $\mathcal{W}$, λ_0 and the leak's prior (through its cell and plan margins) — but **not on** m — such that for every m, every $w \in \mathcal{W}$ with $mw \in \mathbb{N}^{K_0}$, every class map π_0 with those class sizes and every $k_e \geq k_0$:*

(a) *the lift of the weighted K_0-atom orbit Γ_w (Corollary D‴.14) is the unique T_0-periodic orbit of* (CL) *whose parameters satisfy $\sup_t \max_i \|\vartheta_i(t) - \vartheta^*_{\pi_0(i)}(w)\|_{G^{-1}} \leq \rho/2$ and whose plant state stays in Theorem J's tube;*

(b) *every solution with $\max_i \|\vartheta_i(0) - \vartheta^*_{\pi_0(i)}(w)\|_{G^{-1}} \leq \rho/2$ and $\|x(0) - x_m(0)\| \leq R/2$ converges to it, and with $u(t)$ the deviation from the lifted orbit at the times $t = jT_0$,*

$$|||u(jT_0)||| \leq Ce^{-(1-\kappa)\min\{\lambda_{\mathrm{coll}}, \lambda_{\mathrm{sep}}\} jT_0/k_e} \big(|||u(0)||| + \|x(0) - x_p(0)\|/k_e \big);$$

(c) *its Floquet spectrum is Proposition D‴.15's — the K_0-atom orbit's and each class's ghost exponents with multiplicity n_k-1 — and the exponents are $(1/k_e)\lambda_j(H_w)$ and $(1/k_e)\lambda_j(\sigma_0 I - R_k^G(w))$ up to $O(k_e^{-2})$, uniformly in m.*

In particular, for $k_e \geq k_0$ Corollary D‴.14(ii)'s proviso is replaced by a statement uniform in m: for every commensurate m the lift of Γ_q is the unique periodic orbit in the per-particle ball of radius $\rho/2$ around $\Theta_m^*(q)$, and it attracts that ball. (Theorem J's own orbit, unique in its unnormalised ball, coincides with the lift wherever Theorem J applies, but its threshold in that norm grows like $\sqrt{m}$, Remark D‴.19.) Corollary D‴.7's open point — the m-uniformity of Theorem J's thresholds — is settled in the per-particle norm.

Proof. Theorem J's proof with the norm $|||\cdot|||$ in place of the unnormalised $\|\cdot\|_{G_m^{-1}}$; what has to be checked is that every constant of its Steps 1–6 is independent of m in the new norm. Three structural facts do it.

(1) *The plant sees the configuration through the class means, up to second order in the separations.* $\hat{f}_\Theta = \sum_k w_k n_k^{-1} \sum_{i\in k} \tilde{\Phi}(\cdot\,;\vartheta_i)$ and $\sum_{i\in k} u_i^\perp = 0$, so

$$\hat{f}_\Theta = \sum_k w_k \tilde{\Phi}(\cdot\,;\bar{\vartheta}_k) + \tfrac{1}{2}\sum_k w_k n_k^{-1} \sum_{i\in k} \nabla^2\tilde{\Phi}(\cdot\,;\bar{\vartheta}_k)[u_i^\perp, u_i^\perp] + O(\max_i \|u_i^\perp\|^3)\,:$$

the first-order variation of the residual r_Θ, hence of the slaved error $e_\Theta = \varepsilon r_\Theta(x_m) + O(\varepsilon^2)$ of Steps 1–2, is a function of the class means alone, and the separations enter at second order with a coefficient bounded by $\sup\|\nabla^2\tilde{\Phi}\|$, whatever m. Steps 1–2 (the tube, the slaved error, the fast variable ζ) involve Θ through r_Θ and — in the ζ-forcing $-\partial_\Theta e_\Theta \cdot \dot{\Theta}$ and the remainder $\partial_e F \cdot \zeta$ — through the particles' drifts, which are $O(\varepsilon)$ per particle; by (1) and (2) their constants are uniform in m.

(2) *Each particle's drift depends on the others only through the plant.* The particle law is $\dot{\vartheta}_i = F(x, e, \vartheta_i) - \sigma(\vartheta_i - T_i)$: the right-hand side of particle i is a function of (x, e, ϑ_i) and of its target. So the frozen-Θ averaged drift $\langle H\rangle_i(\Theta) = -G\nabla\mathcal{V}[\mu_\Theta](\vartheta_i)$ and the antiderivative $u_i(\Theta, t)$ of Step 3 are functions of $(\vartheta_i, \hat{f}_\Theta)$, bounded and C^2 in ϑ_i with m-independent bounds, and their derivatives in ϑ_j $(j \neq i)$ are $O(1/m)$ and pass through $\hat{f}$ — that is, through the class means at first order. Writing the second-order remainder of the drift at the replicated configuration per particle, $N_i(u) = O(\|u_i\|^2) + O(\|u_i\|\cdot(1/m)\sum_j\|u_j\|) + O((1/m)\sum_j\|u_j\|^2)$ (the second derivatives $\partial^2_{\vartheta_i\vartheta_i}$ are $O(1)$, $\partial^2_{\vartheta_i\vartheta_j}$ and $\partial^2_{\vartheta_j\vartheta_j}$ for $j \neq i$ are $O(1/m)$; $\partial^2_{\vartheta_j\vartheta_l}$ for distinct $j, l \neq i$ vanishes in $\langle H\rangle$, whose residual is linear in the $\tilde{\Phi}$'s, and is $O(1/m^2)$ in the second-order term H_2, which contains products of $\hat{f}$), so the remainder is bounded in $|||\cdot|||$ by an m-independent multiple of $|||u|||^2$; the same holds for the near-identity transformations of Step 3 and for the $O(\varepsilon^3)$ remainders of (N), which are built from the same functions.

(3) *The linearisation at the replicated configuration is block diagonal with an m-independent spectrum, and contracts in $|||\cdot|||$.* By Proposition D‴.5(i) and (1)–(2), $D\langle H\rangle(\Theta_m^*(w))$ acts on each separation $u_i^\perp$ of class k separately, as the matrix $GR_k - \sigma_0 I$ — the same for every particle of the class, similar to $-(\sigma_0 I - R_k^G(w))$ whose eigenvalues are $\leq -\lambda_{\text{sep}}$ — and on the class means as minus the collective block, self-adjoint in the weighted norm $(\sum_k w_k\|\bar{u}_k\|^2_{G^{-1}})^{1/2}$ with eigenvalues $\geq \lambda_{\text{coll}}$; the two families do not mix. Hence $I + \varepsilon T_0 D\langle H\rangle(\Theta_m^*(w))$ has $|||\cdot|||$-operator norm $1 - \varepsilon T_0 \min\{\lambda_{\text{coll}}, \lambda_{\text{sep}}\}$ once εT_0 times the largest eigenvalue is below 1, the same bound for every m; by (2), $D\langle H\rangle$ is Lipschitz on the $|||\cdot|||$-ball around $\Theta_m^*(w)$ in that operator norm with an m-independent constant C_N, the constant used below (the remainder of (2) is then at most $(C_N/2)|||u|||^2$, by integration).

With (1)–(3), Step 4 runs as written in the norm $|||\cdot|||$ for P_Y (P_ζ is unchanged), giving the Lipschitz constant $1 - \varepsilon T_0 \min\{\lambda_{\text{coll}}, \lambda_{\text{sep}}\}(1-\kappa)$ of P_Y on the $|||\cdot|||$-ball of radius ρ, $\kappa = O(\rho + \varepsilon)$, with $\rho := \min\{\lambda_0/(4C_N)$, the cell and plan margins of the leak divided by $C(1 + w_{\min}^{-1/2})\}$ — the $|||\cdot|||$-ball of radius ρ reaches the per-particle distance $(1 + w_{\min}^{-1/2})\rho$, the solutions of (b) reach C times that, and the leak's targets must stay constant there; the coupling constants $\text{Lip}_\zeta P_Y = O(\varepsilon)$

and $\mathrm{Lip}_Y P_\zeta = O(\varepsilon^3)$ are m-independent by (1) and (2). Step 6's 2×2 Lipschitz matrix then makes P a contraction in the adapted product norm on that ball, uniformly in m: its fixed point is unique there, and it is the lift of Γ_w, which is a fixed point by Corollary D‴.14(i) and lies in the ball for $k_e \geq k_0$ because $\sup_t |||\vartheta^\Gamma(t) - \vartheta^*(w)||| = O(1/k_e)$ with an m-independent constant (it is the weighted K_0-atom distance, Theorem J(a) for (CL_w)). This is (a) and (b): the per-particle ball of radius $\rho/2$ lies in the $|||\cdot|||$-ball of radius ρ (since $|||u||| \leq 2\max_i \|u_i\|$), which gives the uniqueness set of (a) and the start set of (b), and the $|||\cdot|||$-ball of radius ρ lies in the per-particle ball of radius $(1 + w_{\min}^{-1/2})\rho$, which is what the margins have to cover. (c) is Proposition D‴.15 with its averaging argument, the ghost matrices A_k being m-independent. □

Remark D‴.19 (which norm)**.** In the unnormalised norm $\|\cdot\|_{G_m^{-1}}$ of Theorem J the basin radius ρ_0 is m-independent too — the Hessian of J_m has the m-independent spectrum of (3) and its Lipschitz constant in that norm is $O(1)$ — but a ball of fixed radius in that norm contains only configurations whose particles are within $\rho_0/\sqrt{m}$ of their atoms on average: it shrinks per particle. And Theorem J's threshold is not m-independent in that norm: the lifted orbit sits $\sqrt{m} \cdot O(1/k_e)$ away from $\Theta_m^*(w)$ there ($\sqrt{m}$ times the weighted K_0-atom distance), and so does the first averaging correction, so the implicit-function step needs $k_e \geq C\sqrt{m}/\rho_0$ — Corollary D‴.14(ii)'s proviso, and the actual reason for the per-particle norm. That norm is the right one because the dynamics is a mean-field one — each particle feels its own position and the class means — and it cannot be weakened to a mean-square (W_2-type) ball for the nearest-atom leak: a single particle pushed into another Voronoi cell changes the cluster weights, and the solution converges to the lift of different weights however small its mean-square displacement — for m large enough that the shifted weights are admissible. At small m they need not be (at $m = 15$ on the matching prior, one particle moved out of a class of five gives the weights $(6, 5, 4)/15$, whose lift is a saddle, numerics below); the solution then splits instead, and either way it does not converge to Γ_q. Under the OT leak the plan restores the masses (a transit, Proposition D‴.12(v)), and the basin at the level of the empirical measure is of a different shape: for the averaged loop Theorem D‴.21 below proves an m-independent energy basin, and its a-posteriori form covers configurations with a fixed fraction of the particles in other cells (the two sets are not nested, and D‴.18 concerns the real loop).

Status. Proved as an adaptation of Theorem J's proof, at Theorem J's level of detail: Steps 1–3 and 5 carry over with m-independent constants by the structural facts (1)–(2), and Steps 4 and 6 are rerun in the norm $|||\cdot|||$ with (3); the m-uniformity rests on the two-sided norm equivalence and on the mean-field structure of every map built in Steps 1–4. The inputs are those of Theorem J, Corollary D‴.14(i), Proposition D‴.15, the margins λ_0 of $\mathcal{W}$ and the leak's cell and plan margins. It replaces Corollary D‴.14(ii)'s proviso by an m-uniform statement and settles the spectral and the basin part of Corollary D‴.7's uniformity question in the per-particle norm. Checked: `tests/test_uniform_basin.py`, on the averaged loop at the prior with centres shifted by $+0.15$ at the matching masses, where the residual is not zero and $\lambda_{\mathrm{sep}} = 0.80\sigma_0$ (so the per-particle terms that vanish at the matching prior are exercised): the second-order remainder of the drift is bounded by $C_N|||u|||^2$ with $C_N < 0.1$ for random per-particle displacements at $m = 6$, 30, 150, 600, not growing with m; for a one-particle displacement the displaced particle's remainder is exactly $N_a + N_b/m$ with N_a the same for every m — non-zero (above 10^{-5}) here and zero (below 10^{-11}) at the matching prior — and every other particle's remainder is exactly $O(1/m)$; a separation perturbation (two particles of one class moved by $\pm v$) moves only those particles at first order, with the response $(GR_k - \sigma_0 I)v$, not the leak's $-\sigma_0 v$, at $m = 6$, 60, 300; and $|||u(s)||| \leq Ce^{-\mu s}|||u(0)|||$ at $\mu = \min\{\lambda_{\mathrm{coll}}, \lambda_{\mathrm{sep}}\}$ itself, with one constant $C \leq 1.05$ for $m = 6$, 30, 150.

Numerics (`exp_uniform_basin.py` → `results/chaos/uniform_basin.md`). The real loop at $k_e = 25$ ($\sigma_0 = 0.15\sigma_{\mathrm{crit}}$; RK4 plant with the control and the reference at the stages, Euler parameters, $\mathrm{d}t = 5 \cdot 10^{-3}$) for 8000 time units — 1273 periods, 320 slow units — with all systems integrated together. *The rates.* The exponents that test (b) and (c) are Floquet exponents (Part

F, by shooting). At the matching prior the 3-atom orbit's parameter exponents are $2.9355 \cdot 10^{-4}$, $3.5579 \cdot 10^{-4}$, $3.9986 \cdot 10^{-4}$, … against $\lambda_j(H_q)/k_e = 2.9521 \cdot 10^{-4}$, $3.6155 \cdot 10^{-4}$, $4.0755 \cdot 10^{-4}$, … (0.56–3.22%), and every ghost exponent is $2.830 \cdot 10^{-4} = \sigma_0/k_e$. At the prior with centres shifted by +0.15 the parameter exponents are within 0.57–3.15% of $\lambda_j(H_q)/k_e$ and the ghost exponents equal $(\sigma_0 - \lambda_j(R_k^G))/k_e$ to $3 \cdot 10^{-4}$ relative, the slowest $2.2701 \cdot 10^{-4}$ (class 2) against $\lambda_{\text{sep}}/k_e = 2.2696 \cdot 10^{-4}$. *The proof's radius* is small: a sampled lower bound $C_N \geq 0.092$ on the $|||\cdot|||$-Lipschitz constant of $D\langle H\rangle$ gives $\rho \leq 0.019$ at the matching prior and $\rho \leq 0.015$ at the shifted one, so the runs below, at per-particle radii 0.3–2.5, start 31–261 times outside the proved start radius $\rho/2$: the clean contraction there is evidence that the basin extends to the cell margins, which the theorem does not claim. (A) The matching prior at the matching masses, where the lift is the exact representation, a constant orbit: sizes $m = 3$, 15, 60, 240, every particle displaced from its class atom by ρ in a random direction of the G^{-1}-unit sphere, the same displacements under both leaks. Inside the cells ($\rho = 0.3$, 1.0, $2.5 < \Delta_\nu/2 = 3.13$) no particle leaves its cell and the two leaks give identical runs; $d_\infty = \max_i \min_k \|\vartheta_i - \vartheta_k^0\|_{G^{-1}}$ decays at the fitted rates $2.69 \cdot 10^{-4}$–$2.97 \cdot 10^{-4}$ for $m \geq 15$ against $\sigma_0/k_e = 2.83 \cdot 10^{-4}$ (ratios 0.95–1.05), with $C(T) := \max_{t \leq T} d_\infty(t) e^{\mu t}/d_\infty(0) = 1.03$–$1.29$ at $T = 8000$ — still growing at the horizon in the $m = 15$ runs at $\rho = 0.3$ and 1 (1.16 and 1.27), constant from $T = 2000$ on in the others. The $m = 3$ runs (one particle per class, no separation modes) are mixtures of collective modes: at $\rho = 1$ the end of the run is carried by the slowest one and the fitted rate is $2.948 \cdot 10^{-4}$ ($\lambda_{\text{coll}}/k_e = 2.952 \cdot 10^{-4}$), at $\rho = 2.5$ by the third (amplitudes 0.029, 0.024, 0.086; fitted $4.06 \cdot 10^{-4}$ against $4.08 \cdot 10^{-4}$), at $\rho = 0.3$ by all three ($3.48 \cdot 10^{-4}$). At $\rho = 4.5$ some particles start in another cell (1, 3, 13 and 55 for $m = 3$, 15, 60, 240). The nearest-atom leak ends at other weights — the counts $(2, 0, 1)$, $(6, 5, 4)$, $(23, 19, 18)$, $(87, 70, 83)$ — and not only through the initial misplacement: at $m = 60$ and 240 the counts change during the transient (in 3 and 12 records, the last at $t = 420$ and 285), and $m = 240$ starts at exactly q. Its clouds have not collapsed at the horizon (within-class rms spreads 0.24–0.66 for $m \geq 15$), but $W^2 = 12.3$, 2.81, 2.30, 1.92 is already that of the lifts of the realised weights for $m = 15$, 60, 240 (3.21, 2.35, 1.93): the cross-cluster transport of the weight mismatch. Two of these weight vectors are inadmissible — $(6, 5, 4)/15$ and $(23, 19, 18)/60$ have splitting margins -0.00348 and -0.00161, so their lifts are saddles — while $(87, 70, 83)/240$ is admissible ($+0.00167$) and $(2, 0, 1)/3$ empties a class (Newton finds no collapsed stationary point near the prior atoms). The OT leak ends at the counts q at every m (the last count change at $t = 2555$, 2695, 1610, 1445), with $W^2 = 0.010$–0.23 still falling at the leak's rate $2\sigma_0$ per slow unit for $m \geq 15$ (log-slopes -0.01411, -0.01408, -0.01417 against $2\sigma_0 = 0.01415$): the transit is over, and what remains is the leak's contraction. The plan relabels — 4, 15 and 53 particles end outside the cell of their initial class — so the OT leak restores the masses q, not the class map π_0. (C) Remark D‴.19's mechanism: the lift of q with one particle of class 0 moved 0.6 of the way to atom 1, a mean-square displacement of 0.311 ($m = 60$) and 0.078 ($m = 240$). Under the nearest-atom leak W^2 grows to 0.79 and 0.20, toward the lifts of the admissible weights $(19, 21, 20)/60$ and $(79, 81, 80)/240$ (W^2 0.8655 and 0.2163); under the OT leak it falls to $3.0 \cdot 10^{-3}$ and $8.2 \cdot 10^{-4}$. (B) A moving orbit: the OT leak toward the prior with centres shifted by +0.15 (a non-zero residual, so Γ_q moves and the tracking error along it is $2.55 \cdot 10^{-4}$), masses q; sizes 15, 60, 240 started at the lift of $\Theta^*(q)$ displaced by $\rho = 0.3$, 1.0 and measured against the lift of the 3-atom loop's trajectory. The second-half fits $2.18 \cdot 10^{-4}$–$2.63 \cdot 10^{-4}$ (ratios 0.96–1.16 to $2.27 \cdot 10^{-4}$) are mixtures of separation modes whose gaps exceed the fit window; they are consistent with $(1 - \kappa) \min\{\lambda_{\text{coll}}, \lambda_{\text{sep}}\}/k_e$, and the Floquet exponents above are the test of the rate. $C(8000) = 1.04$–1.23 (still growing at the horizon at $m = 15$, $\rho = 0.3$), and every m-system's tracking error is 2.55–$2.56 \cdot 10^{-4}$ at the end.

Proposition D‴.20 (the entropic OT leak: a soft mass constraint with a critical excess)**.** *Let* $\nu = \sum_j q_j \delta_{\vartheta_j^0}$, $\varepsilon > 0$, *let* $\pi^\varepsilon(\Theta)$ *be the entropic plan from* μ_Θ *to* ν *for the cost* $\|\cdot\|_{G^{-1}}^2$ *— the unique minimiser of* $\langle C, \pi\rangle + \varepsilon\, \mathrm{KL}(\pi \mid \mu_\Theta \otimes \nu)$ *over the plans with marginals* $(1/m, q)$ *— and*

$T_i^\varepsilon := m\sum_j \pi_{ij}^\varepsilon \vartheta_j^0$ *its barycentric projection, and consider*

$$\frac{\mathrm{d}\vartheta_i}{\mathrm{d}s} = G\langle \nabla_\vartheta (r_\Theta \tilde{\Phi}(\cdot\,;\vartheta_i))\rangle_\rho - \sigma_0(\vartheta_i - T_i^\varepsilon). \tag{$\mathfrak{A}_m^\varepsilon$}$$

(i) (a smooth gradient flow between two leaks) ($\mathfrak{A}_m^\varepsilon$) *is the G-gradient flow of $J_m^\varepsilon := m\bar{L}(\mu_\Theta) + (\sigma_0 m/2)\mathrm{OT}_\varepsilon(\mu_\Theta,\nu)$ — by the envelope theorem $\partial\mathrm{OT}_\varepsilon/\partial\vartheta_i = (2/m)G^{-1}(\vartheta_i - T_i^\varepsilon)$ — and J_m^ε is real-analytic in Θ (the dual potentials are analytic in the cost: the semi-dual is strictly concave once the gauge $g_1 = 0$ is fixed, and the implicit function theorem applies), so there are no plan switches and Łojasiewicz's theorem gives the convergence of every trajectory — all are bounded, T_i^ε lying in* $\operatorname{conv}\operatorname{supp}\nu$ *and the fit's force being bounded — without the nonsmooth machinery of Proposition D‴.17. As $\varepsilon\to 0$, $T^\varepsilon \to \bar{T}$ wherever the OT plan is unique (Proposition D‴.12); as $\varepsilon\to\infty$, $\pi^\varepsilon \to \mu_\Theta\otimes\nu$ and $T_i^\varepsilon \to \sum_j q_j\vartheta_j^0$: the leak becomes the σ-modification toward the barycentre of ν.*

(ii) (collapsed configurations) *At a configuration collapsed by class with weights w the entropic plan is class-symmetric, $\pi_{ij}^\varepsilon = \pi^K_{\pi_0(i)j}/n_{\pi_0(i)}$ with π^K the $K\times K_0$ entropic plan from $\sum_k w_k\delta_{\vartheta_k}$ to ν, so every particle of class k has the class target $T_k = \sum_j(\pi^K_{kj}/w_k)\vartheta_j^0$, and the replicated configuration is a critical point of J_m^ε iff its atoms are a critical point of the weighted K-atom energy with the leak $(\sigma_0/2)\mathrm{OT}_\varepsilon(\sum_k w_k\delta_{\vartheta_k},\nu)$.*

(iii) (the separation block) *The Hessian of J_m^ε at such a configuration splits as in Proposition D‴.5(i), with separation blocks*

$$S_k^\varepsilon = \sigma_0\big(I - (2/\varepsilon)\operatorname{Cov}_k^G\big) - R_k^G, \qquad \operatorname{Cov}_k^G := G^{-1/2}\operatorname{Cov}_{\pi^K(\cdot|k)}(\vartheta^0)G^{-1/2},$$

each of multiplicity $n_k - 1$: a zero-sum perturbation of a class leaves the column marginals, hence the dual potentials, stationary to first order, and the conditional plan of a particle responds to its own displacement through its softmax, which moves its barycentric target by $(2/\varepsilon)\operatorname{Cov}_{\pi^K(\cdot|k)}(\vartheta^0)G^{-1}$ — a positive feedback against the leak's restoring $\sigma_0 I$.

(iv) (the critical excess) *When $w = q$ and $\varepsilon \ll \Delta_\nu^2$, the conditional plans charge the other atoms with exponentially small mass and S_k^ε is the OT leak's $\sigma_0 I - R_k^G$ up to exponentially small terms. Let class k carry an excess, and assume $A_k := \sigma_0 I - R_k^G \succ 0$ at $\Theta_\varepsilon^*(w)$ — the splitting condition of Proposition D‴.5, which fails at large excesses and small ε, where the class is a saddle for the fit's own reason. Then $S_k^\varepsilon = A_k - (2\sigma_0/\varepsilon)\operatorname{Cov}_k^G$ has a negative eigenvalue iff*

$$\varepsilon < 2\sigma_0\lambda_{\max}\big(A_k^{-1/2}\operatorname{Cov}_k^G A_k^{-1/2}\big),$$

twice σ_0 times the largest generalised eigenvalue of the conditional covariance relative to A_k. A_k and Cov_k^G depend on ε through $\Theta_\varepsilon^*(w)$ and its plan, so the critical regularisation ε_c is the root of $\lambda_{\min}(S_k^\varepsilon) = 0$ along the branch $\Theta_\varepsilon^*(w)$. When the conditional plan sends a fraction p to one atom j and $1-p$ to its own, $\operatorname{Cov}_k^G = p(1-p)\hat{\Delta}\hat{\Delta}^\top$ with $\hat{\Delta} := G^{-1/2}(\vartheta_j^0 - \vartheta_k^0)$, and the criterion is the rank-one $\varepsilon < 2\sigma_0 p(1-p)\hat{\Delta}^\top A_k^{-1}\hat{\Delta}$. Below ε_c the over-weighted collapsed class is a saddle and splits; its split-off particles leave toward an atom its conditional plan charges, which need not be an under-weighted one, so a split moves the excess but need not reduce it. Above ε_c the class's separation block is positive, and the replicated configuration is a local minimum when the collective block and the other classes' blocks are positive as well; the excess is then absorbed by a bias of the class atom toward T_k.

(v) (the floor at large m) *For fixed ε the critical excess $\delta_c(\varepsilon)$ is positive. At small ε and $R\approx 0$ the critical conditional fraction is $p_c(\varepsilon)\approx \varepsilon/(2\|\hat{\Delta}\|^2)$, and the plan then sends away exactly the excess, so $\delta_c \approx w_k p_c$. The sampled excess of an i.i.d. warm start has the standard deviation $(q_k(1-q_k)/m)^{1/2}$, so the entropic leak equalises the cluster masses only down to the critical excess, and for $m \gg ((1-q_k)/q_k)(2\|\hat{\Delta}\|^2/\varepsilon)^2$ it does not equalise them at all.*

The masses are then conserved, as under the nearest-atom leak, and partly absorbed by the class biases: the stationary measure depends on the sample again, and the floor returns as Proposition D‴.5's law at the entropic stationary point. Only the hard constraint $\varepsilon \to 0$ removes the floor — on the commensurate sizes; on the others its $K_0 - 1$ bridge particles leave a floor of order m^{-2} (Proposition D‴.12's numerics).

Status. (i) proved (the envelope theorem; the analyticity of the entropic potentials; the limits are the convergence of the entropic plan to an optimal plan as $\varepsilon \to 0$ — the optimal plan of maximal entropy where it is not unique [21] — and the product-measure limit as $\varepsilon \to \infty$). (ii) proved (uniqueness of the entropic plan and the class symmetry). (iii) proved (the first-order stationarity of the potentials under zero-sum class perturbations and the derivative of the conditional softmax); checked against the finite-difference Hessian of J_m^ε at $m = 15$ in `tests/test_entropic_leak.py`. (iv) proved under the hypothesis $A_k \succ 0$ (the generalised-eigenvalue criterion and its rank-one form); the reading of a split as a transport of the excess is not part of it. (v) formal (the scaling of the sampled excess and the identification of the dynamics' end state); numerical below. The structure is that of the critical temperature of deterministic annealing [77], with the same roles — the particles as the codevectors, the prior's atoms as the data, ε as the temperature — and twice the largest eigenvalue of the conditional covariance when $R = 0$. New are the two-sided mass constraint, under which the conditional covariance of a collapsed class is an effect of its mass excess rather than of a data spread, and the fit's curvature in A_k. Checked: `tests/test_entropic_leak.py` (the semi-dual Newton plan's marginals and its agreement with log-domain Sinkhorn; the gradient of J_m^ε against finite differences; the two limits at their rates; the replicated configuration critical with a class-symmetric plan, and the finite-difference Hessian leaving each class's separation subspace invariant and acting there as $I \otimes (\sigma_0(I - (2/\varepsilon)\operatorname{Cov}_k^G) - R_k^G)$, entrywise; the over-weighted class of $(6, 5, 4)/15$, with $A_1 \succ 0$ checked, unstable at $\varepsilon = 16$, 17 and stable at $\varepsilon = 18$, 20, 64; every weighted stationary point to a Newton residual of 10^{-9}).

Numerics (`exp_entropic_leak.py` → `results/chaos/entropic_leak.md`). The instance with the matching prior (masses 1/3) at $\sigma_0 = 0.15\sigma_{\text{crit}}$; the entropic plan by Newton's method on the semi-dual; every weighted stationary point continued from a converged neighbour and checked to a residual of 10^{-9}. *The critical excess.* For $w_\delta = q + \delta(1.5e_1 - 0.5)$ (the excess of class 1 taken from both other atoms) the branch $\Theta_\varepsilon^*(w_\delta)$, continued in ε from 256 down to 1 (residuals $\le 8.7 \cdot 10^{-13}$), changes the sign of the over-weighted class's margin once, at $\varepsilon_c = 5.11$, 13.5, 23.1, 65.0 for $\delta = 0.02$, 0.05, 0.1, 0.2. There $A_1 \succ 0$ ($\lambda_{\min}$ 0.0054, 0.0050, 0.0051, 0.0020), the collective block is positive (0.0073–0.0078) and so are the other classes' margins: above ε_c the replicated configuration is a local minimum. At ε_c the class sends the fraction $p = 0.057$, 0.15, 0.33, 0.57 of its mass away, about equally to the two under-weighted atoms (conditional plans $(0.9434, 0.0277, 0.0289)$, $(0.8485, 0.0764, 0.0751)$, $(0.6655, 0.1850, 0.1496)$, $(0.4257, 0.3074, 0.2669)$), so Cov_1^G has rank two; the exact threshold $2\sigma_0\lambda_{\max}(A_1^{-1/2}\operatorname{Cov}_1^G A_1^{-1/2})$ equals ε_c to the printed digits, and the rank-one formula with all of p sent to one atom gives 1.22–1.31 times ε_c with atom 2 and 0.70–1.22 with atom 3. With the excess taken from atom 3 alone ($w'_\delta = q + \delta(e_1 - e_3)$) $\varepsilon_c = 6.99$, 14.6, 24.3 for $\delta = 0.02$, 0.05, 0.1. At fixed ε the critical excess is $\delta_c = 0.0042$, 0.0083, 0.016, 0.030, 0.062, 0.14, 0.20 at $\varepsilon = 1$, 2, 4, 8, 16, 32, 64 (w_δ; 0.0035–0.18 for w'_δ). Up to $\varepsilon = 8$ it equals $w_1 p_c$, with p_c within 12% (w_δ) and 26% (w'_δ) of the small-ε scale $\varepsilon/(2\|\hat{\Delta}\|^2)$, as (v) says; from $\varepsilon = 16$ on the plan no longer sends away exactly the excess. *Dynamics.* $m = 15$, 60, 240 against $M = 510$ from the warm clouds of Proposition D‴.12's numerics, four seeds, $\varepsilon = 1$, 4, 16, 64, 600 slow units, and the $\varepsilon = 4$, 16, 64 runs continued to 6000 slow units. At $\varepsilon = 1$ every system equalises its masses (all sixteen at q by $s = 600$; fit errors 1.7–$3.0 \cdot 10^{-4}$; function-level distances to the reference $3.7 \cdot 10^{-9}$–$1.6 \cdot 10^{-6}$ per seed on $(450, 600]$, values at the stopping time, still falling): the OT leak's behaviour. At $\varepsilon = 4$ the masses move toward q slowly: at $s = 600$ five of the twelve larger systems sit at unstable weights (margins -0.00035 to -0.00146 at $\Theta_4^*(w)$, e-folding times $1/|\text{margin}|$ of 700–2900 slow units, longer than the run).

Continued, every system ends at stable weights (positive margins at its $s = 6000$ weights), the larger ones with residual excesses up to 1/60, 3/240 and 7/510 ($\|w - q\|_\infty = 0.0167$, 0.0125, 0.0137), of the order of the critical excess at $\varepsilon = 4$ (0.016 for an excess taken from both other atoms, 0.012 from one). At $\varepsilon = 16$ the over-weighted $(8, 4, 3)/15$ reaches $(6, 6, 3)$ within 600 slow units, and in the continuation the three systems at unstable weights move: $(6, 5, 4)/15$ to $(5, 6, 4)$ at $s \approx 3700$ — the split-off particle joins atom 2, which was at q, so the excess is moved, not reduced — $(6, 6, 3)/15$ to $(5, 6, 4)$ as well, and $(25, 17, 18)/60$ to $(23, 19, 18)$; every other system keeps its sampled weights, which are stable. At $\varepsilon = 64$ only $(8, 4, 3)/15$ moves (to $(7, 5, 3)$, at $s \approx 1400$). The floors return as Proposition D‴.5's law at the entropic stationary point $\Theta^*_\varepsilon(q)$, 0.55 ($\varepsilon = 16$) and 0.74 ($\varepsilon = 64$) of the nearest-atom leak's law; the observed floors on the same clouds are 0.23–0.84 ($\varepsilon = 16$) and 0.47–0.91 ($\varepsilon = 64$) of the nearest-atom leak's, per seed; at $\varepsilon = 64$ the bias toward the barycentre of ν dominates the fit (fit errors 0.12–0.14). The soft constraint thus interpolates in two ways at once: in ε between the OT leak and the σ-modification toward the barycentre, and in m between equalisation (small m, large sampled excesses) and conserved sampled masses with a floor of the nearest-atom kind, smaller by the factor of the law (large m).

Theorem D‴.21 (a measure-level basin for the OT leak: a Polyak–Łojasiewicz inequality uniform in m)**.** *Let $\nu = \sum_j q_j \delta_{\vartheta^0_j}$ and let $\mu^* = \sum_j q_j \delta_{\vartheta^*_j}$ be a replicated stationary measure of $(\mathfrak{A}^{\mathrm{OT}}_m)$ — every atom critical for its class potential, $\nabla V_L[\mu^*](\vartheta^*_j) + \sigma_0 G^{-1}(\vartheta^*_j - \vartheta^0_j) = 0$, and the diagonal plan optimal ($\varepsilon < \Delta_\nu/2$ suffices, Proposition D‴.12(iii)) — with residual r^*. Put $\mathcal{J}(\mu) := \bar{L}(\mu) + (\sigma_0/2) W^2_{G^{-1}}(\mu, \nu)$, so that $J^{\mathrm{OT}}_m(\Theta) = m\mathcal{J}(\mu_\Theta)$, and $\mathcal{J}^* := \mathcal{J}(\mu^*)$. Let m be commensurate; every vertex optimal plan is then an assignment π with $|\pi^{-1}(j)| = mq_j$ (Proposition D‴.12(ii)), $J^{\mathrm{OT}}_m = \min_\pi J^\pi_m$ over these assignments with $J^\pi_m(\Theta) := m\bar{L}(\mu_\Theta) + (\sigma_0/2) \sum_i \|\vartheta_i - \vartheta^0_{\pi(i)}\|^2_{G^{-1}}$, and for a configuration Θ and such a π write*

$$u_i := \vartheta_i - \vartheta^*_{\pi(i)}, \qquad D_\pi := \frac{1}{m} \sum_i \|u_i\|^2_{G^{-1}}, \qquad \Delta\hat{f} := \hat{f}_\Theta - \hat{f}[\mu^*], \qquad E := \mathcal{J}(\mu_\Theta) - \mathcal{J}^*,$$

and, along the plan segments $S_i := \{\vartheta^*_{\pi(i)} + t u_i : t \in [0, 1]\}$,

$$\kappa^* := \max_i \sup_{S_i} \lambda_{\max} R^G[\mu^*], \quad \kappa^*_- := \max_i \sup_{S_i} \big(-\lambda_{\min} R^G[\mu^*]\big)_+, \quad \kappa^\Theta := \max_i \sup_{S_i} \lambda_{\max} R^G[\mu_\Theta].$$

(i) (two identities along the plan segments) *For every assignment π with the counts mq, optimal or not,*

$$J^\pi_m(\Theta)/m - \mathcal{J}^* = \tfrac{1}{2}\|\Delta\hat{f}\|^2 + (\sigma_0/2) D_\pi - \frac{1}{m} \sum_i \int_0^1 (1 - t) u_i^\top R[\mu^*](\vartheta^*_{\pi(i)} + t u_i) u_i \, \mathrm{d}t, \tag{E}$$

$$\frac{1}{m} \sum_i \langle \nabla_{\vartheta_i} J^\pi_m(\Theta), u_i \rangle = \|\Delta\hat{f}\|^2 + \sigma_0 D_\pi - \frac{1}{m} \sum_i \int_0^1 u_i^\top \big[(1 - t) R[\mu^*] + t R[\mu_\Theta]\big](\vartheta^*_{\pi(i)} + t u_i) u_i \, \mathrm{d}t, \tag{P}$$

the norms in $L^2(\rho)$. The counts enter through $\Delta\hat{f} = (1/m) \sum_i [\tilde{\Phi}(\cdot\,; \vartheta_i) - \tilde{\Phi}(\cdot\,; \vartheta^*_{\pi(i)})]$, which holds because $|\pi^{-1}(j)| = mq_j$, and through the constant term of the leak. (E) *holds for every coupling of μ_Θ with the column masses q, fractional plans included ($(1/m)\sum_i$ replaced by $\sum_{ij} \gamma_{ij}$);* (P)*, with one test direction per particle, needs an assignment. With counts $N \neq mq$ both acquire a defect — $\sum_k (N_k/m - q_k)\big(-\langle r^*, \tilde{\Phi}(\cdot\,; \vartheta^*_k)\rangle + (\sigma_0/2)\|\vartheta^*_k - \vartheta^0_k\|^2_{G^{-1}}\big)$ in* (E) *and $-\langle \Delta\hat{f}, \hat{f}[\mu_N] - \hat{f}[\mu^*]\rangle$ in* (P)*, $\mu_N := \sum_k (N_k/m) \delta_{\vartheta^*_k}$ — and at incommensurate m no PL inequality relative to $\mathcal{J}^*$ can hold: the stationary configurations of Proposition D‴.12(iv), with their bridge particles, are critical at a positive excess energy.*

(ii) (the Polyak–Łojasiewicz inequality) *For every assignment π with the counts mq, $\frac{1}{2}\|\Delta\hat{f}\|^2 + ((\sigma_0-\kappa^*)/2)D_\pi \le J_m^\pi(\Theta)/m - \mathcal{J}^* \le \frac{1}{2}\|\Delta\hat{f}\|^2 + ((\sigma_0+\kappa_-^*)/2)D_\pi$ (the lower bound is informative when $\kappa^* \le \sigma_0$). If π is optimal at Θ, so that the middle term is E, and the plan-segment margin $\alpha := \sigma_0 - (\kappa^* + \kappa^\Theta)/2$ is positive, then*

$$\|\nabla J_m^\pi(\Theta)\|_{G_m}^2 \ge \lambda \cdot mE, \qquad \lambda := \alpha \cdot \min\{2, 2\alpha/(\sigma_0+\kappa_-^*)\}.$$

(iii) (a-posteriori decay) *Along every vertex-selection solution of $(\mathfrak{A}_m^{\mathrm{OT}})$ (Section 13, Proposition D‴.17(i): an absolutely continuous solution that follows an optimal integral plan $\pi(s)$ for almost every s; their existence is assumed, as in Section 13), E is non-increasing and*

$$E(s) \le E(0)\exp\Big(-\int_0^s \lambda(\sigma)_+ \,\mathrm{d}\sigma\Big),$$

with $\lambda(\sigma)$ computed at $\Theta(\sigma)$ with its plan ($\lambda_+ := 0$ where $\alpha \le 0$); where $\kappa^* < \sigma_0$, moreover $W^2(\mu_{\Theta(s)}, \mu^*) \le D_{\pi(s)} \le 2E(s)/(\sigma_0-\kappa^*)$ and $\|\hat{f}_{\Theta(s)} - \hat{f}[\mu^*]\|^2 \le 2E(s)$.

(iv) (an a-priori basin, uniform in m) *Assume* (H*)*: $\kappa_R[\mu^*](B) \le \sigma_0 - \kappa$ for some $\kappa > 0$ on a convex set B, forward-invariant for $(\mathfrak{A}_m^{\mathrm{OT}})$ and containing the stationary atoms — for instance the whole parameter space, or the a-priori invariant set $N_{R_\Theta}(\operatorname{conv}\operatorname{supp}\nu)$ of Proposition D‴.17 ($\bar{T}_i \in \operatorname{conv}\operatorname{supp}\nu$ and the fit's force is bounded). Then for every commensurate m, $J_m^{\mathrm{OT}} \ge m\mathcal{J}^*$ on B^m, with equality exactly at the replicated configurations (μ^* is the unique global minimiser among the m-point empirical measures in B, up to the labelling); and every vertex-selection solution from B with $E(0) \le E_0 := \kappa^2/(2(L_2^G)^2)$ satisfies $E(s) \le E(0)e^{-\lambda_0 s}$, $\lambda_0 := (\kappa/2)\min\{2, \kappa/(\sigma_0 + \sup_B\|R^G[\mu^*]\|)\}$. The constants depend on κ, σ_0, L_2^G and r^* — not on m.*

(v) (the matching prior; sharpness) *At the matching prior ($r^* = 0$ on $\operatorname{supp}\rho$, $\mu^* = \nu$), $R[\mu^*] \equiv 0$,* (H*) *holds on the whole parameter space with $\kappa = \sigma_0$, $E = \bar{L}(\mu_\Theta) + (\sigma_0/2)W^2(\mu_\Theta, \nu)$, $\alpha = \sigma_0 - \kappa^\Theta/2$ and $\lambda = \alpha \cdot \min\{2, 2\alpha/\sigma_0\}$, which tends to $2\sigma_0$ as $\kappa^\Theta \to 0$. The constant $2\sigma_0$ is attained by the separations (each particle's displacement within its class decays at σ_0, the separation blocks of Theorem D‴.18), so (ii) is sharp there.*

(vi) (what the mass constraint removes) *For the nearest-atom leak the assignment is the Voronoi one, $j(\cdot)$, whose counts N are conserved inside the cells (Proposition D‴.12(i)) and differ from mq in general. The derivation of* (E)–(P) *breaks at the counts: $(1/m)\sum_i \tilde{\Phi}(\cdot\,; \vartheta^*_{j(\vartheta_i)}) = \hat{f}[\mu_N]$ with $\mu_N := \sum_k (N_k/m)\delta_{\vartheta_k^*}$, not $\hat{f}[\mu^*]$. And the nearest-atom energy has a critical point at the replicated configuration of the weights N/m — a strict local minimum when they are admissible, a saddle when a separation margin is negative — for every count vector N in the domain of the weighted stationary point $w \mapsto \Theta^*(w)$ (a neighbourhood of q, Proposition D‴.5(i)) whose atoms lie in their own cells, at a positive excess energy when $N \ne mq$ (under* (H*)*, by (iv)). These spurious minima carry the floor of Proposition D‴.5. Under the mass constraint they are not critical points; under* (H*) *μ^* is the unique global minimiser in B and the only critical point where $\alpha > 0$ (in particular in $\{E \le E_0\}$). Local minima of J_m^{OT} at a positive excess energy are not excluded: by (ii) they need $\kappa^\Theta \ge 2\sigma_0 - \kappa^*$ on their segments.*

Proof. (i) The stationarity of the atoms reads $\sigma_0 G^{-1}(\vartheta_j^* - \vartheta_j^0) = \langle \nabla_\vartheta \tilde{\Phi}(\cdot\,; \vartheta_j^*)^\top r^* \rangle_\rho$. In $J_m^\pi/m - \mathcal{J}^*$ the loss contributes $-\langle r^*, \Delta\hat{f}\rangle + \frac{1}{2}\|\Delta\hat{f}\|^2$ and the leak $(\sigma_0/2)D_\pi + (\sigma_0/m)\sum_i \langle u_i, \vartheta^*_{\pi(i)} - \vartheta^0_{\pi(i)}\rangle_{G^{-1}} = (\sigma_0/2)D_\pi + \langle r^*, (1/m)\sum_i \nabla_\vartheta \tilde{\Phi}(\cdot\,; \vartheta^*_{\pi(i)})u_i\rangle$. With the counts, $\Delta\hat{f} = (1/m)\sum_i[\tilde{\Phi}(\vartheta_i) - \tilde{\Phi}(\vartheta^*_{\pi(i)})]$, and Taylor's formula with integral remainder along S_i turns the two r^*-terms into $-(1/m)\sum_i \int_0^1 (1-t)u_i^\top R[\mu^*](\vartheta^*_{\pi(i)} + tu_i)u_i \,\mathrm{d}t$: this is (E). For (P), $\nabla_{\vartheta_i} J_m^\pi = -\langle \nabla_\vartheta \tilde{\Phi}(\cdot\,; \vartheta_i)^\top r_\Theta \rangle + \sigma_0 G^{-1}(\vartheta_i - \vartheta^0_{\pi(i)})$ with $r_\Theta = r^* - \Delta\hat{f}$, and the stationarity gives $\langle \nabla_{\vartheta_i} J_m^\pi, u_i\rangle = \sigma_0\|u_i\|_{G^{-1}}^2 -$

$\langle r^{*\top}(\nabla\tilde\Phi(\vartheta_i)-\nabla\tilde\Phi(\vartheta^*_{\pi(i)}))u_i\rangle+\langle\Delta\hat f^\top\nabla\tilde\Phi(\vartheta_i)u_i\rangle$. The second term is $-\int_0^1 u_i^\top R[\mu^*](\vartheta^*_{\pi(i)}+tu_i)u_i\,\mathrm{d}t$. Averaged over i, the third is $\|\Delta\hat f\|^2$ plus, by Taylor's formula for $\tilde\Phi(\vartheta^*_{\pi(i)})$ around ϑ_i, $(1/m)\sum_i\int_0^1 t\cdot u_i^\top R_{\Delta\hat f}(\vartheta^*_{\pi(i)}+tu_i)u_i\,\mathrm{d}t$, where $R_g:=\langle\sum_k g^k\nabla^2\tilde\Phi^k\rangle$ and $R_{\Delta\hat f}=R[\mu^*]-R[\mu_\Theta]$ because R is linear in the residual. Collecting the $R[\mu^*]$-terms gives (P).

(ii) The t-weights $1-t$ and t integrate to $\frac12$ each. By (E), $E=J_m^\pi/m-\mathcal{J}^*$ lies between $\frac12\|\Delta\hat f\|^2+((\sigma_0-\kappa^*)/2)D_\pi$ and $\frac12\|\Delta\hat f\|^2+((\sigma_0+\kappa^*_-)/2)D_\pi$, and by (P), $Q:=(1/m)\sum_i\langle\nabla_{\vartheta_i}J_m^\pi,u_i\rangle\ge\|\Delta\hat f\|^2+\alpha D_\pi$. Hence $Q\ge cE$ with $c:=\min\{2,2\alpha/(\sigma_0+\kappa^*_-)\}$, and $Q\ge\alpha D_\pi$. Cauchy–Schwarz in the duality of G and G^{-1}, $mQ\le\|\nabla J_m^\pi\|_{G_m}(mD_\pi)^{1/2}$, gives $\|\nabla J_m^\pi\|^2_{G_m}\ge mQ^2/D_\pi\ge m\alpha Q\ge m\alpha cE$.

(iii) $J_m^{\mathrm{OT}}=\min_{\pi'}J_m^{\pi'}\le J_m^{\pi(s)}$ with equality at $\Theta(s)$. At every s where $\Theta(\cdot)$ and $s\mapsto J_m^{\mathrm{OT}}(\Theta(s))$ are differentiable — almost every s, J_m^{OT} being locally Lipschitz — $\mathrm{d}J_m^{\mathrm{OT}}(\Theta(s))/\,\mathrm{d}s\le\langle\nabla J_m^{\pi(s)},\dot\Theta(s)\rangle=-\|\nabla J_m^{\pi(s)}\|^2_{G_m}\le-m\lambda(s)_+E(s)$, and Grönwall's lemma applies. The distance bounds are (E)'s lower bound and $W^2\le D_\pi$, the assignment being a coupling of μ_Θ and μ^*.

(iv) The segments lie in B by convexity, so $\kappa^*\le\sigma_0-\kappa$, and (E) gives $E\ge\frac12\|\Delta\hat f\|^2+(\kappa/2)D_\pi\ge0$, with equality iff $D_\pi=0$. Since $\|R^G[\mu_\Theta](\vartheta)-R^G[\mu^*](\vartheta)\|\le L_2^G\|\Delta\hat f\|_{L^1(\rho)}\le L_2^G(2E)^{1/2}$, $\kappa^\Theta\le\kappa^*+L_2^G(2E)^{1/2}$, and $\alpha\ge\kappa-L_2^G(E/2)^{1/2}\ge\kappa/2$ when $E\le E_0$; E is non-increasing and the solution stays in B (forward invariance), so this persists, and (iii) gives the rate.

(v) $R[\mu^*]=\langle r^*\nabla^2\tilde\Phi\rangle=0$. For a separation u at μ^* (zero class means), $\Delta\hat f=O(\|u\|^2)$, $\nabla_{\vartheta_i}J_m^\pi=\sigma_0G^{-1}u_i+O(\|u\|^2)$ and $E=(\sigma_0/2)D_\pi+O(\|u\|^3)$, so $\|\nabla J_m^\pi\|^2_{G_m}/(mE)\to2\sigma_0$.

(vi) With counts N the identity behind $\Delta\hat f$ becomes $(1/m)\sum_i\tilde\Phi(\vartheta^*_{j(\vartheta_i)})=\hat f[\mu_N]$; the replicated configuration of the weights N/m is critical for the nearest-atom energy by Proposition D‴.9, a strict local minimum when the weights are admissible and a saddle when a separation margin is negative (Proposition D‴.5), and its excess energy is positive unless $N=mq$. Under (H*) a critical point with $E>0$ has $\alpha\le0$ by (ii), that is $\kappa^*+\kappa^\Theta\ge2\sigma_0$. □

Remark D‴.22 (the mean-field form, propagation of chaos, and the real loop)**.** The identities and their proof hold verbatim for a probability measure μ whose optimal plan to ν is induced by a map π (for instance an absolutely continuous μ: the Laguerre map of Proposition D‴.12), with $(1/m)\sum_i$ replaced by $\int\mathrm{d}\mu$. So formally the Laguerre-cell flow ($\mathfrak{A}_0^{\mathrm{OT}}$) has the same PL inequality, and in the basin both the particle system and the mean-field flow converge to μ^* exponentially in W_2, at the same m-independent rate, so $W_2(\mu_{\Theta(s)},\mu_s)\le W_2(\mu_{\Theta(s)},\mu^*)+W_2(\mu_s,\mu^*)\to0$: for every commensurate m the particle system and the mean-field flow have the same limit — no floor, where Theorem D‴'s synchronous coupling kept the transit's mismatch for ever (Proposition D‴.12(v)). This is not propagation of chaos uniform in time, which needs $\sup_s\mathbb{E}W_2(\mu_{\Theta(s)},\mu_s)\to0$ as $m\to\infty$: the bound is not small in m at any fixed time (an i.i.d. warm start lies in the a-priori basin only at astronomically large m, numerics below), and smallness in m during the transit is not claimed; uniform-in-time statements of that kind are known for entropy-regularised mean-field dynamics under a uniform log-Sobolev inequality [12]. The existence of the mean-field OT flow and its chain rule are not established here. For the real loop: once $E(s)\le(\sigma_0-\kappa^*)(\rho/2)^2/(2m)$, every particle is within $\rho/2$ of its atom ($\max_i\|u_i\|^2\le mD_\pi$), and Theorem D‴.18 takes over; on the averaged dynamics this happens after a time $O(\lambda^{-1}\log m)$. The transit of the real loop is covered only on finite horizons, by averaging with plan switches (formal).

Status. (i), (ii), (iv), (v) proved. (iii) proved for vertex-selection solutions, whose existence is assumed (Section 13); for this leak a two-particle tie is never attracting — with $h:=\langle\vartheta_i-\vartheta_{i'},\vartheta_k^0-\vartheta_j^0\rangle_{G^{-1}}$, $J^{\pi_1}-J^{\pi_2}=\sigma_0h$, and on the two sides of the tie $\mathrm{d}h/\,\mathrm{d}s=\ell-\sigma_0\|\vartheta_k^0-\vartheta_j^0\|^2_{G^{-1}}$ and $\ell+\sigma_0\|\vartheta_k^0-\vartheta_j^0\|^2_{G^{-1}}$ with the same loss term ℓ, so no solution slides along it — while multi-way ties are not analysed; a Clarke solution in the sense of Proposition D‴.17 may stop at a kink-type Clarke-critical point where the active plans' gradients cancel (two particles at one point between two atoms), which a selection solution leaves. (vi) proved for the critical point; its minimality is Proposition D‴.5's admissibility. Remark D‴.22 is formal. The classical case is the leak alone: the metric slope of $\frac12W^2(\cdot,\nu)$ is $W(\cdot,\nu)$ at every μ, so it

satisfies a PL inequality with constant 2 (AGS [2, §10.4], [74]); a particle velocity field realises the slope where the optimal plan is a map. The plan segments are the generalised geodesic from μ^* to μ_Θ based at ν (AGS [2, §9.2]): the diagonal plan $\nu \to \mu^*$ and the assignment $\nu \to \mu_\Theta$ make $(1-t)\vartheta^*_{\pi(i)} + t\vartheta_i$ that curve, along which $\frac{1}{2}W^2(\cdot,\nu)$ is 1-convex — the $(\sigma_0/2)D_\pi$ of (E) — and the passage from convexity along a curve ending at a point to a slope bound is AGS's. PL inequalities in Wasserstein space through optimal maps are known for functionals that are not geodesically convex (barycentre functionals; the distance of uniform point clouds to a density) [7]. Here $\mathcal{J}$ is semiconvex along generalised geodesics only with a negative constant (the loss contributes $-L_2^G \sup\langle|r|\rangle$), so a global Łojasiewicz argument, as for the λ-convex functionals of [7], is unavailable. What is specific is the loss linearised exactly in $\hat{f}$ with the curvature taken at the stationary residual, the explicit local constants, and the counts, which make the generalised geodesic a particle interpolation and the constant independent of m. (ii) is sharp at a matching prior and not in general: it bounds E with $\sigma_0 + \kappa^*_-$ and Q with $(\kappa^* + \kappa^\Theta)/2$ separately (numerics below). Checked: `tests/test_ot_basin.py` ((E) and (P) to 10^{-12} relative for random assignments with the counts mq at both priors, and with other counts the two defects exactly; the PL chain $\|\nabla J\|^2/(mE) \geq Q^2/(D_\pi E) \geq \lambda$; the constant $2\sigma_0$ attained by a separation to 10^{-4}; from the lift with one particle moved into another cell the margin positive, the particle kept in its class, E below the a-posteriori bound and decaying at $2\sigma_0$ at the end, while the nearest-atom leak keeps the floor; the nearest-atom spurious minimum at $(11, 9, 10)/30$, admissible — its separation margins and its K-atom Hessian positive — critical for that leak, not critical for the OT leak, with the PL inequality holding there).

Numerics (`exp_ot_basin.py` → `results/chaos/ot_basin.md`). The averaged slow loop at $\sigma_0 = 0.15\sigma_{\text{crit}}$ from the warm clouds of Proposition D‴.12's numerics: sizes $m = 15$, 60, 240, 510, four seeds, 600 slow units, under both leaks, at the matching prior and at the prior with centres shifted by $+0.15$. The identities (E) and (P) hold along every run to $6 \cdot 10^{-13}$ relative. Under the OT leak the initial optimal plan is never changed (no particle is relabelled between records), and the margin α is positive from $s_+ \leq 300$ on — from $s \leq 25$ in 19 of the 32 runs. From then on the observed ratio $\|\nabla J\|^2/(mE)$ is at least λ at every record (the smallest ratio/λ is 1.00 at the matching prior and 1.61–1.72 at the shifted one, $Q^2/(D_\pi E)/\lambda$ 1.58–1.69), and E falls from $4.1 \cdot 10^{-3}$–$4.0 \cdot 10^{-2}$ to $8.3 \cdot 10^{-7}$–$8.1 \cdot 10^{-6}$ (means over the seeds; $W^2(\mu_\Theta, \mu^*)$ from 0.99–5.25 to $2.3 \cdot 10^{-4}$–$2.3 \cdot 10^{-3}$), below the a-posteriori bound in every run. At the end the observed ratio is 0.01416–0.01417 at the matching prior, against $2\sigma_0 = 0.01415$ and the theorem's $\lambda = 0.0140$–0.0141: the inequality is sharp. At the shifted prior $\lambda = 0.0077$–0.0078 against an observed 0.0137–0.0139. The local PL constant there is $2\min\{\lambda_{\text{sep}}, \lambda_{\text{coll}}\} = 0.01135$ ($\lambda_{\text{sep}} = 0.00567$), so the curvature itself costs a factor 0.802 against $2\sigma_0$ and the inequality (ii) another 0.683 (its slack: the theorem's late λ is 0.548 of $2\sigma_0$); the window still mixes the separation modes. *The nearest-atom contrast.* Under the nearest-atom leak no particle changes cell. At the 22 admissible count vectors (every run at $m = 240$ and 510, and $(5, 4, 6)/15$, $(19, 21, 20)/60$, $(19, 18, 23)/60$ at both priors) the runs sit at the replicated configuration of the sampled weights — $E(600)$ is 0.02–0.24% below its energy and still rising toward it, the classes collapsed (spreads ≤ 0.006) — and that configuration is a critical point of the nearest-atom energy (drift $\leq 6 \cdot 10^{-14}$) and not of the OT energy (drift 2.3–$4.4 \cdot 10^{-2}$): the floor is a spurious minimum. At the other 10, five per prior at $m \leq 60$ — $(6, 5, 4)$, $(6, 3, 6)$, $(8, 4, 3)/15$ and $(24, 15, 21)$, $(25, 17, 18)/60$ — the weights are inadmissible (class-0 separation margins -0.0004 to -0.0061): the replicated configuration is a saddle, class 0 splits (spreads 0.04–0.62), and E ends 0.5–16.5% below it, still falling. *The a-priori basin.* At the matching prior $\kappa = \sigma_0$ on the whole space and $E_0 = 2.0 \cdot 10^{-7}$. At the shifted prior the supremum of $\lambda_{\max} R^G[\mu^*]$ over the whole parameter space is $6.60 \cdot 10^{-3} < \sigma_0$, approached at a saturation corner ($a \to a_{\max}$, $s \to s_{\min}$): (H*) holds on the whole space with $\kappa = 4.7 \cdot 10^{-4}$ — μ^* is the unique global minimiser among all m-point empirical measures for every commensurate m — and $E_0 = 9.1 \cdot 10^{-10}$ (over the box of the runs' end configurations, which is not forward-invariant, the supremum is $4.39 \cdot 10^{-3}$ and $E_0 = 2.9 \cdot 10^{-8}$). These basins are tiny: a particle outside its cell is at G^{-1}-distance at least 3.13 (matching) and 2.95 (shifted)

from its class atom, so $\{E \le E_0\}$ holds a fraction $p \le 5.8 \cdot 10^{-6}$ (matching) and $4.4 \cdot 10^{-7}$ (shifted) of the particles in other cells — none below $m \approx 1.7 \cdot 10^5$ and $2.3 \cdot 10^6$. The runs start at $E(0) = 4.1 \cdot 10^{-3}$–$4.0 \cdot 10^{-2}$, far outside, reach the region $\alpha > 0$ by themselves, and are covered from there a posteriori by (iii). *The measure-level basin.* With a fraction p of all the particles, taken from class 0, moved 0.6 of the way into cell 1, the plan keeps them in their class, and $\kappa^\Theta = 0.0096, 0.019, 0.029, 0.048$ for $p = 1/30, 1/15, 1/10, 1/6$ (segment suprema on 129 points): the margin is $+0.0023$ at $p = 1/30$ and negative from $p = 1/15$ on, the same at $m = 30$ and $m = 300$, which are the same measures. With one particle moved, E decays from $3.3 \cdot 10^{-3}$ to $2.5 \cdot 10^{-5}$ ($m = 30$) and from $2.3 \cdot 10^{-4}$ to $3.1 \cdot 10^{-6}$ ($m = 300$), at the rate $2\sigma_0$ at the end and below $E(0)\exp(-\int \lambda_+)$ with λ evaluated at every step ($9.9 \cdot 10^{-5}$ and $3.7 \cdot 10^{-6}$; $\lambda \ge 0.00148$ throughout), and the particle returns to its own cell. Under the nearest-atom leak it stays in cell 1, and E rises toward the spurious minimum of the counts $(9, 11, 10)/30$ and $(99, 101, 100)/300$, both admissible: $5.72 \cdot 10^{-3}$ and $5.56 \cdot 10^{-4}$ at $s = 300$, $6.304 \cdot 10^{-3}$ and $6.128 \cdot 10^{-4}$ at $s = 1000$, and the minimum's $6.3126 \cdot 10^{-3}$ and $6.1328 \cdot 10^{-4}$ at $s = 3000$.

Theorem D‴.23 (adaptive priors: a learned prior removes the leak's bias, at the rate of the measure-space excitation)**.** *Let q be masses, m commensurate, $\eta \ge 0$, and let the prior's atoms $Z = (z_1, \dots, z_{K_0})$ move toward the barycentres of the particles that the plan assigns to them while the particles follow the OT leak toward $\nu_Z := \sum_j q_j \delta_{z_j}$:*

$$\frac{\mathrm{d}\vartheta_i}{\mathrm{d}s} = G\langle \nabla_\vartheta (r_\Theta \tilde{\Phi}(\cdot\,; \vartheta_i))\rangle_\rho - \sigma_0(\vartheta_i - z_{\pi(i)}), \qquad \frac{\mathrm{d}z_j}{\mathrm{d}s} = -\eta(z_j - \hat{T}_j),$$
$$\hat{T}_j := \frac{1}{m q_j} \sum_{i \in \pi^{-1}(j)} \vartheta_i, \tag{$\mathfrak{A}_m^{\mathrm{OT},\eta}$}$$

π an optimal assignment from μ_Θ to ν_Z ($\eta = 0$ is the fixed prior of Proposition D‴.12).

(i) (a gradient flow whose replicated stationary points carry no leak bias) *For $\eta > 0$, ($\mathfrak{A}_m^{\mathrm{OT},\eta}$) is the vertex-selection gradient flow of $\mathcal{J}(\mu_\Theta, Z) := \bar{L}(\mu_\Theta) + (\sigma_0/2) W^2_{G^{-1}}(\mu_\Theta, \nu_Z)$ in the metric that weighs ϑ_i by $(1/m)G^{-1}$ and z_j by $(\sigma_0 q_j/\eta)G^{-1}$ — by the envelope theorem $\nabla_{z_j} W^2 = 2q_j G^{-1}(z_j - \hat{T}_j)$ — so $\mathcal{J}$ is non-increasing (under any schedule $\eta(s) \ge 0$: $\mathcal{J}$ does not involve η), and bounded solutions converge to Clarke-critical points (Proposition D‴.17's argument: $\mathcal{J}$ is a minimum over the vertices of definable smooth functions of (Θ, Z)). Boundedness is now a hypothesis (Proposition D‴.27 restores it by projecting the prior onto a compact convex set). Proposition D‴.17's a-priori bound rested on the fixed prior's support, while the leak toward an adaptive prior is an internal force between a class and its atom — with the plan held, the leak cancels from $\frac{\mathrm{d}}{\mathrm{d}s}[(1/m)\sum_{i\in\pi^{-1}(j)} \vartheta_i + (\sigma_0 q_j/\eta) z_j] = (1/m)\sum_{i\in\pi^{-1}(j)} G\langle r \nabla \tilde{\Phi}(\vartheta_i)\rangle$ — and solutions can leave every compact set (numerics, Part G(a) and (b)), even from a fit that the network attains exactly. At a stationary point $z_j = \hat{T}_j$, and each particle satisfies $G\langle r\nabla\tilde{\Phi}(\vartheta_i)\rangle = \sigma_0(\vartheta_i - \hat{T}_{\pi(i)})$: the learning forces cancel over each class, but a particle may be held off its class barycentre by the leak. A replicated configuration — the particles of each class $\pi^{-1}(j)$ at one point ϑ_j — is therefore stationary with Z iff $z_j = \vartheta_j$ and $\nabla V_L[\sum_l q_l \delta_{\vartheta_l}](\vartheta_j) = 0$ for every j: the leak force vanishes, and the stationary replicated configurations are the critical points of the unregularised weighted K_0-atom loss. (The classes are the plan's: labelled by the atoms of a given configuration this reads $z_{\sigma(k)} = \vartheta_k$ for the plan's class-to-atom bijection σ, and at an exact representation a prior permuted among atoms of equal mass is stationary; a permutation between atoms of different masses is not, since a replicated configuration with $W^2 = 0$ needs $q_{\sigma(k)} = q_k$.) Non-replicated stationary points — a class held apart against the loss by the leak toward its barycentre — exist (a saddle on the instance), and carry the K_0-cluster bias of (iv)'s regulariser (numerics, Part G(a)). The stationary equations do not involve the prior's initial position: an exact representation with the weights q, with the prior at its atoms, is*

stationary and a global minimum ($\mathcal{J}=0$ there); which stationary point is reached does depend on it ((vi)).

(ii) (the linearisation at an exact representation) *At an exact representation $\mu^*=\sum_k q_k\delta_{\vartheta_k^*}$ ($r^*=0$) with $Z=(\vartheta_k^*)$, the linearisation of $(\mathfrak{A}_m^{\mathrm{OT},\eta})$ splits, for every m, into the separations of each class, which decay at σ_0, and the collective pairs $(\bar u,\zeta)$ of class means and prior offsets. In the whitened weighted coordinates $y_k:=\sqrt{q_k}G^{-1/2}\bar u_k$, $x_k:=\sqrt{q_k}G^{-1/2}\zeta_k$ it reads $\dot y=-H_Ly-\sigma_0(y-x)$, $\dot x=-\eta(x-y)$, with the Gauss–Newton block $H_L:=[\sqrt{q_kq_l}G^{1/2}\langle\nabla_\vartheta\tilde\Phi(\cdot\,;\vartheta_k^*)^\top\nabla_\vartheta\tilde\Phi(\cdot\,;\vartheta_l^*)\rangle_\rho G^{1/2}]_{kl}\succeq 0$ (eigenvalues $h_1\le\cdots\le h_{pK_0}$). On its eigenvectors every pair evolves by $\left[\begin{smallmatrix}-(h_j+\sigma_0) & \sigma_0\\ \eta & -\eta\end{smallmatrix}\right]$, whose rates $\lambda_j^\pm$ are the roots of $\lambda^2-(h_j+\sigma_0+\eta)\lambda+\eta h_j=0$; for $h_j,\eta>0$ the slower one satisfies*

$$\eta h_j/(h_j+\sigma_0+\eta)<\lambda_j^-<\min\{h_j,\eta\},$$

the lower bound being tight as $\eta\to 0$; $\lambda_j^-=0$ if $h_j\eta=0$ and $\lambda_j^-=h_j$ at $\eta=\infty$; and λ^- increases with h ($\mathrm{d}\lambda^-/\,\mathrm{d}h=(\eta-\lambda^-)/(\lambda^+-\lambda^-)>0$), so the slowest collective rate is λ_1^-. So the adaptive prior's convergence in mode j is never faster than the loss's own curvature h_j, while a fixed prior's is $h_j+\sigma_0$; in an unexcited mode ($h_j=0$) $\lambda_j^-=0$, and the network and the prior drift together, as an adaptive law without a leak does without persistent excitation. The pair matrix is self-adjoint in the metric of (i) (in the coordinates $(y_j,(\sigma_0/\eta)^{1/2}x_j)$ it is symmetric, with off-diagonal $(\sigma_0\eta)^{1/2}$).

(iii) (the bias it removes) *With a fixed prior near an exact representation, the stationary point is, to first order in the prior's error x^0, $y^*=\sigma_0(H_L+\sigma_0 I)^{-1}x^0$: mode j keeps the fraction $\sigma_0/(h_j+\sigma_0)$ of the error. The leak buys the rate $h_j+\sigma_0$ at the price of that fraction, and the adaptive prior removes it at the rate λ_j^-. Weakly excited modes ($h_j\ll\sigma_0$) keep almost all of the prior's error under a fixed prior and lose it slowly under an adaptive one; the fit, $\frac12\sum_j h_jy_j^2$ to second order, weighs them by h_j, so the fit is restored faster than the parameters. The spectrum $\{h_j\}$ is the measure-space excitation level of the reference at μ^* (Section 26, branch 7).*

(iv) (infinitely fast priors: a capacity-constrained K_0-means regulariser) *Formally, as $\eta\to\infty$ the prior follows the class barycentres, $z_j=\hat T_j$, and $(\mathfrak{A}_m^{\mathrm{OT},\eta})$ becomes the gradient flow of $\bar L(\mu_\Theta)+(\sigma_0/2)\mathcal{Q}_q(\mu_\Theta)$, with $\mathcal{Q}_q(\mu):=\min_Z W^2(\mu,\nu_Z)$ the capacity-constrained K_0-means distortion of μ — along the plan the flow carries, which is optimal for its own barycentres (Lloyd-consistent) and need not be jointly optimal: the flow follows the selected piece of $\mathcal{Q}_q$, and $\mathcal{Q}_q$ itself need not decrease (Proposition D‴.27(ii)) —: a regulariser that pulls the network toward K_0 clusters of the prescribed masses without biasing where they sit. The separations keep the rate σ_0, and mode j of the class means relaxes at h_j. It does bias how many there are: a target that the network fits exactly only with a class split is fitted with K_0 clusters (numerics, Part G), so K_0 remains a design input.*

(v) (uniformity in m) *The blocks of (ii) and their rates do not depend on m. In the norm*

$$|||(u,\zeta)|||_\eta:=\max\Big\{\Big(\sum_k q_k\big(\|\bar u_k\|_{G^{-1}}^2+(\sigma_0/\eta)\|\zeta_k\|_{G^{-1}}^2\big)\Big)^{1/2},\ \max_i\|u_i^\perp\|_{G^{-1}}\Big\},$$

the per-particle norm of Theorem D‴.18 with the prior's offsets added in the metric of (i), the linearisation acts on the separations as $-\sigma_0 I$ and is self-adjoint on the collective part with the eigenvalues $-\lambda_j^\pm$; the prior is driven by the class barycentres, which are linear in the class means while the plan is diagonal, so Theorem D‴.18's facts (1)–(2) hold for the joint drift with m-independent constants, and for $0<\eta<\infty$ the averaged loop converges,

for every commensurate m, in the $|||\cdot|||_\eta$-ball of radius

$$\rho(\eta) := \min\Big\{\lambda_0(\eta)/(4C_N),\ \text{margin}/\big(C[1+q_{\min}^{-1/2}+(\eta/(\sigma_0 q_{\min}))^{1/2}]\big)\Big\},$$
$$\lambda_0(\eta) := \min\{\sigma_0, \lambda_1^-(\eta)\},$$

at the rate $(1-\kappa)\lambda_0(\eta)$ — C_N the Lipschitz constant of the joint drift's Jacobian in the $|||\cdot|||_\eta$-operator norm (the prior's part of the drift is linear, so it is the particles'), margin *the plan margin of Proposition D‴.12(iii) at the exact representation, C as in Theorem D‴.18: the ball reaches the per-particle distance $(1+q_{\min}^{-1/2})\rho$ and the prior's offsets $\|\zeta_k\|_{G^{-1}} \le \rho(\eta/(\sigma_0 q_k))^{1/2}$, and the plan stays diagonal while both are within the margin. At $\eta=\infty$ the prior is the class barycentres ($\zeta=\bar u$), the norm is Theorem D‴.18's, $\lambda_0 = \min\{\sigma_0, h_1\}$, and the margin term is* margin$/(C(1+2q_{\min}^{-1/2}))$*; for large finite η the bound degenerates like $\eta^{-1/2}$, because this norm does not use the prior's relaxation onto the class means at the rate η (a singular perturbation). As $\eta\to 0$, $\lambda_1^- < \eta$ and the rate and the radius vanish with it; at $\eta=0$ the prior is a parameter, and the statement is Theorem D‴.18's contraction for the fixed prior.*

(vi) (the transit: when the prior may start) *From a sample whose cluster masses differ from q, the plan's classes contain particles of other clusters until the transit of Proposition D‴.12 ends, and their barycentres $\hat T_j$ are pulled toward those clusters: an adaptive prior is dragged along, and afterwards relaxes back only at the rates λ_j^-. At $\eta=\infty$ the collective modes keep no leak curvature at all, and the drag can carry the class atoms onto a plateau of the unregularised loss, which the flow then crosses at that loss's own curvature. Freezing the prior while mass is in transit — $\eta=0$ as long as some particle is assigned to an atom that is not its nearest — keeps the prior out of the drag; $\mathcal J$ stays non-increasing, as under any schedule of η, and at $\eta=\infty$ the opening is a jump of Z to the class barycentres, which lowers $\mathcal J$. But the adaptive prior forgets a perturbation of mode j only at $\lambda_j^- < h_j$, so whatever the transient leaves in the weakly excited modes when the prior starts is kept for a time of order $1/\lambda_1^-$. When the parameters, not only the fit, are wanted at a given horizon, the adaptation should therefore start from the fixed-prior stationary point — after the transit, and after the fixed-prior dynamics has settled, which it does at the fixed prior's rates (of order σ_0 and faster) — and then (ii) and (iii) apply with the prior's error as the only initial displacement. The wait is a delay, which the fit error does not always repay (numerics).*

Proof. (i) $W^2(\mu_\Theta,\nu_Z) = \min_\gamma \sum_{ij}\gamma_{ij}\|\vartheta_i - z_j\|^2_{G^{-1}}$; where the optimal plan is unique, $\nabla_{z_j}W^2 = \sum_i \gamma_{ij} 2G^{-1}(z_j-\vartheta_i) = 2q_jG^{-1}(z_j-\hat T_j)$, so $\mathrm{d}z_j/\mathrm{d}s = -(\eta/(\sigma_0 q_j))G\nabla_{z_j}\mathcal J$ and $\mathrm{d}\vartheta_i/\mathrm{d}s = -mG\nabla_{\vartheta_i}\mathcal J$ (Proposition D‴.12); the vertex-selection argument and the convergence are Proposition D‴.17's with (Θ,Z) in place of Θ, for solutions that stay bounded. With the plan held, $(1/m)\sum_{i\in\pi^{-1}(j)}\sigma_0(\vartheta_i - z_j) = \sigma_0 q_j(\hat T_j - z_j) = (\sigma_0 q_j/\eta)\dot z_j$, which is the conservation law. $\dot z = 0$ forces $z_j = \hat T_j$ for every j, and then $\dot\vartheta_i = 0$ reads $G\langle r\nabla\tilde\Phi(\vartheta_i)\rangle = \sigma_0(\vartheta_i - \hat T_{\pi(i)})$. At a replicated configuration $\hat T_j$ is the class point ϑ_j, so the leak term vanishes and stationarity is $z_j=\vartheta_j$ and $\nabla V_L(\vartheta_j)=0$. At an exact representation $\bar L = 0$ and $W^2(\mu^*,\nu_{Z^*})=0$. (ii) At $r^*=0$, $-G\nabla V_L[\mu](\vartheta_i) = -G\langle \Delta\hat f^\top \nabla_\vartheta\tilde\Phi(\cdot\,;\vartheta_i)\rangle$ with $\Delta\hat f = \sum_l q_l\nabla_\vartheta\tilde\Phi(\cdot\,;\vartheta_l^*)\bar u_l + O(\|u\|^2)$ (the separations enter at second order, Theorem D‴.18's fact (1)), so a separation evolves by $-\sigma_0 u^\perp$ and the class means by $\dot{\bar u}_k = -G\sum_l\langle\nabla\tilde\Phi_k^\top\nabla\tilde\Phi_l\rangle q_l\bar u_l - \sigma_0(\bar u_k - \zeta_k)$, with $\dot\zeta_k = -\eta(\zeta_k-\bar u_k)$ because $\hat T_k - \vartheta_k^* = \bar u_k$; the coordinates y, x symmetrise it. The eigenvalues of the pair matrix are $-\lambda$ with $\lambda^2 - (h+\sigma_0+\eta)\lambda + \eta h = 0$. Their product is ηh and their sum $h+\sigma_0+\eta$; for $h,\eta>0$ both roots are positive, so $\lambda^+ < h+\sigma_0+\eta$ and $\lambda^- = \eta h/\lambda^+ > \eta h/(h+\sigma_0+\eta)$; the polynomial is $-\sigma_0 h<0$ at $\lambda=h$ and $-\sigma_0\eta<0$ at $\lambda=\eta$, so both lie strictly between the roots; and differentiating the polynomial in h at λ^- gives $\mathrm{d}\lambda^-/\mathrm{d}h = (\eta-\lambda^-)/(\lambda^+-\lambda^-) > 0$. (iii) The stationary equations

of the fixed prior, linearised, read $0 = -H_L y - \sigma_0(y - x^0)$; $\bar{L} = \frac{1}{2}\|\Delta\hat{f}\|^2 = \frac{1}{2}y^\top H_L y + O(\|y\|^3)$. (iv) For a fixed assignment the minimiser of W^2 over Z is the class barycentres; by the envelope theorem the particles' drift of $\bar{L} + (\sigma_0/2)\mathcal{Q}_q$ is $-G\nabla V_L - \sigma_0(\vartheta_i - \hat{T}_{\pi(i)})$, the $\eta \to \infty$ limit of $(\mathfrak{A}_m^{\mathrm{OT},\eta})$ (formal: a singular perturbation in which Z is the fast variable). (v) (sketch) The symmetrisation of (ii) makes the collective part self-adjoint in $|||\cdot|||_\eta$, with the eigenvalues $-\lambda_j^\pm$, and the separations decouple at $-\sigma_0$; the rest is Theorem D‴.18's contraction argument (its fact (3) with this norm) for the averaged loop, the diagonal plan being locally constant at commensurate m (Proposition D‴.12(iii)) inside the ball, whose per-particle and prior extents are as stated. The constants κ, C, C_N and the margin are not derived. (vi) $\mathcal{J}$ does not involve η, so it is non-increasing for any schedule $\eta(s) \geq 0$, the gates' included; at $\eta = \infty$ the opening sets Z to the class barycentres of the current plan, the minimiser of $\mathcal{J}$ over Z for that plan, so $\mathcal{J}$ drops there. □

Status. (i)–(iii) proved, (i) for the solutions that stay bounded (a hypothesis now); (iv) formal; (v) sketched, as an adaptation of Theorem D‴.18's contraction argument for the averaged loop in the norm $|||\cdot|||_\eta$ — the structure (the decoupled separations, the self-adjoint collective pairs, the m-independent blocks) verified, the radius stated, its constants not derived; (vi) the monotonicity of $\mathcal{J}$ under the gates proved (it holds under any schedule of η), the drag, the memory of the weak modes and the remedy numerical (below). The prior that tracks the network is the continuous-time form of the proximal point method, or iterated Tikhonov regularisation — the leak toward a moving centre — and the coupled flow of the particles and a centre drawn to their mean is elastic averaging; the fast limit is the K-means form of soft weight-sharing, a regulariser of network compression [90], here with the masses fixed (capacity-constrained, [5]). The per-mode formulas are those of the quadratic analyses: $\sigma_0/(h_j + \sigma_0)$ is the complement $1 - h_j/(h_j + \sigma_0)$ of the Tikhonov filter factor in the Gauss–Newton eigenbasis (the residual factor: the fraction of the prior's error that the regularised solution keeps), iterated Tikhonov contracts mode j by it per step (in continuous time, the gradient flow of the Moreau envelope, whose curvature $h_j\sigma_0/(h_j + \sigma_0)$ gives the slow-prior limit $\eta h_j/(h_j + \sigma_0)$ of λ_j^-), and the pair matrix is the continuous-time form of elastic averaging's worker–centre map on a quadratic [90]. New are their transfer to the mass-constrained particle system — H_L identified as the measure-space excitation spectrum, and the separations decoupled at σ_0, which make the trade-off between the fixed prior's bias and the adaptive prior's rate hold uniformly in m — and the transit's drag with its gate. The limits of the bias removal are the precedents' own properties in this form: the lost a-priori bound is the proximal point method's behaviour without a minimiser (its iterates are bounded iff a zero exists [90], cited from memory), and the K_0-cluster bias is the quantisation error of soft weight-sharing's hard limit [90]; specific here are their particle and mass-constrained form, and the run-off from a fit that the network attains exactly (Part G(a)). Checked: `tests/test_adaptive_prior.py` (the joint drift against finite differences of $\mathcal{J}$ in the stated metric; the exact representation stationary with the prior at its atoms and the fixed prior's stationary point not; the spectrum of the linearisation at $m = 6$ equal to the separation rate σ_0 and the pair rates $\lambda_j^\pm$ for two η, and the linearisation self-adjoint in the gradient-flow metric; the fixed prior's bias at a smaller shift to the first-order formula; the transit gate frozen while mass is in transit and the settled gate opening only when the fixed-prior drift has fallen below its tolerance, with $\mathcal{J}$ non-increasing in both; the limits of (i): a non-replicated stationary saddle held by the leak, a prior permuted among atoms of equal mass stationary at the exact representation, and the adaptive prior's run-off — the force on the amplitude tending to the positive Karush–Kuhn–Tucker multiplier of the loss's minimum on the face $a = a_{\max}$, where a bounded limit would need it to vanish, and the run-off law $(a_{\max} - a)s \to a_{\max}^2(\sigma_0 + \eta)/(4\eta G_\alpha F_\infty)$ of the conservation law — against the fixed prior's convergence).

Numerics (`exp_adaptive_prior.py` → `results/chaos/adaptive_prior.md`). The averaged slow loop at $\sigma_0 = 0.15\sigma_{\mathrm{crit}}$ with the OT leak toward a misplaced prior — the matching prior's centres shifted by $+0.15$, masses $1/3$. *The linear theory.* At the exact representation the whitened weighted Gauss–Newton spectrum is $h_j = 3.06 \cdot 10^{-4}$ … 0.195. A fixed prior keeps the fraction

$\sigma_0/(h_j+\sigma_0)$ of its error, 0.959 in the weakest mode and 0.035 in the strongest, with collective rates $h_j+\sigma_0 \geq 7.38\cdot10^{-3}$; the adaptive prior's slowest rate is $\lambda_1^- = 2.68\cdot10^{-5}$, $1.51\cdot10^{-4}$, $2.78\cdot10^{-4}$, $3.06\cdot10^{-4}$ for $\eta=\sigma_0/10$, σ_0, $10\sigma_0$, ∞. *(v)'s radius.* A sampled lower bound on the Jacobian-Lipschitz constant of the collective block is $C_N \geq 0.341$, so the proof's radius is at most $2.0\cdot10^{-5}$, $1.1\cdot10^{-4}$, $2.0\cdot10^{-4}$, $2.2\cdot10^{-4}$ for the four η — $\lambda_0/\sigma_0 = 0.0038$–$0.043$ times Theorem D‴.18's at the same C_N — while the K_0-atom runs below start at $|||\cdot|||_\eta = 1.17$, 0.37, 0.12, 0.021, 94 to $6\cdot10^4$ times outside, the warm clouds of the m-particle systems $2.2\cdot10^4$–$3.6\cdot10^5$ times and the separated starts of Part F 131–3344 times: they probe the dynamics beyond the proved ball. *The fixed prior's bias.* $\Theta^*(q)$ has the fit error $9.78\cdot10^{-3}$, and the first-order formula $y^* = \sigma_0(H_L+\sigma_0 I)^{-1}x^0$ describes it to 4.8% in norm; per mode the weak modes 1 and 3 hold 0.85 and 0.73 of the prediction, and the error is first order in the shift (halving the shift from 0.15 to 0.01875 takes the norm error through 4.8, 2.3, 1.2, 0.58% and modes 1 and 3 to 0.98 and 0.97 of the prediction). The weakest mode keeps 0.814 of its error, against the predicted 0.959. *The K_0-atom loop* from $\Theta^*(q)$, with the prior at the shifted centres, for the four η over 6000 slow units: every mode amplitude above 10^{-6} — thirteen of the 36 — decays at its predicted rate (log-slopes on $[3000, 6000]$ between 0.981 and 1.003 times λ_j^-), at a ratio to the exact pair solution between 0.82 and 1.12 that changes by at most 1.04% from $s=1000$ on for $\eta \geq \sigma_0$ (6.0% at $\sigma_0/10$; the offset is set in the first, nonlinear phase); below 10^{-6} the faster modes stop at the level of the second-order remainder. The fit error falls from $9.78\cdot10^{-3}$ to $3.6\cdot10^{-4}$, $1.8\cdot10^{-5}$, $8.0\cdot10^{-6}$, $6.7\cdot10^{-6}$ at $s=6000$, while the weakest mode still holds $1.45\cdot10^{-3}$, $6.7\cdot10^{-4}$, $3.0\cdot10^{-4}$, $2.5\cdot10^{-4}$ of its $1.58\cdot10^{-3}$: the fit is restored long before the parameters, as (iii) says. *m-particle systems:* $m=15$, 60, 240 and $M=510$ from the warm clouds of Proposition D‴.12's numerics, four seeds, 3000 slow units. Under the fixed prior every system ends at the lift of $\Theta^*(q)$ ($\|\bar{y}\| = 2.10\cdot10^{-2}$, $W^2(\mu_\Theta, \mu_{\text{exact}}) = 4.4\cdot10^{-4}$, fit error $9.78\cdot10^{-3}$). Ungated, the adaptive prior is dragged by the transit: at $s=500$ $\|\bar{y}\|$ is 0.032–0.32 ($\eta=\sigma_0$) and 0.048–0.71 ($\eta=\infty$) in seed means, against the K_0-atom run's $7.4\cdot10^{-3}$ and $4.3\cdot10^{-3}$, and at $s=3000$ still 0.0094–0.10 and 0.012–0.59, the largest at $m=15$. At $\eta=\infty$ one $m=15$ system is carried onto a plateau of the unregularised loss: its split class re-collapses by $s=7000$, onto atoms where the loss's Hessian has the eigenvalues $-1.8\cdot10^{-5}$ and $7.2\cdot10^{-5}$ (fit error $3.46\cdot10^{-2}$ at $s=23000$, the drift $1.4\cdot10^{-5}$). The collapsed $\eta=\infty$ dynamics from there, the unregularised K_0-atom gradient flow, falls below 0.9 of the plateau's loss $2.52\cdot10^{-4}$ after $4.16\cdot10^4$ more slow units, halves it at $4.78\cdot10^4$ and falls below 10^{-12} at $7.97\cdot10^4$ (the crossings located as events), at the exact representation: the plateau is crossed on the time scale $1/1.84\cdot10^{-5} = 5.4\cdot10^4$ of its negative curvature. The transit gate opens at $s=88$–149 and removes the plateau, but the transient's residue in the weak modes stays: $\|\bar{y}\| = 4.9\cdot10^{-3}$–$3.7\cdot10^{-2}$ ($\eta=\sigma_0$) at $s=3000$, against the K_0-atom run's $1.3\cdot10^{-3}$, decreasing with m. The settled gate opens at $s=1211$–1304, with the systems replicated (rms separations $1.2\cdot10^{-4}$–$7.9\cdot10^{-4}$) and their class means within $6.7\cdot10^{-5}$ of $\Theta^*(q)$; each is then the K_0-atom system started from its own class means, and at $s=3000$ $\|\bar{y}\|$ and the fit error of every system equal the K_0-atom run's at $s-t_{\text{open}}$ — per-seed ratios 0.995–1.000 ($\eta=\sigma_0$) and 0.968–1.001 ($\eta=\infty$) for every m from 15 to 510, as the invariance of the replicated configurations requires, the departures from 1 being the opening offset in the weakest mode — with fit errors $1.6\cdot10^{-4}$ ($\eta=\sigma_0$) and 3.4–$3.5\cdot10^{-5}$ ($\eta=\infty$) against the fixed prior's $9.78\cdot10^{-3}$, $W^2(\mu_\Theta, \mu_{\text{exact}}) = 6.7$–$7.2\cdot10^{-6}$ and 9.3–$9.9\cdot10^{-7}$ against $4.4\cdot10^{-4}$, and the counts at q in every run. The wait costs 1211–1304 of the 3000 slow units: in the seed means the settled gate's $\|\bar{y}\|$ and W^2 are the lower everywhere, and its fit error at $\eta=\infty$ and at $m\leq240$ (at $\eta=\sigma_0$ and $m=510$ the transit gate's fit error is the lower, $1.55\cdot10^{-4}$ against $1.64\cdot10^{-4}$); system by system the transit gate ends lower in all three in 6 of the 32 systems. *The structure behind (v), beyond its ball* (Part F): from the lift of $\Theta^*(q)$ plus zero-mean per-particle separations of rms 0.02 and 0.1 ($m=15$ and 510, ungated, five draws of their direction), $\|\bar{y}\|$ at $s=1750$ departs from the K_0-atom run's, in the draw means, by +0.13% and +3.1% at $\eta=\sigma_0$ and by +0.77% and +18.5% at $\eta=\infty$ for $m=15$ (+0.09%, +2.2%, +0.63% and +15.5%

for $m = 510$): second order in the separations draw by draw (factors 23.8–25.0 for the factor 5), with a spread across the draws 3.3–6.0 times larger at $m = 15$, and consistent with the same expected size at both m (the draw means, higher at $m = 15$ in all four columns, differ by +0.9 to +1.7 combined standard errors; five draws) — the prediction of Theorem D‴.18's facts (1)–(2) at second order, the departure being linear in the separations' class covariances, whose expectation is the same at every m. The starts lie 131–3344 times outside (v)'s ball, so (v) itself asserts nothing there. *The limits of (i)* (Part G). On a four-bump target that six kernels fit exactly with one class split ($K_0 = 3$, $m = 6$), the joint flow has a stationary saddle with the class split — the pair at G^{-1}-distance 1.64, against the exact fit's 3.29 — held by a leak force $5.8 \cdot 10^{-3}$ against an unregularised gradient of the same size (fit error $4.2 \cdot 10^{-2}$); from the exact fit the flow collapses the class within 8000 slow units (at $s = 4 \cdot 10^4$, $\eta = \infty$: the merged width 1.05, fit error $2.23 \cdot 10^{-2}$), and the merged atom then widens toward $s_{\max}$ — continued, its width coordinate runs off like $\ln s$ and the width tends to $s_{\max}$ like $1/s$, as in (b): the solution is unbounded — while its physical image approaches the minimiser of the loss on the face $s = s_{\max}$ (fit error $1.943 \cdot 10^{-2}$, a Karush–Kuhn–Tucker point with $\partial L/\partial s = -1.3 \cdot 10^{-6}$, not a critical point), reached only as $s \to \infty$: the K_0-cluster bias drives the solution from an exact fit to infinity while $\mathcal{J}$ decreases. With a target bump of weighted amplitude $4.8 > a_{\max} = 4$, the fixed prior's K_0-atom solution converges (fit error 0.175), while under the adaptive prior ($\eta = \sigma_0$ and ∞) the amplitude coordinate grows like $2 \ln s$ (its increments per decade $1.99 \ln 10$ by $s = 10^6$) and the amplitude tends to $a_{\max}$ like $1/s$: the solution leaves every compact set. At the exact representation (equal masses) a cyclically permuted prior is stationary (drift $1.1 \cdot 10^{-17}$), with $z_{\sigma(k)} = \vartheta_k$.

Proposition D‴.24 (the finite-size floor is the leak's bias at the sampled weights; an adaptive prior removes it without a mass constraint)**.** *Let the target admit, for the weights w in a neighbourhood W of q, an exact weighted K_0-atom representation $\vartheta^e(w)$ — $f = \sum_k w_k \tilde{\Phi}(\cdot\,; \vartheta^e_k(w))$, a critical point of the unregularised weighted loss whose whitened weighted Gauss–Newton block $H_L(w)$ is invertible, smooth in w — and let the prior be matching, $\vartheta^0 = \vartheta^e(q)$. (On the instance the network's amplitude–mass degeneracy gives $\vartheta^e(w)$ explicitly: the amplitudes A_k/w_k with the bumps' centres and widths, on the branch of $\vartheta^e(q)$ while $|A_k|/w_k < a_{\max}$; these weights are the* representable *ones, W. They are not Proposition D‴.5's admissible set W_{adm} of the fixed prior, where its replicated stationary point is a strict local minimiser; a sample in W_{adm} is called* D‴.5-admissible.*)*

(i) (the floor is the bias) *Under the nearest-atom leak toward the fixed prior, the weighted stationary point $\Theta^*(w)$ of Proposition D‴.5 satisfies, with Theorem D‴.23's whitened weighted coordinates at the weights w,*

$$\hat{f}_{\Theta^*(w)} - f = \hat{J}(w)\sigma_0(H_L(w) + \sigma_0 I)^{-1}x^0(w) + O(\|x^0(w)\|^2),$$
$$x^0_k(w) := \sqrt{w_k}G^{-1/2}(\vartheta^0_k - \vartheta^e_k(w)),$$

$\hat{J}(w) := [\sqrt{w_k}\nabla_\vartheta \tilde{\Phi}(\cdot\,; \vartheta^e_k(w))G^{1/2}]_k$ (so that $\hat{J}^\top \hat{J} = H_L$): the leak's bias at the sampled weights, zero at $w = q$. The function-level weight-sensitivity of Propositions D‴.5 and D‴.9 is its derivative, $F = \hat{J}\sigma_0(H_L + \sigma_0 I)^{-1}\partial_w x^0$ at q — exactly, the remainder's derivative vanishing with $x^0(q)$ — and on the eigenvectors v_j of $H_L(q)$ (eigenvalues h_j) the function-level floor law of Proposition D‴.9 reads

$$\mathbb{E}\|\hat{f}_m - \hat{f}_M\|^2 = (1/m - 1/M) \cdot \sum_j \frac{h_j \sigma_0^2}{(h_j + \sigma_0)^2} \cdot z_j^\top \tilde{C} z_j + o(1/m), \qquad z := V^\top \partial_w x^0, \quad \text{(B)}$$

∂_w taken along the zero-sum directions $e_j - e_{K_0}$ and $\tilde{C}$ the multinomial covariance of $(w_1, \ldots, w_{K_0-1})$ — Proposition D‴.9's law written on the eigenvectors, and like it a law for samples in W_{adm}: at a saddle the replicated stationary point is left by a split

(Proposition D″.5), and the floor is the split configuration's. Each mode is weighted by $h_j\sigma_0^2/(h_j+\sigma_0)^2 \le \sigma_0/4$*, with equality at* $h_j=\sigma_0$*: a band-pass centred at the level of the leak, at least half its peak for* $h_j/\sigma_0 \in [3-2\sqrt{2}, 3+2\sqrt{2}]$ *(a factor 5.8 either side). The law is therefore at most* $(1/m-1/M)(\sigma_0/4)\,\mathrm{tr}(\partial_w x^0 \tilde{C} \partial_w x^{0\top})$*, whatever the excitation spectrum. A weakly excited mode keeps its bias but hardly shows it in the function; a stiff one shows little bias; which modes carry the floor is set first by the gauge's displacement* $\partial_w x^0$ *and then re-weighted toward* $h \approx \sigma_0$*.*

(ii) (the nearest-atom leak toward an adaptive prior: the masses minimised out) *Let the prior's atoms follow the means of their Voronoi cells,* $\mathrm{d}z_j/\,\mathrm{d}s = -\eta(z_j - \hat{T}_j)$ *(*$\eta>0$*) with* $\hat{T}_j$ *the mean of the particles nearest to* z_j*, while the particles follow the nearest-atom leak toward* Z*. Inside the cells this is the gradient flow of*

$$\mathcal{J}_\circ(\Theta, Z) := \bar{L}(\mu_\Theta) + (\sigma_0/2m)\sum_i \min_j \|\vartheta_i - z_j\|^2_{G^{-1}} = \min_{q' \in \Delta} \mathcal{J}(\mu_\Theta, \sum_j q'_j \delta_{z_j})$$

in the metric $(1/m)G^{-1}$ *per particle and* $(\sigma_0 w_j/\eta)G^{-1}$ *per atom,* w *the cell masses: the OT leak of Theorem D″.23 with the prior's masses minimised out (semi-relaxed transport — the infinitely fast limit of a dynamics of the masses, not a learning law for them), which sends every particle to its nearest atom. The counts change only when a particle crosses into another cell; across a crossing the flow is the vertex-selection flow of* $\mathcal{J}_\circ$ *— a minimum over the assignments of smooth functions, in a metric that depends on the active assignment — and* $\mathcal{J}_\circ$ *stays non-increasing, as in Theorem D″.23(i). A cell without particles has a degenerate metric: its atom is frozen, and the cell is refilled only if a particle's own dynamics carries it in (on the instance an empty cell stays empty; tests).*

(iii) (no function-level floor at the representable samples) *At fixed counts* N*,* $w = N/m$*, the stationary replicated configurations are the critical points of the unregularised weighted* K_0*-atom loss at* w *(Theorem D″.23(i) with* w *in place of* q*), and for* $w \in W$ *the exact representation* $\vartheta^e(w)$ *with the prior at its atoms is one, with* $\hat{f} = f$*. At representable sampled weights the function-level sensitivity therefore vanishes identically, and the function-level distance between two systems at their exact representations is zero. There* $r = 0$*, so* $R = 0$ *and Proposition D″.5's splitting margins are the maximal* σ_0 *for every* $w \in W$*: the adaptive prior also removes the fixed prior's splitting instability, whose replicated point is a saddle wherever a cluster overshoots (on the instance at* $w_1 \gtrsim 0.4$*). Outside* W *(on the instance some* $|A_k|/w_k \ge a_{\max}$*) no exact representation exists on the branch of* $\vartheta^e(q)$ *— a permutation of the classes over the bumps can be representable, but the dynamics from the warm clouds follows the branch — and the adaptive prior removes the leak's bias without reaching* f*: the amplitude coordinates run off (Theorem D″.23(i)), and the sample keeps a function-level error. The probability of such samples decays exponentially in* m *(on the instance at most* $\sum_k \exp(-2m(q_k - |A_k|/a_{\max})^2)$*, by Hoeffding's inequality, vacuous at* $m = 15$*), so if every representable sample converges to its exact representation the expected limiting floor is exponentially small in* m *rather than of order* $1/m$*; on the instance this asymptotic statement is not yet in force at* $m \le 60$ *(numerics, Part E). The parameter-level stationary measure* $\sum_k w_k \delta_{\vartheta^e_k(w)}$ *still carries the sample's weights — the counts fix the amplitude–mass gauge — and its coupled distance is the larger one, conditionally on representability (outside* W *the parameters run off and the coupled distance diverges): Proposition D″.5's coupled distance has the sensitivity* $S^e := \partial_w \vartheta^e$ *of the exact representation, where the fixed prior's is, in the whitened weighted coordinates and to first order,* $H_L(H_L + \sigma_0 I)^{-1} S^e$*. Per mode, the weight fluctuation of the exact representation divides into the fraction* $h_j/(h_j+\sigma_0)$ *carried by the parameters and the fraction* $\sigma_0/(h_j+\sigma_0)$ *left as the bias, which the function sees with the weight* h_j*; the adaptive prior moves all of it into the parameters. Since* $S^e = -\partial_w x^0$*, the bound of (i) is* $\sigma_0/4$ *times the adaptive prior's*

coupled-distance law: the fixed prior's function-level law is at most $\sigma_0/4$ times the adaptive prior's parameter-level law. At the measure level the two priors hardly differ: between the stationary measures of two samples, W^2 is the transport of the mass $\frac{1}{2}\|w-w'\|_1$ across the inter-cluster distances, of order $\|w-w'\|_1 \sim m^{-1/2}$ under both priors, plus a within-class part of order $\|w-w'\|^2$, the larger under the adaptive prior.

(iv) (the rates) *At $\vartheta^e(w)$ the linearisation is Theorem D‴.23(ii) with w in place of q: the collective pairs relax at $\lambda_j^-(w) < \min\{h_j(w), \eta\}$, and the fit, weighted by h_j, recovers faster than the parameters. On the boundary of W (an amplitude A_k/w_k equal to the saturation $a_{\max}$) the exact representation sits at infinity in the unconstrained coordinates and the convergence is algebraic: with $a = a_{\max}\tanh(\tilde{a}/a_{\max})$, $a_{\max} - a \approx 2a_{\max}e^{-2\tilde{a}/a_{\max}}$, the loss along the saturating coordinate is of order $e^{-4\tilde{a}/a_{\max}}$, its gradient flow gives $\tilde{a} \approx (a_{\max}/4)\log s$, and the function-level error decays like $a_{\max}^2 w_k(1+\sigma_0/\eta)/(8G_\alpha s)$, the loss's curvature cancelling (formal).*

(v) (the design table) *Fixed prior, nearest-atom leak: the function-level floor* (B) *at the D‴.5-admissible samples (elsewhere a cluster splits) and, if the prior is misplaced, its bias. Fixed prior, OT leak (Proposition D‴.12): on the commensurate sizes no function- or measure-level floor, and the prior's bias. Adaptive prior, nearest-atom leak: at the representable samples no function-level floor and no bias, and the larger coupled distance of the gauge. Adaptive prior, OT leak (Theorem D‴.23): neither function-level floor nor bias, the masses fixed by design. Under both OT leaks the coupled distance keeps the transit's mismatch (Proposition D‴.12's* Open*). The mass constraint fixes the gauge, and the adaptive prior removes the bias — with it the function-level floor, which is its slope in w. This resolves Remark D‴.10's trade-off between the window's margin and the coupling's floor at the function level: at an exact representation the separation margin is the maximal σ_0 and both the bias and the function-level floor vanish. The price is the collective rate, λ_j^- in place of $h_j + \sigma_0$ (Theorem D‴.23(ii)), the leak's a-priori bound (restored by projecting the prior, Proposition D‴.27), and a larger coupled distance.*

(vi) (the real loop: perfect tracking) *With the adaptive prior, the lift of the exact representation of representable sampled weights, with the prior at its atoms and the plant on the reference ($e = 0$), is an invariant set of the real closed loop of Theorem J: $\hat{f} = f$ on the reference orbit — the support of the occupation measure ρ, which is all the error equation sees at $e = 0$ — so $e \equiv 0$ solves the error equation there (its linear stability also uses $\partial_x\hat{f} = \partial_x f$ on the orbit, which the exact representation, an identity of functions, supplies: Corollary D‴.26(c)), and the adaptive law's gradient term (proportional to e), the leak (toward each particle's own position) and the prior's motion (toward its own atom) all vanish: perfect tracking at every representable sample. With a fixed prior the tracking error keeps Theorem J's slaved value $e = (1/k_e)r_{\Theta^*(w)}(x_m) + O(k_e^{-2})$: the leak's bias, divided by the gain.*

Proof. (i) The derivation of Theorem D‴.23(iii) at the weights w: to first order in $u_k := \vartheta_k^*(w) - \vartheta_k^e(w)$, the stationarity of $\Theta^*(w)$ reads $H_L(w)y + \sigma_0(y - x^0(w)) = 0$ in $y_k := \sqrt{w_k}G^{-1/2}u_k$, and $\hat{f}_{\Theta^*(w)} - f = \sum_k w_k\nabla_\vartheta\tilde{\Phi}(\cdot\,;\vartheta_k^e)u_k + O(\|u\|^2) = \hat{J}y + O(\|u\|^2)$. Since $x^0(q) = 0$, the derivative at q keeps only $\partial_w x^0$ (the derivatives of $\hat{J}$, of H_L, of the factors $\sqrt{w_k}$ and of the $O(\|x^0\|^2)$ remainder multiply $x^0(q)$ or $u(q) = 0$), so F is exact at q; the vectors $\hat{J}v_j$ are orthogonal with $\|\hat{J}v_j\|^2 = h_j$, which with Proposition D‴.9's law gives (B); and $h\sigma_0^2/(h+\sigma_0)^2 \le \sigma_0/4$ is $(h+\sigma_0)^2 \ge 4h\sigma_0$, with $h\sigma_0^2/(h+\sigma_0)^2 \ge \sigma_0/8$ iff $h^2 - 6h\sigma_0 + \sigma_0^2 \le 0$. (ii) For fixed Z and μ every coupling of μ with some $\sum_j q_j'\delta_{z_j}$ costs at least $\int \min_j\|\vartheta - z_j\|^2\,\mathrm{d}\mu$, with equality for the nearest-atom map, whose image masses are the cell masses; inside the cells $\nabla_{z_j}(\sigma_0/2m)\sum_{i:j(i)=j}\|\vartheta_i - z_j\|^2_{G^{-1}} = \sigma_0 w_j G^{-1}(z_j - \hat{T}_j)$, and $\nabla_{\vartheta_i}\mathcal{J}_\circ$ is Proposition D‴.9's. Across a crossing, $\mathcal{J}_\circ$ is the minimum over the assignments of smooth functions; a vertex-selection solution follows the gradient flow of an active assignment, whose derivative along it is minus the squared norm of the velocity in that assignment's metric

and bounds the upper Dini derivative of the minimum, so $\mathcal{J}_\circ$ is non-increasing. An empty cell's atom feels no force ($w_j = 0$) and does not move. (iii) As Theorem D‴.23(i); at $\vartheta^e(w)$ the residual vanishes, so $R[\mu](\vartheta) = \langle r\nabla^2_\vartheta\tilde{\Phi}\rangle_\rho = 0$ and the separation blocks $\sigma_0 I - R^G$ of Proposition D‴.5 are $\sigma_0 I$. The parameter level: two replicated measures with the weights w and w' and atoms $O(\|w - w'\|)$ apart differ by the mass $\frac{1}{2}\|w - w'\|_1$ that a coupling must move across inter-cluster distances. The split: $\vartheta^*(w) = \vartheta^e(w) + u(w)$ with $\partial_w y = -\sigma_0(H_L + \sigma_0 I)^{-1} S^e_y$ at q (S^e_y the whitened weighted S^e), so the fixed prior's $S_y = S^e_y + \partial_w y = H_L(H_L + \sigma_0 I)^{-1} S^e_y$; and $S^e_y = -\partial_w x^0$ by the definition of x^0. (iv) As Theorem D‴.23(ii); the boundary case: with $\varepsilon := a_{\max} - a$ and $\alpha - z_\alpha$ slaved, the conservation law $\frac{\mathrm{d}}{\mathrm{d}s}[\alpha + (\sigma_0/\eta)z_\alpha] = G_\alpha\langle r\varphi_k\rangle\partial a/\partial\alpha$ gives $\mathrm{d}\varepsilon/\,\mathrm{d}s = -(\eta/(\sigma_0 + \eta))G_\alpha\langle r\varphi_k\rangle(\partial a/\partial\alpha)^2$, with $\langle r\varphi_k\rangle = w_k\varepsilon\langle\varphi_k^2\rangle$ and $\partial a/\partial\alpha \approx 2\varepsilon/a_{\max}$, so $\varepsilon^2 \approx a^2_{\max}(\sigma_0 + \eta)/(8\eta G_\alpha w_k\langle\varphi_k^2\rangle s)$ and $\langle r^2\rangle = w_k^2\varepsilon^2\langle\varphi_k^2\rangle$. (vi) Substitution into the closed loop; the slaved error is Theorem J's Step 2. □

Status. (i): $F = \hat{J}\sigma_0(H_L + \sigma_0 I)^{-1}\partial_w x^0$ is exact at q, the bias formula at $w \neq q$ is first order in $x^0(w)$, and the law (B) is formal — the delta method, as Proposition D‴.9 — and holds for D‴.5-admissible samples; its value is the asymptotic one (on the instance the exact floor's expectation is 2.8, 1.5 and 1.1 times it at $m = 15$, 60, 240). (ii) proved inside the cells, and across crossings for vertex-selection solutions; (iii) proved for the stationary points, the probability bound by Hoeffding's inequality, the convergence to the exact representations numerical; (iv) as Theorem D‴.23, the boundary case formal; (v) a summary of (i)–(iii) with Proposition D‴.12 and Theorem D‴.23; (vi) the invariance proved, its attractivity in the real loop not proved here (it would follow from (iv) and Theorem D‴.18's argument for the joint system, as sketched in Theorem D‴.23(v)) and checked numerically. The amplitude–mass degeneracy is the homogeneity that the conic (Wasserstein–Fisher–Rao) view of mean-field networks exploits [16], [23]; the finite-size fluctuations of trained mean-field networks around their mean-field limit are known to scale like $1/m$, with central limit theorems for the training dynamics ([89], [91]), and the asymptotic deviation is known to vanish when the mean-field dynamics converges to a measure that interpolates the data (Chen–Rotskoff–Bruna–Vanden-Eijnden, [91]). That is the mechanism by which the adaptive prior removes the function-level floor: its replicated stationary points are unregularised interpolants, and an exact fit is weight-invariant, so the absence of a function-level floor in (iii) follows at once. (i) is the chain rule applied to Theorem D‴.23(iii)'s first-order bias at $x^0(q) = 0$, and its weight is the squared complement of the Tikhonov filter factor times the mode's visibility h_j, bounded by $\sigma_0/4$ by the AM–GM bound familiar from ridge regression's filter factors. New are the identification of the function-level floor with the weight-derivative of the regulariser's bias at the sampled weights, with its mode-by-mode law and the bound's identity with the adaptive prior's coupled-distance law; the split of the weight fluctuation between the parameters and the bias (iii); and a leak toward a learned prior as the way to restore the interpolation property under regularisation, which also removes the fixed prior's splitting instability. Checked: `tests/test_adaptive_floor.py` (the nearest-atom adaptive flow against finite differences of $\mathcal{J}_\circ$ with the cell masses; $\mathcal{J}_\circ$ the minimum of the OT energy over the masses, attained at the cell masses and strictly below the energies with masses moved by $\pm 1/m$; the bias formula against the finite-difference sensitivity, the mode terms summing to the law, the law below the bound, and the weight moving the shares toward $h \approx \sigma_0$ against the unweighted ones at a small and a large σ_0; at the D‴.5-admissible sample $(5, 4, 6)/15$ the exact representation, with the amplitudes A_k/w_k, stationary for the adaptive prior while the fixed prior's stationary point is a strict local minimiser and biased; at the saddle sample $(6, 5, 4)/15$ the fixed prior's splitting margin negative and the exact representation's margins σ_0; the parameter split; $\mathcal{J}_\circ$ decreasing across a cell crossing at $\eta = \sigma_0$ and ∞, and an empty cell staying empty with its atom frozen).

Numerics (`exp_adaptive_floor.py` → `results/chaos/adaptive_floor.md`). The three-bump instance at $\sigma_0 = 0.15\sigma_{\mathrm{crit}}$ (and $0.05\sigma_{\mathrm{crit}}$, $0.5\sigma_{\mathrm{crit}}$ for the law), the matching prior unless stated, the warm clouds of Proposition D‴.12's numerics (width 0.02, four seeds), $m = 15$, 60, 240 against $M = 510$; the checks at the sampled weights from the stored final states. *The*

floor is the bias (Part A). The function-level sensitivity F of the fixed prior's stationary point, by central differences in w, equals the bias form $\hat{J}\sigma_0(H_L + \sigma_0 I)^{-1}\partial_w x^0$ to the finite-difference error ($1.1 \cdot 10^{-6}$, $1.9 \cdot 10^{-7}$, $1.2 \cdot 10^{-8}$ at $\sigma_0/\sigma_{\text{crit}} = 0.05$, 0.15, 0.5, with the step 10^{-4}); the law (B) per unit of $1/m - 1/M$ is $1.356 \cdot 10^{-2}$, $6.166 \cdot 10^{-2}$, 0.2291, equal to the sum of its mode terms, at 0.496, 0.753, 0.839 of the bound, and the bound is $\sigma_0/4$ times the adaptive prior's coupled-distance law 46.32. Modes 2–6 carry 99.4%, 99.0%, 97.9% of the law, and 97.2% of the unweighted sum $\sum_j z_j^\top \tilde{C} z_j$ — the concentration is mostly the gauge's — while the weight moves the shares toward $h \approx \sigma_0$: the modes with $h_j/\sigma_0 \in [1/5, 5]$ carry 42%, 99%, 90% of the law against 21%, 97%, 78% unweighted. The law's value at $m = 15$, 60, 240 is $3.99 \cdot 10^{-3}$, $9.07 \cdot 10^{-4}$, $1.36 \cdot 10^{-4}$. The parameter split $S = H_L(H_L + \sigma_0 I)^{-1} S^e$ holds to $1.4 \cdot 10^{-7}$–$2.9 \cdot 10^{-8}$, and the adaptive prior's coupled-distance law is 1.44, 2.17, 5.27 times the fixed prior's. Every sampled weight vector is representable (margins $\min_k(w_k - |A_k|/a_{\max}) = 0.025$–$0.10$) except seed 3's at $m = 15$, $(8, 4, 3)/15$, which lies on the boundary ($w_3 = |A_3|/a_{\max}$). But 5 of the 12 sampled m-systems are not D‴.5-admissible for the fixed prior: $(6, 5, 4)/15$, $(6, 3, 6)/15$, $(24, 15, 21)/60$, $(8, 4, 3)/15$ and $(25, 17, 18)/60$ have splitting margins -3.48, -1.08, -2.16, -6.14, $-3.98 \cdot 10^{-3}$, all at class 1 with $w_1 \geq 0.4$; the other seven and the four references have margins $\geq 3.78 \cdot 10^{-3}$. *Fixed against adaptive priors* (Part B, the nearest-atom leak, 6000 slow units). Under the fixed prior $\|\hat{f}_m - \hat{f}_M\|^2$ stays at its stationary value, $3.0 \cdot 10^{-4}$–$5.5 \cdot 10^{-3}$ at $m = 15$, $1.1 \cdot 10^{-4}$–$9.9 \cdot 10^{-4}$ at $m = 60$ and $4.8 \cdot 10^{-6}$–$8.2 \cdot 10^{-5}$ at $m = 240$ over the seeds. At the seven D‴.5-admissible samples it equals the exact bias at the sampled weights, $\|\hat{f}_{\Theta^*(w^m)} - \hat{f}_{\Theta^*(w^M)}\|^2$, to four digits (ratio 1.0000), while the delta method's $\|F(w^m - w^M)\|^2$ misses it by the factors 0.86–1.72; the five saddle systems have split (class diameters 0.64–1.50 at the end, class 1 broken into sub-clusters) and sit at 0.81–1.01 of the replicated point's floor. Under the adaptive prior the function-level distance decreases without a floor: at $s = 6000$ to $4.6 \cdot 10^{-9}$–$2.0 \cdot 10^{-6}$ ($\eta = \sigma_0$) and $1.9 \cdot 10^{-10}$–$5.4 \cdot 10^{-8}$ ($\eta = \infty$) at $m = 15$, to $5.4 \cdot 10^{-10}$–$6.2 \cdot 10^{-9}$ and $2.4 \cdot 10^{-10}$–$6.2 \cdot 10^{-10}$ at $m = 60$, and to $2.5 \cdot 10^{-12}$–$1.9 \cdot 10^{-8}$ and $7.1 \cdot 10^{-14}$–$2.2 \cdot 10^{-9}$ at $m = 240$, every adaptive system replicated at the end and the class means converging to the exact representation of the sampled weights. *The test of (iii)* at the parameter level: the coupled distance D of the shared particles is 0.944–1.000 of the replicated $\Theta^*(w)$'s under the fixed prior and 0.951–1.000 of the exact representations' under the adaptive ones (both samples representable), and the adaptive prior's is 1.72–4.81 times the fixed prior's, sample by sample. $W^2(\mu_m, \mu_M)$ is dominated, under both priors, by the cross-class transport of the mass $\frac{1}{2}\|w^m - w^M\|_1$ — 0.82–0.99 of W^2 under the fixed prior, 0.67–0.99 under the adaptive ones — so the two priors' W^2 differ by the factors 1.00–1.32 only, while their within-class parts differ by 1.71–4.84. On the boundary sample the function-level distance decays algebraically ($6000 \cdot \|\hat{f}_m - \hat{f}_M\|^2 = 0.88$ at $\eta = \sigma_0$ and 0.42 at $\eta = \infty$, against the predicted 0.8 and 0.4), and its parameters run off ($D = 6.4$ and 7.8 at $s = 6000$). The counts are conserved in every run. *The design table* (Part C, the misplaced prior, 3000 slow units, seed means at $m = 15$, 60, 240). Nearest-atom leak: the fixed prior's fit error $6.8 \cdot 10^{-2}$, $3.1 \cdot 10^{-2}$, $2.0 \cdot 10^{-2}$ and floor $2.5 \cdot 10^{-3}$, $5.1 \cdot 10^{-4}$, $5.9 \cdot 10^{-5}$ — the same five samples are saddles at the misplaced prior (margins -2.87, -0.38, -1.52, -5.76, $-3.41 \cdot 10^{-3}$), four split by $s = 3000$ and the fifth partly — against the adaptive prior's ($\eta = \sigma_0$) $8.9 \cdot 10^{-3}$, $5.5 \cdot 10^{-4}$, $6.0 \cdot 10^{-4}$ and $8.0 \cdot 10^{-5}$, $1.6 \cdot 10^{-7}$, $1.1 \cdot 10^{-7}$ (at $m = 15$ one sample is on the boundary). OT leak: the fixed prior's $9.78 \cdot 10^{-3}$, the bias, with no floor (below $3 \cdot 10^{-21}$), against the adaptive prior's $1.6 \cdot 10^{-4}$ with the floor below $5 \cdot 10^{-11}$ (settled gate). At the coupling level the OT designs end with the largest D, 6.0, 3.5, 1.0 at $m = 15$, 60, 240 — the transit moves 7, 14, 26 of the m-systems' particles into another cell (over the four seeds) — against the nearest-atom leak's 0.86, 0.25, 0.017 (fixed) and 2.3, 0.48, 0.046 (adaptive), while their W^2 is at most $2.1 \cdot 10^{-8}$. The ranking of the fit errors depends on the horizon: until the settled gate opens ($s = 1211$–1304) the OT design toward the adaptive prior is the fixed one, and at $s = 1250$ the ungated nearest-atom adaptive prior leads at $m = 60$ and 240 ($3.8 \cdot 10^{-3}$ and $2.8 \cdot 10^{-3}$ against 9.2–$9.8 \cdot 10^{-3}$), not at $m = 15$ ($1.75 \cdot 10^{-2}$ in the seed mean); from $s \approx 2000$ the OT design leads (4.0–$4.2 \cdot 10^{-4}$ against

$1.2 \cdot 10^{-2}$, $1.6 \cdot 10^{-3}$, $1.4 \cdot 10^{-3}$). *The real loop* (Part D, $k_e = 25$, 3000 time units, seed 0's sampled weights). From the fixed prior's stationary point the rms tracking error stays at 1.17–$1.18 \cdot 10^{-3}$ ($m = 15$) and 3.43–$3.44 \cdot 10^{-4}$ ($m = 60$), at the end 1.001 and 0.996 times the averaged loop's $(1/k_e)\|f - \hat{f}\|_\rho$ at $s = t/k_e$ — the $m = 15$ run on a saddle orbit (Proposition D‴.15; margin $-3.48 \cdot 10^{-3}$), held there by the exact replication of its start; with the adaptive prior ($\eta = \infty$) it falls to $3.27 \cdot 10^{-4}$ and $1.52 \cdot 10^{-4}$ by $t = 3000$ ($s = 120$), along the averaged loop (ratios 0.987 and 0.992); from the exact representation of the sampled weights with the adaptive prior at its atoms it stays at the integrator's level (the five-period rms $9.0 \cdot 10^{-12}$ at the first window, falling to 1.4–$1.5 \cdot 10^{-12}$): perfect tracking is invariant in the real loop. *The instance's probabilities* (Part E, exact multinomial sums). Outside W: 0.361, 0.074, 0.0021 at $m = 15$, 60, 240, and on its boundary 0.118, 0.053, 0.0012, against Hoeffding's bound 1.87, 0.60, 0.036; a permutation of the classes would be representable for 31%, 87%, 100% of the mass outside W. The fixed prior's replicated point is a saddle with probability 0.369, 0.260, 0.109 — a failure set comparable to the adaptive prior's (outside W or on its boundary: 0.48, 0.13, 0.003) at $m = 15$ and much larger at $m \geq 60$. Taking the constrained minimum of the loss on the branch of $\vartheta^e(q)$ as the limit outside W (the dynamics from the warm clouds follows the branch; a local minimum, the global infimum being 0 wherever a permutation is representable), the adaptive prior's expected limiting floor is at least $7.3 \cdot 10^{-3}$, $2.2 \cdot 10^{-4}$, $6.6 \cdot 10^{-7}$: 1.8, 0.24, 0.005 times the fixed prior's law and 0.66, 0.16, 0.004 times its Monte Carlo expectation. *The law's accuracy* (Part F, 600 nested samples, the exact replicated stationary points). The fixed prior's $\mathbb{E}\|\hat{f}_m - \hat{f}_M\|^2$ is 2.78 ± 0.20, 1.53 ± 0.11, 1.11 ± 0.07 times the law (2.89 ± 0.26, 1.56 ± 0.15, 1.07 ± 0.07 for D‴.5-admissible pairs): the $o(1/m)$ remainder dominates at $m = 15$, where the gauge's derivative varies fast near the saturation (the matching amplitude 3 against $a_{\max} = 4$). The fixed prior's coupled distance follows its law more closely ($\mathbb{E}D$/law 0.68, 0.93, 0.95); the adaptive prior's (0.53, 1.17, 1.26) has a far heavier tail — its gauge diverges at the boundary of W; at $m = 240$ one sample lies at 55.7 times the law, and without it the mean is 1.17 — so its means are not reliable estimates at this sample size; it is 2.9–3.7 times the fixed prior's.

The planar check (`exp_adaptive_floor_2d.py` → `results/chaos/adaptive_floor_2d.md`). The planar plant of `exp_ot_2d.py` — four bumps with vector amplitudes, the Lissajous reference, $p = 5$, $G = \mathrm{diag}(1, 1, 0.25, 0.25, 0.25)$, $\sigma_0 = 0.05$ — with the averaged slow loop on $\mathbb{R}^2$ of `src/core/planar_loop.py`, the matching prior, $m = 16$, 32, 64 against $M = 128$, six seeds, 6000 slow units ($\eta = \infty$: the atoms reset to the cell means at every refresh of the cells, every slow unit, and frozen in between — a lagged prior). The excitation spectrum is $h_j/\sigma_0 = 0.004$–1.41 (twenty modes, all below $1.5\sigma_0$); F equals the bias form to the finite-difference error ($6.0 \cdot 10^{-7}$), and the law per unit of $1/m - 1/M$ is 0.310, at 0.215 of the bound, spread over the modes (the largest share 0.18). Representability needs $w > (0.200, 0.150, 0.175, 0.150)$, whose sum 0.675 leaves little room: 7 of the 18 m-systems are representable, with the six references. Every sample is D‴.5-admissible for the fixed prior (splitting margins $+0.016$ to $+0.047$), and the fixed prior's floor equals the exact bias at the sampled weights in all 18 systems (ratio 1.0000): $3.2 \cdot 10^{-4}$–$4.2 \cdot 10^{-3}$ at the representable systems, $3.0 \cdot 10^{-3}$–$2.1 \cdot 10^{-2}$ at the others. At the representable systems the adaptive prior's falls without a floor, to $1.0 \cdot 10^{-7}$–$4.1 \cdot 10^{-5}$ ($\eta = \sigma_0$) and $1.2 \cdot 10^{-9}$–$1.0 \cdot 10^{-5}$ ($\eta = \infty$) at $s = 6000$ — at most 0.019 and 0.006 of the fixed prior's, still falling in every system — and at the others to 0.023–0.64 of the fixed prior's, still falling, toward the constrained minimum of the loss at their sampled weights — a floor of their own (Proposition D‴.24(iii)) — not toward zero. The counts are conserved in every run. In the real loop ($k_e = 25$, $\mathrm{d}t = 2 \cdot 10^{-3}$, 2500 time units, seed 0, $m = 64$) the fixed prior's rms tracking error stays at the averaged prediction (ratios 0.993–1.009) and the adaptive prior's ($\eta = \infty$) follows the averaged decay (0.979–1.009); from the exact representation of the sampled weights with the adaptive prior at its atoms it stays at the integrator's level, 2.0–$4.7 \cdot 10^{-10}$.

Remark D‴.25 (the design table of the leak)**.** Propositions D‴.9, D‴.12, D‴.20, D‴.24 and

Theorem D‴.23 choose, for the averaged loop near a replicated configuration, what the leak fixes — the prior's atoms and its masses — and what each choice costs. With a matching prior unless a misplaced one is named, at commensurate sizes for the OT leaks, and with the weights w sampled from an i.i.d. warm start ($\delta w := w^m - w^M$; Table 19):

Table 19: Remark D‴.25: the design table of the leak for the averaged loop near a replicated configuration — for each leak and prior, the function-level floor, the coupled distance D of Proposition D‴.5, W^2 at the measure level, the bias of a misplaced prior, the transit, the collective rates and the a-priori bound.

leak, prior	function-level floor	coupled distance D (Proposition D‴.5)	W^2 (measure level)	bias of a misplaced prior	transit	collective rates	a-priori bound
nearest atom, fixed (D‴.9)	$(1/m - 1/M)$ $\sum_j h_j \sigma_0^2/(h_j+\sigma_0)^2$ $z_j^\top \tilde{C} z_j$ at D‴.5-admissible w (D‴.24(i))	D‴.5's law with $H_L(H_L + \sigma_0 I)^{-1} S^e$	the transport of $\frac{1}{2}\|\delta w\|_1$ between the clusters, plus $O(\|\delta w\|^2)$	$\sigma_0(H_L + \sigma_0 I)^{-1} x^0$	none while no particle crosses a cell boundary (D‴.9(i)'s hypothesis)	$h_j + \sigma_0$	yes
nearest atom, adaptive (D‴.24)	0 at representable w	D‴.5's law with S^e (larger), at representable w	likewise, a larger within-class part	0	none (likewise)	$\lambda_j^-(w) < \min\{h_j(w), \eta\}$	no (yes when projected, D‴.27)
OT, fixed (D‴.12)	0	the transit's mismatch	0	$\sigma_0(H_L + \sigma_0 I)^{-1} x^0$	the excess moved by the plan	$h_j + \sigma_0$	yes
OT, adaptive (D‴.23)	0	the transit's mismatch	0	0	gated: adapt after the fixed-prior relaxation	$\lambda_j^-(q)$	no (yes when projected, D‴.27)
entropic OT, fixed (D‴.20)	D‴.9's function-level law at $\Theta_\varepsilon^*(q)$ for sampled excesses below $\delta_c(\varepsilon)$	D‴.5's law there	—	that of $\Theta_\varepsilon^*(q)$	equalises the masses down to $\delta_c(\varepsilon)$	—	yes

Reading. The finite-size floor has three faces: the function level (the sample's weights acting through the leak's bias), the coupling level (the shared particles displaced by the amplitude–mass gauge, or relabelled by the transit), and the measure level (the mass $\frac{1}{2}\|\delta w\|_1$ that has to move between the clusters). The mass constraint removes the function-level and the measure-level floors by fixing the weights; at the coupling level it replaces the gauge's displacement by the transit's mismatch, which relabels different particles in the two systems. The adaptive prior removes the function-level floor by removing the bias at every representable weight, enlarges the coupled distance (the gauge's whole displacement goes into the parameters) and leaves the measure level's leading term unchanged; it also removes the fixed prior's splitting instability, every separation relaxing at σ_0 at an exact representation. The bias of a misplaced prior is removed only by an adaptive prior, and every adaptive prior pays with the rate — $\lambda_j^- < h_j$, so a weakly excited collective mode converges at the loss's own curvature rather than at the leak's — and with the leak's a-priori bound on the parameters (Theorem D‴.23(i); restored by projecting the prior onto a compact convex set, Proposition D‴.27, at the price of a bias where the limit

reaches its boundary). The design without a function-level floor or bias is the OT leak toward an adaptive prior started from the fixed-prior stationary point; the floor-free design when the prior is trusted is the OT leak toward a fixed one; and when the masses cannot be prescribed (the sample decides them), the nearest-atom leak toward an adaptive prior keeps the fitted function exact at the representable samples and leaves the fluctuation in the parameters.

Corollary D‴.26 (perfect tracking with an adaptive prior in the real loop: the Floquet spectrum, uniformly in m)**.** *Let* (CL) *be the real closed loop of Theorem J with the nearest-atom leak toward an adaptive prior — the particles leak at $\sigma = \sigma_0/k_e$ toward the atom of their class, and the atoms follow the class means at the rate η/k_e, $\dot z_j = -(\eta/k_e)(z_j - \bar\vartheta_j)$, $0 < \eta \le \infty$ ($\eta = \infty$: $z_j = \bar\vartheta_j$) — and let $w = N/m$ be representable sampled weights ($w \in W$, Proposition D″.24) with the exact representation $\vartheta^e(w)$. Let Γ_w be the lift of $\vartheta^e(w)$ by the classes, with $Z = \vartheta^e(w)$ and the plant on the reference: a T_0-periodic solution with $e \equiv 0$ (Proposition D″.24(vi)). Its Floquet exponents (the rates $-\log|\lambda|/T_0$ of its monodromy) are:*

(a) *σ_0/k_e exactly, with multiplicity $(m - K_0)p$, on the separation subspaces of the classes;*

(b) *on the collapsed subspace, the exponents of the weighted K_0-atom loop's periodic solution at $\vartheta^e(w)$ — the same for every m with these weights;*

(c) *if moreover $\partial_x \hat f_{\vartheta^e(w)} = \partial_x f$ on the reference orbit ($\hat f = f$ on a neighbourhood of it — on the instances everywhere, the exact representation being an identity of functions) and $x_m \in C^4$: n exponents $k_e + O(1/k_e)$ of the error channel, summing to $nk_e - \operatorname{tr} H_L(w)/k_e + O(k_e^{-3})$, and for the $K_0 p$ collective pairs*

$$(1/k_e)\big[\lambda_j^{\pm}(w) + c_j^{\pm}(w)/k_e^2 + O(k_e^{-4})\big],$$

$\lambda_j^{\pm}(w)$ the rates of Theorem D″.23(ii) at the weights w ($\eta = \infty$: $h_j(w)$) and $c_j^{\pm}(w)$ the explicit coefficients of the proof (for simple eigenvalues; at a degenerate one only the rates are even). The odd orders vanish because the slow block at $-\varepsilon$ ($\varepsilon := 1/k_e$) is the time reversal of the reversed reference's at ε, so with the variational equation's symmetry the slow generator's spectrum is even in ε — an identity (Lemma D″.26a(iv)) that a derivative mismatch $a \not\equiv 0$ breaks; term by term, every term of $\bar A_k$ has the parity $(-1)^k$ under reversal, the first order in particular because the first-order averaging correction is skew. c_j is the sum of the error filter's attenuation $-(\mathrm{d}\lambda_j^{\pm}/\mathrm{d}h)v_j^\top\langle\dot{\hat J}^\top\dot{\hat J}\rangle v_j \le 0$ (at $\eta = \infty$ it is $-h_j\omega_j^2$, $\omega_j^2 = \sum_n n^2|\hat J_n v_j|^2/\sum_n|\hat J_n v_j|^2$ the mode's mean squared frequency in the regressor, by Parseval, independent of η: the averaged rate of a mode holds when $k_e \gg \omega_j$), of the slaving's part from $-(\bar J^\top\bar J)^2$, which speeds the modes up, and of the averaging terms, of either sign (the level repulsion through $\bar A_1$ is ≥ 0 for the slowest mode): its sign is not fixed — on the instances every $c_j < 0$, the filter's part dominating, and with the adaptation gain scaled up some $c_j > 0$.

Hence, where $H_L(w) \succ 0$ (so $\lambda_1^-(w) > 0$) and for k_e above a threshold — which depends on the spectral gaps of the collective block through Lemma D″.26a's ε_0 —, perfect tracking is exponentially stable (locally, for each m) at every representable sample, with the rate $(1/k_e)\min\{\sigma_0, \lambda_1^-(w) + c_1^-(w)/k_e^2 + O(k_e^{-4})\}$; the same holds for the OT leak toward an adaptive prior (Theorem D‴.23) at a commensurate m, with $w = q$ and the diagonal plan (locally constant at the lift). Without the derivative condition the mismatch $a(t) := \partial_x(f - \hat f)(x_m(t))$ enters the error equation and moves the collective rates at first order in $1/k_e$ (the proof).

Proof. Along Γ_w the linearisation of (CL) reads $\delta\dot e = (a(t) - k_e)\delta e - D\hat f[\delta\Theta](x_m(t))$, $\delta\dot\vartheta_i = G\nabla_\vartheta\tilde\Phi(x_m(t);\vartheta^e_{\pi(i)})^\top\delta e - (\sigma_0/k_e)(\delta\vartheta_i - \delta z_{\pi(i)})$, $\delta\dot z_j = -(\eta/k_e)(\delta z_j - \delta\bar\vartheta_j)$, with $a \equiv 0$ under the derivative condition: the second-derivative term $G\nabla^2_\vartheta(e^\top\tilde\Phi)\delta\vartheta_i$ and the x-derivative of the learning law vanish because $e \equiv 0$. (a) A separation (zero class means, $\delta z = 0$, $\delta e = 0$) does not move $\hat f$ at first order — $D\hat f[\delta\Theta] = \sum_k (1/m)\nabla_\vartheta\tilde\Phi(\cdot\,;\vartheta^e_k)\sum_{i\in\pi^{-1}(k)}\delta\vartheta_i = 0$ — so it stays

a separation with $\delta\dot\vartheta_i = -(\sigma_0/k_e)\delta\vartheta_i$: the subspace is invariant and the exponent is σ_0/k_e, whatever a. (b) The collapsed subspace is invariant (Corollary D‴.14(i), with the prior's equation: the class atoms and their prior atoms follow the weighted K_0-atom equations), and the loop on it is the weighted K_0-atom loop with its prior. (c) In the whitened weighted coordinates $y_k = \sqrt{w_k}G^{-1/2}\delta\vartheta_k$ (and ζ_k likewise for δz_k) the collapsed variational equation is the adjoint pair $\delta\dot e = (a-k_e)\delta e - \hat J(t)y$, $\dot y = \hat J(t)^\top\delta e - (\sigma_0/k_e)(y-\zeta)$, $\dot\zeta = -(\eta/k_e)(\zeta-y)$, with $\hat J(t) = [\sqrt{w_k}(G^{1/2}\nabla_\vartheta\tilde\Phi(x_m(t);\vartheta_k^e))^\top]_k$ ($\eta=\infty$: $\zeta = y$, and the leak vanishes on the collapsed subspace). Let $\varepsilon = 1/k_e$, $Y = (y,\zeta)$, $\bar J = \hat J$ on the y block and C the leak and the prior's constant block in slow time. The error channel is slaved: the linear periodic system has the invariant subspace $\delta e = L(t)Y$, $L = \varepsilon L_1 + \varepsilon^2 L_2 + \varepsilon^3 L_3 + \dots$, with $L_1 = -\bar J$, $L_2 = \dot{\bar J} - a\bar J$ and, for $a \equiv 0$, $L_3 = -\ddot{\bar J} + \bar J C - \bar J\bar J^\top\bar J$ (the Chang transformation, [49]: Kokotović–Khalil–O'Reilly). This leaves, in slow time $s = \varepsilon t$, the slow block $\mathrm{d}Y/\mathrm{d}s = (B_0 + \varepsilon B_1 + \varepsilon^2 B_2 + \dots)Y$ with $B_0 = C - \bar J^\top\bar J$, $B_1 = \bar J^\top(\dot{\bar J} - a\bar J)$ and, for $a \equiv 0$, $B_2 = -\bar J^\top\ddot{\bar J} + \bar J^\top\bar J C - (\bar J^\top\bar J)^2$, a system of fast period εT_0. Its near-identity averaging gives the averaged generator $\bar A = \bar A_0 + \varepsilon\bar A_1 + \varepsilon^2\bar A_2 + \dots$, whose eigenvalues are the slow-time exponents: $\bar A_0 = \langle B_0\rangle = C - \langle\bar J^\top\bar J\rangle$, the averaged loop's linearisation at the replicated configuration (Theorem D‴.23(ii) at the weights w, Proposition D‴.24(iv)), self-adjoint in the metric $\mathrm{diag}(I,(\sigma_0/\eta)I)$ of Theorem D‴.23(i); $\bar A_1 = \langle B_1\rangle + \langle\tilde B_0 U_1\rangle$ with $U_1' = \tilde B_0$ (zero means); $\bar A_2 = \langle B_2\rangle + \langle B_1U_1\rangle + \langle B_0U_2\rangle$ with $U_2' = B_1 + B_0U_1 - U_1\bar A_0 - \bar A_1$. With $a \equiv 0$, $\bar A_1$ is skew: the self-adjoint parts of $\langle\bar J^\top\dot{\bar J}\rangle$ and $\langle\tilde B_0U_1\rangle$ are $\frac12\langle(\bar J^\top\bar J)^\cdot\rangle = 0$ and $\frac12\langle(U_1^2)'\rangle = 0$, and a skew perturbation moves no simple eigenvalue of a self-adjoint matrix at first order ($v^\top Kv = 0$; on a degenerate eigenspace it splits the eigenvalue into a pair of unchanged real part). All odd orders vanish: the variational matrix $M(t)$ satisfies $M(t)^\top = DM(t)D^{-1}$ with $D = \mathrm{diag}(-I_n, I, (\sigma_0/\eta)I)$ (at $\eta=\infty$ on $(\delta e, y)$, $D = \mathrm{diag}(-I_n, I)$), so the system with its coefficients reversed in time, $M(-t)$, has the same Floquet multipliers (its monodromy is conjugate to the inverse of the adjoint system's, which is the inverse transpose of the original's); reversing time changes the sign of every fast-time derivative and antiderivative, and a term of $\bar A_k$ carries a number of them of the parity of k (ε is a time, $\hat J$ and $\mathrm{d}/\mathrm{d}t$ are rates, C a rate in slow time), so $\bar A_k \mapsto (-1)^k\bar A_k$, and the exponents, being unchanged, are even in ε — for simple eigenvalues of $\bar A_0$; at a degenerate one the first-order splitting $\pm i\varepsilon\kappa$ is imaginary and only the rates are even. The symmetry $M(t)^\top = DM(t)D^{-1}$ alone does not suffice: it persists with $a \not\equiv 0$, which enters only the $(\delta e, \delta e)$ entry, while a breaks the parity count. (The mechanism is that of the even error expansions of symmetric integrators and of the time symmetry of the Magnus exponent, [99], [95].) Perturbation theory gives $c_j = -\ell_j\bar A_2 r_j - \sum_{i\neq j}(\ell_j\bar A_1 r_i)(\ell_i\bar A_1 r_j)/(\lambda_i - \lambda_j)$ (r_j, ℓ_j the right and left eigenvectors of $-\bar A_0$, $\ell_j r_j = 1$), and $\langle B_2\rangle$ contains $-\langle\bar J^\top\ddot{\bar J}\rangle = \langle\dot{\bar J}^\top\dot{\bar J}\rangle$: the error filter $k_e/(k_e + i\omega)$ attenuates the regressor's harmonics, $H_L - \varepsilon^2\langle\dot{\hat J}^\top\dot{\hat J}\rangle + O(\varepsilon^4)$ in place of H_L; $\langle B_2\rangle$ also contains $-\langle(\bar J^\top\bar J)^2\rangle$, from the slaving's L_3, which enters c_j with the opposite sign, and the level repulsion $-\sum_{i\neq j}(\ell_j\bar A_1 r_i)(\ell_i\bar A_1 r_j)/(\lambda_i - \lambda_j) = \sum_{i\neq j}|\ell_j\bar A_1 r_i|^2/(\lambda_i - \lambda_j)$ ($\bar A_1$ skew) is ≥ 0 for the slowest mode, so the sign of c_j is not fixed. The error channel's exponents are $k_e + O(\varepsilon)$, and by Liouville's formula all the exponents sum to $-\langle\mathrm{tr}\,M\rangle = nk_e + (\sigma_0+\eta)K_0p/k_e$ ($\eta=\infty$: nk_e), the collective ones to $(1/k_e)\,\mathrm{tr}(-\bar A_0) + O(k_e^{-3})$. With $a \not\equiv 0$ the term $-\langle\bar J^\top a\bar J\rangle$ of $\bar A_1$ is self-adjoint and moves the rates at first order, and a — a rate that does not change sign under time reversal — breaks the parity. □

Status. (a) and (b) proved, exactly (no averaging); (c) proved by Lemma D‴.26a for $x_m \in C^4$ — the slaving is an exact block-triangularisation, the expansion has a bounded $O(k_e^{-4})$ remainder at simple eigenvalues (the threshold and the constant depending on the spectral gaps) and the parity is the exact identity $\mathrm{eig}\,\bar A(-\varepsilon) = \mathrm{eig}\,\bar A(\varepsilon)$, the stability statement under $H_L(w) \succ 0$ — with the structure (the skewness of $\bar A_1$, c_j explicit) derived here; the stability consequence local, for each m (the nearest-atom cells are constant near the lift, the particles lying inside their cells, so the field is smooth there and the multipliers decide). (a) and (b) are Corollary D‴.14(ii) and Proposition D‴.15 at zero residual with the prior's

coordinates added; what the adaptive prior brings is the zero-residual lift at every representable w, not only at $w = q$ under the OT leak. The basin uniform in m in the per-particle norm would follow from Theorem D‴.18's contraction argument for the joint system, as sketched in Theorem D‴.23(v), and is not proved here. With a fixed prior the lift of $\vartheta^e(w)$ is a solution only when the prior sits at $\vartheta^e(w)$ — at a matching prior, $w = q$; at the other sampled weights the leak's bias keeps $e = O(1/k_e)$ (Theorem J) — so perfect tracking at every representable sample, with this spectrum, is specific to the adaptive prior. (a) does not depend on the fixed prior's splitting margin at w: at $(6, 5, 4)/15$, where the fixed prior's replicated orbit is a saddle orbit (Proposition D‴.24's numerics), every separation of perfect tracking decays at σ_0/k_e. Checked: `tests/test_adaptive_floquet.py` (at the matching weights, $\eta = \infty$, $m = 6$: the return of one period, the nine separation exponents at σ_0/k_e, the collective exponents the K_0-atom loop's, within the averaging correction of the prediction; at the weights $(2, 1, 1)/4$ with the prior in the state, $\eta = \sigma_0$, $m = 4$: the real loop's Jacobian at the lift mapping the separations to $-(\sigma_0/k_e)$ times themselves and the collapsed subspace into itself as the collapsed variational equation, the monodromy's 3 separation exponents at σ_0/k_e and its 18 collective ones the variational ones, within the second-order prediction and below the first-order one; on an asymmetric reference: $\bar{A}_1$ non-zero and skew, the residual of the second-order prediction falling like k_e^{-4}, the measured second-order coefficient the explicit c_1, the error channel by Liouville's formula).

Numerics (`exp_adaptive_floquet.py` → `results/chaos/adaptive_floquet.md`). The three-bump instance at $\sigma_0 = 0.15\sigma_{\text{crit}}$, $x_m = 2\sin t$, the sampled weights $w = (6, 5, 4)/15$ (the amplitudes $A_k/w_k = 2.5, -2.1, 3.0$; representable, and a saddle of the fixed prior, splitting margin $-3.48 \cdot 10^{-3}$), $\eta = \sigma_0$ and ∞, $k_e = 25$ and 50; the monodromy by central differences of the period map (step 10^{-5} relative, DOP853 at rtol 10^{-13}). Every lift returns after one period to $\le 1.7 \cdot 10^{-15}$. (a) At $m = 15$ and 30 the $(m-3) \cdot 3 = 36$ and 81 separation exponents equal σ_0/k_e to $\le 1.7 \cdot 10^{-6}$ (relative). (b) The other 18 ($\eta = \sigma_0$) or 9 ($\eta = \infty$) exponents equal the weighted K_0-atom loop's to $1.4 \cdot 10^{-5}$–$6.4 \cdot 10^{-4}$ (relative, the resolution of the differences for the slowest of them). (c) Part C integrates the variational equation on the collapsed subspace over a period (its generator equals the central differences of the real loop's field to $\le 5.8 \cdot 10^{-11}$ relative, its exponents the differences' to $\le 4.1 \cdot 10^{-5}$) at $k_e = 12.5$, 25, …, 400. At $\eta = \sigma_0$ the slowest exponent lies 37.5%, 13.6%, 3.84%, 0.99%, 0.25% and 0.063% below $\lambda_1^-(w)/k_e$ — every exponent lies below its first-order prediction at every k_e, a property of the instance — and the largest deviation falls by 2.77, 3.53, 3.86, 3.96, 3.99 per doubling of k_e ($\to 4$), while the residual of the second-order prediction falls by 11.1, 14.0, 15.4, 15.8, 16.0 ($\to 16$: $O(k_e^{-4})$; $4.9 \cdot 10^{-7}$ at $k_e = 400$). Every c_j is negative here (c_j/λ_j from -0.47 to -100.6); the slowest mode's $c_1/\lambda_1 = -100.6$ (-101.7 at $\eta = \infty$), of which the error filter's part is $-102.1 = -0.989\omega_1^2$ ($-103.3 = -\omega_1^2$ at $\eta = \infty$), $\omega_1 = 10.16$ the slowest mode's root-mean-square frequency in the regressor — the weakest-excited mode is carried by the regressor's high harmonics, which the error filter attenuates unless $k_e \gg \omega_1$ — and the measured $(k_e\mu_1 - \lambda_1)k_e^2/\lambda_1$ approaches it (-99.4, -100.3, -100.5 at $k_e = 100$, 200, 400; μ_1 the slowest exponent). On $x_m = 2\sin t$ the first-order correction vanishes identically (in one dimension $\langle \hat{J}^\top \dot{\hat{J}} \rangle$ is the integral of an exact differential, and the commutator term vanishes by the symmetry $t \to \pi - t$), so this reference alone could not test the mechanism; on the asymmetric $x_m = 2\sin t + 0.8\sin 2t$ it does not vanish ($\|\bar{A}_1\| = 5.6 \cdot 10^{-3}$, its self-adjoint part $1.4 \cdot 10^{-16}$), and the deviations still fall by $\to 4$ per doubling and the second-order residuals by $\to 16$ (15.5 and 15.9 at the last two doublings), with $c_1/\lambda_1 = -108.2$ (the filter's part $-109.7 = -0.984\omega_1^2$, $\omega_1 = 10.56$). With the adaptation gain scaled by 30 two coefficients are positive ($c_8/\lambda_8 = +4.17$, $c_9/\lambda_9 = +12.65$) and those exponents lie above their first-order predictions by $c_j/(\lambda_j k_e^2)$ to three digits ($+4.17 \cdot 10^{-4}$ and $+1.27 \cdot 10^{-3}$ at $k_e = 100$), the slowest still below. At $\eta = 10\sigma_0$, where the metric $\mathrm{diag}(I, (\sigma_0/\eta)I)$ is not the identity, $\bar{A}_0$ is self-adjoint in it (to $8.7 \cdot 10^{-18}$; in I its asymmetry is $6.4 \cdot 10^{-2}$) and $\bar{A}_1$ skew (to $1.4 \cdot 10^{-16}$). The error channel's exponent by Liouville's formula is within $O(k_e^{-3})$ of $k_e - \operatorname{tr} H_L(w)/k_e$ (24.98084 against 24.98037 at $k_e = 25$).

Lemma D‴.26a (the exact slow subspace of the perfect-tracking linearisation: Corollary D‴.26(c) and Proposition D‴.30(ii) proved for the linearised loop)**.** *Let $K:\mathbb{R}\to\mathbb{R}^{n\times q}$ be T_0-periodic and C^4, $\Lambda\in\mathbb{R}^{q\times q}$, $a:\mathbb{R}\to\mathbb{R}$ T_0-periodic and C^1, $N_0\ge 0$, and for real $\varepsilon\neq 0$ consider*

$$\begin{aligned}\varepsilon\,\mathrm{d}w &= -(w+K(t)Y)\,\mathrm{d}t-(1+\varepsilon a(t))\varepsilon^{-1}\sqrt{N_0}\,\mathrm{d}W,\\ \mathrm{d}Y &= \varepsilon(K(t)^{\top}w-\Lambda Y)\,\mathrm{d}t+K(t)^{\top}\sqrt{N_0}\,\mathrm{d}W\end{aligned}\tag{$\bigstar$}$$

(W a standard n-dimensional Wiener process). At $\varepsilon=1/k_e>0$ and $N_0=0$ this is the collapsed variational equation of Corollary D‴.26 under its derivative condition — $\delta e=\varepsilon w$, $K=[\hat J,0]$, $Y=(y,\zeta)$, $\Lambda=\left[\begin{smallmatrix}\sigma_0 I & -\sigma_0 I\\ -\eta I & \eta I\end{smallmatrix}\right]$ for $0<\eta<\infty$ (at $\eta=\infty$: $K=\hat J$, $Y=y$, $\Lambda=0$; at $\eta=0$, the fixed prior at the atoms: $K=\hat J$, $Y=y$, $\Lambda=\sigma_0 I$ — with the pair's Λ the prior's block would have a zero rate there) — and at $N_0>0$ Proposition D‴.30(ii)'s linearised loop under a white measurement noise of density N_0 (the error equation's input $-(k_e+a)\nu$, $a=\partial_x\hat f$ on the orbit, the law's $\hat J^{\top}\nu$). $K\in C^4$ holds when $x_m\in C^4$. There is $\varepsilon_0>0$, depending on the C^4 norms of K, on $|\Lambda|$, $\|a\|_{C^1}$ and T_0 — and, for (iv), on the distance of $-\lambda_j$ from the rest of the spectrum of $\bar A_0$ and on the condition of its eigenvector, for (v) on $\bar A_0$'s stability margin, as do the constants of the remainders there — such that for $0<|\varepsilon|\le\varepsilon_0$:

(i) (the slow subspace) *The equation $\varepsilon\dot P+P=-K-\varepsilon^2P(K^{\top}P-\Lambda)$ has a unique T_0-periodic solution $P(\cdot\,;\varepsilon)$ with $\|P+K\|_\infty=O(\varepsilon)$, and $P=\sum_{k\le 3}\varepsilon^k P_k+O(\varepsilon^4)$ uniformly in t, $P_0=-K$, $P_1=\dot K$, $P_2=-(\ddot K+KK^{\top}K+K\Lambda)$, $P_k=-\dot P_{k-1}-\sum_{i+j=k-2}P_iK^{\top}P_j+P_{k-2}\Lambda$. In the coordinates $v:=w-P(t)Y$ the deterministic part of ($\bigstar$) is block-triangular,*

$$\varepsilon\dot v=-(I+\varepsilon^2PK^{\top})v,\qquad \dot Y=\varepsilon A(t)Y+\varepsilon K^{\top}v,$$
$$A:=K^{\top}P-\Lambda=-K^{\top}K-\Lambda+\varepsilon K^{\top}\dot K-\varepsilon^2K^{\top}(\ddot K+KK^{\top}K+K\Lambda)+O(\varepsilon^3).$$

(ii) (the slow multipliers, exactly) *For $\varepsilon>0$, q Floquet multipliers of ($\bigstar$) are those of the slow block $\dot Y=\varepsilon A(t)Y$, of modulus $\ge e^{-\varepsilon T_0\|A\|_\infty}$, and the other n those of the fast block, of modulus $\le e^{-T_0/(2\varepsilon)}$; the fast exponents sum to $n/\varepsilon+\varepsilon\langle\operatorname{tr}PK^{\top}\rangle=nk_e-\langle\operatorname{tr}K^{\top}K\rangle/k_e+O(k_e^{-3})$.*

(iii) (the slow generator) *Let $\Phi(\varepsilon)$ be the slow block's monodromy and $\bar A(\varepsilon):=\log\Phi(\varepsilon)/(\varepsilon T_0)$ (the principal logarithm). Then $\bar A(\varepsilon)=\bar A_0+\varepsilon\bar A_1+\varepsilon^2\bar A_2+\varepsilon^3\bar A_3+O(\varepsilon^4)$ uniformly in $|\varepsilon|\le\varepsilon_0$, with $\bar A_0=-\langle K^{\top}K\rangle-\Lambda$, $\bar A_1=\langle K^{\top}\dot K\rangle+(1/(2T_0))\int_0^{T_0}\int_0^{t_1}[A_0(t_1),A_0(t_2)]\,\mathrm{d}t_2\,\mathrm{d}t_1$ ($A_0:=-K^{\top}K-\Lambda$) and the higher coefficients those of the Magnus series of $\varepsilon A_{(3)}$ collected by powers of ε ($A_{(3)}:=K^{\top}P_{(3)}-\Lambda$, $P_{(3)}:=\sum_{k\le 3}\varepsilon^kP_k$, a polynomial in ε; A itself has only an asymptotic expansion); the full Magnus series of εA converges where $\varepsilon T_0\|A\|_\infty<\pi$, a sufficient form of Moan–Niesen's $\int_0^{T_0}\|\varepsilon A\|_2<\pi$ ([102]). The averaged generator of Corollary D‴.26(c)'s near-identity averaging, carried there to $\bar A_2$, is conjugate to $\bar A(\varepsilon)$ up to $O(\varepsilon^3)$ — a conjugate generator, with the same eigenvalues to that order; its coefficients differ from $\bar A(\varepsilon)$'s Taylor coefficients by commutators with $\bar A_0$.*

(iv) (the rates, and their parity proved) *At a simple real eigenvalue $-\lambda_j$ of $\bar A_0$ the slow-time rate $\mu_j(\varepsilon):=-(\varepsilon T_0)^{-1}\log|\nu_j(\varepsilon)|$ of the continued multiplier ν_j is $\mu_j(\varepsilon)=\lambda_j+\varepsilon\mu_j^1+\varepsilon^2\mu_j^2+\varepsilon^3\mu_j^3+O(\varepsilon^4)$, the coefficients those of perturbation theory for $\bar A(\varepsilon)$ ($\mu_j^1=-\ell_j\bar A_1r_j$ and $\mu_j^2=c_j$ of Corollary D‴.26(c)). If the full matrix $M_\varepsilon(t)$ of ($\bigstar$) satisfies $M_\varepsilon(t)^{\top}=DM_\varepsilon(t)D^{-1}$ for an invertible D — Corollary D‴.26's symmetry, $D=\operatorname{diag}(-\varepsilon^2I,I,(\sigma_0/\eta)I)$ in (w,y,ζ) ($\operatorname{diag}(-\varepsilon^2I,I)$ at $\eta=\infty$ and at $\eta=0$), which persists with a mismatch in the error channel — and a mismatch does not enter the error channel (it breaks the parity, below), then the spectrum of $\bar A(\varepsilon)$ is even: $\operatorname{eig}\bar A(-\varepsilon)=\operatorname{eig}\bar A(\varepsilon)$ as multisets of complex numbers, the slow block at $-\varepsilon$ defined by (i). In particular $\mu_j(-\varepsilon)=\mu_j(\varepsilon)$ exactly, $\mu_j^1=\mu_j^3=0$ and $\mu_j(\varepsilon)=\lambda_j+c_j\varepsilon^2+O(\varepsilon^4)$, with a bounded remainder, and at a degenerate eigenvalue the rates are even.*

(v) (the slow noise) *For $0<\varepsilon\le\varepsilon_0$ the anti-causal equation $S-\varepsilon\dot S=\varepsilon K^\top-\varepsilon^2(SPK^\top+AS)$ has a unique T_0-periodic solution with $S=\varepsilon K^\top+\varepsilon^2\dot K^\top+O(\varepsilon^3)$, and $Y':=Y+\varepsilon S(t)v$ solves exactly*

$$\begin{aligned}\mathrm{d}Y' &= \varepsilon A(t)Y'\,\mathrm{d}t+\varepsilon b(t)\sqrt{N_0}\,\mathrm{d}W,\\ b &:= \varepsilon^{-1}[K^\top-\varepsilon^{-1}(1+\varepsilon a)S-\varepsilon SPK^\top]\\ &= -(\dot K+aK)^\top-\varepsilon(\ddot K+KK^\top K+K\Lambda^\top+a\dot K)^\top+O(\varepsilon^2):\end{aligned}$$

the slow coordinates' drift is decoupled from the fast ones, and their noise is $\varepsilon b(t)\nu$ — at leading order the high-passed $-\varepsilon(\dot K+aK)^\top\nu$; the same Wiener process drives v, so $\mathbb{E}[vY'^\top]\neq0$ — with the diffusion $\varepsilon^2N_0bb^\top$ per unit time (in the slow time $\varepsilon N_0\langle bb^\top\rangle=(N_0/k_e)Q+O(k_e^{-2})$, $Q:=\langle(\dot K+aK)^\top(\dot K+aK)\rangle$). If $\bar A_0$ is Hurwitz, the periodic stationary second moments are $\mathrm{Cov}(v)=(N_0/(2\varepsilon^3))(I+O(\varepsilon))$, $\mathbb{E}[vY'^\top]=-N_0b^\top+O(\varepsilon N_0)$ and $\mathrm{Cov}(Y')=\bar\Sigma+O(\varepsilon^2N_0)$, $\bar\Sigma$ the solution of $\bar A_0\bar\Sigma+\bar\Sigma\bar A_0^\top+\varepsilon N_0\bar Q=0$ ($\bar Q=Q$ on the y block, $\bar\Sigma=O(\varepsilon N_0)$); hence $\mathrm{Cov}(Y)(t)-\mathrm{Cov}(Y')(t)=(\varepsilon N_0/2)K(t)^\top K(t)-(\varepsilon^2N_0/2)(\mathrm{d}(K^\top K)/\mathrm{d}t+2aK^\top K)(t)+O(\varepsilon^3N_0)$ and $\mathrm{Cov}(Y)(t)=\bar\Sigma+(\varepsilon N_0/2)K(t)^\top K(t)+O(\varepsilon^2N_0)$ — the slow part plus the fast jitter — and Proposition D‴.30(ii)'s floor, $(N_0/2k_e)[\sum_j Q_{jj}(h_j+\eta)/(h_j+\eta+\sigma_0)+\mathrm{tr}\,H^2]$, holds for the linearised loop under white noise as a period average, with a relative remainder $O(1/k_e)$ (at $\eta=0$ with $\Lambda=\sigma_0I$, whose filter factor is $h_j/(h_j+\sigma_0)$): at time t the fast part is $(N_0/2k_e)\,\mathrm{tr}(H\hat J(t)^\top\hat J(t))$, whose oscillation over the period is of the floor's own order.

Proof. (i) On the Banach space of continuous T_0-periodic matrix functions (sup norm) the periodic solution of $\varepsilon\dot u+u=g$ is $\mathcal{G}g(t)=\int_0^\infty e^{-\sigma}g(t-\varepsilon\sigma)\,\mathrm{d}\sigma$ — for $\varepsilon<0$ the anti-causal $\int_0^\infty e^{-\sigma}g(t+|\varepsilon|\sigma)\,\mathrm{d}\sigma$ —, so $\|\mathcal{G}g\|\le\|g\|$, and $\mathcal{G}g-g=-\varepsilon\mathcal{G}\dot g$. The equation is $P=\mathcal{G}F(P)$, $F(P):=-K-\varepsilon^2P(K^\top P-\Lambda)$. With $r:=2|\varepsilon|(\|\dot K\|+\beta)$, $\beta:=(\|K\|+1)(\|K\|(\|K\|+1)+|\Lambda|)$, the ball $\|P+K\|\le r\le1$ is mapped into itself ($\|\mathcal{G}F(P)+K\|\le\|\varepsilon\mathcal{G}\dot K\|+\varepsilon^2\beta$) and $\mathcal{G}F$ is a contraction on it with constant $\varepsilon^2L<1$, $L:=|\Lambda|+2\|K\|(\|K\|+1)$: Banach's theorem gives P, unique in the ball. The truncation $P_{(3)}:=\sum_{k\le3}\varepsilon^kP_k$ satisfies $\varepsilon\dot P_{(3)}+P_{(3)}=F(P_{(3)})+\rho$ with $\|\rho\|=O(\varepsilon^4)$ (the recursion cancels the orders ε^0–ε^3; ρ involves K up to its fourth derivative), so $P_{(3)}=\mathcal{G}(F(P_{(3)})+\rho)$ and $\|P-P_{(3)}\|\le\|\rho\|/(1-\varepsilon^2L)=O(\varepsilon^4)$. With $v=w-PY$, $\varepsilon\dot v=-(v+PY)-KY-\varepsilon\dot PY-\varepsilon^2P(K^\top(v+PY)-\Lambda Y)=-(I+\varepsilon^2PK^\top)v$, the coefficient of Y vanishing by P's equation; and $\dot Y=\varepsilon(K^\top(v+PY)-\Lambda Y)$. (ii) The change of coordinates is periodic, so it conjugates the monodromies; in (v,Y) the monodromy is block-triangular, its eigenvalues those of the two diagonal blocks. The fast block's monodromy has norm $\le\exp(-T_0/\varepsilon+\varepsilon T_0\|P\|\|K\|)\le e^{-T_0/(2\varepsilon)}$ (a logarithmic-norm estimate), the slow one's eigenvalues modulus $\ge e^{-\varepsilon T_0\|A\|}$; by Liouville's formula the fast exponents sum to $\varepsilon^{-1}\langle\mathrm{tr}(I+\varepsilon^2PK^\top)\rangle$, and $\langle\mathrm{tr}\,PK^\top\rangle=-\langle\mathrm{tr}\,K^\top K\rangle+\varepsilon\langle\mathrm{tr}\,\dot KK^\top\rangle+O(\varepsilon^2)$ with $\langle\mathrm{tr}\,\dot KK^\top\rangle=\frac12\langle(\mathrm{tr}\,KK^\top)^\cdot\rangle=0$. (iii) $A-A_{(3)}=K^\top(P-P_{(3)})=O(\varepsilon^4)$ for $A_{(3)}:=K^\top P_{(3)}-\Lambda$, a polynomial in ε, so by Grönwall's inequality the monodromies of εA and $\varepsilon A_{(3)}$ differ by $O(\varepsilon^5)$; the latter, $\Phi_{(3)}(\varepsilon)$, is analytic in ε with $\Phi_{(3)}(0)=I$, and the principal logarithm is analytic near I, so $\log\Phi_{(3)}(\varepsilon)/(\varepsilon T_0)$ is analytic and $\log\Phi(\varepsilon)/(\varepsilon T_0)$ differs from it by $O(\varepsilon^4)$. Its Taylor coefficients are obtained by any formal expansion of the logarithm of the monodromy — the Magnus series of $\varepsilon A_{(3)}$ collected by powers of ε ($\Omega_1=\varepsilon T_0\langle A\rangle$, $\Omega_2=\frac12\int\int[\varepsilon A(t_1),\varepsilon A(t_2)]$ gives $\bar A_1$'s commutator) or the near-identity averaging: a periodic transformation $I+\varepsilon U_1+\dots$ conjugates the system to an autonomous one up to $O(\varepsilon^4)$, so the monodromy is conjugate, by $I+\varepsilon U(0)+\dots$, to $\exp(\varepsilon T_0(\bar A_0+\dots+\varepsilon^3\bar A_3)+O(\varepsilon^5))$. The convergence of the full Magnus series for $\int_0^{T_0}\|\varepsilon A\|_2<\pi$ is Moan and Niesen's theorem ([102]). (iv) A simple real eigenvalue of $\bar A_0$ continues to a simple real eigenvalue of the analytic family $\log\Phi_{(3)}(\varepsilon)/(\varepsilon T_0)$, analytic in ε (Kato, [102]), and simple eigenvalues are Lipschitz under the $O(\varepsilon^4)$ perturbation to $\bar A(\varepsilon)$: the expansion of μ_j with a bounded remainder, the coefficients by standard perturbation theory. Parity. Let $\tilde K(t):=K(-t)$, $\tilde a(t):=a(-t)$. The full matrix of ($\bigstar$) for $(\tilde K,\tilde a)$ is $M_\varepsilon(-t)$, and the system with the coefficients $M_\varepsilon(-t)$ has the multipliers of $M_\varepsilon(t)$: its monodromy is conjugate to the transpose of M_ε's, by the symmetry ($\dot z=-M(s)z$ is D-conjugate

to the adjoint system, whose monodromy is the inverse transpose). The fast multipliers being exponentially small and the slow ones near 1, the slow multipliers of $\tilde{K}$ at ε are those of K at ε. On the other hand $P[K](-t;-\varepsilon)$ solves the equation of (i) for $\tilde{K}$ at ε (the substitution $t \mapsto -t$, $\varepsilon \mapsto -\varepsilon$ leaves $\varepsilon\dot{P} + P = F(P)$ invariant when a does not enter it), so by uniqueness $A[\tilde{K}](t;\varepsilon) = A[K](-t;-\varepsilon)$: the slow block of $\tilde{K}$ at ε is the time reversal of the slow block of K at $-\varepsilon$, and its monodromy is the inverse, $\Phi_{\tilde{K}}(\varepsilon) = \Phi_K(-\varepsilon)^{-1}$, whence $\bar{A}[\tilde{K}](\varepsilon) = \bar{A}[K](-\varepsilon)$. Together, the slow-time exponents of K at $-\varepsilon$ are those at ε, and the continued simple eigenvalue is even: $\mu_j(-\varepsilon) = \mu_j(\varepsilon)$ — more, $\Phi_{\tilde{K}}(\varepsilon) = \Phi_K(-\varepsilon)^{-1}$ as matrices and $\tilde{K}$'s slow multipliers are K's, so $\operatorname{eig}\bar{A}(-\varepsilon) = \operatorname{eig}\bar{A}(\varepsilon)$ as multisets (a shorter route needs no P: the full system at $-\varepsilon$ is the time reversal of $\tilde{K}$'s at ε, so with the D-symmetry its slow multipliers are the inverses of K's at ε); the expansion of (iv), valid for both signs with the same coefficients (the recursion of (i) does not depend on the sign), has no odd terms. A mismatch a in the error channel ($\varepsilon\dot{w} = -(1-\varepsilon a)w - KY$) adds $\varepsilon a P$ to $F(P)$, which the substitution changes in sign: it breaks the parity, at first order. (v) In the coordinates (v, Y), $\mathrm{d}v = -\varepsilon^{-1}(I + \varepsilon^2 PK^\top)v\,\mathrm{d}t - (\varepsilon^{-2}(1+\varepsilon a) + PK^\top)\sqrt{N_0}\,\mathrm{d}W$ and $\mathrm{d}Y = (\varepsilon AY + \varepsilon K^\top v)\,\mathrm{d}t + K^\top\sqrt{N_0}\,\mathrm{d}W$; for $Y' = Y + \varepsilon S v$ the drift's coefficient of v is $\varepsilon K^\top + \varepsilon\dot{S} - S - \varepsilon^2(SPK^\top + AS)$, zero by S's equation, whose periodic solution is the anti-causal contraction as in (i); the noise of Y' is $K^\top - \varepsilon S(\varepsilon^{-2}(1+\varepsilon a) + PK^\top) = K^\top - (K^\top + \varepsilon\dot{K}^\top)(1+\varepsilon a) + O(\varepsilon^2) = -\varepsilon(\dot{K} + aK)^\top + O(\varepsilon^2)$ times $\sqrt{N_0}\,\mathrm{d}W$. The second moments solve periodic Lyapunov equations: v's, $\dot{V} = -\varepsilon^{-1}(\Gamma V + V\Gamma^\top) + N_0 gg^\top$ with $\Gamma = I + \varepsilon^2 PK^\top$, $g = \varepsilon^{-2}(1+\varepsilon a) + PK^\top$, quasi-static at the fast rate ($V = N_0(1+\varepsilon a)^2/(2\varepsilon^3) + O(N_0/\varepsilon)$); the cross moment $C = \mathbb{E}[vY'^\top]$, $\dot{C} = -\varepsilon^{-1}\Gamma C + \varepsilon C A^\top - \varepsilon N_0 g b^\top$, likewise $C = -N_0 b^\top + O(\varepsilon N_0)$; and Y''s, $\dot{\Sigma}' = \varepsilon(A\Sigma' + \Sigma' A^\top) + \varepsilon^2 N_0 bb^\top$, a linear periodic equation of the slow rate ε: with $\bar{\Sigma}$ as stated, $Z := \Sigma' - \bar{\Sigma}$ is driven by $\varepsilon[(A - \bar{A}_0)\bar{\Sigma} + \bar{\Sigma}(A - \bar{A}_0)^\top] + \varepsilon^2 N_0(bb^\top - \bar{Q})$, of period average $O(\varepsilon^3 N_0)$ and oscillating part $O(\varepsilon^2 N_0)$; the oscillating part's zero-mean antiderivative is $O(\varepsilon^2 N_0)$, and the slow system's periodic response to the rest is $O(\varepsilon^{-1}\cdot\varepsilon^3 N_0)$ since its Lyapunov operator is invertible at rate ε ($\bar{A}_0$ Hurwitz): $Z = O(\varepsilon^2 N_0)$. Then $\operatorname{Cov}(Y) = \operatorname{Cov}(Y') - \varepsilon(SC + C^\top S^\top) + \varepsilon^2 SVS^\top$, where $-\varepsilon(SC + C^\top S^\top) = -\varepsilon^2 N_0(\mathrm{d}(K^\top K)/\mathrm{d}t + 2aK^\top K) + O(\varepsilon^3 N_0)$ and, with $S = \varepsilon K^\top + \varepsilon^2\dot{K}^\top + O(\varepsilon^3)$ and $V = N_0(1+\varepsilon a)^2/(2\varepsilon^3) + O(N_0/\varepsilon)$, $\varepsilon^2 SVS^\top = (\varepsilon N_0/2)K^\top K + (\varepsilon^2 N_0/2)(\mathrm{d}(K^\top K)/\mathrm{d}t + 2aK^\top K) + O(\varepsilon^3 N_0)$: $\operatorname{Cov}(Y) = \bar{\Sigma} + (\varepsilon N_0/2)K^\top K + O(\varepsilon^2 N_0)$. The floor proved is the period average of $\operatorname{tr}(H\operatorname{Cov}(y))$: the slow part by D‴.30(ii)'s pair blocks (at $\eta = 0$ by $-(H + \sigma_0 I)$) and the fast part $(\varepsilon N_0/2)\operatorname{tr}(H\langle\hat{J}^\top\hat{J}\rangle) = (N_0/2k_e)\operatorname{tr}H^2$; at time t the fast part is $(\varepsilon N_0/2)\operatorname{tr}(H\hat{J}(t)^\top\hat{J}(t))$. □

Status. Proved (the constants not made explicit). It upgrades Corollary D‴.26(c) from a formal to a proved expansion, with a bounded $O(k_e^{-4})$ remainder and the odd orders vanishing exactly (for $x_m \in C^4$ at simple eigenvalues; the stability where $H_L(w) \succ 0$), and Proposition D‴.30(ii) to proved for the linearised loop under white noise, for the floor's period average, with the relative remainder $O(1/k_e)$ — the order of round 21's observation, whose first-order term comes from the slope a in the noise input (with $a = 0$ in the noise the numerics show an even expansion; (v) does not prove it) — at $\eta = 0$ with the identification $K = \hat{J}$, $\Lambda = \sigma_0 I$. The real loop's noise at second order (D‴.30(iii)), the OU noise of the numerics and the real nonlinear loop stay as stated there. Chang's decoupling transformation of linear singularly perturbed systems — the pair (P, S), here periodic —, its time-varying and stochastic versions (Kokotović, Khalil and O'Reilly's Ch. 4–5 and App. A, with the slow subsystem's transformed noise and its covariance; Haddad and Kokotović; Yang and Zhu's contraction for slowly time-varying systems), the slow manifolds of stochastic slow–fast systems (Berglund and Gentz) and the convergence of the Magnus series are classical ([102], [49]), and so is the reversal argument for even expansions ([99]); what the lemma adds is their application to the perfect-tracking linearisation of the transport-leak loop: the slaving of Corollary D‴.26 made an exact block-triangularisation, the parity made an exact identity $\operatorname{eig}\bar{A}(-\varepsilon) = \operatorname{eig}\bar{A}(\varepsilon)$ through the slow block's definition at negative ε, and the slow noise identified exactly as $\varepsilon b(t)\nu$, $b = -(\dot{K} + aK)^\top - \varepsilon(\ddot{K} + KK^\top K + K\Lambda^\top + a\dot{K})^\top + O(\varepsilon^2)$ — the high-passed noise at leading order, the drift decoupled, the noise shared with the fast block. Checked by `exp_adaptive_floquet.py` Part D: the asymmetric reference at $\eta = 0$, σ_0 and ∞, $k_e = 12.5$–1600 — P's invariance residual, the

truncations against ε^3 and ε^4, the slow block's rates against the full monodromy's, the rates at $\pm\varepsilon$ per mode, a mismatch breaking their equality at first order, the Magnus parameter; under white noise the decoupling residual, b against b_0 and $b_0+\varepsilon b_1$, the exact periodic covariance in the coordinates (v, Y') with the weakest mode's deviation split into its two sources, the jitter with its next term, and the floor over the period. Checked separately by `tests/test_adaptive_floquet.py`: the slow block's rates against the monodromy's and $P-P_2$ against ε^3 on the instance; on a synthetic block with an $O(1)$, non-symmetric Λ, $P-P_2$ against ε^3 and $P-P_{(3)}$ against ε^4, with a flipped Λ and a dropped cubic term caught; the parity and its breaking by a mismatch at $\eta=\sigma_0$ and ∞; and b_0 against $\dot K$ by the chain rule and the plant's own slope by differences, b_1 against ε^2, and the covariance's orders.

Numerics (`exp_adaptive_floquet.py` Part D → `results/chaos/adaptive_floquet.md`). Corollary D‴.26's instance ($w=(6,5,4)/15$) on the asymmetric reference $x_m=2\sin t+0.8\sin 2t$, $\eta=0$ ($K=\hat J$, $\Lambda=\sigma_0 I$), σ_0 and ∞, $k_e=12.5$–1600, a 512-point Fourier grid. The contraction for P converges in 3–9 iterations to an invariance residual $\le 3.8\cdot10^{-15}$ (at $\eta=0$ and $k_e=12.5$ it stalls at rounding: 499 iterations, the residual $3.3\cdot10^{-15}$). $|P-P_2|/\varepsilon^3$ tends to $3.34\cdot10^3$ and $|P-P_{(3)}|/\varepsilon^4$ to $5.8\cdot10^4$ (bounded: the $O(\varepsilon^3)$ of (i) and the third term of (iii), their constants set by the regressor's harmonics). The slow block's slow-time rates equal the full variational equation's to $\le 1.0\cdot10^{-8}$ (the monodromies' accuracy). The rates at $\pm\varepsilon$ agree to $\le 5.9\cdot10^{-12}$ of the largest rate and $\le 1.1\cdot10^{-9}$ per mode, while their deviation from the first-order prediction is $9.8\cdot10^{-2}$–$8.9\cdot10^{-6}$ of the largest rate and, per mode, up to 29.6% — the slowest mode's at $\eta=\infty$: -29.6%, -12.5%, -4.0%, -1.1% at $k_e=12.5$, 25, 50, 100; at $\eta=0$ the slowest mode's is -1.7% and the largest 12.0% at $k_e=12.5$. The parity of (iv) is an identity, not an asymptotic statement. With the plant's slope as a mismatch in the error channel the rates at $\pm\varepsilon$ differ by $1.1\cdot10^{-2}$–$7.8\cdot10^{-5}$ of the largest rate (per mode $6.1\cdot10^{-2}$–$3.5\cdot10^{-4}$), halving per doubling of k_e: the parity broken at first order. The Magnus parameter is 0.48 at $k_e=12.5$, ≤ 0.25 from $k_e=25$ and 0.004 at 1600, so the full Magnus series converges throughout. Under white measurement noise the second transformation decouples Y' to a residual $\le 3\cdot10^{-16}$. The slow noise b differs from $b_0=-(\dot K+aK)^\top$ by 63%, 40%, 22%, 12%, 5.9%, 3.0%, 1.5% and 0.74% ($O(\varepsilon)$, with the large constant of the regressor's harmonics: $\varepsilon\omega_1\approx0.4$ at $k_e=25$), and $|b-b_0-\varepsilon b_1|/\varepsilon^2$ tends to $3.34\cdot10^3$ (bounded: b_1 of (v)). At $t=0$ the exact periodic covariance gives $\mathrm{Cov}(v)\cdot2\varepsilon^3-1$ from $-9.6\cdot10^{-2}$ to $-2.4\cdot10^{-5}$ and $\|\mathbb{E}[vY'^\top]+b^\top\|/\|b\|$ from 0.15 to $1.2\cdot10^{-3}$ ($O(\varepsilon)$). $\|\mathrm{Cov}(Y')-\bar\Sigma\|/\|\bar\Sigma\|$ falls from 0.35 to $9.4\cdot10^{-5}$ ($\eta=\infty$; to $1.7\cdot10^{-4}$ at $\eta=\sigma_0$ and $2.4\cdot10^{-4}$ at $\eta=0$), by 3.6–4 per doubling from $k_e=50$ up to 200–400 and by 2–2.9 beyond: the $O(\varepsilon)$ of (v), with a small constant. In the weakest mode the deviation is $O(\varepsilon^2)$ (at $\eta=\infty$: -34%, -17%, -6.0%, -1.7% at $k_e=12.5$–100, $-6.6\cdot10^{-5}$ at 1600). It is the sum of two parts of opposite sign: the slow noise's attenuation, $\langle bb^\top\rangle$ below $\bar Q$ (Σ with $\langle bb^\top\rangle$ in place of $\bar Q$: -56%, …, $-1.1\cdot10^{-4}$), and the rate's correction (Σ with the exact slow generator in place of $\bar A_0$: $+49\%$, …, $+4.6\cdot10^{-5}$). The attenuation is 2.4–2.5 times the correction from $k_e=100$ at $\eta=\infty$ (1.2 times at $k_e=12.5$) and 12–23 times at $\eta=0$. $\mathrm{Cov}(Y)-\mathrm{Cov}(Y')$ equals the jitter $(\varepsilon/2)K(0)^\top K(0)$ to a relative 0.32, 0.16, 0.081, 0.041 and 0.020 at $k_e=100$–1600 ($O(\varepsilon)$), and with the next term $-(\varepsilon^2/2)(\mathrm{d}(K^\top K)/\mathrm{d}t+2aK^\top K)(0)$ to 0.30, 0.079, 0.020, 0.005 and 0.001 ($O(\varepsilon^2)$). The floor at $t=0$ is 0.763, 0.905, 0.971, 0.992, 0.998, 0.9995, 0.9999 and 1.0000 of the lemma's slow-plus-fast value ($\eta=\infty$). Over the period the floor's average is 0.766, 0.907, 0.972, 0.9925, 0.9981, 0.99951, 0.99987 and 0.99997 of Proposition D‴.30(ii)'s formula. Pointwise it ranges over $[0.9952, 1.0091]$ of the formula at $k_e=1600$, which is the range of $\mathrm{tr}(H(\bar\Sigma+(\varepsilon/2)\hat J(t)^\top\hat J(t)))$: the fast block's jitter, ±1%. So the formula holds as the period average (v), not pointwise.

Proposition D‴.27 (a projected adaptive prior restores the leak's a-priori bound)**.** *Let $\mathcal{Z}\subset\mathbb{R}^p$ be a compact convex set — on the instance a bounded box in the unconstrained coordinates — and let the prior's atoms of Theorem D‴.23 (or of Proposition D‴.24(ii)) start in it, $Z(0)\in\mathcal{Z}^{K_0}$, and be confined to it: $\mathrm{d}z_j/\mathrm{d}s=\Pi_{T_{\mathcal{Z}}(z_j)}(-\eta(z_j-\hat T_j))$, the velocity projected onto the tangent cone of $\mathcal{Z}$ at z_j in the metric G^{-1} (for a box and diagonal G: the outward components set to*

zero on the active faces); at $\eta=\infty$, $z_j=\Pi_{\mathcal{Z}}(\hat T_j)$, the G^{-1}-projection of the class barycentre. Let $B:=\sup_{\Theta,\vartheta}\|G\langle r_\Theta\nabla_\vartheta\tilde\Phi(\cdot\,;\vartheta)\rangle_\rho\|_{G^{-1}}\le(\max_\rho|f|+a_{\max})\cdot L^G_\vartheta$, with $L^G_\vartheta:=\sup_{x,\vartheta}\|G^{1/2}\nabla_\vartheta\tilde\Phi(x;\vartheta)\|$ ($|\hat f|\le a_{\max}$ for the saturated network).

(i) (the a-priori bound) *Every particle satisfies* $\operatorname{dist}_{G^{-1}}(\vartheta_i(s),\mathcal{Z})\le B/\sigma_0+(\operatorname{dist}_{G^{-1}}(\vartheta_i(0),\mathcal{Z})-B/\sigma_0)_+e^{-\sigma_0 s}$, *at most* $\max\{\operatorname{dist}_{G^{-1}}(\vartheta_i(0),\mathcal{Z}),B/\sigma_0\}$, *for all* $s\ge0$, *and the prior stays in* $\mathcal{Z}$; $\mathcal{Z}$ *being bounded, every solution is bounded, for every* m, *every* $\eta\in(0,\infty]$ *and every target. (For an unbounded* $\mathcal{Z}$ *the distance bound still holds but bounds nothing:* $\mathcal{Z}=\mathbb{R}^p$ *is Theorem D‴.23's prior, which runs off.)*

(ii) (the energy) $\mathcal{J}=\bar L+(\sigma_0/2)W^2$ *— for the nearest-atom prior of Proposition D‴.24(ii) its energy* $\mathcal{J}_\circ$, *with the cell masses — is non-increasing along the projected flow, which is the vertex-selection gradient flow of* $\mathcal{J}+\iota_{\mathcal{Z}^{K_0}}(Z)$ *in the metric of Theorem D‴.23(i) (for* $\mathcal{J}_\circ$ *with the cell masses) — at* $\eta=\infty$ *in its selected form* $\mathcal{J}(\Theta,\Pi_{\mathcal{Z}}\hat T(\gamma))$, γ *the plan the flow carries, optimal for its own projected barycentres (Lloyd-consistent) but not necessarily jointly optimal, so that the reduced energy* $\min_{Z\in\mathcal{Z}^{K_0}}\mathcal{J}$ *itself need not decrease. At* $\eta=\infty$ *the bounded solutions are expected to converge, by a Kurdyka–Łojasiewicz argument on the finitely many selected pieces, to a critical point of a Lloyd-consistent piece, which need not be Clarke-critical for the reduced energy; at finite* η *the solutions are expected to converge to Clarke-critical points of* $\mathcal{J}+\iota$. *At a critical point with a unique plan the prior is a stationary point of Theorem D‴.23(i) where it is interior, and each boundary atom satisfies the Karush–Kuhn–Tucker condition* $z_j=\Pi_{\mathcal{Z}}(\hat T_j)$, $\hat T_j-z_j$ *in the normal cone of* $\mathcal{Z}$ *at* z_j; *at a tie of plans the condition holds for a convex combination of the active pieces, as in Proposition D‴.17(iii).*

(iii) (nothing changes inside) *As long as the prior stays in the interior of* $\mathcal{Z}$ *the projected flow is Theorem D‴.23's. In particular, at an exact representation whose atoms lie in* $\operatorname{int}\mathcal{Z}$ *the stationary point and its linearisation are Theorem D‴.23(i)–(ii)'s, and the bias removal of (iii) and the local convergence of (v) hold unchanged for starts close enough that the prior stays in* $\operatorname{int}\mathcal{Z}$.

(iv) (the price outside) *If the limit has an atom on* $\partial\mathcal{Z}$ *with its class barycentre outside* $\mathcal{Z}$ *— necessarily so when no stationary point of Theorem D‴.23(i) has its prior in* $\mathcal{Z}$ *— the particles of that class leak toward the face, and the limit keeps a bias, bounded by (i), in place of the unprojected prior's run-off (an atom on* $\partial\mathcal{Z}$ *with* $\hat T_j=z_j$ *is a stationary point of Theorem D‴.23(i) whose prior lies on the boundary, with no bias). At* $\eta=\infty$ *on a collapsed class the leak is* $-\sigma_0(\vartheta-\Pi_{\mathcal{Z}}\vartheta)$, *the gradient of the exterior quadratic penalty* $(\sigma_0/2)\operatorname{dist}^2_{G^{-1}}(\vartheta,\mathcal{Z})$ *— zero on* $\mathcal{Z}$ *and growing linearly outside: a switching* σ*-modification toward* $\mathcal{Z}$, *the leak off on* $\mathcal{Z}$ *([47], [95]).*

Proof. (i) $z_{\pi(i)}\in\mathcal{Z}$ at all times, and so is the leak target $m\sum_j\gamma_{ij}z_j$ of a split row (incommensurate m), a convex combination. With $\Pi:=\Pi_{\mathcal{Z}}$ and $d:=\operatorname{dist}_{G^{-1}}(\vartheta_i,\mathcal{Z})$, $\frac12d^2$ is C^1 with gradient $\vartheta-\Pi\vartheta$ in the metric G^{-1}, so $(\mathrm{d}/\mathrm{d}s)\frac12d^2=\langle\vartheta_i-\Pi\vartheta_i,G\langle r\nabla\tilde\Phi(\vartheta_i)\rangle\rangle_{G^{-1}}-\sigma_0\langle\vartheta_i-\Pi\vartheta_i,\vartheta_i-z\rangle_{G^{-1}}$, z the leak target. The projection inequality $\langle\vartheta-\Pi\vartheta,z-\Pi\vartheta\rangle_{G^{-1}}\le0$ for $z\in\mathcal{Z}$ gives $\langle\vartheta_i-\Pi\vartheta_i,\vartheta_i-z\rangle_{G^{-1}}\ge d^2$, so $(\mathrm{d}/\mathrm{d}s)\frac12d^2\le Bd-\sigma_0d^2$, i.e. $d'\le B-\sigma_0d$ where $d>0$, and the comparison principle gives the bound; boundedness follows from that of $\mathcal{Z}$. The bound on B: $\|G\langle r\nabla\tilde\Phi\rangle\|_{G^{-1}}=\|\langle rG^{1/2}\nabla\tilde\Phi\rangle\|\le\sup|r|\cdot L^G_\vartheta$ and $|r|\le|f|+|\hat f|$. (ii) The particles' part of $\mathrm{d}\mathcal{J}/\mathrm{d}s$ is Theorem D‴.23(i)'s, minus the squared norm of their velocity; the prior's is $\langle\nabla_Z\mathcal{J},\Pi_T(-c\nabla_Z\mathcal{J})\rangle=-\|\Pi_T(-c\nabla_Z\mathcal{J})\|^2/c$ (c the metric's factor; Moreau's decomposition of $-c\nabla_Z\mathcal{J}$ into its tangent-cone and normal-cone parts, which are orthogonal), so $\mathcal{J}$ is non-increasing. At $\eta=\infty$ the prior minimises the W^2 part over $\mathcal{Z}$ for the current plan: for each plan γ, $\min_Z\mathcal{J}=\bar L+(\sigma_0/2)\sum_j\big[\sum_i\gamma_{ij}\|\vartheta_i-\hat T_j\|^2_{G^{-1}}+q_j\operatorname{dist}^2_{G^{-1}}(\hat T_j,\mathcal{Z})\big]$ (minimising $\sum_i\gamma_{ij}\|\vartheta_i-z\|^2=q_j\|z-\hat T_j\|^2+\text{const}$ over $z\in\mathcal{Z}$), each piece $C^{1,1}$ and definable (the squared distance to a box is piecewise quadratic), and its gradient in ϑ_i is the learning term and the leak

toward $\Pi_{\mathcal{Z}}\hat{T}_j$ (the derivative of $q_j \operatorname{dist}^2(\hat{T}_j, \mathcal{Z})$ adds $\gamma_{ij}(\hat{T}_j - \Pi_{\mathcal{Z}}\hat{T}_j)$ to the within-class term). Between plan refreshes the particles follow the gradient flow of the piece $E_\gamma(\Theta) := \min_Z \mathcal{J}(\Theta, Z; \gamma)$ of the plan γ they carry, and at a refresh the new plan is optimal for the current prior and the prior then re-minimises, so the selected energy does not increase there either. But γ is Lloyd-consistent — optimal for ν_Z with $Z = \Pi_{\mathcal{Z}}\hat{T}(\gamma)$ — and need not attain $\min_\gamma E_\gamma = \min_Z \mathcal{J}$: the flow is then not a subgradient trajectory of the reduced energy, and Proposition D‴.17's argument, whose fixed prior makes every selected piece the active one, does not apply as it stands (Part H(e): the reduced energy rises by 26% while the selected energy falls). A Kurdyka–Łojasiewicz argument on the finitely many definable pieces that share the limiting value should give the bounded solutions finite length, with a critical point of a Lloyd-consistent piece as the limit; it is not carried out here. At finite η the joint energy carries the indicator of $\mathcal{Z}^{K_0}$, and the Kurdyka–Łojasiewicz argument with it is not carried out here. At a critical point the projected velocity of every atom vanishes, which is the stated condition. (iii) The projection is inactive in $\operatorname{int}\mathcal{Z}$, and Theorem D‴.23(v)'s ball keeps the prior within $\rho(\eta/(\sigma_0 q_k))^{1/2}$ of the atoms. (iv) If no stationary point of Theorem D‴.23(i) has its prior in $\operatorname{int}\mathcal{Z}$, a critical point of (ii) has an atom on $\partial\mathcal{Z}$. At $\eta = \infty$ with a class collapsed at ϑ, $\hat{T} = \vartheta$ and $z = \Pi_{\mathcal{Z}}\vartheta$. □

Status. (i) proved; (ii) the monotonicity of the selected energy and the characterisation of the critical points proved, the convergence argued at $\eta = \infty$ (a Kurdyka–Łojasiewicz argument on the selected pieces, not carried out) and formal at finite η; (iii) proved, its local convergence as Theorem D‴.23(v) (sketched); (iv) the identification at $\eta = \infty$ on collapsed classes proved, the boundary atoms and the bias outside numerical. The bound of (i) is crude (it uses the worst-case force; $L^G_\vartheta \le 4.658$ is the closed-form bound, 4.044 by search), and its content is qualitative: boundedness is no longer a hypothesis. Parameter projection onto a convex set, the exterior quadratic penalty and the switching σ-modification are classical robustness modifications of adaptive laws, and the switching σ-modification keeps the ideal (unbiased) properties of the adaptive law when the true parameter lies in its set ([47], [95]); on collapsed classes at $\eta = \infty$ the construction is that law. New is its transfer to the learned centre of a mass-constrained K_0-atom prior at finite η — a projected adaptive law for the prior's atoms, whose velocity jumps at the faces while the particles' law keeps its smooth gradient form — and its compatibility with Theorem D‴.23's bias removal wherever the target is representable inside $\mathcal{Z}$. Checked: `tests/test_adaptive_prior.py` (beyond the saturation, the projected prior at $\eta = \infty$ (LSODA) keeps the particles within 2 of the box and converges, while without the box the prior's coordinate passes 11 by $s = 10^4$; at $\eta = \sigma_0$ the projected midpoint rule at $\mathrm{d}t = 0.125$ keeps the prior in the box, brings its amplitude atom onto the face, keeps the particles bounded and $\mathcal{J}$ non-increasing; at the misplaced prior near the exact representation the projected and unprojected runs are identical).

Numerics (`exp_adaptive_prior.py` Part H → `results/chaos/adaptive_prior.md`). The box $\mathcal{Z}$ holds the amplitudes and centres to 0.9 of $a_{\max}$ and $c_{\max}$ ($\alpha \in [-5.889, 5.889]$, $\gamma \in [-4.417, 4.417]$) and the width coordinate in $[-6, 6]$; the instance's exact atoms are inside. (a) Part G(b)'s saturated target (a weighted amplitude $4.8 > a_{\max}$), the K_0-atom system from the exact atoms with the prior there: under the projected prior both $\eta = \sigma_0$ (the projected midpoint rule — positions projected inside the step — at $\mathrm{d}t = 0.25$ and 0.125, to $s = 10^5$) and $\eta = \infty$ (LSODA, rtol 10^{-10}) converge to the same bounded stationary point (largest drift $1.8 \cdot 10^{-15}$, $3.5 \cdot 10^{-15}$ and $1.2 \cdot 10^{-16}$; the two steps' end states within $3.4 \cdot 10^{-12}$), the first prior atom's amplitude coordinate on the face of $\mathcal{Z}$ ($z_a = 5.889$) and the particle's at 6.728 (amplitude 3.7325), with the fit error 0.1415 — the two steps' agreement shows their shared fixed point; the scheme's consistency: at $s = 300$ the trajectory lies within $6.3 \cdot 10^{-7}$, $1.9 \cdot 10^{-7}$ and $3.8 \cdot 10^{-8}$ of a reference at $\mathrm{d}t = 1/64$ for $\mathrm{d}t = 0.25$, 0.125, 0.0625. The particles' largest G^{-1}-distance from $\mathcal{Z}$ along the run is 0.92 ($\eta = \sigma_0$) and 0.89 ($\eta = \infty$), against the bound $B/\sigma_0 \le 3686$ ($B \le 26.08$ with the closed-form L^G_ϑ). (b) Part C's K_0-atom loop at the misplaced prior ($\eta = \sigma_0$, 6000 slow units) is identical with and without $\mathcal{Z}$ (largest difference 0.0): the box is never approached — the

prior and the particles keep at least 16.9% of the box's width to every face — so the two runs coincide by construction. (c) The box trades the fit error against the parameters' bound: with the amplitude bound 0.75, 0.8, 0.9, 0.95, 0.99 of $a_{\max}$ — boxes that contain the instance's exact atoms — the limit's fit error is 0.1676, 0.1611, 0.1415, 0.1268, 0.1101, decreasing between the fixed prior's 0.1750 and the unprojected adaptive prior's 0.1050 at $s = 10^6$ (which that prior approaches only as its amplitude coordinate runs off), the prior's amplitude atom on the face in every case; the bound 0.7 (0.1736) already excludes the exact atoms (the start is projected at once), and the tighter boxes 0.65, 0.6, 0.5 are worse than the fixed prior (0.1792, 0.1845, 0.1949): a box should contain the representation. (e) At $\eta = \infty$ on six particles in a Lloyd-consistent, non-optimal pairing (a seeded search; zero amplitudes, a small three-bump target, the box) the selected energy falls from $4.52 \cdot 10^{-3}$ to $4.95 \cdot 10^{-5}$ (its largest increment $-3.4 \cdot 10^{-6}$) while the reduced energy $\min_Z \mathcal{J}$ rises from $1.251 \cdot 10^{-3}$ to $1.574 \cdot 10^{-3}$ before the plan becomes optimal at $s = 80$. (d) On Part G(a)'s four-bump target, where without the box the merged atom's width coordinate grows like $\ln s$ and the solution leaves every compact set, the box stops it: from the exact split fit at $\eta = \infty$ (the joint flow for 40000 slow units, then the collapsed dynamics) κ passes 2.159, 3.843, 5.529 at $s = 10^6$, 10^7, 10^8 as before and stays at the face $\kappa = 6$ from $s = 10^9$ (the atom within $6.6 \cdot 10^{-7}$ of it, drift $5.3 \cdot 10^{-17}$), with the fit error $1.943376 \cdot 10^{-2}$, within $9.3 \cdot 10^{-7}$ of the minimiser's on the face $s = s_{\max}$ that the unprojected solution approaches only as $s \to \infty$.

Proposition D‴.28 (the adaptive prior's floor removal is a gauge property: it survives an approximation error)**.** *Let $\vartheta^b(q)$ be a nondegenerate critical point of the unregularised weighted loss $L_q(\Theta) := \frac{1}{2}\langle |f - \sum_k q_k \tilde{\Phi}(\cdot\,; \vartheta_k)|^2 \rangle_\rho$ — not necessarily an exact representation: $r_b := f - \hat{f}_b$ may be non-zero — and let the masses enter the network only through the products $w_k a(\alpha_k)$, as they do for the saturated network. The amplitude–mass gauge $G_w : \alpha_k \mapsto a^{-1}(q_k a(\alpha_k)/w_k)$, the other coordinates unchanged, is defined on $W_g := \{w : q_k |a_k^b| < w_k a_{\max} \text{ for every } k\}$ and satisfies $\hat{f}_w \circ G_w = \hat{f}_q$, hence $L_w \circ G_w = L_q$.*

(i) (the stationary points carry no function-level floor) *For every $w \in W_g$, $\vartheta^b(w) := G_w \vartheta^b(q)$ is a nondegenerate critical point of L_w with the same function $\hat{f}_b$, the same residual r_b and the same Hessian signature. With the prior at its atoms it is a stationary replicated configuration of the nearest-atom leak toward an adaptive prior (Proposition D‴.24(iii)), so two systems at their samples' gauge transforms carry the same function whatever the approximation error, with the fit error $\|r_b\|_\rho$. The function-level floor — the limit of $\mathbb{E}\|\hat{f}_m - \hat{f}_M\|^2$ along the dynamics — therefore vanishes at the samples of W_g with positive margins in (iii) when $\vartheta^b(q)$ is a strict local minimum and the systems converge to the gauge transforms (numerical); W_g shrinks as the approximation error grows (its thresholds $q_k|a_k^b|/a_{\max}$ rise with the amplitudes). (At an exact representation W_g is Proposition D‴.24's W.)*

(ii) (the fixed prior's floor) *At the matching prior $\vartheta^0 = \vartheta^b(q)$ the function-level weight-sensitivity of the fixed prior's stationary point is*

$$F = \sigma_0 \hat{J}(\tilde{A} + \sigma_0 I)^{-1} \tilde{g}, \qquad \tilde{A} := H_L - \tilde{R},$$

$\tilde{A}$ the whitened weighted Hessian of L_q at $\vartheta^b(q)$ ($\tilde{R}$ the whitened weighted residual curvature) and $\tilde{g}$ the whitened gauge displacement, the direct effect of the weights on $\hat{f}$ being $\hat{J}\tilde{g}$: Proposition D‴.24(i) with the full Hessian in place of the Gauss–Newton block. Along the same derivation the adaptive prior's sensitivity is $F(\sigma)$ at $\sigma = 0$, which vanishes, as (i) says.

(iii) (the splitting margins under the adaptive prior) *For a scalar amplitude per kernel (one output), at a critical point the residual curvature $R_k = \langle r \nabla^2_\vartheta \tilde{\Phi}(\cdot\,; \vartheta_k) \rangle_\rho$ has a zero amplitude row and column, and its block in the centre and width coordinates is $a(\alpha_k)$ times a function of those coordinates alone. Hence $R_k(\vartheta^b(w)) = (q_k/w_k) R_k(\vartheta^b(q))$ exactly, and Proposition*

D‴.5's splitting margins at the adaptive prior's replicated stationary point are

$$\sigma_0 - (q_k/w_k)\lambda_k^+, \qquad \lambda_k^+ := \lambda_{\max}(G^{1/2} R_k(\vartheta^b(q)) G^{1/2}) \geq 0 :$$

a class sampled below $q_k\lambda_k^+/\sigma_0$ can split. At an exact representation $\lambda^+ = 0$ and every margin is σ_0 (Proposition D‴.24(iii)); with an approximation error, and $\vartheta^b(q)$ a strict local minimum, the adaptive prior's admissible set is $\{w \in W_g : w_k > q_k\lambda_k^+/\sigma_0\}$. For vector amplitudes ($n \geq 2$ outputs, $a(\alpha) \in \mathbb{R}^n$ saturated componentwise) only the centre–width block scales by q_k/w_k: stationarity in the amplitudes gives $\Psi_r = 0$ for every output r but stationarity in the centres only $\sum_r a_r \partial_\gamma \Psi_r = 0$, so the amplitude–centre block $a_r'(\alpha_r)\partial_\gamma \Psi_r$ survives, is gauge-invariant in the physical amplitudes, and no closed margin formula follows.

Proof. (i) G_w is a diffeomorphism from $\{\alpha : |q_k a(\alpha)/w_k| < a_{\max}\}$ onto its image with $\hat{f}_w(G_w\Theta) = \hat{f}_q(\Theta)$, so $L_w \circ G_w = L_q$, and it maps critical points to critical points with the same Hessian signature; the function and the residual are unchanged by construction. The leak toward an adaptive prior at its atoms vanishes (Theorem D‴.23(i), Proposition D‴.24(ii)). (ii) Let $u := \partial_w(G_w\vartheta^b(q))$ at $w = q$, $D := \operatorname{diag}(q_k G^{-1})$, $A := \nabla^2 L_q(\vartheta^b)$, $B := \partial_w \nabla_\Theta L_w$ at fixed Θ, $J := \partial_\Theta \hat{f}$, $\Phi := \partial_w \hat{f}$ at fixed Θ. Differentiating $\hat{f}_w(G_w\Theta) = \hat{f}_q(\Theta)$ gives $\Phi = -Ju$, and differentiating $\nabla L_w(G_w\vartheta^b) = 0$ gives $B = -Au$. The fixed prior's stationarity $\nabla L_w(\Theta) + \sigma_0 D(\Theta - \vartheta^0) = 0$, linearised at (q, ϑ^b), reads $(A + \sigma_0 D)\delta\Theta = -B\delta w = Au\delta w$, so $\delta\hat{f} = \Phi\delta w + J\delta\Theta = -J(A + \sigma_0 D)^{-1}[(A+\sigma_0 D) - A]u\delta w = -\sigma_0 J(A+\sigma_0 D)^{-1} D u \delta w$; in the whitened coordinates ($\tilde{J} = JD^{-1/2}$, $\tilde{A} = D^{-1/2}AD^{-1/2}$, $\tilde{g} := -D^{1/2}u$) this is $\sigma_0 \tilde{J}(\tilde{A} + \sigma_0 I)^{-1}\tilde{g}$. With $\sigma_0 = 0$, $\delta\Theta = u\delta w$ and $\delta\hat{f} = 0$. At an exact representation $\tilde{R} = 0$, $\tilde{A} = H_L$ and $\tilde{g} = \partial_w x^0$. (iii) With a scalar amplitude and $\Psi(\gamma, \kappa) := \langle r\varphi(\cdot\,; c(\gamma), s(\kappa))\rangle_\rho$, $\langle r\tilde{\Phi}(\cdot\,; \vartheta)\rangle_\rho = a(\alpha)\Psi$, whose gradient is $(a'\Psi, a\Psi_\gamma, a\Psi_\kappa)$ and whose Hessian is $\begin{bmatrix} a''\Psi & a'\Psi_\gamma & a'\Psi_\kappa \\ a'\Psi_\gamma & a\Psi_{\gamma\gamma} & a\Psi_{\gamma\kappa} \\ a'\Psi_\kappa & a\Psi_{\gamma\kappa} & a\Psi_{\kappa\kappa} \end{bmatrix}$. At a critical point the gradient vanishes, so $\Psi = \Psi_\gamma = \Psi_\kappa = 0$ ($a' > 0$, $a \neq 0$) and only the block $a\Psi_{(\gamma,\kappa)}$ remains; r and Ψ are unchanged by the gauge and a scales by q_k/w_k. The α-direction contributes the eigenvalue 0, so $\lambda_k^+ \geq 0$. □

Status. (i) and (iii) proved, exactly, as statements about the stationary points and their linearisation (no linearisation in w: the gauge maps critical points to critical points); (ii) as Proposition D‴.24(i) — exact at q, first order in w; the convergence of the dynamics to the gauge transform numerical, and with it the floor's vanishing. The amplitude–mass degeneracy is the homogeneity of the conic view [16], [23], and the rescaling symmetries of networks, which map critical points to critical points with the same function while changing their curvature, are classical ([98]); what is new is that this symmetry, not exact representability, is the mechanism of Proposition D‴.24's floor removal — so at a strict local minimum in the interior the adaptive prior keeps the fitted function independent of the sampled weights without an exact representation — and, for a scalar amplitude per kernel, the exact scaling of the residual curvature under the gauge, which gives the adaptive prior's splitting margins in closed form. Its hypotheses are a strict local minimum in the interior and samples in W_g with positive margins: on the instance $P(w^m \in W_g)$ is only 0.52 at $m = 15$ (0.87 at $m = 60$), and the branch ends at the saturation near $\varepsilon = 0.68$; targets whose best approximant lies on the saturation boundary (a sharp friction step fitted by a dipole of two kernels; the cogging servo of Proposition D‴.29, globally; Proposition D‴.27's beyond-the-saturation example) are outside it. Checked: `tests/test_gauge_floor.py` (at $\varepsilon = 0.3$, a critical point with a 1.7% fit error: the gauge transform a critical point at two sampled weight vectors, the critical point continued in w from $\vartheta^b(q)$ landing on it; the adaptive prior's sensitivity, from critical points reached by Newton from $\vartheta^b(q)$, below 10^{-8} of the fixed prior's, the fixed prior's equal to $\sigma_0\tilde{J}(\tilde{A} + \sigma_0 I)^{-1}\tilde{g}$ to 10^{-6} and not to the Gauss–Newton form; the residual curvature's zero amplitude row, its scaling (q_k/w_k), the margins $\sigma_0 - (q_k/w_k)\lambda_k^+$ and the dynamics' separation eigenvalues $-\sigma_0 + (q_k/w_k)\operatorname{eig}(G^{1/2}R_k G^{1/2})$ to 10^{-9}; adaptive-prior runs at $\eta = \infty$ and σ_0 converging to the gauge transform, 1.2 away from $\vartheta^b(q)$, with the approximant's function).

Numerics (`exp_gauge_floor.py` → `results/chaos/gauge_floor.md`). The three-bump instance with a fourth bump $\varepsilon \cdot \exp(-(x-0.8)^2/(2\cdot 0.3^2))$ that $K_0 = 3$ kernels cannot represent, $\varepsilon = 0$–0.5, the masses $q = 1/3$, $\sigma_0 = 0.15\sigma_{\text{crit}}$; $\vartheta^b(q)$ the weighted approximant, a critical point of the unregularised loss continued in ε from the exact representation (relative fit error $3.6\cdot 10^{-3}$, $6.8\cdot 10^{-3}$, $1.2\cdot 10^{-2}$, $1.7\cdot 10^{-2}$, $2.5\cdot 10^{-2}$ at $\varepsilon = 0.05$, 0.1, 0.2, 0.3, 0.5; gradient residual $\le 1.6\cdot 10^{-15}$; nondegenerate, the smallest eigenvalue of $\tilde{A}$ $4.6\cdot 10^{-5}$–$3.1\cdot 10^{-4}$). (i) At six weight vectors 0.01 inside W_g the unregularised gradient at $G_w\vartheta^b(q)$ is $\le 1.6\cdot 10^{-15}$, and Newton's method continued in w from $\vartheta^b(q)$ (20 steps) lands on $G_w\vartheta^b(q)$ to $\le 1.7\cdot 10^{-11}$ with $\hat{f}$ unchanged to $\le 5.5\cdot 10^{-13}$ — plain Newton from $\vartheta^b(q)$ at w does so in 3–5 of the 6 cases, elsewhere reaching other critical points of L_w; the adaptive prior's function-level sensitivity, from the critical points reached from $\vartheta^b(q)$, is $1.4\cdot 10^{-11}$–$2.7\cdot 10^{-10}$ of the fixed prior's. The samples fall in W_g with the probability 0.52, 0.87–0.88, 0.997 at $m = 15$, 60, 240 ($\varepsilon = 0.05$–0.3; 0.26, 0.71, 0.98 at $\varepsilon = 0.5$), and the branch continued to $\varepsilon = 0.68$ reaches $\max_k|a_k^b|/a_{\max} = 0.999$, the smallest eigenvalue of $\tilde{A}$ $1.2\cdot 10^{-8}$ and W_g's threshold $0.333 = q_2$ for its class: there $w = q$ leaves W_g. (ii) The fixed matching prior's sensitivity equals $\sigma_0\hat{J}(\tilde{A}+\sigma_0 I)^{-1}\tilde{g}$ to $\le 2.2\cdot 10^{-9}$ and misses Proposition D‴.24's Gauss–Newton form by $7.0\cdot 10^{-4}$–$5.6\cdot 10^{-3}$, growing with ε as $\|\tilde{R}\|/\|H_L\|$ does ($5.3\cdot 10^{-4}$–$2.8\cdot 10^{-3}$); its floor law per unit of $1/m - 1/M$ rises from $6.17\cdot 10^{-2}$ ($\varepsilon = 0$) to $8.52\cdot 10^{-2}$ ($\varepsilon = 0.5$). (iii) The residual curvature's amplitude row is $\le 7.4\cdot 10^{-14}$ (its largest entry $2.6\cdot 10^{-4}$–$2.9\cdot 10^{-3}$), $\lambda_k^+ \le 7.2\cdot 10^{-4}$ (at most $0.10\sigma_0$), and over every count vector of $m = 15$ and 60 in W_g the adaptive prior's splitting margins equal $\sigma_0 - (q_k/w_k)\lambda_k^+$ to $\le 2.1\cdot 10^{-14}$, none negative (the smallest $+6.23\cdot 10^{-3}$); the separation eigenvalues of the dynamics at a replicated configuration (counts $(6,4,5)$, the Jacobian of the averaged drift at $\eta = \sigma_0$ and ∞) equal $-\sigma_0 + (q_k/w_k)\,\mathrm{eig}(G^{1/2}R_kG^{1/2})$ to $\le 5.8\cdot 10^{-12}$. *The dynamics* (Part B: warm clouds of width 0.02 around $\vartheta^b(q)$, $m = 15$, 60, 240 against $M = 510$, four seeds, 6000 slow units, $\varepsilon = 0.1$ and 0.3, the nearest-atom leak). Under the fixed prior $\|\hat{f}_m - \hat{f}_M\|^2$ stays within 8% of its value at $s = 600$ (the ratio at $s = 6000$ 0.917–1.003; at $\varepsilon = 0.1$: $3.1\cdot 10^{-4}$–$5.9\cdot 10^{-3}$, $1.1\cdot 10^{-4}$–$1.0\cdot 10^{-3}$ and $5.0\cdot 10^{-6}$–$8.2\cdot 10^{-5}$ at $m = 15$, 60, 240) and the fit error above the approximant's ($9.0\cdot 10^{-3}$–$1.2\cdot 10^{-1}$ against $6.84\cdot 10^{-3}$), the classes splitting at the samples where the fixed prior's point is a saddle (diameters up to 1.6). Under the adaptive priors the classes stay replicated, their means approach the gauge transform $G_w\vartheta^b(q)$, $\|\hat{f}_m - \hat{f}_M\|^2$ falls (at $m = 60$, $\varepsilon = 0.1$, from $6.9\cdot 10^{-6}$–$1.2\cdot 10^{-4}$ at $s = 600$ to $1.0\cdot 10^{-9}$–$1.4\cdot 10^{-8}$ at $\eta = \sigma_0$ and to $3.2\cdot 10^{-11}$–$3.8\cdot 10^{-9}$ at $\eta = \infty$ by $s = 6000$), and the fit error tends to the approximant's: at the samples at least 0.02 inside W_g it exceeds it at $s = 6000$ by a median $1.2\cdot 10^{-4}$, $1.2\cdot 10^{-4}$, $1.6\cdot 10^{-5}$ (relative; at most $1.1\cdot 10^{-1}$, $1.2\cdot 10^{-2}$, $1.3\cdot 10^{-3}$) at $m = 15$, 60, 240. At the $m = 15$ sample $(0.533, 0.267, 0.200)$, outside W_g (margins -0.0016 and -0.0097 at $\varepsilon = 0.1$ and 0.3), $\vartheta^b(q)$'s branch has no gauge transform: the adaptive priors' fit errors at $s = 6000$ ($2.2\cdot 10^{-2}$ and $1.6\cdot 10^{-2}$ at $\varepsilon = 0.1$, $3.7\cdot 10^{-2}$ and $3.3\cdot 10^{-2}$ at $\varepsilon = 0.3$, for $\eta = \sigma_0$ and ∞) are still falling toward the minimum of L_w over the closed amplitude box on the branch, 1.107 and 1.586 times the approximant's fit error (one amplitude at the bound, its coordinate running off), and a gauge transform with two classes exchanged exists there, which the warm-cloud dynamics does not reach; at $(0.400, 0.200, 0.400)$, $6\cdot 10^{-4}$ inside W_g at $\varepsilon = 0.3$, the gauge transform's amplitude lies near the saturation and the approach is slow (2.8–3.1 from it at $s = 6000$), as at the boundary of Proposition D‴.24's W.

Proposition D‴.29 (second-order plants: the sliding-variable reduction)**.** *Let the plant be $\ddot{x} = f(x,\dot{x}) + u$, x_m a T_0-periodic C^3 reference, $e := x - x_m$, $s := \dot{e} + \lambda e$ ($\lambda > 0$), and let the controller and the adaptive law be*

$$u = \ddot{x}_m - \lambda\dot{e} - k_s s - \hat{f}_\Theta(z), \qquad \dot{\vartheta}_i = G\nabla_\vartheta(s\tilde{\Phi}(z;\vartheta_i)) - (\sigma_0/k_s)(\vartheta_i - T_i),$$

z the coordinates of $(x,\dot{x})$ on which f depends and T_i any of the leak targets of this section (a fixed prior, the nearest-atom or the OT plan, an adaptive or a projected prior).

(i) (the error channel) $\dot{s} = -k_s s + r_\Theta(z)$ *and* $\dot{e} = -\lambda e + s$, *with* $z = z_m + (e, \dot{e})$ *the regressor at the plant's state: Theorem J's error channel with* s *in place of* e.

(ii) (the averaged slow loop) *As* $k_s \to \infty$ *at fixed* λ, s *and* e *are slaved —* $s = (\mathrm{d}/\mathrm{d}t + k_s)^{-1}[r_\Theta(z_m)] + O(k_s^{-2}) = r_\Theta(z_m)/k_s + O(k_s^{-2})$ *and* $e = (\mathrm{d}/\mathrm{d}t + \lambda)^{-1}s$, *the periodic solutions of the two filters — and in the slow time* $\tau = t/k_s$ *the parameters follow Theorem J's averaged slow loop along the regressor's orbit* $z_m(t)$,

$$\frac{\mathrm{d}\vartheta_i}{\mathrm{d}\tau} = G\langle r_\Theta \nabla_\vartheta \tilde{\Phi}(\cdot\,; \vartheta_i)\rangle_\rho - \sigma_0(\vartheta_i - T_i), \quad \rho \text{ the occupation measure of } z_m \text{ over one period,}$$

up to $O(1/k_s)$, *the prior's law at finite* η *being* $\mathrm{d}z_j/\mathrm{d}t = -(\eta/k_s)(z_j - \hat{T}_j)$ *in real time. Both forms of the slaved error — the exact filter* $(\mathrm{d}/\mathrm{d}t + k_s)^{-1}$ *and* r/k_s *— are accurate to* $O(k_s^{-2})$ *in* s, *i.e. to* $O(1/k_s)$ *relatively, the parameters' within-period motion and the nonlinearity* $r(z) - r(z_m)$ *entering at the same order; the exact filter keeps the error filter's attenuation of the residual's harmonics, which* r/k_s *drops at the relative order* $n^2\omega_0^2/(2k_s^2)$ *in the rms of the* n*-th harmonic, and is the better prediction where that term dominates (a small residual of high frequency).*

(iii) (the transfer) *Every statement of this section about the averaged slow loop holds for the second-order plant with this* ρ *— Theorems D‴, D‴.21 and D‴.23, Propositions D‴.5, D‴.9, D‴.12, D‴.17, D‴.24(i)–(v), D‴.27 and D‴.28 — and the tracking error of the position is the filtered residual of (ii); the real-loop statements (Proposition D‴.24(vi), Corollaries D‴.14 and D‴.26) would need the variational equation of (i).*

(iv) (where the kernels live) *When* f *depends on one coordinate (a cogging torque on the position, a friction law on the velocity), kernels on that coordinate see the orbit as an interval, and the instance is this section's one-dimensional one with another target. Isotropic Gaussian kernels on the phase plane over a circular orbit (*$x_m = R\sin\omega_0 t$ *with* $\omega_0 = 1$ *in the units where the kernel is isotropic in* $(x, \dot{x})$*) are over-parametrised: on the circle* $z = R(\sin\theta, \cos\theta)$ *a kernel restricts to* $a \cdot e^{-(R^2+|c|^2)/(2s^2)} \cdot e^{(R|c|/s^2)\cos(\theta-\theta_c)}$, *a von Mises function of three effective parameters for four, so in general position the Gauss–Newton block has a zero eigenvalue for every kernel (more at degenerate atoms, as for kernels centred on the orbit's centre) — a direction that the fixed prior's leak pins at the rate* σ_0, *that an adaptive prior leaves neutral (*$\lambda^- = 0$, *Theorem D‴.23(ii)), and that a projected prior keeps bounded (Proposition D‴.27). On other orbits the same directions are weakly excited rather than unexcited: the classical partial excitation of Gaussian networks along an orbit ([92, 93, 94]).*

Proof. (i) $\ddot{e} = f + u - \ddot{x}_m = r_\Theta(z) - \lambda\dot{e} - k_s s$, so $\dot{s} = \ddot{e} + \lambda\dot{e} = -k_s s + r_\Theta(z)$, and $\dot{e} = s - \lambda e$ by the definition of s. (ii) The error subsystem (s, e) is linear, exponentially stable (eigenvalues $-k_s$ and $-\lambda$) and driven by $r_\Theta(z)$, while the parameters move at the rate $O(1/k_s)$ (their velocity is $G\nabla_\vartheta(s\tilde{\Phi})$ with $s = O(1/k_s)$). Theorem J's argument — for the smooth fixed-prior leak near a hyperbolic critical point, using the error channel's exponential stability, the regressor's smoothness and its Step 3's normal form — carries over with the fast pair (s, e) once its error-channel steps are redone for it (s slaved at k_s, e filtered at the fixed λ: the error channel's multipliers are $e^{-k_s T_0}$ and $e^{-\lambda T_0}$), the separation being between λ and the parameters' rate; for switching targets (plan and cell switches, adaptive and projected priors) not even the first-order loop has more than Theorem D‴.18's locally constant case. At the slaved state $z = z_m + O(1/k_s)$, and the period average of $G\nabla_\vartheta(s\tilde{\Phi}(z_m; \vartheta_i))$ with $s = r_\Theta(z_m)/k_s$ is $(1/k_s)$ times the averaged learning term. For the n-th harmonic of r, $(\mathrm{d}/\mathrm{d}t + k_s)^{-1}$ has the gain $(k_s^2 + n^2\omega_0^2)^{-1/2} = (1/k_s)(1 - n^2\omega_0^2/(2k_s^2) + \dots)$. (iii) The statements of this section use the averaged loop only through ρ, the target on its support and the network. (iv) $|z - c|^2 = R^2 + |c|^2 - 2R|c|\cos(\theta - \theta_c)$ on the circle; the map $(a, |c|, \theta_c, s) \mapsto (ae^{-(R^2+|c|^2)/(2s^2)}, R|c|/s^2, \theta_c)$ has a one-dimensional fibre, along which the kernel's restriction, and hence $\hat{f}$ on ρ's support, is constant. □

Status. (i) and (iv) exact; (ii) and (iii) formal — first-order averaging with the fast pair (s, e), Theorem J's steps carrying over for the smooth fixed-prior leak near a hyperbolic critical point once the error-channel steps are redone, and switching targets beyond even the first-order theory; (iii) covers the statements about the averaged loop, not the real-loop ones (Proposition D‴.24(vi) and Corollaries D‴.14 and D‴.26 would need the variational equation of (i)). The filtered error is the classical construction of adaptive control for higher-order plants, used with Gaussian networks by Sanner and Slotine ([79]); what is added is the transfer of the transport leaks, their floors and the adaptive and projected priors to such plants, and the exact fibre of isotropic phase-plane kernels on a circular orbit (the weak excitation of the same directions on other orbits being the classical partial excitation [92, 93, 94]). Checked: `tests/test_sliding.py` (the phase-plane loop as the gradient flow of $\bar{L}$; on a circular orbit one zero Gauss–Newton eigenvalue per kernel in general position, the next above 10^{-11} of the largest; the servo's parameter displacement over ten periods within 3% of the averaged loop's at $k_s = 25$ and closer at $k_s = 50$, and its position error within 1% of both predictions of (ii) at $k_s = 50$).

Numerics (`exp_servo_cogging.py` → `results/servo/servo_cogging.md`). The servo $\ddot{x} = -\tau_{\mathrm{cog}}(x) + u$ with the cogging torque $\tau_{\mathrm{cog}}(x) = 0.4\sin(3x + 0.3) + 0.1\sin 7x$, $x_m = \sin t$, $\lambda = 5$; $K_0 = 4$ saturated Gaussian kernels on the position ($a_{\max} = 4$, $c_{\max} = 1.6$, widths in $[0.05, 1]$), masses $1/4$, $\sigma_0 = 0.01$. (A) $\vartheta^b(q)$, the nondegenerate local minimum of the unregularised weighted loss that BFGS reaches from a fixed start, has the relative fit error $1.46 \cdot 10^{-2}$, and Proposition D‴.28 is its local analysis: the whitened unregularised Hessian $\tilde{A}$ spans $2.9 \cdot 10^{-4}\sigma_0$–$30\sigma_0$ (four modes below $0.1\sigma_0$: the adaptive prior's slowest rates $1.4 \cdot 10^{-4}\sigma_0$ at $\eta = \sigma_0$ and $2.9 \cdot 10^{-4}\sigma_0$ at $\eta = \infty$, against the fixed prior's $\approx \sigma_0$), the residual curvature's $\lambda_k^+ \le 0.0098\sigma_0$ (the adaptive prior's margins positive on all of $W_g = \{w > (0.061, 0.085, 0.119, 0.008)\}$), the fixed matching prior's floor law $1.29 \cdot 10^{-2}$ per unit of $1/m - 1/M$ and its margin $+9.9 \cdot 10^{-3}$ at $w = q$. Globally the loss has its infimum at the saturation: of 120 BFGS runs from random starts one ends at $\vartheta^b(q)$ and 33 below its fit error, all with saturated amplitudes; the best reaches $6.39 \cdot 10^{-3}$, and with its two saturated amplitudes held at $|\alpha| = 6$, 8, 10, 14 ($|a| = 3.62$–3.99) and the rest re-optimised the fit error is $7.14 \cdot 10^{-3}$, $6.65 \cdot 10^{-3}$, $6.49 \cdot 10^{-3}$, $6.40 \cdot 10^{-3}$ — approached, not attained. Globally the servo is Proposition D‴.27's case (critical points at infinity), and the runs below are statements about the basin of $\vartheta^b(q)$ and about an engineer's model. A misplaced prior — the approximant of the cogging model with the phase 0.5 in place of 0.3, the engineer's model, its own fit error 0.200 — has a replicated stationary point (the same from its atoms and from $\vartheta^b(q)$) with the fit error $5.45 \cdot 10^{-2}$ and the splitting margin $-3.97 \cdot 10^{-3}$: a D‴.5 saddle. (B) The averaged loop ($m = 16$, 64, 128 against $M = 256$, three seeds, 2000 slow units). From warm clouds around $\vartheta^b(q)$ — computed from the true torque — with the matching prior $\vartheta^b(q)$: the fixed prior's nearest-atom leak keeps a finite-size floor ($\|\hat{f}_m - \hat{f}_M\|^2$ $1.85 \cdot 10^{-3}$, $2.24 \cdot 10^{-4}$, $4.49 \cdot 10^{-5}$; fit error 0.174, 0.066, 0.038 against the approximant's 0.0146); the adaptive prior's floor is 38–780 times lower ($\eta = \infty$: $4.85 \cdot 10^{-5}$, $2.88 \cdot 10^{-7}$, $6.36 \cdot 10^{-8}$), with the fit error 0.0298 (the classes not yet replicated at $m = 16$ by $s = 2000$), 0.0148, 0.0147; the OT leak toward the matching prior sits at the approximant (0.01462, no floor); the σ-modification toward $\vartheta^0 = 0$ at the same σ_0 keeps a large bias (0.328) — and no leak at all ($\sigma_0 = 0$) does as well as the best of them (0.0180, 0.0147, 0.0146; floors $7.7 \cdot 10^{-6}$, $2.4 \cdot 10^{-7}$, $6.6 \cdot 10^{-8}$). In this noise-free simulation the floors and biases that the adaptive and OT leaks remove are those a fixed leak creates; the leaks' purpose — robustness to noise and disturbances, Propositions D‴.27, D‴.30, D‴.31 — is not exercised here. With the misplaced prior from that start, at $\eta = \infty$ the prior never acts (its initial position enters only through the first cells: the runs are the matching ones), at $\eta = \sigma_0$ its bias decays (0.035, 0.020, 0.019 at $s = 2000$), and the fixed designs keep a bias — their m-systems leave the saddle by splitting their classes (diameters 0.32–1.07) and end at 0.048 (OT) and 0.10, 0.060, 0.048 (the nearest atom; above the replicated point's 0.0545 at $m = 16$ and 64). From warm clouds around the engineer's model, with the prior at it — the start an engineer has — the fixed designs stay biased (0.102, 0.060, 0.048; OT 0.048), the adaptive prior removes

most of the bias but slowly ($\eta=\infty$: 0.027, 0.025, 0.022 at $s=2000$; $\eta=\sigma_0$: 0.032, 0.028, 0.028), and no leak reaches the basin's fit (0.0152, 0.0147, 0.0147): its m particles spread and explore directions that the K_0-atom structure the leaks impose leaves weakly excited. The collapsed K_0-atom flows give the time scale: from the engineer's model at $\eta=\infty$ — where the collapsed flow is the unregularised one — the fit error is 0.0297, 0.0213, 0.0169, 0.0155 at $s=600$, 2000, 10^4, 10^5, still 0.55 from $\vartheta^b(q)$ in the parameters (10^5 slow units are $2.5\cdot10^6$ time units of the real servo at $k_s=25$), at $\eta=\sigma_0$ 0.0345, 0.0259, 0.0181, 0.0160; from $\vartheta^b(q)$ toward the misplaced prior at $\eta=\sigma_0$ 0.0240, 0.0183, 0.0154, 0.0149. (C) The real servo ($k_s=25$, RK4 at $\mathrm{d}t=2\cdot10^{-3}$, 1000 time units, $m=16$ and 64, seed 0) from the oracle-start designs' end states, with each design's leak (its targets refreshed, and the error recorded, every 0.02 time units; the finite-η prior relaxed in real time): the position error's rms is 1.7–$2.1\cdot10^{-5}$ for the adaptive designs, the matching OT design and no leak, $7.8\cdot10^{-5}$ for the misplaced OT design, $9.1\cdot10^{-5}$–$1.8\cdot10^{-4}$ for the fixed nearest-atom prior and 5.4–$5.5\cdot10^{-4}$ for the σ-modification; it equals the prediction of (ii) with the exact filter to 1.0001–1.0029 at $t=500$ and 1000 in every design (0.990–1.005 over the first five periods, where the parameters still move), and with the slaved $s=r/k_s$ to 0.931–0.996, the deficit largest for the adaptive designs, whose residual is the approximation error's broad fine scale; over these 40 slow units no design drifts, no leak included. (D) Phase-plane kernels (five isotropic kernels in general position, $\sigma_0=0.01$): on the circle 5 exact zero Gauss–Newton eigenvalues (one per kernel) and 8 below $10^{-6}\sigma_0$, the $(K+1)$-th eigenvalue $2.4\cdot10^{-10}$ of the largest; on ellipses with the velocity axis scaled by 1.1 and 2, 8 and 6 below $10^{-6}\sigma_0$ and none exact; on the orbit of $1.2\sin t+0.3\sin 2t$ 3; with the five centres at the origin 16 exact zeros of 20.

Proposition D‴.30 (measurement noise at perfect tracking: a coherent, high-passed noise, and the errors in variables)**.** *Let* (CL) *be the real loop of Corollary D‴.26 ($n=1$) at the lift Γ_w of an exact representation of representable weights w, with the prior at its atoms — fixed there ($\eta=0$) or adaptive ($0<\eta\le\infty$) — and $\partial_x\hat f=\partial_x f$, $\partial_x^2\hat f=\partial_x^2 f$ on the orbit (an exact representation, an identity of functions, supplies both), and let the controller and the adaptive law use the measurement $y=x+\nu$ in place of x: $u=\dot x_m-k_e(y-x_m)-\hat f_\Theta(y)$, $\dot\vartheta_i=G\nabla_\vartheta((y-x_m)^\top\tilde\Phi(y;\vartheta_i))-(\sigma_0/k_e)(\vartheta_i-z_{\pi(i)})$, ν a stationary zero-mean noise, independent of the rest, with the spectral density N_0 at the frequencies of the regressor and of the error channel. Let $H:=H_L(w)=\langle\hat J^\top\hat J\rangle$ (the whitened weighted Gauss–Newton matrix, Corollary D‴.26's $\hat J$) with the eigenpairs (h_j,v_j), $a(t):=\partial_x\hat f(x_m(t))$, $Q:=\langle(\dot{\hat J}+a\hat J)^\top(\dot{\hat J}+a\hat J)\rangle$ and $Q_{jj}:=v_j^\top Q v_j$.*

(i) (coherence) *The particles of a class see the same y, the same error and the same prior atom, so a replicated configuration stays replicated under every path of ν: on the replicated subspace the m-system is the weighted K_0-atom loop driven by the same noise, for every m with these weights, at every order in ν. Measurement noise does not average out with m, as the per-particle dither of Theorem J′ does. This is invariance, not attraction. A separation $u=G^{-1/2}\delta\vartheta_i$ of class k obeys $\dot u=(G^{1/2}\varepsilon_m\nabla_\vartheta^2\tilde\Phi(y;\vartheta_k)G^{1/2}-\sigma_0/k_e)u$ along the noisy class trajectory, and to second order in ν its mean drift is $\mathbb{E}[\varepsilon_m^2]G^{1/2}\partial_x\nabla_\vartheta^2\tilde\Phi G^{1/2}+\mathbb{E}[e]G^{1/2}\nabla_\vartheta^2\tilde\Phi G^{1/2}$ (the measured regressor; the parameters' cross term negligible), $\mathbb{E}[e]$ the mean tracking error. At the lift — the parameters at their atoms, $\mathbb{E}[e]$ the noise's own mean error at fixed parameters — its period average S_k gives the decay rate $\sigma_0/k_e-\mathbb{E}\nu^2\lambda_{\max}(S_k)$ to leading order, so near the lift the replicated configuration attracts — and an m-system started near it follows the K_0-atom loop — only below the noise level $\mathbb{E}\nu^2=(\sigma_0/k_e)/\max_k\lambda_{\max}(S_k)$. Along the noisy trajectory $\mathbb{E}[e]$ also carries the mean tracking error $-\hat J\bar y(t)/k_e$ of the developing errors-in-variables deviation $\bar y(t)$ of (iii), of the same order in $\mathbb{E}\nu^2$, so the drift $S_k(t)$ changes on $\bar y$'s time scale and to this order the separation follows $\dot u=(\mathbb{E}\nu^2S_k(t)-\sigma_0/k_e)u$ (the first-order multiplicative noise, high-passed by the loop, adds a negligible cumulant to its exponent); the long-time threshold — where $\bar y$ is $O(1)$ at these noise levels and the expansion no longer applies — is open.*

(ii) (the floor at first order) *To first order in ν the measured error's noise is* $((\mathrm{d}/\mathrm{d}t-a)/(\mathrm{d}/\mathrm{d}t+$

$k_e))\nu$ — *the loop high-passes it — and in the slow time the collective coordinates diffuse with the matrix* $(N_0/k_e)Q$*; their stationary covariance solves* $\bar{A}_0\Sigma+\Sigma\bar{A}_0^\top+(N_0/k_e)\operatorname{diag}(Q,0)=0$ *(*$\bar{A}_0$ *Corollary D‴.26's averaged collective block), and the function-level floor, averaged over a period, is*

$$\mathbb{E}\|\hat{f}-f\|_\rho^2=(N_0/2k_e)\Big[\sum_j Q_{jj}(h_j+\eta)/(h_j+\eta+\sigma_0)+\operatorname{tr}H^2\Big]+O(k_e^{-2})$$

(at time t *its fast part is* $(N_0/2k_e)\operatorname{tr}(H\hat{J}(t)^\top\hat{J}(t))$*, whose oscillation over the period is of the floor's own order): the parameters' part, each mode's share multiplied by* $(h_j+\eta)/(h_j+\eta+\sigma_0)$ — *the fixed prior's Tikhonov filter factor* $h_j/(h_j+\sigma_0)$ *at* $\eta=0$*, the factor 1 at* $\eta=\infty$*, where the part is* $(N_0/2k_e)\operatorname{tr}Q$ *whatever the spectrum — and the jitter of the frequencies that the loop passes, independent of the prior (the compensation error at the plant's current point has* $\langle|\hat{J}|^4\rangle$ *in place of* $\operatorname{tr}H^2$*). Mode* j*'s parameter variance is* $(N_0/2k_e)Q_{jj}(h_j+\eta)/(h_j(h_j+\eta+\sigma_0))$ — *from* $Q_{jj}/(2(h_j+\sigma_0))$ *to* $Q_{jj}/(2h_j)$ *in units of* N_0/k_e — *plus the fast part* $(N_0/2k_e)h_j$*.*

(iii) (the errors in variables) *To second order the noise inside the regressor* $\tilde{\Phi}(y)$ *exerts a mean force on the collective coordinates. Its part carried by the noise's own second moments, at fixed parameters, is* k_eF *in the slow time with* $F:=\langle\hat{J}'^\top\mathbb{E}[\varepsilon_m^2]+\hat{J}^\top\mathbb{E}[e]\rangle=O(\mathbb{E}\nu^2)$ *(*$\hat{J}':=\partial_x\hat{J}$*,* $\varepsilon_m:=y-x_m$ *the measured error; for an Ornstein–Uhlenbeck noise of rate* B*,* $\mathbb{E}[\varepsilon_m^2]=\mathbb{E}\nu^2(a^2+k_eB)/(k_e(k_e+B))\to\mathbb{E}\nu^2$ *as* $B/k_e\to\infty$*); the parameters' first-order fluctuation adds a force of the same order in* N_0 — *the compensation's curvature and slope–parameter terms in the error equation, the correlation* $\nabla_\vartheta^2\tilde{\Phi}\mathbb{E}[\varepsilon_m\delta\vartheta]$ *in the law — negligible against* F *for the measured regressor, not for the desired one (Proposition D‴.31(iv)). The stationary response is* $k_e(H+\sigma_0 I)^{-1}F$ *under the fixed prior and* $k_eH^{-1}F$ *under the adaptive prior, for every* $\eta>0$*: the adaptive prior gives up the leak's bound on the response to a persistent force in the weakly excited modes — the robustness half of the classical trade-off, here against the noise, and the lack-of-excitation drift of adaptive control ([100]) — and its function-level bias* $k_e^2F^\top H^{-1}F$ *grows like* $k_e^2(\mathbb{E}\nu^2)^2$ *while the noise is white well beyond the loop's bandwidth (*$k_e\ll B$*), and saturates for* $k_e\gg B$*, where* $\mathbb{E}[\varepsilon_m^2]\to\mathbb{E}\nu^2B/k_e$*. The part of* F *that the adaptive prior amplifies is the regressor's own,* $\langle\hat{J}'^\top\mathbb{E}[\varepsilon_m^2]\rangle$*; evaluating the regressor at the reference removes it (Proposition D‴.31).*

Hence η *buys the bias-removal rate* $\lambda_j^-(\eta)$ *at the price of the factor* $(h_j+\eta)/(h_j+\eta+\sigma_0)$ *in the floor and of the* H^{-1} *response to the errors in variables, and* k_e *lowers the floor like* $1/k_e$ *while it raises the errors-in-variables bias like* k_e^2 *(*$k_e\ll B$*).*

Proof. (i) The equations of the particles of a class are identical — the same y, the same measured error and the same prior atom (the class mean at $\eta=\infty$, the same z_k otherwise) enter them — and so are their initial values; the field is locally Lipschitz and the cells are constant near the lift, so the trajectories coincide, and the class means follow the weighted K_0-atom loop with the same noise. (ii) Linearised at Γ_w in Corollary D‴.26's whitened coordinates: $\delta\dot{e}=-k_e\delta e-\hat{J}y-(k_e+a)\nu$ (from $f(x)-\hat{f}(x+\nu)=-D\hat{f}[\delta\Theta]-a\nu$), $\dot{y}=\hat{J}^\top(\delta e+\nu)-(\sigma_0/k_e)(y-\zeta)$, $\dot{\zeta}=-(\eta/k_e)(\zeta-y)$ (the regressor's own noise is of second order). The measured error $\varepsilon_m=\delta e+\nu$ obeys $\dot{\varepsilon}_m=-k_e\varepsilon_m-\hat{J}y+\dot{\nu}-a\nu$, so $\varepsilon_m=-(\mathrm{d}/\mathrm{d}t+k_e)^{-1}[\hat{J}y]+n$, $n:=(\mathrm{d}/\mathrm{d}t+k_e)^{-1}[\dot{\nu}-a\nu]$ (written $((\mathrm{d}/\mathrm{d}t-a)/(\mathrm{d}/\mathrm{d}t+k_e))\nu$): the first part is Corollary D‴.26's slaving, the second drives $\dot{y}=\hat{J}^\top n$. With $\xi:=(\mathrm{d}/\mathrm{d}t+k_e)^{-1}\nu$, of variance $N_0/(2k_e)$ for white noise, $n=\dot{\xi}-a\xi+(\mathrm{d}/\mathrm{d}t+k_e)^{-1}[\dot{a}\xi]$ and $\hat{J}^\top n=\mathrm{d}(\hat{J}^\top\xi)/\mathrm{d}t-(\dot{\hat{J}}+a\hat{J})^\top\xi+O(\xi/k_e)$: $y=\hat{J}^\top\xi+u$, the fast jitter $\hat{J}^\top\xi$ and the slow part u driven by $-(\dot{\hat{J}}+a\hat{J})^\top\xi$, white at the slow scale with the intensity $(N_0/k_e^2)Q$ per unit time, $(N_0/k_e)Q$ in the slow time; the averaged slow equation $\mathrm{d}Y=\bar{A}_0Y\,\mathrm{d}s+(N_0/k_e)^{1/2}(Q^{1/2}\,\mathrm{d}W,0)$ (stochastic averaging, [96]; exactly, for the linearised loop, Lemma D‴.26a(v)) has the stated stationary covariance. In the eigenbasis of H the block $\bar{A}_0$ is

a direct sum of the pair blocks $\left[\begin{smallmatrix} -(h+\sigma_0) & \sigma_0 \\ \eta & -\eta \end{smallmatrix}\right]$ ($\eta = 0$: $-(h+\sigma_0)$; $\eta = \infty$: $-h$), whose Lyapunov equation with the noise q on y gives $\Sigma_{\zeta\zeta} = \Sigma_{y\zeta} = \eta\Sigma_{yy}/(h+\eta)$ and $\Sigma_{yy} = q(h+\eta)/(2h(h+\eta+\sigma_0))$; Q's off-diagonal entries in that basis enter only the covariances of different modes, which $\operatorname{tr}(H\Sigma_{yy}) = \sum_j h_j \Sigma_{yy,jj}$ does not see. The fast jitter gives $\mathbb{E}[(\hat{J}^\top\xi)^\top H(\hat{J}^\top\xi)] = (N_0/2k_e)\operatorname{tr} H^2$ over a period, and $\mathbb{E}[(\hat{J}(t)\hat{J}(t)^\top\xi)^2] = (N_0/2k_e)\langle|\hat{J}|^4\rangle$. (iii) At fixed parameters the second-order part of $\mathbb{E}[\varepsilon_m \nabla_\vartheta \tilde{\Phi}(y;\vartheta)]$ at the lift is $\partial_x \nabla_\vartheta \tilde{\Phi} \cdot \mathbb{E}[\varepsilon_m^2] + \nabla_\vartheta \tilde{\Phi} \cdot \mathbb{E}[e]$, $\mathbb{E}[e]$ from the error equation's second-order terms $\frac{1}{2}\partial_x^2 f \cdot e^2 - \frac{1}{2}\partial_x^2 \hat{f} \cdot (e+\nu)^2$ — one second derivative for f and $\hat{f}$, by the hypothesis (otherwise $\frac{1}{2}(\partial_x^2 f - \partial_x^2 \hat{f})\operatorname{Var}(e)$ remains, of the same order); the parameters' fluctuation adds $\nabla_\vartheta^2 \tilde{\Phi} \cdot \mathbb{E}[\varepsilon_m \delta\vartheta]$ in the law and $-\partial_x D\hat{f}[\mathbb{E}[(\delta e + \nu)\delta\Theta]] - \frac{1}{2}\mathbb{E}[D_\Theta^2 \hat{f}[\delta\Theta, \delta\Theta]]$ in the error equation, of the same order in N_0 (computed in the numerics from the periodic covariance: the full second-order mean). For the Ornstein–Uhlenbeck noise, quasi-statically along the orbit, $\mathbb{E}[\nu e] = -(k_e + a)\mathbb{E}\nu^2/(k_e + B)$ and $\operatorname{Var}(e) = (k_e + a)^2\mathbb{E}\nu^2/(k_e(k_e + B))$, which give $\mathbb{E}[\varepsilon_m^2] = \mathbb{E}\nu^2 + 2\mathbb{E}[\nu e] + \operatorname{Var}(e)$ as stated and $\mathbb{E}[e] = -(\partial_x^2 \hat{f}/(2k_e))\mathbb{E}\nu^2(B - k_e - 2a)/(k_e + B)$. A constant slow force F_s on the pair block of a mode has the stationary point $\zeta = y$ and $(h+\sigma_0)y - \sigma_0\zeta = F_s$, i.e. $y = F_s/h$ for every $\eta > 0$, and $y = F_s/(h+\sigma_0)$ under the fixed prior. □

Status. (i) proved, exactly, as invariance; the threshold formal at the lift (the separations' averaged second-order drift there), the drift along the noisy trajectory computed from the second-order transient and checked against the real loop's separation rates on antithetic noise pairs, the long-time threshold open; (ii) proved for the linearised loop under white noise, for the floor's period average (Lemma D‴.26a(v): the slow coordinates' drift decoupled exactly, their noise $\varepsilon b\nu$, at leading order the high-passed one; at $\eta = 0$ with $K = \hat{J}$, $\Lambda = \sigma_0 I$), with the relative remainder $O(1/k_e)$, the closed form exact for the averaged system; the OU noise of the numerics and the real loop formal — and checked against the exact periodic covariance of the linearised loop: the deficit 2.4%, 0.61%, 0.14%, 0.028% at $k_e = 25$–200 ($\eta = 0$) is dominated there by a term in k_e^{-2}, but the relative error is $O(1/k_e)$ — the slope a in the noise input $-(k_e + a)\nu$, a rate that does not change sign under time reversal, breaks the parity of Corollary D‴.26(c) (with the noise input $-k_e\nu$ the deviation is numerically even in $1/k_e$, to $k_e = 800$) — and the ratio crosses 1 near $k_e = 500$; (iii) formal — a quasi-static second-moment computation for the noise's own moments, the parameters' part computed numerically — its stationary response exact for the averaged system, the exact linearised loop's 10.7% (adaptive) and 2.6% (fixed) higher at $k_e = 25$, an $O(k_e^{-2})$ excess; the real nonlinear loop checked by Monte Carlo against the sum of the two. Classical: measurement noise in adaptive control and the errors-in-variables bias of estimates from noisy regressors (bias-compensated least squares and instrumental variables in identification; bounded disturbances and robust modifications, [47]); using the reference in the regressor, as the desired compensation adaptation law does ([96]); the Lyapunov covariance of a constant-gain adaptive algorithm as its diffusion approximation (Benveniste–Métivier–Priouret), the LMS misadjustment's dependence on the input's spectrum through a trace only (Widrow et al.: the $\eta = \infty$ floor's $\operatorname{tr} Q$), the leaky LMS's steady state (Mayyas–Aboulnasr: the fixed prior's factor $h_j/(h_j + \sigma_0)$), the common noise of mean-field theory (Carmona–Delarue–Lacker), of which the coherence is an instance, and the drift of the weakly excited directions under a persistent disturbance (Anderson), of which the H^{-1} response is ([100]). New, in the transport-leak setting: the interpolation $(h_j + \eta)/(h_j + \eta + \sigma_0)$ of the floor in the prior's mobility, the driving matrix $Q = \langle(\dot{\hat{J}} + a\hat{J})^\top(\dot{\hat{J}} + a\hat{J})\rangle$ — the sensitivity-filtered measurement the law sees, built on Corollary D‴.26's derivative Gram — and their combination with the coherence of the K_0-atom identification and its noise threshold at the lift. Checked: `tests/test_measurement_noise.py` (the closed form against the Lyapunov solution, the trace identity at $\eta = \infty$ and the monotonicity in η; the exact periodic floors of the linearised loop at $k_e = 100$ for $\eta = 0$, σ_0 and ∞; a lift $m = 6$ under a noise path staying exactly replicated and following the K_0-atom loop driven by the same path; a separation of the weakest class decaying at the noise std 0.01 and growing at 0.05 — a sign test, the rates within 10^{-4} of the lift's prediction —; the separations' drift against an independent quadrature of its two moments, and along the second-order transient the mean parameters moved by the averaged force and the mean error carrying the tracking error of the developing deviation; the OU noise's stationary variance and correlation, and the law seeing the measured error at the measured

point; the responses $(H+\sigma_0 I)^{-1}$ and H^{-1} to a constant force, the quasi-static force against the full second-order mean's noise part to $2\cdot10^{-3}$, the parameters' part below 10^{-4} of the bias; the exact periodic mean against the filtered-Gram response under both priors, the fixed prior's branch and its leak in the bias; and the mean function's curvature term against an independent contraction).

Numerics (`exp_measurement_noise.py` → `results/chaos/measurement_noise.md`). The three-bump instance ($\sigma_0 = 0.15\sigma_{\mathrm{crit}}$, $x_m = 2\sin t$, $k_e = 25$) at the lift of the exact representation of the matching weights $q = 1/3$, the prior at its atoms. (A) The linear theory: H's nine eigenvalues span $3.06\cdot10^{-4}$–0.195; $\operatorname{tr} Q = 8.784$, of which 8.105 is the derivative Gram $\langle\dot{\hat{J}}^\top\dot{\hat{J}}\rangle$ and 0.679 the slope term $\langle a^2\hat{J}^\top\hat{J}\rangle$ (the cross term vanishes, as on every one-dimensional reference: an exact differential); Q_{jj}/h_j ranges from 104 (mode 1) to 5.5 (mode 6). The parameters' floor per unit N_0 is 0.1597 under the fixed prior, 0.1627 at $\eta=\sigma_0$, 0.1690 at $10\sigma_0$ and $0.1757 = \operatorname{tr} Q/(2k_e)$ at $\eta=\infty$ — the adaptive prior costs 10% here, the strong modes carrying the floor — while the weakest mode's parameter variance grows from 0.0865 to 2.09, ×24.1 in the averaged system (×28.0 in the exact linearised loop at $k_e = 25$, Part B). The fast parts are $\operatorname{tr} H^2 = 0.0657$ and $\langle|\hat{J}|^4\rangle = 0.299$. The second-order force of the noise's own moments (OU noise, $B = 100$) has the averaged function-level stationary bias 491 (fixed prior) and 2397 (adaptive, ×4.88) per unit $(\mathbb{E}\nu^2)^2$, equal to the floor of the parameters at the noise std 0.0026 and 0.0012 respectively and dominant above; the full second-order mean (the periodic covariance contracted with the true field's second derivatives) reproduces the force to $1.7\cdot10^{-4}$, the parameters' fluctuation changing the bias by less than 10^{-5} (2397.66 → 2397.68; the mean function's 2397.65), and the exact periodic second-order mean of the linearised loop gives 504 and 2653 (×5.27). The averaged bias grows in k_e with the local exponent 1.93–1.99 for $B \geq 20k_e$ but 1.69 and 1.47 at $B = 8k_e$ and $4k_e$, and saturates beyond. At the lift the separations' second-order drift has $\lambda_{\max}(S_k) = 0.0148$, 0.0101 and 0.298 per unit $\mathbb{E}\nu^2$ (real time) in the three classes: there the third class's separations stop decaying at the noise std 0.031 (1.5% of the reference's amplitude), and at Part C's stds 0.003 and 0.01 the margin σ_0/k_e is used to 0.9% and 10.5%. (B) The exact periodic stationary covariance of the linearised loop (the monodromy and a discrete Lyapunov equation) against the prediction, at $\eta = 0$, σ_0, ∞ and $k_e = 25$–800: the function-level floor's ratio 0.976–0.980 at $k_e = 25$, 0.994–0.995 at 50, 0.9986–0.9988 at 100, 0.9997–0.9998 at 200, and above 1 at 800 ($+1.2\cdot10^{-5}$ at $\eta=0$, $+1.3\cdot10^{-5}$ at $\eta=\infty$); the deviation's successive ratios per doubling are 3.87, 4.38, 5.26, 11.4 and a change of sign ($\eta=\infty$) — a k_e^{-2} term dominating to $k_e = 200$ over a first-order one of the opposite sign — while with the noise input $-k_e\nu$ (the slope removed, the control) they are 3.67, 3.90, 3.98, 3.99, 4.00: without the slope the expansion is numerically even (to $k_e = 800$; not proved). The weakest mode's variance is 0.845 (fixed prior) and 0.980 (adaptive) of its slow-plus-fast prediction at $k_e = 25$, tending to 1. The exact stationary second-order mean of the linearised loop lies 10.7%, 2.65%, 0.66%, 0.17% above the averaged bias under the adaptive prior and 2.58%, 0.64%, 0.16%, 0.04% under the fixed one at $k_e = 25$, 50, 100, 200 — the weak modes' stiffness being the filtered Gram $H - k_e^{-2}\langle\dot{\hat{J}}^\top\dot{\hat{J}}\rangle$ of Corollary D‴.26(c). (C) The real loop (RK4, $\mathrm{d}t = 10^{-3}$, an OU noise exact at the steps, 12 noise paths of 300 time units common to the four cases) against the transient covariance of the linearised loop plus the second-order mean deviation: at the noise std 0.003 within one standard error at every record (z from −0.99 to 0.99), the mean part 42% of the prediction at $t = 300$; at std 0.01 within 1.75 standard errors (z from −0.3 to 1.75, the records of a run correlated), the mean part 89% at $t = 300$ and the covariance part alone 8.8–8.9 times below the Monte Carlo — the second-order force is the dominant effect of the noise there. The adaptive and the fixed prior drift apart, by 6.0% and 6.4% at $t = 300$ (predicted 5.3% and 6.1%): the weak modes, where the two responses part, equilibrate over $k_e/h_1 \approx 8\cdot10^4$ time units, so over 300 time units the transient checks the force (the weak modes still grow linearly), and the stationary response is checked against the exact linearised loop (B). The lift $m = 15$ (classes of five) driven by the same noise path as the K_0-atom run gives the same $\hat{f}$ to rounding (identical in the stored float32, $5.6\cdot10^{-16}$ apart in float64): the noise is

coherent, as (i) states. (D) But invariance is not attraction: from a separation of size 10^{-6} along S_3's top eigenvector, on eight antithetic pairs of noise paths of 300 time units per std (ν and $-\nu$, whose pair means cancel the first-order part of the log-growth), the lift's third class separates at $-2.50 \cdot 10^{-4} \pm 3.8 \cdot 10^{-7}$ per time unit at the std 0.01, $+2.74 \cdot 10^{-6} \pm 3.3 \cdot 10^{-6}$ at 0.03 and $+4.53 \cdot 10^{-4} \pm 8.4 \cdot 10^{-6}$ at 0.05. The noise-induced part $\lambda_{\text{eff}} := (\text{rate} + \sigma_0/k_e)/\mathbb{E}\nu^2$ is 0.3271 ± 0.0038, 0.3175 ± 0.0037 and 0.2943 ± 0.0034, against the lift's drift $\lambda_{\max}(S_3) = 0.2981$ and the drift along the second-order transient from the lift, 0.3264, 0.3270 and 0.3281 (the same estimator applied to $\dot{u} = (\mathbb{E}\nu^2 S_3(t) - \sigma_0/k_e)u$). Along the transient the mean error's rms per unit $\mathbb{E}\nu^2$ grows from 0.121 to 0.602 over the six 50-unit windows to $t = 300$ and $\lambda_{\max}(S_3(t))$ from 0.307 to 0.349: the tracking error of the developing errors-in-variables deviation. The weighted fit $\lambda_{\text{eff}} = c_0 + c_1 \mathbb{E}\nu^2$ has the small-noise limit $c_0 = 0.3291 \pm 0.0033$ — 10.4% above the lift's drift, within its standard error of the transient's 0.3264 — and the next order $c_1 = -13.8 \pm 2.0$; over this window the measured rate vanishes at the std 0.0299, against 0.0308 by the lift's drift and 0.0294 by the transient's. Above it the replicated configuration repels, its within-class spread staying exactly 0.0 from a replicated start only because each particle's update is the same arithmetic.

Proposition D‴.31 (the adaptive prior is a washout leak; the desired regressor makes the noise's persistent force a least-squares force)**.** *Let the leak be Theorem D‴.23's toward an adaptive prior,* $\mathrm{d}z_j/\,\mathrm{d}s = -\eta(z_j - \hat{T}_j)$ *with* $0 < \eta < \infty$ *(the nearest-atom form of Proposition D‴.24(ii) with the class barycentres likewise), in the averaged loop or in the real loop* (CL) *of Corollary D‴.26* ($s = t/k_e$)*; at an exact representation of representable weights* w *let* H, $\hat{J}$, a *and the mean* $\langle\cdot\rangle$ *over a period be Proposition D‴.30's,* $\|g\|_\rho^2 := \langle g^2\rangle$ *for a function* g *along the orbit, and* P *the* L^2_ρ*-projection onto the span of* $\hat{J}$*'s components.*

(i) (the washout) *Along every solution and every path of the plan with* $\hat{T}_j$ *of bounded variation,* $\hat{T}_j(s) - z_j(s) = \int_0^s e^{-\eta(s-\tau)}\,\mathrm{d}\hat{T}_j(\tau) + e^{-\eta s}(\hat{T}_j(0) - z_j(0))$ *— and for every measurable* $\hat{T}_j$ *(a chattering plan included) in the integrated form* $z_j(s) = e^{-\eta s}z_j(0) + \eta\int_0^s e^{-\eta(s-\tau)}\hat{T}_j(\tau)\,\mathrm{d}\tau$*: the prior lags the barycentre by the barycentre's exponentially forgotten increments. For integral plans (the nearest-atom cells, the OT plan at a commensurate* m*) the leak on a particle,* $-\sigma_0(\vartheta_i - z)$*, is therefore the full* σ*-modification of its offset from the barycentre plus, on the barycentre, the* σ*-modification passed through the washout filter* $(\mathrm{d}/\mathrm{d}s)(\mathrm{d}/\mathrm{d}s + \eta)^{-1}$ *— zero at zero frequency, the full leak above* η*; with split rows (an incommensurate* m*) the barycentres also feel* $-\sigma_0(I - M)z$*,* $M_{jl} := (m/q_j)\sum_i \gamma_{ij}\gamma_{il}$*, which is not washed out. Linearised at the exact representation, mode* j *of the collective block has at the slow frequency* ω *the stiffness* $h_j + \sigma_0 i\omega/(i\omega + \eta)$*: the fixed prior's* $h_j + \sigma_0$ *for* $\omega \gg \eta$*, the bare* h_j *at* $\omega = 0$*. This is Theorem D‴.23(iii)'s bias removal — feedback through a washout filter leaves the equilibria where they are ([97]), here the replicated ones — and its price: no leak opposes a persistent force on the class barycentres (the separations keep the full leak).*

(ii) (the stationary identity) *Let the prior be an affine time-invariant filter of the barycentre and of a fixed anchor* z_a *— linearised,* $\zeta = \mathcal{K}[y] + (I - \mathcal{K})[\delta]$ *in the whitened weighted coordinates,* δ *the anchor's misplacement from the representation; for example* $\mathrm{d}z/\,\mathrm{d}s = -\eta(z - \hat{T}) - \kappa(z - z_a)$*,* $\kappa \geq 0$*, with* $\mathcal{K}(0) = \eta/(\eta + \kappa)$ *(*$\kappa = 0$ *the adaptive prior,* $\kappa = \infty$ *the fixed prior at* z_a*) — and* $S_0 := \sigma_0(I - \mathcal{K}(0))$ *the leak's zero-frequency gain (the anchor entering through* $I - \mathcal{K}$*: the filter returns the anchor when the barycentre sits there). A persistent force* F *on the collective coordinates and the misplacement* δ *give the stationary deviation* $y = RF + B\delta$ *with* $R = (H + S_0)^{-1}$ *and* $B = (H + S_0)^{-1}S_0$*, so*

$$RH + B = I, \qquad \textit{i.e.} \qquad R = (I - B)H^{-1}:$$

the response is the unregularised one times the share not taken by the anchor ($HR + B = I$ *as well where* S_0 *commutes with* H*, as for the family; not for a prior fixed on some coordinates or a projected prior with an active face). On every mode, whatever* S_0*,* $h_j r_{jj} + b_{jj} = 1$

($r_{jj} := v_j^\top R v_j$, $b_{jj} := v_j^\top B v_j$): a vanishing bias forces the response $1/h_j$. So no such prior both removes the leak's bias and keeps its bound on the response to persistent forces — the trade binding on the modes with $h_j < \sigma_0$, where $1/h_j$ exceeds the fixed prior's $1/\sigma_0$ — and a remedy for the robustness loss of Proposition D‴.30(iii) has to change the force.

(iii) (least-squares forces) *For $H \succ 0$ and every S with $S+S^\top \succeq 0$, $F^\top (H+S)^{-\top} H (H+S)^{-1} F \le F^\top H^{-1} F = \min\{\|g\|_\rho^2 : \langle \hat{J}^\top g\rangle = F\}$, with equality at $S = 0$: the function-level stationary response to a persistent force is at most the squared norm of the smallest target perturbation that exerts it ($\langle \hat{J}^\top g\rangle$ is the learning force of the target $f+g$), and the adaptive prior attains the bound. A least-squares force $\langle \hat{J}^\top g\rangle$ therefore moves the learned function by at most $\|g\|_\rho$ under every prior — under the adaptive prior by exactly Pg, so that the stationary point's projected residual $P(g - \hat{J}y)$ vanishes.*

(iv) (the desired regressor) *Let the adaptive law evaluate its regressor at the reference, $\dot\vartheta_i = G\nabla_\vartheta((y-x_m)^\top \tilde\Phi(x_m;\vartheta_i)) - (\sigma_0/k_e)(\vartheta_i - z_{\pi(i)})$ (DR), with the compensation $\hat f_\Theta(y)$ as before or also at the reference, $u = \dot x_m - k_e(y-x_m) - \hat f_\Theta(x_m)$ (DR*, the desired compensation adaptation law [96]). Then (a) without noise the lift Γ_w is a solution and the linearisation at it is Corollary D‴.26's — for DR exactly (the regressor's variation multiplies $e \equiv 0$), for DR* with $-(k_e - a(t))$ in place of $-k_e$ in the error channel — and the averaged loop is unchanged at leading order in $1/k_e$ (the regressor moves by $x - x_m = O(1/k_e)$); (b) to first order in ν, DR is Proposition D‴.30(ii)'s linearised loop driven by the same noise, path by path; DR* has the noise input $-k_e\nu$ in place of $-(k_e+a)\nu$, and the same floor up to a relative $O(1/k_e)$; (c) to second order the part of the force carried by the noise's own second moments (at fixed parameters) is the least-squares force $\langle \hat{J}^\top g\rangle$ (slow time) of $g := k_e\mathbb{E}[e]$, the rescaled mean tracking error at the lift; for an Ornstein–Uhlenbeck noise of rate B, quasi-statically along the orbit, $g = -\frac12 \partial_x^2 \hat f \cdot \mathbb{E}\nu^2 (B - k_e - 2a)/(k_e + B)$ for DR and $g = \frac12 \partial_x^2 f \cdot \mathbb{E}\nu^2 k_e^3/((k_e-a)^2(k_e - a + B))$ for DR* ($k_e > \max a$), so $|g| \le \frac12 |\partial_x^2 f| \mathbb{E}\nu^2 (1 + O(|a|/k_e))$ for all k_e and B. By (iii) its function-level response is at most $\|g\|_\rho^2 \le \frac14 \|\partial_x^2 f\|_\rho^2 (\mathbb{E}\nu^2)^2 (1 + O(1/k_e))$ under every prior — uniformly in k_e, in B and in the spectrum of H — and under the adaptive prior it leaves no projected mean tracking error. The parameters' first-order fluctuation adds a second-order force of the same order in N_0: in the error equation the compensation's curvature $\frac12 D_\Theta^2 \hat f[\Sigma_{\Theta\Theta}]$ and, for MR and DR, its slope–parameter term (DR*'s compensation, at x_m, has none) — a least-squares force again, of a perturbation set by the parameters' covariance; the curvature also shifts the mean function, and the loop removes its tangent part —, in the learning law the correlation $\nabla_\vartheta^2 \tilde\Phi \mathbb{E}[\varepsilon_m \delta\vartheta]$ of the regressor's parameter dependence with the measured error — an errors-in-variables term in the parameters, which the reference in the regressor does not remove and which is not the learning force of a mean tracking error, so no bound in the noise's moments applies to it. Its size is set by the parameters' stationary covariance (mode j's variance $(N_0/2k_e) Q_{jj}(h_j+\eta)/(h_j(h_j+\eta+\sigma_0))$, unbounded as $h_j \to 0$ at $\eta > 0$), so the bound $\|g\|_\rho^2$ is the noise part's, not the total's. With the measured regressor the noise part is $F_1 + \langle \hat{J}^\top g\rangle$, $F_1 := k_e \langle \hat{J}'^\top \mathbb{E}[\varepsilon_m^2]\rangle$ — the regressor's errors in variables proper, the learning force of no mean tracking error — and the adaptive prior's stationary point keeps the projected mean tracking error $P\mathbb{E}[e] = -\hat J H^{-1} F_1/k_e$: what grows like k_e^2 while $k_e \ll B$ is k_e times it, of squared norm $F_1^\top H^{-1} F_1$, the function-level bias; the mean tracking error itself, of squared norm $F_1^\top H^{-1} F_1/k_e^2 \propto (B/(k_e+B))^2$, stays bounded.*

Hence with the desired regressor the first-order floor still falls like $1/k_e$ while the noise part of the errors-in-variables bias stays bounded, and on the instance the parameters' part's own response falls like k_e^{-2} — a one-dimensional reference, where the law's leading term, the period average of a function of x_m times $\dot x_m$ (a line integral $\oint M \partial_x \hat J \,\mathrm{d}x$ along the orbit), vanishes; on a curve in $\mathbb{R}^n$, $n > 1$, it need not: the trade-off in k_e of Proposition D‴.30 disappears, and what is left is the plant's state jitter $\mathbb{E}[e^2] = \mathbb{E}\nu^2 (k_e+a)^2/(k_e(k_e+B))$ (DR; DR: $k_e^2 \mathbb{E}\nu^2/((k_e-a)(k_e-a+B))$)*

— *the classical price of a high gain under measurement noise.*

Proof. (i) The integrated form is the variation of constants for $\mathrm{d}z_j/\,\mathrm{d}s = -\eta(z_j - \hat{T}_j)$; for $\hat{T}_j$ of bounded variation (it jumps where the plan does, finitely often on bounded intervals), $u := \hat{T}_j - z_j$ satisfies $\mathrm{d}u = \mathrm{d}\hat{T}_j - \eta u\,\mathrm{d}s$, whence $u(s) = e^{-\eta s}u(0) + \int_0^s e^{-\eta(s-\tau)}\,\mathrm{d}\hat{T}_j(\tau)$; and $\vartheta_i - z = (\vartheta_i - \hat{T}) + u$ for an integral plan (for split rows the targets are $m\sum_j \gamma_{ij}z_j$, and averaging the leak over a class gives $-\sigma_0(\hat{T} - Mz)$). Linearised, $\zeta = \eta(i\omega+\eta)^{-1}y$ on each mode (the prior and the leak are scalars in the whitened weighted coordinates, so they commute with H), and $\sigma_0(y-\zeta) = \sigma_0 i\omega(i\omega+\eta)^{-1}y$. (ii) At zero frequency $\zeta = \mathcal{K}(0)y + (I - \mathcal{K}(0))\delta$, so the leak on y is $-\sigma_0(y-\zeta) = -S_0(y-\delta)$, and the stationary equation $Hy + S_0(y-\delta) = F$ gives $y = (H+S_0)^{-1}(F + S_0\delta)$; then $RH + B = (H+S_0)^{-1}(H+S_0) = I$, and on the diagonal in H's eigenbasis $h_j r_{jj} + b_{jj} = 1$. For the family, the prior's stationary equation $\eta(y-\zeta) = \kappa(\zeta - \delta)$ gives $\mathcal{K}(0) = \eta/(\eta+\kappa)$. (iii) By congruence with $H+S$, $(H+S)^{-\top}H(H+S)^{-1} \le H^{-1} \Leftrightarrow H \le (H+S)^\top H^{-1}(H+S) = H + S + S^\top + S^\top H^{-1}S$, true for $S + S^\top \succeq 0$. The map $g \mapsto \langle \hat{J}^\top g\rangle$ has the adjoint $F \mapsto \hat{J}F$, so the smallest g with $\langle \hat{J}^\top g\rangle = F$ is $\hat{J}H^{-1}F$, of squared norm $F^\top H^{-1}F$; for $F = \langle \hat{J}^\top g\rangle$, $F^\top H^{-1}F = \|Pg\|_\rho^2 \le \|g\|_\rho^2$, and at $S = 0$, $\hat{J}y = \hat{J}H^{-1}\langle \hat{J}^\top g\rangle = Pg$. (iv) (a) At $e = 0$ both laws vanish, and their variations, $\delta e\nabla_\vartheta\tilde{\Phi}(x;\vartheta) + e(\cdots)$, coincide at $x = x_m$; for DR*, $f(x) - \hat{f}(x_m) = a\delta e - \hat{J}y$ to first order. Off the lift e is slaved, $e = r(x_m)/k_e + O(k_e^{-2})$, so the two regressors differ by $O(1/k_e)$. (b) The regressor's dependence on y enters the law multiplied by ε_m, at second order; for DR* the compensation no longer carries ν. (c) At fixed parameters the mean of $\varepsilon_m\nabla_\vartheta\tilde{\Phi}(x_m;\vartheta)$ is $\nabla_\vartheta\tilde{\Phi}(x_m)\mathbb{E}[e]$, since $\mathbb{E}\nu = 0$; the parameters' fluctuation adds $\nabla_\vartheta^2\tilde{\Phi}(x_m)\mathbb{E}[\varepsilon_m\delta\vartheta]$ in the law and the compensation's curvature and slope–parameter terms in the error equation (Proposition D‴.30's proof). For DR, $\mathbb{E}[e]$ is Proposition D‴.30's (the same compensation); for DR*, $\dot{e} = (a-k_e)e + \frac{1}{2}\partial_x^2 f\, e^2 - k_e\nu + O(3)$, whose linear part has $\mathrm{Var}(e) = k_e^2\mathbb{E}\nu^2/((k_e-a)(k_e-a+B))$ for the OU noise, and the mean balance is $(k_e-a)\mathbb{E}[e] = \frac{1}{2}\partial_x^2 f\,\mathrm{Var}(e)$. The bounds: $|B - k_e - 2a| \le (k_e+B)(1+2|a|/(k_e+B))$ and $k_e^3/((k_e-a)^2(k_e-a+B)) \le (k_e/(k_e-a))^3$. With the measured regressor, at the adaptive prior's stationary point $\langle \hat{J}^\top(g - \hat{J}y)\rangle + F_1 = 0$ and the mean of the error equation gives $k_e\mathbb{E}[e] = g - \hat{J}y$ at second order, so $k_e P\mathbb{E}[e] = -\hat{J}H^{-1}F_1$. □

Status. (i) proved, exactly (every solution of the real or the averaged loop; the differential form for $\hat{T}$ of bounded variation, the integrated one always; the leak's decomposition for integral plans); (ii) proved as $RH + B = I$ and per mode — linear algebra on the averaged linearised block; $HR + B = I$ where S_0 commutes with H; (iii) proved ($S + S^\top \succeq 0$, $H \succ 0$); (iv)(a) proved at the lift and for its linearisation, the leading-order equality of the averaged loops by Theorem J's averaging argument (not rewritten); (iv)(b) the first-order identity proved for the linearised equations, the floors themselves with Proposition D‴.30(ii)'s status; (iv)(c) formal and split — the noise part by Proposition D‴.30(iii)'s quasi-static second-moment computation, its consequences exact for the averaged system; the parameters' part computed numerically (the full second-order mean), not in closed form. Classical: reference regressors and the desired compensation adaptation law ([96]; DR* without Sadegh and Horowitz's nonlinear damping is a local statement), washout filters ([97]), the adaptive law kept near a low-pass filtered copy of itself — Yucelen and Haddad's low-frequency learning, which has the washout structure ([101]) — and elastic averaging's moving-average centre ([90]); the identity, per mode the Tikhonov filter factor plus the residual factor (the Kalman-gain split of a regularised estimate); the bound (iii), the minimum-norm solution with the contraction of the regularised fit; and the split of the noise's force into the regressor's errors in variables proper and the output's, of which only the mean biases; and the parameters' own errors in variables, the classical second-order bias of a constant-gain stochastic approximation from its own fluctuation — the mean field's curvature in the parameters contracted with their stationary covariance, plus the noise–parameter correlation (Benveniste–Métivier–Priouret; for constant-step stochastic gradient descent Dieuleveut–Durmus–Bach, who remove it by Richardson–Romberg extrapolation; the curvature bias of nonlinear least squares, Box's, the analogue of $\frac{1}{2}D_\Theta^2\hat{f}[\Sigma_{\Theta\Theta}]$: [100]). New are their transfer to the transport leaks — the washout reading of the adaptive prior in the mass-constrained, mean-field setting, uniform in m — and the instance's quantification, with the observation that the desired regressor does

not remove the parameters' part. Checked: `tests/test_eiv_remedy.py` (the washout identity along a real-loop solution at finite η, the barycentre's velocity integrated alongside, and on the averaged loop across a plan switch, the jump's share large; the collective block's transfer function against the washout formula, $RH + B = I$ along the prior family and, R and B from their dynamic blocks, for priors whose zero-frequency leak does not commute with H — fixed on the amplitudes, one active face, a matrix anchor gain, a random one with $K + K^\top \succeq 0$ — where $HR + B = I$ fails; the response bound under random leaks $S \geq 0$, the smallest equivalent perturbation and the projection; the lift exact under DR and DR*, on a common noise path DR's run within the second order of MR's, and the regressor's point in the law; DR's and DR*'s noise forces least-squares, their pointwise bounds on g, MR's F_1 three orders of magnitude above; the quasi-static forces against the full second-order mean's noise part, the parameters' part above 1% for DR and DR* and below 10^{-5} for MR).

Numerics (`exp_eiv_remedy.py` → `results/chaos/eiv_remedy.md`). The instance of Proposition D‴.30 with an Ornstein–Uhlenbeck noise of rate B; MR the measured regressor and compensation, DR the regressor at the reference, DR* both at the reference. (A) The averaged system's biases per unit $(\mathbb{E}\nu^2)^2$. From the noise's own moments, the adaptive prior's function-level bias at $k_e = 25$, $B = 100$ is 2397 (MR), 0.471 (DR) and 0.0526 (DR*); with the parameters' fluctuation (the full second-order mean) 2398, 0.486 (+3.3%) and 0.0481 (−8.7%), the mean function's 2398, 0.476 and 0.0511. Over $k_e = 25$–200 and $B = 100$, 1000 the parameters' part changes MR's bias by at most 0.03%, DR's by −0.75% to +3.3% except at $k_e = B = 100$ — where DR's noise part nearly cancels ($4.5 \cdot 10^{-5}$) and the parameters' part raises it by 56% — and DR*'s by −8.7% to −0.15%; its own response is $3.9 \cdot 10^{-4}$, $1.1 \cdot 10^{-4}$, $2.9 \cdot 10^{-5}$ and $7.4 \cdot 10^{-6}$ at $k_e = 25$, 50, 100, 200 ($B = 100$), like k_e^{-2} — the cancellation of the law's leading term on the one-dimensional reference —, while its share of DR's bias does not fall monotonically. MR over DR (full) is $4.9 \cdot 10^3$ at $k_e = 25$, $B = 100$, and $3.0 \cdot 10^3$–$2.1 \cdot 10^8$ over the grid. MR's bias grows like $(k_e B/(k_e + B))^2$ (×46.6 from $k_e = 25$ to 200 at $B = 1000$, the law ×46.7, k_e^2 ×64), while DR's and DR*'s stay at most 1.18, below $\frac{1}{4}\|\partial_x^2 f\|_\rho^2 = 1.31$. DR's noise part attains its bound $\|g\|_\rho^2$ to 0.01% at $k_e = 25$ — g lies almost in the tangent space, the second derivative of a Gaussian bump being in the span of its amplitude and width derivatives. The rescaled projected mean tracking error $k_e^2 \|P\mathbb{E}[e]\|_\rho^2$ under the adaptive prior is 2394 for MR (0 for DR and DR* by construction), the mean tracking error itself 3.83 at $k_e = 25$ and 0.66 at 200 ($B = 100$). The noise level at which the adaptive prior's full bias reaches its floor rises from the std 0.0012 (MR) to 0.085 (DR) and 0.27 (DR*) at $B = 100$ (stds above 0.1 lie outside the small-noise regime). (B) Along the prior family at $\eta = \sigma_0$ ($\kappa/\eta = 0$, 0.1, 1, 10 and the fixed prior; S_0 from 0 to σ_0) the weakest mode's response falls from 3269 to 136 while its bias factor rises from 0 to 0.959, with $h_1 r_1 + b_1 = 1$ and $\max|RH + B - I| \leq 1.1 \cdot 10^{-14}$; the trade binds on the 3 of 9 modes with $h_j < \sigma_0$; MR's errors-in-variables bias falls only from 2397 to 491, and DR's from 0.471 to 0.211 — MR over DR $2.3 \cdot 10^3$–$5.1 \cdot 10^3$ along the family. For priors whose zero-frequency leak does not commute with H, R and B again from their dynamic blocks — the prior fixed on the amplitudes or on the centres and adaptive at $\eta = \sigma_0$ elsewhere, one active face of the projected adaptive prior ($\eta = \infty$), the anchor gain $K = \eta P_{\text{amp}}$, a random anchor gain with $K + K^\top \succeq 0$ — the block's R and B equal $(H + S_0)^{-1}$ and $(H + S_0)^{-1} S_0$, S_0 by elimination of the prior, to $1.9 \cdot 10^{-15}$: $RH + B = I$ holds to $1.4 \cdot 10^{-14}$ and per mode to $8.9 \cdot 10^{-15}$, while $\max|HR + B - I|$ is 0.47–0.79. (C) The real loop ($k_e = 25$, $B = 100$, std 0.01, $\eta = \infty$, RK4 $\mathrm{d}t = 10^{-3}$, 12 runs of 300 time units, the three laws on common noise paths): at $t = 300$, $\mathbb{E}\|\hat{f} - f\|_\rho^2$ is $2.43 \cdot 10^{-6}$ under MR against $2.80 \cdot 10^{-7}$ under DR and $2.75 \cdot 10^{-7}$ under DR*, a ratio of 8.7 (predicted 9.2); DR and DR* sit at their covariance part, their mean parts 0.07% and 0.008% of it. Against the prediction the z-scores run from −2.04 to −0.04 (MR; Proposition D‴.30's runs of the same loop, on other paths, lay 7% above it) and from −1.8 to 0.82 (DR, DR*). The run mean of $\hat{f}_{\text{MR}} - \hat{f}_{\text{DR}}$, in which the first-order responses cancel path by path, equals the predicted $\hat{J}(\bar{y}_{\text{MR}} - \bar{y}_{\text{DR}})$ to a relative L_ρ^2 error of 0.5–5.3% at every record, 0.43–1.83 times the run mean's relative sampling

error; the averaged prediction carries its own $O(1/k_e)$ error, and the test checks F_1 (through MR − DR), not DR's own force, whose mean part is $7 \cdot 10^{-4}$ of its covariance part.

Numerics (`exp_chaos_learning.py` → `results/chaos/learning_table.md`). The instance: $n = 1$, saturated features with $a_{\max} = 4$, $c_{\max} = 3$, $s \in [0.3, 2]$, prior $\vartheta^0 = 0$, $G = \mathrm{diag}(1, 0.25, 0.25)$; a plant of three Gaussian bumps, $(A, C, S) = (1, -1.5, 0.5)$, $(-0.7, 0, 0.4)$, $(0.8, 1.5, 0.5)$, which three kernels of weights $3A_k$ represent exactly; reference $x_m = 2\sin t$, ρ on a grid of 256 times; slow time, midpoint rule with step 0.02; $m \in \{16, 64, 256\}$ sharing their particles with an $M = 2048$ reference (the stand-in for the mean-field flow, as in Theorems D′ and D″); horizon 900 slow units; four seeds per start. Proposition K gives $\sigma_{\mathrm{crit}} = 0.04716$ (collapsed atom $a = 0.331$, $c = -0.057$, $s = 1.218$); the runs use $\sigma_0 = 0.15\sigma_{\mathrm{crit}} = 0.00707$ (leak time 141 slow units). Whitened constants by search: $L_\vartheta^G = 4.044$ (closed form 4.658), $L_2^G = 11.11$ (closed form $\lambda_G a_{\max}/s_{\min}^2 = 44.44$); $K_\Phi = 4$, $F_\rho = 0.999$.

The hypothesis at the stationary measure, and the window (table 1). The critical point of J_3 reached from the exact representation, atoms

$$(a, c, s) = (2.4504, -1.4161, 0.8287), \quad (-2.2879, 0.0120, 0.7009), \quad (2.0478, 1.3427, 0.8810),$$

has $\|\nabla J_3\|_\infty = 4 \cdot 10^{-17}$ (a stationary measure μ^* of the mean-field flow), fit error $\|f - \hat{f}\|/\|f\| = 0.244$ on the orbit (the leak's bias: amplitudes 2.0–2.5 against $3A_k = 2.1$–3, widths 0.70–0.88 against 0.4–0.5), and a positive whitened Hessian (eigenvalues 0.0057–0.2003). κ_R at the atoms is 0.00004, −0.00227, 0.00411; over the spheres of radius ϱ' around them (400 directions, radii ϱ' and $\varrho'/2$) it is 0.00461 ($\varrho' = 0.05$), 0.00512 (0.1), 0.00619 (0.2) and 0.00730 (0.3), so (H) holds at μ^* with $\varrho' = 0.1$ and margin $\kappa = \sigma_0 - 0.00512 = 0.00195$ (relaxation time $1/\kappa = 512$) and fails from $\varrho' \approx 0.3$ on. The window of Remark D‴.1 is realised: $\kappa_R[\mu^*](N_{0.1}) = 0.00512 < \sigma_0 = 0.00707 < \sigma_{\mathrm{crit}} = 0.04716$. At $\sigma_0 = 0.3\sigma_{\mathrm{crit}}$, by contrast, the three-atom critical point (fit error 0.406) fails the splitting condition at its third atom ($\kappa_R = 0.01696 > \sigma_0 = 0.01415$): there the mean-field limit has more atoms, and the hypothesis has to be checked at that configuration. The theorem's constants for the warm start ($\Lambda_1 = 0$, $\kappa := \frac{1}{2}\min\{\kappa(\varrho'), \min_s a(s)\} = 0.00070$, the factor $\frac{1}{2}$ leaving room for the sampling of the supremum by finitely many directions): $R_\Theta = 2857$, $\varrho = 1.1 \cdot 10^{-6}$, $\theta = 4.6 \cdot 10^{-11}$, $C_1 = 2.2 \cdot 10^9$, $m_0 = 1.2 \cdot 10^{23}$ — the worst-case envelope of Remark D‴.6.

The hypothesis along the flow (table 2). On the warm start (μ_0 the mixture of Gaussians of width 0.02 around the atoms) the margin $a(s) = \sigma_0 - \kappa_R[\mu^*]$(spheres of radius 0.1 around 256 reference particles, 24 directions) stays between +0.00139 and +0.00167 over the whole run, so (H) holds with $s_1 = 0$; the same statistic with the reference's own residual is larger by at most 0.00013 (its sampling fluctuation; the proof's bound (0.2) allows $L_2^G\|r_{\mathrm{ref}} - r^*\|_{L^1(\rho)} \le 0.214$), the reference's rms drift decays from $6 \cdot 10^{-3}$ to $3 \cdot 10^{-6}$ and its fit error stays at 0.245–0.248: the reference is stationary. On the cold start ($a \sim \mathcal{N}(0, 0.8^2)$, $c \sim \mathcal{U}(-2, 2)$, $s \sim \mathcal{U}(0.4, 1)$, a connected cloud) the margin is negative over the whole run, −0.196 at the first record (−0.179 at $s = 1$) and −0.0025 at $s = 900$, while the fit error falls from 0.948 to 0.244 — the same fit as the warm limit — and the rms drift from 0.15 to $4.7 \cdot 10^{-4}$: the cold reference converges to a configuration of the same quality whose support reaches the ridges (Remark D‴.2).

Warm start (table 3). Window maxima of D_m over $(0, 112.5]$, $(112.5, 225]$, $(225, 450]$, $(450, 900]$ (Table 20):

Table 20: Theorem D‴ on the three-bump instance, warm start: window maxima of the coupled distance D_m over four windows of slow time, the mean on the last window, the theorem's constant C_1/m, the learned-function distance and the per-particle maximum (`results/chaos/learning_table.md`, table 3).

m	$(0, 112.5]$	$(112.5, 225]$	$(225, 450]$	$(450, 900]$	mean on $(450, 900]$	C_1/m	mean $\|\hat{f}_\Theta - \hat{f}_{\mathrm{ref}}\|$, last window	$\max_i \|\delta_i\|_{G^{-1}}$ at $s = 900$
16	0.396 ± 0.24	0.423 ± 0.26	0.441 ± 0.28	0.448 ± 0.28	0.446 ± 0.28	$1.4 \cdot 10^8$	0.0307 ± 0.0097	1.36
64	0.122 ± 0.046	0.131 ± 0.049	0.136 ± 0.050	0.138 ± 0.050	0.138 ± 0.051	$3.4 \cdot 10^7$	0.0154 ± 0.0033	0.62
256	0.0119 ± 0.0050	0.0127 ± 0.0052	0.0134 ± 0.0055	0.0137 ± 0.0056	0.0136 ± 0.0055	$8.4 \cdot 10^6$	0.00467 ± 0.0012	0.21

The distance saturates: last-to-first window ratios 1.13, 1.13, 1.15 and last-to-previous 1.02, 1.01, 1.02. Its m-law has slope -1.28 ± 0.30 (per-seed fits; the theorem's mean square: -1), the learned function's -0.69 ± 0.16 ($m^{-1/2}$: -0.5). The per-particle maxima 1.36, 0.62, 0.21 are far above $3\varrho'/4 = 0.075$: the localisation of part (c) is not in force at these m (Remark D‴.6), and the values lie below the vacuous C_1/m by 8.4–8.8 orders of magnitude.

Proposition D‴.5 (the same table). The sensitivities of the three atoms to the weights have whitened norms $\|S_k\| = 8.38$, 4.66, 8.32. The multinomial expectation $\mathbb{E}D_\infty = (1/m - 1/M) \sum_k \bar{w}_k \operatorname{tr}(\tilde{S}_k \tilde{C} \tilde{S}_k^\top)$ is 0.823, 0.201, 0.0454 against the four-seed means 0.446 ± 0.28, 0.138 ± 0.051, 0.0136 ± 0.0055 (ratios 0.54, 0.69, 0.30: a mean over four seeds of a heavy-tailed quantity, $\mathbb{E}|w^m - w^M|^2$ being carried by the rare large fluctuations). The per-seed law is the sharp test: $\sum_k w_k^m \|\tilde{S}_k(w^m - w^M)\|^2$ computed from the realised cluster weights predicts the measured D_m at $s = 900$ within 25% in 9 of the 12 (seed, m) pairs (for instance $0.123 \to 0.112$, $0.187 \to 0.186$, $0.206 \to 0.205$ at $m = 16$; $0.0303 \to 0.0261$, $0.00087 \to 0.00085$, $0.0081 \to 0.0087$, $0.0187 \to 0.0190$ at $m = 256$); the largest deviation, a ratio 0.57 at $m = 16$, is the seed with the largest weight fluctuation $|w^m - w^M| = 0.275$, where the linearisation's remainder and the saturation $a \le a_{\max}$ enter. The exact first equality of (ii) — $\sum_k w_k^m \|\vartheta_k^*(w^m) - \vartheta_k^*(w^M)\|^2$ with the weighted stationary points themselves — reproduces the measured D_m within 2.3% in all 12 pairs ($1.292 \to 1.292$ for the seed just mentioned): every m-system sits at the replicated weighted stationary configuration. The splitting margin of $\Theta^*(w^m)$ is positive in 11 of the 12 pairs ($+0.0006$ to $+0.0050$); it is negative, -0.0022, for $m = 16$, seed 3, whose weights $(9, 4, 3)/16$ lie outside the admissible set: there the replicated configuration is a saddle of J_{16} and the plateau at $s = 900$ is the saddle's — referee round 9 continued that run to $s = 3600$ and saw D grow from 1.29 to 5.13 as two particles changed basins and the weights re-sorted to $(7, 5, 4)/16$ (Proposition D‴.5(iv); the root-mean-square weight fluctuation is 0.204, 0.102, 0.051 at $m = 16$, 64, 256 against admissible margins of 0.06–0.36 in $\|\cdot\|_\infty$). Per cluster (seed 0), the measured mean displacements of the m-system's particles point along $S_k(w^m - w^M)$ with cosines 0.92–1.000 and norms within 0.8–20% ($m = 16$: 0.434 against 0.455, 0.130 against 0.129, 0.338 against 0.358): the particles of a cluster move together, by the predicted vector. The ratios 0.54, 0.69, 0.30 of the four-seed means to the multinomial expectation are the sampling of a heavy-tailed quantity at $m = 16$ and 64 and, at $m = 256$, four seeds whose realised fluctuations happen to be small.

Cold start (table 4). Window maxima of D_m: $1.81 \pm 0.28 \to 3.13 \to 4.50 \to 5.33 \pm 0.16$ ($m = 16$), $0.505 \to 0.900 \to 1.42 \to 1.79 \pm 0.93$ (64), $0.0744 \to 0.162 \to 0.347 \to 0.583 \pm 0.19$ (256): the distance grows over the whole horizon (last-to-first 2.95, 3.54, 7.84; last-to-previous 1.19, 1.26, 1.68), with slope -0.87 ± 0.14 in m (the learned function -0.66 ± 0.06, steeper than the -0.5 of the warm start). The per-particle deviations are bimodal: medians 0.40, 0.14, 0.067 (the $m^{-1/2}$ law of the particles that stayed with their references) against maxima 6.7, 6.3, 6.2 for every m (the separation between basins), and the fraction of pairs more than one whitened unit apart

is 0.203 ± 0.047, 0.082 ± 0.021, 0.037 ± 0.010 — a factor 2.5 and 2.2 per factor 4 in m, a decay like $m^{-0.64\pm0.08}$ (per-seed fits; the medians decay like $m^{-0.62\pm0.07}$) — steeper than the $m^{-1/2}$ of Remark D‴.2's boundary-layer count, which is a lower-order prediction (the flips contribute $p_m \cdot O(R^2) \approx 0.2 \cdot 38 = 7.8$, 3.1, 1.4 against the measured 5.3, 1.8, 0.58). This is Theorem D″'s negative finding reproduced and diagnosed on the averaged system: a connected cold start that splits between basins is outside (H), and its coupling distance is dominated by the pairs that chose different basins.

The real loop (table 5). The closed loop at $k_e = 25$ ($\sigma = \sigma_0/k_e = 2.8 \cdot 10^{-4}$, $\tau = 0$, $\mathrm{d}t = 10^{-3}$, three seeds) from the same warm start: window maxima over $(0, 500]$, $(500, 1000]$, $(1000, 2000]$ real-time units are $0.328 \pm 0.11 \to 0.467 \to 0.590 \pm 0.17$ ($m = 16$), $0.0605 \to 0.0801 \to 0.0909 \pm 0.045$ (64), $0.0188 \to 0.0253 \to 0.0294 \pm 0.011$ (256), slope -1.13 ± 0.15 in m; the reference's fit error stays at 0.245–0.246. The horizon is 80 slow units, a sixth of the relaxation time $1/\kappa = 512$, and the growth (last-to-first 1.80, 1.50, 1.56) is the early transient of the averaged dynamics: the averaged flow started from the real loop's own initial clouds (reproduced from their seeds) gives $D_{\mathrm{real}}(t)/D_{\mathrm{averaged}}(t/k_e)$ between 0.993 and 1.004 at every window end, for every seed and every m — the real loop at $k_e = 25$ follows its averaged slow dynamics to within 0.7% (across different clouds the last-window maxima are 1.57, 0.78 and 2.58 times the averaged runs' 0.377, 0.117, 0.0114, within the seed-to-seed spread). Two of the three $m = 16$ clouds have inadmissible or marginal weights (splitting margins -0.0021, -0.00001; the third $+0.0026$), so their eventual configuration is a saddle's; at $m = 64$ and 256 all margins are positive ($+0.0029$ to $+0.0049$). The saturation at the orbit distance of Corollary D‴.7 lies beyond the horizon run.

Reading. In the window $\kappa_R[\mu^*] < \sigma_0 < \sigma_{\mathrm{crit}}$ the network learns (the collapsed configuration is unstable, the three kernels fit the plant to 0.244) and propagation of chaos is uniform in time on a warm start with admissible weights: the coupled distance saturates with the $1/m$ law at the stationary distance of the replicated weighted configuration, which the exact form of Proposition D‴.5(ii) gives within 2.3% — the level is the weight-sensitivity law, not the theorem's constant. At $m = 16$ the multinomial weights leave the admissible set with appreciable probability; the one such seed sits on a saddle and later splits a cluster. The cold start of a connected cloud shows what the hypothesis excludes: growth over the horizon and a floor of pairs in different basins whose fraction decays like $m^{-0.64}$, while the learned function still converges at a rate between $m^{-1/2}$ and $m^{-2/3}$ in both cases. The real loop at $k_e = 25$ follows its averaged slow dynamics to within 0.7% over the horizon run.

Checked by: `tests/test_chaos_learning.py`: ($\mathfrak{A}_m$)'s drift equals $-G\nabla J_m$ of `periodic.ClosedLoop` to machine precision (`test_averaged_slow_loop_is_the_G_gradient_flow_of_J_m`); the three-term identity of Step 1 holds to rounding on random configurations with a mean-field stand-in, the interaction term equals $-\|\hat{f}_\Theta - \hat{f}_{\bar{\Theta}}\|^2$ up to a relative $O(D^{1/2})$ correction (ratio $0.98 \to 0.9999$ as δ is scaled by $1 \to 0.01$) and obeys (3.1) at every scale
(`test_coupling_identity_three_terms_and_the_sign_of_the_interaction`); κ_R at the atoms of a collapsed configuration with the collapsed residual equals Proposition K's $\lambda_{\max}(G^{1/2}R_2G^{1/2})$, which equals σ_0 at $\sigma_0 = \sigma_{\mathrm{crit}}$, the collapsed critical point has zero drift, and the deterministic neighbourhood search over the sphere of radius ϱ reproduces the first-order expansion $\lambda_{\max} + \varrho\|G^{1/2}\nabla\lambda_{\max}\|$ with a relative error $O(\varrho)$ (`test_residual_curvature_...`); the searched whitened constants reproduce the document's values and lie below their closed forms, `theorem_constants` satisfies the four constraints the bootstrap needs, and Λ_1 integrates exactly to s_1 (`test_whitened_constants_...`); at $\sigma_0 = 0.15\sigma_{\mathrm{crit}}$ the three-atom critical point of J_3 has zero drift and (H) holds at it with $\varrho' = 0.1$ inside the learning regime, the replicated weighted stationary point is a critical point of J_{16} to 10^{-10}, the linearisation in the weights has a second-order remainder, and on a coupled run from the jittered mixture ($m = 16$, 64 against $M = 512$, 60 slow units) $\kappa_R[\mu^*]$ over the neighbourhood of the reference's support stays below σ_0, the two curvature statistics differ by at most the bound (0.2), the per-cluster displacements point along $S_k(w^m - w^M)$ (cosine ≥ 0.8) with D within a factor 2 of $\sum_k w_k^m \|S_k(w^m - w^M)\|^2$, $\|\hat{f}_\Theta - \hat{f}_{\bar{\Theta}}\| \leq L_\vartheta^G \cdot \mathrm{mean}\|\delta_i\|$, $D_{64} < D_{16}$ and the (vacuous) bound C_1/m holds

(`test_hypothesis_H_at_the_three_atom_stationary_measure_and_the_weight_sensitivity_law`).

19 Proposition G: the slow geometry

This section concerns joint descent, and variable projection (VarPro) in the induced metric.

Setting.

- $f = f(x)$ is time-invariant and C^2.
- x_m is T_0-periodic, with $\omega_{\text{ref}} := 2\pi/T_0$ and occupation measure $\rho := \frac{1}{T_0}\int_0^{T_0} \delta_{x_m(t)}\,\mathrm{d}t$.
- Law (L) acts in the interior, with gains $(g_a, \delta\gamma_c, \delta\gamma_s)$. Put $\Gamma := \mathrm{diag}(\gamma_c I_n, \gamma_s)$, acting per kernel, and $G_\xi := \delta\Gamma$. Write the geometry as $\xi := (c, s)$.
- $\bar{L}(a, c, s) := \frac{1}{2}\int \|f - \frac{1}{m}\sum_i a_i\varphi(\cdot\,; c_i, s_i)\|^2\,\mathrm{d}\rho$, and $M(c, s) := \frac{1}{m^2}\int \varphi\varphi^\top\,\mathrm{d}\rho \otimes I_n$ is the Hessian of $\bar{L}$ in a.
- $\mathcal{K}$ is the set of geometries ξ on which (H1) holds, and $\mathcal{K}' \Subset \mathcal{K}$ is compact. (The set $\mathcal{K}$ is not the functional $\mathcal{G}(\mu)$ of Section 20.)
- $a^*(\xi) := \mathrm{argmin}_a\,\bar{L}$, and $L^*(\xi) := \bar{L}(a^*(\xi), \xi)$ is the profile (variable-projection) loss.
- Three dimensionless parameters:
  - the rate ratio $R := g_a m_0 \big/ \big(\delta \cdot \sup_{\mathcal{K}'}\|\Gamma^{1/2}\nabla^2 L^*\Gamma^{1/2}\|\big)$;
  - the slope $S := (\delta/g_a)\cdot \sup_{\mathcal{K}'}\|D_\xi a^* \cdot \Gamma^{1/2}\|^2$;
  - the averaging parameter $\varepsilon_{\text{avg}} := (m/k_e)\,\lambda_{\max}(G^{1/2}\nabla^2\bar{L}\,G^{1/2})/\omega_{\text{eff}}$, the second ratio in (H2a′). The result tables in `results/variable_projection` report its inverse, with ω_{ref} in place of ω_{eff}.

Hypotheses.

(H0) On the interval considered, the closed-loop and reduced geometries stay in $\mathcal{K}'$. The parameters stay in $\mathrm{int}\,\Theta_K^\varepsilon$, and $\sup_{\mathcal{K}'}\max_i\|a_i^*\| < a_{\max} - \varepsilon_K$.

(H1) $M \succeq m_0 I$ on $\mathcal{K}$.

(H2a′) *Two scales.* $k_e \gg \omega_{\text{eff}}$ and $\varepsilon_{\text{avg}} \ll 1$, where

$$\omega_{\text{eff}} := \max\Big\{\omega_{\text{ref}},\ \frac{\sup_t\|\dot{x}_m\|}{s_{\min}},\ \sup_X\|\partial_x(f - \hat{f})\|,\ \frac{\sup_t\|\frac{\mathrm{d}}{\mathrm{d}t}r(x_m(t))\|}{\sup\|r\|}\Big\}.$$

The last entry is the bandwidth of the residual along the reference. Also $\sup\|r\|/k_e \ll s_{\min}$.

(H2b) *Rate separation.* $R \gg 1$: the slowest weight rate $g_a m_0 m/k_e$ is much larger than the geometry rate $(m\delta/k_e)\|\Gamma^{1/2}\nabla^2 L^*\Gamma^{1/2}\|$.

(H2c) *Flat slow manifold.* $S \ll 1$. Since $Da^* = -M^{-1}\partial^2_{a,\xi}\bar{L}$, a checkable sufficient condition is $\delta\cdot\max(\gamma_c, \gamma_s)\cdot\|\partial^2_{a,\xi}\bar{L}\|^2 \ll g_a m_0^2$.

Three remarks on the hypotheses.

- **(H2b) is about $\lambda_{\min}$.** It is a condition on $\lambda_{\min}$ of the feature Gram matrix. Since $0 < \varphi \le 1$, $m_0 \le 1/m^2$. So the slowest weight rate is at most $g_a/(k_e m)$, and overlapping kernels make (H2b) fail.

- **The ratios are gain choices.** R and S contain neither k_e nor ω_{ref}. The separation between weights and geometry is therefore set by the gains.
- **(H2b) does not imply (H2c).** The product

$$R \cdot S = m_0 \cdot \sup\|Da^*\Gamma^{1/2}\|^2 \big/ \sup\|\Gamma^{1/2}\nabla^2 L^*\Gamma^{1/2}\|$$

does not depend on g_a, δ or k_e. It is fixed by the problem and by the shape of Γ: in the slope configuration below, a numerical check gives values from 166 to 1149 over $\gamma_s/\gamma_c \in [10^{-4}, 10^4]$. It is large whenever refitting the weights largely compensates a change of geometry.

Proposition G (formal reduction)**.**

(a) Two scales, under (H0) and (H2a′). *To leading order in $\omega_{\mathrm{eff}}/k_e$ and $\varepsilon_{\mathrm{avg}}$, the period-averaged closed loop is the joint gradient flow*

$$\dot{\vartheta}_i = -\frac{m}{k_e}\, G\, \nabla_{\vartheta_i} \bar{L}(\vartheta_1, \dots, \vartheta_m). \tag{19.1}$$

Its measure form is $-\frac{1}{k_e} G \nabla (\delta \bar{L}/\delta\mu)$.

- *This is the drift of $(\mathfrak{A})$ at $\sigma = \tau = 0$, with Φ in place of $\tilde{\Phi}$ and (a, c, s) in place of (α, γ, κ), by the same averaging computation as Theorem H(i).*
- *It is not literally a special case of H(i): the saturated coordinates there induce a different, state-dependent metric.*
- *$\bar{L}$ is non-increasing along the averaged flow. Along the closed loop, $\bar{L}$ at period-averaged parameters decreases while the averaged dissipation $\frac{m}{k_e}\|\nabla \bar{L}\|_G^2$ dominates the $O(\omega_{\mathrm{eff}}/k_e)$ corrections of Proposition H.7 and Remark H.7a. Near equilibrium it can increase by $O((\omega_{\mathrm{eff}}/k_e)^2)$, as the weights settle at their shifted equilibrium (in a numerical check: $+8 \cdot 10^{-7}$ over 35 periods at $k_e = 20$, and $3.7 \cdot 10^{-8}$, about 22 times less, at $k_e = 80$).*

(b′) Three scales, for the averaged system (a), under (H0), (H1), (H2a′) and (H2b). *After a boundary layer, the weights follow the slow manifold:*

$$\begin{aligned} a - a^*(\xi) &= -\frac{k_e}{m g_a} M^{-1} Da^* \cdot \dot{\xi} + \text{h.o.t.} \\ &= \frac{\delta}{g_a} M^{-1} Da^* \Big(\Gamma^{-1} + \frac{\delta}{g_a}(Da^*)^\top Da^*\Big)^{-1} \nabla L^*(\xi) + \text{h.o.t.} \end{aligned} \tag{19.2}$$

This reduces to $\frac{\delta}{g_a} M^{-1} Da^* \,\Gamma\, \nabla L^*(\xi)$ when $S \ll 1$. The geometry follows, to leading order in $1/R$ and for every S,

$$\dot{\xi} = -\frac{m\delta}{k_e}\Big(\Gamma^{-1} + \frac{\delta}{g_a}(Da^*)^\top Da^*\Big)^{-1} \nabla L^*(\xi). \tag{19.3}$$

- *This is the gradient flow of $(m/k_e)L^*$ in the metric that the G^{-1} metric induces on the graph $\{(a^*(\xi), \xi)\}$.*
- *Its metric tensor is $G_\xi^{-1} + g_a^{-1}(Da^*)^\top Da^*$, and it couples all kernels through Da^*.*
- *In the closed loop, the weight equilibrium is further shifted by $O(\omega_{\mathrm{eff}}/k_e)$ (Proposition H.7). This shift can exceed the offset displayed above: in a numerical check on the* `satisfying` *run it is 3–6 $\cdot 10^{-3}$, against an offset of $4 \cdot 10^{-5}$–$3 \cdot 10^{-4}$.*

(b) Variable projection, under (H2c) in addition. *Let $(\bar{c}, \bar{s})$ solve $\bar{\xi}' = -\frac{m\delta}{k_e}\Gamma \nabla L^*(\bar{\xi})$, that is, $\bar{c}_i' = -\frac{m\delta\gamma_c}{k_e}\nabla_{c_i} L^*$ and $\bar{s}_i' = -\frac{m\delta\gamma_s}{k_e}\partial_{s_i} L^*$, with $(\bar{c}, \bar{s})(0) = (c, s)(0)$. Then*

$$\sup_{t \le \tau_1 k_e/(m\delta)} \|(c,s)(t) - (\bar{c},\bar{s})(t)\| \to 0 \quad \text{as } (\omega_{\mathrm{eff}}/k_e,\ \varepsilon_{\mathrm{avg}},\ S,\ 1/R) \to 0, \text{ with } \tau_1 \text{ fixed.} \tag{19.4}$$

- *Formally the error is* $O(\omega_{\text{eff}}/k_e + \varepsilon_{\text{avg}} + S + 1/R)$.
- L^* *is non-increasing along* $\bar{\xi}$*, and the weights track* a^*.
- *This flow is the* $S \to 0$ *limit of* (b′).

Status. Formal (a derivation without error control); checked directly, including a counterexample to plain VarPro at $S = O(1)$ (see the numerics below). Novelty (literature audit): a fixed-m, closed-loop analogue of the two-timescale/VarPro training of [59], [6].

No mean-field form. For fixed m, on configurations of m distinct geometries in $\mathcal{K}$, the flow (b) is the gradient flow of L^*/k_e in the metric $(1/m)\sum_i \|\dot{\xi}_i\|^2_{G_\xi^{-1}}$. Locally, that metric is the Wasserstein metric restricted to uniform m-atom measures.

- L^* is the unregularised profile loss. It is the $\lambda \to 0$ limit of $\lambda\mathcal{L}^\lambda$ of [6] (time rescaled by λ). It is not the $\lambda \to 0$ functional $\mathcal{L}^0$ of [6], which is a min-norm interpolation functional whose flow is an ultra-fast diffusion.
- As a functional of the geometry marginal, L^* depends only on the support, because the free weights absorb the masses. So it is not W-continuous, and the statement has no mean-field form. This is consistent with $m_0 \le 1/m^2$.

Derivation (formal). Each step is a standard theorem on its own time scale. Their composition on the $O(k_e/(m\delta))$ horizon is formal.

1. *Error channel (exact identity).* For $w := e - r(x,\vartheta)/k_e$, $\dot{w} = -k_e w - \dot{r}/k_e$. So $\|w(t)\| \le e^{-k_e t}\|w(0)\| + \sup\|\dot{r}\|/k_e^2$, and
$$e = r(x_m)/k_e + O\big(\sup\|\dot{r}\|/k_e^2 + \sup\|\partial_x r\| \cdot \sup\|r\|/k_e^2\big).$$
This is $O(\omega_{\text{eff}} \cdot \sup\|r\|/k_e^2)$, with the bandwidth term in ω_{eff}.
2. *Period averaging of the joint system* (Khalil [48, Thm. 10.4]; Sastry–Bodson [81, Ch. 4]). This gives (a), on $[0, b/\varepsilon_{\text{avg}}]$.
3. *Weights against geometry.* Use Tikhonov in slow time (Kokotović–Khalil–O'Reilly [49]) with the uniformly exponentially stable averaged weights on $\mathcal{K}'$ (H1).
   - On the slow manifold $a = h(\xi)$, $\dot{a} = Dh\,\dot{\xi} = -\frac{mg_a}{k_e} M(h - a^*)$, so $h - a^* = -\frac{k_e}{mg_a} M^{-1} Dh\,\dot{\xi}$. This must be evaluated with the (b′) velocity, not the VarPro velocity.
   - Since $\bar{L}$ is quadratic in a and $\partial^2_{\xi a}\bar{L} = -(Da^*)^\top M$, we get $\nabla_\xi \bar{L}(h,\xi) = \nabla L^* - (Da^*)^\top M(h - a^*) + O((\delta/g_a)^2)$.
   - Substituting into $\dot{\xi} = -\frac{m\delta}{k_e}\Gamma\nabla_\xi \bar{L}(h,\xi)$, with $Dh = Da^* + O(1/R)$, gives (b′). The resummed form solves the invariance equation exactly once $Dh - Da^*$ and the quadratic term are dropped, so the remaining corrections are in $1/R$ (they involve $D^2 a^*$ and $\partial_\xi M$), not in S.
4. *Envelope theorem.* $\nabla_\xi \bar{L}(a^*(\xi),\xi) = \nabla L^*(\xi)$. When $S \ll 1$, the metric correction in (b′) drops out, which gives (b).

The composition requires the averaged system to remain valid up to $t \sim \tau_1 k_e/(m\delta)$, which is much longer than $b/\varepsilon_{\text{avg}}$. Step 2 does not certify this. A proof would combine:

- a near-identity averaging transform;
- uniform exponential stability of the averaged weights on $\mathcal{K}'$;
- Tikhonov in slow time with an $O(\varepsilon_{\text{avg}})$ perturbation;
- Grönwall in ξ (Sastry–Bodson [81, §4.4]; Sanders–Verhulst–Murdock [78, Ch. 5]).

□

Numerics. The results are in `results/variable_projection` of the accompanying repository; Table 21 summarises the comparison.

- The first two rows use centred differences of period means.
- The slope rows use the displacement between the first and last period means. They start on the slow manifold of (b′), so no weight boundary layer enters the comparison.
- (b′) was evaluated only in the slope rows.
- k_e/ω_{eff} is 2.0 in the first two rows with the box's $s_{\min}$, and 2.6–4.8 with the actual widths, so (H2a′) holds only marginally there. In the `satisfying` run the residual bandwidth, about 7.7, then dominates.
- k_e/ω_{eff} is about 16 in the slope rows.

configuration	(H2b) ratio	S	G(a): closed loop against the joint flow (a)	G(b) and the induced-metric flow (b′)
`violating` ($m = 4$, overlapping)	0.10–0.23	5.4–25	joint cosine 0.9994 / 0.9985, size ratio 1.00 / 0.94	(b) fails: cosine −0.12 / 0.39, and L^* *increases* from $3.8 \cdot 10^{-3}$ to $5.9 \cdot 10^{-3}$
`satisfying` ($m = 2$)	79–87	0.006	joint cosine 0.985 / 0.962, size 0.96 / 0.94; geometry block 0.999 / 0.998; the weight block fails (see below)	(b) holds: cosine 0.999, ratio 0.97; L^* decreases from $1.20 \cdot 10^{-2}$ to $4.11 \cdot 10^{-3}$
`slope`, $\delta = 0.47$ ($m = 1$)	81–178	2.0–2.8	cosine 0.9992, size 1.10	(b) fails: ratio 0.42, cosine 0.93, and the centre moves the opposite way; (b′): cosine 0.9986, ratio 1.12
`slope`, $\delta = 0.47$, all gains $\div 4$	81–178	2.0–2.8	cosine 1.0000, size 1.05	(b) fails: ratio 0.40, cosine 0.92; (b′): cosine 0.9999, ratio 1.065
`slope`, $\delta = 0.047$	654–1779	0.20–0.32	cosine 1.0000, size 1.04	(b): ratio 0.85, cosine 0.9986; (b′): ratio 1.05, cosine 1.0000
`slope`, $\delta = 0.0047$	13162–17790	0.020–0.023	cosine 1.0000, size 1.04	(b): ratio 1.02; (b′): ratio 1.04; both cosines 1.0000

Table 21: Proposition G: the closed loop against the averaged predictions. In the first two rows, an entry 'p / q' gives the medians over the first quarter (p) and over the last quarter (q) of the run.

- **Joint descent.** In all runs, $\bar{L}$ at the period-mean parameters is non-increasing, sampled once per period.
- **(H2b) is not enough.** The `slope` rows satisfy (H2b) by a wide margin, yet (b) fails until S is small. Meanwhile (b′) tracks the closed loop at every S. So (H2c) is the hypothesis missing from an earlier formulation of Proposition G(b), and (b′) is the correct leading-order slow flow. The configuration is a counterexample to that earlier formulation, made reproducible in `experiments/exp_slow_manifold_slope.py`.
- **Error decomposition** (`slope`, $\delta = 0.47$, started on the slow manifold):
  - $\|(\text{a})\|/\|(\text{b}')\| = 1.015$. This is the intrinsic $O(1/R)$ error of (b′).
  - $\|\text{closed}\|/\|(\text{a})\| = 1.10$. It falls to 1.05 when all gains are divided by 4, which leaves R and S unchanged and divides $\varepsilon_{\text{avg}} \approx 0.08$ by 4. So this part is the averaging error.

  - The remaining $\approx 4.4\%$ is the $O(1/k_e)$ error-channel factor at $k_e = 20$. It is the same at $\delta = 0.047$ and 0.0047.

- **The size ratio in the `satisfying` run.** The centred ratio of 0.97 is the error-channel factor times a +1.4% averaging correction from the weights' oscillation within each period, the analogue of Remark H.7a.
  - The frozen-parameter ratio is 0.951–0.956 along the run, and 0.996 at $k_e = 80$.
  - The fastest weight rate over ω_{ref} is about 0.14.
  - In a further numerical check, with the weights running at frozen geometry the ratio is 0.969–0.970; with $g_a/4$ it is 0.956–0.960.
  - The slope correction is below 0.01% here, because ∇L^* is nearly orthogonal to the range of $(Da^*)^\top$. An earlier estimate of 0.94–0.95 contained a 2.5% forward-difference bias.

- **The weight block.** In the `satisfying` run the weights sit at $a^* + O(1/k_e)$, within 0.1% of a^*.
  - Their actual drift is tiny: 0.001–0.008 of the geometry speed in the G^{-1} norm.
  - The prediction $-(m/k_e)g_a\nabla_a\bar{L}$ at the period-mean weights is 0.17–0.30 of the geometry speed, 40–200 times the actual drift. It is the restoring velocity of the $O(1/k_e)$ offset, which (a) does not contain.
  - So the weight block fails (cosine -0.21 / 0.50, carrying 92–95% of the residual), while the geometry block matches.

Checked by: `experiments/exp_variable_projection.py`, which writes `table_violating.md` and `table_satisfying.md`; and `experiments/exp_slow_manifold_slope.py`, which writes `table_slope.md` (all in `results/variable_projection`).

Status.

- **What the loop supplies.** The error-channel and averaging scales, and the data distribution (the occupation measure of the reference), come from the closed loop.
- **What the designer supplies.** The separation between weights and geometry is a gain choice, as in [59], [6]: $\delta \ll g_a \cdot \min\big(m_0/\|\Gamma^{1/2}\nabla^2 L^*\Gamma^{1/2}\|,\ 1/\|Da^*\Gamma^{1/2}\|^2\big)$. Equivalently, $\|G_\xi^{1/2}\nabla^2 L^* G_\xi^{1/2}\| \ll g_a m_0$ and $\|Da^* G_\xi^{1/2}\|^2 \ll g_a$.
- **What is new here.**
  - The learner's data distribution is its own excitation, and at finite k_e it moves with the parameters (Proposition H.7(c)).
  - The slow flow is variable projection in the *induced* metric, valid for every S. The slope S measures the departure from plain VarPro.

20 Theorem H: Langevin adaptive control and global convergence to the regularised optimum

Why something new is needed. Theorems B–G are local, or describe the dynamics. A global statement about kernel placement needs either the Chizat–Bach theory or entropy.

- **Boundary conditions.** The Chizat–Bach theory for noiseless Wasserstein flows (partially 1-homogeneous case, [14]) requires boundary conditions: either Neumann conditions on a

bounded convex domain, or C^1-uniform behaviour at infinity. Gaussian kernels with free centres and widths satisfy neither.

- **Separation.** Its Theorem 3.5 also needs the support of μ_0 to separate $\{-r_0\} \times \Theta$ from $\{r_0\} \times \Theta$. A Dirac start violates this, and that is an obstruction distinct from the boundary conditions.
- **The collapsed configuration is invariant.** It remains a moving Dirac mass and never separates without noise, by permutation symmetry and uniqueness of solutions. The test `test_noiseless_symmetric_initialisation_never_separates_but_dither_does` in the accompanying repository checks this on the real closed loop with $m = 6$. The rows stay bit-identical at every snapshot, the common particle moves, and dither separates the rows within 0.1 s. (Whether the collapsed configuration is a strict local minimum of the regularised loss, so that nearby non-collapsed starts return to it, is decided by the leak gain against the critical leak σ_{crit} of Proposition K, Section 23.)

Entropy removes the obstruction: Theorem H(ii) covers Dirac starts after any positive time.

Design. The particles are unconstrained, $\vartheta = (\alpha, \gamma, \kappa) \in \mathbb{R}^p$, and the features are saturated coordinate by coordinate:

$$a = a_{\max} \tanh(\alpha/a_{\max}), \qquad c = c_{\max} \tanh(\gamma/c_{\max}), \qquad s = s_{\min} + (s_{\max} - s_{\min}) \cdot \operatorname{sig}(\kappa).$$

Here $a_{\max}$ bounds each coordinate of a, unlike Section 9, where it bounds $\|a\|$. So $\tilde{\Phi}(x; \vartheta) = a\varphi(x; c, s)$ satisfies $\|\tilde{\Phi}\| \le K_\Phi := a_{\max}\sqrt{n}$, and every ϑ-derivative of $\tilde{\Phi}$ is bounded uniformly in x. The law is

$$\mathrm{d}\vartheta_i = \big[\, G\nabla_\vartheta g_t(\vartheta_i) - \sigma(\vartheta_i - \vartheta^0) \,\big]\,\mathrm{d}t + \sqrt{2\tau}\, G^{1/2}\,\mathrm{d}B_i, \qquad g_t(\vartheta) := e(t)^\top \tilde{\Phi}(x(t); \vartheta). \tag{L_τ}$$

It has three parts: the classical gradient law, σ-modification written in the G^{-1} metric, and parameter dither of temperature τ.

Averaged mean-field system. Assume $f = f(x)$ is continuous and time-invariant, as in Proposition G (the functions `occupation_loss` and `first_variation` of the accompanying code assume this). Let ρ be the occupation measure of a T_0-periodic reference, and $F_\rho := \sup_{\operatorname{supp}\rho} \|f\|$. Define

$$r_\mu := f - \int \tilde{\Phi}(\cdot\,; \vartheta')\,\mathrm{d}\mu(\vartheta'), \qquad \bar{L}(\mu) := \frac{1}{2}\int \|r_\mu\|^2\,\mathrm{d}\rho,$$
$$V_L[\mu](\vartheta) := -\int r_\mu(x)^\top \tilde{\Phi}(x; \vartheta)\,\mathrm{d}\rho(x), \qquad V[\mu] := \frac{1}{k_e} V_L[\mu] + \frac{\sigma}{2}\|\vartheta - \vartheta^0\|^2_{G^{-1}},$$

and the averaged system

$$\partial_t \mu = \nabla \cdot \big(\mu G \nabla V[\mu]\big) + \tau \nabla \cdot (G \nabla \mu). \tag{$\mathfrak{A}$}$$

Remark H.0 (from the closed loop to ($\mathfrak{A}$): a formal reduction with a measured error)**.**

- **An algebraic identity.** Substitute $e := r_\mu(x_m)/k_e$ and $x := x_m$ in the drift of (L_τ), and average over one reference period at frozen μ. The result is exactly $-G\nabla V[\mu]$. The test `test_averaged_drift_is_the_mean_field_langevin_drift` checks this identity to 10^{-7}. It does not test averaging.
- **The formal reduction.** ($\mathfrak{A}$) is the leading-order description of the mean-field closed loop under two conditions:
  - *Tikhonov:* $k_e \gg \max\big(\omega_{\text{ref}},\ \operatorname{Lip}(f) + K_\Phi/s_{\min},\ (F_\rho + K_\Phi)/s_{\min}\big)$, so that the shift $x - x_m = O(1/k_e)$ is small on the kernel scale.

– *Period averaging:* $T_0 \cdot \sup_\vartheta \|(1/k_e) G \nabla V_L[\mu](\vartheta)\| \ll \ell$, $\sigma T_0 \ll 1$ and $\tau \lambda_G T_0 \ll \ell^2$, where ℓ is the ϑ-scale of $\tilde{\Phi}$. The left-hand side is at most $(T_0/k_e)\lambda_G(F_\rho + K_\Phi) \cdot \sup\|\nabla_\vartheta \tilde{\Phi}\|$. The confinement drift does not depend on time and needs no averaging.

- **The first corrections.**
  – At frozen μ (and frozen ϑ) they are $O(1/k_e)$ relative to the drift, do not depend on the gains, and are given exactly by Proposition H.7.
  – Along the loop, the within-period oscillation of the parameters adds the second-order averaging term b_{osc} of Remark H.7a. Its size relative to the frozen terms is $O(|G|T_0)$, so it does depend on the gains.
  – In the formal limit below, both are $O(k_e^{-2})$.

  Measured relative errors of the frozen-μ averaged drift against $(\mathfrak{A})$, with b_{osc} excluded:

instance	$k_e = 10$	20	40	80	$k_e \times$ error
$n = 1$	4.2%	2.0%	0.91%	0.48%	≈ 0.4
$n = 2$, ellipse	43%	21%	10%	5.1%	≈ 4.2

  Tests: `test_averaging_error_of_the_formal_reduction_decreases_like_one_over_ke` (instance $n = 1$) and `test_averaged_drift_error_decays_like_one_over_ke_for_an_orbit_enclosing_area` (instance $n = 2$; $k_e = 10, 20, 40, 80$).

  – $k_e = 10$ is outside the averaging regime for the second instance.
  – Its larger error does not come from the enclosed area. Proposition H.7 attributes it to the orientation-even state-shift term: it predicts $k_e \times$ error ≈ 4.05, of which U_{odd} alone would give 1.5, and a numerical check on the reversed orbit gives 4.0.
  – A numerical check shows that at the gains of these instances b_{osc} is not small. k_e times its relative size is 0.92 ($n = 1$, $G = \mathrm{diag}(4, 1, 1)$) and 4.4 ($n = 2$, $G = I$), against 0.42 and 4.05 for the frozen terms.

- **The exact formal limit** is $k_e \to \infty$ with G, $k_e\sigma$ and $k_e\tau$ fixed, in slow time t/k_e (Conjecture H.6).
- **Not proved here.** The mean-field limit of (L_τ) itself is not proved; Theorem D covers the deterministic projected law (L).

Status. Formal reduction, with a measured error.

Theorem H. *Assume the setting above, and $\sigma > 0$, $\tau > 0$ where stated.*

(i) Identification. *$V[\mu]$ is the first variation of $\mathcal{G}(\mu) := \frac{1}{k_e}\bar{L}(\mu) + \frac{\sigma}{2}\int \|\vartheta - \vartheta^0\|^2_{G^{-1}}\,\mathrm{d}\mu$. Hence $(\mathfrak{A})$ is the mean-field Langevin dynamics, in the G^{-1} metric, of the free energy $F_\tau := \mathcal{G} + \tau H$, with $H(\mu) = \int \mu \log \mu$. Equivalently, with $\gamma_\tau := \mathcal{N}(\vartheta^0, (\tau/\sigma)G)$,*

$$F_\tau = \frac{1}{k_e}\bar{L} + \tau H(\cdot \,|\, \gamma_\tau) + \mathrm{const.}$$

F_τ is well defined on all of $\mathcal{P}(\mathbb{R}^p)$, and finite exactly when $H(\mu \,|\, \gamma_\tau) < \infty$.

(ii) Global exponential convergence ($\sigma, \tau > 0$). *F_τ has a unique minimiser, $\mu_\tau^* \propto e^{-V[\mu_\tau^*]/\tau}$. For every $\mu_0 \in \mathcal{P}_2(\mathbb{R}^p)$ with $F_\tau(\mu_0) < \infty$,*

$$F_\tau(\mu_t) - F_\tau(\mu_\tau^*) \le e^{-2\tau\rho_\tau t}\big(F_\tau(\mu_0) - F_\tau(\mu_\tau^*)\big),$$
$$W_{G^{-1}}(\mu_t, \mu_\tau^*)^2 \le \frac{2}{\tau\rho_\tau}\, e^{-2\tau\rho_\tau t}\big(F_\tau(\mu_0) - F_\tau(\mu_\tau^*)\big).$$

Here ρ_τ is a log-Sobolev constant, for the whitened measures, of every proximal Gibbs measure $\hat{\mu} \propto e^{-V[\mu]/\tau}$. In ϑ-coordinates this means Fisher information in the G metric. It satisfies

$$\rho_\tau \geq \frac{\sigma}{\tau} \cdot \exp\Big(-\frac{2K_\Phi(F_\rho + K_\Phi)}{k_e \tau}\Big).$$

Suppose μ_0 is only in $\mathcal{P}_2(\mathbb{R}^p)$, for example a Dirac mass or an empirical measure, so that $H(\mu_0) = +\infty$. Then $F_\tau(\mu_{t_0}) < \infty$ for every $t_0 > 0$, and the estimates hold with (μ_0, t) replaced by $(\mu_{t_0}, t - t_0)$.

(iii) Loss and tracking. *Let $\bar{e}_t := r_{\mu_t}/k_e$. Then $\langle \|\bar{e}_t\|^2 \rangle_\rho = 2\bar{L}(\mu_t)/k_e^2$, and*

$$\begin{aligned} |\bar{L}(\mu_t) - \bar{L}(\mu_\tau^*)| &\leq (F_\rho + K_\Phi) L_\Phi \cdot W_{G^{-1}}(\mu_t, \mu_\tau^*) \\ &\leq (F_\rho + K_\Phi) L_\Phi \Big(\frac{2}{\tau \rho_\tau}\Big)^{1/2} e^{-\tau \rho_\tau t} \big(F_\tau(\mu_0) - F_\tau(\mu_\tau^*)\big)^{1/2}, \end{aligned}$$

where $L_\Phi := \sup_{x,\vartheta,\|u\|=1} \|G^{1/2} \nabla_\vartheta (u^\top \tilde{\Phi})\|$. The regularisation bias satisfies

$$\bar{L}(\mu_\tau^*) \leq \inf_\nu \big\{ \bar{L}(\nu) + k_e \tau H(\nu \,|\, \gamma_\tau) \big\}.$$

The link $e \approx \bar{e}$ between $(\mathfrak{A})$ and the closed loop is formal (Remark H.0).

(iv) Annealing. *Let $\mu_0 \in \mathcal{P}_2$ with $F_{\tau_{\text{init}}}(\mu_0) < \infty$. Let (τ_t) be smooth and decreasing from τ_{init}, with $\tau_t = \alpha_a / \log t$ for $t \geq t_1 > 1$, where $\alpha_a > \alpha^* := 2K_\Phi(F_\rho + K_\Phi)/k_e$. Then $\mathcal{G}(\mu_t) - \inf \mathcal{G} \leq C' \cdot \log\log t / \log t$, where $\inf \mathcal{G}$ is the σ-regularised optimum. This is Theorem 4.1 of [15]. Its hypotheses hold:*

- *$\mathcal{G} \geq 0$;*
- *Assumptions 1–3 hold for every $\tau > 0$;*
- *for $\tau \in (0, \tau_{\text{init}}]$, (ii) gives $\rho_\tau \geq C_0 e^{-\alpha^*/\tau}$ with $C_0 = \sigma/\tau_{\text{init}}$, which is all its proof uses.*

Status. Proved for $(\mathfrak{A})$; the reduction to $(\mathfrak{A})$ is formal (Remark H.0); real loop: Conjecture H.6. Novelty (literature audit): a bridge to [15], [64]; new in adaptive control.

Checked by: `tests/test_langevin.py`:
`test_first_variation_bound_used_in_holley_stroock` (the oscillation bound in (ii));
`test_lsi_lower_bound_formula` (formula transcription only);
`experiments/exp_langevin_global.py`, `exp_langevin_tradeoff.py` and `exp_gibbs_reference.py` (global behaviour, speed–bias trade-off, reference values).

Proof. (i) We have

$$\frac{\mathrm{d}}{\mathrm{d}\varepsilon} \bar{L}\big((1-\varepsilon)\mu + \varepsilon\nu\big)\Big|_{\varepsilon=0} = -\int r_\mu^\top \Big(\int \tilde{\Phi}\, \mathrm{d}(\nu - \mu)\Big)\, \mathrm{d}\rho = \int V_L\, \mathrm{d}(\nu - \mu).$$

This is exact, because $\bar{L}$ is quadratic in μ. The confinement is linear in μ. In the G^{-1} metric, $\nabla \cdot \big(\mu G \nabla(V + \tau \log \mu)\big) = \nabla \cdot (\mu G \nabla V) + \tau \nabla \cdot (G \nabla \mu)$. Since $\log \gamma_\tau = -(\sigma/(2\tau))\|\vartheta - \vartheta^0\|^2_{G^{-1}} + \text{const}$, we get $\tau H(\mu \,|\, \gamma_\tau) = \tau H(\mu) + (\sigma/2) \int \|\vartheta - \vartheta^0\|^2_{G^{-1}}\, \mathrm{d}\mu + \text{const}$.

(ii) Whiten with $\zeta := G^{-1/2}\vartheta$. The metric becomes Euclidean, the confinement becomes $(\sigma/2)\|\zeta - \zeta^0\|^2$, $W_{G^{-1}}$ becomes W_2, and H shifts by $\frac{1}{2}\log\det G$. $(\mathfrak{A})$ becomes equation (12) of [15]. We verify Assumptions 1–3 of [15].

- *Assumption 1.* $\nabla V[\mu](\zeta)$ is Lipschitz in (ζ, μ). The constants do not depend on μ, because $\|r_\mu\| \leq F_\rho + K_\Phi$ on $\operatorname{supp}\rho$ for every $\mu \in \mathcal{P}(\mathbb{R}^p)$.
  - In ζ, this holds because $\nabla^2 \tilde{\Phi}$ is bounded (saturation, and $s \geq s_{\min}$).

– In μ, it holds because $\int \tilde{\Phi}\,\mathrm{d}\mu$ is W_1-Lipschitz, $W_1 \le W_2$, and $\nabla\tilde{\Phi}$ is bounded.

- *Assumption 2.* $\mathcal{G}$ is convex: a linear map followed by a convex quadratic, plus a linear term. F_τ has a minimiser by confinement and lower semicontinuity (proof of [15, Prop. 5.1]).
- *Assumption 3.* $e^{-V[\mu]/\tau} = e^{-(\sigma/(2\tau))\|\zeta-\zeta^0\|^2} \cdot e^{-V_L[\mu]/(k_e\tau)}$.
  - The Gaussian factor satisfies LSI(σ/τ) (Bakry–Émery [4]).
  - $|V_L[\mu]| \le \int \|r_\mu\|\|\tilde{\Phi}\|\,\mathrm{d}\rho \le (F_\rho + K_\Phi)K_\Phi$, so the perturbation has oscillation at most $2K_\Phi(F_\rho + K_\Phi)/(k_e\tau)$. Here F_ρ may be replaced by $\int\|f\|\,\mathrm{d}\rho$.
  - Holley–Stroock [43] gives ρ_τ.

Theorem 3.2 and Corollary 3.3 of [15] (the latter with Otto–Villani [65]) give the two inequalities. Uniqueness follows from strict convexity of H.

$\mathcal{P}_2$ *starts.* Work in ζ-coordinates and fix the solution μ_t. Compare the particle SDE with the Ornstein–Uhlenbeck process $\mathrm{d}Y = -\sigma(Y-\zeta^0)\,\mathrm{d}t + \sqrt{2\tau}\,\mathrm{d}B$ started from the same law.

- The two drifts differ by $b_t = -(1/k_e)G^{1/2}\nabla V_L[\mu_t]$, with $\|b_t\| \le B := (1/k_e)\lambda_G^{1/2}(F_\rho + K_\Phi)\cdot \sup\|\nabla_\vartheta\tilde{\Phi}\|$.
- Girsanov and data processing give $H(\mu_{t_0}\,|\,\mathrm{OU}_{t_0}) \le B^2 t_0/(4\tau)$.
- OU_{t_0} has a density at most $\big(2\pi(\tau/\sigma)(1-e^{-2\sigma t_0})\big)^{-p/2}$, so $H(\mu_{t_0}) < \infty$.
- $\mu_{t_0} \in \mathcal{P}_2$, because the drift grows linearly.

Hence $F_\tau(\mu_{t_0}) < \infty$.

(iii) $\bar{L}(\mu) - \bar{L}(\nu) = \frac{1}{2}\int (r_\mu - r_\nu)^\top (r_\mu + r_\nu)\,\mathrm{d}\rho$. Here $\|r_\mu + r_\nu\| \le 2(F_\rho + K_\Phi)$, and $\|r_\mu - r_\nu\| = \|\int \tilde{\Phi}\,\mathrm{d}(\nu-\mu)\| \le L_\Phi \cdot W_{1,G^{-1}} \le L_\Phi \cdot W_{G^{-1}}$. Conclude with (ii). For the bias: $(1/k_e)\bar{L}(\mu_\tau^*) + \tau H(\mu_\tau^*\,|\,\gamma_\tau) \le (1/k_e)\bar{L}(\nu) + \tau H(\nu\,|\,\gamma_\tau)$ for every ν, and $H(\mu_\tau^*\,|\,\gamma_\tau) \ge 0$.

(iv) This is Theorem 4.1 of [15] with the stated hypotheses. □

Remark H.1 (dither adds no output noise in the mean-field limit; its cost is the entropic bias)**.** In the limit, $\hat{f} = \int \tilde{\Phi}\,\mathrm{d}\mu$ is deterministic by the law of large numbers. The only effect of dither is the entropic bias, which Theorem H(iii) quantifies.

In the experiment ($\sigma = 0.02$, $\tau = 0.004$, $G = \mathrm{diag}(4,1,1)$, $k_e = 10$), a grid computation gives $\bar{L}(\mu_\tau^*) = 0.11118$. It is grid-converged (`exp_gibbs_reference.py` → `results/langevin_global/gibbs_reference.md`). At $(\sigma,\tau) = (0.002, 5\cdot 10^{-4})$ the same computation gives 0.0085. The unregularised offline optimum over 32 kernels is about $7.6\cdot 10^{-6}$ (best of 5 L-BFGS restarts, an upper bound), and $\bar{L}(0) = 0.311$.

At finite m, the fluctuation of $\hat{f}$ is $O(K_\Phi/\sqrt{m})$ at each fixed t (finite-horizon propagation of chaos; Theorem D′ of Section 17 gives the rate $m^{-1/2}$ for the plant and the learned function of the dithered loop). It is uniform in time only under Remark H.5.

Remark H.2 (the control knobs, and what the numbers say)**.**

- **The knobs.** σ is the strong-convexity constant of the confinement, τ is the temperature, and $a_{\max}$ sets K_Φ.
- **The role of k_e.** Since $k_e F_\tau = \bar{L} + (k_e\sigma/2)\int\|\vartheta - \vartheta^0\|^2_{G^{-1}}\,\mathrm{d}\mu + k_e\tau H$, μ_τ^* and ρ_τ depend on (σ,τ,k_e) only through $(k_e\sigma, k_e\tau)$. Raising k_e at fixed (σ,τ) has three effects:
  - it improves ρ_τ and the guaranteed rate $2\sigma e^{-\alpha^*/\tau}$, since $\alpha^* \propto 1/k_e$;
  - it moves μ_τ^* toward the prior $\mathcal{N}(\vartheta^0, (\tau/\sigma)G)$, which increases the regularisation bias;
  - it shrinks the loss-driven drift like $1/k_e$.
- **The guaranteed rate** is exponentially small in $1/\tau$, a known feature of Holley–Stroock bounds [43]:

  - $2\tau\rho_\tau \geq e^{-8401}\,\mathrm{s}^{-1}$ at $(\sigma, \tau) = (0.002, 5 \cdot 10^{-4})$;
  - $2\tau\rho_\tau \geq e^{-1053}\,\mathrm{s}^{-1}$ at $(0.02, 0.004)$.

  The experiments do not test this rate.

- **The temperature sweep** (`results/langevin_tradeoff`; $m = 32$, $\sigma = 0.002$, $k_e = 10$, $T = 1500\,\mathrm{s}$, 2 seeds):

τ	escape time of the collapsed start (s), per seed	loss floor of the spread start
0	never, never	$7.0 \cdot 10^{-3}$
$5 \cdot 10^{-4}$	never, 1050	$1.1 \cdot 10^{-2}$
$1 \cdot 10^{-3}$	1480, 1005	$1.5 \cdot 10^{-2}$
$2 \cdot 10^{-3}$	1115, 800	$2.3 \cdot 10^{-2}$
$4 \cdot 10^{-3}$	970, 430	$3.6 \cdot 10^{-2}$
$8 \cdot 10^{-3}$	390, 295	$6.5 \cdot 10^{-2}$
annealed, hot start	990, 530	$3.2 \cdot 10^{-2}$

- **How escape is measured.** Escape is the first sample at which the collapsed run's loss falls to at most twice the spread run's final loss, at the same τ and seed. Two caveats follow:
  - the threshold grows with τ, from 0.022 to 0.13, and the escape time includes convergence after escape;
  - the $\tau = 5 \cdot 10^{-4}$ point is censored and is excluded from the fit.

- **The fit.** The seed-averaged escape time fits $t_{\mathrm{esc}} \approx -1693 + 427 \cdot \log(1/\tau)$ (4.5% relative residual) better than an Arrhenius law (25%). On the eight per-seed values the two laws fit equally poorly (27% and 33%; `langevin_tradeoff/table.md`).

- **No saddle explanation.** The noiseless collapsed end point is a saddle of the averaged landscape, but, as a numerical check shows, a weak one. Its unstable rate is about $5 \cdot 10^{-5}\,\mathrm{s}^{-1}$ for $(\mathfrak{A})$, and about $2 \cdot 10^{-4}\,\mathrm{s}^{-1}$ for the exactly averaged loop at $k_e = 10$. A saddle-escape law would therefore need $\tau \lesssim 10^{-5}$, and would have a $\log(1/\tau)$ slope of 10^3–$10^4\,\mathrm{s}$.

- **What the data show.** At the tested temperatures escape is governed by diffusion, by the approach transient and by the τ-dependent threshold. The data do not identify the escape mechanism. An earlier "saddle, not barrier" reading of these data is withdrawn.

- **The annealed runs** use α_a between about $2 \cdot 10^{-3}$ and $2 \cdot 10^{-2}$, far below $\alpha^* = 4.2$. They are not instances of H(iv).

Remark H.3 (why saturation and not projection)**.** The mean-field Langevin results are on $\mathbb{R}^d$. Projecting the drift with the Pomet–Praly operator [76], while keeping additive noise, leaves the noise unconfined. Reflected (Skorokhod) dynamics on the convex body Θ_K would confine it.

- Such dynamics satisfy a log-Sobolev inequality with constant of order $1/\operatorname{diam}^2$, even for $\sigma = 0$.

- The argument is a diameter-based log-Sobolev inequality for the uniform measure on a convex body, followed by Holley–Stroock [43]. The diameter-based inequality comes from $\mathrm{CD}(0, \infty)$ with a convex Neumann boundary plus a diameter bound (F.-Y. Wang 1997 [87]), or from Frieze–Kannan (1999) [31] for log-concave measures on convex bodies. Together these give $\mathrm{LSI}\big(c \cdot \operatorname{diam}^{-2} \cdot e^{-2K_\Phi(F_\rho + K_\Phi)/(k_e\tau)}\big)$.

- Bakry–Émery [4] alone gives nothing at $\sigma = 0$.
- That route connects to [69] and is left open.

Remark H.4 (the role of σ-modification)**.** In the classical theory, σ-modification is a robustness device with a bias. In the saturated design on $\mathbb{R}^p$ it supplies the confinement that makes $e^{-V/\tau}$ integrable and log-Sobolev, so the global result rests on it. Its bias is the regularisation term of the variational problem that μ_τ^* solves. Under reflection on Θ_K, compactness would play that role instead.

Remark H.5 (finite m; conditional)**.** The m-particle averaged system is

$$\mathrm{d}\vartheta_i = -G\nabla V[\mu_t^m](\vartheta_i)\,\mathrm{d}t + \sqrt{2\tau}\,G^{1/2}\,\mathrm{d}B_i, \qquad \mu_t^m := \frac{1}{m}\sum_i \delta_{\vartheta_i}.$$

Uniform-in-time propagation of chaos would follow from [11], under its hypotheses.

- *Available:* convexity of $\mathcal{G}$; a uniform LSI, from (ii); and a bounded, smooth second variation $(1/k_e)\int \tilde{\Phi}(x;\vartheta)^\top \tilde{\Phi}(x;\vartheta')\,\mathrm{d}\rho$.
- *Still to check:* the remaining regularity conditions of [11], verbatim (open question 4 of `LITERATURE_TRANSPORT.md` in the accompanying repository).

Status. Conditional on the hypotheses of [11]; their remaining regularity conditions are still to be checked.

Conjecture H.6 (the real closed loop, in the scaling where $(\mathfrak{A})$ is the limit)**.** *Let $f = f(x)$ satisfy the tube condition of Theorem D(b), with $a_{\max}$ replaced by K_Φ. Fix $G \succ 0$ and $\sigma_0, \tau_0 > 0$, and set $\sigma := \sigma_0/k_e$ and $\tau := \tau_0/k_e$.*

- *Then μ_τ^*, the minimiser of $\bar{L} + (\sigma_0/2)\int \|\vartheta - \vartheta^0\|_{G^{-1}}^2\,\mathrm{d}\mu + \tau_0 H$, does not depend on k_e.*
- *In slow time $s = t/k_e$, $(\mathfrak{A})$ is the formal limit of the closed loop.*

Conjecture: *for the mean-field limit of the closed loop under (L_τ), for every $\mu_0 \in \mathcal{P}_2$ and every $e(0)$, there are $S(\mu_0)$ and $r(k_e) \to 0$ such that $W_{G^{-1}}(\mu_t, \mu_\tau^*) \le r(k_e)$ for all $t \ge k_e \cdot S$.*

Status. Conjectural.

At fixed k_e the analogous statement is false. The limit of small gains is the exactly averaged system, with the true periodic error e_μ and $x = x_m + e_\mu$.

- Its stationary state is offset from μ_τ^*: $\bar{L} = 0.11415$ against $\bar{L}(\mu_\tau^*) = 0.11118$ at $k_e = 10$ (`results/langevin_global/gibbs_reference.md`).
- In the scaling of Conjecture H.6, the $L^2(\rho)$ distance between the two learned functions is $6.8 \cdot 10^{-3}$, $1.4 \cdot 10^{-3}$ and $9.7 \cdot 10^{-4}$ at $k_e = 10$, 20 and 40. It is small and decreasing, but a clean $1/k_e$ law is not resolved in this range.
- By Proposition H.7(e) it is not the $W_{G^{-1}}$-gradient flow of any free energy. The cyclic sums of its antisymmetric second variation are non-zero in the checked instances, including the network of the experiments; by Theorem I(iii) (Section 21) they are non-zero at every $n = 1$ measure whose residual is non-zero on the orbit, and by Theorem I(vi) on no open set of measures do they all vanish.

21 Finite gain: semi-gradient and non-reciprocal structure

This section continues Section 20 at finite feedback gain k_e. The notation is that of Section 9 and Section 20: the saturated features $\tilde{\Phi}$, the residual r_μ, the potential $V_L[\mu]$, the averaged system $(\mathfrak{A})$ and the law (L_τ). We write $\tilde{\Phi}_\vartheta := \tilde{\Phi}(\cdot;\vartheta)$.

21.1 Frozen-measure structure of the exactly averaged loop

Proposition H.7 (frozen-measure structure of the exactly averaged closed loop)**.** *Assume the following:*

- $f \in C^2(\mathbb{R}^n)$ *is autonomous;*
- $x_m \in C^2$ *is T_0-periodic, with oriented orbit $\mathcal{O}$;*
- *the features are the saturated features of the design (Section 20);*
- $\mu \in \mathcal{P}(\mathbb{R}^p)$ *is frozen.*

Then for every $R > 0$ there is $k_0(R)$, independent of μ, such that for $k_e \geq k_0(R)$ the following hold.

(a) Periodic error. *The frozen-μ error equation $\dot{e} = r_\mu(x_m + e) - k_e e$ has exactly one T_0-periodic solution e_μ with $\sup_t \|e_\mu\| \leq R$. It attracts exponentially every solution with $\|e(0)\| \leq R$, and*

$$e_\mu = \frac{r_\mu(x_m)}{k_e} + \frac{\nabla r_\mu(x_m)\, r_\mu(x_m) - \frac{\mathrm{d}}{\mathrm{d}t} r_\mu(x_m(t))}{k_e^2} + O(k_e^{-3}), \tag{21.1}$$

uniformly in t and μ.

- *Here ∇r_μ is the Jacobian $(\partial r_k / \partial x_l)$, and $\partial_x \tilde{\Phi} = (\partial \tilde{\Phi}_k / \partial x_l)$.*
- *If f is one-sided Lipschitz, $\langle f(x) - f(y), x - y \rangle \leq L_f \|x - y\|^2$, then for $k_e > L_f + a_{\max}\sqrt{n}\, e^{-1/2}/s_{\min}$ the periodic solution is unique and attracts every solution.*
- *Without such a condition uniqueness is only local. For $f(x) = x^2$ there is a second, repelling periodic solution near $e \approx k_e$, and solutions above it blow up.*

(b) Expansion. *The period average of the drift of (L_τ) along e_μ is $G \nabla_\vartheta P_\mu(\vartheta) - \sigma(\vartheta - \vartheta^0)$, where $P_\mu(\vartheta) := \langle e_\mu^\top \tilde{\Phi}(x_m + e_\mu; \vartheta) \rangle_t$ satisfies*

$$P_\mu(\vartheta) = -\frac{1}{k_e} V_L[\mu](\vartheta) + \frac{1}{k_e^2}\big(U_{\mathrm{even}}[\mu](\vartheta) + U_{\mathrm{odd}}[\mu](\vartheta)\big) + O(k_e^{-3}), \tag{21.2}$$

$$U_{\mathrm{even}}[\mu](\vartheta) := \int (r_\mu \cdot \nabla_x)\big(r_\mu^\top \tilde{\Phi}(\cdot; \vartheta)\big)\, \mathrm{d}\rho, \qquad U_{\mathrm{odd}}[\mu](\vartheta) := -\frac{1}{T_0} \oint_{\mathcal{O}} \tilde{\Phi}(\cdot; \vartheta)^\top \, \mathrm{d}r_\mu, \tag{21.3}$$

uniformly in ϑ and μ, together with ϑ-gradients.

(c) Semi-gradient form.

$$-\frac{1}{k_e} V_L[\mu] + \frac{1}{k_e^2} U_{\mathrm{even}}[\mu] = \frac{1}{k_e} \int r_\mu^\top \tilde{\Phi}(\cdot; \vartheta)\, \mathrm{d}\rho_\mu^{\mathrm{cl}} + O(k_e^{-3}). \tag{21.4}$$

Here ρ_μ^{cl} is the occupation measure of the closed-loop state $x_m + e_\mu$.

- *So, apart from the lag term U_{odd}, the averaged law follows the gradient of the closed-loop loss $\frac{1}{2}\int \|r_\nu\|^2 \, \mathrm{d}\rho_\mu^{\mathrm{cl}}$ at $\nu = \mu$, with the closed-loop state distribution ρ_μ^{cl} held fixed.*
- *This is a semi-gradient: the update freezes part of its own parameter dependence. In TD learning the frozen part is the bootstrapped target, and the data distribution does not depend on the parameters [85, 88].*
- *Here the frozen part is the data distribution ρ_μ^{cl}. This is the structure of performative prediction [71]: steps taken with the distribution held at its current value converge to performatively stable points, not to optima.*

- ***The performative defect carries the even non-reciprocity.*** *With* $J^{\mathrm{cl}}(\mu) := \frac{1}{2}\int \|r_\mu\|^2 \,\mathrm{d}\rho_\mu^{\mathrm{cl}}$,

$$\begin{aligned} &-\frac{1}{k_e}V_L[\mu] + \frac{1}{k_e^2}U_{\mathrm{even}}[\mu] \\ &\qquad = -\frac{1}{k_e}\frac{\delta J^{\mathrm{cl}}}{\delta \mu} - \frac{1}{k_e^2}\int (\tilde{\Phi}_\vartheta \cdot \nabla)\big(\tfrac{1}{2}\|r_\mu\|^2\big)\,\mathrm{d}\rho + O(k_e^{-3}). \end{aligned} \tag{21.5}$$

  The first term is a true first variation, hence reciprocal. The second is the defect: it has the same order as U_{even}*, and it carries all of* A_{even} *in (e).*

(d) Circulation kernel. $U_{\mathrm{odd}}[\mu](\vartheta) = -\frac{1}{T_0}\oint_{\mathcal{O}} \tilde{\Phi}(\cdot;\vartheta)^\top \,\mathrm{d}f + \int K(\vartheta,\vartheta')\,\mathrm{d}\mu(\vartheta')$*, with*

$$K(\vartheta,\vartheta') := \frac{1}{T_0}\oint_{\mathcal{O}} \tilde{\Phi}(\cdot;\vartheta)^\top \,\mathrm{d}\tilde{\Phi}(\cdot;\vartheta') = \Big\langle \tilde{\Phi}(x_m;\vartheta)^\top \frac{\mathrm{d}}{\mathrm{d}t}\tilde{\Phi}(x_m;\vartheta')\Big\rangle_t. \tag{21.6}$$

- K *is antisymmetric.*
- *By Stokes,* $K(\vartheta,\vartheta') = \frac{1}{T_0}\int_\Sigma \sum_k \mathrm{d}\tilde{\Phi}_k(\cdot;\vartheta)\wedge \mathrm{d}\tilde{\Phi}_k(\cdot;\vartheta')$ *for any 2-chain* Σ *with* $\partial\Sigma = \mathcal{O}$*.*
- *For* $n = 2$ *and a simple, positively oriented orbit bounding* D*,*

$$K(\vartheta,\vartheta') = \frac{1}{T_0}\sum_k \int_D \det\big[\nabla_x\tilde{\Phi}_k(\cdot;\vartheta), \nabla_x\tilde{\Phi}_k(\cdot;\vartheta')\big]\,\mathrm{d}A.$$

- K *is an integral over the enclosed region, not the area itself. For Gaussian features,* $\sum_k \det[\nabla\tilde{\Phi}_k(\vartheta), \nabla\tilde{\Phi}_k(\vartheta')] = (a^\top a')\cdot\det[\nabla\varphi, \nabla\varphi']$*, so* $K = 0$ *whenever* $a^\top a' = 0$ *or* $c = c'$*, whatever the area.*
- *Hence* $U_{\mathrm{odd}} \equiv 0$ *for* $n = 1$*.*
- *Reversing the orbit preserves* ρ*, flips* U_{odd}*, and leaves* U_{even} *unchanged.*

(e) Non-reciprocity. *Write* $S(\vartheta,\vartheta') := \delta P_\mu(\vartheta)/\delta\mu(\vartheta')$ *for the derivative in the direction* $\delta_{\vartheta'}$*, and call* $S(\vartheta,\vartheta') - S(\vartheta',\vartheta)$ *its antisymmetric part (without the factor* $\frac{1}{2}$*). Adding a function of* ϑ *alone, or a constant* $c(\mu)$*, to* P_μ *changes the antisymmetric part by a coboundary, which the criterion below allows.*

- *At order* $1/k_e$*,* S *is symmetric: it equals* $-\frac{1}{k_e}\int \tilde{\Phi}(\cdot;\vartheta)^\top\tilde{\Phi}(\cdot;\vartheta')\,\mathrm{d}\rho$*.*
- *At order* $1/k_e^2$ *it is not. The second variation of the* k_e^{-2} *coefficient* $U_{\mathrm{even}} + U_{\mathrm{odd}}$ *has antisymmetric part* $A_{\mathrm{even}} + 2K$*, with*

$$A_{\mathrm{even}}(\vartheta,\vartheta') := -\int \Big[(\tilde{\Phi}_{\vartheta'}\cdot\nabla)(r_\mu^\top\tilde{\Phi}_\vartheta) - (\tilde{\Phi}_\vartheta\cdot\nabla)(r_\mu^\top\tilde{\Phi}_{\vartheta'})\Big]\,\mathrm{d}\rho. \tag{21.7}$$

- P_μ*, up to* $O(k_e^{-3})$*, can be a first variation of a functional of* μ *only if* $A_{\mathrm{even}} + 2K$ *is a coboundary* $a(\vartheta) - a(\vartheta')$*, that is, only if all its cyclic sums vanish.*
- *They do not vanish in the checked instances, for* $n = 1$ *(from* A_{even} *alone) and for* $n = 2$*. Theorem I shows that the cyclic condition is equivalent to* $A_{\mathrm{even}} + 2K \equiv 0$*, and characterises when that happens: for* $n = 1$ *exactly when the residual vanishes on the orbit; for* $n \geq 2$ *at perfect fit exactly when the orbit carries no circulation and the residual's Jacobian is symmetric along the orbit. It also shows (Theorem I(vi)) that no open set of measures satisfies the condition, for any* n*.*
- *So at finite* k_e*, the exactly averaged loop is a non-reciprocal McKean–Vlasov dynamics: it is not the* $W_{G^{-1}}$*-gradient flow of any free energy on any open set of measures. A different,* μ*-dependent metric is not excluded by this argument.*

Status. The frozen-measure part is proved. Non-reciprocity of the exact second variation is checked to 3% (asserted; 1–1.5% measured) and characterised by Theorem I. The Stokes identity and the parity under orbit reversal hold to machine precision. The expansion is checked through its $O(1/k_e)$ convergence. Novelty (literature audit): no precedent found (targeted search); ingredients known (higher-order averaging; performative prediction [71]; semi-gradient TD [85, 88]).

Proof. (a) Write $e = r_\mu(x_m)/k_e + w$. Then w solves $\dot w = -k_e w + g(t, w)$ with $g = \nabla r_\mu(x_m)e - \dot r_\mu/k_e + O(|e|^2)$.

- The tube $\{\|e\| \le R\}$ is forward invariant once $k_e R > \sup_{\|e\|\le R}\|r_\mu(x_m + e)\|$.
- Inside the tube, solutions contract at rate $k_e - \mathrm{Lip}_R(r_\mu)$, which gives existence, uniqueness and attraction there. A contraction on T_0-periodic functions gives the expansion.
- The bounds on r_μ, ∇r_μ and $\nabla^2 r_\mu$ on the R-neighbourhood of the orbit are uniform in μ, by the saturation and $f \in C^2$.
- The global statement uses L_f in place of the local Lipschitz constant of f.

(b) Taylor-expand $e^\top \tilde\Phi(x_m + e; \vartheta) = e^\top \tilde\Phi(x_m; \vartheta) + e^\top \partial_x \tilde\Phi(x_m; \vartheta)e + O(|e|^3)$, and insert (a).

- The period averages of the time-symmetric terms are integrals against ρ.
- $(\nabla r\, r)^\top \tilde\Phi + r^\top(\partial_x \tilde\Phi)r = (r \cdot \nabla)(r^\top \tilde\Phi)$.
- ϑ-derivatives commute with the average.
- The remainders are uniform, because the x- and ϑ-derivatives of $\tilde\Phi$ of the orders needed are bounded.

(c) We have

$$\int r_\mu^\top \tilde\Phi_\vartheta \,\mathrm{d}\rho_\mu^{\mathrm{cl}} = \big\langle r_\mu(x_m + e_\mu)^\top \tilde\Phi(x_m + e_\mu; \vartheta)\big\rangle = \int r_\mu^\top \tilde\Phi_\vartheta \,\mathrm{d}\rho + \big\langle \nabla_x (r_\mu^\top \tilde\Phi_\vartheta)\cdot e_\mu\big\rangle + O(k_e^{-2}).$$

By (a) this equals $\int r_\mu^\top \tilde\Phi_\vartheta \,\mathrm{d}\rho + (1/k_e)U_{\mathrm{even}} + O(k_e^{-2})$.

(d) $\frac{\mathrm{d}}{\mathrm{d}t} r_\mu(x_m) = \frac{\mathrm{d}}{\mathrm{d}t} f(x_m) - \int \frac{\mathrm{d}}{\mathrm{d}t}\tilde\Phi(x_m; \vartheta')\,\mathrm{d}\mu(\vartheta')$.

- Antisymmetry: $K(\vartheta, \vartheta') + K(\vartheta', \vartheta) = \frac{1}{T_0}\oint \mathrm{d}\big(\tilde\Phi_\vartheta^\top \tilde\Phi_{\vartheta'}\big) = 0$.
- The area form is Stokes, or Green's theorem for $n = 2$.
- For $n = 1$, every 1-form on $\mathbb{R}$ is exact, so its loop integrals vanish.
- Reversal changes the sign of oriented loop integrals and preserves ρ.

(e) The second variation of U_{even} is

$$S_{\mathrm{even}}(\vartheta, \vartheta') = -\int \Big[(\tilde\Phi_{\vartheta'} \cdot \nabla)(r_\mu^\top \tilde\Phi_\vartheta) + (r_\mu \cdot \nabla)(\tilde\Phi_{\vartheta'}^\top \tilde\Phi_\vartheta)\Big]\,\mathrm{d}\rho,$$

whose second term is symmetric. The second variation of U_{odd} is K. A kernel's antisymmetric part A is a coboundary if and only if $A(\vartheta_0, \vartheta_1) + A(\vartheta_1, \vartheta_2) + A(\vartheta_2, \vartheta_0) = 0$ for all triples.

The remainder of (b) is not differentiated in μ. Instead the argument uses a circulation around a small triangle of measures.

1. Fix $\vartheta_0, \vartheta_1, \vartheta_2$ and $\varepsilon \in (0, 1)$. For (a, b) in $T_\varepsilon := \{a, b \ge 0,\ a + b \le \varepsilon\}$, let $\mu_{a,b} := (1 - \varepsilon)\mu + (\varepsilon - a - b)\delta_{\vartheta_0} + a\delta_{\vartheta_1} + b\delta_{\vartheta_2}$.
2. The circulation $\oint_{\partial T_\varepsilon} \langle Q_{\mu_{a,b}},\, \mathrm{d}\mu_{a,b}\rangle$ vanishes when Q_μ is a first variation plus a constant $c(\mu)$, because $\mathrm{d}\mu_{a,b}$ has zero mass. In particular it vanishes for $-(1/k_e)V_L$.
3. $U_{\mathrm{even}} + U_{\mathrm{odd}}$ is quadratic in μ, so its circulation equals $-(\varepsilon^2/2)\cdot\sum_{\mathrm{cyc}}(A_{\mathrm{even}} + 2K)(\vartheta_i, \vartheta_{i+1})$ at the centroid of T_ε.

4. The $O(k_e^{-3})$ remainder of (b) contributes at most $6\varepsilon \cdot \sup|R| = O(\varepsilon k_e^{-3})$.

5. Suppose that, for all large k_e, P_μ were a first variation up to $O(k_e^{-3})$, with constants uniform in k_e. Then $(\varepsilon^2/(2k_e^2))|\mathrm{cyc}| \le C\varepsilon k_e^{-3}$, that is, $|\mathrm{cyc}| \le 2C/(\varepsilon k_e)$. The cyclic sum does not depend on k_e, so letting $k_e \to \infty$ at fixed ε gives $\mathrm{cyc} = 0$ at the centroid measure of the triangle; letting $\varepsilon \to 0$, since $A_{\mathrm{even}} + 2K$ is affine in the measure, gives $\mathrm{cyc} = 0$ at μ.

This is checked numerically: for $\varepsilon = 0.05$ and $n = 2$ the circulation equals the centroid formula to all printed digits. □

Checked by: `tests/test_averaging.py` in the accompanying repository, using `src/core/averaging.py`; seven tests: four for Proposition H.7, one for Remark H.0, two for Remark H.7a. The checks of Proposition H.7 are listed in Table 22.

Table 22: Checks of Proposition H.7 (`tests/test_averaging.py`).

check	result
expansion: relative error of $k_e^2(P_\mu - P_0)$ against $U_{\mathrm{even}} + U_{\mathrm{odd}}$	halves when k_e doubles. At $k_e = 40 \to 80$ it goes $0.54 \to 0.25$ for $n = 1$, $0.038 \to 0.019$ for $n = 2$ counter-clockwise, and $0.022 \to 0.011$ clockwise. A Richardson step at $k_e = 80$ and 160 brings $n = 1$ within 3% (asserted)
Stokes: loop-integral K against the area integral over the enclosed ellipse	agreement to $7 \cdot 10^{-15}$ relative (asserted at 10^{-6}); K antisymmetric; $K \equiv 0$ for $n = 1$
parity under orbit reversal (same ρ)	U_{even} unchanged and U_{odd} flipped, to $2 \cdot 10^{-15}$ relative (asserted at 10^{-10})
non-reciprocity: the *exact* second variation S ($k_e = 80$) against the prediction	the full matrix $k_e^2 S + k_e S_0$ is within 5% of $S_{\mathrm{even}} + K$. The cyclic sums of the antisymmetric part agree to 3%: $n = 1$ gives -0.0711 against -0.0704, all from the state shift; $n = 2$ gives -0.0471 against -0.0464 (state shift -0.089, circulation $+0.043$)

21.2 The oscillation correction and the orientation of the reference orbit

Remark H.7a (the full first-order correction, and a test in the real closed loop)**.** Proposition H.7 freezes the measure over a period. In the closed loop the parameters also oscillate within each period, by $O(1/k_e)$.

The parameter equation. In the scaling $\sigma = \sigma_0/k_e$, write

$$\dot\Theta = \frac{1}{k_e} H(\Theta, t) + \frac{1}{k_e^2} H_2(\Theta, t) + O(k_e^{-3}), \tag{21.8}$$
$$H_i(\Theta, t) := G\nabla_\vartheta \big[r_\Theta(x_m(t))^\top \tilde\Phi(x_m(t); \vartheta_i) \big] - \sigma_0(\vartheta_i - \vartheta^0).$$

Second-order averaging (Sanders–Verhulst–Murdock [78, §2.9]) gives the dynamics of the period mean Y:

$$\dot Y = \frac{1}{k_e}\langle H\rangle(Y) + \frac{1}{k_e^2}\Big[\langle H_2\rangle(Y) + \big\langle D_\Theta H(Y,t)\cdot u(Y,t)\big\rangle\Big] + O(k_e^{-3}). \tag{21.9}$$

- u is the zero-mean antiderivative in t of $H - \langle H\rangle$.
- $\langle H_2\rangle$ is the frozen-measure correction of Proposition H.7.
- This is formal. The averaging theorem gives it on finite slow-time horizons.
- `averaging.oscillation_correction` computes the oscillation term $b_{\mathrm{osc}} := \langle D_\Theta H \cdot u\rangle$.

Three properties follow.

- **Parity and value.** The oscillation term b_{osc} is odd under reversal of the orbit (`test_oscillation_correction_is_odd_under_orbit_reversal`). Its value matches an independent one-period near-identity flow at $k_e = 80$: the test asserts 3%, and 0.7% is measured (`test_oscillation_correction_matches_a_one_period_flow`).
- **Size.** Relative to the frozen-measure terms it scales like $|G|T_0$, the parameter displacement per period times k_e. It is negligible when adaptation is slow within a period, which is the small-gain limit in the fixed-k_e discussion of Conjecture H.6. It is not negligible as $k_e \to \infty$ at fixed G. Both terms are $O(k_e^{-2})$ and uniformly bounded, so the order of the route to Conjecture H.6 is unchanged.
- **Non-reciprocity survives.** The only orientation-even part of the full correction is U_{even}, and its antisymmetric second variation is not a coboundary (Proposition H.7(e)). Suppose the full first-order dynamics were reciprocal for both orientations. Then the average of the two antisymmetric parts, A_{even}, would be a coboundary, which it is not. So the full first-order averaged dynamics is non-reciprocal for at least one orientation.

A prediction tested in the real closed loop (`exp_orbit_orientation.py` → `results/orbit_orientation/`).

- **The prediction.** Traverse the same elliptic reference both ways. The occupation measure is the same, so ($\mathfrak{A}$), and any functional of ρ, cannot distinguish the two runs. The first-order theory predicts that the learned kernels differ by $\Theta_+ - \Theta_- \approx -2J^{-1}(b_{\mathrm{odd}} + b_{\mathrm{osc}}) = O(1/k_e)$. Here J is the Jacobian of the leading averaged drift at the equilibrium.
- **The setting.** $n = 2$, $m = 6$, $G = \mathrm{diag}(4, 4, 1, 1, 1)$, $\sigma = 2/k_e$, no dither, and the same initial parameters for both runs. The runs converged to machine precision. The six kernels start spread out (kernel spread 1.05) and converge to a common configuration (spread at most $5.6 \cdot 10^{-17}$ at the end of every run), so the learned network is effectively a single kernel: the test probes the orientation effect at a collapsed equilibrium of the $m = 6$ system.

Table 23: Orientation dependence of the learned kernels in the real closed loop (`results/orbit_orientation/table_sigma2_slow16.md`). The size ratio is measured over predicted.

		frozen measure only (H.7)		full (H.7 + oscillation)		full
k_e	measured $\|\Theta_+ - \Theta_-\|$	size ratio	cosine	size ratio	cosine	vector error
10	$5.7 \cdot 10^{-3}$	2.67	0.76	0.94	0.93	0.36
20	$2.8 \cdot 10^{-3}$	2.41	0.83	0.97	0.98	0.20
40	$1.4 \cdot 10^{-3}$	2.29	0.86	0.97	0.994	0.11

- The measured difference scales like $k_e^{-1.01}$.
- At $k_e = 40$ the full first-order theory predicts it to within 3% in size and 11% as a vector (cosine 0.994).
- The vector error halves with each doubling of k_e: 36%, 20% and 11% at $k_e = 10$, 20 and 40.
- The frozen-measure terms alone account for less than half of it.
- A first attempt with a weaker leak ($\sigma = 0.5/k_e$) and a shorter horizon had not converged: its residual drift was a third of the difference. It is not used.

The direction in which the reference orbit is traversed is therefore a property of the excitation that the learned kernels retain, at order $1/k_e$. The first-order theory predicts its effect.

Status. The second-order averaged equation (21.9) is formal; the averaging theorem gives it on finite slow-time horizons. The oscillation term is checked against an independent one-period near-identity flow (within 3% at $k_e = 80$) and is odd under orbit reversal (to 10^{-6}). The full first-order theory predicts the orientation dependence of the learned kernels in the real closed loop to 3% in size and 11% as a vector (cosine 0.994, $k_e = 40$).

Checked by: `tests/test_averaging.py` (the two tests named above); `exp_orbit_orientation.py` → `results/orbit_orientation/table_sigma2_slow16.md`.

21.3 Reciprocity at order k_e^{-2}: a pairwise criterion and its characterisation

Theorem I (reciprocity at order k_e^{-2}: a pairwise criterion and its characterisation)**.** *Assume the setting of Proposition H.7: saturated Gaussian features $\tilde\Phi(x;\vartheta) = a\varphi_\xi(x)$ with $\xi := (c, s)$, $f \in C^2(\mathbb{R}^n)$ autonomous, $x_m \in C^2$ T_0-periodic with oriented orbit $\mathcal{O}$ and occupation measure ρ, and a frozen $\mu \in \mathcal{P}(\mathbb{R}^p)$. Let S be the k_e^{-2} coefficient of the second variation of P_μ, and $A_\mu := S - S^\top = A_{\mathrm{even}} + 2K$ its antisymmetric part, as in H.7(e). Call μ* reciprocal at order k_e^{-2} *if $A_\mu \equiv 0$ (the 1-form that the k_e^{-2} drift defines is closed at μ). Note that A_μ is affine in μ, because r_μ is and K does not depend on μ.*

(i) Pairwise criterion. *A_μ is a coboundary $a(\vartheta) - a(\vartheta')$ if and only if $A_\mu \equiv 0$. Consequently, by Proposition H.7(e), whose hypothesis concerns a neighbourhood of μ (its triangle of measures), P_ν can be a first variation of a functional of ν up to $O(k_e^{-3})$, with constants uniform in k_e and in ν near μ, only if $A_\nu \equiv 0$ for every ν in a neighbourhood of μ.*

(ii) Structure. *With Dr the Jacobian of r_μ, $(Dr)_{kl} = \partial_l r_k$, and $[X, Y] := (X \cdot \nabla)Y - (Y \cdot \nabla)X$ the Lie bracket of vector fields on $\mathbb{R}^n$,*

$$A_{\mathrm{even}}(\vartheta, \vartheta') = -\int \Big\{ \tilde\Phi_\vartheta^\top (Dr_\mu - Dr_\mu^\top)\tilde\Phi_{\vartheta'} + r_\mu^\top [\tilde\Phi_{\vartheta'}, \tilde\Phi_\vartheta] \Big\}\, \mathrm{d}\rho. \tag{21.10}$$

For the Gaussian features A_μ is a bilinear form in the weights: $A_\mu(\vartheta, \vartheta') = a^\top B_\mu(\xi, \xi')a'$ with

$$\begin{aligned} B_\mu(\xi, \xi') := & -\int (Dr_\mu - Dr_\mu^\top)\varphi_\xi \varphi_{\xi'}\, \mathrm{d}\rho - \int r_\mu (\nabla\varphi_\xi)^\top \varphi_{\xi'}\, \mathrm{d}\rho \\ & + \int (\nabla\varphi_{\xi'}) r_\mu^\top \varphi_\xi\, \mathrm{d}\rho + 2\kappa(\xi, \xi') I_n, \end{aligned} \tag{21.11}$$

$$\kappa(\xi, \xi') := \Big\langle \varphi_\xi(x_m(t)) \cdot \frac{\mathrm{d}}{\mathrm{d}t}\varphi_{\xi'}(x_m(t)) \Big\rangle_t, \tag{21.12}$$

so $A_\mu \equiv 0$ if and only if $B_\mu(\xi, \xi') = 0$ for all admissible ξ, ξ'. The first term is the curl of the residual paired with the features, the next two are the residual paired with the Lie bracket of the feature fields, and the last is the circulation kernel $K(\vartheta, \vartheta') = (a^\top a')\kappa(\xi, \xi')$. The performative part is bounded by the residual and by the antisymmetric part of its Jacobian along the orbit:

$$|A_{\mathrm{even}}(\vartheta, \vartheta')| \le |a||a'| \Big[\int \|Dr_\mu - Dr_\mu^\top\|\, \mathrm{d}\rho + \frac{2e^{-1/2}}{s_{\min}} \int |r_\mu|\, \mathrm{d}\rho \Big].$$

(iii) $n = 1$. *$A_\mu \equiv 0$ if and only if $r_\mu = 0$ ρ-a.e. on the orbit. So the one-dimensional loop is reciprocal at order k_e^{-2}, at a frozen μ, exactly when μ interpolates f along the orbit, and its non-reciprocity is bounded by the residual: $|A_\mu(\vartheta, \vartheta')| \le (2e^{-1/2}/s_{\min})|a||a'| \int |r_\mu|\, \mathrm{d}\rho$.*

(iv) $n \ge 2$ at perfect fit. *Suppose $r_\mu = 0$ ρ-a.e. Then the Lie-bracket terms vanish and*

$$B_\mu(\xi, \xi') = -\int (Dr_\mu - Dr_\mu^\top)\varphi_\xi \varphi_{\xi'}\, \mathrm{d}\rho + 2\kappa(\xi, \xi') I_n,$$

an antisymmetric matrix plus a multiple of the identity. Hence $A_\mu \equiv 0$ if and only if both

(a) $K \equiv 0$, *which holds if and only if the* 1*-current of the oriented orbit vanishes,* $T_{\mathcal{O}} := \int_0^{T_0} \delta_{x_m(t)}\, \dot{x}_m(t)\, \mathrm{d}t = 0$ *(an* $\mathbb{R}^n$*-valued measure;* $\int v \cdot \mathrm{d}T_{\mathcal{O}} = \oint_{\mathcal{O}} v \cdot \mathrm{d}x$*); for* $n = 2$*, iff the winding-number function* $w_{\mathcal{O}}$ *of the orbit vanishes a.e.; and*

(b) $Dr_\mu = Dr_\mu^\top$ ρ*-a.e.: the Jacobian of the residual has no antisymmetric part along the orbit.*

Vanishing of r_μ *on the orbit constrains only its tangential derivative, so (b) is a genuine extra condition: for* $f := \hat{f}_\mu + h e_1$ *with* $h = 0$ *on the orbit and* $\partial_\perp h \neq 0$*,* $A_\mu - 2K = -\int \varphi_\xi \varphi_{\xi'} \partial_\perp h \,\mathrm{d}\rho \cdot (e_1 e_\perp^\top - e_\perp e_1^\top) \neq 0$*. When* r_μ *vanishes on a neighbourhood of the orbit,* $Dr_\mu = 0$ *there and* $A_\mu = 2K$*: the semi-gradient defect has vanished and the loop is non-reciprocal exactly when the orbit carries circulation, through the orbit-area kernel alone.*

(v) General $n \geq 2$. *In any connected finite-dimensional real-analytic family of measures, for example atomic measures with* J *atoms parametrised by their weights and lifted positions, the set of reciprocal measures is either the whole family or a closed real-analytic subvariety of measure zero. The instances of* `tests/test_averaging.py` *(*$n = 1$ *and the* $n = 2$ *ellipse) have* $A_\mu \not\equiv 0$*, with cyclic sums* -0.0704 *and* -0.0464 *of the predicted kernel.*

(vi) Nowhere a gradient flow. *For every* n*, every* (f, x_m) *and every* μ*, every neighbourhood of* μ *contains measures* ν *with* $A_\nu \not\equiv 0$*: along the segments* $\nu_\delta := (1-\delta)\mu + \delta\delta_{\vartheta'}$ *either the zero feature* $\vartheta' = \vartheta_z$ *or a feature with* $a' \neq 0$ *gives* $A_{\nu_\delta} \not\equiv 0$ *for all small* $\delta > 0$*. Hence, by (i), there is no open set of measures on which* P_ν *is a first variation up to* $O(k_e^{-3})$ *with uniform constants: the exactly averaged closed loop is nowhere the* $W_{G^{-1}}$*-gradient flow of a free energy at order* k_e^{-2}*. The reciprocal measures of (iii) and (iv) are coincidences without interior.*

Status. Proved; every formula checked against the generic second-variation code and, by the round-5 referee, against the exact second variation; the characterisations checked on perfect-fit, curl and zero-current instances. Novelty (literature audit): new; the analytic tools (injectivity of Gaussian convolution; Stokes with winding numbers) are classical.

Proof. (i) Let $\vartheta_z := (\alpha, \gamma, \kappa)$ with $\alpha = 0$, an admissible point of $\mathbb{R}^p$, so $a = 0$ and $\tilde{\Phi}(\cdot; \vartheta_z) \equiv 0$ as a function of x; its x-derivatives vanish too. Every term of $A_{\mathrm{even}}(\vartheta, \vartheta_z)$ contains $\tilde{\Phi}_{\vartheta_z}$ or $(\tilde{\Phi}_{\vartheta_z} \cdot \nabla)$, and $K(\vartheta, \vartheta_z) = \langle \tilde{\Phi}_\vartheta^\top \frac{\mathrm{d}}{\mathrm{d}t} \tilde{\Phi}_{\vartheta_z} \rangle = 0$; so $A_\mu(\vartheta, \vartheta_z) = 0$ for every ϑ. If $A_\mu(\vartheta, \vartheta') = a(\vartheta) - a(\vartheta')$, then $a(\vartheta) = a(\vartheta_z)$ for every ϑ, so $A_\mu \equiv 0$. The converse is trivial. For the consequence, apply H.7(e) at any ν near μ with $\vartheta_2 := \vartheta_z$ in its triangle of measures: the cyclic sum over $(\vartheta_0, \vartheta_1, \vartheta_z)$ is $A(\vartheta_0, \vartheta_1)$, evaluated at the centroid measure of the triangle, which tends to ν as the triangle's size $\varepsilon \to 0$; A is affine in the measure, so $A_\nu(\vartheta_0, \vartheta_1) = 0$.
(ii) Expand $(\tilde{\Phi}' \cdot \nabla)(r^\top \tilde{\Phi}) = \sum_{k,l} \tilde{\Phi}'_l (\partial_l r_k) \tilde{\Phi}_k + \sum_{k,l} \tilde{\Phi}'_l r_k \partial_l \tilde{\Phi}_k = \tilde{\Phi}^\top (Dr) \tilde{\Phi}' + r^\top (\tilde{\Phi}' \cdot \nabla) \tilde{\Phi}$, and subtract the expression with ϑ and ϑ' exchanged. For $\tilde{\Phi} = a\varphi$, $(\tilde{\Phi}' \cdot \nabla)\tilde{\Phi} = a(a'^\top \nabla\varphi)\varphi'$, so $r^\top[\tilde{\Phi}', \tilde{\Phi}] = (r^\top a)(a'^\top \nabla\varphi)\varphi' - (r^\top a')(a^\top \nabla\varphi')\varphi$, and $K(\vartheta, \vartheta') = (a^\top a')\kappa(\xi, \xi')$. Collect the bilinear form in (a, a'). Since a ranges over the open cube $(-a_{\max}, a_{\max})^n$, $A_\mu \equiv 0$ iff $B_\mu \equiv 0$. The bound uses $\varphi \leq 1$ and $|\nabla\varphi| \leq e^{-1/2}/s_{\min}$.
(iii) For $n = 1$, $Dr - Dr^\top = 0$, and for equal widths s, $\varphi_{c'} D\varphi_c - \varphi_c D\varphi_{c'} = \varphi_c \varphi_{c'} (c - c')/s^2$, so $B_\mu(\xi, \xi') = -((c - c')/s^2) \int r_\mu \varphi_c \varphi_{c'} \,\mathrm{d}\rho$. Now $\varphi_c \varphi_{c'} = e^{-(c-c')^2/(4s^2)} \cdot e^{-(x-m)^2/s^2}$ with $m := (c + c')/2$. Put $\nu := r_\mu \rho$, a finite signed measure with compact support. The function $m \mapsto \int e^{-(x-m)^2/s^2} \,\mathrm{d}\nu(x)$ extends to an entire function. If $B_\mu \equiv 0$ on admissible pairs, it vanishes for all $m \in (-c_{\max}, c_{\max})$ (take $c = m - \delta$, $c' = m + \delta$ with $0 < \delta$ small), hence everywhere; its Fourier transform is the product of the Gaussian's, which has no zeros, with $\hat{\nu}$, so $\nu = 0$, that is $r_\mu = 0$ ρ-a.e. Conversely, if $\nu = 0$ then $A_{\mathrm{even}}(\vartheta, \vartheta') = -\int (\varphi' D\varphi - \varphi D\varphi') a a' \,\mathrm{d}\nu = 0$ for all widths.
(iv) If $r_\mu = 0$ ρ-a.e. the two Lie-bracket integrals vanish, which gives the displayed B_μ; the first term is an antisymmetric matrix and the second a multiple of I_n, so $B_\mu \equiv 0$ iff both vanish (take

the trace, which is $2n\kappa$, and the antisymmetric part). The first vanishes identically iff the matrix-valued measure $(Dr_\mu - Dr_\mu^\top)\rho$ is zero, by the argument of (iii) applied entrywise with equal widths (the entries of $\int (Dr - Dr^\top)\varphi_c\varphi_{c'}\,\mathrm{d}\rho$ are Gaussian convolutions of the entries of that measure). For (a): with equal widths, $\varphi_\xi\varphi_{\xi'} = e^{-|c-c'|^2/(4s^2)}e^{-|x-m|^2/s^2}$ and $\nabla\varphi_{\xi'} = -((x-c')/s^2)\varphi_{\xi'}$, so

$$\kappa(\xi,\xi') = -\frac{1}{s^2}e^{-|c-c'|^2/(4s^2)}\big\langle e^{-|x_m-m|^2/s^2}(x_m-c')\cdot\dot{x}_m\big\rangle_t.$$

Write $x_m - c' = (x_m - m) + (c-c')/2$. The first part integrates to zero, being the time derivative of $-(s^2/2)e^{-|x_m-m|^2/s^2}$ along a period. Hence

$$\kappa(\xi,\xi') = -\frac{1}{2s^2}e^{-|c-c'|^2/(4s^2)}(c-c')\cdot F_s(m),$$
$$F_s(m) := \big\langle e^{-|x_m(t)-m|^2/s^2}\dot{x}_m(t)\big\rangle_t = \frac{1}{T_0}\int e^{-|x-m|^2/s^2}\,\mathrm{d}T_{\mathcal{O}}(x),$$

the Gaussian convolution of the vector measure $T_{\mathcal{O}}$. If $\kappa \equiv 0$ on admissible pairs, then $F_s(m)\cdot v = 0$ for every m with $\|m\| < c_{\max}$ and every direction v (take $c = m - \delta v$, $c' = m + \delta v$), so $F_s \equiv 0$ on that ball; F_s is entire, so $F_s \equiv 0$ on $\mathbb{R}^n$, and the Fourier argument gives $T_{\mathcal{O}} = 0$. Conversely, $\kappa(\xi,\xi') = \frac{1}{T_0}\int \varphi_\xi\nabla\varphi_{\xi'}\cdot\mathrm{d}T_{\mathcal{O}} = 0$ for all widths when $T_{\mathcal{O}} = 0$. For $n = 2$, $\oint_{\mathcal{O}} v\cdot\mathrm{d}x = \int w_{\mathcal{O}}\,\mathrm{curl}\,v\,\mathrm{d}A$ for every smooth v (Stokes with winding numbers), and every smooth compactly supported g is $\mathrm{curl}\,v$ for some smooth v, so $T_{\mathcal{O}} = 0$ iff $w_{\mathcal{O}} = 0$ a.e. ($T_{\mathcal{O}}$ is compactly supported, so v need not be). For the example, $r = he_1$ with $h = 0$ on $\mathcal{O}$ gives $(Dr - Dr^\top)_{1l} = \partial_l h$ and the only non-zero entries are the normal derivative; the Lie-bracket terms vanish because $r = 0$ on $\mathcal{O}$.
(v) The saturated features are real-analytic in the lifted coordinates and extend holomorphically to a complex neighbourhood of $\mathbb{R}^p$ (tanh and the logistic function are holomorphic on the strips $|\mathrm{Im}| < \pi a_{\max}/2$, $\pi c_{\max}/2$, π, where s stays away from 0), uniformly for x on the compact orbit; integrals over the orbit of such integrands are real-analytic in the parameters, and r_μ is affine in the weights. The zero set of a real-analytic function on a connected real-analytic manifold is either the whole manifold or a closed set of Lebesgue measure zero. The instances are those of `test_exactly_averaged_dynamics_is_non_reciprocal_for_n1_and_n2`.
(vi) Since r_ν is affine in ν, $B_{\nu_\delta} = B_\mu + \delta B^{\mathrm{even}}[g]$ with $g := \hat{f}_\mu - \tilde{\Phi}_{\vartheta'}$, where $B^{\mathrm{even}}[g]$ denotes the three residual terms of B with g in place of r_μ. The trace of $B^{\mathrm{even}}[g]$ at equal widths is $-(1/s^2)(c-c')\cdot\int g\varphi_c\varphi_{c'}\,\mathrm{d}\rho$ (the curl term is traceless). If $A_{\nu_\delta} \equiv 0$ for all small δ, then $B^{\mathrm{even}}[g] \equiv 0$, so $(c-c')\cdot F[g\rho]((c+c')/2) = 0$ for all admissible c, c', where $F[g\rho]$ is the Gaussian convolution of the vector measure $g\rho$; as in (iv), $g\rho = 0$, that is $g = 0$ ρ-a.e. For $\vartheta' = \vartheta_z$ this says $\hat{f}_\mu = 0$ on the orbit; for ϑ' with $a' \neq 0$ it then says $\tilde{\Phi}_{\vartheta'} = 0$ on the orbit, which is false since a Gaussian has no zeros. So one of the two segments has $A_{\nu_\delta} \not\equiv 0$ for all small $\delta > 0$. □

Checked by: `tests/test_nonreciprocity.py`, using `src/core/nonreciprocity.py` against `src/core/averaging.py`: the bilinear form $a^\top B_\mu a'$ reproduces the antisymmetric part of the generic second-variation coefficient (10^{-10}, $n = 1$ and 2); $A_\mu(\vartheta,\vartheta_z) = 0$ (10^{-14}) and the cyclic sum over $(\vartheta_0,\vartheta_1,\vartheta_z)$ equals $A_\mu(\vartheta_0,\vartheta_1)$; for $n = 1$ with $f := \hat{f}_\mu$ (perfect fit) $A_\mu \equiv 0$ (below 10^{-13}) while a non-zero residual gives $A_\mu \neq 0$, equal to the closed form; the closed form of κ (10^{-10}), the second component of $F_s(m)$ at $m = (0.5, 0)$, $s = 0.7$ is positive on the ellipse, and $\kappa \equiv 0$ on a segment traversed back and forth (below 10^{-12}); with $f \equiv \hat{f}_\mu$ on $\mathbb{R}^2$ (so $Dr_\mu \equiv 0$), $B_\mu = 2\kappa I$ (10^{-13}), $A_\mu \neq 0$ on the ellipse and $A_\mu \equiv 0$ on the segment; with $f := \hat{f}_\mu + he_1$, $h = (x_1/1.2)^2 + (x_2/0.8)^2 - 1$ (zero on the ellipse, non-zero normal derivative), $B_\mu - 2\kappa I$ equals $-\int\varphi\varphi'\partial_2 h\,\mathrm{d}\rho\cdot(e_1e_2^\top - e_2e_1^\top)$ (10^{-12}) and is not small; the positivity witness $f \equiv 0$, $\mu = \delta_{\bar{\vartheta}}$, for which $A_\mu(\vartheta_0,\vartheta_1) = a_0a_1\bar{a}\,((c_0-c_1)/s^2)\int\varphi_{\bar{c}}\varphi_{c_0}\varphi_{c_1}\,\mathrm{d}\rho$ has a strictly positive integral. The round-5 referee re-derived every formula and checked $a^\top B_\mu a'$ against the exact second variation of the averaged potential (Richardson in k_e) to $2\cdot10^{-4}$–$9\cdot10^{-4}$ relative, including parameters at the saturation limits.

Remark I.1 (what Theorem I changes in the reading of Proposition H.7)**.**

- The non-reciprocity of H.7(e) is no longer an instance-by-instance check. Pointwise, for $n = 1$ it holds at every frozen μ that does not interpolate f along the orbit; for $n \geq 2$ at perfect fit it holds unless the orbit carries no circulation and the residual's Jacobian is symmetric along the orbit; in general it fails only on a null set of any connected analytic family that contains one non-reciprocal member. Globally, (vi) shows that no open set of measures is reciprocal, for any n: the averaged loop is nowhere a gradient flow at order k_e^{-2}.
- The two defects scale differently. The performative (semi-gradient) defect is bounded by the residual and the antisymmetric part of its Jacobian along the orbit, so it is small at a well-fitting equilibrium whose residual is smooth across the orbit; the circulation defect does not depend on the fit at all.
- For the $n = 1$ Langevin experiments and Conjecture H.6: the non-gradient correction that falsifies H.6 at fixed k_e is, by (iii), bounded by the residual, which at the regularised optimum μ_τ^* is the regularisation bias. Both the leading drift and the correction are proportional to the residual there, so the correction is $O(1/k_e)$ relative to the leading drift and $O(k_e^{-2})$ absolute, as in Remark H.0.
- The pairwise criterion also simplifies the test of reciprocity: a single pair (ϑ, ϑ') with $B_\mu(\xi, \xi') \neq 0$ suffices.

21.4 Input-to-state stability of $(\mathfrak{A})$ and the route to Conjecture H.6

Proposition H.8 (input-to-state stability of $(\mathfrak{A})$)**.** *Assume $\sigma, \tau > 0$ and the setting of Theorem H. Let μ_t solve the perturbed equation*

$$\partial_t \mu = \nabla \cdot \big(\mu(G\nabla V[\mu] - b_t)\big) + \tau \nabla \cdot (G\nabla\mu) \tag{21.13}$$

from $\mu_0 \in \mathcal{P}_2$ with $F_\tau(\mu_0) < \infty$. Here b_t is a velocity perturbation with $\beta(t)^2 := \int \|b_t\|_{G^{-1}}^2 \,\mathrm{d}\mu_t \leq \bar{\beta}^2$. Assume the chain rule of [15, Lemma 2.2] holds along this solution:

$$\frac{\mathrm{d}}{\mathrm{d}t} F_\tau(\mu_t) = \int \langle \nabla(\delta F_\tau/\delta\mu)(\mu_t),\, v_t\rangle \,\mathrm{d}\mu_t,$$

in whitened coordinates, with v_t the velocity. It is expected for smooth bounded perturbations (parabolic regularity and the entropy-dissipation estimate), but it is not verified here. Then

$$\frac{\mathrm{d}}{\mathrm{d}t}\big(F_\tau(\mu_t) - F^*\big) \leq -\tau\rho_\tau\big(F_\tau(\mu_t) - F^*\big) + \tfrac{1}{2}\beta(t)^2, \tag{21.14}$$

$$F_\tau(\mu_t) - F^* \leq e^{-\tau\rho_\tau t}\big(F_\tau(\mu_0) - F^*\big) + \frac{\bar{\beta}^2}{2\tau\rho_\tau}, \tag{21.15}$$

$$\limsup_{t\to\infty} W_{G^{-1}}(\mu_t, \mu_\tau^*) \leq \frac{\bar{\beta}}{\tau\rho_\tau}. \tag{21.16}$$

Status. Proved, assuming the chain rule for the perturbed flow. A proof only; not checked in the repository. Novelty (literature audit): elementary given [15].

Proof. Work in $\zeta = G^{-1/2}\vartheta$, and let $\xi_t := \nabla(\delta F_\tau/\delta\mu)(\mu_t) = \nabla\tilde{V}[\mu_t] + \tau\nabla\log\mu_t$. The velocity is $-\xi_t + \tilde{b}_t$, with $\|\tilde{b}_t\| = \|b_t\|_{G^{-1}}$.

1. By the chain rule and Young's inequality,

$$\frac{\mathrm{d}}{\mathrm{d}t} F_\tau(\mu_t) = -\int \|\xi_t\|^2 \,\mathrm{d}\mu_t + \int \langle \xi_t,\, \tilde{b}_t\rangle \,\mathrm{d}\mu_t \leq -\frac{1}{2}\int \|\xi_t\|^2 \,\mathrm{d}\mu_t + \frac{1}{2}\beta^2.$$

2. With $\hat{\mu}_t \propto e^{-\tilde{V}[\mu_t]/\tau}$,

$$\int \|\xi_t\|^2 \,\mathrm{d}\mu_t = \tau^2 I(\mu_t|\hat{\mu}_t) \geq 2\rho_\tau \tau^2 H(\mu_t|\hat{\mu}_t) \geq 2\rho_\tau \tau \big(F_\tau(\mu_t) - F^*\big),$$

by the LSI of Theorem H(ii) and the entropy sandwich [15, Lemma 3.4].

3. Grönwall gives the first two lines.

4. For the third, $F_\tau(\mu) - F^* \geq \tau H(\mu|\mu_\tau^*)$ [15, Lemma 3.4]. μ_τ^* satisfies LSI(ρ_τ), hence Talagrand's inequality $W^2 \leq (2/\rho_\tau) H(\cdot|\mu_\tau^*)$ (Otto–Villani [65]). So $W^2 \leq (2/(\tau\rho_\tau))(F_\tau - F^*)$, and the limsup follows from the second line. □

The gain $1/(\tau\rho_\tau) \leq e^{\alpha^*/\tau}/\sigma$ is exponentially large in $1/\tau$, like the inverse of the rate.

Route to Conjecture H.6. In the scaling of Conjecture H.6:

- $\tau\rho_\tau \geq (\sigma_0/k_e) \cdot e^{-2K_\Phi(F_\rho + K_\Phi)/\tau_0}$, so $1/(\tau\rho_\tau) = O(k_e)$.
  - In this scaling the whitened proximal Gibbs measures do not depend on k_e, so $\tau\rho_\tau = \Theta(1/k_e)$.
  - At the experiment's $\tau_0 = 0.04$ the constant is e^{1049}.
- Define $b := (1/k_e^2)\big[G\nabla(U_{\mathrm{even}} + U_{\mathrm{odd}}) + b_{\mathrm{osc}}\big] + O(k_e^{-3})$. The first part is the frozen-measure correction of Proposition H.7; the second is the oscillation term of Remark H.7a. Both are bounded uniformly in μ and ϑ, so $\bar{\beta} = O(k_e^{-2})$.
- Proposition H.8 then confines every solution of $(\mathfrak{A}) + b$ to an asymptotic W-ball around μ_τ^* of radius $O(1/k_e) \cdot e^{2K_\Phi(F_\rho + K_\Phi)/\tau_0}/\sigma_0$. This needs $F_\tau(\mu_0) < \infty$, or a $\mathcal{P}_2$ start, after the Girsanov regularisation of Theorem H(ii), which applies because b is bounded.

Three things are still missing for Conjecture H.6:

1. control of the frozen-measure approximation along the moving μ_t, that is, a near-identity averaging transformation for the stochastic McKean–Vlasov loop, uniformly in time;
2. the mean-field limit of (L_τ), since Theorem D covers only (L);
3. the chain rule that Proposition H.8 assumes for the perturbed flow.

21.5 Evidence from the real closed loop

The evidence comes from the real closed loop with $m = 32$ and $n = 1$, not from the averaged system.

- **Dither is necessary.** Without dither the collapsed start is trapped forever. Its loss stays at 0.166, against 0.007 from the spread start (`results/langevin_tradeoff`, 1500 s).
- **Equalisation at moderate temperature.** At $\sigma = 0.02$ and $\tau = 0.004$ (3 seeds, `results/langevin_global/table_sigma0.02_tau0.004.md`):
  - collapsed and spread starts reach the same loss (ratio 1.03) and tracking RMS (0.0465 against 0.0452);
  - their final 32-particle configurations, driven by the same noise realisation, are 4.2× closer in W_2 over the physical coordinates (a, c, s) than without dither (0.40 against 1.69);
  - a numerical check shows that two independent 32-point samples of μ_τ^* are typically about 0.6 apart, so 0.40 is at the sampling floor;

  - the common loss of about 0.11 matches the grid value $\bar{L}(\mu_\tau^*) = 0.11118$. It is dominated by the regularisation bias.

- **Metastability at low temperature.** At $\sigma = 0.002$ and $\tau = 5 \cdot 10^{-4}$ (table for these values in `results/langevin_global/`), dither frees the collapsed start ($0.166 \to 0.048$). But the starts have not equalised within 900 s (loss ratio 5.2). The spread start's $9.2 \cdot 10^{-3}$ is close to the regularised optimum $\bar{L}(\mu_\tau^*) = 8.5 \cdot 10^{-3}$.

- **Limits of the evidence.**
  - The Langevin runs have $n = 1$. By Proposition H.7, the averaged dynamics is already non-reciprocal for $n = 1$, through the state shift.
  - The only $n \geq 2$ closed-loop runs are the deterministic orientation runs of Remark H.7a. They confirm the first-order theory quantitatively. No $n \geq 2$ run with dither has been made.
  - The runs use fixed k_e, not the scaling of Conjecture H.6.
  - All Langevin runs use $k_e = 10$, below the Tikhonov scale of Remark H.0 for this network ($(F_\rho + K_\Phi)/s_{\min} \approx 35$). The measured frozen-$\mu$ drift error at $k_e = 10$ is 4.2% ($n = 1$), so the condition is conservative, but the agreement with ($\mathfrak{A}$) is empirical.
  - So the evidence supports Conjecture H.6 qualitatively, but the equalisation time can exceed practical horizons at low τ.

22 Theorem J: the finite-m closed loop as a perturbed gradient flow

This section proves the first step of Section 26, branch 1: at finite m, without dither, in the scaling $\sigma = \sigma_0/k_e$, the real closed loop has, near every hyperbolic critical point of the m-particle regularised occupation loss, a unique periodic orbit with the same stability type, at distance $O(1/k_e)$, whose first-order location is the one Proposition H.7 and Remark H.7a predict. It turns Remark H.7a's "formal" second-order averaging into a theorem for this system.

Setting. The closed loop of Theorem H's design with $\tau = 0$, at finite m, in the scaling $\sigma = \sigma_0/k_e$:

$$\begin{aligned} \dot{x} &= f(x) + \dot{x}_m - k_e e - \hat{f}_\Theta(x), \qquad e := x - x_m(t), \qquad \hat{f}_\Theta := \frac{1}{m}\sum_j \tilde{\Phi}(\cdot\,;\vartheta_j), \\ \dot{\vartheta}_i &= G\nabla_\vartheta\big[e^\top \tilde{\Phi}(x;\vartheta_i)\big] - \frac{\sigma_0}{k_e}(\vartheta_i - \vartheta^0), \qquad i = 1,\dots,m, \end{aligned} \tag{CL}$$

with $\Theta := (\vartheta_1,\dots,\vartheta_m) \in \mathbb{R}^{mp}$, $G_m := I_m \otimes G$, saturated features, $x_m \in C^2$ T_0-periodic with occupation measure ρ, and $f \in C^2$ on a neighbourhood of the tube $\mathcal{T}_R := \{x : \operatorname{dist}(x, x_m([0,T_0])) \leq R\}$ for some $R > 0$. Put

$$r_\Theta(x) := f(x) - \hat{f}_\Theta(x), \qquad J_m(\Theta) := m\bar{L}_m(\Theta) + \frac{\sigma_0}{2}\sum_i \|\vartheta_i - \vartheta^0\|^2_{G^{-1}},$$

$$\bar{L}_m(\Theta) := \frac{1}{2}\int \|r_\Theta\|^2 \,\mathrm{d}\rho,$$

$$H_i(\Theta,t) := G\nabla_\vartheta\big[r_\Theta(x_m(t))^\top \tilde{\Phi}(x_m(t);\vartheta)\big]\Big|_{\vartheta=\vartheta_i} - \sigma_0(\vartheta_i - \vartheta^0),$$

$$\langle H\rangle(\Theta) = -G_m \nabla J_m(\Theta).$$

The averaged system of (CL) is $\dot{\Theta} = \frac{1}{k_e}\langle H\rangle(\Theta) = -\frac{1}{k_e}G_m\nabla J_m(\Theta)$: the gradient flow of J_m in the G^{-1} metric, which is ($\mathfrak{A}$) at $\tau = 0$ for the m-atom measure (Remark H.0); in slow time

$s := t/k_e$ it does not depend on k_e. Its equilibria are the critical points of J_m. A critical point Θ^* is *hyperbolic* if $\nabla^2 J_m(\Theta^*)$ is non-singular; its index is the number of negative eigenvalues; it is a *non-degenerate minimum* if $\nabla^2 J_m(\Theta^*) \succ 0$. Write $\nu_1 \le \cdots \le \nu_{mp}$ for the eigenvalues of $G_m^{1/2}\nabla^2 J_m(\Theta^*)G_m^{1/2}$ (those of $G_m\nabla^2 J_m(\Theta^*)$), and $\lambda_* := \nu_1$ when Θ^* is a minimum.

Let $u(\Theta,t)$ be the zero-mean antiderivative in t of $H(\Theta,t) - \langle H\rangle(\Theta)$ (T_0-periodic), and define, as in Proposition H.7 and Remark H.7a, at the atomic measure $\mu^* := \frac{1}{m}\sum_j \delta_{\vartheta_j^*}$:

$$J := D_\Theta\big[\tfrac{1}{k_e}\langle H\rangle\big](\Theta^*) = -\frac{1}{k_e}G_m\nabla^2 J_m(\Theta^*),$$

$$b_{\mathrm{frozen},i} := \frac{1}{k_e^2}G\nabla_\vartheta\big(U_{\mathrm{even}} + U_{\mathrm{odd}}\big)[\mu^*](\vartheta)\Big|_{\vartheta=\vartheta_i^*}, \qquad b_{\mathrm{osc}} := \frac{1}{k_e^2}\big\langle D_\Theta H(\Theta^*,t)\cdot u(\Theta^*,t)\big\rangle_t.$$

Theorem J. *Let Θ^* be a hyperbolic critical point of J_m. There are $k_0, \rho_0, C > 0$, depending on (f, x_m, G, σ_0, R, the feature bounds, Θ^*) but not on k_e, such that for every $k_e \ge k_0$:*

(a) Existence, uniqueness, location. (CL) *has a T_0-periodic solution (x_p, Θ_p), unique among the T_0-periodic solutions with $\sup_t\|\Theta_p(t) - \Theta^*\|_{G^{-1}} \le \rho_0$ and $\sup_t\|x_p(t) - x_m(t)\| \le R$. It satisfies*

$$\sup_t\|x_p(t) - x_m(t)\| \le C/k_e, \qquad \sup_t\|\Theta_p(t) - \Theta^*\|_{G^{-1}} \le C/k_e,$$

$$\Theta_p(t) - \bar\Theta_p = \frac{1}{k_e}u(\Theta^*,t) + O(k_e^{-2}),$$

with $\bar\Theta_p$ the period mean of Θ_p.

(b) Stability type. *The monodromy matrix of (x_p, Θ_p) has n eigenvalues of modulus at most $e^{-(k_e - L)T_0}$, with L independent of k_e (the error channel), and mp eigenvalues*

$$\lambda_j = \exp\big(-(T_0/k_e)\nu_j + O(k_e^{-2})\big), \qquad j = 1,\dots,mp.$$

Hence the orbit is hyperbolic with the index of Θ^*: exponentially stable if $\nabla^2 J_m(\Theta^*) \succ 0$, with parameter Floquet exponents $(1/k_e)\nu_j + O(k_e^{-2})$, and unstable otherwise.

(c) Basin (non-degenerate minimum). *If $\nabla^2 J_m(\Theta^*) \succ 0$, every solution of* (CL) *with $\|\Theta(0) - \Theta^*\|_{G^{-1}} \le \rho_0$ and $\|x(0) - x_m(0)\| \le R$ converges to the periodic orbit, and*

$$\|\Theta(t) - \Theta_p(t)\|_{G^{-1}} + \|x(t) - x_p(t)\| \le Ce^{-\mu t}\big(\|\Theta(0) - \Theta_p(0)\|_{G^{-1}} + \|x(0) - x_p(0)\|\big),$$

$$\mu := \frac{\lambda_*}{k_e}(1 - \kappa),$$

with $\kappa = \kappa(\rho_0, 1/k_e) \to 0$ as $(\rho_0, 1/k_e) \to 0$. The basin radius ρ_0 does not depend on k_e; the rate is λ_*/k_e to leading order, that is, λ_* in slow time.

(d) First-order location. *The period mean satisfies*

$$\begin{aligned}\bar\Theta_p &= \Theta^* - J^{-1}(b_{\mathrm{frozen}} + b_{\mathrm{osc}}) + O(k_e^{-2})\\ &= \Theta^* + \frac{1}{k_e}\big(G_m\nabla^2 J_m(\Theta^*)\big)^{-1}\cdot k_e^2(b_{\mathrm{frozen}} + b_{\mathrm{osc}}) + O(k_e^{-2}).\end{aligned}$$

The offset is $O(1/k_e)$. Its orientation-odd part is the quantity measured in Remark H.7a, so the prediction $\Theta_+ - \Theta_- = -2J^{-1}(b_{\mathrm{odd}} + b_{\mathrm{osc}}) + O(k_e^{-2})$ of that remark is now a statement with error control.

Status. Proved (tube and slaving, periodic Riccati block-triangularisation, second-order averaging, Poincaré map); checked by Newton shooting for the exact orbit, its monodromy matrix and its period mean against the averaged Hessian and the first-order prediction, at $k_e = 10$–160 (`tests/test_periodic.py`, `exp_periodic_orbit.py`). Novelty (literature audit): the first step of §8 branch 1; the tools (two-time-scale averaging under attraction) are classical.

Proof. Write $\varepsilon := 1/k_e$. All constants below are uniform in $\varepsilon \le 1/k_0$ and in Θ in the ball $\mathcal{K} := \{\|\Theta - \Theta^*\| \le 2\rho_0\}$, because the saturated features and their derivatives are bounded uniformly in Θ and $f \in C^2$ on the compact tube $\mathcal{T}_R$.

Step 1 (the tube and the slaved error). Let $M_r := \sup\{\|r_\Theta(x)\| : x \in \mathcal{T}_R,\ \Theta \in \mathcal{K}\}$ and $L_r := \sup\|\partial_x r_\Theta\|$ on the same set. For $k_e > M_r/R$ the tube $\{\|e\| \le R\}$ is forward invariant: on $\|e\| = R$, $\frac{\mathrm{d}}{\mathrm{d}t}\frac{1}{2}\|e\|^2 = e^\top r_\Theta(x) - k_e\|e\|^2 < 0$. Put $w := e - \varepsilon r_\Theta(x)$ with x the actual state. Since $r_\Theta(x) - k_e e = -k_e w$ and $\dot{x} = \dot{x}_m - k_e w$,

$$\dot{w} = -\big(k_e I - \partial_x r_\Theta(x)\big)w - \varepsilon\,\partial_x r_\Theta(x)\dot{x}_m - \varepsilon\,\partial_\Theta r_\Theta(x)\cdot\dot{\Theta},$$

and $\dot{\Theta} = F(e,\Theta,t) := \big(G\nabla_\vartheta[e^\top\tilde{\Phi}(x;\vartheta_i)] - \varepsilon\sigma_0(\vartheta_i - \vartheta^0)\big)_i$ is $O(\|e\| + \varepsilon) = O(\|w\| + \varepsilon)$. So for $k_e \ge 2L_r + C$, $\frac{\mathrm{d}}{\mathrm{d}t}\|w\| \le -(k_e/2)\|w\| + C_1\varepsilon$ on $\{\|w\| \le 1\} \cap \{\|e\| \le R\}$, whence $\|w(t)\| \le e^{-k_e t/2}\|w(0)\| + 2C_1\varepsilon^2$. The set $W_\varepsilon := \{\|w\| \le 3C_1\varepsilon^2\}$ is forward invariant and absorbs every solution of the tube within $t_1 = O(\varepsilon\log(1/\varepsilon))$, during which Θ moves by $O(\varepsilon\|e(0)\| + \varepsilon^2\log(1/\varepsilon))$ ($\dot{\Theta} = O(\|e\| + \varepsilon)$ and $\|e\|$ decays at rate $k_e/2$). On W_ε, $e = \varepsilon r_\Theta(x_m(t)) + \varepsilon[r_\Theta(x_m + e) - r_\Theta(x_m)] + w = \varepsilon r_\Theta(x_m(t)) + O(\varepsilon^2)$.

Step 2 (the slow equation and the fast variable). Let $e_\Theta(t)$ be the frozen-Θ periodic error of Proposition H.7(a): the unique T_0-periodic solution of $\dot{e} = r_\Theta(x_m(t) + e) - k_e e$, exponentially stable with rate at least $k_e/2$, with $e_\Theta = \varepsilon r_\Theta(x_m) + \varepsilon^2(\partial_x r_\Theta\, r_\Theta - \dot{r}_\Theta)(x_m) + O(\varepsilon^3)$ in sup norm (Taylor expansion of r_Θ in x to second order; $f \in C^2$). Because Θ enters $r_\Theta = f - \hat{f}_\Theta$ only through the smooth $\hat{f}_\Theta$, $e_\Theta(t)$ is C^2 in Θ as soon as $f \in C^1$, and differentiating the frozen equation (gain $1/k_e$) gives $\partial_\Theta e_\Theta = \varepsilon\,\partial_\Theta r_\Theta(x_m) + O(\varepsilon^2)$ and $\partial_\Theta^2 e_\Theta = O(\varepsilon)$, uniformly on $\mathcal{K}$. Let $\zeta := e - e_{\Theta(t)}(t)$ be the fast variable. Then

$$\dot{\zeta} = \big[r_\Theta(x_m + e) - r_\Theta(x_m + e_\Theta)\big] - k_e\zeta - \partial_\Theta e_\Theta(t)\cdot\dot{\Theta},$$

where the bracket is Lipschitz in ζ with constant L_r and the last term is $O(\varepsilon)\cdot O(\varepsilon) = O(\varepsilon^2)$ ($\dot{\Theta} = F = O(\|e\| + \varepsilon)$). So for $k_e \ge 2L_r + C$ the set $Z_\varepsilon := \{\|\zeta\| \le C_2\varepsilon^3\}$ is forward invariant and absorbs every solution of the tube within $O(\varepsilon\log(1/\varepsilon))$; on W_ε of Step 1 one has $\|\zeta\| = O(\varepsilon^2)$ at once and $O(\varepsilon^3)$ after this further transient. On Z_ε, since F is C^1 in e with $\partial_e F = O(1)$,

$$\dot{\Theta} = F(e_\Theta(t),\Theta,t) + R(\Theta,\zeta,t) = \varepsilon H(\Theta,t) + \varepsilon^2 H_2(\Theta,t) + O(\varepsilon^3), \qquad \|R\| \le C\|\zeta\| = O(\varepsilon^3), \tag{S}$$

where $\varepsilon H + \varepsilon^2 H_2$ is the expansion of the frozen-Θ drift along the frozen periodic error, and by Proposition H.7(b), $\langle H_2\rangle(\Theta)_i = G\nabla_\vartheta(U_{\mathrm{even}} + U_{\mathrm{odd}})[\mu_\Theta](\vartheta_i)$, that is, $\varepsilon^2\langle H_2\rangle(\Theta^*) = b_{\mathrm{frozen}}$. H and H_2 are T_0-periodic and C^2 in Θ on $\mathcal{K}$, with bounds uniform in ε.

Regularity of the remainders (used in Steps 4 and 6). The second-order terms of (S), and of (N) below, are used in sup norm only, for the location (d). The implicit-function and contraction arguments of Steps 4 and 6 need first-order C^1 closeness, which holds: $F(e_\Theta(t),\Theta,t)$ is C^2 in Θ with $\partial_\Theta[F(e_\Theta(t),\Theta,t)] = \varepsilon\,\partial_\Theta H(\Theta,t) + O(\varepsilon^2)$ (from $\partial_\Theta e_\Theta = \varepsilon\,\partial_\Theta r_\Theta(x_m) + O(\varepsilon^2)$ and the smooth dependence of F on (e,Θ) through $\tilde{\Phi}$); the remainder R is Lipschitz in ζ with constant $\sup\|\partial_e F\| = O(1)$ and Lipschitz in Θ with constant $O(\|\zeta\|) = O(\varepsilon^3)$; and the forcing $-\partial_\Theta e_\Theta\cdot\dot{\Theta}$ of the ζ-equation is Lipschitz in Θ with constant $O(\varepsilon^2)$ (from $\partial_\Theta^2 e_\Theta = O(\varepsilon)$ and $\partial_\Theta F = O(\varepsilon)$). None of this uses more than $f \in C^2$.

Step 3 (second-order averaging normal form). Let u be the zero-mean antiderivative of $H - \langle H\rangle$ and $\Theta = Y + \varepsilon u(Y,t)$. Then $\dot{Y} = \varepsilon\langle H\rangle(Y) + \varepsilon^2\big[H_2(Y,t) + D_\Theta H(Y,t)u(Y,t) - D_Y u(Y,t)\langle H\rangle(Y)\big] + O(\varepsilon^3)$. Let u_2 be the zero-mean antiderivative of the fluctuating part of the bracket and $Y = Y_2 + \varepsilon^2 u_2(Y_2,t)$. Since $\langle D_Y u\rangle = D_Y\langle u\rangle = 0$,

$$\dot{Y}_2 = \varepsilon\langle H\rangle(Y_2) + \varepsilon^2\Phi_2(Y_2) + O(\varepsilon^3), \qquad \Phi_2 := \langle H_2\rangle + \langle D_\Theta H\cdot u\rangle, \tag{N}$$

the second-order averaged equation of Remark H.7a (Sanders–Verhulst–Murdock [78, §2.9]), now with a remainder that is uniformly $O(\varepsilon^3)$ on $Z_\varepsilon \times \mathcal{K}$ because (S) holds there. The transformation

$\Theta \mapsto Y_2$ is a T_0-periodic near-identity diffeomorphism from $\mathcal{K}$ onto its image, $O(\varepsilon)$-close to the identity with $O(\varepsilon)$ derivative, and the period mean of Θ along any curve $Y_2(t) = Y_2^0 + O(\varepsilon^2)$ equals the period mean of Y_2 up to $O(\varepsilon^3)$, because u and u_2 have zero mean.

Step 4 (the Poincaré map in the slow coordinates). Let $P = (P_Y, P_\zeta)$ be the time-T_0 map of (CL) from $t = 0$, written in the coordinates (Y_2, ζ) on $\mathcal{K} \times Z_\varepsilon$. By (N) and Step 2,

$$P_Y(Y_2, \zeta) = Y_2 + \varepsilon T_0 \tilde{A}(Y_2) + O(\varepsilon^3), \qquad \tilde{A} := \langle H \rangle + \varepsilon \big[\Phi_2 + (T_0/2) D\langle H \rangle \cdot \langle H \rangle\big],$$
$$\|P_\zeta(Y_2, \zeta)\| \le e^{-k_e T_0/2} \|\zeta\| + O(\varepsilon^3),$$

the $\varepsilon^2 (T_0^2/2) D\langle H \rangle \langle H \rangle$ term being the second-order term of the time-T_0 flow of (N). The Lipschitz constants of the coupling are $\mathrm{Lip}_\zeta P_Y = O(\varepsilon)$ (a perturbation of ζ acts on $\dot\Theta$ through $\partial_e F = O(1)$ while it decays at rate $k_e/2$, so its integrated effect over a period is $O(\varepsilon)\|\zeta\|$) and $\mathrm{Lip}_Y P_\zeta = O(\varepsilon^3)$ (the forcing of the ζ-equation is $O(\varepsilon^2)$-Lipschitz in Θ and is integrated against the fast decay). Hence $\zeta \mapsto P_\zeta(Y_2, \zeta)$ has a unique fixed point $\zeta^*(Y_2) = O(\varepsilon^3)$, Lipschitz in Y_2 with constant $O(\varepsilon^3)$, and the fixed points of P are the zeros of

$$\tilde{A}_\varepsilon(Y) := \frac{P_Y(Y, \zeta^*(Y)) - Y}{\varepsilon T_0} = \tilde{A}(Y) + O(\varepsilon^2).$$

By the regularity paragraph, $D_Y P_Y = I + \varepsilon T_0 D\langle H \rangle(Y) + O(\varepsilon^2)$ (the variational equation of the slow flow over one period, whose coefficient is $\varepsilon\, \partial_\Theta H + O(\varepsilon^2)$), and the ζ^*-dependence contributes $\mathrm{Lip}_\zeta P_Y \cdot \mathrm{Lip}_Y \zeta^* = O(\varepsilon^4)$, so $\tilde{A}_\varepsilon$ is Lipschitz with $\mathrm{Lip}(\tilde{A}_\varepsilon - \langle H \rangle) = O(\varepsilon)$ on $\mathcal{K}$. Since $D\langle H \rangle(\Theta^*) = -G_m \nabla^2 J_m(\Theta^*)$ is invertible, $Y \mapsto Y - D\langle H \rangle(\Theta^*)^{-1} \tilde{A}_\varepsilon(Y)$ maps a ball of ε-independent radius ρ_0 around Θ^* into itself and is a contraction there for ε small (the Lipschitz form of the implicit function theorem), which gives a unique zero $Y_2^* = \Theta^* + O(\varepsilon)$, hence a unique fixed point of P with Θ-component within $O(\varepsilon)$ of Θ^*, that is, a unique T_0-periodic solution in the stated set; the bound on $x_p - x_m$ is Step 1. Expanding $\tilde{A}_\varepsilon(Y_2^*) = 0$ to first order, $D\langle H \rangle(\Theta^*)(Y_2^* - \Theta^*) + \varepsilon \Phi_2(\Theta^*) = O(\varepsilon^2)$ (the term $(T_0/2) D\langle H \rangle \langle H \rangle(Y_2^*)$ is $O(\varepsilon)$ because $\langle H \rangle(Y_2^*) = O(\varepsilon)$), so

$$Y_2^* = \Theta^* - \varepsilon D\langle H \rangle(\Theta^*)^{-1} \Phi_2(\Theta^*) + O(\varepsilon^2) = \Theta^* - J^{-1}(b_{\mathrm{frozen}} + b_{\mathrm{osc}}) + O(\varepsilon^2),$$

using $J = \varepsilon D\langle H \rangle(\Theta^*)$ and $\varepsilon^2 \Phi_2(\Theta^*) = b_{\mathrm{frozen}} + b_{\mathrm{osc}}$. Along the periodic orbit $Y_2(t) = Y_2^* + O(\varepsilon^2)$ (since $\langle H \rangle(Y_2^*) = O(\varepsilon)$), so by Step 3 the period mean of Θ_p is $Y_2^* + O(\varepsilon^2)$: this is (d), and $\Theta_p(t) - \bar{\Theta}_p = \varepsilon u(\Theta^*, t) + O(\varepsilon^2)$ is (a).

Step 5 (the monodromy matrix). Linearise (CL) along the periodic orbit: with δe and $\delta\Theta$,

$$\delta\dot{e} = (\partial_x r - k_e I)\delta e + \partial_\Theta r \cdot \delta\Theta, \qquad \delta\dot\Theta = \partial_e F \cdot \delta e + \partial_\Theta F \cdot \delta\Theta,$$

all coefficients T_0-periodic and evaluated along the orbit; $\partial_x r$, $\partial_\Theta r$, $\partial_e F$ are $O(1)$ and $\partial_\Theta F = O(\varepsilon)$ (F is linear in e with coefficients smooth in Θ, plus the leak $-\varepsilon\sigma_0(\vartheta_i - \vartheta^0)$, so $\partial_\Theta F = O(\|e\| + \varepsilon)$, and $e = O(\varepsilon)$ on the orbit). Let $S(t)$ be a T_0-periodic solution of the Riccati equation

$$\dot{S} = (\partial_x r - k_e I)S + \partial_\Theta r - S\, \partial_e F\, S - S\, \partial_\Theta F.$$

For k_e large it exists, is unique with $\|S\| \le C/k_e$, and depends smoothly on the data (contraction of the periodic solution operator of the dominant part $\dot{S} = -k_e S + g$; cf. Proposition H.7(a)). In the coordinates $\eta := \delta e - S\delta\Theta$ the linearisation is block triangular:

$$\dot\eta = \big[(\partial_x r - k_e I) - S\, \partial_e F\big]\eta, \qquad \delta\dot\Theta = \partial_e F \cdot \eta + \big[\partial_e F\, S + \partial_\Theta F\big]\delta\Theta,$$

so the monodromy matrix is similar to a block-triangular matrix whose diagonal blocks are the monodromies of the two diagonal equations. The η-block has norm at most $e^{-(k_e - L)T_0}$ with $L := L_r + \sup\|S\, \partial_e F\| = O(1)$ (Grönwall): these are the n error-channel multipliers.

The $\delta\Theta$-block is the linearisation along the orbit of the slow equation (S) in Θ-coordinates, $\varepsilon D_\Theta H(\Theta_p(t),t)+O(\varepsilon^2) = \varepsilon D_\Theta H(\Theta^*,t)+O(\varepsilon^2)$, whose monodromy is $I+\varepsilon T_0 D\langle H\rangle(\Theta^*)+O(\varepsilon^2) = I-\varepsilon T_0 G_m\nabla^2 J_m(\Theta^*)+O(\varepsilon^2)$. Its eigenvalues are $1-\varepsilon T_0\nu_j+O(\varepsilon^2) = \exp(-\varepsilon T_0\nu_j+O(\varepsilon^2))$, those of $G_m\nabla^2 J_m(\Theta^*)$ being the ν_j by similarity with $G_m^{1/2}\nabla^2 J_m(\Theta^*)G_m^{1/2}$; the $O(\varepsilon^2)$ perturbation moves them by $O(\varepsilon^2)$ because $I-\varepsilon T_0 G_m\nabla^2 J_m(\Theta^*)$ is diagonalisable (similar to a symmetric matrix through $G_m^{1/2}$), so the Bauer–Fike theorem applies with a condition number independent of ε — this covers the repeated eigenvalues of a collapsed configuration. This is (b).

Step 6 (the basin). Let $\nabla^2 J_m(\Theta^*) \succ 0$. In the norm $\|v\|_{G^{-1}} := \|G_m^{-1/2}v\|$, the matrix $I-\varepsilon T_0 G_m\nabla^2 J_m(\Theta^*)$ is similar to a symmetric contraction with norm exactly $1-\varepsilon T_0\lambda_*$ once $\varepsilon T_0\nu_{mp}<1$. By Step 4, $D_Y P_Y = I-\varepsilon T_0 G_m\nabla^2 J_m(Y)+O(\varepsilon^2)$ on $\mathcal{K}$, and $\nabla^2 J_m$ is Lipschitz, so on the ball $\{\|Y_2-Y_2^*\|_{G^{-1}}\le\rho_0\}$ the Lipschitz constant of P_Y in Y_2 is at most $1-\varepsilon T_0\lambda_*(1-\kappa)$ with $\kappa = O(\rho_0+\varepsilon)$; the coupling constants are $\mathrm{Lip}_\zeta P_Y = O(\varepsilon)$ and $\mathrm{Lip}_Y P_\zeta = O(\varepsilon^3)$, and P_ζ contracts ζ by $e^{-k_e T_0/2}$ (Step 4). The 2×2 matrix of Lipschitz constants

$$\begin{pmatrix} 1-\varepsilon T_0\lambda_*(1-\kappa) & C\varepsilon \\ C\varepsilon^3 & e^{-k_e T_0/2}\end{pmatrix}$$

has spectral radius $1-\varepsilon T_0\lambda_*(1-\kappa')$ with $\kappa' = O(\rho_0+\varepsilon)$, so on $\{\|Y_2-Y_2^*\|_{G^{-1}}\le\rho_0\}\times Z_\varepsilon$, with ρ_0 small but independent of ε, P is a contraction in an adapted product norm with that factor: its iterates converge to the fixed point geometrically, uniformly in ε. Solutions started in the tube with $\|e(0)-e_p(0)\|\le 1$ enter Z_ε within $O(\varepsilon\log(1/\varepsilon))$, during which Θ moves by $O(\varepsilon)\|e(0)-e_p(0)\|$ (Step 1), which is absorbed by halving ρ_0. Between the sampling times the solutions stay within a factor C of each other (Grönwall on one period, with the fast variable absorbing $\|e(0)-e_p(0)\|$ into an $O(\varepsilon)\|e(0)-e_p(0)\|$ displacement of Θ, Step 1). Transporting back through the near-identity transformations gives (c). □

Numerics (`exp_periodic_orbit.py` → `results/periodic_orbit/table.md`; `tests/test_periodic.py`). Two instances, $k_e = 10$, 20, 40, 80, 160. For each k_e the periodic orbit of (CL) is found by Newton shooting on the Poincaré map (DOP853 at relative tolerance 10^{-11}; residual below 10^{-11} in every case), its monodromy matrix is the finite-difference Jacobian of the map, and its period mean is compared with Θ^* and with the first-order prediction $\Theta^* - J^{-1}(b_{\mathrm{frozen}}+b_{\mathrm{osc}})$ of (d). Distances and cosines are in the labelled G^{-1} metric.

- **Instance A** ($n=2$, $m=3$, the plant and elliptic reference of Remark H.7a, $\sigma_0=0.5$, $G=\mathrm{diag}(4,4,1,1,1)$). The minimiser of J_m is a collapsed configuration (kernel spread $4\cdot 10^{-14}$) with $\lambda_{\min}(\nabla^2 J_m) = 0.0148$: a non-degenerate minimum with permutation symmetry.

k_e	$\|\bar\Theta_p-\Theta^*\|$	full prediction: size ratio, cosine, vector error	frozen-measure part only: vector error	parameter Floquet exponents against $(1/k_e)\nu_j$: max, median relative error
10	$4.09\cdot 10^{-2}$	1.149, 0.9972, 0.169	0.398	$8.3\cdot 10^{-2}$, $1.2\cdot 10^{-2}$
20	$1.91\cdot 10^{-2}$	1.074, 0.9993, 0.083	0.360	$7.3\cdot 10^{-2}$, $7.3\cdot 10^{-3}$
40	$9.22\cdot 10^{-3}$	1.036, 0.9998, 0.041	0.348	$4.5\cdot 10^{-2}$, $4.0\cdot 10^{-3}$
80	$4.53\cdot 10^{-3}$	1.018, 1.0000, 0.020	0.343	$2.5\cdot 10^{-2}$, $2.1\cdot 10^{-3}$
160	$2.24\cdot 10^{-3}$	1.009, 1.0000, 0.010	0.342	$1.3\cdot 10^{-2}$, $1.1\cdot 10^{-3}$

The offset scales like $k_e^{-1.05}$ (fitted exponent; (a) predicts -1), the vector error of the full prediction halves with every doubling of k_e (the $O(k_e^{-2})$ remainder of (d) against an $O(1/k_e)$ offset), and the frozen-measure part alone has a 35% vector error at every k_e: the oscillation

term b_{osc} of Remark H.7a is an $O(1)$ fraction of the first-order location. The median relative error of the Floquet exponents is $O(1/k_e)$, as (b) states; the largest relative error sits on the $(m-1)$-fold degenerate separation pair with the smallest exponent (0.0018 at $k_e = 20$), where an $O(k_e^{-2})$ absolute error is a large fraction of a small exponent. The two error-channel multipliers are below the resolution of the finite-difference monodromy (exponents of order k_e); their product is recovered from Liouville's identity $\sum \log|\lambda| = \int_0^{T_0} \operatorname{tr} \partial F/\partial z \, \mathrm{d}t$ along the orbit, which gives a mean error-channel exponent of $k_e - 0.16$, $k_e - 0.11$, $k_e - 0.086$, $k_e - 0.074$, $k_e - 0.068$ at $k_e = 10 \ldots 160$ (instance B: $k_e + 0.05 \ldots k_e + 0.09$): the multipliers are $e^{-(k_e - L)T_0}$ with $|L| < 0.2$, as (b) states.

- **Instance B** ($n = 1$, $m = 2$, a two-bump plant $f(x) = 1.6e^{-(x-1)^2/0.4} - 1.2e^{-(x+1)^2/0.4} + 0.1x$ on $x_m = 2\sin t$, $\sigma_0 = 0.05$, $G = \operatorname{diag}(2, 1, 1)$). The minimiser has two distinct kernels of opposite sign (spread 2.1, $\lambda_{\min}(\nabla^2 J_m) = 0.066$). Here $b_{\mathrm{osc}} = 0$ exactly: for a one-dimensional sinusoidal sweep the reflection $t \mapsto \pi - t$ maps x_m to itself and u to $-u$, so $\langle D_\Theta H \cdot u \rangle = 0$, and the full and frozen predictions coincide. The reference is fast on the kernel scale ($\sup\|\dot{x}_m\|/s$ is $2/s$: between 1.3 and 10 over the admissible widths, about 2.6 at the minimiser's width; the residual bandwidth is of the same order), so $k_e = 10$–20 is outside the averaging regime:

k_e	$\|\bar{\Theta}_p - \Theta^*\|$	prediction: size ratio, cosine, vector error	Floquet: max, median relative error
10	$6.44 \cdot 10^{-2}$	2.818, 0.880, 1.995	$9.5 \cdot 10^{-2}$, $4.0 \cdot 10^{-2}$
20	$2.22 \cdot 10^{-2}$	1.942, 0.929, 1.078	$3.2 \cdot 10^{-2}$, $1.4 \cdot 10^{-2}$
40	$8.37 \cdot 10^{-3}$	1.464, 0.970, 0.551	$1.2 \cdot 10^{-2}$, $5.4 \cdot 10^{-3}$
80	$3.51 \cdot 10^{-3}$	1.227, 0.990, 0.277	$4.6 \cdot 10^{-3}$, $2.2 \cdot 10^{-3}$
160	$1.59 \cdot 10^{-3}$	1.111, 0.9969, 0.139	$2.0 \cdot 10^{-3}$, $9.5 \cdot 10^{-4}$

  The vector error and the Floquet error halve with each doubling of k_e, as the theorem's $O(k_e^{-2})$ remainders predict, but the constants are large: the asymptotic regime starts around $k_e \approx 80$ here. This instance shows what the theorem does not claim: the $O(1/k_e)$ location is sharp only once k_e exceeds the effective bandwidth ω_{eff} of Proposition G's (H2a′) (Section 19).

Checked by: `tests/test_periodic.py`: ∇J_m against finite differences (10^{-7}) and the non-degenerate two-kernel minimum of instance B; at $k_e = 20$ and 40 the shooting residual is below 10^{-10}, $\|\bar{\Theta}_p - \Theta^*\|$ and $\max\|e\|$ fall by more than 35% from 20 to 40, the vector error of the first-order location falls by more than 40% and is below 0.6 at $k_e = 40$, and the maximal relative Floquet error falls by more than 40% and is below 2% at $k_e = 40$.

Remark J.1 (what Theorem J says about the program of Section 26)**.**

- It is the deterministic, finite-m first step of branch 1: near every hyperbolic critical point of the m-particle regularised loss $k_e \cdot \mathcal{G}$ at $\tau = 0$ the real closed loop has exactly one stationary state (a periodic orbit), an $O(1/k_e)$-perturbation of it with the same stability type, and the convergence rate is the one of the averaged gradient flow in slow time. (The theorem does not say that every periodic orbit of (CL) arises this way.) The two missing pieces of Conjecture H.6 remain: dither ($\tau > 0$, where the stationary state is a measure) and uniformity in m.
- It makes the stationary (period-mean) prediction of Remark H.7a a theorem for this system: the first-order location of the orbit, hence the orientation difference of the period-mean kernels, is $\Theta^* - J^{-1}(b_{\mathrm{frozen}} + b_{\mathrm{osc}})$ with an $O(k_e^{-2})$ error. When the leading offset is itself small, the $O(k_e^{-2})$ remainder can dominate its *direction* at moderate k_e: a collapsed saddle instance examined in the round-5 review (`review/REVIEW_round5_J.md`) has a first-order

location nearly orthogonal to the true offset at $k_e = 20$ and aligned with it at $k_e = 80$, while the sizes obey the theorem's orders throughout.

- The leak is not needed for the structure, only for the hyperbolicity of Θ^*. At a collapsed critical point the separation Hessian is $\sigma_0 G^{-1} - R_2$ (Proposition K, Section 23); for $n = 1$ and $\sigma_0 = 0$ the amplitude row of R_2 vanishes at a leak-free critical point ($\langle r\varphi\rangle = \partial_c\langle r\varphi\rangle = \partial_s\langle r\varphi\rangle = 0$ there), so the $m-1$ amplitude-redistribution directions are exactly flat and the theorem does not apply, while for $n \geq 2$ only the amplitude block of R_2 vanishes and a collapsed critical point is generically non-degenerate even without a leak. Where $\nabla^2 J_m$ is singular the theorem does not apply.
- At a collapsed critical point (all ϑ_i^* equal) the permutation symmetry makes the separation modes $(m-1)$-fold degenerate; the Floquet exponents come in such multiplets (`results/periodic_orbit/table.md`); Proposition K (Section 23) works out this case.

23 Proposition K: collapsed configurations, the critical leak, and the orientation of the orbit as a design input

Branch 2 of Section 26 asks whether the circulation of the reference orbit can be designed. Theorem J (Section 22) makes the question precise for the collapsed configuration, the trap that Theorem H's dither was introduced to escape: its stability is decided by the eigenvalues of a Hessian that the orientation shifts at order k_e^{-2}.

Setting. As in Theorem J. A configuration is *collapsed* if $\vartheta_i = \bar\vartheta$ for all i. Then $\hat f_\Theta = \tilde\Phi(\cdot\,;\bar\vartheta)$, so Θ is a critical point of J_m if and only if $\bar\vartheta$ is a critical point of the single-kernel objective $J_1(\vartheta) := \frac{1}{2}\int \|f - \tilde\Phi(\cdot\,;\vartheta)\|^2\,\mathrm{d}\rho + \frac{\sigma_0}{2}\|\vartheta - \vartheta^0\|^2_{G^{-1}}$. Write $\bar r := f - \tilde\Phi(\cdot\,;\bar\vartheta)$ and

$$M_2 := \int \nabla_\vartheta \tilde\Phi(\cdot\,;\bar\vartheta)^\top \nabla_\vartheta \tilde\Phi(\cdot\,;\bar\vartheta)\,\mathrm{d}\rho \succeq 0,$$

$$R_2 := \int \sum_k \bar r_k \nabla^2_\vartheta \tilde\Phi_k(\cdot\,;\bar\vartheta)\,\mathrm{d}\rho = \nabla^2_\vartheta \big\langle \bar r^\top \tilde\Phi(\cdot\,;\vartheta)\big\rangle_\rho\Big|_{\bar\vartheta} \quad \text{(residual frozen)}.$$

R_2 is the Hessian of $-V_L[\bar\mu](\vartheta)$ at the atom: the second-order cost of moving a small amount of mass off the atom at frozen residual.

Proposition K (collapsed configurations, the critical leak, and the orientation as a design input)**.** *Let Θ^* be a collapsed critical point of J_m with atom $\bar\vartheta$.*

(i) Symmetry splitting. $\nabla^2 J_m(\Theta^*) = I_m \otimes (\sigma_0 G^{-1} - R_2) + \frac{1}{m}(\mathbf{1}\mathbf{1}^\top) \otimes M_2$. *On the separation subspace $\{(u_1,\dots,u_m) : \sum_i u_i = 0\}$, of dimension $(m-1)p$, it acts as $\sigma_0 G^{-1} - R_2$; on the collective subspace $\{u_i$ all equal$\}$ as $\nabla^2 J_1(\bar\vartheta) = \sigma_0 G^{-1} - R_2 + M_2$. The same splitting holds for the Jacobian of any permutation-equivariant particle drift at a collapsed configuration, in particular for the second-order averaged drift of Remark H.7a (Section 21): it has the form $I_m \otimes D + \frac{1}{m}(\mathbf{1}\mathbf{1}^\top) \otimes E$.*

(ii) Separation stability at leading order. *Θ^* is a non-degenerate local minimum of J_m iff $\nabla^2 J_1(\bar\vartheta) \succ 0$ and $\sigma_0 G^{-1} \succ R_2$, that is, iff $\bar\vartheta$ is a non-degenerate minimiser of J_1 and $\sigma_0 > \sigma_{\mathrm{crit}}(\bar\vartheta) := \lambda_{\max}(G^{1/2} R_2 G^{1/2})$. The separation Floquet exponents of the collapsed periodic orbit of* (CL) *are then, by Theorem J(b), $(1/k_e)\lambda_j(\sigma_0 I - G^{1/2} R_2 G^{1/2}) + O(k_e^{-2})$, each with multiplicity $m-1$, and they do not depend on m. The condition $\sigma_0 G^{-1} \succ R_2$ is the statement that the atom is a strict local minimiser of the first-variation potential $V_L[\bar\mu] + \frac{\sigma_0}{2}\|\cdot\|^2_{G^{-1}}$ on its support, the local form of the Chizat–Bach optimality condition for measures.*

(iii) The orientation-odd shift. *Let $\mathcal{O}$ be traversed in either direction (the same ρ, hence the same Θ^* and the same leading exponents). Write the second-order normal form of Theorem J's proof as*

$$\dot{\Theta} = -\frac{1}{k_e} G_m \nabla J_m(\Theta) + b(\Theta) + O(k_e^{-3}), \qquad b := b_{\text{frozen}} + b_{\text{osc}} = O(k_e^{-2}), \tag{N}$$

with $b_{\text{frozen}} = \frac{1}{k_e^2} G_m \nabla (U_{\text{even}} + U_{\text{odd}})$ (Proposition H.7) and b_{osc} of Remark H.7a. (N) *has a unique equilibrium $\bar{\Theta}_2$ near Θ^*, which is collapsed, with $\bar{\Theta}_2 - \Theta^* = -J_1^{-1} b(\Theta^*) + O(k_e^{-2})$ for $J_1 := -\frac{1}{k_e} G_m \nabla^2 J_m(\Theta^*)$ (the first-order location of Theorem J(d)); let J_2 be the Jacobian of* (N) *at $\bar{\Theta}_2$. The parameter Floquet exponents of the collapsed orbit are the eigenvalues of $-J_2$ up to $O(k_e^{-3})$, and*

$$J_2 = J_1 + \partial b(\Theta^*) + \partial J_1(\Theta^*)[\bar{\Theta}_2 - \Theta^*] + O(k_e^{-3}),$$

where the third term — the third derivative of J_m contracted with the first-order offset — is $O(k_e^{-2})$, of the same order as ∂b, and must be kept: the Jacobian of (N) *at Θ^* is not the right linearisation. Under orbit reversal ρ, Θ^*, J_1 and U_{even} are unchanged while U_{odd} and b_{osc} change sign (Proposition H.7(d), Remark H.7a), so with $b_{\text{even}} := \frac{1}{k_e^2} G_m \nabla U_{\text{even}}$ and $b_{\text{odd}} := \frac{1}{k_e^2} G_m \nabla U_{\text{odd}} + b_{\text{osc}}$,*

$$J_2 = J_1 + E \pm O,$$
$$E := \partial b_{\text{even}}(\Theta^*) - \partial J_1(\Theta^*)\big[J_1^{-1} b_{\text{even}}(\Theta^*)\big], \qquad O := \partial b_{\text{odd}}(\Theta^*) - \partial J_1(\Theta^*)\big[J_1^{-1} b_{\text{odd}}(\Theta^*)\big],$$

for the two orientations. J_1, E and O are permutation-equivariant and preserve the separation subspace; by (i) *their restrictions to one separation block are $p \times p$ matrices, written $J_1|_{\text{sep}} = -\frac{1}{k_e}(\sigma_0 I - G R_2)$, D_{even}, D_{odd}. The separation Floquet exponents of the two orientations differ by*

$$\lambda_j^{(+)} - \lambda_j^{(-)} = -2\,\frac{w_j^\top D_{\text{odd}} v_j}{w_j^\top v_j} + O(k_e^{-3})$$

(simple eigenvalue λ_j of $-J_1|_{\text{sep}}$, right and left eigenvectors v_j, w_j), an $O(k_e^{-2})$ quantity that scales like k_e^{-2} at fixed (G, σ_0), while the exponents themselves are $O(1/k_e)$. The difference is a property of the excitation's orientation alone: the occupation measure, the averaged system $(\mathfrak{A})$ and all of Theorems B–H are blind to it.

(iv) The critical leak as a design point. *At $\sigma_0 = \sigma_{\text{crit}}(\bar{\vartheta})$ the softest separation exponent vanishes at leading order and equals the k_e^{-2} value $-(w^\top D_{\text{even}} v)/(w^\top v) \mp (w^\top D_{\text{odd}} v)/(w^\top v) + O(k_e^{-3})$. If $|w^\top D_{\text{odd}} v| > |w^\top D_{\text{even}} v|$, the collapsed configuration is stable for one orientation and unstable for the other: the direction in which the reference is traversed decides whether the kernels separate. More generally, for σ_0 in a window of width $O(1/k_e)$ around σ_{crit} — the window where the leading exponent $(\sigma_0 - \sigma_{\text{crit}})(1 + o(1))/k_e$ lies between the two k_e^{-2} values — the orientation decides the sign, and the centre of the window, $\sigma_0 = \sigma_{\text{crit}} + k_e (w^\top D_{\text{even}} v)/(w^\top v) + O(k_e^{-1})$, can be computed from the averaged data alone.*

Status. (i) and (ii) are proved (linear algebra at the collapsed point and Theorem J(b)). (iii) and (iv) are proved given the second-order normal form (N) with an $O(k_e^{-3})$ remainder, which is Theorem J's proof carried one order further (periodic averaging to second order, Sanders–Verhulst–Murdock [78, Ch. 2–3]; the first-order step is written out in the proof of Theorem J, the second-order step is Remark H.7a's derivation and is not written out here). The orientation-odd shift, its k_e^{-2} scaling, the prediction and the design window are checked numerically in one instance (`exp_orientation_stability.py`).

Proof. *(i)* $\partial^2 J_m/\partial\vartheta_i\partial\vartheta_j = \frac{1}{m}\int \nabla\tilde\Phi_i^\top\nabla\tilde\Phi_j\,\mathrm{d}\rho - \delta_{ij}\int\sum_k r_k\nabla^2\tilde\Phi_{i,k}\,\mathrm{d}\rho + \delta_{ij}\sigma_0 G^{-1}$ (the first term from the quadratic dependence of $\frac12\int\|r\|^2\,\mathrm{d}\rho$ on the m features with weights $1/m$, the second from the ϑ_i-curvature of the residual term: with $r = f - \frac{1}{m}\sum\tilde\Phi_j$, $\partial_{\vartheta_i}(m\bar L_m) = -\int r^\top\nabla\tilde\Phi_i\,\mathrm{d}\rho$ and $\partial_{\vartheta_j}\partial_{\vartheta_i}(m\bar L_m) = \frac{1}{m}\int\nabla\tilde\Phi_j^\top\nabla\tilde\Phi_i\,\mathrm{d}\rho - \delta_{ij}\int\sum_k r_k\nabla^2\tilde\Phi_{i,k}\,\mathrm{d}\rho$). At a collapsed point all blocks coincide, giving the Kronecker form; the two subspaces are the isotypic components of the permutation action, on which $\frac{1}{m}(\mathbf{1}\mathbf{1}^\top)$ acts as 0 and 1 respectively. The second-order averaged drift is permutation-equivariant (each particle moves by a function of its own position and of the measure), so its Jacobian at a collapsed point has the same form with D the self-derivative and E the derivative through the measure.

(ii) Positive definiteness of a block-diagonal form; the exponents are Theorem J(b) applied to the separation block, whose eigenvalues of $G_m^{1/2}\nabla^2 J_m G_m^{1/2}$ restricted to the separation subspace are those of $\sigma_0 I - G^{1/2}R_2G^{1/2}$, each repeated $m-1$ times. For the last sentence, the first variation of $\mathcal{G} := \bar L + \frac{\sigma_0}{2}\int\|\cdot-\vartheta^0\|^2_{G^{-1}}\,\mathrm{d}\mu$ at $\bar\mu = \delta_{\bar\vartheta}$ is $V_L[\bar\mu](\vartheta) + \frac{\sigma_0}{2}\|\vartheta-\vartheta^0\|^2_{G^{-1}} = -\langle\bar r^\top\tilde\Phi(\cdot\,;\vartheta)\rangle_\rho + \frac{\sigma_0}{2}\|\vartheta-\vartheta^0\|^2_{G^{-1}}$, whose Hessian at $\bar\vartheta$ is $-R_2+\sigma_0G^{-1}$.

(iii) The proof of Theorem J carries one order further: the near-identity change of variables of its Step 3 removes the oscillating terms through order k_e^{-2}, which turns (CL) into (N) with an $O(k_e^{-3})$ remainder (this is the normal form of Remark H.7a's derivation, with b_{osc} the term that the frozen-measure expansion misses). (N) is a regular perturbation of the averaged system near the hyperbolic point Θ^*, so it has a unique equilibrium $\bar\Theta_2$ within $O(1/k_e)$ of Θ^*, obtained by the implicit function theorem; by the first-order expansion, $\bar\Theta_2 - \Theta^* = -J_1^{-1}b(\Theta^*) + O(k_e^{-2})$ ($J_1^{-1} = O(k_e)$, $b = O(k_e^{-2})$). (N) is permutation-equivariant and Θ^* is collapsed, so the equivariant Newton iterates stay collapsed and $\bar\Theta_2$ is collapsed. Theorem J(b) applied to (N) (same proof, remainder $O(k_e^{-3})$) gives the Floquet exponents as the eigenvalues of $-J_2 + O(k_e^{-3})$. Expanding the Jacobian of (N) about Θ^*: $J_2 = J_1 + \partial b(\Theta^*) + \partial J_1(\Theta^*)[\bar\Theta_2-\Theta^*] + O(\|\bar\Theta_2-\Theta^*\|^2/k_e) + O(k_e^{-2}\|\bar\Theta_2-\Theta^*\|)$, and both error terms are $O(k_e^{-3})$; substituting the offset gives E and O, which are the even and odd parts under reversal because ∂J_1 is even and b_{even}, b_{odd} have the parities named. All three operators are permutation-equivariant, hence preserve the separation subspace, and by the Kronecker structure of (i) each acts on it as $I_{m-1}\otimes$ (a $p\times p$ block). First-order eigenvalue perturbation of the simple eigenvalue λ_j of $-J_1|_{\mathrm{sep}}$ gives

$$\lambda_j^{(\pm)} = \lambda_j - \frac{w_j^\top D_{\mathrm{even}}v_j}{w_j^\top v_j} \mp \frac{w_j^\top D_{\mathrm{odd}}v_j}{w_j^\top v_j} + O(k_e^{-3}),$$

and the difference as stated. (For the $(m-1)$-fold degenerate separation modes the degeneracy is the trivial one of the Kronecker structure, which the restriction to one p-dimensional block removes; a further degeneracy within the block would call for degenerate perturbation theory.)

(iv) is (iii) at a vanishing leading eigenvalue; the window follows because the leading exponent is $(\sigma_0-\sigma_{\mathrm{crit}})/k_e\cdot(1+o(1))$ near σ_{crit} ($\bar\vartheta$, R_2 and the eigenvectors depend smoothly on σ_0) while the shifts are $O(k_e^{-2})$; its centre is where the leading exponent cancels the even shift. □

Remark K.1 (reading)**.** Three things.

- The collapsed configuration's stability is a single-kernel question at leading order (the Hessian $\sigma_0G^{-1}-R_2$ does not depend on m), which explains why the $m=6$ and $m=32$ experiments of Section 20 collapsed alike: a leak above σ_{crit} traps every network size.
- Dither (Theorem H) and the orientation (this proposition) are two different ways to leave the trap: the first adds entropy, the second tilts the Hessian by $O(k_e^{-2})$. The second is free but small; it matters only near σ_{crit}, that is, when the leak is tuned to the edge of trapping, or for nearly flat separation modes such as the weak saddle of Remark H.2.
- The sign of the shift depends on the data $(f,\mathcal{O},\bar\vartheta)$ through D_{odd} and has no general sign: the design rule is to compute D_{odd} (from the averaged data: U_{odd}, b_{osc} and the third

derivative of J_m) and choose the orientation with the smaller softest separation exponent (or the negative one); the window of (iv) is where the choice matters.

Numerics (`exp_orientation_stability.py` → `results/orientation_stability/table.md`). Instance A of Theorem J's numerics (Section 22; $n = 2$, $m = 3$, the elliptic reference traversed in both directions, $G = \mathrm{diag}(4, 4, 1, 1, 1)$). For each orientation the collapsed periodic orbit is found by Newton shooting (residual $\le 4 \cdot 10^{-13}$), the parameter block of its monodromy matrix is restricted to the separation and collective subspaces, and the exponents are compared with the leading order $(\sigma_0 I - G^{1/2} R_2 G^{1/2})/k_e$ and with the second-order prediction, the eigenvalues of $-J_2$ at the equilibrium $\bar\Theta_2$ of (N) (Newton from Θ^*; every $\bar\Theta_2$ found is collapsed to 10^{-8}). The second-order equilibrium is $\|\bar\Theta_2 - \Theta^*\|_{G^{-1}} = 3.7 \cdot 10^{-2}$, $1.8 \cdot 10^{-2}$, $9.0 \cdot 10^{-3}$, $4.5 \cdot 10^{-3}$ away from Θ^* at $k_e = 10$, 20, 40, 80: the $O(1/k_e)$ offset of Theorem J(d).

- *The shift and its prediction* ($\sigma_0 = 0.5$, a stable collapse). The softest separation exponents of the two orientations differ by $9.0 \cdot 10^{-4}$, $1.9 \cdot 10^{-4}$, $4.1 \cdot 10^{-5}$, $9.7 \cdot 10^{-6}$ at $k_e = 10$, 20, 40, 80: fitted exponent -2.18 against k_e ((iii): -2). Scaled by k_e^2, the measured corrections to the leading order are -0.032, -0.057, -0.070, -0.077 for $(+)$ and -0.123, -0.132, -0.136, -0.139 for $(-)$, approaching the predicted -0.083 and -0.143 with a gap that halves at every doubling of k_e (the $O(k_e^{-3})$ remainder); the predicted difference at $\bar\Theta_2$ — $7.8 \cdot 10^{-4}$, $1.7 \cdot 10^{-4}$, $3.9 \cdot 10^{-5}$, $9.4 \cdot 10^{-6}$ — is within 14%, 9%, 5%, 3% of the measured one. The prediction with the Jacobian taken at Θ^* instead of $\bar\Theta_2$ (the third-derivative term omitted) reads $9.0 \cdot 10^{-4}$, $2.1 \cdot 10^{-4}$, $5.0 \cdot 10^{-5}$, $1.2 \cdot 10^{-5}$: it agrees at $k_e = 10$ by accident and its k_e^2-scaled value converges to 0.078 while the measured one converges to 0.060, the 30% error that the numerics exposed (`review/REVIEW_LOG.md`, round 5, K-1). The separation exponents come in exact pairs (multiplicity $m - 1 = 2$) and the collective exponents differ between the orientations by 2% at $k_e = 10$ and 0.2% at $k_e = 80$, as (i)–(iii) describe.
- *At the critical leak* ($\sigma_0 = \sigma_{\mathrm{crit}} = 0.4606$). The leading exponent vanishes and the k_e^{-2} terms decide: the softest separation exponent is $-4.1 \cdot 10^{-4}$ for $(+)$ and $-1.27 \cdot 10^{-3}$ for $(-)$ at $k_e = 10$, $-4.8 \cdot 10^{-5}$ and $-8.7 \cdot 10^{-5}$ at $k_e = 40$ — unstable for both orientations in this instance, the even part dominating the odd one by a factor 2–3; the predictions at $\bar\Theta_2$ are within 14% and 6% of the measured values at $k_e = 40$.
- *The orientation decides* (σ_0 designed from the averaged data). Solving $\mathrm{pred}_+(\sigma_0) + \mathrm{pred}_-(\sigma_0) = 0$ for the predicted softest exponents gives $\sigma_0 - \sigma_{\mathrm{crit}} = 0.0125$ at $k_e = 10$ and 0.0061 at $k_e = 20$, the centre of the window of (iv); there the measured softest separation exponents are $+8.5 \cdot 10^{-4}$ and $-2.5 \cdot 10^{-5}$ ($k_e = 10$), $+1.4 \cdot 10^{-4}$ and $-3.7 \cdot 10^{-5}$ ($k_e = 20$): the same reference orbit, traversed one way, keeps the collapsed kernels together, and traversed the other way makes them separate. The window is narrow — the leading exponent at these σ_0 is $1.2 \cdot 10^{-3}$ and $3.0 \cdot 10^{-4}$ — and the design lands inside it although the k_e^{-3} terms move both measured exponents by about $+4.7 \cdot 10^{-4}$ and $+6 \cdot 10^{-5}$ against the prediction.

Checked by: `tests/test_periodic.py`: the Hessian splitting (i) against finite differences (10^{-5} relative) at the collapsed minimiser of instance A, the isotypic eigenvalue structure, the gap equation of σ_{crit} (10^{-8}), and the two-fold zero of the separation block at σ_{crit} with a positive definite collective block (`test_proposition_K_hessian_splitting_and_critical_leak`); the experiment above for (iii)–(iv) (`exp_orientation_stability.py` → `results/orientation_stability/table.md`).

24 Theorem J′: the stationary law of the dithered finite-m closed loop

This section proves the second step of Section 26, branch 1 — the finite-m closed loop *with dither* — and answers the first "next step" of branch 2 (the orientation effect when the stationary state is a measure). It also opens a new angle: the stationary state of the dithered loop is a non-equilibrium steady state, with a stationary probability current of order $1/k_e$ that vanishes exactly when the first-order averaged drift is reciprocal (Theorem I), and the stationary parameter process is not reversible in time.

Setting. The closed loop of Theorem H's design at finite m, in the scaling $\sigma = \sigma_0/k_e$, $\tau = \tau_0/k_e$ (the formal limit of Remark H.0: $k_e \to \infty$ with G, $k_e\sigma$, $k_e\tau$ fixed), $\varepsilon := 1/k_e$:

$$\begin{aligned} \dot{x} &= f(x) + \dot{x}_m - k_e e - \hat{f}_\Theta(x), \qquad e := x - x_m(t), \\ \mathrm{d}\vartheta_i &= \big[G\nabla_\vartheta\big(e^\top \tilde{\Phi}(x;\vartheta_i)\big) - (\sigma_0/k_e)(\vartheta_i - \vartheta^0)\big]\,\mathrm{d}t + \sqrt{2\tau_0/k_e}\,G^{1/2}\,\mathrm{d}B_i, \qquad i = 1..m, \end{aligned} \tag{CL_τ}$$

with the notation of Section 22: saturated features, $\Theta \in \mathbb{R}^{mp}$, $G_m := I_m \otimes G$, $x_m \in C^\infty$ T_0-periodic with occupation measure ρ, $f \in C^\infty$ on a neighbourhood of the tube $\mathcal{T}_R$, $F_R := \sup_{\mathcal{T}_R}\|f\|$, and r_Θ, J_m, H, u, $\langle H\rangle = -G_m\nabla J_m$ as there. Assume $k_e \geq k_T := (F_R + K_\Phi)/R$, so that the ball $\{\|e\| \leq R\}$ is forward invariant for every parameter path ($\|\dot{e}\| \leq F_R + K_\Phi - k_e\|e\|$ on it): the plant never sees f outside the tube, and f may be modified outside a neighbourhood of $\mathcal{T}_R$ to be smooth with bounded derivatives on $\mathbb{R}^n$ without changing the law of the loop.

Slow time and the three scales. Put $s := \varepsilon t$, $\theta := t \bmod T_0 \in \mathbb{T} := \mathbb{R}/T_0\mathbb{Z}$, and $\eta := e/\varepsilon = k_e e$. Then $\mathrm{d}W_s := \sqrt{\varepsilon}\,\mathrm{d}B_t$ is a Brownian motion in slow time and (CL_τ) reads

$$\begin{aligned} \frac{\mathrm{d}\eta}{\mathrm{d}s} &= \varepsilon^{-2}\big[r_\Theta(x_m(\theta) + \varepsilon\eta) - \eta\big], \\ \mathrm{d}\vartheta_i &= \big[G\nabla_\vartheta\big(\eta^\top\tilde{\Phi}(x_m(\theta)+\varepsilon\eta;\vartheta_i)\big) - \sigma_0(\vartheta_i - \vartheta^0)\big]\,\mathrm{d}s + \sqrt{2\tau_0}\,G^{1/2}\,\mathrm{d}W_i, \qquad \frac{\mathrm{d}\theta}{\mathrm{d}s} = \varepsilon^{-1}: \end{aligned}$$

the error relaxes at rate ε^{-2} to the slaved value $\eta^*(\Theta,\theta) := r_\Theta(x_m(\theta))$, the phase turns at rate ε^{-1}, and the parameters move at rate one. The generator of the Markov process $Z := (\Theta,\eta,\theta)$ on $\mathcal{S} := \mathbb{R}^{mp} \times \{\|\eta\| \leq F_R + K_\Phi\} \times \mathbb{T}$ is

$$\begin{aligned} L_\varepsilon &= \varepsilon^{-2}A_\varepsilon + \varepsilon^{-1}\partial_\theta + B_\varepsilon, \qquad A_\varepsilon := \big(r_\Theta(x_m(\theta)+\varepsilon\eta) - \eta\big)\cdot\nabla_\eta, \\ B_\varepsilon &:= \sum_i \big[G\nabla_\vartheta\big(\eta^\top\tilde{\Phi}(x_m(\theta)+\varepsilon\eta;\vartheta_i)\big) - \sigma_0(\vartheta_i - \vartheta^0)\big]\cdot\nabla_{\vartheta_i} + \tau_0\sum_i \mathrm{tr}\big(G\nabla^2_{\vartheta_i}\big), \end{aligned}$$

with $A_\varepsilon = A_0 + \varepsilon A_1 + \ldots$, $A_0 = (\eta^* - \eta)\cdot\nabla_\eta$, $A_1 = (D_x r_\Theta(x_m)\eta)\cdot\nabla_\eta$, and $B_\varepsilon = B_0 + \varepsilon B_1 + \ldots$, where B_0 has the drift $\sum_i[G\nabla_\vartheta(\eta^\top\tilde{\Phi}(x_m;\vartheta_i)) - \sigma_0(\vartheta_i - \vartheta^0)]$ (equal to $H(\Theta,\theta)$ at $\eta = \eta^*$) and $B_1 = \sum_i G\nabla_\vartheta\big(\eta^\top D_x\tilde{\Phi}(x_m;\vartheta_i)\eta\big)\cdot\nabla_{\vartheta_i}$.

The averaged objects. $\bar{L}_0 := -G_m\nabla J_m\cdot\nabla + \tau_0\,\mathrm{tr}(G_m\nabla^2)$ is the Langevin generator in the G^{-1} metric with invariant measure the Gibbs measure $\pi := Z^{-1}\exp(-J_m/\tau_0)$ (Theorem H(ii) at finite m; π has Gaussian tails because $J_m \geq (\sigma_0/2)\sum\|\vartheta_i - \vartheta^0\|^2_{G^{-1}} - C$). The first-order drift is the vector field on $\mathbb{R}^{mp}$

$$\begin{aligned} b(\Theta) &:= b_{\mathrm{fr}}(\Theta) + b_{\mathrm{osc}}(\Theta), \qquad b_{\mathrm{fr},i} := G\nabla_\vartheta(U_{\mathrm{even}} + U_{\mathrm{odd}})[\mu_\Theta](\vartheta)\big|_{\vartheta=\vartheta_i}, \\ b_{\mathrm{osc}} &:= \big\langle (D_\Theta H)(\Theta,\theta)\cdot u(\Theta,\theta)\big\rangle_\theta, \end{aligned}$$

with $\mu_\Theta := \frac{1}{m}\sum_j \delta_{\vartheta_j}$ and U_{even}, U_{odd} the functionals of Proposition H.7 (so $b = k_e^2(b_{\mathrm{frozen}} + b_{\mathrm{osc}})$ of Theorem J, now at every Θ and not only at critical points; `periodic.second_order_drift` and

`dither.SingleKernelInstance.averaged` compute it). The *first-order correction* $g\colon \mathbb{R}^{mp} \to \mathbb{R}$ is the solution with $\pi(g) = 0$ of the weighted Poisson equation

$$\tau_0 \nabla \cdot (\pi G_m \nabla g) = \nabla \cdot (\pi b)$$

(equivalently $\bar{L}_0 g = -\tau_0 \pi^{-1} \nabla \cdot (\pi b) = \nabla J_m \cdot b - \tau_0 \nabla \cdot b$, which has zero π-mean),

the *first-order current* is $j_1 := \pi(b - \tau_0 G_m \nabla g) = \pi b_\perp$ with $b_\perp := b - \tau_0 G_m \nabla g$, so that $\nabla \cdot j_1 = 0$ and $b = G_m \nabla(\tau_0 g) + b_\perp$ is the Hodge decomposition of b in $L^2(\pi; G_m^{-1})$ into a G_m-gradient and a π-divergence-free field. For an observable $\varphi \in C^{3,\alpha}(\mathbb{R}^{mp})$ with polynomially bounded derivatives, ψ_φ denotes the solution of $\bar{L}_0 \psi_\varphi = \varphi - \pi(\varphi)$, $\pi(\psi_\varphi) = 0$.

Theorem J′ (the stationary law of the dithered finite-m closed loop)**.** *Assume the setting above, with G diagonal (as in every instance of this document; for a general symmetric G replace $G_l \partial_{\vartheta_l}$ by the contraction with G below). There are $\varepsilon_0 > 0$ and, for every observable $\varphi \in C^{3,\alpha}(\mathbb{R}^{mp})$ (three derivatives, the third Hölder continuous) with polynomially bounded derivatives up to order three, a constant C_φ (depending on φ, f, x_m, G, σ_0, τ_0, R, m, the feature bounds, and not on ε) such that for every $\varepsilon = 1/k_e \in (0, \varepsilon_0]$:*

(a) Invariant law. *The process Z has an invariant probability measure m_ε on $\mathcal{S}$, every invariant measure is supported in $\{\|\eta\| \le F_R + K_\Phi\}$ and has moments of all orders in Θ bounded uniformly in ε, and its θ-marginal is uniform. If moreover $a_{\max} > F_R$ componentwise and the tube lies in the box of the saturated centres, the invariant measure is unique, so the law of Z_t converges to m_ε and time averages along a single trajectory converge to m_ε-expectations.*

(b) The Gibbs measure at leading order. *For every invariant measure m_ε, $|m_\varepsilon(\varphi) - \pi(\varphi)| \le C_\varphi \varepsilon$.*

(c) First-order correction. $m_\varepsilon(\varphi) = \pi(\varphi) - \varepsilon\, \pi(b \cdot \nabla \psi_\varphi) + O(\varepsilon^2) = \pi(\varphi) + \varepsilon \int \varphi g \,\mathrm{d}\pi + O(\varepsilon^2)$, *with $|O(\varepsilon^2)| \le C_\varphi \varepsilon^2$. In particular the stationary parameter means are $\pi(\Theta) + \varepsilon \int \Theta g \,\mathrm{d}\pi + O(\varepsilon^2)$.*

(d) Observables of the full state. *For $\varphi(\Theta, \eta, \theta)$ smooth, T_0-periodic in θ, with polynomially bounded derivatives in Θ ($\bar{\varphi}$ below in the class of* (b)*), $m_\varepsilon(\varphi) = \pi(\bar{\varphi}) + O(\varepsilon)$, where $\bar{\varphi}(\Theta) := \langle \varphi(\Theta, r_\Theta(x_m(\theta)), \theta) \rangle_\theta$. In particular the stationary mean-square tracking error is*

$$\mathbb{E}_{m_\varepsilon} \|e\|^2 = \frac{2}{k_e^2}\, \pi(\bar{L}_m) + O(k_e^{-3}), \qquad \bar{L}_m(\Theta) = \tfrac{1}{2} \int \|r_\Theta\|^2 \,\mathrm{d}\rho :$$

the Gibbs average of the occupation loss, divided by $k_e^2/2$.

(e) Circulation and irreversibility. *Let*

$$V(\Theta, \eta, \theta) := k_e \cdot (\textit{drift of } \Theta \textit{ in physical time}) = \sum_i \big[G \nabla_\vartheta \big(\eta^\top \tilde{\Phi}(x_m(\theta) + \varepsilon\eta; \vartheta_i) \big) - \sigma_0(\vartheta_i - \vartheta^0) \big]$$

(so $\dot{\Theta} = \varepsilon V + \text{noise}$). For every pair of coordinates (k, l) of $\mathbb{R}^{mp}$,

$$\Lambda_\varepsilon^{kl} := \mathbb{E}_{m_\varepsilon}[\Theta_k V_l - \Theta_l V_k] = \varepsilon \Big[\int (\Theta_k j_{1,l} - \Theta_l j_{1,k}) \,\mathrm{d}\Theta + \pi\big(\langle u_k \dot{u}_l - u_l \dot{u}_k \rangle_\theta \big) \Big] + O(\varepsilon^2),$$

where $\dot{u} := \partial_\theta u = H - \langle H \rangle$. If the bracket is non-zero for some (k, l), then for ε small the stationary parameter process $(\Theta_t)_{t \ge 0}$ under m_ε is not reversible in time: $\mathbb{E}[\Theta_{k,t} \Theta_{l,t+h}] \ne \mathbb{E}[\Theta_{l,t} \Theta_{k,t+h}]$ for small $h > 0$.

Status. Proved. The proof imports three standard theorems, named where they are used: the existence, growth and regularity theory of the Poisson equation for the Langevin generator with a confining potential (Pardoux–Veretennikov [66], with interior Schauder estimates [35] for the higher derivatives), and, for the

uniqueness in (a) only, Hörmander's theorem for the smoothness of the transition densities of a hypoelliptic time-periodic diffusion [44], [10], the Stroock–Varadhan support theorem [83] and the Doob–Khasminskii and Harris theorems [22], [40], [41]. Parts (b)–(e) hold for every invariant measure and do not use the uniqueness. The first-order drift b is the one of Theorem J, so (c) is the thermal version of Theorem J(d): at low temperature it reduces to the $O(1/k_e)$ offset of the periodic orbit (Remark J′.1). The orientation effect of Remark H.7a and Proposition K appears in (c) as the orientation-odd part of $\int \Theta g \,\mathrm{d}\pi$, and in the numerics it is measured to within 7–8% at $k_e = 10$ and within the Monte Carlo resolution (10–30%) at $k_e = 20$–80. Checked numerically on the single-kernel instance below (`exp_dither_stationary.py`, `tests/test_dither.py`), where the first-order prediction removes the $O(1/k_e)$ discrepancy between the real loop and the Gibbs measure; the circulation of (e) is below the Monte Carlo resolution of the real loop at the affordable cost, and the current machinery behind it is checked on the first-order averaged SDE instead (Proposition J′.2).

Proof. Throughout, C denotes a constant independent of ε, and "polynomially bounded" means bounded by $C(1 + \|\Theta\|^q)$ on $\mathcal{S}$ for some q.

Step 0 (invariance, moments, existence). By the choice $k_e \geq k_T$, $\{\|e\| \leq R\}$ is forward invariant and $\|e(t)\| \leq e^{-k_e t}\|e(0)\| + (F_R + K_\Phi)/k_e$, so the set $\{\|\eta\| > F_R + K_\Phi + \delta\}$ is left in finite time and never re-entered: every invariant measure is supported in $\{\|\eta\| \leq F_R + K_\Phi\}$, and on this set the Θ-drift of (CL_τ) is bounded by $\lambda_G(F_R+K_\Phi)L_\vartheta + \sigma\|\Theta - \Theta^0\|$. For $W := \|\Theta - \Theta^0\|^2$ (slow time) $L_\varepsilon W \leq -2\sigma_0 W + 2\lambda_G(F_R + K_\Phi)L_\vartheta\sqrt{mW} + 2\tau_0 \operatorname{tr} G_m \leq -\sigma_0 W + C$, and likewise $L_\varepsilon W^q \leq -c_q W^q + C_q$ for every $q \geq 1$ (Itô's formula and Young's inequality). Hence $\mathbb{E}_z W^q(Z_s) \leq e^{-c_q s} W^q(z) + C_q/c_q$: the Cesàro averages of the laws of Z_s are tight (W^q is coercive in Θ, η and θ live in compact sets), the semigroup is Feller (the coefficients are smooth with bounded derivatives in η and θ and of linear growth in Θ), and Krylov–Bogoliubov gives an invariant probability measure m_ε. For any invariant measure, $0 = m_\varepsilon(L_\varepsilon W^q) \leq -c_q\, m_\varepsilon(W^q) + C_q$ gives the uniform moment bounds; the θ-marginal is uniform because θ moves at constant speed. Dynkin's formula with these moment bounds shows $m_\varepsilon(L_\varepsilon \psi) = 0$ for every $\psi \in C^2(\mathcal{S})$, T_0-periodic in θ, with polynomially bounded derivatives up to order two: the local martingale in Itô's formula is a martingale, and the stationary expectation of $\psi(Z_s) - \psi(Z_0)$ vanishes.

Step 1 (uniqueness). The noise acts on all of Θ; the x-equation carries no noise. Write the SDE for (x, Θ) in physical time with the vector fields X_0 (drift, time-periodic) and $X_{i,k} := \sqrt{2\tau_0/k_e}\,(G^{1/2})_{\cdot k} \cdot \partial_{\vartheta_i}$. The bracket $[X_{i,\alpha_k}, X_0]$ contains, in the x-direction, $-\frac{1}{m}(\partial_{\alpha_k}\tilde{\Phi})(x;\vartheta_i) \cdot \partial_{x_k} = -\frac{1}{m}(\,\mathrm{d}a_k/\,\mathrm{d}\alpha_k)\varphi(x; c_i, s_i)\partial_{x_k}$ with $\,\mathrm{d}a_k/\,\mathrm{d}\alpha_k > 0$ and $\varphi > 0$: together with the $X_{i,k}$ these brackets span $\mathbb{R}^n \times \mathbb{R}^{mp}$ at every point and every time, so the parabolic Hörmander condition holds everywhere. By Hörmander's theorem for time-dependent hypoelliptic diffusions with smooth coefficients of bounded derivatives ([44]; the non-homogeneous case is [10]), the transition probabilities $P_{s,t}(z, \cdot)$ have densities that are smooth in z, so the period map $P := P_{0,T_0}$ is strong Feller. Accessibility: fix Θ_1 and let $z^* := (e^*, \Theta_1)$ be the state at time T_0 of the deterministic skeleton that follows, on $[0, T_0/2]$, a C^1 parameter path along which $\hat{f}_{\Theta(t)}(x(t)) = f(x(t)) - w(t)$ with $w(t) := \lambda \cdot \beta(t)$ for a fixed bump β supported in $(0, T_0/2)$ and $\lambda \in \mathbb{R}$ chosen so that $e(T_0/2) = 0$ (possible since $\dot{e} = -k_e e + w$ is controllable and the required λ is $O(e^{-k_e T_0/2})$; such parameter paths exist because $a_{\max} > F_R$: m kernels centred at $x(t)$ with amplitude $f(x(t)) - w(t)$ realise $\hat{f} = f - w$, and the saturated centre box contains the tube), and on $[T_0/2, T_0]$ any C^1 path from that configuration to Θ_1. From every initial (e_0, Θ_0) in the invariant set the same construction (with λ depending on e_0) reaches $e(T_0/2) = 0$ at the same parameter configuration, hence z^* at time T_0. By the Stroock–Varadhan support theorem [83] (the skeleton paths are realised by controls $h \in L^2$: the Θ-noise is non-degenerate and the skeleton's Θ-path is C^1), $z^* \in \operatorname{supp} P(z, \cdot)$ for every z. The supports of two distinct ergodic invariant measures of a strong Feller kernel are disjoint, and both contain z^* ([22, Prop. 4.1.1 and Thm 4.2.1]; [40, Cor. 1.14–Thm 1.16]), so P has at most one invariant measure. The periodic family $P^*_{0,t} m_\varepsilon$ is then the unique invariant law of the augmented process; the strong Feller property with the

accessible point gives a minorisation of P^2 on the sublevel sets of the Lyapunov function W ([40, Lemma 1.13]), and Harris' theorem ([41]) gives the convergence of the law of Z_{nT_0} to m_ε in a W-weighted total variation norm, hence the convergence of time averages along a single trajectory.

Step 2 (the fast operator). At frozen (Θ,θ), A_0 is the generator of the flow $\dot\eta=\eta^*-\eta$, which contracts to η^* at rate one. For g smooth with $g(\Theta,\eta^*,\theta)=0$ the function

$$(A_0^{-1}g)(\Theta,\eta,\theta):=-\int_0^\infty g\big(\Theta,\eta^*+e^{-s}(\eta-\eta^*),\theta\big)\,\mathrm{d}s$$

is smooth, satisfies $A_0(A_0^{-1}g)=g$, and its derivatives in (Θ,θ) of any order are bounded by those of g; if g is linear in $\eta-\eta^*$, $g=(\eta-\eta^*)^\top w(\Theta,\theta)$, then $A_0^{-1}g=-(\eta-\eta^*)^\top w$. Conversely $A_0\psi=0$ with ψ bounded in η forces $\psi=\psi(\Theta,\theta)$.

Step 3 (the expansion). Let $\varphi\in C^5(\mathbb{R}^{mp})$ with polynomially bounded derivatives up to order five. Define

$$\psi_1:=-u\cdot\nabla\varphi,\qquad w_\varphi:=\sum_{i,l}G_l\,\partial_{\vartheta_l}\tilde\Phi(x_m(\theta);\vartheta_i)\,\partial_{(i,l)}\varphi\in\mathbb{R}^n$$

$$\text{(for a symmetric } G\text{: } (w_\varphi)_k=\sum_i\big(\nabla_\vartheta\tilde\Phi_k(x_m;\vartheta_i)\big)^\top G\nabla_{\vartheta_i}\varphi),$$

$$\psi_2:=(\eta-\eta^*)^\top w_\varphi+\psi_2'(\Theta,\theta),$$

where ψ_2' is the zero-mean antiderivative in θ of $b\cdot\nabla\varphi-[-\dot r^\top w_\varphi+(D_xr_\Theta\,r)^\top w_\varphi+(B_0\psi_1)|_{\eta^*}+(B_1\varphi)|_{\eta^*}]$ with $\dot r:=\partial_\theta[r_\Theta(x_m(\theta))]$ and $r:=r_\Theta(x_m(\theta))$, and $\psi_3:=A_0^{-1}$ of the η-dependent remainder defined below. Then $\psi_\varepsilon:=\varphi+\varepsilon\psi_1+\varepsilon^2\psi_2+\varepsilon^3\psi_3$ satisfies, order by order:

- ε^{-2}: $A_0\varphi=0$.
- ε^{-1}: $A_0\psi_1+\partial_\theta\varphi+A_1\varphi=0$, since ψ_1 does not depend on η.
- ε^0: $A_0\psi_2+\partial_\theta\psi_1+A_1\psi_1+(B_0+A_2)\varphi$. Here $A_1\psi_1=A_2\varphi=0$ and $A_0\psi_2=-(\eta-\eta^*)^\top w_\varphi=(B_0\varphi)|_{\eta^*}-B_0\varphi$, so the sum equals $\partial_\theta\psi_1+(B_0\varphi)|_{\eta^*}=-\dot u\cdot\nabla\varphi+H\cdot\nabla\varphi+\tau_0\,\mathrm{tr}(G_m\nabla^2\varphi)=\langle H\rangle\cdot\nabla\varphi+\tau_0\,\mathrm{tr}(G_m\nabla^2\varphi)=\bar L_0\varphi$, because $\dot u=H-\langle H\rangle$. This is Remark H.0's algebraic identity: the slaved drift averages to the Langevin drift.
- ε^1: $A_0\psi_3+\partial_\theta\psi_2+A_1\psi_2+(B_0+A_2)\psi_1+(B_1+A_3)\varphi$. At $\eta=\eta^*$: $\partial_\theta\psi_2=-\dot r^\top w_\varphi+\partial_\theta\psi_2'$ (the lag of the error behind its slaved value), $A_1\psi_2=(D_xr_\Theta(x_m)r)^\top w_\varphi$ (the state shift in the error equation), $B_0\psi_1=-H\cdot\nabla(u\cdot\nabla\varphi)-\tau_0\,\mathrm{tr}(G_m\nabla^2(u\cdot\nabla\varphi))$, $A_2\psi_1=A_3\varphi=0$, and $B_1\varphi=\sum_i G\nabla_\vartheta\big(r^\top D_x\tilde\Phi(x_m;\vartheta_i)r\big)\cdot\nabla_{\vartheta_i}\varphi$ (the state shift in the feature). By the choice of ψ_2' the sum at $\eta=\eta^*$ equals $b\cdot\nabla\varphi$ pointwise in θ, once the θ-averages are identified: $\langle-\dot r^\top w_\varphi\rangle=G\nabla_\vartheta\langle-\dot r^\top\tilde\Phi(x_m;\vartheta)\rangle\cdot\nabla\varphi=(G_m\nabla U_{\mathrm{odd}})\cdot\nabla\varphi$ at the particles, $\langle(D_xr\,r)^\top w_\varphi+(B_1\varphi)|_{\eta^*}\rangle=\langle(r\cdot\nabla_x)(r^\top\tilde\Phi)\rangle$-gradient $\cdot\nabla\varphi=(G_m\nabla U_{\mathrm{even}})\cdot\nabla\varphi$ (the two state shifts combine into U_{even}), so these three give $b_{\mathrm{fr}}\cdot\nabla\varphi$; and $\langle B_0\psi_1|_{\eta^*}\rangle=-\langle(D_\Theta u)H\rangle\cdot\nabla\varphi-\langle H\otimes u\rangle:\nabla^2\varphi-\tau_0\,\mathrm{tr}(G_m\nabla^2(\langle u\rangle\cdot\nabla\varphi))=\langle(D_\Theta H)u\rangle\cdot\nabla\varphi=b_{\mathrm{osc}}\cdot\nabla\varphi$, because $\langle u\rangle=0$, $\langle(D_\Theta u)\dot u\rangle=-\langle(D_\Theta\dot u)u\rangle=-\langle(D_\Theta H)u\rangle$ (integration by parts in θ, $\langle D_\Theta u\rangle\langle H\rangle=0$), and $\langle H_ju_k\rangle+\langle H_ku_j\rangle=\langle\partial_\theta(u_ju_k)\rangle=0$ so the symmetric contraction with $\nabla^2\varphi$ vanishes (the same cancellation as $N_2=0$ in Lemma E.0′). Writing $T_1(\Theta,\eta,\theta)$ for the sum of the order-ε terms other than $A_0\psi_3$, one has $T_1(\Theta,\eta^*,\theta)=b\cdot\nabla\varphi$, so $\psi_3:=A_0^{-1}(b\cdot\nabla\varphi-T_1)$ is well defined (Step 2) and $A_0\psi_3+T_1=b\cdot\nabla\varphi$: the order-$\varepsilon$ term is exactly $\bar L_1\varphi:=b\cdot\nabla\varphi$.
- ε^2: $R_\varepsilon:=\varepsilon^{-2}[L_\varepsilon\psi_\varepsilon-\bar L_0\varphi-\varepsilon\bar L_1\varphi]$ is a finite sum of products of derivatives of f on the tube (orders ≤3), of $\tilde\Phi$ in (x,ϑ) (bounded), of x_m, of u, w_φ, ψ_2', ψ_3 and of derivatives of φ: ψ_2' contains $\nabla^3\varphi$ (through $\tau_0\,\mathrm{tr}(G_m\nabla^2(u\cdot\nabla\varphi))$ in $(B_0\psi_1)|_{\eta^*}$) and ψ_3 contains $\nabla^2\varphi$, so $B_0\psi_2\ni\tau_0\,\mathrm{tr}(G_m\nabla^2\psi_2')$ carries $\nabla^5\varphi$ and $B_0\psi_3$ carries $\nabla^4\varphi$; using the exact operators A_ε, B_ε and Taylor's theorem in ε rather than their formal series, $|R_\varepsilon|\le C(1+\|\Theta\|^q)$ on $\mathcal{S}$

uniformly in $\varepsilon \in (0, \varepsilon_0]$ for $\varphi \in C^5$ with polynomially bounded derivatives up to order five. (For the zeroth-order statement (b) alone ψ_2' can be dropped — it only serves to make the order-ε term exactly $b \cdot \nabla\varphi$ — and C^3 suffices.)

Step 4 (the Poisson equation). $\bar{L}_0$ is uniformly elliptic with smooth coefficients, ∇J_m has linear growth with bounded derivatives of all orders, and $\langle -G_m \nabla J_m(\Theta),\, \Theta - \Theta^0\rangle \le -\sigma_0\|\Theta - \Theta^0\|^2 + C\|\Theta - \Theta^0\|$: the recurrence condition of Pardoux–Veretennikov [66, Thm 1] holds with every polynomial Lyapunov function. For $\varphi \in C^{3,\alpha}(\mathbb{R}^{mp})$ with polynomially bounded derivatives up to order three there is a unique $\psi_\varphi \in C^2(\mathbb{R}^{mp})$ with $\bar{L}_0\psi_\varphi = \varphi - \pi(\varphi)$, $\pi(\psi_\varphi) = 0$, and ψ_φ, $\nabla\psi_\varphi$ are polynomially bounded ([66, Thm 1 and 2]); interior Schauder estimates on unit balls [35, Thm 6.2 and 6.17], which gain two derivatives and whose constants depend polynomially on $\|\Theta\|$ through the local Hölder norms of the coefficients, give $\psi_\varphi \in C^{5,\alpha}$ with polynomially bounded derivatives up to order five.

Step 5 ((b) and (c)). Apply Step 3 to $\psi := \psi_\varphi$ and Step 0's identity to ψ_ε: $0 = m_\varepsilon(L_\varepsilon\psi_\varepsilon) = m_\varepsilon(\bar{L}_0\psi) + \varepsilon m_\varepsilon(b\cdot\nabla\psi) + \varepsilon^2 m_\varepsilon(R_\varepsilon) = m_\varepsilon(\varphi) - \pi(\varphi) + \varepsilon m_\varepsilon(b\cdot\nabla\psi) + O(\varepsilon^2)$, the $O(\varepsilon^2)$ by the moment bounds of Step 0. Since $b\cdot\nabla\psi$ is polynomially bounded, $|m_\varepsilon(\varphi) - \pi(\varphi)| \le C\varepsilon$, which is (b). Now $b\cdot\nabla\psi \in C^4(\mathbb{R}^{mp})$ with polynomially bounded derivatives (b is smooth with bounded derivatives in Θ, $\psi \in C^{5,\alpha}$), so (b) applies to it (its Poisson solution is in C^3, which the zeroth-order expansion needs): $m_\varepsilon(b\cdot\nabla\psi) = \pi(b\cdot\nabla\psi) + O(\varepsilon)$, and $m_\varepsilon(\varphi) = \pi(\varphi) - \varepsilon\pi(b\cdot\nabla\psi_\varphi) + O(\varepsilon^2)$. The second form of (c) is Green's identity for the weighted Poisson equation:

$$\begin{aligned}\pi(b\cdot\nabla\psi) = \int b\cdot\nabla\psi\,\pi = -\int \psi\nabla\cdot(\pi b) = -\tau_0\int \psi\nabla\cdot(\pi G_m\nabla g) = -\tau_0\int g\nabla\cdot(\pi G_m\nabla\psi)\\ = -\int g\pi\bar{L}_0\psi = -\int g(\varphi - \pi(\varphi))\pi = -\int \varphi g\,\mathrm{d}\pi,\end{aligned}$$

using $\tau_0\pi^{-1}\nabla\cdot(\pi G_m\nabla\psi) = \bar{L}_0\psi$ and $\pi(g) = 0$; the boundary terms vanish by the Gaussian tails of π and the polynomial growth of g, ψ and their gradients (the same regularity for g as for ψ_φ, with φ replaced by the polynomially bounded $\nabla J_m\cdot b - \tau_0\nabla\cdot b$).

Step 6 ((d)). Write $\varphi = \bar{\varphi} + \varphi_\theta + \varphi_\eta$ with $\varphi_\theta(\Theta,\theta) := \varphi(\Theta,\eta^*,\theta) - \bar{\varphi}(\Theta)$ (zero θ-mean) and $\varphi_\eta := \varphi(\Theta,\eta,\theta) - \varphi(\Theta,\eta^*,\theta)$ (zero at η^*). With $\chi_1 := -$(zero-mean antiderivative in θ of φ_θ) and $\chi_2 := -A_0^{-1}\varphi_\eta$ (Step 2), $L_\varepsilon(\varepsilon\chi_1 + \varepsilon^2\chi_2) = \partial_\theta\chi_1 + A_0\chi_2 + \varepsilon[B_0\chi_1 + (\partial_\theta + A_1)\chi_2] + O(\varepsilon^2) = -(\varphi_\theta + \varphi_\eta) + \varepsilon[\dots] + O(\varepsilon^2)$, all terms polynomially bounded. Step 0's identity gives $m_\varepsilon(\varphi) = m_\varepsilon(\bar{\varphi}) + \varepsilon m_\varepsilon(B_0\chi_1 + (\partial_\theta + A_1)\chi_2) + O(\varepsilon^2) = \pi(\bar{\varphi}) + O(\varepsilon)$ by (b). For $\varphi = \|\eta\|^2$, $\bar{\varphi} = \langle\|r_\Theta(x_m(\theta))\|^2\rangle_\theta = 2\bar{L}_m(\Theta)$, and $\mathbb{E}\|e\|^2 = \varepsilon^2\mathbb{E}\|\eta\|^2$.

Step 7 ((e)). Expand $V = V^0 + \varepsilon V^1 + O(\varepsilon^2)$ with V^0 the B_0-drift and V^1 the B_1-drift, and apply Step 6 to $\varphi := \Theta_k V_l - \Theta_l V_k$ keeping the order-ε terms. For $\varphi^0 := \Theta\wedge V^0$ ($\wedge$ on the pair (k,l)): $\bar{\varphi}^0 = \Theta\wedge\langle H\rangle = \Theta\wedge(-G_m\nabla J_m)$, and $\pi(\bar{\varphi}^0) = 0$ because $-\pi G_m\nabla J_m = \tau_0 G_m\nabla\pi$ and $\int\Theta\wedge(G_m\nabla\pi) = 0$ for a symmetric G_m (integration by parts: the integrand is a divergence plus $\pi\cdot(G_{lk} - G_{kl})$). Next, $\varphi^0_\theta = \Theta\wedge(H - \langle H\rangle) = \Theta\wedge\dot{u}$, so $\chi_1 = -\Theta\wedge u$ and $\langle(B_0\chi_1)|_{\eta^*}\rangle = -\langle H\cdot\nabla_\Theta(\Theta\wedge u)\rangle - \tau_0\langle\operatorname{tr} G_m\nabla^2(\Theta\wedge u)\rangle = -\langle H\wedge u\rangle - \langle\Theta\wedge(D_\Theta u\cdot H)\rangle = -\langle\dot{u}\wedge u\rangle + \Theta\wedge\langle(D_\Theta H)u\rangle = \langle u\wedge\dot{u}\rangle + \Theta\wedge b_{\mathrm{osc}}$ (the τ_0 term and $\langle H\rangle\wedge\langle u\rangle$ vanish since $\langle u\rangle = 0$, and $\langle(D_\Theta u)H\rangle = -\langle(D_\Theta H)u\rangle$ as in Step 3). Further, $\varphi^0_\eta = \Theta\wedge(V^0(\eta) - V^0(\eta^*))$ is linear in $\eta - \eta^*$, so $\chi_2 = \varphi^0_\eta$ and $\langle((\partial_\theta + A_1)\chi_2)|_{\eta^*}\rangle = \Theta\wedge\big\langle\sum_i G\nabla_\vartheta\big((-\dot{r} + D_x r\, r)^\top\tilde{\Phi}(x_m;\vartheta_i)\big)\big\rangle = \Theta\wedge$(lag and error-shift parts of b_{fr}); with $\bar{\varphi}^1 = \Theta\wedge\langle V^1|_{\eta^*}\rangle = \Theta\wedge$(feature-shift part of b_{fr}) these complete $\Theta\wedge b_{\mathrm{fr}}$. Collecting, and using (c) for $\bar{\varphi}^0$ ($m_\varepsilon(\bar{\varphi}^0) = \varepsilon\int\Theta\wedge(-G_m\nabla J_m)g\,\mathrm{d}\pi + O(\varepsilon^2)$) and (b) for the order-$\varepsilon$ observables,

$$\Lambda_\varepsilon = \varepsilon\Big[\int\Theta\wedge(b - gG_m\nabla J_m)\,\mathrm{d}\pi + \pi\big(\langle u\wedge\dot{u}\rangle\big)\Big] + O(\varepsilon^2),$$

and $b - gG_m\nabla J_m = (b - \tau_0 G_m\nabla g) + (\tau_0 G_m\nabla g - gG_m\nabla J_m) = b_\perp + \tau_0\pi^{-1}G_m\nabla(g\pi)$, whose second term integrates to zero against $\Theta\wedge$ as above. This is the formula of (e). For the irreversibility:

under m_ε the Θ-process is stationary; reversibility would give $\mathbb{E}[\Theta_{k,t}\Theta_{l,t+h}] = \mathbb{E}[\Theta_{l,t}\Theta_{k,t+h}]$ for all h, i.e. $\mathbb{E}[\Theta_t \wedge \Theta_{t+h}] = 0$, whereas $\mathbb{E}[\Theta_t \wedge (\Theta_{t+h} - \Theta_t)] = \mathbb{E}[\Theta_t \wedge \int_t^{t+h} \varepsilon V \,\mathrm{d}s] + 0$ (the martingale term has mean zero) $= h\varepsilon\Lambda_\varepsilon + o(h)$ by stationarity and the continuity of the integrand; so $\Lambda_\varepsilon \neq 0$ excludes reversibility. □

Remark J′.1 (reading)**.**

- *From the periodic orbit to the thermal state.* As $\tau_0 \to 0$ with Θ^* the non-degenerate global minimiser of J_m, $\pi \to \delta_{\Theta^*}$, $\psi_{\Theta_k} \to$ the linear function $v_k^\top(\Theta - \Theta^*)$ with $-(G_m\nabla^2 J_m)^\top v_k = e_k$, and (c) gives $m_\varepsilon(\Theta) \to \Theta^* + \varepsilon(G_m\nabla^2 J_m(\Theta^*))^{-1}b(\Theta^*)$: Theorem J(d)'s period mean of the periodic orbit. Theorem J′(c) is the finite-temperature version: the correction is a Gibbs average of the same first-order drift, weighted by the Poisson solution instead of the inverse Hessian.

- *The orientation effect with dither (branch 2).* Reversing the direction of traversal of the reference flips U_{odd} and b_{osc} and leaves U_{even}, J_m and π unchanged, so the stationary means of the two traversals differ at order ε by $2\varepsilon\int \Theta g_{\mathrm{odd}}\,\mathrm{d}\pi$ with g_{odd} the correction driven by the odd part of b: the quantity measured in Remark H.7a for the periodic orbit is now a statement about stationary measures, and it is checked below (to within 7–8% at $k_e = 10$, and within the resolution of the runs at larger k_e) in the single-kernel instance (where, in one state dimension, $U_{\mathrm{odd}} \equiv 0$ — a 1-form integrated over a path that retraces itself — and the whole odd effect comes from b_{osc}; a reference with the time-reversal symmetry $x_m(\theta_0 + \theta) = x_m(\theta_0 - \theta)$, such as a sinusoid, has no odd effect at all, since its reversal is a time shift). As a design rule: a time-symmetric reference (one whose Fourier series carries only cosines after a time shift, such as $x_m = X[\cos\theta + r_2\cos 2\theta]$) removes b_{osc} and U_{odd} identically, so its first-order correction is the performative term $G_m\nabla U_{\mathrm{even}}$ alone, which is reciprocal exactly when Theorem I's $A_{\mathrm{even}} \equiv 0$; a reference traversed one way carries both, and the odd part can be used (Proposition K) or cancelled by reversing the traversal. Checked: for the cosine reference the code's b_{osc} and $\langle u \wedge \dot{u}\rangle$ are 10^{-17} at every point tested.

- *Non-equilibrium.* (e) says that the dithered learning loop, at stationarity, is not in detailed balance even to first order: the periodic forcing alone (the term $\pi\langle u \wedge \dot{u}\rangle$, the signed area swept by the fast parameter loop) breaks the time-reversal symmetry, and the slow dynamics carries in addition the current j_1 of Proposition J′.2, nonzero exactly when the first-order drift is not reciprocal. The first is a property of any periodically forced system; the second is the signature of closed-loop learning (Theorem I).

- *Hypotheses.* Smoothness of f on the tube and of the features is used through the expansion (three x-derivatives of f, five Θ-derivatives of the Poisson solution, i.e. three Hölder derivatives of the observable) and, for the uniqueness only, through Hörmander's theorem; $k_e \geq k_T$ keeps the plant in the tube for every parameter path. The constants are not explicit: the result is an asymptotic expansion of the invariant measure, not a quantitative bound at a given k_e, and the numerics show where it applies.

Proposition J′.2 (the first-order current, entropy production, and reciprocity)**.** *Consider the first-order averaged dynamics* $\mathrm{d}\Theta = (-G_m\nabla J_m + \varepsilon b)\,\mathrm{d}s + \sqrt{2\tau_0}\,G_m^{1/2}\,\mathrm{d}W$ *on* $\mathbb{R}^{mp}$*, with* b *as above.*

(i) *Its stationary law* $m_\varepsilon^{(1)}$ *satisfies* $m_\varepsilon^{(1)}(\varphi) = \pi(\varphi) + \varepsilon\int \varphi g\,\mathrm{d}\pi + O(\varepsilon^2)$ *for every* φ *as in Theorem J′ (the expansion* $m_\varepsilon^{(1)} = \pi(1 + \varepsilon g) + O(\varepsilon^2)$ *in the weak sense; on the grid it holds in* L^1 *with an* ε^2 *remainder, Numerics below), and its stationary current* $j := m_\varepsilon^{(1)}(-G_m\nabla J_m + \varepsilon b) - \tau_0 G_m\nabla m_\varepsilon^{(1)}$ *equals* $\varepsilon j_1 + O(\varepsilon^2)$ *in the same sense, with* $j_1 = \pi b_\perp$*;*

its entropy production rate (in slow time) is

$$\mathrm{EPR}_\varepsilon := \int \frac{\|j\|^2_{G_m^{-1}}}{\tau_0 m_\varepsilon^{(1)}}\,\mathrm{d}\Theta = \varepsilon^2\tau_0^{-1}\pi\big(\|b_\perp\|^2_{G_m^{-1}}\big) + O(\varepsilon^3)$$

(formal in the remainder: the ε^2 term uses only the first-order current, the $O(\varepsilon^3)$ presumes the expansion of j in $L^2(1/\pi)$).

(ii) *The following are equivalent: $j_1 \equiv 0$; $b_\perp \equiv 0$; b is a G_m-gradient field, $b = -G_m\nabla J_1$; $D_\Theta(G_m^{-1}b)$ is a symmetric matrix at every Θ. In that case the first-order averaged dynamics* is *the Langevin dynamics of $J_m+\varepsilon J_1$, with stationary density exactly $\propto \exp(-(J_m+\varepsilon J_1)/\tau_0)$, in detailed balance.*

(iii) *The antisymmetric part of $D_\Theta(G_m^{-1}b)$ is, block by block ((i,j) blocks of size $p\times p$),*

$$\tfrac12\big[D(G_m^{-1}b) - D(G_m^{-1}b)^\top\big]_{ij} = \frac{1}{2m}\nabla_\vartheta\nabla_{\vartheta'}A_{\mu_\Theta}(\vartheta_i,\vartheta_j) + \tfrac12\big\langle[\nabla^2_\Theta\mathcal{J},\nabla^2_\Theta\mathcal{U}]_{G_m}\big\rangle_{\theta,ij},$$

where $A_\mu = S - S^\top$ is the antisymmetric kernel of Theorem I at the atomic measure μ_Θ (derivatives in the kernel arguments, μ_Θ frozen), $\mathcal{J}(\Theta,\theta) := \frac{m}{2}\|r_\Theta(x_m(\theta))\|^2 + \frac{\sigma_0}{2}\sum_i\|\vartheta_i - \vartheta^0\|^2_{G^{-1}}$ is the instantaneous loss (so $H = -G_m\nabla\mathcal{J}$, $\langle\mathcal{J}\rangle = J_m$), $\mathcal{U}$ its zero-mean antiderivative in θ, and $[X,Y]_G := XGY - YGX$. The first term is the finite-m form of the reciprocity defect of Theorem I; the second is the failure of the instantaneous Hessians at different phases to commute, the non-gradient part of second-order averaging of a time-periodic gradient flow.

(iv) *In the setting of Theorem I (saturated Gaussian features, $f\in C^2$, $x_m\in C^2$ periodic), for $m\ge 3$, b_{fr} is never a G_m-gradient on $\mathbb{R}^{mp}$ (b_{fr} does not depend on σ_0 and depends on G through the factor G only); consequently, along the gain scale $t\mapsto tG$, the full drift $b = b_{\mathrm{fr}} + b_{\mathrm{osc}}$ is a $(tG)_m$-gradient for at most one value of $t>0$. For $m = 1,\,2$ the condition of* (ii) *is the explicit vanishing of* (iii)*, checked to fail in the single-kernel instance below (non-gradient fraction $\pi(\|b_\perp\|^2)/\pi(\|b\|^2)\approx 0.3$–$0.5$).*

Proof. (i) Stationary Fokker–Planck: $\nabla\cdot(mG_m\nabla J_m) - \varepsilon\nabla\cdot(mb) + \tau_0\nabla\cdot(G_m\nabla m) = 0$; at order ε^0 the Gibbs measure, at order ε^1 with $\pi_1 = g\pi$: $\nabla\cdot(g\pi G_m\nabla J_m) + \tau_0\nabla\cdot(G_m\nabla(g\pi)) = \tau_0\nabla\cdot(\pi G_m\nabla g)$ (using $\tau_0\nabla\pi = -\pi\nabla J_m$), so the equation is $\tau_0\nabla\cdot(\pi G_m\nabla g) = \nabla\cdot(\pi b)$, the defining equation of g; the expansion is justified by the same Poisson-equation argument as in Theorem J′ (the generator of this SDE is $\bar L_0 + \varepsilon b\cdot\nabla$, and Steps 4–5 apply verbatim without the fast variables). The current: $j = \pi(-G_m\nabla J_m) - \tau_0 G_m\nabla\pi + \varepsilon[\pi b - g\pi G_m\nabla J_m - \tau_0 G_m\nabla(g\pi)] + O(\varepsilon^2) = \varepsilon\,\pi(b - \tau_0 G_m\nabla g) + O(\varepsilon^2)$. The entropy production rate of a diffusion with mobility G_m and temperature τ_0 in a stationary state m with current j is $\int j^\top(\tau_0 G_m)^{-1}j/m$ [82], which gives the formula.

(ii) $j_1\equiv 0$ iff $b = \tau_0 G_m\nabla g$ iff b is a G_m-gradient (then $\tau_0 g = -J_1 + \text{const}$ and the equation for g is satisfied); on the simply connected $\mathbb{R}^{mp}$ a smooth field $G_m^{-1}b$ is a gradient iff its Jacobian is symmetric (Poincaré lemma). When $b = -G_m\nabla J_1$ the drift $-G_m\nabla(J_m+\varepsilon J_1)$ is a gradient drift, so the SDE is the Langevin dynamics of $J_m+\varepsilon J_1$ with the exact Gibbs law $\propto\exp(-(J_m+\varepsilon J_1)/\tau_0)$ and $j\equiv 0$; consistently, $g = -J_1/\tau_0 + \text{const}$ solves the defining equation.

(iii) $G^{-1}b_{\mathrm{fr},i} = \nabla_\vartheta P_2[\mu_\Theta](\vartheta)|_{\vartheta_i}$ with $P_2 := U_{\mathrm{even}} + U_{\mathrm{odd}}$, a functional of μ (quadratic in μ through r_μ) and of the evaluation point. Differentiating with respect to ϑ_j moves the atom ϑ_j of μ_Θ, which gives $\frac1m\nabla_\vartheta\nabla_{\vartheta'}S(\vartheta_i,\vartheta_j)$ with $S(\vartheta,\vartheta') := \delta P_2[\mu](\vartheta)/\delta\mu(\vartheta')$ (Theorem I's k_e^{-2} second-variation kernel), and, for $j = i$, additionally the symmetric $\nabla^2_\vartheta P_2(\vartheta_i)$. The antisymmetric part of the (i,j) block is therefore $\frac{1}{2m}\nabla\nabla'[S(\vartheta_i,\vartheta_j) - S(\vartheta_j,\vartheta_i)] = \frac{1}{2m}\nabla\nabla' A_\mu(\vartheta_i,\vartheta_j)$. For b_{osc}: $H = -G_m\nabla\mathcal{J}$ and $u = -G_m\nabla\mathcal{U}$, so $G_m^{-1}b_{\mathrm{osc}} = \langle\nabla^2\mathcal{J}\,G_m\nabla\mathcal{U}\rangle$ and $\partial_j(G_m^{-1}b_{\mathrm{osc}})_k = \langle\partial_j\partial_k\partial_l\mathcal{J}\,G_l\partial_l\mathcal{U}\rangle + \langle(\nabla^2\mathcal{J}G_m\nabla^2\mathcal{U})_{kj}\rangle$; the first is symmetric in (j,k), the antisymmetric part of the second is $\frac12\langle\nabla^2\mathcal{J}G_m\nabla^2\mathcal{U} - \nabla^2\mathcal{U}G_m\nabla^2\mathcal{J}\rangle$.

(iv) Suppose b_{fr} is a G_m-gradient for some $m \geq 3$. By (iii), $\nabla_\vartheta \nabla_{\vartheta'} A_{\mu_\Theta}(\vartheta_i, \vartheta_j) = 0$ for all Θ and all i, j. A_μ is affine in μ (Theorem I): $A_\mu = A_f + B(\int \tilde{\Phi}\,\mathrm{d}\mu)$ with B linear in the function $\int \tilde{\Phi}(\cdot\,;\vartheta)\,\mathrm{d}\mu(\vartheta)$, so for $\Theta = (\vartheta, \vartheta', \vartheta_3, \dots)$, $\nabla_\vartheta \nabla_{\vartheta'}[A_f + \frac{1}{m}\sum_j B(\tilde{\Phi}(\cdot\,;\vartheta_j))](\vartheta, \vartheta') = 0$ with the atom positions inside B frozen. Varying ϑ_3 alone shows that $\nabla_\vartheta \nabla_{\vartheta'} B(\tilde{\Phi}(\cdot\,;\vartheta_3))(\vartheta, \vartheta')$ does not depend on ϑ_3; at the zero feature $\vartheta_3 = \vartheta_z$ it vanishes ($B(0) = 0$), so it vanishes for all ϑ_3, and then $\nabla\nabla' A_f(\vartheta, \vartheta') = 0$ as well: $\nabla_\vartheta \nabla_{\vartheta'} A_\mu(\vartheta, \vartheta') = 0$ for every μ and all (ϑ, ϑ'). A smooth antisymmetric kernel with vanishing mixed derivative on the connected $\mathbb{R}^p \times \mathbb{R}^p$ has the form $a(\vartheta) - a(\vartheta')$, a coboundary, hence $A_\mu \equiv 0$ by Theorem I(i) — for every μ, contradicting Theorem I(vi). For the gain scale: under $G \to tG$, $b_{\mathrm{fr}} \to t\, b_{\mathrm{fr}}$ and $b_{\mathrm{osc}} \to t^2 b_{\mathrm{osc}}$, so the antisymmetric part of $D((tG)^{-1}b)$ is $a_{\mathrm{fr}} + t\, a_{\mathrm{osc}}$ with $a_{\mathrm{fr}} \not\equiv 0$; it vanishes identically in Θ for at most one t. □

Corollary J′.3 (the stationary law in the iterated limit $k_e \to \infty$, $m \to \infty$)**.** *Let π_m be the Gibbs measure of Theorem J′ on $\mathbb{R}^{mp}$, and μ_τ^* the unique minimiser of Theorem H's free energy F_τ at $\sigma = \sigma_0/k_e$, $\tau = \tau_0/k_e$; since $k_e F_\tau = \bar{L} + \frac{\sigma_0}{2}\int \|\vartheta - \vartheta^0\|^2_{G^{-1}}\,\mathrm{d}\mu + \tau_0 H(\mu) = \bar{L} + \tau_0 H(\cdot\,|\,\gamma_\tau) + \mathrm{const}$ with $\gamma_\tau = \mathcal{N}(\vartheta^0, (\tau_0/\sigma_0)G)$, μ_τ^* does not depend on k_e. Then:*

(i) Chaoticity of the Gibbs measures. *Under π_m the empirical measure $\mu_\Theta = \frac{1}{m}\sum_i \delta_{\vartheta_i}$ converges in probability to μ_τ^* as $m \to \infty$, and for every k the k-particle marginal of π_m converges weakly to $(\mu_\tau^*)^{\otimes k}$.*

(ii) The iterated limit of the real loop. *For every $\varphi \in C^2(\mathbb{R}^p)$ bounded with bounded derivatives up to order two,*

$$\lim_{m\to\infty} \lim_{k_e\to\infty} \mathbb{E}_{m_\varepsilon}\Big[\frac{1}{m}\sum_i \varphi(\vartheta_i)\Big] = \int \varphi\,\mathrm{d}\mu_\tau^*,$$

and the inner limit is attained at the rate $O(1/k_e)$ of Theorem J′(b), with a constant that depends on m.

Status. Proved (Sanov's theorem and Varadhan's lemma, [24, Thms 6.2.10 and 4.3.1]; Theorem H(i)–(ii); Theorem J′(b)). The joint limit, with k_e and m related, is Conjecture H.6 and remains open; the corollary places the stationary law of the finite loop between Theorem J′ and Theorem H. Numerics: `exp_gibbs_m.py` → `results/dither/gibbs_m.md` (below).

Proof. $J_m(\Theta) = m[\bar{L}(\mu_\Theta) + \frac{\sigma_0}{2}\int \|\vartheta - \vartheta^0\|^2_{G^{-1}}\,\mathrm{d}\mu_\Theta]$, so $\pi_m(\,\mathrm{d}\Theta) \propto \exp(-m\bar{L}(\mu_\Theta)/\tau_0)\,\gamma_\tau^{\otimes m}(\,\mathrm{d}\Theta)$. The map $\mu \mapsto \bar{L}(\mu) = \frac{1}{2}\int \|f - \int \tilde{\Phi}(\cdot\,;\vartheta)\,\mathrm{d}\mu(\vartheta)\|^2\,\mathrm{d}\rho$ is bounded and continuous for the weak topology of $\mathcal{P}(\mathbb{R}^p)$ (the features are bounded and continuous in ϑ, and ρ is a probability measure). Under $\gamma_\tau^{\otimes m}$ the empirical measure satisfies Sanov's large-deviation principle with rate function $H(\cdot\,|\,\gamma_\tau)$; by Varadhan's lemma (tilting by the bounded continuous $-m\bar{L}/\tau_0$), under π_m it satisfies a large-deviation principle with rate function $I(\mu) = \bar{L}(\mu)/\tau_0 + H(\mu\,|\,\gamma_\tau) - \inf = (k_e/\tau_0)(F_\tau(\mu) - F_\tau(\mu_\tau^*))$, by Theorem H(i). F_τ is strictly convex ($\bar{L}$ is convex, the entropy strictly convex) with the unique minimiser μ_τ^* (Theorem H(ii)), so I has the unique zero μ_τ^* and $\mu_\Theta \to \mu_\tau^*$ in probability. Exchangeability of π_m then gives the convergence of the k-marginals to $(\mu_\tau^*)^{\otimes k}$ (Sznitman's equivalence of chaoticity and the convergence of the empirical measure). For (ii), Theorem J′(b) applied to the observable $\frac{1}{m}\sum_i \varphi(\vartheta_i) \in C^2(\mathbb{R}^{mp})$ gives $\lim_{k_e\to\infty} \mathbb{E}_{m_\varepsilon}[\frac{1}{m}\sum \varphi(\vartheta_i)] = \pi_m(\frac{1}{m}\sum \varphi(\vartheta_i)) = \pi_m^{(1)}(\varphi)$, the one-particle marginal, which converges to $\int \varphi\,\mathrm{d}\mu_\tau^*$ by (i). □

Remark J′.4 (what the corollary says about Conjecture H.6)**.** The conjecture asks for the limit of the real loop's law when $k_e \to \infty$ with m fixed or growing, uniformly in time. Theorem J′ settles the stationary part at fixed m, and the corollary shows that the resulting Gibbs measures are chaotic with limit μ_τ^*, the stationary state of $(\mathfrak{A})$: the stationary law of the real dithered loop is the minimiser of Theorem H's free energy in the iterated limit. What is missing for the conjecture is the uniformity in m of Theorem J′'s $O(1/k_e)$ (its constant grows with m through the Poisson equation on $\mathbb{R}^{mp}$) and the time-dependent statement.

Numerics (`exp_dither_stationary.py` → `results/dither/table.md`). The single-kernel instance: $n = 1$, $m = 1$, plant $f(x) = e^{-(x-0.5)^2/0.5} + 0.4x$, reference $x_m(t) = 1.5[\sin t' + \frac{1}{3}\sin 2t']$ with $t' = \pm t$ (the two orientations), saturated amplitude and centre $\vartheta = (\alpha, \gamma)$ ($a_{\max} = 3$, $c_{\max} = 2.5$), frozen width $s = 0.6$, $G = \mathrm{diag}(2, 2)$, $\sigma_0 = 0.4$, $\vartheta^0 = 0$, $\tau_0 = 0.1$, $k_e \in \{10, 20, 40, 80\}$. The averaged data on a 201^2 grid: π, g, j_1 (rms $b_{\mathrm{fr}} = 0.143$, rms $b_{\mathrm{osc}} = 0.225$; non-gradient fraction $\pi(\|b_\perp\|^2)/\pi(\|b\|^2) = 0.32$ and 0.46 for the two orientations; EPR coefficient $\tau_0^{-1}\pi(\|b_\perp\|^2_{G^{-1}}) = 0.144$ and 0.118; circulation coefficient $\int \Theta \wedge j_1 + \pi\langle u \wedge \dot{u}\rangle = +0.0117 - 0.0167 = -0.0051$ and $-0.0103 + 0.0167 = +0.0065$). The real loop: $2 \cdot 10^4$ independent replicas per (k_e, orientation), started from the first-order law $\pi(1 + \varepsilon g)$, Strang splitting with $k_e h = 1/2$, burn-in ≥ 10 and averaging ≥ 40 slow-time units (whole numbers of reference periods), standard errors from the spread of the per-replica time averages (the slowest mode of $\bar{L}_0$ relaxes in 3.5 slow-time units on this instance, so batch means in time would be correlated; referee round 6).

The first-order law (b), (c). The stationary means against the Gibbs value and the first-order prediction, orientation +1 (ε-coefficients $\int \varphi g \, \mathrm{d}\pi$: -0.255 for α, -0.426 for γ); Table 24.

Table 24: Theorem J′(b), (c) on the single-kernel instance: the stationary means of the real loop against the Gibbs value and the first-order prediction (`results/dither/table.md`).

k_e	$m_\varepsilon(\alpha) - \pi(\alpha)$ (s.e. 0.0013)	$\times k_e$	$m_\varepsilon(\alpha)$ − first order	$m_\varepsilon(\gamma) - \pi(\gamma)$ (s.e. 0.0018)	$\times k_e$	$m_\varepsilon(\gamma)$ − first order
10	−0.0276	−0.28	−0.0021	−0.0441	−0.44	−0.0015
20	−0.0179	−0.36	−0.0051	−0.0269	−0.54	−0.0056
40	−0.0089	−0.36	−0.0025	−0.0127	−0.51	−0.0020
80	−0.0046	−0.37	−0.0015	−0.0083	−0.67	−0.0030

The discrepancy between the real loop and the Gibbs measure falls like $k_e^{-0.87}$ (α), $k_e^{-0.83}$ (γ), $k_e^{-0.72}$ (α^2) and $k_e^{-1.09}$ ($\alpha\gamma$) (γ^2 has a first-order coefficient 0.03, below resolution), and the first-order prediction removes most of it: 93% and 97% of the α and γ discrepancies at $k_e = 10$, 70–85% at $k_e = 20$–40; at $k_e = 80$ the residuals (-0.0015 ± 0.0013 and -0.0030 ± 0.0018) are at the Monte Carlo floor. The second-order decay of the residual is *not* resolved: the residuals at $k_e = 20$–80 are 1–3 standard errors with a common sign, compatible with an $O(\varepsilon^2)$ coefficient of order -1 to -2 and with noise; at $k_e = 10$ the even part of the α^2 and γ^2 shifts exceeds its prediction by a factor 1.5–1.8 (an $O(\varepsilon^2)$ term of the size seen in Theorem J's instance A, where $c_2/c_1 \approx 1.4$).

The orientation-odd part (Remark J′.1). Measured $(m_+ - m_-)/2$ over predicted $(p_{1+} - p_{1-})/(2k_e)$; Table 25.

Table 25: Theorem J′ on the single-kernel instance: the orientation-odd part of the stationary means, measured over predicted (`results/dither/table.md`).

k_e	α	γ	$\alpha\gamma$
10	0.92 ± 0.04	0.93 ± 0.04	0.89 ± 0.06
20	1.23 ± 0.09	1.10 ± 0.07	1.28 ± 0.13
40	1.28 ± 0.17	1.15 ± 0.15	1.03 ± 0.26
80	1.25 ± 0.34	1.38 ± 0.29	1.11 ± 0.51

The dither version of Remark H.7a's test: the direction in which the reference is traversed shifts the stationary kernel by $2\varepsilon \int \Theta g_{\mathrm{odd}} \, \mathrm{d}\pi$ (here entirely from b_{osc}), measured to within 7–8% at $k_e = 10$ (where the odd shift is 0.02–0.03 against standard errors of 0.001) and consistent with the prediction within the 10–30% resolution of the runs at $k_e = 20$–80 (the ratios sit 10–30% above one with error bars of the same size: a second-order odd term of relative size $O(\varepsilon)$ is not excluded).

The tracking error (d). $k_e^2\mathbb{E}\|e\|^2/(2\pi(\bar{L})) = 1.054$, 1.033, 1.017, 1.009 (orientation $+1$) and 1.023, 1.014, 1.008, 1.004 (-1) at $k_e = 10$, 20, 40, 80, with standard errors 0.001: the deviation from one is 0.54–0.74/k_e and 0.23–0.35/k_e, the $O(1/k_e)$ of (d).

The circulation (e) and Proposition J′.2. The predicted first-order circulation $\varepsilon[\int \Theta \wedge j_1 + \pi\langle u \wedge \dot{u}\rangle]$ is $-5 \cdot 10^{-4}$ ($+1$) and $+6.5 \cdot 10^{-4}$ (-1) at $k_e = 10$ and scales like $1/k_e$, against a Monte Carlo standard error of $5 \cdot 10^{-4}$ at every k_e: the real loop's circulation is not resolved at first order for $k_e \geq 20$ (measured $-4 \cdot 10^{-4} \cdots + 1.2 \cdot 10^{-3}$, all within 1.7 standard errors of the prediction), and at $k_e = 10$ the measurement ($+3.7 \cdot 10^{-3} \pm 0.5 \cdot 10^{-3}$ for $+1$, $-0.6 \cdot 10^{-3} \pm 0.5 \cdot 10^{-3}$ for -1) is dominated by higher orders (an even $O(\varepsilon^2)$ term of coefficient $\approx +0.2$ and an odd one of the opposite sign to the first-order odd part). The current machinery is therefore checked where the signal is clean, on the first-order averaged SDE itself: its stationary density m_ε and current j_ε, computed on the grid (the discrete current at $\varepsilon = 0$, $-6.6 \cdot 10^{-5}$, a residual of the central scheme, subtracted), give $\int \Theta \wedge (j_\varepsilon - j_0)/(\varepsilon \int \Theta \wedge j_1) = 1.47$, 1.21, 1.09, 1.03 at $\varepsilon = 0.2$, 0.1, 0.05, 0.025 — one plus a term linear in ε — and $\|m_\varepsilon - m_0(1 + \varepsilon g)\|_1/\varepsilon^2 = 0.307$, 0.308, 0.307, 0.305: the first-order expansion of Proposition J′.2(i) with a clean ε^2 remainder.

Corollary J′.3 (`exp_gibbs_m.py` → `results/dither/gibbs_m.md`). On the same instance, μ_τ^* is computed on the 201^2 grid as the fixed point of the first-variation condition (59 damped iterations, residual 10^{-11}): $\int \alpha \, \mathrm{d}\mu^* = 0.4963$, $\int \gamma \, \mathrm{d}\mu^* = 0.1873$, $\int \alpha^2 \, \mathrm{d}\mu^* = 0.9353$, $\bar{L}(\mu_\tau^*) = 0.1394$. The Gibbs measures π_m for $m = 1$, 2, 4, 8, 16, 32 are sampled by MALA (4000 replicas, acceptance 0.98–0.996, standard errors from the independent replicas; the $m = 1$ row agrees with the quadrature of π_1 within one standard error); Table 26.

Table 26: Corollary J′.3 on the single-kernel instance: the one-particle marginals of the Gibbs measures π_m against the mean-field minimiser μ_τ^*, with the rescaled deviations (`results/dither/gibbs_m.md`).

m	$\mathbb{E}_{\pi_m}[\alpha_1^2]$	$\mathbb{E}_{\pi_m}[\gamma_1]$	$\mathbb{E}_{\pi_m}[\bar{L}(\mu_\Theta)]$	$(\mathbb{E}[\alpha_1^2] - \int \alpha^2 \, \mathrm{d}\mu^*) \cdot m$	$(\mathbb{E}[\gamma_1] - \int \gamma \, \mathrm{d}\mu^*) \cdot m$	$(\mathbb{E}[\bar{L}] - \bar{L}(\mu^*)) \cdot m$
1	0.536	0.251	0.1960	−0.40	+0.064	+0.057
2	0.729	0.225	0.1681	−0.41	+0.075	+0.057
4	0.832	0.204	0.1540	−0.41	+0.067	+0.058
8	0.881	0.194	0.1470	−0.43	+0.050	+0.061
16	0.909	0.191	0.1432	−0.42	+0.051	+0.060
32	0.922	0.190	0.1413	−0.42	+0.084	+0.059

The one-particle marginals converge to μ_τ^* with a clean $1/m$ law (slopes -1.01 for γ and -0.99 for the Gibbs-averaged occupation loss, whose rescaled deviation is 0.058 ± 0.002 at every m; the second moment of α moves from 0.54 at $m = 1$ to 0.92 at $m = 32$ against 0.935, with rescaled deviation -0.41 ± 0.01 throughout: the single kernel sits in a narrow well, while in the mean field each particle sees the wide one-body potential $V[\mu_\tau^*]$); the mean of α is within 0.01 of its limit already at $m = 1$ (a small $1/m$ coefficient, its slope not resolved). The stationary law of the finite dithered loop thus approaches the minimiser of Theorem H's free energy in the iterated limit at the rate $1/m$, as the corollary states.

Checked by: `tests/test_dither.py`: the single-kernel averaged data (J, ∇J, b_{fr}, b_{osc}) against the general implementation of Proposition H.7 and Remark H.7a (`averaging.expansion_terms`, `averaging.oscillation_correction`, `periodic.ClosedLoop`) at three parameter points ($2 \cdot 10^{-5}$ relative); the weighted Poisson solver against the Ornstein–Uhlenbeck closed forms ($g = G^{-1}b \cdot \Theta/\tau_0$, mean shift $(G\lambda)^{-1}b$, the Green identity $\int \varphi g \, \mathrm{d}\pi = -\pi(b \cdot \nabla\psi_\varphi)$, zero current and entropy production for a gradient drift); the stationary Fokker–Planck solver of the first-order SDE (OU exact; on the instance the current ratio $\to 1$ and $\|m - \pi(1 + \varepsilon g)\|_1 \propto \varepsilon^2$); the real loop at $k_e = 20$ (8000 replicas: the stationary means of α and γ closer to the first-order prediction than to π, the γ shift resolved at ≥ 3 standard errors, the tracking error within 10% of $2\pi(\bar{L})$); and the Corollary J′.3 numerics (the mean-field fixed point to 10^{-9}, the Gibbs sampler against the quadrature of π_1)

(test_single_kernel_averaged_data_matches_the_general_H7_implementation,
test_poisson_solver_reproduces_the_ornstein_uhlenbeck_closed_forms,
test_first_order_stationary_density_and_current_of_the_averaged_sde,
test_real_loop_means_are_closer_to_the_first_order_prediction_than_to_gibbs,
test_corollary_J3_mean_field_minimiser_and_gibbs_sampler_consistency).

Part III
Evidence and outlook

25 Numerical evidence

Every number in this section is produced by the named script from raw files, and no table is edited by hand. The scripts are in the directory `experiments/` of the accompanying repository and write their output to subdirectories of `results/`; "`script.py` → `dir/`" below names a script and its output directory or file.

- The test files check the inequalities at every sample of simulated trajectories, or on sampled states: `tests/test_transport.py`, `tests/test_langevin.py`, `tests/test_averaging.py`, `tests/test_excitation.py`, `tests/test_nonreciprocity.py`, `tests/test_periodic.py` and `tests/test_barycentric.py`.
- They are run with `venv\Scripts\python -m pytest tests`. The current number of tests is given in the repository's `README.md`.

25.1 Theorem B: the label-free Lyapunov function

Script: `exp_transport_permutation.py` → `transport_permutation/`.

Each network starts as a random relabelling of the ideal configuration plus noise 0.05, with mean-field gains and 3 seeds per m. "Excess" is $|(1/m)\sum$ Taylor remainders$|$, the part that actually enters $\dot{V}$. Table 27 and Figure 4 show the results.

Table 27: Theorem B, permuted-ideal start, mean-field gains; means over 3 seeds. The two bounds are the same bound, evaluated with the matched (OT) error and with the labelled error, along the same trajectory.

m	labelled error ($t = 0$)	OT error W ($t = 0$)	RMS$\|e\|$	actual excess	bound, matched error	same bound, labelled error
8	2.675	0.1070	0.0113	$1.96 \cdot 10^{-5}$	$8.27 \cdot 10^{-2}$	2.93
16	2.532	0.1044	0.0065	$9.58 \cdot 10^{-6}$	$4.37 \cdot 10^{-2}$	1.86
32	2.684	0.1122	0.0079	$2.69 \cdot 10^{-5}$	$7.07 \cdot 10^{-2}$	2.11
64	2.764	0.1136	0.0094	$1.23 \cdot 10^{-5}$	$6.80 \cdot 10^{-2}$	2.43

- The bound is at least the actual excess at every sample of every run.
- The matched error is about 1/25 of the labelled one.
- The bound is 30–45× smaller when evaluated with the matched error.
- Nothing grows with m.
- The bounds remain far above the actual excess, because the Hessian supremum is global.

Corollary B.5 along (L_σ)**.** This is checked by the test `test_corollary_B5_fails_along_lsigma_and_its_leak_variant_holds`. The test asserts every hypothesis of C(iv) and measures $\sup W \le \bar{\eta}$. There, the plain B.5 bound is exceeded by more than 3×, while the leak variant with $\sigma_m b^2/4$ holds.

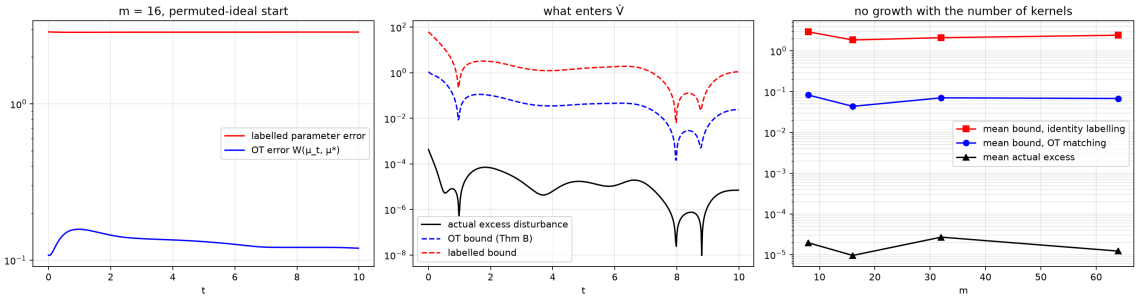


Figure 4: Theorem B, permuted-ideal start. Left: labelled parameter error and OT error $W(\mu_t, \mu^*)$ for $m = 16$. Centre: the actual excess disturbance that enters $\dot{V}$, the OT bound of Theorem B and the labelled bound, for $m = 16$. Right: the mean bound with identity labelling, the mean bound with OT matching and the mean actual excess, against m.

25.2 Theorem C: OT σ-modification

Scripts: `exp_ot_sigma_mod.py` and `exp_sigma_mod_transient.py` → `ot_sigma_mod/`.

The two starts share plant, prior, geometry and x_0. They differ only in particle labels (Table 28, Figure 5). Table 29 shows the reshuffling transient from the relabelled start (5 seeds).

Table 28: Theorem C: OT σ-modification against labelled σ-modification (oracle plant, $m = 6$, 5 seeds, $\sigma_m = 0.3$). RMS$\|e\|$ is taken over 20–40 s and reported as mean ± 95% confidence interval over seeds; W_{end} is the OT error at the end.

start	mechanism	RMS$\|e\|$ (20–40 s)	W_{end}	labelled error at end
aligned	projection	0.0051 ± 0.0010	0.2953	0.2953
aligned	labelled σ-mod.	0.0091 ± 0.0018	0.3040	0.3040
aligned	OT σ-mod.	0.0091 ± 0.0018	0.3040	0.3040
relabelled	projection	0.0051 ± 0.0010	0.2953	0.2953
relabelled	labelled σ-mod.	0.0091 ± 0.0018	0.3040	**2.0590**
relabelled	OT σ-mod.	0.0091 ± 0.0018	0.3040	0.3040

Table 29: Reshuffling transient from the relabelled start ($m = 6$, 5 seeds, $\sigma_m = 0.3$): RMS$\|e\|$ over time windows, and the particle path length.

leak	$[0, 5]$ s	$[5, 10]$ s	$[10, 20]$ s	$[20, 40]$ s	particle path length
labelled	0.0771	0.0086	0.0093	0.0091	2.2585
OT	0.0743	0.0072	0.0093	0.0091	0.3917

- The labelled leak's longer path costs only about 4% of early RMS$\|e\|$. Theorem F does not price leak-driven transport (Proposition C.5).
- The C(iv) instance in the tests satisfies every hypothesis, asserted. V_{OT} stays below the envelope $e^{-\alpha t}V(0) + \beta/\alpha$ at every sample and contracts by more than 5×.
- The C(ii) test asserts $C \le b^2 + 4\eta R$, $\sqrt{C} \le b + 2\eta$, and $C = b^2$ under both gap conditions, on 300 configurations.

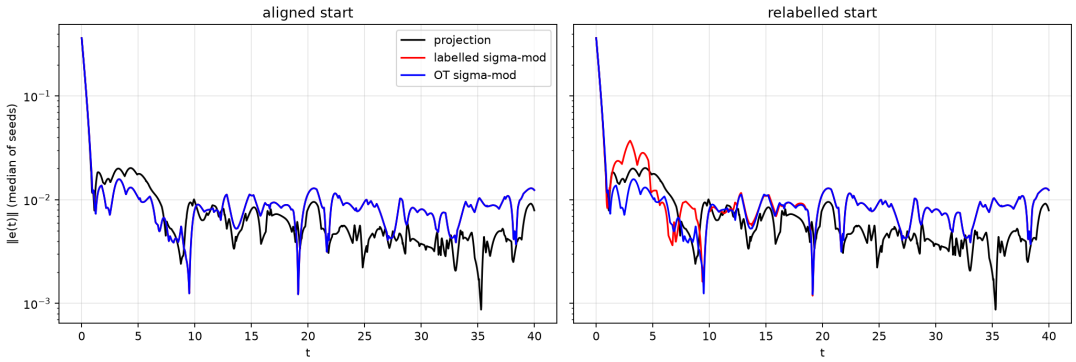


Figure 5: Theorem C: $\|e(t)\|$ (median of seeds) for projection, labelled σ-modification and OT σ-modification, from the aligned start (left) and from the relabelled start (right).

25.3 Theorem C′: the barycentric OT leak

Script: `exp_barycentric_leak.py` → `barycentric_leak/`. Table 30 shows the pure leak toward a prior with more atoms than the network, and Table 31 the closed loop with three leaks of equal gain toward three priors.

Table 30: Theorem C′: pure barycentric leak toward a 48-atom prior ($m = 6$, $n = 1$, lifted dimension 3, $G = \mathrm{diag}(2, 1, 0.5)$, $\sigma_m = 1$, $\mathrm{d}t = 0.01$, 3000 steps). The last column is the best of 20 equal-mass Lloyd runs.

seed	$W^2(\mu_0, \nu)$	$W^2(\mu_T, \nu)$	V^2_{bar} at T	Var_γ at T	W^2 non-increasing	plan switches	best Lloyd
0	2.5526	1.1642	$1.1 \cdot 10^{-26}$	1.1642	yes	3	1.1708
1	2.3032	1.2128	$4.8 \cdot 10^{-26}$	1.2128	yes	2	1.1708
2	2.5807	1.2030	$2.2 \cdot 10^{-25}$	1.2030	yes	6	1.1708

Table 31: Theorem C′: closed loop with three leaks of equal gain ($\sigma_m = 0.5$) toward three priors (representable plant, $m = 4$, $k_e = 4$): (A) the OT leak to the ideal, (B) the barycentric leak to a 32-atom cloud, (C) the labelled leak to a random 4-subset. Each entry of (A)–(C) is RMS$\|e\|$ on $[20, 40]$ s, $W(\mu_T, \mu^*)$. Here $b = W(\mu^*, \mathrm{cloud})$, b_{bar} is the centroid defect of the ideal with respect to the cloud, and "labelled" is the labelled bias to the subset.

seed	b	b_{bar}	$W(\mu^*, \mathrm{subset})$	labelled	(A)	(B)	(C)
0	0.446	0.168	0.647	1.987	0.0000, 0.000	0.0035, 0.167	0.0431, 0.614
1	0.328	0.097	1.027	1.479	0.0000, 0.000	0.0024, 0.097	0.0430, 1.016
2	0.401	0.159	0.395	1.765	0.0000, 0.000	0.0069, 0.157	0.0293, 0.352

- The pure leak stops exactly where $W^2 = \mathrm{Var}_\gamma$ (every particle at the barycentre of its cell; $V^2_{\mathrm{bar}} \le 2.2 \cdot 10^{-25}$): an equal-mass centroidal configuration of the prior. Like Lloyd's algorithm it finds a local minimiser of the quantisation energy, not always the global one (compare the last two columns of Table 30).
- With $m_0 = m$ the same leak gives the exact decay of Theorem C(i): the maximal relative deviation of $W(\mu_t, \mu^0)$ from $(1 - \sigma_m \, \mathrm{d}t)^k W_0$ over 500 Euler steps is $5.1 \cdot 10^{-16}$.
- In the closed loop, the bias of the barycentric leak is governed by b_{bar}, the distance of the ideal from the centroids of the prior's cells: $W(\mu_T, \mu^*) = 0.167, 0.097, 0.157$ against $b_{\mathrm{bar}} = 0.168, 0.097, 0.159$, not by $b = W(\mu^*, \nu) = 0.45, 0.33, 0.40$ (Theorem C′(ii)–(iii)).

The labelled leak to the subset ends at $W(\mu_T, \mu^*) \approx W(\mu^*, \text{subset})$. A rich prior whose centroidal quantisation is close to the ideal is a better anchor than a crude prior with the right number of atoms.

- Tests (`tests/test_barycentric.py`): the plan reduces to the assignment for $m_0 = m$ and $W^2 = V_{\text{bar}}^2 + \text{Var}_\gamma$; the pure leak dissipates at every step with $D^+ Z \le -\sigma_m V_{\text{bar}}^2$ and ends at a Lloyd fixed point; $\sqrt{C_{\text{bar}}} \le b + 2\eta$ on 90 configurations, with $C_{\text{bar}} = b_{\text{bar}}^2$ at small η; the Lyapunov inequality (iii) along a closed loop with a 24-atom prior.

25.4 Theorem D: convergence in m

Script: `exp_mean_field_m.py` → `mean_field_m/`. Table 32 and Figure 6 show the results.

Table 32: Theorem D: mean-field consistency in m (Van der Pol plant, mean-field gains, smooth initial measure quantised at m points). The ratio is $W(\mu_T^m, \mu_T^{2m}) / W(\mu_0^m, \mu_0^{2m})$.

m	$\sup_t \|e^m - e^{2m}\|$	$W(\mu_0^m, \mu_0^{2m})$	$W(\mu_T^m, \mu_T^{2m})$	ratio
16 vs 32	0.0046	0.2179	0.3658	1.68
32 vs 64	0.0009	0.1091	0.1788	1.64
64 vs 128	0.0001	0.0545	0.0920	1.69
128 vs 256	0.0000	0.0273	0.0459	1.68

This is a consistency check in m, not a check of the constant.

- The guaranteed factor $e^{\Lambda T}$ is at least e^{496} on this box.
- The Van der Pol plant fails the tube condition of Theorem D(b), so the run is covered by the localised D(d).

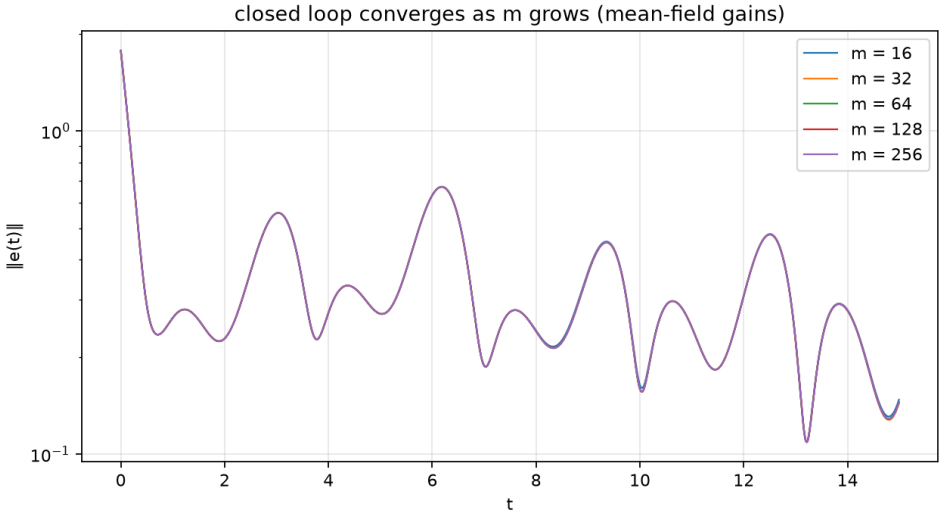


Figure 6: Theorem D: tracking error $\|e(t)\|$ of the closed loop (Van der Pol plant, mean-field gains) for $m = 16$, 32, 64, 128 and 256.

25.5 Theorem D′: propagation of chaos for the dithered loop

Script: `exp_chaos.py` → `chaos/` (raw per-seed values in `raw_sup_D.npz`). The m-particle loop ($n = 1$, $k_e = 10$, $\sigma = 0.02$, $G = \text{diag}(4, 1, 1)$, $T = 20\,\text{s}$, $\text{d}t = 0.001$; Euler–Maruyama for the particles and RK4 for the plant) is coupled synchronously to an $M = 4096$ reference that shares its

first m particles and Brownian motions, and $\sup_t D_m$ with $D_m := \|x^m - x^M\| + \frac{1}{m}\sum_i \|\vartheta_i^m - \vartheta_i^M\|$ is averaged over 24 seeds (Table 33).

Table 33: Theorem D′: synchronous coupling of the m-particle dithered loop to an $M = 4096$ reference, 24 seeds. Slopes are log-log fits, with bootstrap standard errors over seeds.

	$\tau = 0$	$\tau = 2 \cdot 10^{-3}$
mean $\sup_t D_m$, $m = 16 \to 1024$	$1.49 \cdot 10^{-1} \to 2.17 \cdot 10^{-2}$	$1.52 \cdot 10^{-1} \to 2.32 \cdot 10^{-2}$
slope of the mean against m ($m \le 256$)	-0.47 ± 0.06	-0.46 ± 0.05
slope of the median	-0.46	-0.45
slope against $(1/m - 1/M)^{1/2}$ (predicted 1)	0.94 ± 0.08	0.92 ± 0.08

- The $m^{-1/2}$ rate of Theorem D′(b) holds within one standard error at both temperatures; the shared-particle coupling predicts the mean $\propto (1/m - 1/M)^{1/2}$, which the data follow up to $m = M/4$.
- Single seeds differ by more than an order of magnitude at fixed m; a three-seed version of the experiment gave slopes -0.25 and -0.21 and was withdrawn (review round 5). The constant of the theorem ($TK_\Phi e^{\Lambda T}$) is not checked.
- There is no test: the experiment is the check.
- Theorem D″ (`exp_chaos_uniform.py` → `chaos/uniform_table.md`; $n = 1$, $k_e = 25$, $G = 0.05 \cdot \mathrm{diag}(4, 1, 1)$, a two-bump plant that is not odd, $M = 2048$): exact Lipschitz constants $L_x = L_\vartheta = 8.087$, $L_2 = L_{x\vartheta} = 44.4$ and the a-priori tube $E_X = 0.228$ give $\sigma_{\mathrm{cert}} = 4.04$, against Proposition K's critical leak $\sigma_K = 0.00144$ of the collapsed critical point (ratio 2800). At $1.2\sigma_{\mathrm{cert}}$ with dither $\tau = 0.02$ (80 s, 8 seeds) the coupled distance is $6.7 \cdot 10^{-4} \to 7.9 \cdot 10^{-4}$ ($m = 16$) and $1.6 \cdot 10^{-4} \to 1.9 \cdot 10^{-4}$ ($m = 256$) between the first and the last window, slope -0.51 ± 0.02, bound 0.55 and 0.14; the network stays at its prior (fit error 1.000, rms spread 0.033 = the dither cloud). Without dither the reference collapses (spread 10^{-14}) and the distance is zero; at $\sigma_{\mathrm{cert}}/8$ it decays to 10^{-11}. In the learning regime ($\sigma_K/3$, 1000 s, 6 seeds) the network stays spread (rms 1.53) and fits the plant (fit error 0.632), and the distance follows $m^{-1/2}$ (slope -0.62 ± 0.16) but grows threefold between the windows $(100, 200]$ and $(500, 1000]$: no empirical uniformity there (the first draft's "spread regime" run was a slow collapse; review round 6).

25.6 Theorem D‴: uniform-in-time propagation of chaos in the learning regime

Script: `exp_chaos_learning.py` → `chaos/learning_table.md`.

- Instance: $n = 1$, $G = \mathrm{diag}(1, 0.25, 0.25)$, three-bump plant $(1, -1.5, 0.5)$, $(-0.7, 0, 0.4)$, $(0.8, 1.5, 0.5)$, $x_m = 2\sin t$, $\sigma_{\mathrm{crit}} = 0.04716$ (Proposition K), $\sigma_0 = 0.15\sigma_{\mathrm{crit}} = 0.00707$; $L_\vartheta^G = 4.044$, $L_2^G = 11.11$, $K_\Phi = 4$, $F_\rho = 0.999$. Averaged slow loop ($\mathfrak{A}_m$), $m = 16$, 64, 256 against $M = 2048$, 900 slow units, 4 seeds per start; real loop at $k_e = 25$ over 2000 time units, 3 seeds.
- The hypothesis at the three-atom stationary measure (fit error 0.244): κ_R at the atoms 0.00004, -0.00227, 0.00411; over spheres of radius 0.05, 0.1, 0.2, 0.3: 0.00461, 0.00512, 0.00619, 0.00730 — (H) holds with $\varrho' = 0.1$ and margin 0.00195 ($1/\kappa = 512$), the window $\kappa_R[\mu^*](N_{0.1}) = 0.00512 < \sigma_0 = 0.00707 < \sigma_{\mathrm{crit}} = 0.04716$ is realised; at $0.3\sigma_{\mathrm{crit}}$ the three-atom measure fails splitting ($\kappa_R = 0.01696 > 0.01415$). Theorem constants: $C_1 = 2.2 \cdot 10^9$, $m_0 = 1.2 \cdot 10^{23}$ (vacuous envelopes).

- Along the warm run (jittered mixture around the atoms) the margin $a(s)$ stays in $[+0.00139, +0.00167]$; the own-residual statistic exceeds the r^*-one by ≤ 0.00013 ((0.2) allows 0.214); rms drift $6 \cdot 10^{-3} \to 3 \cdot 10^{-6}$. Along the cold run (connected cloud) the margin is negative throughout ($-0.196 \to -0.0025$) while the fit converges to 0.244 as well.
- Warm start: window maxima of D_m 0.396 $\to$ 0.448 ($m = 16$), 0.122 $\to$ 0.138 (64), 0.0119 $\to$ 0.0137 (256) over $(0, 112.5] \to (450, 900]$ (last/first 1.13–1.15, last/previous 1.01–1.02: saturated); slope -1.28 ± 0.30 (mean square; theorem -1), learned function -0.69 ± 0.16; per-particle maxima 1.36, 0.62, 0.21 $\gg 3\varrho'/4 = 0.075$ (Remark D‴.6).
- Proposition D‴.5: $\|S_k\| = 8.38$, 4.66, 8.32; $\mathbb{E}D_\infty$ predicted 0.823, 0.201, 0.0454 against means 0.446 ± 0.28, 0.138 ± 0.051, 0.0136 ± 0.0055 (four seeds, heavy-tailed); per seed the linearised prediction is within 25% of the measured D in 9 of 12 pairs (worst ratio 0.57 at $|w^m - w^M| = 0.275$) and the exact stationary distance of the replicated weighted configuration within 2.3% in all 12; splitting margins positive in 11 of 12 (+0.0006 to +0.0050), negative (-0.0022) for $m = 16$, seed 3 (weights $(9, 4, 3)/16$: a saddle, left later by particle migration — review round 9); per-cluster displacements along $S_k(w^m - w^M)$ with cosines 0.92–1.000 and norms within 0.8–20%.
- Cold start: D grows 1.81 $\to$ 5.33 (16), 0.505 $\to$ 1.79 (64), 0.0744 $\to$ 0.583 (256) (last/first 2.95, 3.54, 7.84), slope -0.87 ± 0.14; bimodal deviations (medians 0.40, 0.14, 0.067; maxima 6.2–6.7); fraction of pairs > 1 whitened unit apart 0.203 ± 0.047, 0.082 ± 0.021, 0.037 ± 0.010, decaying like $m^{-0.64 \pm 0.08}$ (medians $m^{-0.62 \pm 0.07}$): the floor of Remark D‴.2.
- Real loop ($k_e = 25$, warm start): window maxima 0.328 $\to$ 0.590 (16), 0.0605 $\to$ 0.0909 (64), 0.0188 $\to$ 0.0294 (256) over $(0, 500] \to (1000, 2000]$ (slow time $\le 80 = 0.16/\kappa$), slope -1.13 ± 0.15; on the same initial clouds the averaged flow gives $D_{\text{real}}/D_{\text{averaged}}$ between 0.993 and 1.004 at every window end (across different clouds the last-window ratios are 1.57, 0.78, 2.58): the real loop follows its averaged slow dynamics to within 0.7% (Corollary D‴.7); two of the three $m = 16$ clouds have inadmissible or marginal weights (margins -0.0021, -0.00001).

25.7 Proposition D‴.9: measure-valued priors and the design of the window

Script: `exp_prior_design.py` → `chaos/prior_design.md`.

- Same instance and leak as Theorem D‴'s numerics ($\sigma_0 = 0.15\sigma_{\text{crit}} = 0.00707$), nearest-atom leak toward three prior atoms. Matching prior (exact representation): fit error 0, $\kappa_R = 0$, margin $\sigma_0 = 0.00707$ against the σ-modification's 0.00196 (the theorem's relaxation time $1/\kappa$ 141 against 511 slow units; observed late-time e-folding of the coupled distance 48–84 against 107–306); $\|S_k\| = 13.01$, 6.18, 8.91 against 8.38, 4.66, 8.32; $\mathbb{E}D_\infty(m = 64)$ 0.323 against 0.201 (1.61×); function-level floor $9.33 \cdot 10^{-4}$ against $4.29 \cdot 10^{-4}$ (2.18×).
- Misplaced priors: centres +0.15 / +0.3 / +0.45 / +0.6 → margins +0.0055 / +0.0037 / +0.0017 / −0.0006; amplitudes ×0.5 / ×0.7 / ×1.2 → +0.0074 / +0.0073 / −0.0317; widths ×0.7 / ×1.5 / ×2 → −0.0261 / +0.0058 / +0.0040; the shrunk priors (×0.5, ×0.7) have the largest margins and lower floors ($\mathbb{E}D_\infty$ 0.199, 0.230; function level $7.1 \cdot 10^{-4}$, $7.7 \cdot 10^{-4}$) at fit errors 0.09, 0.06.
- Admissible weights (splitting margin of $\Theta^*(w)$): $(7, 5, 4)/16$: +0.0026 / −0.0054 / +0.0073; $(9, 4, 3)/16$: −0.0022 / −0.0058 / +0.0057; $(10, 3, 3)/16$: −0.0038 / −0.0042 / +0.0054; $(0.5, 0.3, 0.2)$: −0.0006 / −0.0065 / +0.0066 (σ-modification / matching / amplitudes ×0.7): the shrunk prior keeps all four admissible (above σ_0 only at $(7, 5, 4)/16$; its over-weighted atom overshoots by 1–3% at the other three), and the matching prior's set is smallest toward the largest bump (an excess 0.04 on atom 1 alone is inadmissible, 0.08 on atom 2 or 3 alone is not).

- Matching-prior coupled runs ($m = 16$, 64, 256 vs $M = 2048$, 300 slow units, 3 seeds): D saturates (last/previous 1.04–1.05), slope -1.23 ± 0.40; per-seed linearised-law ratios 0.81–1.11 (parameter level) and 0.61–1.69 (function level), exact law 0.987–0.998 in all nine pairs; two of nine realised weight vectors are saddles (margins -0.0005, -0.0030), left by a local split inside the cell with the coupled distance on its plateau (round 10); observed e-folding times 48–84 slow units (median 59) against 107–306 (median 195) for the σ-modification.
- Function-level law for the σ-modification against Theorem D‴'s warm runs: predicted RMS 0.0419, 0.0207, 0.0098 vs measured means 0.0307 ± 0.0097, 0.0154 ± 0.0033, 0.0047 ± 0.0012 (ratios 0.73, 0.74, 0.48; RMS over seeds/predicted 0.84, 0.79, 0.52; per-seed law from the realised weights 0.60–1.57).

25.8 Proposition D‴.12: the OT leak's mass constraint against the weight fluctuation

Script: `exp_ot_leak.py` → `chaos/ot_leak.md`.

- Same instance and matching prior as Proposition D‴.9's numerics ($\sigma_0 = 0.15\sigma_{\text{crit}}$, $q = (1/3, 1/3, 1/3)$, $\Delta_\nu = 6.26$); the nearest-atom and the barycentric OT leak on the same warm clouds, 600 slow units ($4.2/\sigma_0$), 4 seeds; $m = 15$, 60, 240 vs $M = 510$ (commensurate) and $m = 16$, 64, 256 vs $M = 512$ (incommensurate).
- Cluster weights: conserved exactly under the nearest-atom leak (max deviation 0 in 32 systems); under the OT leak the mass in transit $\frac{1}{2}|w(0) - q|_1 = 0.117$, 0.058, 0.027, 0.024 (seed means; 1–21 particles per system) reaches zero and the weights equal q from $s_{\text{eq}} = 93$–164 slow units ($\ln 2/\sigma_0 = 98$). Fit error at the end (seed means): 0.067, 0.029, 0.017, 0.012 (nearest) $\to 2.8 \cdot 10^{-4}$, $1.2 \cdot 10^{-4}$, $1.2 \cdot 10^{-4}$, $0.8 \cdot 10^{-4}$ (OT; the migrated particles still 0.06–0.16 from their atoms, the OT finals not yet stationary).
- Commensurate sizes, last window: function-level distance^2 $2.5 \cdot 10^{-3}$, $5.2 \cdot 10^{-4}$, $5.9 \cdot 10^{-5}$ (nearest; law predicts $4.0 \cdot 10^{-3}$, $9.1 \cdot 10^{-4}$, $1.4 \cdot 10^{-4}$) vs $4.9 \cdot 10^{-7}$, $4.9 \cdot 10^{-8}$, $7.5 \cdot 10^{-9}$ (OT, still falling $\times 0.02$–0.035 per 150 slow units); W^2 5.1, 2.7, 0.81 (nearest; cross-cluster transport $\frac{1}{2}|w^m - w^M|_1 \Delta_\nu^2 = 4.9$, 2.5, 0.70; slope -0.67 ± 0.13) vs $7.9 \cdot 10^{-3}$, $2.3 \cdot 10^{-3}$, $6.9 \cdot 10^{-4}$ (OT, $\times 0.12$ per 150 slow units); synchronous D 0.87, 0.25, 0.017 (nearest) vs 5.8, 3.3, 0.94 (OT: the reassigned particles; at 0.95–0.96 of the mismatch plateau and still rising). Exact $K \times K$ transport between the weighted stationary measures at the realised weights: 0.997–0.999 of the measured W^2 at the seven admissible pairs.
- Incommensurate sizes: one or two split (bridge) particles per system at 0.28–0.58 Δ_ν from the nearest atom; against ν (the mean-field stationary measure): $W^2(\mu_m, \nu) = 1.21$, 0.33, 0.079, 0.038, $m \cdot W^2 = 19.4$–21.0 (slope -0.98 ± 0.03), fit error 0.028, 0.0093, 0.0023, 0.0011, $m \cdot$ fit = 0.45–0.60 (squared slope -1.81 ± 0.05); against the $M = 512$ reference (two bridge particles of its own) OT $3.4 \cdot 10^{-4}$, $3.1 \cdot 10^{-5}$, $1.1 \cdot 10^{-6}$ (function level; $m = 16$ still relaxing, plateau $2.8 \cdot 10^{-4}$ in round 11's continued run) and 1.16, 0.29, 0.041 (W^2) vs nearest $2.5 \cdot 10^{-3}$, $5.0 \cdot 10^{-4}$, $4.4 \cdot 10^{-5}$ and 4.8, 2.7, 0.69; nearest fit errors 0.061, 0.035, 0.013, 0.012.
- Masses as a design input: $q_\delta = (1/3 + \delta, 1/3 - \delta/2, 1/3 - \delta/2)$ gives fit error 0.0069–0.049 for $\delta = 0.01$–0.1, splitting margin $+0.0048$, $+0.0028$, $+0.0011$, -0.0015, -0.0049 (inadmissible between $\delta = 0.03$ and 0.05), diagonal plan unique at every row ($\varepsilon \leq 1.04 < 3.13$).
- Tests (`tests/test_ot_leak.py`): the plan is a vertex with the prescribed masses (integral when commensurate, $\leq K_0 - 1$ split rows otherwise) and the single-atom prior is the σ-modification; the leak is the G-gradient flow of J_m^{OT} against finite differences; the replicated weighted configuration is stationary for three commensurate sizes with the diagonal plan

and the same learned function ($W^2 = 0$ between sizes), for matching and mismatched masses; conservation against constraint on a short run; the forest bound; the exact W^2 by replication against the transportation LP.

25.9 Corollary D‴.14: the real loop with the OT leak

Script: `exp_ot_real_loop.py` → `chaos/ot_real_loop.md`.

- Real loop at $k_e = 25$ ($\sigma = \sigma_0/k_e$, RK4 plant with the control and the reference evaluated at the stages, Euler parameters, $\mathrm{d}t = 10^{-3}$, 100 time units $= 16$ periods) from the clouds at the end of Proposition D‴.12's averaged runs (contracted where the weights are admissible) (seeds 0, 1; $m = 15$, 60, $M = 510$) and from the collapsed lifts; last five periods on the reference orbit.
- OT leak: $\|\hat{f}_m - \hat{f}_M\|^2_{L^2(\mathrm{orbit})} = 7.5 \cdot 10^{-9}$–$2.7 \cdot 10^{-8}$, against the 3-atom lift $5.2 \cdot 10^{-10}$–$3.6 \cdot 10^{-8}$; fit error along the orbit $3.5 \cdot 10^{-5}$–$2.9 \cdot 10^{-4}$; rms tracking error $0.9 \cdot 10^{-6}$–$7.3 \cdot 10^{-6}$ (the lift $8 \cdot 10^{-12}$); $W^2(\mu_m, \mu_M)$ $3.3 \cdot 10^{-4}$–$1.4 \cdot 10^{-3}$; within-class spreads 0.013–0.033.
- Nearest-atom leak: $\|\hat{f}_m - \hat{f}_M\|^2 = 1.1 \cdot 10^{-4}$–$2.5 \cdot 10^{-3}$; fit errors 0.012–0.075; rms tracking errors $3.2 \cdot 10^{-4}$–$1.9 \cdot 10^{-3}$; $W^2(\mu_m, \mu_M)$ 1.6–5.5. Three of six systems have inadmissible realised weights (margins -0.0035, -0.0011, -0.0022): their lifts are saddle orbits and the systems have split (spread 0.04–0.11; lift-to-lift ratios 0.960, 0.998, 0.961); at the one pair with both weights admissible the disagreement is the lifted orbits' distance to 1.0003, the admissible systems within $2.5 \cdot 10^{-11}$ of their lifts (spread 0.0006–0.0013).
- Tests (`tests/test_ot_leak.py`): the collapsed 15-system and the 3-atom loop integrate identically under the OT leak at masses q (10^{-12} over 300 steps); the collapsed system with weights $(6, 5, 4)/15$ stays collapsed under the nearest-atom leak and loses a particle under the OT leak; two commensurate lifts keep $W^2 = 0$ and the same learned function; the lifted orbit's class separation subspaces are invariant and carry the ghost spectrum (Proposition D‴.15, $m = 6$).

25.10 Corollary D‴.14 in two dimensions

Script: `exp_ot_2d.py` → `chaos/ot_2d.md`.

- Planar plant (four Gaussian bumps with vector amplitudes), Lissajous reference, $p = 5$ parameters per kernel, matching prior (masses $\frac{1}{4}$), $k_e = 25$, $\sigma_0 = 0.05$, 2500 time units $= 100$ slow units, sizes 8, 32 vs 128 from the same jittered clouds, two seeds.
- OT leak: weights equalised at 13–14 slow units ($\ln 2/\sigma_0 = 14$); $\|\hat{f}_m - \hat{f}_M\|^2 = 5.0 \cdot 10^{-8}$–$1.6 \cdot 10^{-7}$; fit error along the orbit $2.1 \cdot 10^{-4}$–$8.7 \cdot 10^{-4}$; rms tracking error $3.9 \cdot 10^{-6}$–$1.6 \cdot 10^{-5}$; $W^2(\mu_m, \mu_M)$ $2.0 \cdot 10^{-4}$–$4.4 \cdot 10^{-4}$.
- Nearest-atom leak: weights conserved; $\|\hat{f}_m - \hat{f}_M\|^2 = 5.5 \cdot 10^{-3}$–$2.3 \cdot 10^{-2}$; fit errors 0.11–0.39; rms tracking errors $1.9 \cdot 10^{-3}$–$6.7 \cdot 10^{-3}$; $W^2(\mu_m, \mu_M)$ 4.7–11.8.

25.11 Proposition D‴.15: the transverse spectrum of a lifted orbit

Script: `exp_lifted_orbit.py` → `chaos/lifted_orbit.md`.

- Real loop at $k_e = 25$; monodromies by central differences (DOP853, rtol 10^{-11}). OT leak at the matching masses: the 3-atom orbit (the exact representation, zero tracking error) lifted to $m = 6$, 9, 15 — return residuals $\leq 1.7 \cdot 10^{-15}$; separation exponents $\sigma_0/k_e = 2.830 \cdot 10^{-4}$ in every class (= ghosts; invariance defects $\leq 1.8 \cdot 10^{-8}$); collective exponents $2.935 \cdot 10^{-4} \ldots 7.934 \cdot 10^{-3}$ identical across m to 10^{-4} relative, Theorem J(b)'s prediction $2.952 \cdot 10^{-4} \ldots 8.091 \cdot 10^{-3}$ (0.6–3.2%).

- σ-modification at $(7,5,4)/16$ (margin $+0.00257$) and $(9,4,3)/16$ (-0.00219), lifts to $m = 16$ (residuals $\le 4.9 \cdot 10^{-14}$): separation exponents = ghosts to $2 \cdot 10^{-5}$ relative, $= (\sigma_0 - \lambda_j(R_k^G))/k_e$ within 0.2–4.1% (defects $\le 5.7 \cdot 10^{-8}$); the saddle's class 0 exponent $-9.13 \cdot 10^{-5}$ (predicted $-8.77 \cdot 10^{-5}$, multiplicity 8); period-start offsets from $\Theta^*(w)$ $1.1 \cdot 10^{-2}$, $1.6 \cdot 10^{-2}$ (Theorem J(a)).

25.12 Remark D‴.16: dither and the mass constraint

Script: `exp_dither_ot.py` → `chaos/dither_ot.md`.

- Dithered averaged slow loop (Langevin in slow time, Gibbs law $\exp(-J_m/\tau_0)$), $m = 15$ from the counts $(6,5,4)$, matching prior, $\sigma_0 = 0.15\sigma_{\text{crit}}$; $\tau_0 = 0.005$, 0.01, 0.02 over 50 000, 200 000, 50 000 slow units; two seeds.
- Nearest-atom leak: record-level label changes $1.0 \cdot 10^{-5}$, $5.1 \cdot 10^{-4}$, $5.6 \cdot 10^{-3}$ per slow unit (about half undone at the next change; Arrhenius barrier 0.048, 0.043–0.052 at 95%, among the ridge minima 0.040–0.074); time at the balanced counts 0.00, 0.54, 0.25 against the energy–multinomial law 0.86, 0.58, 0.33 (standard errors ≈ 0.1 at $\tau_0 = 0.01$; the law under-weights the saddle count vectors 2.7–3.2 times).
- OT leak: balanced counts in 100%, 99.4%, 88% of the records after the initial transit (99.6%, 99.2%, 88% of the whole runs); the excursions are returns of the crossing particle (121 of 121 at $\tau_0 = 0.01$, 298 of 336 at 0.02, the rest exchanges); off-balance fraction $6.3 \cdot 10^{-3}$, 0.117, effective barrier 0.058: the mass constraint makes a crossing reversible without raising its barrier.

25.13 Theorem D‴.18: Theorem J uniformly in m

Script: `exp_uniform_basin.py` → `chaos/uniform_basin.md`.

- Real loop at $k_e = 25$ ($\sigma_0 = 0.15\sigma_{\text{crit}}$, $\mathrm{d}t = 5 \cdot 10^{-3}$), 8000 time units, many systems integrated together.
- The test of the rates (Part F, by shooting): the 3-atom orbit's parameter Floquet exponents within 0.56–3.22% (matching prior) and 0.57–3.15% (prior centres $+0.15$) of $\lambda_j(H_q)/k_e$; the ghosts' exponents $\sigma_0/k_e = 2.830 \cdot 10^{-4}$ (matching) and $(\sigma_0 - \lambda_j(R_k^G))/k_e$ to $3 \cdot 10^{-4}$ relative (shifted; slowest $2.2701 \cdot 10^{-4}$ against $2.2696 \cdot 10^{-4}$). The proof's radius is at most 0.019 (sampled $C_N \ge 0.092$): the runs start 31–261 times outside the proved start radius.
- Matching prior at the matching masses (constant lift), sizes 3, 15, 60, 240, per-particle random displacements ρ: inside the cells ($\rho = 0.3$, 1.0, 2.5) no particle leaves its cell and both leaks are identical; fitted decay rates of the per-particle distance 2.69–$2.97 \cdot 10^{-4}$ for $m \ge 15$ ($\sigma_0/k_e = 2.83 \cdot 10^{-4}$; ratios 0.95–1.05), constants $C(8000) = 1.03$–1.29 (still growing at the horizon for $m = 15$ at $\rho = 0.3$, 1); at $m = 3$ (no separation modes) the runs are mixtures of collective modes ($2.948 \cdot 10^{-4}$ when the slowest carries the end, $\lambda_{\text{coll}}/k_e = 2.952 \cdot 10^{-4}$; $4.06 \cdot 10^{-4}$ when the third does).
- Across a cell boundary ($\rho = 4.5$; 1, 3, 13, 55 particles start in another cell): the nearest-atom leak ends at counts $(2,0,1)$, $(6,5,4)$, $(23,19,18)$, $(87,70,83)$ — at $m = 60$, 240 through crossings during the transient — with W^2 12.3, 2.81, 2.30, 1.92, already that of the lifts of the realised weights (3.21, 2.35, 1.93 for $m = 15$, 60, 240) though the clouds have not collapsed; $(6,5,4)/15$ and $(23,19,18)/60$ are inadmissible (margins -0.00348, -0.00161). The OT leak ends at q for every m (last count change at $t \le 2695$), W^2 0.010–0.23 falling at the leak's rate $2\sigma_0$ per slow unit (the transit is over); the plan relabels 4, 15, 53 particles out of their initial class's cell.

- One particle across a boundary (Remark D‴.19; mean-square displacement 0.311, 0.078 at $m = 60, 240$): W^2 grows to 0.79, 0.20 under the nearest-atom leak (the lifts of $(19, 21, 20)/60$ and $(79, 81, 80)/240$: 0.8655, 0.2163) and falls to $3.0 \cdot 10^{-3}$, $8.2 \cdot 10^{-4}$ under the OT leak.
- Moving orbit (OT leak, prior centres +0.15): second-half fits 2.18–$2.63 \cdot 10^{-4}$ against $2.27 \cdot 10^{-4}$ (ratios 0.96–1.16; mixtures of separation modes, consistent with the bound $(1 - \kappa)\mu$), constants 1.04–1.23.
- Tests (`tests/test_uniform_basin.py`, at the prior with centres shifted by +0.15, where the residual is not zero and $\lambda_{\text{sep}} = 0.80\sigma_0$): the drift's second-order remainder $\leq C_N |||u|||^2$ with $C_N < 0.1$ for random per-particle displacements at $m = 6$, 30, 150, 600, not growing with m; the displaced particle's remainder exactly $N_a + N_b/m$ with N_a the same for every m (non-zero here, zero at the matching prior), the others' exactly $O(1/m)$; a separation perturbation moves only its own particles at first order, with the response $(GR_k - \sigma_0 I)v$ at $m = 6$, 60, 300; the averaged loop contracts $|||\cdot|||$ at $\mu = \min\{\lambda_{\text{coll}}, \lambda_{\text{sep}}\}$ with one constant $C \leq 1.05$ for $m = 6$, 30, 150.

25.14 Proposition D‴.20: the entropic OT leak

Script: `exp_entropic_leak.py` → `chaos/entropic_leak.md`.

- Matching prior, $\sigma_0 = 0.15\sigma_{\text{crit}}$; the entropic plan by semi-dual Newton; every weighted stationary point continued from a converged neighbour and checked to 10^{-9}.
- Critical excess at $w_\delta = q + \delta(1.5e_1 - 0.5)$, by continuation in ε (residuals $\leq 8.7 \cdot 10^{-13}$): $\varepsilon_c = 5.11$, 13.5, 23.1, 65.0 for $\delta = 0.02$, 0.05, 0.1, 0.2; at ε_c the outflow $p = 0.057$, 0.15, 0.33, 0.57 splits about equally between the two under-weighted atoms (Cov_1^G of rank two); the exact threshold $2\sigma_0\lambda_{\max}(A^{-1/2}\,\text{Cov}_1^G\,A^{-1/2})$ equals ε_c; the rank-one formula 1.22–1.31 (atom 2) and 0.70–1.22 (atom 3) times ε_c. Excess from atom 3 alone: $\varepsilon_c = 6.99$, 14.6, 24.3 for $\delta = 0.02$, 0.05, 0.1. At fixed ε: $\delta_c = 0.0042$, 0.0083, 0.016, 0.030, 0.062, 0.14, 0.20 at $\varepsilon = 1$, 2, 4, 8, 16, 32, 64, equal to $w_1 p_c$ up to $\varepsilon = 8$.
- Dynamics ($m = 15$, 60, 240 vs $M = 510$, 600 slow units, 4 seeds; $\varepsilon = 4$, 16, 64 continued to 6000): $\varepsilon = 1$ — all equalised, fit errors 1.7–$3.0 \cdot 10^{-4}$; $\varepsilon = 4$ — five of the twelve larger systems at unstable weights at $s = 600$, every system at stable weights at $s = 6000$ with residual excesses up to 1/60, 3/240, 7/510 ($\delta_c(4) = 0.012$–0.016); $\varepsilon = 16$ — the stable systems keep their sampled weights, three unstable ones move ($(6, 5, 4)/15 \to (5, 6, 4)$: the excess moved to an atom at q, not reduced); $\varepsilon = 64$ — only $(8, 4, 3)/15$ moves, fit errors 0.12–0.14 (the barycentric bias).
- Floors: Proposition D‴.5's law at $\Theta_\varepsilon^*(q)$ is 0.55 ($\varepsilon = 16$) and 0.74 ($\varepsilon = 64$) of the nearest-atom law; observed 0.23–0.84 and 0.47–0.91 of the nearest-atom leak's floors on the same clouds, per seed.
- Tests (`tests/test_entropic_leak.py`): marginals and the agreement of the two solvers; the gradient of J_m^ε against finite differences; the limits $\varepsilon \to 0$ and $\varepsilon \to \infty$ at their rates (the OT plan's uniqueness checked); the separation blocks of the finite-difference Hessian at a collapsed configuration equal to $I \otimes (\sigma_0(I - (2/\varepsilon)\,\text{Cov}^G) - R^G)$ entrywise, the separation subspace invariant; the over-weighted class of $(6, 5, 4)/15$, continued in ε with $A \succ 0$ checked and every residual below 10^{-9}, unstable at $\varepsilon = 16$, 17 and stable at 18, 20, 64.

25.15 Theorem D‴.21: a measure-level basin for the OT leak

Script: `exp_ot_basin.py` → `chaos/ot_basin.md`.

- Averaged slow loop at $\sigma_0 = 0.15\sigma_{\text{crit}}$, the warm clouds of Proposition D‴.12, $m = 15$, 60, 240, 510, four seeds, 600 slow units, the matching prior and the prior with centres $+0.15$, both leaks. The identities (E), (P) hold along every run to $6 \cdot 10^{-13}$ relative.

- OT leak: no particle relabelled; the margin α positive from $s \le 300$ on (from $s \le 25$ in 19 of 32 runs); the observed ratio at least λ at every record with $\alpha > 0$ (smallest ratio/λ 1.00 at the matching prior, 1.61–1.72 at the shifted one); E from $4.1 \cdot 10^{-3}$–$4.0 \cdot 10^{-2}$ to $8.3 \cdot 10^{-7}$–$8.1 \cdot 10^{-6}$ ($W^2(\mu_\Theta, \mu^*)$ from 0.99–5.25 to $2.3 \cdot 10^{-4}$–$2.3 \cdot 10^{-3}$), below the a-posteriori bound; at the end the observed rate is 0.01416–0.01417 against $2\sigma_0 = 0.01415$ and $\lambda = 0.0140$–0.0141 (matching prior: sharp); at the shifted prior $\lambda = 0.0077$–0.0078 against the observed 0.0137–0.0139, the local PL constant $2\lambda_{\text{sep}} = 0.01135$ (curvature cost 0.802, the slack of (ii) 0.683).

- Nearest-atom leak: no particle changes cell. At the 22 admissible count vectors $E(600)$ is 0.02–0.24% below the energy of the replicated configuration of the sampled weights and rising toward it — a critical point of the nearest-atom energy (drift $\le 6 \cdot 10^{-14}$) but not of the OT energy (drift 2.3–$4.4 \cdot 10^{-2}$); at the 10 inadmissible ones ($m \le 60$; class-0 margins -0.0004 to -0.0061) class 0 splits and E ends 0.5–16.5% below it, still falling.

- The a-priori basin: $E_0 = 2.0 \cdot 10^{-7}$ at the matching prior ($\kappa = \sigma_0$); at the shifted prior (H*) holds on the whole parameter space ($\sup \lambda_{\max} R^G[\mu^*] = 6.60 \cdot 10^{-3} < \sigma_0$, at a saturation corner; $\kappa = 4.7 \cdot 10^{-4}$, $E_0 = 9.1 \cdot 10^{-10}$). $\{E \le E_0\}$ holds a fraction $\le 5.8 \cdot 10^{-6}$ (matching) and $4.4 \cdot 10^{-7}$ (shifted) of the particles in other cells.

- The measure-level basin (matching prior): a fraction p of the particles moved 0.6 of the way into another cell stays in its class; $\kappa^\Theta = 0.0096$, 0.019, 0.029, 0.048 and $\alpha = +0.0023$ at $p = 1/30$, negative from $p = 1/15$ (the same at $m = 30$ and $m = 300$: the same measures). One particle moved: E $3.3 \cdot 10^{-3} \to 2.5 \cdot 10^{-5}$ ($m = 30$) and $2.3 \cdot 10^{-4} \to 3.1 \cdot 10^{-6}$ ($m = 300$), at $2\sigma_0$ at the end, below $E(0) \exp(-\int \lambda_+)$, the particle back in its cell; under the nearest-atom leak E rises to the spurious minima's $6.3126 \cdot 10^{-3}$ and $6.1328 \cdot 10^{-4}$ by $s = 3000$.

- Tests (`tests/test_ot_basin.py`): (E) and (P) to 10^{-12} for random assignments with the counts mq at both priors, and their exact defects at other counts; the PL chain; the constant $2\sigma_0$ attained by a separation to 10^{-4}; the decay from a particle in another cell, with a positive margin, below the a-posteriori bound, at $2\sigma_0$ at the end, against the nearest-atom floor; the spurious minimum $(11, 9, 10)/30$ of the nearest-atom energy, admissible (separation margins and K-atom Hessian positive), not critical for the OT leak.

25.16 Theorem D‴.23: adaptive priors

Script: `exp_adaptive_prior.py` → `chaos/adaptive_prior.md`.

- Averaged slow loop at $\sigma_0 = 0.15\sigma_{\text{crit}}$, the OT leak toward the prior with centres $+0.15$ (masses 1/3); the prior's atoms move toward the class barycentres at the rate η.

- Linear theory at the exact representation: $h_j = 3.06 \cdot 10^{-4} \ldots 0.195$; a fixed prior keeps $\sigma_0/(h_j + \sigma_0) = 0.959 \ldots 0.035$ of its error and relaxes at $h_j + \sigma_0 \ge 7.38 \cdot 10^{-3}$; the adaptive prior's slowest rate $\lambda_1^- = 2.68 \cdot 10^{-5}$, $1.51 \cdot 10^{-4}$, $2.78 \cdot 10^{-4}$, $3.06 \cdot 10^{-4}$ for $\eta = \sigma_0/10$, σ_0, $10\sigma_0$, ∞. (v)'s radius: $C_N \ge 0.341$ (sampled), so $\rho \le 2.0 \cdot 10^{-5}$, $1.1 \cdot 10^{-4}$, $2.0 \cdot 10^{-4}$, $2.2 \cdot 10^{-4}$, while the K_0-atom runs start 94–$6 \cdot 10^4$ times outside, the warm clouds $2.2 \cdot 10^4$–$3.6 \cdot 10^5$ times and the separated starts 131–3344 times. The fixed prior's bias: fit error $9.78 \cdot 10^{-3}$, within 4.8% of the first-order formula in norm, 0.85 and 0.73 of it in the weak modes 1 and 3; the error is first order in the shift (0.58% in norm, 0.98 and 0.97 in modes 1 and 3, at 1/8 of the shift).

- K_0-atom loop from $\Theta^*(q)$, 6000 slow units: every mode above 10^{-6} (13 of 36) decays at λ_j^- (log-slopes 0.981–1.003 of it; ratios to the pair solutions 0.82–1.12, constant to 1.04% from $s = 1000$ on for $\eta \geq \sigma_0$); fit error $9.78 \cdot 10^{-3} \to 3.6 \cdot 10^{-4}$, $1.8 \cdot 10^{-5}$, $8.0 \cdot 10^{-6}$, $6.7 \cdot 10^{-6}$.
- $m = 15$, 60, 240, 510 from the warm clouds, four seeds, 3000 slow units. Fixed prior: the lift of $\Theta^*(q)$ ($\|\bar{y}\| = 2.10 \cdot 10^{-2}$, $W^2 = 4.4 \cdot 10^{-4}$). Ungated: the prior dragged by the transit ($\|\bar{y}\|$ 0.032–0.32 at $\eta = \sigma_0$, 0.048–0.71 at $\eta = \infty$, at $s = 500$); one $m = 15$ system at $\eta = \infty$ on a plateau of the unregularised loss (Hessian eigenvalues $-1.8 \cdot 10^{-5}$, $7.2 \cdot 10^{-5}$; fit error $3.46 \cdot 10^{-2}$ at $s = 23000$), which the unregularised gradient flow crosses to the exact representation (below 0.9 of the plateau's loss after $4.16 \cdot 10^4$, half at $4.78 \cdot 10^4$, below 10^{-12} at $7.97 \cdot 10^4$ more slow units, the crossings located as events). Transit gate (opens at $s = 88$–149): $\|\bar{y}\|$ $4.9 \cdot 10^{-3}$–$3.7 \cdot 10^{-2}$ at $\eta = \sigma_0$ against the K_0-atom run's $1.3 \cdot 10^{-3}$. Settled gate (opens at $s = 1211$–1304, the systems replicated: rms separations $\leq 7.9 \cdot 10^{-4}$): $\|\bar{y}\|$ and the fit error equal to the K_0-atom run's at $s - t_{\text{open}}$ for every m (per-seed ratios 0.995–1.000 at $\eta = \sigma_0$, 0.968–1.001 at $\eta = \infty$: the invariance of the replicated configurations); fit errors $1.6 \cdot 10^{-4}$ and 3.4–$3.5 \cdot 10^{-5}$, W^2 6.7–$7.2 \cdot 10^{-6}$ and 9.3–$9.9 \cdot 10^{-7}$; in the seed means the settled gate is the lower except the fit error at $\eta = \sigma_0$ and $m = 510$ ($1.64 \cdot 10^{-4}$ against the transit gate's $1.55 \cdot 10^{-4}$), and system by system the transit gate ends lower in all three of $\|\bar{y}\|$, W^2 and the fit error in 6 of 32.
- Separated starts (the structure behind (v), beyond its ball; five draws): $\|\bar{y}\|$ at $s = 1750$ departs from the K_0-atom run's by +0.13% and +3.1% (rms separation 0.02 and 0.1) at $\eta = \sigma_0$ and +0.77% and +18.5% at $\eta = \infty$ for $m = 15$ in the draw means ($m = 510$: +0.09%, +2.2%, +0.63%, +15.5%): second order in the separations draw by draw (factors 23.8–25.0), consistent with the same expected size at both m (the means, higher at $m = 15$, within 0.9–1.7 combined standard errors; five draws), the spread 3.3–6.0 times larger at $m = 15$.
- The limits of (i): a non-replicated stationary saddle on a four-bump target (one class split, held by a leak force $5.8 \cdot 10^{-3}$); from the exact split fit the flow collapses the class (width 1.05, fit error $2.23 \cdot 10^{-2}$ at $s = 4 \cdot 10^4$), and the merged width coordinate then runs off like $\ln s$, the width tending to $s_{\max}$ like $1/s$: the solution is unbounded, its physical image approaching the minimiser of the loss on the face $s = s_{\max}$ (fit error $1.943 \cdot 10^{-2}$, a Karush–Kuhn–Tucker point with $\partial L/\partial s = -1.3 \cdot 10^{-6}$); beyond the saturation (weighted amplitude $4.8 > a_{\max}$) the fixed prior's solution converges while the adaptive prior's amplitude coordinate grows like $2 \ln s$; a prior permuted among atoms of equal mass is stationary at the exact representation.
- Tests (`tests/test_adaptive_prior.py`): the joint drift is the gradient flow of $\mathcal{J}$ in the stated metric (finite differences); the exact representation stationary, the fixed prior's stationary point not; the linearisation's spectrum (σ_0 and the pair rates $\lambda_j^{\pm}$) and its self-adjointness in the gradient-flow metric; the first-order bias; the transit gate and the settled gate, with $\mathcal{J}$ non-increasing; the non-replicated saddle, the permuted prior and the adaptive prior's run-off (the force on the amplitude tending to the positive multiplier of the loss's face minimum, and the run-off law $(a_{\max} - a)s \to a_{\max}^2(\sigma_0 + \eta)/(4\eta G_\alpha F_\infty)$).

25.17 Proposition D‴.24: the floor is the leak's bias; an adaptive prior without a mass constraint

Script: `exp_adaptive_floor.py` → `chaos/adaptive_floor.md`.

- Averaged slow loop on the three-bump instance at $\sigma_0 = 0.15\sigma_{\text{crit}}$ ($0.05\sigma_{\text{crit}}$ and $0.5\sigma_{\text{crit}}$ for the law), the nearest-atom leak toward the matching prior unless stated; the warm clouds (four seeds), $m = 15$, 60, 240 against $M = 510$; the checks at the sampled weights from the stored final states.

- The floor as the bias (no dynamics): F equals $\hat{J}\sigma_0(H_L+\sigma_0 I)^{-1}\partial_w x^0$ to the finite-difference error ($1.1\cdot10^{-6}$, $1.9\cdot10^{-7}$, $1.2\cdot10^{-8}$; exact at q); the law (B) per unit of $1/m-1/M$ is $1.356\cdot10^{-2}$, $6.166\cdot10^{-2}$, 0.2291, the sum of its mode terms, at 0.496, 0.753, 0.839 of the bound $(\sigma_0/4)\,\mathrm{tr}(\partial_w x^0\tilde{C}\partial_w x^{0\top})$, itself $\sigma_0/4$ times the adaptive prior's coupled-distance law 46.32. Modes 2–6 carry 99.4%, 99.0%, 97.9% of the law and 97.2% unweighted; the weight (a band-pass centred at $h=\sigma_0$) moves the shares of the modes with $h_j/\sigma_0\in[1/5,5]$ from 21%, 97%, 78% to 42%, 99%, 90%. The law's value at $m=15$, 60, 240: $3.99\cdot10^{-3}$, $9.07\cdot10^{-4}$, $1.36\cdot10^{-4}$. The parameter split $S=H_L(H_L+\sigma_0 I)^{-1}S^e$ holds to $\le1.4\cdot10^{-7}$; the adaptive prior's coupled-distance law is 1.44, 2.17, 5.27 times the fixed prior's.

- Two notions of admissibility. Representable ($w>(0.25,0.175,0.2)$): every sample but $(8,4,3)/15$, on the boundary. D‴.5-admissible for the fixed prior: 7 of the 12 m-systems; $(6,5,4)/15$, $(6,3,6)/15$, $(24,15,21)/60$, $(8,4,3)/15$, $(25,17,18)/60$ are saddles (margins -3.48, -1.08, -2.16, -6.14, $-3.98\cdot10^{-3}$, class 1, $w_1\ge0.4$); the references and the others $\ge3.78\cdot10^{-3}$.

- Fixed against adaptive priors, 6000 slow units: the fixed prior's $\|\hat{f}_m-\hat{f}_M\|^2$ stays ($3.0\cdot10^{-4}$–$5.5\cdot10^{-3}$, $1.1\cdot10^{-4}$–$9.9\cdot10^{-4}$, $4.8\cdot10^{-6}$–$8.2\cdot10^{-5}$ at $m=15$, 60, 240); at the seven D‴.5-admissible samples it is the exact bias at the sampled weights (ratio 1.0000), the delta method missing it by 0.86–1.72; the five saddles have split (class diameters 0.64–1.50) at 0.81–1.01 of the replicated floor. The adaptive prior's decreases without a floor (at $s=6000$ at most $2.0\cdot10^{-6}$ at $\eta=\sigma_0$ and $5.4\cdot10^{-8}$ at $\eta=\infty$ over the representable samples); the boundary sample decays like $1/s$ ($6000\cdot\|\hat{f}_m-\hat{f}_M\|^2=0.88$ and 0.42 against the predicted 0.8 and 0.4); the counts are conserved.

- The parameter level: the coupled distance D is 0.944–1.000 of the replicated stationary points' (fixed: $\Theta^*(w)$; adaptive: $\vartheta^e(w)$), and 1.72–4.81 times larger under the adaptive prior; W^2 is dominated by the cross-class transport of $\frac{1}{2}\|w^m-w^M\|_1$ (0.82–0.99 of it fixed, 0.67–0.99 adaptive), so the two priors' W^2 differ by 1.00–1.32 and their within-class parts by 1.71–4.84.

- The design table (the misplaced prior, 3000 slow units, $m=15$, 60, 240): nearest atom and fixed, fit error $6.8\cdot10^{-2}$–$2.0\cdot10^{-2}$ and floor $2.5\cdot10^{-3}$–$5.9\cdot10^{-5}$ (the same five samples saddles, margins -0.38 to $-5.76\cdot10^{-3}$); nearest atom and adaptive, $8.9\cdot10^{-3}$–$5.5\cdot10^{-4}$ and $8.0\cdot10^{-5}$–$1.1\cdot10^{-7}$; OT and fixed, $9.78\cdot10^{-3}$ and no floor; OT and adaptive (settled gate), $1.6\cdot10^{-4}$ and below $5\cdot10^{-11}$. The OT designs' coupled distance is the largest (6.0, 3.5, 1.0: the transit's mismatch) with $W^2\le2.1\cdot10^{-8}$. At $s=1250$, before the settled gate's effect, the nearest-atom adaptive prior leads at $m=60$ and 240 (fit error $3.8\cdot10^{-3}$ and $2.8\cdot10^{-3}$ against 9.2–$9.8\cdot10^{-3}$), not at $m=15$ ($1.75\cdot10^{-2}$); from $s\approx2000$ the OT design leads.

- The real loop ($k_e=25$, 3000 time units, seed 0's weights): the fixed prior's rms tracking error 1.17–$1.18\cdot10^{-3}$ ($m=15$, a saddle orbit) and 3.43–$3.44\cdot10^{-4}$ ($m=60$), at the averaged prediction (ratios 1.001, 0.996 at the end); the adaptive prior's ($\eta=\infty$) $1.16\cdot10^{-3}\to3.27\cdot10^{-4}$ and $3.40\cdot10^{-4}\to1.52\cdot10^{-4}$ along the averaged loop (0.987, 0.992); from the exact representation the five-period rms is at most $9.0\cdot10^{-12}$.

- The instance's probabilities (exact): outside W 0.361, 0.074, 0.0021 and on its boundary 0.118, 0.053, 0.0012 at $m=15$, 60, 240, against Hoeffding's 1.87, 0.60, 0.036; a permutation of the classes representable for 31%, 87%, 100% of the mass outside W; the fixed prior's saddles 0.369, 0.260, 0.109 (comparable to the adaptive prior's failure set at $m=15$, much larger at $m\ge60$). The adaptive prior's expected limiting floor (the constrained minima on the branch of $\vartheta^e(q)$ outside W; the global infimum is 0 where a permutation is representable) $\ge7.3\cdot10^{-3}$, $2.2\cdot10^{-4}$, $6.6\cdot10^{-7}$: 1.8, 0.24, 0.005 times the fixed prior's law.

- Monte Carlo (600 nested samples, the exact replicated stationary points): the fixed prior's floor is 2.78 ± 0.20, 1.53 ± 0.11, 1.11 ± 0.07 times the law (2.89, 1.56, 1.07 for D‴.5-admissible pairs); $\mathbb{E}D$/law 0.68, 0.93, 0.95 (fixed) and 0.53, 1.17, 1.26 (adaptive: a far heavier tail, one sample at 55.7 times the law at $m = 240$, 1.17 without it — the means not reliable), the adaptive prior's 2.9–3.7 times the fixed prior's.

- In two dimensions (`exp_adaptive_floor_2d.py` → `chaos/adaptive_floor_2d.md`; four bumps with vector amplitudes, $p = 5$, $\sigma_0 = 0.05$, $m = 16$, 32, 64 against $M = 128$, six seeds): F equals the bias form to $6.0 \cdot 10^{-7}$, the law 0.310 per unit of $1/m - 1/M$ at 0.215 of the bound; every sample D‴.5-admissible (margins +0.016 to +0.047) and the fixed prior's floor equal to the exact bias at the sampled weights in all 18 systems (ratio 1.0000); 7 of the 18 systems representable ($w > (0.200, 0.150, 0.175, 0.150)$), where the adaptive prior's function-level distance falls to $1.0 \cdot 10^{-7}$–$4.1 \cdot 10^{-5}$ ($\eta = \sigma_0$) and $1.2 \cdot 10^{-9}$–$1.0 \cdot 10^{-5}$ ($\eta = \infty$: the atoms reset to the cell means every slow unit), at most 0.019 and 0.006 of the fixed prior's, elsewhere to 0.023–0.64 of it (still falling, toward the constrained minimum at the sampled weights); the real loop ($k_e = 25$, $m = 64$; the adaptive prior at $\eta = \infty$) at the averaged prediction (ratios 0.979–1.009) and invariant from the exact representation (2.0–$4.7 \cdot 10^{-10}$).

- Tests (`tests/test_adaptive_floor.py`): the nearest-atom adaptive flow is the gradient flow of $\mathcal{J}_\circ$ with the cell masses (finite differences); $\mathcal{J}_\circ$ is the minimum of the OT energy over the masses, strictly below the energies with masses moved by $\pm 1/m$; the floor law as the bias derivative, below the bound, the weight moving the shares toward $h \approx \sigma_0$ against the unweighted ones at two σ_0; at the D‴.5-admissible sample $(5, 4, 6)/15$ the exact representation stationary under the adaptive prior and the fixed prior's stationary point a strict local minimiser, biased; at the saddle $(6, 5, 4)/15$ the exact representation's margins σ_0; the parameter split; $\mathcal{J}_\circ$ decreasing across a cell crossing; an empty cell staying empty with its atom frozen. `tests/test_planar_loop.py`: the planar averaged loop (the scalar one at $n = 1$; the learning term against finite differences of $\bar{L}$ and the orbit average of the real law), the exact representation of representable weights stationary for the adaptive prior, the floor law as the bias derivative in two dimensions, and the finite-η branch of its integrator second order against DOP853.

25.18 Corollary D‴.26: perfect tracking with an adaptive prior in the real loop

Script: `exp_adaptive_floquet.py` → `chaos/adaptive_floquet.md`.

- The three-bump instance at $\sigma_0 = 0.15\sigma_{\mathrm{crit}}$, $x_m = 2\sin t$, the sampled weights $w = (6, 5, 4)/15$ (amplitudes $A_k/w_k = 2.5$, -2.1, 3.0; representable, and a saddle of the fixed prior), the nearest-atom leak toward an adaptive prior in the real loop with the prior in the state; $\eta = \sigma_0$ and ∞, $k_e = 25$ and 50; the monodromy by central differences of the period map (step 10^{-5}, DOP853 at rtol 10^{-13}).

- Every lift returns after one period to $\le 1.7 \cdot 10^{-15}$. Separations: at $m = 15$ and 30 the 36 and 81 exponents equal σ_0/k_e to $\le 1.7 \cdot 10^{-6}$ (relative). Collective exponents: the other 18 ($\eta = \sigma_0$) or 9 ($\eta = \infty$) equal the weighted K_0-atom loop's to $1.4 \cdot 10^{-5}$–$6.4 \cdot 10^{-4}$.

- Part C, the variational equation on the collapsed subspace (its generator the real loop's linearisation to $5.8 \cdot 10^{-11}$, its exponents the differences' to $4.1 \cdot 10^{-5}$), $k_e = 12.5$–400, on $x_m = 2\sin t$ (where the first-order averaging correction vanishes identically) and on $2\sin t + 0.8\sin 2t$ (where it does not: $\|\bar{A}_1\| = 5.6 \cdot 10^{-3}$, skew): every exponent below its first-order prediction $\lambda_j^{\pm}(w)/k_e$ here — the slowest mode by 13.6% and 3.84% at $k_e = 25$ and 50 ($\eta = \sigma_0$) — the deviations falling by $\to 4$ per doubling of k_e ($O(k_e^{-2})$ relative), the residuals of the second-order prediction by $\to 16$ ($O(k_e^{-4})$: the odd orders vanish; $4.9 \cdot 10^{-7}$

at $k_e = 400$). The second-order coefficients are all negative here, $c_1/\lambda_1 = -100.6$ in the slowest mode, of which the error filter's part $-102.1 = -0.989\omega_1^2$, $\omega_1 = 10.16$ (measured -100.5 at $k_e = 400$); with the adaptation gain scaled by 30 two are positive ($+4.17$, $+12.65$) and those exponents lie above their first-order prediction by $c_j/(\lambda_j k_e^2)$; at $\eta = 10\sigma_0$ $\bar{A}_0$ is self-adjoint in $\mathrm{diag}(I, (\sigma_0/\eta)I)$ only. The error channel's exponent by Liouville's formula within $O(k_e^{-3})$ of $k_e - \operatorname{tr} H_L(w)/k_e$.

- Tests (`tests/test_adaptive_floquet.py`): at the matching weights, $\eta = \infty$, $m = 6$: the return of one period; the nine separation exponents at σ_0/k_e; the collective exponents equal to the weighted K_0-atom loop's, within the averaging correction of the prediction. At the weights $(2, 1, 1)/4$ with the prior in the state, $\eta = \sigma_0$, $m = 4$: the real loop's Jacobian at the lift maps the separations to $-\sigma_0/k_e$ times themselves and the collapsed subspace into itself as the collapsed variational equation; the monodromy's 3 separation exponents at σ_0/k_e and its 18 collective ones the variational ones, within the second-order prediction and below the first-order one (an instance property). On an asymmetric reference: $\bar{A}_1$ non-zero and skew, $\ell_j \bar{A}_1 r_j = 0$, the second-order residual falling like k_e^{-4}, the measured second-order coefficient the explicit c_1, the error channel by Liouville's formula; at $\eta = 10\sigma_0$ $\bar{A}_0$ self-adjoint in the metric and not in I, $\bar{A}_1$ skew in it.

25.19 Lemma D‴.26a: the exact slow subspace: Corollary D‴.26(c) and Proposition D‴.30(ii) proved for the linearised loop

Script: `exp_adaptive_floquet.py` Part D → `chaos/adaptive_floquet.md`.

- Corollary D‴.26's instance on the asymmetric reference $x_m = 2\sin t + 0.8\sin 2t$, $\eta = 0$ ($K = \hat{J}$, $\Lambda = \sigma_0 I$), σ_0 and ∞, $k_e = 12.5$–1600.

- Chang's slow subspace: the contraction in 3–9 iterations to an invariance residual $\leq 3.8 \cdot 10^{-15}$ (at $\eta = 0$, $k_e = 12.5$ it stalls at rounding, 499 iterations); $|P - P_2|/\varepsilon^3 \to 3.34 \cdot 10^3$ and $|P - P_{(3)}|/\varepsilon^4 \to 5.8 \cdot 10^4$; the slow block's rates the full variational equation's to $\leq 1.0 \cdot 10^{-8}$.

- The parity as an identity: the rates at $\pm\varepsilon$ equal to $\leq 5.9 \cdot 10^{-12}$ of the largest rate and $\leq 1.1 \cdot 10^{-9}$ per mode, while their first-order deviation is $9.8 \cdot 10^{-2}$–$8.9 \cdot 10^{-6}$ of the largest rate and per mode up to 29.6% (the slowest mode at $\eta = \infty$: -29.6%, -12.5%, -4.0%, -1.1% at $k_e = 12.5$–100); a mismatch in the error channel breaks the equality at $O(1/k_e)$ ($1.1 \cdot 10^{-2} \to 7.8 \cdot 10^{-5}$); the Magnus parameter ≤ 0.48 (0.004 at $k_e = 1600$).

- The slow noise: the decoupling residual $\leq 3 \cdot 10^{-16}$; b against $b_0 = -(\dot{K} + aK)^\top$ at $O(1/k_e)$ ($63\% \to 0.74\%$) and against $b_0 + \varepsilon b_1$ at $O(1/k_e^2)$ ($|b - b_0 - \varepsilon b_1|/\varepsilon^2 \to 3.34 \cdot 10^3$); at $t = 0$ $\mathrm{Cov}(v) \cdot 2\varepsilon^3 \to 1$, $\mathbb{E}[vY'^\top] \to -b^\top$ at $O(1/k_e)$, $\mathrm{Cov}(Y') \to \bar{\Sigma}$ ($0.35 \to 9.4 \cdot 10^{-5}$ at $\eta = \infty$; $O(\varepsilon^2)$ to $k_e = 200$–400, $O(\varepsilon)$ beyond). The weakest mode's deviation ($-34\% \to -6.6 \cdot 10^{-5}$, $O(\varepsilon^2)$) is the slow noise's attenuation, $\langle bb^\top \rangle$ below $\bar{Q}$ ($-56\% \to -1.1 \cdot 10^{-4}$), plus the rate's correction ($+49\% \to +4.6 \cdot 10^{-5}$): 2.4–2.5 to 1 and of opposite sign at $\eta = \infty$ (12–23 to 1 at $\eta = 0$). $\mathrm{Cov}(Y) - \mathrm{Cov}(Y')$ is the jitter $(\varepsilon/2)K(0)^\top K(0)$ at $O(\varepsilon)$ and with its next term at $O(\varepsilon^2)$ ($0.30 \to 0.001$ at $k_e = 100$–1600).

- The floor: at $t = 0$ $0.763 \to 1.0000$ of the slow-plus-fast value; over the period its average $0.766 \to 0.99997$ of Proposition D‴.30(ii)'s formula, pointwise in $[0.9952, 1.0091]$ of it at $k_e = 1600$ — the predicted range of the fast jitter $(\varepsilon/2)\operatorname{tr}(H\hat{J}(t)^\top \hat{J}(t))$: the formula holds as the period average.

- Tests (`tests/test_adaptive_floquet.py`): Chang's P with the exact slow multipliers and the ε^3 truncation on the instance; on a synthetic block with an $O(1)$, non-symmetric Λ the truncations at ε^3 and ε^4, a flipped Λ and a dropped cubic term caught; the rates exactly even in ε at $\eta = \sigma_0$ and ∞, a mismatch breaking it; the slow noise decoupled, b_0 against $\dot{K}$

by the chain rule and the plant's own slope by differences, b_1 at ε^2, the covariances at their orders.

25.20 Proposition D‴.27: a projected adaptive prior

Script: `exp_adaptive_prior.py` Part H → `chaos/adaptive_prior.md`.

- The box $\mathcal{Z}$ (unconstrained coordinates): $\alpha \in [-5.889, 5.889]$, $\gamma \in [-4.417, 4.417]$ (amplitudes and centres to 0.9 of $a_{\max}$ and $c_{\max}$), $\kappa \in [-6, 6]$; the instance's exact atoms inside.
- Beyond the saturation (Part G(b)'s target, weighted amplitude $4.8 > a_{\max}$; the K_0-atom system from the exact atoms with the prior there): the projected prior converges at $\eta = \sigma_0$ (the projected midpoint rule, positions projected inside the step, $\mathrm{d}t = 0.25$ and 0.125, to $s = 10^5$; the two end states within $3.4 \cdot 10^{-12}$, their shared fixed point; the trajectory at $s = 300$ within $6.3 \cdot 10^{-7}$, $1.9 \cdot 10^{-7}$, $3.8 \cdot 10^{-8}$ of a reference for $\mathrm{d}t = 0.25$, 0.125, 0.0625) and at $\eta = \infty$ (LSODA) to the same bounded stationary point (drift $\leq 3.5 \cdot 10^{-15}$): the prior's amplitude atom on the face (5.889), the particle's at 6.728, the fit error 0.1415.
- The particles' largest G^{-1}-distance from $\mathcal{Z}$ along the run: 0.92 ($\eta = \sigma_0$), 0.89 ($\eta = \infty$), against the a-priori bound $B/\sigma_0 \leq 3686$ ($B \leq 26.08$ with the closed-form bound 4.658 of L^G_ϑ; 4.044 by search).
- The box's size: the amplitude bound 0.75, 0.8, 0.9, 0.95, 0.99 of $a_{\max}$ (boxes containing the exact atoms) gives the fit errors 0.1676, 0.1611, 0.1415, 0.1268, 0.1101 — between the fixed prior's 0.1750 and the unprojected adaptive prior's 0.1050, reached only as its coordinate runs off; 0.7 (0.1736) excludes the exact atoms, and the tighter 0.65, 0.6, 0.5 are worse than the fixed prior (0.1792, 0.1845, 0.1949): a box should contain the representation.
- At $\eta = \infty$ the plan is Lloyd-consistent, not jointly optimal: on six particles in a consistent, non-optimal pairing the selected energy falls (largest increment $-3.4 \cdot 10^{-6}$) while the reduced energy $\min_Z \mathcal{J}$ rises from $1.251 \cdot 10^{-3}$ to $1.574 \cdot 10^{-3}$ before the plan becomes optimal.
- Part G(a)'s four-bump target: the box stops the merged atom's width coordinate at the face $\kappa = 6$ from $s = 10^9$ (unprojected: $\kappa = 9.75$ at $s = 10^{10}$ and growing like $\ln s$), the fit error within $9.3 \cdot 10^{-7}$ of the face minimiser's.
- Part C's K_0-atom loop at the misplaced prior ($\eta = \sigma_0$, 6000 slow units) is identical with and without $\mathcal{Z}$: the box is never approached (margins at least 16.9% of its width), so the runs coincide by construction.
- Tests (`tests/test_adaptive_prior.py`): beyond the saturation the projected prior at $\eta = \infty$ keeps the particles within 2 of the box and converges, while without the box the prior's coordinate passes 11 by $s = 10^4$; at $\eta = \sigma_0$ and $\mathrm{d}t = 0.125$ the prior stays in the box with its amplitude atom on the face (the step that projects the velocity at the midpoint stalled it inside), the particles bounded and $\mathcal{J}$ non-increasing; at the misplaced prior near the exact representation the projected and unprojected runs are identical.

25.21 Proposition D‴.28: the floor removal is a gauge property

Script: `exp_gauge_floor.py` → `chaos/gauge_floor.md`.

- The three-bump instance plus a fourth bump $\varepsilon \cdot \exp(-(x - 0.8)^2/(2 \cdot 0.3^2))$ that three kernels cannot represent, $\varepsilon = 0$–0.5, masses 1/3, $\sigma_0 = 0.15\sigma_{\mathrm{crit}}$: the weighted approximant $\vartheta^b(q)$ continued in ε, a strict local minimum of the unregularised loss with the relative fit error $3.6 \cdot 10^{-3}$–$2.5 \cdot 10^{-2}$ ($\varepsilon = 0.05$–0.5).

- The gauge: at six weight vectors of W_g the critical point continued in w from $\vartheta^b(q)$ is the gauge transform (to $\le 1.7 \cdot 10^{-11}$, $\hat{f}$ unchanged to $\le 5.5 \cdot 10^{-13}$); the adaptive prior's sensitivity $1.4 \cdot 10^{-11}$–$2.7 \cdot 10^{-10}$ of the fixed prior's; $P(w^m \in W_g) = 0.52$, 0.87, 0.997 at $m = 15$, 60, 240 ($\varepsilon \le 0.3$); the branch reaches the saturation near $\varepsilon = 0.68$.
- The fixed prior's sensitivity equals $\sigma_0 \hat{J}(\tilde{A} + \sigma_0 I)^{-1} \tilde{g}$ (the full Hessian) to $\le 2.2 \cdot 10^{-9}$, against $7.0 \cdot 10^{-4}$–$5.6 \cdot 10^{-3}$ for the Gauss–Newton form; its floor law $6.17 \cdot 10^{-2}$ ($\varepsilon = 0$) to $8.52 \cdot 10^{-2}$ ($\varepsilon = 0.5$) per unit of $1/m - 1/M$.
- The residual curvature: the amplitude row $\le 7.4 \cdot 10^{-14}$; the margins $\sigma_0 - (q_k/w_k)\lambda_k^+$ exact to $\le 2.1 \cdot 10^{-14}$ over every count vector of $m = 15$ and 60 in W_g, none negative; the dynamics' separation eigenvalues at a replicated configuration equal to them to $\le 5.8 \cdot 10^{-12}$.
- The dynamics ($m = 15$, 60, 240 against $M = 510$, four seeds, 6000 slow units, $\varepsilon = 0.1$ and 0.3): the fixed prior's floor within 8% of its value at $s = 600$, its fit error above the approximant's (its bias; split classes at its saddles); the adaptive priors' floor falling ($m = 60$, $\varepsilon = 0.1$: to $1.0 \cdot 10^{-9}$–$1.4 \cdot 10^{-8}$ at $\eta = \sigma_0$, $3.2 \cdot 10^{-11}$–$3.8 \cdot 10^{-9}$ at $\eta = \infty$) and the fit error tending to the approximant's (median relative excess $1.2 \cdot 10^{-4}$, $1.2 \cdot 10^{-4}$, $1.6 \cdot 10^{-5}$ at $m = 15$, 60, 240 for the samples 0.02 inside W_g); at the sample outside W_g the systems approach the box minimum on the branch (1.107 and 1.586 times the approximant's fit error), at the sample $6 \cdot 10^{-4}$ inside it the approach is slow.
- Tests (`tests/test_gauge_floor.py`): at $\varepsilon = 0.3$, the gauge transform a critical point at two sampled weight vectors with the same function, reached by continuation from $\vartheta^b(q)$; the adaptive prior's sensitivity negligible, the fixed prior's the full-Hessian formula and not the Gauss–Newton form; the residual curvature's zero amplitude row, its scaling by q_k/w_k, the margins and the dynamics' separation eigenvalues; adaptive-prior runs ($\eta = \infty$ and σ_0) converging to the gauge transform with the approximant's function.

25.22 Proposition D‴.29: second-order plants: the sliding-variable reduction; a cogging-torque servo

Script: `exp_servo_cogging.py` → `servo/servo_cogging.md`.

- The servo $\ddot{x} = -\tau_{\mathrm{cog}}(x) + u$, $\tau_{\mathrm{cog}}(x) = 0.4 \sin(3x + 0.3) + 0.1 \sin 7x$, $x_m = \sin t$, $s = \dot{e} + 5e$; $K_0 = 4$ saturated Gaussian kernels on the position, masses 1/4, $\sigma_0 = 0.01$.
- $\vartheta^b(q)$, the nondegenerate local minimum BFGS reaches from a fixed start (no exact representation): fit error $1.46 \cdot 10^{-2}$, $\tilde{A}$ spanning $2.9 \cdot 10^{-4}\sigma_0$–$30\sigma_0$ (four modes below $0.1\sigma_0$; the adaptive prior's slowest rates $1.4 \cdot 10^{-4}\sigma_0$ and $2.9 \cdot 10^{-4}\sigma_0$ against the fixed prior's $\approx \sigma_0$), $\lambda_k^+ \le 0.0098\sigma_0$, the floor law $1.29 \cdot 10^{-2}$. Globally the loss's infimum lies at the saturation: of 120 random starts 1 ends at $\vartheta^b(q)$ and 33 below its fit error, all saturated; the best $6.39 \cdot 10^{-3}$, approached with the saturated amplitudes held at $|\alpha| = 6$–14 ($7.14 \cdot 10^{-3}$–$6.40 \cdot 10^{-3}$), not attained. A misplaced prior (phase 0.5, the engineer's model, fit error 0.200) whose replicated stationary point is a D‴.5 saddle (margin $-3.97 \cdot 10^{-3}$, fit error 0.0545).
- The averaged loop ($m = 16$, 64, 128 against $M = 256$, three seeds, 2000 slow units) from clouds around $\vartheta^b(q)$: the fixed prior's floor $1.85 \cdot 10^{-3}$, $2.24 \cdot 10^{-4}$, $4.49 \cdot 10^{-5}$ (fit error 0.174, 0.066, 0.038); the adaptive prior's ($\eta = \infty$) 38–780 times lower, the fit error 0.0298, 0.0148, 0.0147 against the approximant's 0.0146; the OT leak at the approximant with no floor; the σ-modification toward 0 biased (0.328); no leak ($\sigma_0 = 0$) as good: 0.0180, 0.0147, 0.0146 (floors $7.7 \cdot 10^{-6}$, $2.4 \cdot 10^{-7}$, $6.6 \cdot 10^{-8}$) — noise-free, the floors and biases the leaks remove are those a fixed leak creates. Misplaced: the fixed designs split their classes and end at 0.048 (OT) and 0.10–0.048 (nearest atom); the adaptive prior is unaffected at $\eta = \infty$ (its prior never acts from this start), slower at $\eta = \sigma_0$ (0.035–0.019).

- From the engineer's model (clouds around it, the prior at it): the fixed designs 0.102, 0.060, 0.048 (OT 0.048); the adaptive prior 0.027, 0.025, 0.022 ($\eta = \infty$) and 0.032, 0.028, 0.028 ($\eta = \sigma_0$) at $s = 2000$; no leak 0.0152, 0.0147, 0.0147. The collapsed K_0-atom flow from the engineer's model at $\eta = \infty$: 0.0297, 0.0213, 0.0169, 0.0155 at $s = 600$, 2000, 10^4, 10^5.
- Phase-plane kernels (five isotropic kernels): on the circle 5 exact zero Gauss–Newton eigenvalues and 8 below $10^{-6}\sigma_0$; ellipses $\times 1.1$ and $\times 2$: 8 and 6 below $10^{-6}\sigma_0$, none exact; a two-harmonic orbit 3; centred kernels 16 exact of 20.
- The real servo ($k_s = 25$, 1000 time units; records and leak targets every 0.02 time units): position-error rms 1.7–$2.1 \cdot 10^{-5}$ (adaptive designs, matching OT, no leak), $7.8 \cdot 10^{-5}$ (misplaced OT), $9.1 \cdot 10^{-5}$–$1.8 \cdot 10^{-4}$ (fixed nearest atom), 5.4–$5.5 \cdot 10^{-4}$ (σ-modification), equal to the prediction $(\mathrm{d}/\mathrm{d}t + \lambda)^{-1}(\mathrm{d}/\mathrm{d}t + k_s)^{-1} r(x_m)$ to 1.0001–1.0029 at $t = 500$ and 1000 (with the slaved r/k_s: 0.931–0.996, the filter's attenuation of the residual's broad fine scale).
- Tests (`tests/test_sliding.py`): the phase-plane averaged loop is the gradient flow of $\bar{L}$; on a circular orbit isotropic phase-plane kernels in general position have one zero Gauss–Newton eigenvalue each; the servo's parameters follow the averaged loop over ten periods (within 3% at $k_s = 25$, closer at 50), its position error within 1% of both predictions.

25.23 Proposition D‴.30: measurement noise at perfect tracking: coherent, high-passed, and the errors in variables

Script: `exp_measurement_noise.py` → `chaos/measurement_noise.md`.

- The three-bump instance at the lift of the exact representation of $q = 1/3$ ($\sigma_0 = 0.15\sigma_{\mathrm{crit}}$, $x_m = 2\sin t$, $k_e = 25$), the measurement $y = x + \nu$ in the controller and the adaptive law.
- The linear theory: $\operatorname{tr} Q = 8.784$ (the derivative Gram 8.105, the slope term 0.679); the parameters' floor per unit N_0 0.1597 (fixed prior), 0.1627 ($\eta = \sigma_0$), 0.1690 ($10\sigma_0$), $0.1757 = \operatorname{tr} Q/(2k_e)$ ($\eta = \infty$); the weakest mode's parameter variance 0.0865 → 2.09 ($\times 24.1$; $\times 28.0$ in the exact linearised loop at $k_e = 25$); the fast parts $\operatorname{tr} H^2 = 0.0657$, $\langle |\hat{J}|^4 \rangle = 0.299$.
- The errors in variables (OU noise, $B = 100$): the averaged stationary function-level bias 491 (fixed) and 2397 (adaptive, every $\eta > 0$) per unit $(\mathbb{E}\nu^2)^2$, above the floor of the parameters beyond the noise std 0.0026 and 0.0012; the parameters' fluctuation changes it by less than 10^{-5} (2397.68); the exact periodic second-order mean of the linearised loop 504 and 2653, the excess $O(k_e^{-2})$ (10.7%, 2.65%, 0.66%, 0.17% at $k_e = 25$–200, adaptive); the bias's growth in k_e at the local exponent 1.93–1.99 for $B \geq 20k_e$, 1.69 and 1.47 at $B = 8k_e$ and $4k_e$.
- The exactly linearised loop: its periodic stationary covariance at the prediction to 0.976–0.980 ($k_e = 25$), 0.994–0.995 (50), 0.9986–0.9988 (100), 0.9997–0.9998 (200), above 1 at 800 ($+1.3 \cdot 10^{-5}$): the relative error is $O(1/k_e)$, its first-order term from the slope a in the noise input — with the noise input $-k_e\nu$ the deviation is numerically even in $1/k_e$ (successive ratios 3.67 → 4.00 to $k_e = 800$; not proved).
- Invariance, not attraction: at the lift the separations' second-order drift $\lambda_{\max}(S_k) = 0.0148$, 0.0101, 0.298 per $\mathbb{E}\nu^2$ in the three classes, the third's separations growing there above the noise std 0.031; along the noisy trajectory the mean error also carries the tracking error of the developing errors-in-variables deviation (rms 0.121 → 0.602 per $\mathbb{E}\nu^2$, $\lambda_{\max}(S_3)$ 0.307 → 0.349 over 300 time units). In the real loop (the lift $m = 15$, eight antithetic pairs of noise paths of 300 time units per std) the separations' rates are $-2.50 \cdot 10^{-4}$, $+2.74 \cdot 10^{-6}$ and $+4.53 \cdot 10^{-4}$ per time unit at std 0.01, 0.03, 0.05; $\lambda_{\mathrm{eff}} = (\mathrm{rate} + \sigma_0/k_e)/\mathbb{E}\nu^2$ is 0.3271 ± 0.0038, 0.3175 ± 0.0037, 0.2943 ± 0.0034 against the lift's 0.2981 and the transient's

0.3264, 0.3270, 0.3281; the fit's small-noise limit 0.3291 ± 0.0033, the next order -13.8 ± 2.0; the rate vanishes at std 0.0299 over the window (0.0308 by the lift's drift, 0.0294 by the transient's); the long-time threshold is open.

- The real loop (12 noise paths of 300 time units, common to the four cases): within 1.75 standard errors of the linear prediction plus the mean part at both noise levels, the mean part 42% (std 0.003) and 89% (std 0.01) at $t = 300$, the priors 6.0–6.4% apart (predicted 5.3–6.1%); the lift $m = 15$ on a common noise path identical to the K_0-atom run to rounding ($5.6 \cdot 10^{-16}$ in float64).

- Tests (`tests/test_measurement_noise.py`): the closed form against the Lyapunov solution, the trace identity at $\eta = \infty$ and the floor increasing in η; the exact periodic floors of the linearised loop at $k_e = 100$ for $\eta = 0$, σ_0, ∞; a lift $m = 6$ under a noise path staying replicated and following the K_0-atom loop; a separation of the weakest class decaying at std 0.01 and growing at 0.05 (a sign test); the separations' drift against an independent quadrature of its two moments; along the second-order transient the mean parameters moved by the averaged force and the mean error carrying the tracking error of the developing deviation; the OU noise's stationary variance and correlation, and the law seeing the measurement; the responses $(H + \sigma_0 I)^{-1}$ and H^{-1} to a constant force, and the quasi-static force against the full second-order mean's noise part; the exact periodic mean against the filtered-Gram response under both priors, the fixed prior's branch and its leak in the bias; the mean function's curvature term against an independent contraction.

25.24 Proposition D‴.31: the adaptive prior is a washout leak; the desired regressor makes the noise's own persistent force a least-squares force

Script: `exp_eiv_remedy.py` → `chaos/eiv_remedy.md`.

- The instance of Proposition D‴.30 with an Ornstein–Uhlenbeck noise of rate B; MR the measured regressor (Proposition D‴.30), DR the regressor at the reference, DR* the regressor and the compensation at the reference (the desired compensation adaptation law).

- The adaptive prior's function-level errors-in-variables bias per unit $(\mathbb{E}\nu^2)^2$ at $k_e = 25$, $B = 100$: from the noise's own moments 2397 (MR), 0.471 (DR), 0.0526 (DR*); with the parameters' fluctuation (the full second-order mean) 2398, 0.486, 0.0481 — the parameters' part $\leq 0.03\%$ for MR, -8.7% to $+3.3\%$ for DR and DR* over the grid except $+56\%$ for DR at $k_e = B = 100$, where its noise part nearly cancels; its own response $3.9 \cdot 10^{-4}$, $1.1 \cdot 10^{-4}$, $2.9 \cdot 10^{-5}$, $7.4 \cdot 10^{-6}$ at $k_e = 25$–200 ($\propto k_e^{-2}$: on the one-dimensional reference the law's leading term, a line integral along the orbit, vanishes), its share of DR's bias not falling monotonically. This part is the classical second-order bias of a constant-gain stochastic approximation (Benveniste–Métivier–Priouret, Dieuleveut–Durmus–Bach, Box's curvature bias); the desired regressor does not remove it. MR over DR $3.0 \cdot 10^3$–$2.1 \cdot 10^8$ over $k_e = 25$–200 and $B = 100$, 1000; MR's growing like $(k_e B/(k_e + B))^2$, DR's and DR*'s at most 1.18, below $\frac{1}{4}\|\partial_x^2 f\|_\rho^2 = 1.31$; DR's noise part at its least-squares bound to 0.01%; $k_e^2 \|P\mathbb{E}[e]\|_\rho^2$ 2394 for MR (zero by construction for DR, DR*), the mean tracking error itself $3.83 \to 0.66$ ($k_e = 25 \to 200$). The noise std at which the bias reaches the floor: 0.0012 (MR), 0.085 (DR), 0.27 (DR*).

- The prior family $\mathrm{d}z/\,\mathrm{d}s = -\eta(z - \hat{T}) - \kappa(z - z_a)$: $RH + B = I$ to $1.1 \cdot 10^{-14}$; the weakest mode's response $3269 \to 136$ as its bias factor goes $0 \to 0.959$, the trade binding on the 3 of 9 modes with $h_j < \sigma_0$; MR's bias $2397 \to 491$, DR's $0.471 \to 0.211$. Priors whose zero-frequency leak does not commute with H (fixed amplitudes or centres with $\eta = \sigma_0$ elsewhere, an active face at $\eta = \infty$, the anchor gain ηP_{amp}, a random anchor gain with

$K + K^\top \succeq 0$), R and B from their dynamic blocks: equal to $(H + S_0)^{-1}$ and $(H + S_0)^{-1} S_0$, S_0 by elimination, to $1.9 \cdot 10^{-15}$; $RH + B = I$ to $1.4 \cdot 10^{-14}$, $HR + B - I$ 0.47–0.79.

- The real loop on common noise paths ($k_e = 25$, std 0.01, $\eta = \infty$, 12 runs of 300 time units): $\mathbb{E}\|\hat{f} - f\|^2$ at $t = 300$ $2.43 \cdot 10^{-6}$ (MR) against $2.80 \cdot 10^{-7}$ (DR) and $2.75 \cdot 10^{-7}$ (DR*), DR and DR* at their covariance part; the MR − DR run mean at its prediction to 0.5–5.3%, 0.43–1.83 times the run mean's sampling error.
- Tests (`tests/test_eiv_remedy.py`): the washout identity along a real-loop solution at finite η and across a plan switch of the averaged loop; the collective block's transfer function against the washout formula, $RH + B = I$ along the prior family and, R and B from their dynamic blocks, for non-commuting leaks ($HR + B = I$ failing there); the response bound under random leaks $S \geq 0$, the smallest equivalent target perturbation and the projection; the lift exact under DR and DR*, on a common noise path DR's run within the second order of MR's (the difference $\propto$ the noise's variance), the regressor's point in the law; DR's and DR*'s noise forces least-squares with their pointwise bounds on g, MR's F_1 three orders of magnitude above; the quasi-static forces against the full second-order mean, the parameters' part above 1% for DR and DR*, below 10^{-5} for MR.

25.25 Theorem E: label-free convergence under excitation

Script: `exp_pe_convergence.py` → `pe_convergence/`. Table 34 and Figure 7 show the results.

Table 34: Theorem E: tangential excitation and label-free parameter convergence ($m = 3$, $n = 2$, permuted start, $T = 200\,\mathrm{s}$). $\lambda_{\min}$ is that of the excitation Gramian.

reference	seed	$\lambda_{\min}$, first 60 s window	min over sliding 60 s windows	$W(T)/W(0)$	fitted rate of W, $t \geq 100\,\mathrm{s}$ (1/s)	labelled(T) / labelled(0)
rich (6 tones)	0	$8.60 \cdot 10^{-3}$	$4.12 \cdot 10^{-3}$	0.544	$2.31 \cdot 10^{-3}$	0.992
rich (6 tones)	1	$5.52 \cdot 10^{-3}$	$1.02 \cdot 10^{-3}$	0.409	$2.32 \cdot 10^{-3}$	0.962
rich (6 tones)	2	$1.08 \cdot 10^{-3}$	$5.91 \cdot 10^{-4}$	0.369	$3.07 \cdot 10^{-3}$	1.007
single sine	0	$2.54 \cdot 10^{-2}$	$2.41 \cdot 10^{-2}$	0.274	$4.05 \cdot 10^{-3}$	0.988
single sine	1	$2.96 \cdot 10^{-2}$	$2.71 \cdot 10^{-2}$	0.314	$4.26 \cdot 10^{-3}$	0.967
single sine	2	$6.74 \cdot 10^{-3}$	$6.61 \cdot 10^{-3}$	0.226	$2.91 \cdot 10^{-3}$	1.008

- The matched error decays in all six runs. The labelled error stays within 4% of its initial value.
- The rank correlation between the uniform excitation level and the fitted rate is 0.60 over the six runs. The PE level is a hypothesis of Theorem E, not a rate predictor (Remark E.1).
- The proof's linearisation is checked exactly: the Jacobian equals $A(t)$, and $QA + A^\top Q = \mathrm{diag}(-k_e I, 0)$.

25.26 Lemma E.0 and Theorem E with explicit constants

Script: `exp_explicit_rate.py` → `explicit_rate/`. The explicit rate of Lemma E.0 is compared with the exact Floquet rate of the linearised loop (Tables 35 and 36). It lies below the exact rate in every row; the ordering is numerical (Ψ_M, Ψ_D are grid maxima, α a converged quadrature and ε a grid optimum), with margins of at least 17×.

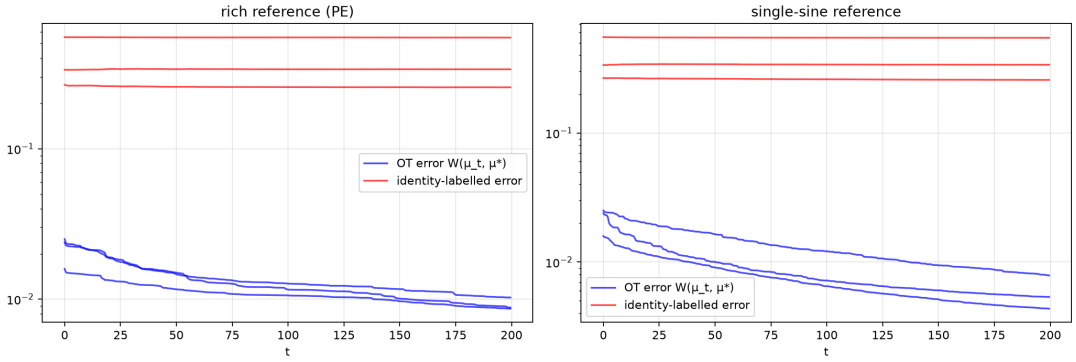


Figure 7: Theorem E: OT error $W(\mu_t, \mu^*)$ and identity-labelled error for the three seeds, with the rich reference (left) and the single-sine reference (right).

Table 35: Lemma E.0: the explicit rate against the exact Floquet rate of the linearised loop (`explicit_rate/table.md`).

instance	regime	exact / explicit
Remark E.1 ($m = n = 1$, $G = I$, $k_e = 3$), $\omega = 1, \dots, 64$	$\Psi_M^2/k_e \approx 1$	$954 \to 17.6$ (decreasing in ω)
the ideal of the Theorem E tests ($m = 3$, $n = 2$, $k_e = 6$), two-tone reference of period 20π, $G = g \cdot \mathrm{diag}(2, 2, 1)$, $g = 30, \dots, 0.01$	fast $\to$ slow	$6.56 \cdot 10^{10} \to 4.22 \cdot 10^5$ (optimised ε; $8.74 \cdot 10^{10} \to 5.63 \cdot 10^5$ with the closed-form ε)

Table 36: Lemma E.0 in Remark E.1's instance, $x_m = 1.5 \sin \omega t$ ($m = n = 1$, $G = I$, $k_e = 3$): PE window T, level $\alpha(T)$, Ψ_D ($\Psi_M = 1.78$ throughout), the exact Floquet rate and the explicit rate with the closed-form and with the optimised ε.

ω	T	$\alpha(T)$	Ψ_D	exact rate	explicit, closed-form ε	explicit, optimised ε
1	6.28	1.13	3.69	$4.660 \cdot 10^{-2}$	$3.688 \cdot 10^{-5}$	$4.886 \cdot 10^{-5}$
2	3.14	0.566	7.38	$2.158 \cdot 10^{-2}$	$7.858 \cdot 10^{-5}$	$1.040 \cdot 10^{-4}$
4	1.57	0.283	14.8	$6.834 \cdot 10^{-3}$	$8.696 \cdot 10^{-5}$	$1.152 \cdot 10^{-4}$
8	0.785	0.142	29.5	$1.831 \cdot 10^{-3}$	$4.736 \cdot 10^{-5}$	$6.301 \cdot 10^{-5}$
16	0.393	0.0708	59	$4.662 \cdot 10^{-4}$	$1.650 \cdot 10^{-5}$	$2.199 \cdot 10^{-5}$
32	0.196	0.0354	118	$1.171 \cdot 10^{-4}$	$4.720 \cdot 10^{-6}$	$6.286 \cdot 10^{-6}$
64	0.0982	0.0177	236	$2.931 \cdot 10^{-5}$	$1.249 \cdot 10^{-6}$	$1.665 \cdot 10^{-6}$

- In Remark E.1's instance the exact rate falls like ω^{-2} (log-log slope -1.97 over $\omega \geq 4$); the explicit rate approaches that law (slope -1.56 over $\omega \geq 4$, -1.92 over the last octave $\omega = 32 \to 64$). Lemma E.0(iii) derives the ω^{-2} law from $\Psi_D \propto \omega$.

- In the slow-adaptation limit ($T\Psi_M^2 \ll k_e$) the ratio decreases towards a constant of order $\kappa_\Psi(1 + \Psi_D/(k_e\Psi_M))^2$ for the optimised ε ($3\kappa_\Psi(\cdots)^2$ for the closed-form bound (ii)), with κ_Ψ the condition number of the averaged Gramian ($1.27 \cdot 10^5$ for this ideal; formula values $4.9 \cdot 10^5$ and $1.47 \cdot 10^6$; measured $4.22 \cdot 10^5$ with the optimised ε and $5.63 \cdot 10^5$ with the closed-form ε at $g = 0.01$). On this instance the exact rate is 0.66 of $\lambda_{\min}\langle\Psi^\top\Psi\rangle/k_e$, because $k_e = 6$ is not large against the reference's bandwidth and the $T^2\Psi_M^3$ term is still 13% of c_1; the PE level is evaluated with 800 Gauss–Legendre nodes, converged over the 20π window. The exact rate pays for ill-conditioning once, the bound twice (Remark E.0a).

- Lemma E.0′ (the slow regime; table 4 of the same file): on the $g = 0.01$ ideal ($\kappa(\bar{M}) = 7.8{\cdot}10^3$; peak factors $\rho = 69.9$, $\rho_D = 3.7 \cdot 10^3$, $\rho_{DD} = 3.7 \cdot 10^5$; threshold $k_0 = 226$ for $\delta = 0.2$, 321 for $\delta = 0.1$) the guaranteed rate is within a factor 1.26 ($\delta = 0.2$) and 1.12 ($\delta = 0.1$) of the exact Floquet rate at $k = 400$ and 600, where Lemma E.0's optimised rate is $1.3 \cdot 10^5$ times smaller; the certificate $\sup \dot{V}/V \leq -\gamma$ holds on the grid wherever the conditions hold; exact/$(\lambda_{\min}/k) = 0.9885$, 0.9971, 0.9993, 0.9997, 0.9998, 0.9999 at $k = 50, \ldots, 600$ (the k^{-2} law of $N_2 = 0$).

- Corollary E.0′.2 (Theorem E in the slow regime; table 5): on the single-kernel instance at $k_e = 40$ the nonlinear rate from Lemma E.0′'s V is $4.53 \cdot 10^{-4}$ ($\delta = 0.2$) against $2.63 \cdot 10^{-5}$ from Lemma E.0 and an exact linear rate of $1.35 \cdot 10^{-3}$; the explicit ball radius is $1.53 \cdot 10^{-6}$ against $6.15 \cdot 10^{-6}$ (the anisotropy of the quadratic form), the overshoot 11.8 against 1.43.

- Tests (`tests/test_excitation.py`): the pointwise inequalities of Lemma E.0 on random states for four regressors, the envelope along trajectories, the ordering explicit $\leq$ exact, the ω^{-2} law, the no-growth of the ratio as the gains shrink, and the explicit nonlinear envelope of Theorem E on the implemented closed loop (single-kernel instance with $G = 0.3\,I$, where $r_0 \approx 10^{-5}$ in the w-coordinates and the explicit nonlinear rate is $3.7 \cdot 10^{-5}$: the inequality holds at every sample over 8000 s, about 0.3 e-folds of the explicit envelope, while the state itself decays by far more). Lemma E.0′: the certificate and the sharpness on $\Psi = (\sin t, \cos t)$ ($k = 20$, 50, 200; ratio $\to 1/(1-\delta)$) and on the slow-adaptation ideal at $k = 400$ (ratio ≤ 1.5; more than 10^4 above Lemma E.0's rate). Corollary E.0′.2: the explicit envelope on the implemented single-kernel loop at $k_e = 40$ over 1.5 e-folds ($\mathrm{d}t = 0.005$, below the fixed-step discretisation floor).

- Corollary E.0′.3 (table 6): the explicit bound on the stationary parameter spread of the dithered (LTV), with $\Sigma = (2 \cdot 10^{-3})^{1/2} I$, against the exact periodic stationary covariance: on $\Psi = (\sin t, \cos t)$ the ratio bound/exact is 2.69, 1.58, 1.31 ($\delta = 0.2$) and 2.52, 1.41, 1.17 ($\delta = 0.1$) at $k = 20$, 50, 200, falling toward $1/(1-\delta)$; on the Theorem E ideal ($\kappa = 7.8 \cdot 10^3$) it is 15.4–17.5, the price of $q = 15$ directions charged at the worst excitation; the exact value equals the averaged Langevin covariance $k \cdot \mathrm{tr}\, X$ within $3 \cdot 10^{-4}$ (ideal) and $1.25/k^2$ (sinusoid), and $\mathbb{E}\|e\|^2 = \|\Sigma\|_F^2/(2k) = \mathrm{tr}(\bar{M}X)/k$ to five digits.

- Lemma E.0″ (`exp_pe_general.py` → `explicit_rate/general_pe.md`): the exponential-average construction on two non-periodic regressors. Quasi-periodic $\Psi = (\sin t, \cos\sqrt{2}\,t)$: $\lambda_T = 0.297$, 0.391, 0.444 for windows 2π, 4π, 8π; conditions from $k = 20$; exact/guaranteed ($\delta = 0.2$ / 0.1) 2.49/2.13 $\to$ 1.63/1.45 at $T = 4\pi$ and 2.10/1.81 $\to$ 1.43/1.27 at $T = 8\pi$ from $k = 20$ to 200; the exact rate is $0.5/k$ to within 1%. Chaotic $\Psi = (x_1/20, (x_3 - 25)/20)$ along a Lorenz orbit traversed five times slower than standard time: $\lambda_T = 0.015$, 0.024, 0.051 for windows 5, 10, 20 against an effective excitation 0.15 (exact rates $0.00759/k \cdot 20$ …); conditions from $k = 100$; exact/guaranteed 5.0/4.3 ($k = 100$) $\to$ 4.2/3.7 ($k = 200$) at

$T = 20$, dominated by the window loss $\lambda_\infty/\lambda_T \approx 3$; the certificate holds at every admissible row.

25.27 Theorem F: transport cost

Theorem F is checked by tests.

- Along (L), the particle path length (which is at least the Wasserstein length) stays below $\lambda_G^{1/2} L_1 \int \|e\|$ at every sample.
- Along (L_σ) with the OT leak, $W(\mu_{t_1}, \mu_{t_0}) \le \int (\lambda_G^{1/2} L_1 \|e\| + \sigma_m W(\mu_r, \mu^0))$ on a grid of pairs.
- Lemma P.1 holds on 2000 adversarial states, and a tight instance of F(i) reaches ratio ≥ 0.99.

25.28 Proposition G: slow geometry

Scripts: `exp_variable_projection.py` and `exp_slow_manifold_slope.py`, both → `variable_projection/`.

In Table 37, ranges are along the run (5 samples), and "a / b" means first-quarter / last-quarter medians. Velocities are centred differences of period means.

Table 37: Proposition G: the two configurations, `violating` ($m = 4$, $k_e = 20$, $g_a = 2$, $\delta = 0.1$, $T = 800$ s) and `satisfying` ($m = 2$, $k_e = 20$, $g_a = 13$, $\delta = 0.004$, $T = 3000$ s). In the rows for L^* and $\bar{L}$, "$x \to y$" is first sample → last sample.

quantity	violating ($m = 4$)	satisfying ($m = 2$)
(H2a) ω_{ref} / joint fastest	15.3–25.6	6.53–7.27
(H2b) slowest weight / geometry	0.0993–0.231	78.8–87.2
(H2c) S	5.44–25	0.00601–0.00602
G(a) direct check (cosine, G^{-1} metric)	0.9994 / 0.9985	0.9848 / 0.9624 (geometry block 0.9992 / 0.9982)
G(b) cosine	−0.118 / 0.385	0.999 / 0.999
G(b) size ratio	4.207 / 0.472	0.969 / 0.970
L^*	**increases** ($3.8136 \cdot 10^{-3} \to 5.8866 \cdot 10^{-3}$)	decreases ($1.1987 \cdot 10^{-2} \to 4.1073 \cdot 10^{-3}$)
$\bar{L}$ at period-mean weights	non-increasing ($0.21892 \to 0.0080127$)	non-increasing

Slope counterexample. Table 38 (`table_slope.md`). Here $m = 1$, $k_e = 20$, gains (g_a, δ, δ), and (H2b) holds in every row. The closed loop and the joint flow start on the slow manifold of (b′).

- (a) / (b′) isolates the $O(1/R)$ error of the induced-metric flow.
- closed / (a) isolates the averaging error, which shrinks with the gains, plus a 4.4% $O(1/k_e)$ error-channel factor.

What the tables show:

- G(b) holds in the satisfying configuration, where (H2b) and $S \ll 1$ both hold. The violating configuration fails both.

Table 38: Proposition G(b): closed loop against the averaged predictions. "closed / (b)" compares the closed loop with plain variable projection (VarPro) G(b), and "closed / (b′)" with the induced-metric flow G(b′); size is the ratio of displacement norms.

case	(H2b)	S	closed / (b)		closed / (b′)		(a) /	closed /
	ratio		size	cosine	size	cosine	(b′)	(a)
$\delta = 0.47$, $g_a = 20$	81–178	2–2.81	0.422	0.9303†	1.121	0.9986	1.015	1.104
$\delta = 0.47/4$, $g_a = 5$ (all gains ÷4)	81–178	2–2.81	0.401	0.9156	1.065	0.9999	1.015	1.049
$\delta = 0.047$, $g_a = 20$	654–1779	0.2–0.319	0.850	0.9986	1.047	1.0000	1.003	1.044
$\delta = 0.0047$, $g_a = 20$	13162–17790	0.02–0.0229	1.023	1.0000	1.044	1.0000	1.000	1.044

†The centre moves the opposite way.

- The slope rows show that (H2b) alone is not sufficient.
- The induced-metric flow G(b′) tracks the closed loop at every S.
- In the satisfying configuration, the G(a) weight block is dominated by the $O(1/k_e)$ shift of the weight equilibrium: cosine -0.21 / 0.50, carrying 92–95% of the residual. The geometry block matches.

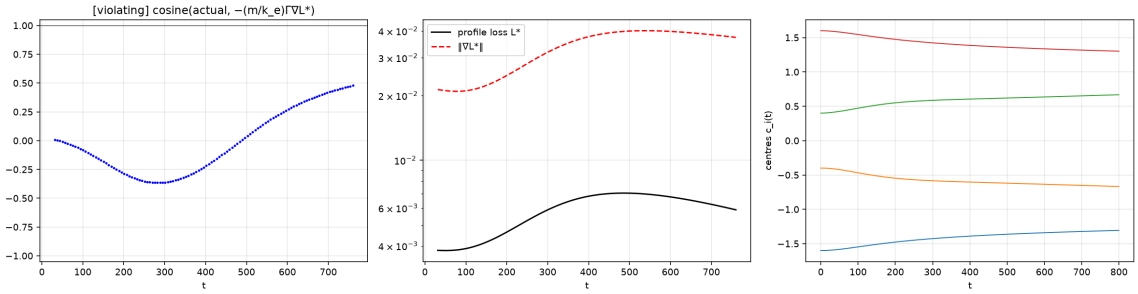


Figure 8: Proposition G, `violating` configuration. Left: the G(b) cosine between the actual geometry velocity and the VarPro direction $-(m\delta/k_e)\Gamma\nabla L^*$, along the run. Centre: the profile loss L^* and $\|\nabla L^*\|$. Right: the centres $c_i(t)$.

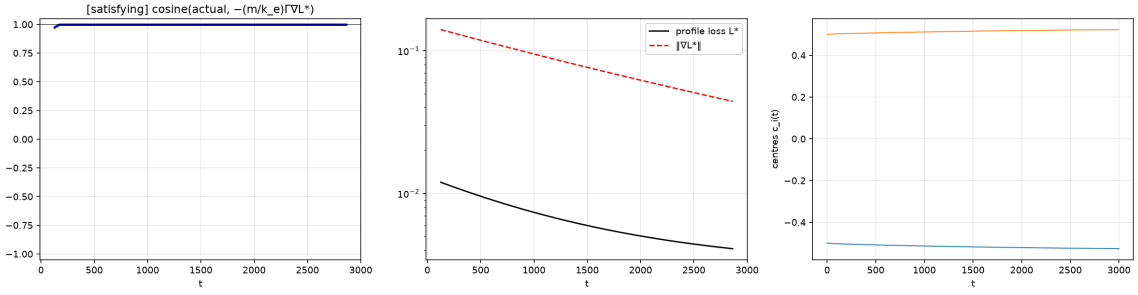


Figure 9: Proposition G, `satisfying` configuration; panels as in Figure 8.

25.29 Theorem H: Langevin adaptive control

Scripts: `exp_langevin_global.py` → `langevin_global/`, `exp_langevin_tradeoff.py` → `langevin_tradeoff/`, and `exp_gibbs_reference.py` → `langevin_global/gibbs_reference.md`.

The setting is the real closed loop with $m = 32$, a scalar plant and a periodic reference.

- The unregularised offline optimum of the occupation loss is about $7.6 \cdot 10^{-6}$ (best of 5 L-BFGS restarts, an upper bound). One run gave $7.0 \cdot 10^{-6}$, because of floating-point summation order.
- The regularised optimum $\bar{L}(\mu_\tau^*)$ is 0.11118 at $(\sigma, \tau) = (0.02, 0.004)$ and 0.00850 at $(0.002, 5 \cdot 10^{-4})$, grid-converged (`exp_gibbs_reference.py` → `langevin_global/gibbs_reference.md`).
- In the scaling $\sigma = 0.2/k_e$, $\tau = 0.04/k_e$ of Conjecture H.6, $\bar{L}(\mu_\tau^*) = 0.11118$ for every k_e. The exactly averaged stationary state gives $\bar{L} = 0.11415$, 0.11148 and 0.11101 at $k_e = 10$, 20 and 40. The $L^2(\rho)$ distance between the two learned functions is $6.81 \cdot 10^{-3}$, $1.43 \cdot 10^{-3}$ and $9.7 \cdot 10^{-4}$.

Speed–bias trade-off. Table 39 and Figure 10 ($\sigma = 0.002$, 2 seeds, `langevin_tradeoff/table.md`).

Table 39: Theorem H: speed–bias trade-off of parameter dither ($m = 32$, $\sigma = 0.002$, $k_e = 10$, $T = 1500$ s, 2 seeds). The escape time is the first sample at which the collapsed run's loss is at most twice the spread run's final loss (same seed); per-seed values are listed, and ∞ means not within the horizon. The losses are the final values of $\bar{L}$.

τ	escape of the collapsed start (s)	loss, collapsed	loss, spread
0	∞, ∞	0.166	0.00696
0.0005	∞, 1050	0.0249	0.0109
0.001	1480, 1005	0.0216	0.0148
0.002	1115, 800	0.0288	0.0233
0.004	970, 430	0.0392	0.0362
0.008	390, 295	0.0603	0.0653
annealed (hot start)	990, 530	0.0317	0.0320

- **Fit.** The seed-averaged escape time fits $A + B \log(1/\tau)$ with an RMS relative residual of 0.045, against 0.253 for an Arrhenius fit. On the eight per-seed values both laws fit poorly (0.274 and 0.330). The data do not identify the escape mechanism, and the "saddle" reading of an earlier version of this analysis is withdrawn (Remark H.2).
- **Caveats.** The threshold grows with τ, and the censored $\tau = 5 \cdot 10^{-4}$ point is excluded from the fit.
- **Guaranteed rate.** The Holley–Stroock [43] rate $2\tau\rho_\tau$ is at least $e^{-8401}\,\mathrm{s}^{-1}$ at $(0.002, 5 \cdot 10^{-4})$ and at least $e^{-1053}\,\mathrm{s}^{-1}$ at $(0.02, 0.004)$.

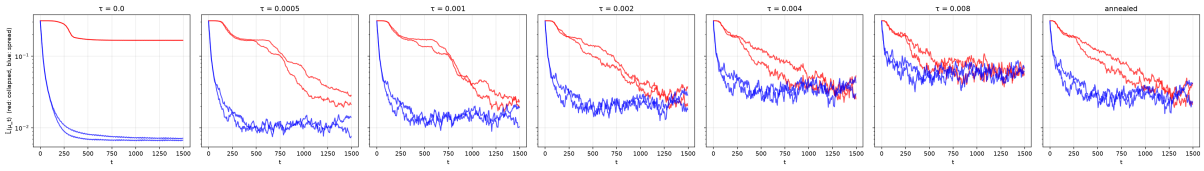
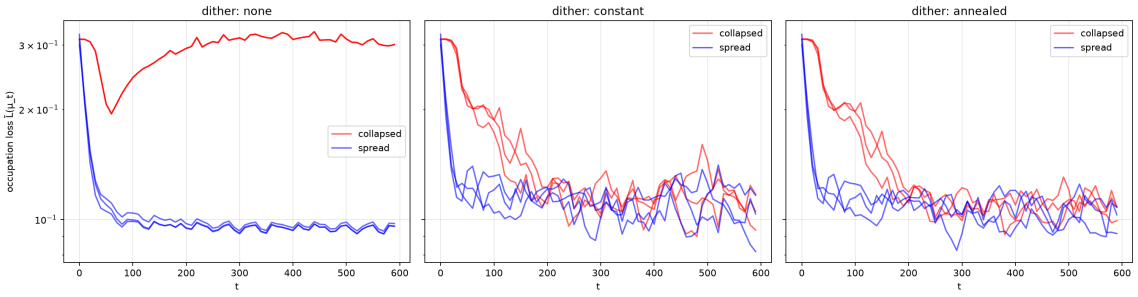


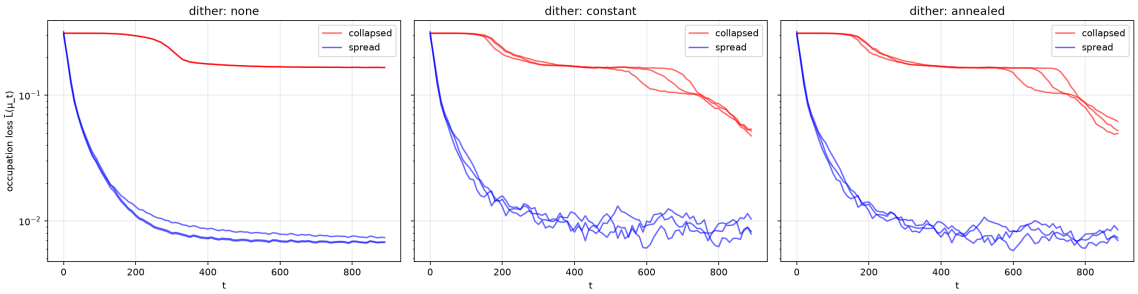
Figure 10: Theorem H, speed–bias trade-off: $\bar{L}(\mu_t)$ of the collapsed start (red) and of the spread start (blue), 2 seeds, for $\tau = 0$, 0.0005, 0.001, 0.002, 0.004, 0.008 and the annealed (hot-start) schedule.

Two fixed settings. Three seeds (`langevin_global/table_*.md`; Figure 11).

- $\sigma = 0.02$, $\tau = 0.004$. Both starts equalise: the loss ratio is 1.03, and the final measures are 4.2× closer in W than without dither. The common loss is about 0.11, which matches the grid value $\bar{L}(\mu_\tau^*) = 0.111$. It is dominated by the regularisation bias.
- $\sigma = 0.002$, $\tau = 5 \cdot 10^{-4}$. Dither frees the collapsed start ($0.166 \to 0.0480$), but the starts do not equalise within 900 s (loss ratio 5.23).



(a) $\sigma = 0.02$, $\tau = 0.004$



(b) $\sigma = 0.002$, $\tau = 5 \cdot 10^{-4}$

Figure 11: Theorem H, global kernel placement ($m = 32$, scalar plant, $k_e = 10$, 3 seeds): the occupation loss $\bar{L}(\mu_t)$ of the collapsed and spread starts without dither, with constant dither and with annealed dither.

25.30 Remark H.0 and Proposition H.7: the exactly averaged closed loop

These are checked by tests. Table 40 gives the relative error of the frozen-μ averaged drift against $(\mathfrak{A})$, and Table 41 the structure of the first correction (`test_averaging.py`).

Table 40: Remark H.0: relative error of the frozen-μ averaged drift against $(\mathfrak{A})$.

instance	$k_e = 10$	20	40	80	$k_e \times$ error
$n = 1$ (`test_langevin.py`)	4.2%	2.0%	0.91%	0.48%	≈ 0.4
$n = 2$, ellipse (`test_averaging.py`)	43%	21%	10%	5.1%	≈ 4.2

Table 41: Proposition H.7: structure of the first correction (`test_averaging.py`).

check	$n = 1$	$n = 2$ (ellipse)
expansion error of $k_e^2(P_\mu - P_0)$ against $U_{\text{even}} + U_{\text{odd}}$, $k_e = 40 \to 80$	$0.54 \to 0.25$	$0.038 \to 0.019$
circulation kernel K	$\equiv 0$ (asserted at 10^{-12})	antisymmetric; equals the enclosed-area integral to $7 \cdot 10^{-15}$ (asserted at 10^{-6})
orbit reversal (same ρ)	$U_{\text{odd}} \equiv 0$	U_{even} unchanged, U_{odd} flipped, to $2 \cdot 10^{-15}$ (asserted at 10^{-10})
$k_e^2 \cdot$ cyclic sum of the antisymmetric second variation, exact ($k_e = 80$) against predicted (asserted at 3%; the full matrix at 5%)	-0.0711 against -0.0704	-0.0471 against -0.0464

25.31 Remark H.7a: the orientation of the reference orbit

Script: `exp_orbit_orientation.py` → `orbit_orientation/`.

The setup is deterministic: $n = 2$, $m = 6$, $G = \text{diag}(4, 4, 1, 1, 1)$, $\sigma = 2/k_e$, no dither. The same elliptic reference is traversed both ways, so the occupation measure is the same, and so is the initial configuration. The runs converged to machine precision: the residual drift over the last 10 periods is about 10^{-16} (between $1.0 \cdot 10^{-16}$ and $5.0 \cdot 10^{-16}$). The six kernels converge to a common configuration (kernel spread 1.05 at the start, at most $5.6 \cdot 10^{-17}$ at the end of every run), so the learned network is effectively a single kernel and the test probes the effect at a collapsed equilibrium. Predictions are linear responses at the (+) equilibrium (Table 42).

Table 42: Remark H.7a: orientation dependence of the learned kernels in the real closed loop. Size ratios are measured / predicted.

		frozen-measure prediction (H.7)		full prediction (H.7 + oscillation term)	
k_e	measured $\|\Theta_+ - \Theta_-\|$	size ratio	cosine	size ratio	cosine
10	$5.729 \cdot 10^{-3}$	2.670	0.7577	0.937	0.9314
20	$2.843 \cdot 10^{-3}$	2.414	0.8302	0.965	0.9809
40	$1.403 \cdot 10^{-3}$	2.294	0.8631	0.969	0.9944

- The vector error $\|\text{measured} - \text{predicted}\|/\|\text{predicted}\|$ of the full prediction is 0.364, 0.195 and 0.109 at $k_e = 10$, 20 and 40.
- The measured difference scales like $k_e^{-1.01}$, against a predicted -1.
- The full first-order theory accounts for it. The frozen-measure terms alone account for less than half.
- A first attempt with $\sigma = 0.5/k_e$ and half the horizon had not converged (residual drift about a third of the difference). It was discarded and rerun.

25.32 Theorem I: reciprocity at order k_e^{-2}

There is no new experiment. Every statement of Theorem I is checked against the generic second-variation code of `src/core/averaging.py` by the tests of `tests/test_nonreciprocity.py` (Table 43).

Table 43: Theorem I: the checks of `tests/test_nonreciprocity.py`.

check	result
$A(\vartheta,\vartheta') = a^\top B(\xi,\xi')a'$ against the antisymmetric part of $S_{\text{even}} + K$, $n = 1$ and 2	agreement to 10^{-10}
$A(\vartheta,\vartheta_z) = 0$ for the zero feature; the cyclic sum over $(\vartheta_0,\vartheta_1,\vartheta_z)$ equals $A(\vartheta_0,\vartheta_1)$	10^{-14}
$n = 1$, $f := \hat{f}_\mu$ (perfect fit on the orbit)	$\max\|A\| < 10^{-13}$ over 20 random pairs
$n = 1$, non-zero residual	$\|A\| > 10^{-3}$ found; equals the closed form $-aa' \frac{c-c'}{s^2} \int r\varphi_c\varphi_{c'}\,\mathrm{d}\rho$ to 10^{-12}
$\kappa(\xi,\xi')$ against its closed form for $n = 2$: the Gaussian factor $-\frac{1}{2s^2}e^{-\lvert c-c'\rvert^2/(4s^2)}$ times $(c-c')\cdot F_s$ at the midpoint of c and c'	10^{-10}
$F_2((0.5,0)) > 0$ on the ellipse; $\kappa \equiv 0$ on a segment traversed back and forth	$F_2 > 10^{-3}$; $\|\kappa\| < 10^{-12}$
$n = 2$ with $f \equiv \hat{f}_\mu$ on $\mathbb{R}^2$ (residual and its Jacobian vanish on the orbit): $B = 2\kappa I$	10^{-13}; $A \neq 0$ on the ellipse, $A \equiv 0$ ($< 10^{-13}$) on the segment
$n = 2$ with $f = \hat{f}_\mu + he_1$, h zero on the ellipse (residual zero on the orbit, Jacobian not symmetric): $B - 2\kappa I = -\int \varphi\varphi'\partial_2 h\,\mathrm{d}\rho \cdot (e_1 e_2^\top - e_2 e_1^\top)$	10^{-12}, and not small (the round-5 correction of Theorem I(iv))
positivity witness $f \equiv 0$, $\mu = \delta_{\bar{\vartheta}}$	integral > 0; the generic code equals the closed form to 10^{-12}

25.33 Theorem J: periodic orbits of the finite-m closed loop

Script: `exp_periodic_orbit.py` → `periodic_orbit/`. Newton shooting on the Poincaré map finds the exact periodic orbit of the real closed loop near the critical point Θ^* of J_m (residual $< 10^{-11}$); its period mean $\bar{\Theta}_p$ and its monodromy matrix are compared with the averaged system (Table 44). Instance A has $n = 2$, $m = 3$, $\sigma_0 = 0.5$, $G = \mathrm{diag}(4,4,1,1,1)$ and a collapsed minimiser ($J_m(\Theta^*) = 0.762775$, $\lambda_{\min}(\nabla^2 J_m) = 0.0148$); instance B has $n = 1$, $m = 2$, $\sigma_0 = 0.05$, $G = \mathrm{diag}(2,1,1)$, two distinct kernels ($J_m(\Theta^*) = 0.258249$, $\lambda_{\min}(\nabla^2 J_m) = 0.0661$) and a fast sweep. Distances and cosines are in the labelled G^{-1} metric.

Table 44: Theorem J: the exact periodic orbit against the averaged system. "vec. error" is the vector error of the first-order location $\Theta^* - J^{-1}(b_{\text{frozen}} + b_{\text{osc}})$ of Proposition H.7 and Remark H.7a; "frozen only" omits b_{osc} (in instance B, $b_{\text{osc}} = 0$ by symmetry). "Floquet" is the maximal relative error of the parameter Floquet exponents against $\frac{1}{k_e}\lambda(G_m^{1/2}\nabla^2 J_m G_m^{1/2})$.

	instance A (collapsed minimiser)				instance B (two distinct kernels)		
k_e	$\|\bar{\Theta}_p - \Theta^*\|$	vec. error	frozen only	Floquet	$\|\bar{\Theta}_p - \Theta^*\|$	vec. error	Floquet
10	$4.09 \cdot 10^{-2}$	0.169	0.398	$8.31 \cdot 10^{-2}$	$6.44 \cdot 10^{-2}$	1.995	$9.48 \cdot 10^{-2}$
20	$1.91 \cdot 10^{-2}$	0.083	0.360	$7.34 \cdot 10^{-2}$	$2.22 \cdot 10^{-2}$	1.078	$3.21 \cdot 10^{-2}$
40	$9.22 \cdot 10^{-3}$	0.041	0.348	$4.50 \cdot 10^{-2}$	$8.37 \cdot 10^{-3}$	0.551	$1.16 \cdot 10^{-2}$
80	$4.53 \cdot 10^{-3}$	0.020	0.343	$2.46 \cdot 10^{-2}$	$3.51 \cdot 10^{-3}$	0.277	$4.63 \cdot 10^{-3}$
160	$2.24 \cdot 10^{-3}$	0.010	0.342	$1.28 \cdot 10^{-2}$	$1.59 \cdot 10^{-3}$	0.139	$2.03 \cdot 10^{-3}$

- Fitted exponent of the offset $\|\bar{\Theta}_p - \Theta^*\|$ against k_e: -1.05 (A) and -1.33 (B; not yet asymptotic below $k_e \approx 80$); Theorem J gives -1.
- The vector error of the first-order location halves with every doubling of k_e in both instances. In A the frozen-measure terms alone leave a 35% error at every k_e, so the oscillation term of Remark H.7a is an $O(1)$ fraction of the offset.

- The median relative error of the parameter Floquet exponents falls from $1.2 \cdot 10^{-2}$ to $1.1 \cdot 10^{-3}$ (A) and from $4.0 \cdot 10^{-2}$ to $9.5 \cdot 10^{-4}$ (B) between $k_e = 10$ and 160.
- Error channel: the error-channel multipliers are below the resolution of the finite-difference monodromy (exponents of order k_e), so their product is recovered from Liouville's identity $\sum \log|\lambda| = \int_0^{T_0} \operatorname{tr} \partial F/\partial z \, \mathrm{d}t$ along the orbit. The mean error-channel exponent is $k_e - L$ with $L = 0.160$, 0.111, 0.086, 0.074, 0.068 (A) and $L = -0.054$, -0.074, -0.084, -0.089, -0.092 (B) at $k_e = 10$, 20, 40, 80, 160, as Theorem J(b) states ($L = O(1)$).
- Tests (`tests/test_periodic.py`): ∇J_m against finite differences; shooting residual $< 10^{-10}$; the offset, the vector error and the Floquet error all fall by the required factors from $k_e = 20$ to 40 on instance B.

25.34 Proposition K: collapsed configurations and the orientation of the orbit

Script: `exp_orientation_stability.py` → `orientation_stability/`. The setting is instance A of Theorem J ($n = 2$, $m = 3$, $G = \operatorname{diag}(4, 4, 1, 1, 1)$) with the ellipse $(1.2\cos t, \pm 0.8 \sin t)$ traversed both ways. The Floquet exponents of the collapsed periodic orbit of the real closed loop are computed by Newton shooting and restricted to the separation subspace $\sum_i u_i = 0$; an exponent is $-\log|\lambda|/T_0$, so positive means stable. The leading-order separation exponents are the eigenvalues of $(\sigma_0 I - G^{1/2} R_2 G^{1/2})/k_e$, identical for both orientations. The second-order prediction is the linearisation of $\frac{1}{k_e}\langle H \rangle + b_{\text{frozen}} + b_{\text{osc}}$ at its own equilibrium $\bar{\Theta}_2$ (every $\bar{\Theta}_2$ found is collapsed, kernel spread below $1.2 \cdot 10^{-8}$); the prediction "at Θ^*" uses the Jacobian at the collapsed critical point instead, which omits the third-derivative term (Table 45).

Table 45: Proposition K: the softest separation Floquet exponent of the collapsed orbit for the two orientations at $\sigma_0 = 0.5$, and its orientation difference, measured against the second-order prediction (`orientation_stability/table.md`).

	softest separation exponent		difference (+) − (−)			$k_e^2 \times$ difference	
k_e	(+)	(−)	measured	pred. at $\bar{\Theta}_2$	pred. at Θ^*	measured	pred. at $\bar{\Theta}_2$
10	$3.563 \cdot 10^{-3}$	$2.660 \cdot 10^{-3}$	$9.03 \cdot 10^{-4}$	$7.77 \cdot 10^{-4}$	$9.01 \cdot 10^{-4}$	0.090	0.078
20	$1.800 \cdot 10^{-3}$	$1.614 \cdot 10^{-3}$	$1.86 \cdot 10^{-4}$	$1.69 \cdot 10^{-4}$	$2.08 \cdot 10^{-4}$	0.075	0.068
40	$9.277 \cdot 10^{-4}$	$8.863 \cdot 10^{-4}$	$4.14 \cdot 10^{-5}$	$3.93 \cdot 10^{-5}$	$4.98 \cdot 10^{-5}$	0.066	0.063
80	$4.737 \cdot 10^{-4}$	$4.640 \cdot 10^{-4}$	$9.71 \cdot 10^{-6}$	$9.44 \cdot 10^{-6}$	$1.22 \cdot 10^{-5}$	0.062	0.060

- Fitted exponent of the difference against k_e: -2.18 (Proposition K(iii): -2). The prediction at $\bar{\Theta}_2$ converges to the measured k_e^2-scaled limit; the prediction at Θ^*, which omits the third-derivative term, converges to a value 30% too large.
- At $\sigma_{\text{crit}} = 0.4606$ both orientations are unstable ($-4.1 \cdot 10^{-4}$ and $-1.27 \cdot 10^{-3}$ at $k_e = 10$); the even part dominates. The two separation exponents coincide (multiplicity $m - 1 = 2$, Proposition K(ii)).
- Designed σ_0, chosen so that the predicted softest exponents of the two orientations are opposite ($\text{pred}_+ + \text{pred}_- = 0$): $\sigma_0 - \sigma_{\text{crit}} = 0.0125$ ($k_e = 10$) and 0.0061 ($k_e = 20$). The measured softest exponents are $+8.5 \cdot 10^{-4} / -2.5 \cdot 10^{-5}$ and $+1.4 \cdot 10^{-4} / -3.7 \cdot 10^{-5}$: one orientation of the same reference orbit (same occupation measure, same averaged system) leaves the collapsed configuration stable and the other makes it unstable.
- Test `test_proposition_K_hessian_splitting_and_critical_leak` in `tests/test_periodic.py`: the Hessian splitting at the collapsed point (10^{-5}), the gap equation of σ_{crit} (10^{-8}), the two-fold zero separation mode at σ_{crit}.

25.35 Theorem J′: the stationary law of the dithered single-kernel loop

Script: `exp_dither_stationary.py` → `dither/`. Single-kernel instance ($n = 1$, $m = 1$, $\vartheta = (\alpha, \gamma)$, frozen width, $G = \mathrm{diag}(2, 2)$, $\sigma_0 = 0.4$, $\tau_0 = 0.1$), reference traversed both ways; $2 \cdot 10^4$ independent replicas per (k_e, orientation) started from the first-order law, $k_e = 10$, 20, 40, 80; the Gibbs measure, the first-order correction g, the current j_1 and the circulation coefficient on a 201^2 grid. Full tables in `results/dither/table.md` (raw in `raw.npz`); Table 46.

Table 46: Theorem J′: the stationary law of the dithered single-kernel loop against the Gibbs measure and the first-order prediction (`dither/table.md`).

	$k_e = 10$	20	40	80
$(m_\varepsilon(\alpha) - \pi(\alpha)) \cdot k_e$, prediction -0.255	-0.28	-0.36	-0.36	-0.37
$(m_\varepsilon(\gamma) - \pi(\gamma)) \cdot k_e$, prediction -0.426	-0.44	-0.54	-0.51	-0.67
residual $m_\varepsilon(\alpha)$ − first order (s.e. 0.0013)	-0.0021	-0.0051	-0.0025	-0.0015
residual $m_\varepsilon(\gamma)$ − first order (s.e. 0.0018)	-0.0015	-0.0056	-0.0020	-0.0030
orientation-odd shift of γ, measured/predicted	0.93 ± 0.04	1.10 ± 0.07	1.15 ± 0.15	1.38 ± 0.29
$k_e^2 \mathbb{E}e^2/(2\pi(\bar{L}))$, orientation +1	1.054	1.033	1.017	1.009
circulation Λ_ε (+1), measured ± 0.0005 / predicted	$+0.0037$ / -0.0005	$+0.0006$ / -0.0003	-0.0004 / -0.0001	$+0.0003$ / -0.0001

- The Gibbs discrepancy decays like $k_e^{-0.87}$ (α), $k_e^{-0.83}$ (γ), $k_e^{-1.09}$ ($\alpha\gamma$), and the first-order prediction removes 93–97% of it at $k_e = 10$ and 70–85% at $k_e = 20$–40; the residual is at the Monte Carlo floor at $k_e = 80$ and its second-order decay is not resolved.
- The orientation-odd shift (the dither version of Remark H.7a) is predicted to 7–8% at $k_e = 10$ and within the 10–30% resolution at larger k_e.
- The tracking error is the Gibbs-averaged occupation loss over $k_e^2/2$ with an $O(1/k_e)$ correction of 0.5–$0.7/k_e$.
- The first-order circulation is below the Monte Carlo resolution for $k_e \geq 20$ and dominated by higher orders at $k_e = 10$; the current machinery of Proposition J′.2 is checked on the first-order averaged SDE on the grid: $\int \Theta \wedge (j_\varepsilon - j_0)/(\varepsilon \int \Theta \wedge j_1) = 1.47$, 1.21, 1.09, 1.03 at $\varepsilon = 0.2$, 0.1, 0.05, 0.025 and $\|m_\varepsilon - m_0(1 + \varepsilon g)\|_1/\varepsilon^2 = 0.31$ constant.
- Referee round 6 (`review/REVIEW_round6_Jprime.md`): the first run's batch standard errors were biased low by 1.4–1.9 and its burn-in too short (JP-4–JP-6); the run reported here is the corrected one.
- Corollary J′.3 (`exp_gibbs_m.py` → `dither/gibbs_m.md`): μ_τ^* on the grid ($\int \alpha \, \mathrm{d}\mu^* = 0.4963$, $\int \gamma \, \mathrm{d}\mu^* = 0.1873$, $\int \alpha^2 \, \mathrm{d}\mu^* = 0.9353$, $\bar{L}(\mu^*) = 0.1394$); the Gibbs measures π_m for $m = 1$–32 by MALA (4000 replicas): $\mathbb{E}[\bar{L}(\mu_\Theta)] = 0.1960$, 0.1681, 0.1540, 0.1470, 0.1432, 0.1413 with $(\mathbb{E}[\bar{L}] - \bar{L}(\mu^*)) \cdot m = 0.057$–$0.061$ (slope -0.99), $\mathbb{E}[\gamma_1]$ slope -1.01, $\mathbb{E}[\alpha_1^2]$ from 0.536 to 0.922 with rescaled deviation -0.41 ± 0.01; the $m = 1$ row matches the quadrature of π_1 within one standard error.

26 Research program

Each branch states what is proved, what is open, and the first concrete step.

1. **Infinite-horizon averaging for McKean–Vlasov closed loops (Conjecture H.6).**
   - *Proved:* the averaged system converges globally and exponentially (H(ii)); the exact first correction at frozen measure (H.7); input-to-state stability of the averaged system (H.8).
   - *Open:* a near-identity averaging transformation for the stochastic McKean–Vlasov loop that keeps the frozen-measure correction valid along the moving measure, uniformly in time.
   - *Done (Theorem J, Section 22):* the deterministic case ($\tau = 0$, $\sigma = \sigma_0/k_e$) at fixed m: near every hyperbolic critical point of J_m the real loop has a unique periodic orbit with the same stability type, a k_e-uniform basin, and the first-order location of Proposition H.7 and Remark H.7a.
   - *Done (Theorem J′, Section 24):* the dithered case at fixed m. The invariant law of the loop has the Gibbs measure $\exp(-J_m/\tau_0)$ as its parameter marginal up to $O(1/k_e)$, with the first-order correction driven by the same drift b as Theorem J; the fast error and the phase are eliminated by a Poisson-equation expansion of the generator rather than a transformation of the Fokker–Planck equation.
   - *Next steps:* uniformity in m — the mean-field limit of the stationary law (the Gibbs measures of J_m converge to the minimiser of the mean-field free energy F_τ of Theorem H, a propagation-of-chaos question for Gibbs measures which is where Conjecture H.6 now lives), the second-order correction, and quantitative constants in place of the $O(\varepsilon^2)$.
2. **Non-reciprocal mean-field dynamics from closed-loop learning (new; Proposition H.7).**
   - *Proved at a frozen measure:* the averaged adaptive dynamics is a semi-gradient flow plus a lag term (Proposition H.7). Along the real loop a second-order averaging term is added (Remark H.7a).
   - Its interaction has two antisymmetric parts:
     - the performative defect A_{even} of the state shift, which is present for every n and is the only part for $n = 1$;
     - the circulation kernel K, an integral over the region that the reference orbit encloses ($n \geq 2$).
   - *Proved (Theorem I, Section 21):* reciprocity at order k_e^{-2} is equivalent to $A_\mu \equiv 0$; for $n = 1$ it holds exactly at perfect fit on the orbit, and for $n \geq 2$ at perfect fit exactly when the orbit carries no circulation and the residual's Jacobian is symmetric along the orbit; no open set of measures is reciprocal, for any n. The performative defect is bounded by the residual and the antisymmetric part of its Jacobian; the circulation defect depends on neither.
   - *Open questions:*
     - Are stationary states unique beyond the perturbative regime?
     - Can circulation be *designed*, choosing the orbit's orientation and area, to break symmetric traps or to speed escape from the collapsed configuration? (Partly answered by Proposition K, below: the orientation decides the stability of the collapsed configuration in a window around the critical leak.)
     - Is there a Lyapunov functional other than F_τ for the non-reciprocal flow?
   - *Context:* the closest analogues are performative prediction [71], semi-gradient TD learning [88], [85] and non-reciprocal phase transitions [32].
   - *Done (Remark H.7a):* deterministic $n = 2$ closed-loop runs with the reference traversed both ways (same ρ). The full first-order theory predicts the $O(1/k_e)$ difference of the

learned kernels to 3% in size and 11% as a vector.

- *Done (Proposition K, Section 23):* at a collapsed configuration the orientation-odd k_e^{-2} terms shift the separation Floquet exponents of the real loop by $O(k_e^{-2})$, with a closed-form prediction from the averaged data; at the critical leak σ_{crit} the direction of traversal decides whether the kernels separate, and the σ_0 at which it does can be computed in advance.
- *Done (Theorem J′):* the test with dither. The orientation-odd part of the stationary parameter means is $2\varepsilon \int \Theta g_{\mathrm{odd}}\, \mathrm{d}\pi$, with g_{odd} driven by the odd part of b; measured on the single-kernel instance of Section 24 (where, in one state dimension, the whole odd effect is b_{osc}).
- *New (Proposition J′.2): the stationary state is a non-equilibrium steady state.* The first-order averaged dynamics carries a stationary current $j_1 = \pi(b - \tau_0 G \nabla g)$ exactly when b is not a G-gradient, which for $m \geq 3$ is always the case; its entropy production rate is $\varepsilon^2 \tau_0^{-1} \pi(\|b_\perp\|^2_{G^{-1}})$; and the real loop's stationary parameter process is not reversible. The Lyapunov-functional question for the non-reciprocal flow is therefore a question about a non-equilibrium steady state (its free energy is not a Lyapunov functional of the real loop at order ε).
- *Next steps:* the design of the orbit's area (not only its orientation) as a control input; the orientation effect at non-collapsed configurations with several kernels; measuring the current of the real loop (below the Monte Carlo resolution in Section 24's instance; a variance-reduced estimator or a stronger instance is needed); the mean-field limit of the current.

3. **Uniform-in-time propagation of chaos with a plant in the loop.**

- Theorems D and D′ are finite-horizon; D′ gives the $m^{-1/2}$ rate for the dithered loop by synchronous coupling, with constants independent of the temperature and an exponent independent of the dimension for the state and the learned function. Uniform-in-time results exist for mean-field Langevin dynamics [11] and for shallow networks [36].
- *Done (Theorem D″, Section 17):* uniform in time when the leak dominates the loop gain of the adaptive law (a small-gain condition between the plant channel and the particle channel, explicit in Lipschitz constants). The certified regime is leak-dominated: the network does not learn (without dither it collapses to one kernel; with dither it stays in the cloud of the prior, where the bound has content: the coupled distance is of order $m^{-1/2}$ and bounded in time). The result marks where the contraction method stops rather than settling the question.
- *Open, and now with a negative indication:* the learning regime (kernels separated, leak below the critical value of Proposition K), where the plant couples all particles through e and (by H.7) the averaged interaction is not reciprocal. The long-horizon run of `exp_chaos_uniform.py` ($\sigma = \sigma_K/3$, 1000 s) shows the coupled distance growing threefold between the windows $(100, 200]$ and $(500, 1000]$ while keeping the $m^{-1/2}$ law at each time: on that horizon only the finite-horizon Theorem D′ is supported. Whether the growth saturates (the leak's time scale is $1/\sigma = 2000$ s), and whether a uniform statement holds at all in the learning regime, is the open question; the first draft's claim of empirical uniformity was based on a run that was in fact a slow collapse and is withdrawn (review round 6).
- *Done (Theorem D‴, Section 18):* for the averaged slow loop at $\tau = 0$ the question is settled where the fitted configuration is non-degenerately stable: the interaction term of the synchronous coupling is a negative square at linear order (flat convexity), the self term carries the residual-weighted curvature at the actual residual, and uniform-in-time propagation of chaos at the rate $1/m$ (mean square) holds under the strict second-order

optimality condition along the mean-field flow, with explicit constants carrying the transient's $e^{2\Lambda_1}$; the window $\kappa_R[\mu_\infty] < \sigma_0 < \sigma_{\mathrm{crit}}$ is where learning and uniformity coexist. The cold start of Theorem D‴'s experiment was outside it on two counts (the expanding transient, and a connected support threading the ridges, where the coupling has a floor of pairs in different basins). At finite m the stationary coupled distance is the weight-sensitivity law of Proposition D‴.5, and Corollary D‴.7 carries the uniform statement to the real loop on a warm start through Theorem J.

- *Done (Corollary D‴.14, Proposition D‴.15):* in the real loop the collapsed-by-class subspace is invariant and carries the weighted K_0-atom loop, so Theorem J's orbit at a replicated configuration is, under its uniqueness-ball proviso, the lift of one K_0-atom orbit — with the OT leak the same orbit for every commensurate size (the real loop's finite-size floor is zero), with the nearest-atom leak the orbit of the realised weights (Corollary D‴.7's floor is exactly the distance between two weighted K_0-atom orbits); the Floquet spectrum of a lifted orbit is the K_0-atom orbit's together with each class's single-particle (ghost) spectrum of multiplicity $n_k - 1$, independent of m — the spectral part of Corollary D‴.7's uniformity question.
- *Done (Theorem D‴.18):* Theorem J uniformly in m at a replicated configuration with admissible weights (margins λ_0 on a compact set $\mathcal{W}$), in the per-particle norm — one threshold k_0, one basin radius ρ and one rate for every size; for $k_e \geq k_0$ the proviso of Corollary D‴.14(ii) is replaced by this m-uniform statement. In the mean-square norm no m-independent basin exists for the nearest-atom leak (Remark D‴.19).
- *Done (Theorem D‴.21):* the measure-level basin of the OT leak, for the averaged loop. The OT energy satisfies a Polyak–Łojasiewicz inequality whose constant is set by the residual-weighted curvature along the plan segments and does not depend on m (sharp, $2\sigma_0$, at a matching prior); the excess energy decays exponentially from an m-independent sublevel set (tiny on the instance) and a posteriori wherever the plan-segment margin is positive, which covers configurations with a fixed fraction of the particles in other cells; under a curvature hypothesis at the stationary residual the replicated configuration is the unique global minimum among the empirical measures, while the nearest-atom energy has a spurious minimum at every admissible count vector. *Open:* the transit of the real loop (averaging with plan switches) and the existence of the mean-field OT flow (Remark D‴.22); the dithered real loop with the OT leak (Remark D‴.16, checked on the averaged loop: under the OT leak a ridge crossing is reversible — the particle stays assigned to its own atom — while under the nearest-atom leak dither equalises the masses only on the Arrhenius time scale) — Theorem J′'s numerics with the two leaks.
- *Open:* the measure-weighted form of hypothesis (H) (a bad set of small μ_s-mass, the bootstrap on the good particles, the flip fraction p_m and its m-law); the transfer to the real loop at finite k_e through Theorem J′'s $O(\varepsilon)$ corrections, uniformly in time; dither (the entropy method of [11], whose hypotheses Theorem H verifies, against the coupling's floor).

4. **Geometry of two-time-scale adaptation (Proposition G(b′)).**

- *Proved formally, and checked:* the slow geometry follows the gradient flow of the profile loss in the metric induced on the slow manifold, and the slope S measures the departure from plain variable projection.
- *Open:* a rigorous long-horizon statement; the $O(1/R)$ corrections, which involve D^2a^* and $\partial_\xi M$; gain designs that shape the induced metric.

5. **Noiseless global convergence.**

- *Done (Proposition K):* the local criterion for the collapsed trap: a collapsed configuration

is a strict local minimum of the regularised loss iff $\sigma_0 > \sigma_{\mathrm{crit}}(\bar{\vartheta}) = \lambda_{\max}(G^{1/2} R_2 G^{1/2})$, where R_2 is the Hessian of the first-variation potential at the atom; below σ_{crit} the noiseless flow leaves the trap along the softest separation mode, without dither, and the residual at the atom is what drives the separation (R_2 is linear in it).

- *Open:* a closed-loop separation condition under which the noiseless flow avoids spurious stationary points, despite the failure of the Chizat–Bach [14] boundary conditions. Candidates are a compactified width coordinate and reflecting boundaries (Remark H.3).

6. **Unbalanced transport, meaning online structure adaptation.**

   - Treat the weights as masses (Hellinger–Kantorovich / Wasserstein–Fisher–Rao) to obtain birth and death of kernels with Lyapunov guarantees. The conic particle descent of [16] and [23] is the static analogue.

7. **Excitation theory in measure space.**

   - *Done (Lemma E.0):* a quantitative, uniform UES estimate for the skew-symmetric class, explicit in $(k, \alpha, T, \Psi_M, \Psi_D)$, which reproduces the ω^{-2} law of Remark E.1.
   - *Done (Lemma E.0′):* a strict Lyapunov function whose rate is the averaged rate $\lambda_{\min}\langle \Psi^\top \Psi \rangle / k$ up to a factor $1 - \delta$, for k above a threshold set by the peak factors of Ψ, $\dot{\Psi}$, $\ddot{\Psi}$ (second-order slaving of the error to its slow manifold and a second-order averaging correction, whose mean part vanishes identically); the ill-conditioning of the averaged Gramian is paid once, as in the exact rate.
   - *Done (Corollary E.0′.3):* the same Lyapunov function bounds the stationary parameter spread of the dithered linearised loop, $\limsup \mathbb{E}\|\zeta\|^2 \le (1 + O(\varepsilon)) k \|\Sigma\|_F^2 / (2(1-\delta)\lambda_{\min})$, within $1/(1-\delta)$ of the exact (averaged Langevin) value on isotropic regressors and paying the condition number in the trace otherwise: an explicit design formula for how much dither a given tracking budget tolerates.
   - *Done (Lemma E.0″):* the non-periodic case. The period mean is replaced by a forward exponential average over a window T (whose derivative identity $T\dot{\bar{M}} = \bar{M} - M$ gives the antiderivatives in closed form), the peak factors become time-local, and one residual second-order term $\bar{N}_2$ appears, costing $\delta_2 = O(1/k)$ in the rate; the guaranteed rate is $(1 - \delta - \delta_2)\lambda_T / k$ with λ_T the exponentially weighted PE level of the window. Checked on a quasi-periodic regressor and on a regressor driven by a Lorenz orbit, where no periodic construction applies.
   - *Done (Theorem D″.23(ii)):* at an exact representation the measure-space excitation spectrum h_j — the whitened weighted Gauss–Newton spectrum of the K_0-atom loss — sets both a fixed prior's bias, the fraction $\sigma_0/(h_j + \sigma_0)$ of its error kept in mode j, and an adaptive prior's rates $\lambda_j^- < \min\{h_j, \eta\}$ ($h_j, \eta > 0$): removing the leak's bias costs the excitation of every collective mode, and the leak's a-priori bound on the parameters (the leak toward an adaptive prior is an internal force, and the solutions can run off where no exact representation exists; projecting the prior onto a compact convex set restores the bound, Proposition D‴.27) — the measure-space form of the classical trade-off between a leak's robustness and persistent excitation (an unexcited mode, $h_j = 0$, drifts with the prior).
   - *Done (Proposition D″.24(i)):* the same spectrum sets the finite-size floor of a fixed prior, each mode weighted by the band-pass $h_j \sigma_0^2 / (h_j + \sigma_0)^2 \le \sigma_0/4$, centred at $h = \sigma_0$ — the floor sits in the modes that carry the gauge's displacement, re-weighted toward the level of the leak.

- *Open:* the continuum form of (PE); the trajectory kernel

$$K_T(\vartheta,\vartheta') = \int \Phi(x(s);\vartheta)\Phi(x(s);\vartheta')\,\mathrm{d}s;$$

identifiability of measures up to symmetry; the relation to [39]. Remark E.1 shows that the PE level alone does not predict the rate. For Lemmas E.0′ and E.0″: lowering the threshold k_0 (a third-order slaving trades ρ_{DD} for the peak factor of $\ddot{\Psi}$ one order higher); the choice of the window T as an optimisation; the nonlinear Theorem E with the constants of E.0″ (Corollary E.0′.2's steps carry over verbatim, with λ_T in place of $\lambda_{\min}(\bar{M})$).

8. **Transport-regularised adaptive laws.**

- Wasserstein proximal (JKO) discrete-time adaptive laws.
- *Done (Theorem C′, Section 13):* the OT leak toward priors with a different number of atoms, using optimal plans with barycentric projection; it is Lloyd's algorithm in continuous time, and its bias is the centroid defect of the ideal.
- *Open (in part):* its mean-field form — done for an atomic prior (the Laguerre-cell leak, Proposition D‴.12), open for a general prior ν and a general μ (where the barycentric projection is the superdifferential of $\frac{1}{2}W^2(\cdot,\nu)$; Ambrosio–Gigli–Savaré [2, Thm. 10.4.12]) — and adaptive priors (ν itself learned: its atoms, Theorem D‴.23; its masses, Proposition D‴.24).
- *Done (Proposition D‴.9, Section 18):* the nearest-atom leak toward a prior measure inherits Theorem D‴, Proposition D‴.5 and Corollary D‴.7 inside the Voronoi cells; the prior is a design input of the window (maximal margin $\kappa = \sigma_0$ at a matching prior, the window closing as the prior is misplaced) and of the floors (which grow with the stationary amplitudes).
- *Done (Proposition D‴.12):* the barycentric OT leak's mass constraint against the weight fluctuation. The nearest-atom leak conserves the cluster masses, and the finite-m floor of Proposition D‴.5 is the memory of the initial cloud that this conservation carries; the OT leak constrains the masses to the prior's, so on the sizes commensurate with the masses the finite-m stationary measure is the mean-field one exactly (the replicated weighted configuration is a critical point for every such m, by total unimodularity and c-cyclical monotonicity), and otherwise differs from it by at most $K_0 - 1$ split particles; for an atomic prior the mean-field form of the barycentric leak is the Laguerre-cell leak. *Open:* a coupling theorem for the OT dynamics from an i.i.d. start (Theorem D‴'s synchronous coupling needs a balanced sample, of probability $\approx 0.83/m$, and otherwise keeps the transit's mismatch for ever; the convergence of every trajectory to a Clarke-critical point is Proposition D‴.17, without a rate; Theorem D‴.21 gives the rate in an m-independent energy basin, with the plan as the test direction, and formally (Remark D‴.22) propagation of chaos with no floor there; smallness in m during the transit from an i.i.d. start remains open); a penalised (unbalanced) mass constraint — the entropic one is Proposition D‴.20, which equalises the masses only down to a critical excess and conserves smaller ones, so that its floor returns at large m (Proposition D‴.5's law at the entropic stationary point); the mean-field form for a general prior ν (the barycentric projection as the superdifferential of $\frac{1}{2}W^2(\cdot,\nu)$, Ambrosio–Gigli–Savaré [2, Thm. 10.4.12]) and adaptive priors with learned masses beyond the cell masses (Proposition D‴.24).
- *Done (Theorem D‴.23):* adaptive priors with the masses fixed. The prior's atoms move toward the class barycentres of the plan; the joint flow is the gradient flow of $\bar{L} + (\sigma_0/2)W^2(\mu_\Theta,\nu_Z)$, its replicated stationary points are the unregularised K_0-atom critical points (no leak bias; the non-replicated ones carry the K_0-cluster bias), and

at an exact representation every collective mode relaxes at $\lambda_j^- < \min\{h_j, \eta\}$, for every commensurate m; the fast limit is a capacity-constrained K_0-means regulariser. The transit drags the prior and its weak modes remember the drag, so the adaptation starts from the fixed-prior stationary point (a gate). *Open:* the real loop with an adaptive prior over the adaptation's own time scale beyond the linearisation (Corollary D‴.26); the plateaus of the unregularised loss that the fast limit exposes; the constants of (v).

- *Done (Corollary D‴.26):* perfect tracking with an adaptive prior is an exponentially stable periodic solution of the real loop (locally, for each m) at every representable sample: its Floquet exponents are σ_0/k_e exactly on the separations and the weighted K_0-atom loop's on the collapsed subspace, for every m, and the latter are $(1/k_e)[\lambda_j^{\pm}(w) + c_j^{\pm}(w)/k_e^2 + O(k_e^{-4})]$ — the odd orders vanish (the time-reversal parity of the averaged terms), c_j explicit — the error filter slowing, the slaving speeding up, its sign not fixed. *Open:* its basin uniformly in m (Theorem D‴.18's contraction for the joint system).
- *Done (Lemma D‴.26a):* the slow Floquet expansion proved for the linearised loop. Chang's periodic decoupling makes the slaving exact, so the slow multipliers are exactly the slow block's, with an expansion to $O(k_e^{-4})$ whose remainder is bounded; defining the slow block at negative ε by the same contraction, the time-reversal symmetry becomes the identity $\operatorname{eig} \bar{A}(-\varepsilon) = \operatorname{eig} \bar{A}(\varepsilon)$ — the odd orders vanish exactly; and Chang's second transformation decouples the slow coordinates' drift exactly, their noise $\varepsilon b \nu$ with b the high-passed $-(\dot{K} + aK)^\top$ at leading order and its next term explicit, which proves Proposition D‴.30(ii)'s floor for the linearised loop under white noise as a period average (pointwise the fast jitter oscillates by $\pm 1\%$ on the instance). *Open:* the constants explicit; the evenness of the noise floor without the slope (observed to $k_e = 800$, not proved); the real loop's nonlinear noise response.
- *Done (Proposition D‴.28):* the adaptive prior's floor removal is a property of the amplitude–mass gauge, not of exact representability: at any nondegenerate critical point of the unregularised loss the gauge transform is a critical point at the sampled weights, with the same function, so at a strict local minimum the function-level floor vanishes at the samples of the gauge set W_g with positive margins, whatever the approximation error (W_g shrinking as the error grows); the fixed prior's floor is the bias formula with the full Hessian; for a scalar amplitude the residual curvature scales exactly by q_k/w_k under the gauge, so the adaptive prior's splitting margins are $\sigma_0 - (q_k/w_k)\lambda_k^+$. *Open:* critical points at infinity (the best approximant on the saturation boundary, as for a sharp friction step, a cogging function that two kernels fit as a dipole, or Proposition D‴.29's servo globally), where only the projected prior keeps the parameters bounded; vector amplitudes, where the margins have no closed form.
- *Done (Proposition D‴.31):* the adaptive prior is a washout leak, and the remedy for the errors in variables. The prior lags the class barycentre by its exponentially forgotten increments, so the leak acts in full on the separations and on the barycentre through the high-pass $i\omega/(i\omega + \eta)$ — the bias removal and the lost robustness to persistent forces in one transfer function; along every affine time-invariant prior the stationary response R and the bias B satisfy $RH + B = I$, so no prior gives both; the response to a force is at most the norm of its smallest equivalent target perturbation, under every leak; and the desired regressor (the adaptive law's regressor at the reference) keeps the lift, the linearisation and the first-order floor while turning the noise's own persistent force into a least-squares one, bounded uniformly in k_e and in the spectrum — the parameters' fluctuation adding a force that is not the learning force of a mean tracking error (the classical second-order bias of a constant-gain stochastic approximation): small in absolute terms on the instance ($\leq 3.9 \cdot 10^{-4}$ per $(\mathbb{E}\nu^2)^2$), -8.7% to $+3.3\%$ of DR's and DR*'s biases and $+56\%$ of DR's at $k_e = B = 100$, its own response falling like k_e^{-2} on a

one-dimensional reference, where the law's leading term averages out — which on a curve in $\mathbb{R}^n$ it need not. *Open:* the second-order computation made rigorous, the parameters' part in closed form, and in $\mathbb{R}^n$; the stationary biases in the real loop (the weak modes need $\sim 10^5$ time units); a sensor model with quantisation and delay.

- *Done (Proposition D‴.30):* measurement noise at perfect tracking. The noise is coherent — a replicated start stays replicated, so its floor does not average out with m (invariance, not attraction: near the lift the separations grow above a noise level, which drifts along the trajectory with the developing errors-in-variables deviation); at first order the loop high-passes the measured error's noise and the function-level floor is $(N_0/2k_e)[\sum_j Q_{jj}(h_j+\eta)/(h_j+\eta+\sigma_0) + \operatorname{tr} H^2]$, the adaptive prior costing at most the filter factor; at second order the noise inside the regressor exerts a mean force (the errors in variables) whose stationary response is $(H+\sigma_0 I)^{-1}$ under the fixed prior and H^{-1} under the adaptive prior for every $\eta > 0$ — the robustness half of the leak, lost in the weakly excited modes. *Open:* the real loop's noise response beyond the linearisation (the linearised loop under white noise: Lemma D‴.26a); a projected or partially fixed prior on the weak modes; the optimal k_e and η for a given noise level.
- *Done (Proposition D‴.29):* second-order plants. With the sliding variable $s = \dot{e} + \lambda e$ the error channel is Theorem J's with s in place of e, so the averaged slow loop — and with it every averaged-loop statement of Section 18 — holds (formally) along the regressor's orbit, the position error being the residual filtered by $(\mathrm{d}/\mathrm{d}t + \lambda)^{-1}(\mathrm{d}/\mathrm{d}t + k_s)^{-1}$. On a cogging-torque servo whose torque four kernels cannot represent — the loss's infimum lies at the saturation, and its local minimum $\vartheta^b(q)$ is Proposition D‴.28's regime — the adaptive prior removes the fixed prior's floor and, from an engineer's misplaced model, most of its bias, at the rate of the weakest excitation; the real servo follows the exactly filtered prediction to 0.3%. Noise-free, no leak at all does as well: the floors and biases the leaks remove there are those a fixed leak creates. *Open:* the real-loop statements (Corollaries D‴.14, D‴.26) for second-order plants; the servo with measurement noise and disturbances, where the leaks' robustness is the point, against the classical baselines and a Fourier regressor of the known harmonics; velocity-dependent friction with the projected prior in the real loop.
- *Done (Proposition D‴.27):* the a-priori bound restored. Confining the adaptive prior's atoms to a compact convex set keeps every particle within B/σ_0 of it (eventually) whatever the target and the energy non-increasing (at $\eta = \infty$ in its selected form, the plan being Lloyd-consistent, not jointly optimal); inside the set the bias removal of Theorem D‴.23 is unchanged, and where the limit has an atom on the boundary a bounded bias remains — the box trading the fit error against the parameters' bound, as long as it contains the representation; at $\eta = \infty$ on collapsed classes it is a switching σ-modification toward the set. *Open:* the convergence (at $\eta = \infty$ the Kurdyka–Łojasiewicz argument on the selected pieces, at finite η with the indicator), the choice of the set as a design input (how large a box keeps every representable target inside), and a sharper bound than the worst-case force B.
- *Done (Proposition D‴.24):* the masses minimised out of the leak's own energy are the cell masses ($\min_q W^2(\mu, \nu_{Z,q})$ is the nearest-atom distortion), so the nearest-atom leak toward an adaptive prior is the OT leak with free masses — and the function-level floor does not return at the representable samples (those outside W, of probability exponentially small in m but 0.36 at $m = 15$ on the instance, keep an error): it is the weight-derivative of the leak's bias at the sampled weights, which the adaptive atoms remove. What remains is the coupled distance of the amplitude–mass gauge, conditional on representability, which the mass constraint replaces by the transit's mismatch, and at the measure level the cross-class transport common to both priors; the adaptive prior also removes the fixed

prior's splitting instability. With an adaptive prior the real loop has perfect tracking as an invariant set at every representable sample. *Open:* the gauge as a design input (the masses set the excitation spectrum $H_L(q)$); the floor law and its removal for a misplaced prior beyond first order.

- Transfer: transporting a learned measure to a new plant.

9. **Quantisation and the curse of dimensionality (Corollary B.6).**
   - *Done (Theorem C′):* an adaptive law whose leak tracks the equal-mass centroidal quantisers of a prior measure (centroidal Voronoi tessellations [57]; coverage control [19], [45]), and Theorem D′'s observation that the state and the learned function converge at the rate $m^{-1/2}$, whose exponent is dimension-free: the curse of dimensionality of Corollary B.6 lives in the W-distance of the parameter measures, which the loop never evaluates.
   - *Open:* second-order (centroid) quantisation rates for the ideal measure on the lifted space; approximation rates that do not go through W (Maurey-type).
10. **Symmetric systems beyond kernels.** The OT quotient applies verbatim to swarms of identical agents, mixtures and multi-head models ([27] and [45] are the control-side precedents).
11. **Deep approximators.** Hierarchical (multi-layer) mean-field measures, with a layer-wise OT Lyapunov function.

A Verification map

Table 47 lists the tests and experiments of the accompanying repository that check each statement; statements not checked are listed after it. Test files are abbreviated: `T` is `tests/test_transport.py`, `Lg` is `tests/test_langevin.py`, `Av` is `tests/test_averaging.py`, `Ex` is `tests/test_excitation.py`, `NR` is `tests/test_nonreciprocity.py`, `Pe` is `tests/test_periodic.py` and `Ba` is `tests/test_barycentric.py`.

Table 47: Verification map.

statement	what is checked	where
constants (1) of Section 9, closed forms	$\nabla^2_{(c,s)}\varphi \le H_\varphi$ by brute force	`T::test_gaussian_constants_are_suprema`
Lemma P.1	$\|\mathrm{Proj}_K(\vartheta, Gy)\|_{G^{-1}} \le \|y\|_G$ on 2000 adversarial states, with equality when inactive; a tight instance of F(i) with ratio ≥ 0.99	`T::test_lemma_P1_projection_is_nonexpansive_and_tight_velocity_bound`
A(i) raw rates	lifted raw rates equal classical raw rates under the gain map (10^{-10})	`T::test_identification_with_classical_law`
A(i) projected rates, only-if	per-kernel projection with rescaled layer matches to 10^{-10} with active projections; an unrescaled layer does not; five joint balls are each distinguished (two-case construction: four active against an inactive per-kernel field, one inactive against an active one)	`T::test_identification_projected_rates_per_kernel`
B with constant ideal and $\varepsilon = 0$	the identity to floating-point precision; cross term ≤ 0; remainder $\le$ Lemma B.2; $\dot V_\pi \le$ (B) at every sample	`T::test_theorem_B_inequality_along_trajectories`
B with a moving ideal, $\varepsilon \ne 0$, switches and active projection	the same plus $V_* W$; at least one switch; projection active at ≥ 10 samples; Dini quotient $\le$ (B)	`T::test_theorem_B_moving_ideal_nonrepresentable_switches_and_active_projection`

continued on the next page

Table 47, continued

statement	what is checked	where
B, Dini form	forward-difference quotient of $V_{\mathrm{OT}} \leq$ (B)	`T::test_dini_derivative_of_V_OT_bounded_by_active_branch`
B against labelling	matched error $\ll$ labelled error; the bound is 30–45$\times$ smaller	`T::test_permuted_ideal_labelled_vs_OT`, `exp_transport_permutation.py`
B.5: failure of the plain bound along (L_σ); the (L_σ) variant holds	with every C(iv) hypothesis asserted, the plain bound fails by $> 3\times$ and the leak variant holds	`T::test_corollary_B5_fails_along_lsigma_and_its_leak_variant_holds`
C(i), Lemma C.3 (unprojected pure leak)	zero switches and exact $e^{-\sigma t}$ decay (explicit Euler, pure leak)	`T::test_theorem_C_i_exact_dissipation_pure_leak`
C(ii)	$C \leq b^2 + 4\eta R$; $\sqrt{C} \leq b + 2\eta$; $C = b^2$ under both gap conditions (300 configurations)	`T::test_lemma_C_ii_induced_coupling_bound`
C(iv)	an instance satisfying **every** hypothesis, asserted, including the gap condition $4\bar{\eta}(b + \bar{\eta}) < \Delta_{\mathrm{gap}}$; the envelope holds at every sample, with actual contraction	`T::test_theorem_C_iv_exponential_estimate`
C(v), Proposition C.5	label invariance of the OT leak; the reshuffling transient of the labelled leak	`exp_ot_sigma_mod.py`, `exp_sigma_mod_transient.py`
C′(i)	the plan reduces to the assignment for $m_0 = m$ (10^{-10}) and $W^2 = V_{\mathrm{bar}}^2 + \mathrm{Var}_\gamma$ (10^{-10}); along the pure leak W^2 is non-increasing at every step and $D^+ Z \leq -\sigma_m V_{\mathrm{bar}}^2$ holds at every state; the flow reaches $V_{\mathrm{bar}}^2 < 10^{-8}$ and ends at a fixed point of the equal-mass Lloyd iteration	`Ba::test_plan_reduces_to_assignment_and_pythagorean_split`, `Ba::test_pure_barycentric_leak_dissipates_to_a_centroidal_configuration`, `exp_barycentric_leak.py` → `results/barycentric_leak/table.md`
C′(ii)	$\sqrt{C_{\mathrm{bar}}} \leq b + 2\eta$ on 90 random configurations; $C_{\mathrm{bar}} = b_{\mathrm{bar}}^2$ at small η (plan stability)	`Ba::test_induced_coupling_bound_and_plan_stability`
C′(iii)	the Lyapunov inequality at every recorded state of a closed loop with the barycentric leak toward a 24-atom prior	`Ba::test_lyapunov_inequality_along_the_barycentric_leak_closed_loop`
D	consistency in m (nearly constant amplification 1.64–1.69; not the constant $e^{\Lambda T}$)	`exp_mean_field_m.py`
D′(b), (d)	the $m^{-1/2}$ rate of the synchronous coupling to a 4096-particle reference: slopes -0.47 ± 0.06 ($\tau = 0$) and -0.46 ± 0.05 ($\tau = 2 \cdot 10^{-3}$) over 24 seeds, 0.94 ± 0.08 and 0.92 ± 0.08 against $(1/m - 1/M)^{1/2}$; not the constant	`exp_chaos.py` → `results/chaos/table.md`
D″	with exact Lipschitz constants and the a-priori tube, $\sigma_{\mathrm{cert}} = 4.04$ against Proposition K's $\sigma_K = 0.00144$; at $1.2\sigma_{\mathrm{cert}}$ with dither the coupled distance is of order $m^{-1/2}$ (slope -0.51 ± 0.02), bounded in time and 700$\times$ below the bound, with the network at its prior (fit error 1.000); without dither the reference collapses (spread 10^{-14}) and the distance is zero; at $\sigma_{\mathrm{cert}}/8$ it decays to 10^{-11}; in the learning regime ($\sigma_K/3$, 1000 s) the network stays spread and fits the plant (fit error 0.63) and the distance follows $m^{-1/2}$ (-0.62 ± 0.16) but grows threefold between the windows $(100, 200]$ and $(500, 1000]$	`exp_chaos_uniform.py` → `results/chaos/uniform_table.md`

continued on the next page

Table 47, continued

statement	what is checked	where
D‴, D‴.5, D‴.7	the window realised at $\sigma_0 = 0.15\sigma_{\text{crit}}$ ($\kappa_R[\mu^*](N_{0.1}) = 0.0051 < 0.0071 < 0.047$), (H) along the warm run (margin ≥ 0.0014), saturation of the coupled distance (last/first window ratios 1.13–1.15 over 900 slow units), slope -1.28 ± 0.30 in m; the weight-sensitivity law per seed within 25% in 9 of 12 pairs, per-cluster cosines 0.92–1.00; the exact stationary distance of Proposition D‴.5(ii) within 2.3% in all 12 pairs, splitting margins positive in 11 (one $m = 16$ seed a saddle); cold start: growth (2.9–7.8×) and the floor's fraction 0.20, 0.08, 0.04 decaying like $m^{-0.64\pm0.08}$; real loop at $k_e = 25$ within 0.7% of its averaged slow dynamics on the same clouds; not the constants C_1, m_0 (vacuous: $2 \cdot 10^9$, 10^{23}) nor the saturation of Corollary D‴.7	`exp_chaos_learning.py` → `results/chaos/learning_table.md`
D‴.9, D‴.11	the matching prior: zero residual, $\kappa_R = 0$, margin $\sigma_0 = 0.00707$ against the σ-modification's 0.00196, floors 1.61× and 2.18× higher; misplacement closes the window (centres +0.6, amplitudes ×1.2, widths ×0.7); shrunk amplitudes ×0.5–0.7 give margins above σ_0 at the uniform weights and lower floors; admissible-weight table under three priors (overshoot splits, undershoot stabilises; the shrunk prior keeps all four tested weights admissible, +0.0054 to +0.0073); matching-prior coupled runs saturate within 2.1 of the theorem's relaxation times (observed e-folding 48–84 slow units against 107–306 for the σ-modification) with per-seed linearised-law ratios 0.81–1.11 (parameter) and 0.61–1.69 (function) and the exact law within 0.987–0.998 in all nine pairs; the two saddles among them are left by a local split with the coupled distance on its plateau; the function-level law for the σ-modification within ratios 0.48–0.74 of the measured means (RMS over seeds 0.52–0.84; per-seed law from the realised weights 0.60–1.57)	`exp_prior_design.py` → `results/chaos/prior_design.md`

continued on the next page

Table 47, continued

statement	what is checked	where
D‴.12	the same clouds under the nearest-atom and the OT leak: cluster weights conserved (max deviation 0 in 32 systems) against equalised ($s_{\mathrm{eq}} = 93$–164 slow units; $\ln 2/\sigma_0 = 98$; 1–21 particles moved); commensurate sizes: function-level distance^2 $4.9 \cdot 10^{-7}$–$7.5 \cdot 10^{-9}$ against $2.5 \cdot 10^{-3}$–$5.9 \cdot 10^{-5}$ and W^2 $7.9 \cdot 10^{-3}$–$6.9 \cdot 10^{-4}$ against 5.1–0.81 (the exact $K \times K$ transport between the weighted stationary measures within 0.997–0.999 at admissible weights; the formula $\frac{1}{2}\|w^m - w^M\|_1 \Delta_\nu^2$ within -7% to $+28\%$), both OT distances still falling (0.02–0.035 and 0.12 per 150 slow units) while the synchronous D rises to 0.95–0.96 of its mismatch plateau; incommensurate sizes: one or two bridge particles at 0.28–0.58 Δ_ν from the atoms; against ν: $m \cdot W^2 = 19.4$–21.0 (slope -0.98 ± 0.03) and $m \cdot \mathrm{fit} = 0.45$–$0.60$ (squared slope -1.81 ± 0.05); against the $M = 512$ reference slopes -2.07 ± 0.05 and -1.21 ± 0.04 (the reference carries its own bridge particles; the $m = 16$ value still relaxing); fit error of the reference $0.012 \to 8 \cdot 10^{-5}$; the designed masses inadmissible between $\delta = 0.03$ and 0.05	`exp_ot_leak.py` → `results/chaos/ot_leak.md`; `tests/test_ot_leak.py` (the vertex plan, the gradient flow, the m-independent stationary configuration, conservation against constraint, the forest bound, the exact W^2)
D‴.14	the real loop at $k_e = 25$ from the clouds at the end of `exp_ot_leak.py` (contracted where the weights are admissible) and from the collapsed lifts: under the OT leak $\|\hat{f}_m - \hat{f}_M\|^2_{L^2(\mathrm{orbit})} = 7.5 \cdot 10^{-9}$–$2.7 \cdot 10^{-8}$ ($m = 15$, 60 against 510) and $5.2 \cdot 10^{-10}$–$3.6 \cdot 10^{-8}$ against the 3-atom lift, rms tracking error $0.9 \cdot 10^{-6}$–$7.3 \cdot 10^{-6}$ (the lift $8 \cdot 10^{-12}$); under the nearest-atom leak $1.1 \cdot 10^{-4}$–$2.5 \cdot 10^{-3}$ with tracking errors $3.2 \cdot 10^{-4}$–$1.9 \cdot 10^{-3}$; at the one pair with both weights admissible the disagreement equals the lifted orbits' distance to 1.0003 and the admissible systems sit within $2.5 \cdot 10^{-11}$ of their lifts; three systems at saddle weights (margins -0.0035, -0.0011, -0.0022) have split (spread 0.04–0.11) and sit off their lifts	`exp_ot_real_loop.py` → `results/chaos/ot_real_loop.md`; `tests/test_ot_leak.py` (the collapsed subspace's invariance, the lifts' equality)
D‴.15	lifts of the 3-atom orbit to $m = 6$, 9, 15 (OT leak, matching masses) and to $m = 16$ (σ-modification at $(7, 5, 4)/16$ and $(9, 4, 3)/16$): fixed points of the Poincaré map to 10^{-14}–10^{-15}; class separation subspaces invariant to $\leq 6 \cdot 10^{-8}$; separation exponents equal to the ghosts' to $2 \cdot 10^{-5}$ relative and to $(\sigma_0 - \lambda_j(R_k^G))/k_e$ within 0.2–4.1%; collective exponents equal to the K_0-atom orbit's to 10^{-4} relative for every m and to Theorem J(b)'s prediction within 0.6–3.2%; the saddle's exponent $-9.13 \cdot 10^{-5}$ (multiplicity 8) at $(9, 4, 3)/16$	`exp_lifted_orbit.py` → `results/chaos/lifted_orbit.md`; `tests/test_ot_leak.py` (the $m = 6$ lift against the ghosts and the 3-atom loop)

continued on the next page

Table 47, continued

statement	what is checked	where
D‴.16	the dithered averaged slow loop, $m = 15$ from the counts $(6, 5, 4)$: nearest-atom record-level label changes $1.0 \cdot 10^{-5}$, $5.1 \cdot 10^{-4}$, $5.6 \cdot 10^{-3}$ per slow unit at $\tau_0 = 0.005$, 0.01, 0.02 (Arrhenius barrier 0.048, among the ridge minima 0.040–0.074); time at the balanced counts 0.00, 0.54, 0.25 against the energy–multinomial law 0.86, 0.58, 0.33 (standard errors ≈ 0.1 at $\tau_0 = 0.01$) and against the OT leak 0.996, 0.992, 0.88, whose excursions are returns of the crossing particle (121 of 121, 298 of 336) with the effective barrier 0.058	`exp_dither_ot.py` → `results/chaos/dither_ot.md`
D‴.14 ($n = 2$)	a planar plant (four vector-valued bumps, a Lissajous reference, $p = 5$), $k_e = 25$, $\sigma_0 = 0.05$, sizes 8, 32 against 128 from the same clouds: the OT leak equalises the weights at 13–14 slow units ($\ln 2/\sigma_0 = 14$), the sizes agree to $5.0 \cdot 10^{-8}$–$1.6 \cdot 10^{-7}$ with rms tracking errors $3.9 \cdot 10^{-6}$–$1.6 \cdot 10^{-5}$, against $5.5 \cdot 10^{-3}$–$2.3 \cdot 10^{-2}$ and $1.9 \cdot 10^{-3}$–$6.7 \cdot 10^{-3}$ under the nearest-atom leak (fit errors along the orbit 2–$9 \cdot 10^{-4}$ against 0.11–0.39)	`exp_ot_2d.py` → `results/chaos/ot_2d.md`
D‴.17	J_m^{OT} non-increasing along 300 steps of a run with reassigned particles (strong leak; the plan does not switch there, so this is a smooth gradient flow), $\mathrm{d}J/\,\mathrm{d}s = -\lVert\mathrm{drift}\rVert^2_{G^{-1}}$ where the plan is unique, and at a constructed tie between two optimal plans (mirror-image particles) J^{OT} equals both frozen energies and decreases along either plan's drift	`tests/test_ot_leak.py`
D‴.18	the real loop at $k_e = 25$, 8000 time units, sizes 3–240: the 3-atom orbit's Floquet exponents within 0.56–3.22% of $\lambda_j(H_q)/k_e$ and the ghosts' equal to $(\sigma_0 - \lambda_j(R_k^G))/k_e$ to $3 \cdot 10^{-4}$ (matching and shifted priors); inside the cells the per-particle distance to the lift decays at 2.69–$2.97 \cdot 10^{-4}$ for $m \geq 15$ (prediction $2.83 \cdot 10^{-4}$) with constants 1.03–1.29 under both leaks, the $m = 3$ runs mixtures of collective modes; across a cell boundary ($\rho = 4.5$) the nearest-atom leak ends at other counts (W^2 1.9–12.3, that of the lifts of the realised weights; two of them inadmissible) and the OT leak at q (W^2 0.010–0.23, falling at $2\sigma_0$); one particle across a boundary: W^2 grows to 0.79, 0.20 (nearest) and falls to $3.0 \cdot 10^{-3}$, $8.2 \cdot 10^{-4}$ (OT); on a moving orbit (shifted prior) 2.18–$2.63 \cdot 10^{-4}$ against $2.27 \cdot 10^{-4}$, constants 1.04–1.23; the proof's radius at most 0.019	`exp_uniform_basin.py` → `results/chaos/uniform_basin.md`; `tests/test_uniform_basin.py` (at the shifted prior: the remainder constant and the exact $N_a + N_b/m$ structure of the per-particle term, the separation response $(GR_k - \sigma_0 I)v$, the contraction at $\mu = \min\{\lambda_{\mathrm{coll}}, \lambda_{\mathrm{sep}}\}$ with one constant, the same for every m)

continued on the next page

Table 47, continued

statement	what is checked	where
D‴.20	the critical excess by continuation in ε (Newton residuals $\le 8.7 \cdot 10^{-13}$): $\varepsilon_c = 5.11$, 13.5, 23.1, 65.0 for $\delta = 0.02$–0.2 (the exact threshold equal to ε_c; the rank-one formula 1.22–1.31 and 0.70–1.22 times ε_c with the two atoms), 6.99, 14.6, 24.3 for an excess taken from one atom; $\delta_c(\varepsilon) = 0.0042$–$0.20$ for $\varepsilon = 1$–64; dynamics for $m = 15$–510 at $\varepsilon = 1$, 4, 16, 64 over 600 slow units, continued to 6000: equalised at $\varepsilon = 1$; at $\varepsilon = 4$ stable end weights with residual excesses up to 1/60, 3/240, 7/510; at $\varepsilon = 16$ and 64 the stable systems keep their sampled weights and the four unstable ones move (one moving its excess to an atom that was at q); floors 0.55 and 0.74 of the nearest-atom law, 0.23–0.84 and 0.47–0.91 of the observed nearest-atom floors; fit errors 0.12–0.14 at $\varepsilon = 64$	`exp_entropic_leak.py` → `results/chaos/entropic_leak.md`; `tests/test_entropic_leak.py` (the solvers, the gradient flow, the limits at their rates, the separation block entrywise with the invariance of the subspace, the critical regularisation by continuation in ε with $A \succ 0$ checked and every residual below 10^{-9})
D‴.21	the averaged loop from the warm clouds ($m = 15$–510, four seeds, matching and shifted priors, both leaks, 600 slow units): (E), (P) to $6 \cdot 10^{-13}$; under the OT leak no relabelling, $\alpha > 0$ from $s \le 300$, ratio/$\lambda \ge 1.00$ (matching) and ≥ 1.61 (shifted), E from $4.1 \cdot 10^{-3}$–$4.0 \cdot 10^{-2}$ to $8.3 \cdot 10^{-7}$–$8.1 \cdot 10^{-6}$ below the a-posteriori bound, the end rate 0.01416–0.01417 against $2\sigma_0 = 0.01415$ and $\lambda = 0.0140$–0.0141, at the shifted prior the curvature's cost 0.802 and (ii)'s slack 0.683; the nearest-atom E at the spurious minima's energy at the 22 admissible count vectors (within 0.24%), split and still falling 0.5–16.5% below it at the 10 inadmissible ones ($m \le 60$); (H*) on the whole space at the shifted prior ($\kappa = 4.7 \cdot 10^{-4}$, $E_0 = 9.1 \cdot 10^{-10}$), the a-priori basins holding no particle in another cell below $m \approx 1.7 \cdot 10^5$ (matching) and $2.3 \cdot 10^6$ (shifted); the margin +0.0023 with 1/30 of the particles moved into another cell, negative from 1/15	`exp_ot_basin.py` → `results/chaos/ot_basin.md`; `tests/test_ot_basin.py` (the identities for every assignment with the counts and their exact defects at other counts, the PL chain and its sharp constant, the decay from a particle in another cell below the a-posteriori bound with a positive margin, the admissible nearest-atom spurious minimum)

continued on the next page

Table 47, continued

statement	what is checked	where
D‴.23	the linear theory at the exact representation ($h_j = 3.06 \cdot 10^{-4}$–0.195; $\lambda_1^- = 2.68 \cdot 10^{-5}$–$3.06 \cdot 10^{-4}$ for $\eta = \sigma_0/10 \ldots \infty$; (v)'s radius at most $2.0 \cdot 10^{-5}$–$2.2 \cdot 10^{-4}$, 94–$6 \cdot 10^4$ times below the K_0-atom runs' starts, $2.2 \cdot 10^4$–$3.6 \cdot 10^5$ below the warm clouds'); the fixed prior's bias within 4.8% of the first-order formula in norm (0.85 and 0.73 of it in modes 1 and 3, the error first order in the shift); the K_0-atom loop: every mode above 10^{-6} decays at λ_j^- (log-slopes 0.981–1.003 of it), fit error $9.78 \cdot 10^{-3} \to 1.8 \cdot 10^{-5}$ ($\eta = \sigma_0$) and $6.7 \cdot 10^{-6}$ ($\eta = \infty$) by $s = 6000$; separated starts (five draws): departures from the K_0-atom run second order in the separations, consistent with the same expected size at $m = 15$ and 510; $m = 15$–510 from the warm clouds: ungated the prior is dragged by the transit ($\|\bar{y}\|$ 0.032–0.71 at $s = 500$) and one $m = 15$ system at $\eta = \infty$ is carried onto a plateau of the unregularised loss (Hessian eigenvalues $-1.8 \cdot 10^{-5}$, $7.2 \cdot 10^{-5}$; still there at $s = 23000$, crossed by the gradient flow, below 10^{-12} after $7.97 \cdot 10^4$ more slow units); the transit gate leaves a residue in the weak modes ($\|\bar{y}\|$ $4.9 \cdot 10^{-3}$–$3.7 \cdot 10^{-2}$ against $1.3 \cdot 10^{-3}$, $\eta = \sigma_0$); the settled gate ($t_{\text{open}} = 1211$–1304) opens replicated and follows the K_0-atom run for every m (per-seed ratios 0.995–1.000 at $\eta = \sigma_0$, 0.968–1.001 at $\eta = \infty$), fit errors $1.6 \cdot 10^{-4}$ and 3.4–$3.5 \cdot 10^{-5}$ against $9.78 \cdot 10^{-3}$; the limits of (i): a non-replicated saddle held by the leak, and unbounded adaptive solutions, beyond the saturation and from an exact six-kernel fit (the width coordinate running off toward a Karush–Kuhn–Tucker point of the box-constrained loss)	`exp_adaptive_prior.py` → `results/chaos/adaptive_prior.md`; `tests/test_adaptive_prior.py` (the gradient structure, stationarity, the linearisation's rates and its self-adjointness in the gradient-flow metric, the first-order bias, the two gates, the limits of (i))

continued on the next page

Table 47, continued

statement	what is checked	where
D‴.24	the floor as the bias: F equal to the bias form to the finite-difference error ($1.1 \cdot 10^{-6}$–$1.2 \cdot 10^{-8}$); the law (B) $1.36 \cdot 10^{-2}$, $6.17 \cdot 10^{-2}$, 0.229 per unit of $1/m - 1/M$ at $\sigma_0/\sigma_{\text{crit}} = 0.05$, 0.15, 0.5 (0.50–0.84 of the bound, which is $\sigma_0/4$ times the adaptive prior's coupled-distance law 46.3), modes 2–6 carrying 98–99% of it and 97% unweighted, the weight moving the shares toward $h \approx \sigma_0$; the parameter split to $1.4 \cdot 10^{-7}$, the adaptive prior's coupled-distance law 1.4–5.3 times the fixed prior's; 5 of the 12 sampled m-systems are fixed-prior saddles (class 1, $w_1 \geq 0.4$) and split (floors 0.81–1.01 of the replicated point's), the exact bias at the sampled weights reproduces the floor at the other seven (ratio 1.0000), the delta method misses it by 0.86–1.72; the nearest-atom leak over 6000 slow units: the fixed prior's floor stays, the adaptive prior's decreases without a floor (at most $2.0 \cdot 10^{-6}$ at $\eta = \sigma_0$ and $5.4 \cdot 10^{-8}$ at $\eta = \infty$ over the representable samples at $s = 6000$); D 0.94–1.00 of the replicated stationary points' under both priors, the adaptive prior's 1.72–4.81 times the fixed prior's; W^2 the cross-class transport (0.67–0.99 of it), the two priors' W^2 within 1.00–1.32; the boundary sample $(8, 4, 3)/15$ decays like $1/s$ ($6000 \cdot \|\hat{f}_m - \hat{f}_M\|^2 = 0.88$ and 0.42 against 0.8 and 0.4); the design table at the misplaced prior (the fit error and the floor of the four designs; the OT designs' D the largest, 6.0–1.0, the transit's mismatch; the ranking at $s = 1250$ and 2000); the real loop: the fixed prior at the averaged floor (ratios 0.996–1.001; the $m = 15$ run on a saddle orbit), the adaptive prior along the averaged decay (0.987–0.992), the exact representation invariant (five-period rms $\leq 9.0 \cdot 10^{-12}$); the instance's probabilities (outside W 0.361, 0.074, 0.0021; Hoeffding 1.87, 0.60, 0.036; the fixed prior's saddles 0.369, 0.260, 0.109) and the adaptive prior's expected limiting floor ($\geq 7.3 \cdot 10^{-3}$, $2.2 \cdot 10^{-4}$, $6.6 \cdot 10^{-7}$: 1.8, 0.24, 0.005 times the fixed prior's law); Monte Carlo (600 nested samples): the exact floor 2.78, 1.53, 1.11 times the law; in two dimensions (four bumps, $p = 5$, $\sigma_0 = 0.05$, six seeds): every sample D‴.5-admissible and the fixed prior's floor the exact bias at the sampled weights (ratio 1.0000 in all 18 systems), the adaptive prior's at most 0.019 ($\eta = \sigma_0$) and 0.006 ($\eta = \infty$, the atoms reset every slow unit) of it at the 7 representable systems and still falling, the real loop ($\eta = \infty$) along the averaged prediction (0.979–1.009) and invariant from the exact representation ($\leq 4.7 \cdot 10^{-10}$)	`exp_adaptive_floor.py` → `results/chaos/adaptive_floor.md`; `tests/test_adaptive_floor.py` (the nearest-atom adaptive flow as a gradient flow, the free-mass energy as the minimum of the OT energy over the masses, also against masses moved by $\pm 1/m$, the floor law as the bias derivative with the weight's band-pass against the unweighted shares, the exact representation of a D‴.5-admissible sample stationary under the adaptive prior and the margins σ_0 at a fixed-prior saddle, the parameter split, the energy across a cell crossing and an empty cell)

continued on the next page

Table 47, continued

statement	what is checked	where
D‴.26	the three-bump instance, $w = (6, 5, 4)/15$ (representable; a fixed-prior saddle), the real loop with the adaptive prior in the state, $\eta = \sigma_0$ and ∞, $k_e = 25$ and 50, the monodromy by central differences (step 10^{-5}, DOP853 at rtol 10^{-13}): return residuals $\le 1.7 \cdot 10^{-15}$; at $m = 15$ and 30 the 36 and 81 separation exponents at σ_0/k_e to $\le 1.7 \cdot 10^{-6}$ (relative), the other 18 ($\eta = \sigma_0$) or 9 ($\eta = \infty$) equal to the weighted K_0-atom loop's to $1.4 \cdot 10^{-5}$–$6.4 \cdot 10^{-4}$; the collapsed variational equation (its generator the real loop's linearisation to $5.8 \cdot 10^{-11}$, its exponents the differences' to $4.1 \cdot 10^{-5}$) at $k_e = 12.5$–400 for $x_m = 2\sin t$ and $2\sin t + 0.8\sin 2t$ ($\|\bar{A}_1\| = 0$ and $5.6 \cdot 10^{-3}$, skew): the first-order deviations falling by $\to 4$ per doubling (the slowest mode 13.6% and 3.84% below at $k_e = 25$, 50; every mode below on the instance), the second-order residuals by $\to 16$ ($4.9 \cdot 10^{-7}$ at $k_e = 400$), $c_1/\lambda_1 = -100.6$ (the filter's part $-102.1 = -0.989\omega_1^2$, $\omega_1 = 10.16$), measured -100.5 at $k_e = 400$; the gain scaled by 30: $c_8/\lambda_8 = +4.17$, $c_9/\lambda_9 = +12.65$, those exponents above their prediction by $c_j/(\lambda_j k_e^2)$ to three digits; at $\eta = 10\sigma_0$ $\bar{A}_0$ self-adjoint in $\mathrm{diag}(I, (\sigma_0/\eta)I)$ to $8.7 \cdot 10^{-18}$ ($6.4 \cdot 10^{-2}$ in I); the error channel by Liouville's formula within $O(k_e^{-3})$ of $k_e - \mathrm{tr}\, H_L(w)/k_e$	`exp_adaptive_floquet.py` (Part C: `--only-C`) → `results/chaos/adaptive_floquet.md`; `tests/test_adaptive_floquet.py` (at $w = q$, $\eta = \infty$, $m = 6$: the return of one period, the nine separation exponents, the K_0-atom loop's spectrum within the averaging correction; at $(2, 1, 1)/4$, $\eta = \sigma_0$, $m = 4$, the prior in the state: the real loop's Jacobian splitting into the separations and the collapsed variational equation, the monodromy's 3 separation and 18 collective exponents, the second-order prediction; on an asymmetric reference: $\bar{A}_1$ skew, $\ell_j \bar{A}_1 r_j = 0$, the residual $O(k_e^{-4})$, c_1 measured, Liouville's formula; at $\eta = 10\sigma_0$ $\bar{A}_0$ self-adjoint in the metric and not in I, $\bar{A}_1$ skew in it)

continued on the next page

Table 47, continued

statement	what is checked	where
D‴.26a	Corollary D‴.26's instance on $x_m = 2\sin t + 0.8\sin 2t$, $\eta = 0$ ($K = \hat{J}$, $\Lambda = \sigma_0 I$), σ_0 and ∞, $k_e = 12.5$–1600, a 512-point Fourier grid: the contraction for P in 3–9 iterations (499 at $\eta = 0$, $k_e = 12.5$: stalled at rounding), invariance residual $\leq 3.8 \cdot 10^{-15}$, $\lvert P - P_2\rvert/\varepsilon^3 \to 3.34 \cdot 10^3$, $\lvert P - P_{(3)}\rvert/\varepsilon^4 \to 5.8 \cdot 10^4$; the slow block's rates the full variational equation's to $\leq 1.0 \cdot 10^{-8}$; the rates at $\pm\varepsilon$ equal to $\leq 5.9 \cdot 10^{-12}$ of the largest, $\leq 1.1 \cdot 10^{-9}$ per mode (first-order deviation $9.8 \cdot 10^{-2}$–$8.9 \cdot 10^{-6}$ of the largest, per mode up to 29.6%: the slowest mode −29.6%, −12.5%, −4.0%, −1.1% at $k_e = 12.5$–100, $\eta = \infty$), with a mismatch $1.1 \cdot 10^{-2}$–$7.8 \cdot 10^{-5}$ (per mode $6.1 \cdot 10^{-2}$–$3.5 \cdot 10^{-4}$); the Magnus parameter 0.48–0.004; the noise: decoupling residual $\leq 3 \cdot 10^{-16}$, $\lVert b - b_0\rVert/\lVert b_0\rVert$ 0.63–0.0074, $\lvert b - b_0 - \varepsilon b_1\rvert/\varepsilon^2 \to 3.34 \cdot 10^3$, $\mathrm{Cov}(v) \cdot 2\varepsilon^3 - 1$ from $-9.6 \cdot 10^{-2}$ to $-2.4 \cdot 10^{-5}$, $\lVert\mathbb{E}[vY'^{\top}] + b^{\top}\rVert/\lVert b\rVert$ 0.15–$1.2 \cdot 10^{-3}$, $\lVert\mathrm{Cov}(Y') - \bar{\Sigma}\rVert/\lVert\bar{\Sigma}\rVert$ 0.35–$9.4 \cdot 10^{-5}$ ($\eta = \infty$; $O(\varepsilon^2)$ to $k_e = 200$–400, $O(\varepsilon)$ beyond), the weakest mode's deviation $-34\% \to -6.6 \cdot 10^{-5}$ the sum of the noise's attenuation ($-56\% \to -1.1 \cdot 10^{-4}$) and the rate's correction ($+49\% \to +4.6 \cdot 10^{-5}$), the jitter with its next term at $O(\varepsilon^2)$ (0.30 → 0.001 at $k_e = 100$–1600), the floor at $t = 0$ at 0.763–1.0000 of the prediction, its period average at 0.766–0.99997 of D‴.30(ii)'s formula and pointwise in $[0.9952, 1.0091]$ at $k_e = 1600$, the jitter's predicted range	`exp_adaptive_floquet.py` (`--add-D`) → `results/chaos/adaptive_floquet.md`; `tests/test_adaptive_floquet.py` (Chang's P and the exact slow multipliers, the ε^3 truncation on the instance; on a synthetic block with an $O(1)$, non-symmetric Λ, $P - P_2$ at ε^3 and $P - P_{(3)}$ at ε^4, a flipped Λ and a dropped cubic term caught; the rates even in ε at $\eta = \sigma_0$ and ∞, a mismatch breaking it; the slow noise decoupled, b_0 against $\dot{K}$ by the chain rule and the plant's own slope by differences, b_1 at ε^2, the covariances' orders)

continued on the next page

Table 47, continued

statement	what is checked	where
D‴.27	the box $\mathcal{Z}$ (amplitudes and centres to 0.9 of $a_{\max}$, $c_{\max}$; the width coordinate in $[-6, 6]$): beyond the saturation (Part G(b)'s target) the projected prior converges at $\eta = \sigma_0$ (the projected midpoint rule, positions projected, $\mathrm{d}t = 0.25$ and 0.125, to $s = 10^5$; end states within $3.4 \cdot 10^{-12}$, the trajectory at $s = 300$ within $6.3 \cdot 10^{-7}$, $1.9 \cdot 10^{-7}$, $3.8 \cdot 10^{-8}$ of a reference for $\mathrm{d}t = 0.25$, 0.125, 0.0625) and $\eta = \infty$ (LSODA) to the same bounded stationary point (drift $\leq 3.5 \cdot 10^{-15}$), the prior's amplitude atom on the face, fit error 0.1415; the particles' largest distance from $\mathcal{Z}$ 0.92 and 0.89 against $B/\sigma_0 \leq 3686$; the amplitude bound 0.5–0.99 of $a_{\max}$: fit errors 0.1949–0.1101, between the fixed prior's 0.1750 and the unprojected 0.1050 for 0.7–0.99, above the fixed prior's for 0.5–0.65, the exact atoms outside the boxes up to 0.7; a Lloyd-consistent, non-optimal plan at $\eta = \infty$ along which the reduced energy rises from $1.251 \cdot 10^{-3}$ to $1.574 \cdot 10^{-3}$ while the selected energy falls (largest increment $-3.4 \cdot 10^{-6}$); on Part G(a)'s four-bump target the width coordinate stopped at the face $\kappa = 6$ from $s = 10^9$ (fit error within $9.3 \cdot 10^{-7}$ of the face minimiser's); the misplaced-prior run identical with and without $\mathcal{Z}$, the box never approached (margins $\geq 16.9\%$ of its width)	`exp_adaptive_prior.py` Part H → `results/chaos/adaptive_prior.md`; `tests/test_adaptive_prior.py` (the projected prior bounded and convergent at $\eta = \infty$, while the unprojected coordinate passes 11 by $s = 10^4$; at $\eta = \sigma_0$ and $\mathrm{d}t = 0.125$ the prior in its box with its amplitude atom on the face, the particles bounded and $\mathcal{J}$ non-increasing; inactive near the exact representation)

continued on the next page

Table 47, continued

statement	what is checked	where
D‴.28	the three-bump instance plus $\varepsilon \cdot \exp(-(x-0.8)^2/(2 \cdot 0.3^2))$, $\varepsilon = 0$–0.5: the approximant's fit error $3.6 \cdot 10^{-3}$–$2.5 \cdot 10^{-2}$ ($\varepsilon = 0.05$–0.5), the gauge check (the gradient at the gauge transform $\leq 1.6 \cdot 10^{-15}$; Newton continued in w from $\vartheta^b(q)$ at it to $\leq 1.7 \cdot 10^{-11}$, $\hat{f}$ to $\leq 5.5 \cdot 10^{-13}$; plain Newton there in 3–5 of 6 cases), the adaptive prior's sensitivity $1.4 \cdot 10^{-11}$–$2.7 \cdot 10^{-10}$ of the fixed prior's, $P(w^m \in W_g) = 0.52$–0.64, 0.87–0.88, 0.997 at $m = 15$, 60, 240 (0.26, 0.71, 0.98 at $\varepsilon = 0.5$), the branch at the saturation near $\varepsilon = 0.68$, the fixed prior's within $2 \cdot 10^{-9}$ of $\sigma_0 \hat{J}(\tilde{A} + \sigma_0 I)^{-1} \tilde{g}$ and $7.0 \cdot 10^{-4}$–$5.6 \cdot 10^{-3}$ of the Gauss–Newton form, the margins $\sigma_0 - (q_k/w_k)\lambda_k^+$ exact to 10^{-14} over all count vectors of $m = 15$, 60 (none negative: $\lambda^+ \leq 0.04\sigma_0$ at $\varepsilon \leq 0.3$, $\leq 0.10\sigma_0$ at $\varepsilon = 0.5$), the dynamics' separation eigenvalues equal to them to $\leq 5.8 \cdot 10^{-12}$; the dynamics ($m = 15$, 60, 240 against 510, four seeds, $\varepsilon = 0.1$ and 0.3, 6000 slow units): the fixed prior's floor within 8% of its value at $s = 600$ ($3.1 \cdot 10^{-4}$–$5.9 \cdot 10^{-3}$ at $m = 15$, $\varepsilon = 0.1$) and its fit error above the approximant's, the adaptive priors' floor falling ($m = 60$, $\varepsilon = 0.1$: to $1.0 \cdot 10^{-9}$–$1.4 \cdot 10^{-8}$ at $\eta = \sigma_0$, $3.2 \cdot 10^{-11}$–$3.8 \cdot 10^{-9}$ at $\eta = \infty$) and the fit error to the approximant's (median excess $1.2 \cdot 10^{-4}$, $1.2 \cdot 10^{-4}$, $1.6 \cdot 10^{-5}$ at $m = 15$, 60, 240 for the samples 0.02 inside W_g); at the sample outside W_g the box minimum on the branch 1.107 and 1.586 times the approximant's fit error, a permuted gauge transform existing	`exp_gauge_floor.py` (`--only-A` recomputes Part A, the branch's end and the outside samples) → `results/chaos/gauge_floor.md`; `tests/test_gauge_floor.py` (the gauge's critical points and function, the continuation from $\vartheta^b(q)$, the two sensitivities, the residual curvature's structure, the margins and the dynamics' separation eigenvalues, the adaptive prior's convergence to the gauge transform at $\eta = \infty$ and σ_0)

continued on the next page

Table 47, continued

statement	what is checked	where
D‴.29	the servo $\ddot{x} = -\tau_{\mathrm{cog}}(x) + u$, $\tau_{\mathrm{cog}} = 0.4\sin(3x + 0.3) + 0.1\sin 7x$, $x_m = \sin t$, $\lambda = 5$, $K_0 = 4$ kernels on the position, $\sigma_0 = 0.01$: $\vartheta^b(q)$ (the local minimum BFGS reaches from a fixed start) fit error $1.46 \cdot 10^{-2}$, $\tilde{A}$ spanning $2.9 \cdot 10^{-4}\sigma_0$–$30\sigma_0$ (four modes below $0.1\sigma_0$), $\lambda_k^+ \le 0.0098\sigma_0$, $W_g = \{w > (0.061, 0.085, 0.119, 0.008)\}$, the floor law $1.29 \cdot 10^{-2}$; 120 random starts: 1 at $\vartheta^b(q)$, 33 below it, all saturated, the best $6.39 \cdot 10^{-3}$, with its saturated amplitudes held at $\lvert\alpha\rvert = 6$–14 $7.14 \cdot 10^{-3}$–$6.40 \cdot 10^{-3}$; the misplaced prior's (the engineer's model, fit error 0.200) replicated point a saddle (margin $-3.97 \cdot 10^{-3}$, fit error 0.0545); the averaged loop ($m = 16$, 64, 128 against 256, three seeds, 2000 slow units) from clouds around $\vartheta^b(q)$: fixed nearest atom floor $1.85 \cdot 10^{-3}$, $2.24 \cdot 10^{-4}$, $4.49 \cdot 10^{-5}$ and fit error 0.174–0.038, adaptive ($\eta = \infty$) $4.85 \cdot 10^{-5}$, $2.88 \cdot 10^{-7}$, $6.36 \cdot 10^{-8}$ and 0.0298–0.0147, OT 0.01462 with no floor, σ-modification 0.328, no leak 0.0180, 0.0147, 0.0146 (floors $7.7 \cdot 10^{-6}$–$6.6 \cdot 10^{-8}$); misplaced: the fixed designs split and end at 0.048–0.10, the adaptive prior unaffected at $\eta = \infty$, 0.035–0.019 at $\eta = \sigma_0$; from the engineer's model: fixed 0.102, 0.060, 0.048, OT 0.048, adaptive 0.027–0.022 ($\eta = \infty$) and 0.032–0.028 ($\eta = \sigma_0$), no leak 0.0152–0.0147, the K_0-atom flow at $\eta = \infty$ 0.0297, 0.0213, 0.0169, 0.0155 at $s = 600$, 2000, 10^4, 10^5; phase-plane kernels: the circle 5 exact zeros and 8 below $10^{-6}\sigma_0$, ellipses $\times 1.1$ and $\times 2$ 8 and 6, a two-harmonic orbit 3, centred kernels 16 exact of 20; the real servo ($k_s = 25$, 1000 time units, records and targets every 0.02): position-error rms 1.7–$2.1 \cdot 10^{-5}$ (adaptive, matching OT, no leak), $7.8 \cdot 10^{-5}$ (misplaced OT), $9.1 \cdot 10^{-5}$–$1.8 \cdot 10^{-4}$ (fixed), 5.4–$5.5 \cdot 10^{-4}$ (σ-modification), at the exactly filtered prediction to 1.0001–1.0029 at $t \ge 500$ (the slaved 0.931–0.996)	`exp_servo_cogging.py` (`--only-A` recomputes Parts A and D, `--add-B` adds the missing Part B jobs, `--only-C [designs]` reruns Part C) → `results/servo/servo_cogging.md`; `tests/test_sliding.py` (the phase-plane loop as the gradient flow of $\bar{L}$, the zero Gauss–Newton eigenvalue per kernel on a circular orbit in general position, the servo following the averaged loop with the position error at both predictions)

continued on the next page

Table 47, continued

statement	what is checked	where
D‴.30	the three-bump instance at the lift of the exact representation of $q = 1/3$, $k_e = 25$: $h_j = 3.06 \cdot 10^{-4}$–0.195, $\operatorname{tr} Q = 8.784 = 8.105$ $(\langle \dot{\hat{J}}^\top \dot{\hat{J}} \rangle) + 0.679$ $(\langle a^2 \hat{J}^\top \hat{J} \rangle)$, $\operatorname{tr} H^2 = 0.0657$, $\langle \lvert \hat{J} \rvert^4 \rangle = 0.299$; the parameters' floor per unit N_0 0.15967, 0.16271, 0.16897, 0.17567 at $\eta = 0$, σ_0, $10\sigma_0$, ∞, the weakest mode's variance 0.0865–2.09 ($\times$24.1); the second-order force (OU, $B = 100$) and its averaged stationary biases 491 and 2397 per $(\mathbb{E}\nu^2)^2$, equal to the floor at std 0.0026 and 0.0012, the full second-order mean's 2397.68, the exact periodic mean's 504 and 2653; the bias's local exponents in k_e 1.93–1.99 ($B \geq 20k_e$), 1.69, 1.47 ($B = 8k_e$, $4k_e$); the separations' drift at the lift $\lambda_{\max}(S_k) = 0.0148$, 0.0101, 0.298 per $\mathbb{E}\nu^2$, the threshold std 0.031 there; along the second-order transient the mean error's rms 0.121 $\to$ 0.602 and $\lambda_{\max}(S_3)$ 0.307 $\to$ 0.349 over 300 time units; the exact periodic covariance of the linearised loop ($k_e = 25, \ldots,$ 800; $\eta = 0$, σ_0, ∞) at the prediction to 0.9762–1.00001, the control (noise input $-k_e\nu$) numerically even in $1/k_e$ (successive ratios $\to$ 4.00), the weakest mode's variance $\times$28.0 at $k_e = 25$, the exact stationary mean 10.7%–0.17% above the averaged (adaptive, $k_e = 25$–200); the real loop (RK4 $\mathrm{d}t = 10^{-3}$, OU $B = 100$, 12 common noise paths of 300 time units, std 0.003 and 0.01, $\eta = 0$ and ∞): z-scores -0.99 to 1.75 against the transient covariance plus the mean part, the mean part 42% and 89% at $t = 300$, the priors 6.0–6.4% apart (predicted 5.3–6.1%); the lift $m = 15$ on a common noise path identical to rounding ($5.6 \cdot 10^{-16}$ in float64); the separations of the lift's third class on eight antithetic pairs at $-2.50 \cdot 10^{-4}$, $+2.74 \cdot 10^{-6}$, $+4.53 \cdot 10^{-4}$ per time unit at std 0.01, 0.03, 0.05 (λ_{eff} 0.3271, 0.3175, 0.2943; the lift's drift 0.2981, the transient's 0.3264, 0.3270, 0.3281), the fit's small-noise limit 0.3291 ± 0.0033, the rate vanishing at std 0.0299 (0.0308 by the lift's drift, 0.0294 by the transient's)	`exp_measurement_noise.py` (`--only-A`, `--only-B`, `--only-D` recompute a part) $\to$ `results/chaos/measurement_noise.md`; `tests/test_measurement_noise.py` (the closed form against the Lyapunov solution, the trace identity at $\eta = \infty$ and the monotonicity in η; the exact periodic floors at $k_e = 100$, $\eta = 0$, σ_0, ∞; a lift $m = 6$ staying replicated under a noise path and following the K_0-atom loop; a separation decaying at std 0.01 and growing at 0.05 (a sign test); the separations' drift against an independent quadrature; along the second-order transient the mean parameters at the averaged force and the mean error carrying the developing deviation's tracking error; the exact periodic mean against the filtered Gram, the fixed prior's branch and its leak in the bias; the mean function's curvature term; the OU noise's variance and correlation, the law seeing the measurement; the responses $(H + \sigma_0 I)^{-1}$ and H^{-1} to a constant force, the quasi-static force against the full second-order mean's noise part)

continued on the next page

Table 47, continued

statement	what is checked	where
D‴.31	the instance of D‴.30 with an OU noise: the adaptive prior's bias per $(\mathbb{E}\nu^2)^2$ for MR, DR, DR* at $k_e = 25$, 50, 100, 200 and $B = 100$, 1000, from the noise's own moments and with the parameters' fluctuation (MR 2397–$1.66 \cdot 10^5$, its parameters' part $\le 0.03\%$; DR and DR* $\le 1.18 < \frac{1}{4}\|\partial_x^2 f\|_\rho^2 = 1.31$, their parameters' part -8.7% to $+3.3\%$ except $+56\%$ for DR at $k_e = B = 100$, its own response $3.9 \cdot 10^{-4}$, $1.1 \cdot 10^{-4}$, $2.9 \cdot 10^{-5}$, $7.4 \cdot 10^{-6}$ at $k_e = 25$–200 ($\propto k_e^{-2}$ on the one-dimensional reference); DR's noise part at its bound $\|g\|_\rho^2$ to 0.01% at $k_e = 25$; $k_e^2\|P\mathbb{E}[e]\|_\rho^2$ 2394 for MR, 0 for DR and DR* by construction, $\|P\mathbb{E}[e]\|_\rho^2$ 3.83 → 0.66; the bias reaching the floor at the std 0.0012, 0.085, 0.27); the prior family at $\eta = \sigma_0$, $\kappa/\eta = 0$, 0.1, 1, 10 and fixed (the weakest mode's response 3269–136, its bias factor 0–0.959, $RH + B = I$ to $1.1 \cdot 10^{-14}$; MR's bias 2397–491, DR's 0.471–0.211) and five priors with non-commuting leaks, R and B from their dynamic blocks (the definitions with S_0 by elimination to $1.9 \cdot 10^{-15}$, $RH + B = I$ to $1.4 \cdot 10^{-14}$, $HR + B - I$ 0.47–0.79); the real loop (RK4 $\mathrm{d}t = 10^{-3}$, $B = 100$, std 0.01, $\eta = \infty$, 12 runs of 300 time units, common noise paths): $\mathbb{E}\|\hat{f} - f\|^2$ at $t = 300$ $2.43 \cdot 10^{-6}$ (MR), $2.80 \cdot 10^{-7}$ (DR), $2.75 \cdot 10^{-7}$ (DR*), z-scores -2.04 to 0.82; the MR − DR run mean at $\hat{J}(\bar{y}_{\mathrm{MR}} - \bar{y}_{\mathrm{DR}})$ to 0.5–5.3%, 0.43–1.83 sampling errors	`exp_eiv_remedy.py` (`--only-AB` recomputes Parts A and B, `--only-B` Part B) → `results/chaos/eiv_remedy.md`; `tests/test_eiv_remedy.py` (the washout identity along a real-loop solution and across a plan switch of the averaged loop; the block's transfer function, $RH + B = I$ along the prior family and, from their dynamic blocks, for non-commuting leaks; the response bound under random leaks, the smallest equivalent perturbation and the projection; the lift exact under DR and DR*, DR's run within the second order of MR's on a common path, the regressor's point; DR's and DR*'s noise forces least-squares with their pointwise bounds, MR's F_1 three orders of magnitude above; the quasi-static forces against the full second-order mean)
F(ii) along (L)	particle path length ($\ge$ Wasserstein length) $\le \lambda_G^{1/2} L_1 \int \|e\|$ at every sample	`T::test_theorem_F_wasserstein_length_bounded_by_integrated_error`
F(ii′) along (L_σ)	$W(\mu_{t_1}, \mu_{t_0}) \le \int (\lambda_G^{1/2} L_1 \|e\| + \sigma_m W(\mu_r, \mu^0))$ on a grid of pairs; a tight instance with $e \equiv 0$ (ratio ≥ 0.99), where the motion is $W \approx 1.05$	`T::test_theorem_F_leak_clause_along_ot_sigma_mod`, `T::test_theorem_F_leak_clause_is_tight_when_the_leak_pays_for_all_motion`
E linearisation	Jacobian of the implemented loop equals $A(t)$; $QA + A^\top Q = \mathrm{diag}(-k_e I, 0)$	`T::test_theorem_E_linearisation_and_lyapunov_identity`
E convergence	the matched error decays and the labelled error does not (qualitative); sliding-window excitation and fitted rates	`T::test_theorem_E_matched_error_converges_labelled_does_not`, `exp_pe_convergence.py`
Lemma E.0	$p_{\mathrm{lo}}\|w\|^2 \le V \le p_{\mathrm{hi}}\|w\|^2$ and $\dot{V} \le -\gamma\|w\|^2$ at random (t, w) for four regressors, closed-form and optimised ε; the envelope along trajectories; explicit rate $\le$ exact Floquet rate	`Ex::test_lemma_E0_pointwise_inequalities_hold_on_random_states`, `Ex::test_lemma_E0_envelope_and_floquet_ordering`
Lemma E.0(iii), Remark E.0a	the ω^{-2} law of the explicit rate and of the exact rate; the saturation of exact/explicit in the slow-adaptation limit; the tables	`Ex::test_remark_E1_omega_scaling_of_the_explicit_rate`, `Ex::test_slow_adaptation_limit_ratio_is_bounded`, `exp_explicit_rate.py` → `results/explicit_rate/table.md`

continued on the next page

Table 47, continued

statement	what is checked	where
E, explicit constants	on the implemented loop, from a start inside the explicit ball, $\|w(t)\| \le (p_{\mathrm{hi}}/p_{\mathrm{lo}})^{1/2} e^{-\lambda t}\|w(0)\|$ at every sample (L_J estimated from samples)	`Ex::test_theorem_E_explicit_nonlinear_envelope_on_the_implemented_loop`
Corollary E.0′.2	the explicit envelope of Theorem E with Lemma E.0′'s constants on the implemented single-kernel loop at $k_e = 40$: 1.5 e-folds inside the envelope from a start in the explicit ball ($\mathrm{d}t = 0.005$, below the discretisation floor); $\lambda' > 10 \times$ Theorem E's rate from Lemma E.0 and $< 4 \times$ the exact linear rate; the constants table	`Ex::test_corollary_E0prime_2_envelope_on_the_implemented_loop`, `exp_explicit_rate.py` (table 5)
Corollary E.0′.3	the explicit bound on the stationary parameter spread of the dithered (LTV) dominates the exact periodic covariance (sinusoid: ratios $2.7 \to 1.3$ ($\delta = 0.2$) and $2.5 \to 1.2$ ($\delta = 0.1$) at $k = 20 \to 200$, falling toward $1/(1-\delta)$; the Theorem E ideal: 15–18, the price of $q = 15$ directions at the worst excitation); the exact value equals the averaged $k \cdot \mathrm{tr}\, X$ within $2/k^2$	`Ex::test_corollary_E0prime_3_dither_covariance_bound_against_the_periodic_lyapunov_solution`, `exp_explicit_rate.py` → `results/explicit_rate/table.md` (table 6)
Lemma E.0″	on $\Psi = (\sin t, \cos\sqrt{2}\,t)$ (not periodic) with window 4π: the exponential-average identities on the grid (10^{-4}), the conditions at $k = 20$, 50, 200 with $\delta_2 \le 5\%$, the certificate on 400 times, the guaranteed rate below the exact decay rate (top Lyapunov exponent) within a factor 2.6 and improving with k; the experiment: exact/guaranteed $2.5 \to 1.3$ on the quasi-periodic regressor and $5.0 \to 3.7$ on a slowed Lorenz-driven regressor ($k = 100$–400), the certificate holding at every admissible row	`Ex::test_lemma_E0dprime_quasi_periodic_regressor_certificate_and_rate`, `exp_pe_general.py` → `results/explicit_rate/general_pe.md`
Lemma E.0′	the certificate $\sup_{t,x} \dot{V}/V \le -\gamma$ (a generalised eigenvalue problem on a time grid) on $\Psi = (\sin t, \cos t)$ at $k = 20$, 50, 200 and on the slow-adaptation ideal at $g = 0.01$, $k = 400$; $N_2 = 0$ numerically; the guaranteed rate below the exact Floquet rate and within a factor 2 (sinusoid) and 1.5 (ideal) of it, above Lemma E.0's by more than 10^4 on the ideal; the table of k and δ	`Ex::test_lemma_E0prime_sinusoidal_regressor_certificate_and_sharpness`, `Ex::test_lemma_E0prime_pays_for_ill_conditioning_once_on_the_theorem_E_ideal`, `exp_explicit_rate.py` → `results/explicit_rate/table.md` (table 4)
G(a), two scales	direct check (cosine ≥ 0.998 in the G^{-1} metric, `violating`; the weight block fails in `satisfying`, as explained in Section 19); $\bar{L}$ at period-mean parameters non-increasing	`exp_variable_projection.py` → `table_violating.md`, `table_satisfying.md`
G(b), (b′), (H2c)	VarPro holds when (H2b) and $S \ll 1$ hold; it fails when $S = O(1)$ although (H2b) holds; the induced-metric flow tracks the closed loop at every S (started on the slow manifold; gains $\div 4$ isolates the averaging error)	`exp_variable_projection.py` → `table_satisfying.md`; `exp_slow_manifold_slope.py` → `table_slope.md`
H(i), algebraic identity	with $e := r/k_e$ and $x := x_m$, the averaged drift equals $-G\nabla V$ (code gradients against finite differences, 10^{-7}); does not test averaging	`Lg::test_averaged_drift_is_the_mean_field_langevin_drift`
Remark H.0	the frozen-μ averaging error decays like $1/k_e$, $n = 1$ and $n = 2$	`Lg::test_averaging_error_of_the_formal_reduction_decreases_like_one_over_ke`, `Av::test_averaged_drift_error_decays_like_one_over_ke_for_an_orbit_enclosing_area`

continued on the next page

Table 47, continued

statement	what is checked	where
H, need for entropy	finite-m real loop: a noiseless symmetric start stays bit-identical (and moves); dither separates it	`Lg::test_noiseless_symmetric_initialisation_never_separates_but_dither_does`
H(ii), oscillation bound	$\lvert V_L[\mu](\vartheta)\rvert \le K_\Phi(F_\rho + K_\Phi)$ for random and adversarial μ, ϑ	`Lg::test_first_variation_bound_used_in_holley_stroock`
H(ii) constant	formula transcription only	`Lg::test_lsi_lower_bound_formula`
H, global behaviour	collapsed against spread starts, with and without dither, two (σ, τ) settings	`exp_langevin_global.py`
H, speed–bias trade-off	escape time and loss floor against τ; $\log(1/\tau)$ and Arrhenius fits, seed-averaged and per seed (neither identifies the mechanism)	`exp_langevin_tradeoff.py`
H, reference values	$\bar{L}(\mu_\tau^*)$ by a concave dual on a grid; the stationary state of the exactly averaged loop; their offset in the scaling of H.6	`exp_gibbs_reference.py` → `results/langevin_global/gibbs_reference.md`
H.7	expansion, Stokes area identity, parity under reversal, non-reciprocity of the exact second variation	`Av::test_frozen_measure_expansion_converges_at_rate_one_over_ke`, `Av::test_circulation_kernel_antisymmetric_vanishes_in_1d_and_equals_the_enclosed_area_integral`, `Av::test_shift_correction_is_even_and_lag_correction_is_odd_under_orbit_reversal`, `Av::test_exactly_averaged_dynamics_is_non_reciprocal_for_n1_and_n2`
H.7a, oscillation term	value against an independent one-period near-identity flow (within 3% at $k_e = 80$); odd under orbit reversal (10^{-6})	`Av::test_oscillation_correction_matches_a_one_period_flow`, `Av::test_oscillation_correction_is_odd_under_orbit_reversal`
I(i), (ii)	$a^\top B(\xi, \xi')a'$ equals the antisymmetric part of $S_{\text{even}} + K$ (10^{-10}); $A(\vartheta, \vartheta_z) = 0$ and the cyclic sum over $(\vartheta_0, \vartheta_1, \vartheta_z)$ equals $A(\vartheta_0, \vartheta_1)$ (10^{-14})	`NR::test_bilinear_form_B_reproduces_the_antisymmetric_second_variation`, `NR::test_zero_feature_collapses_the_cyclic_criterion_to_a_pairwise_one`
I(iii)	$n = 1$: $A \equiv 0$ at perfect fit ($< 10^{-13}$); $A \neq 0$ with a residual, equal to the closed form; the positivity witness	`NR::test_n1_reciprocal_iff_residual_vanishes_on_the_orbit`, `NR::test_positivity_witness_n1`
I(iv)	the closed form of κ (10^{-10}); the second component of $F_s((0.5, 0))$ is positive on the ellipse; $\kappa \equiv 0$ on a zero-current orbit; with $f \equiv \hat{f}_\mu$ on $\mathbb{R}^2$ ($Dr_\mu \equiv 0$) $B = 2\kappa I$, $A \neq 0$ on the ellipse and $A \equiv 0$ on the segment; with $f = \hat{f}_\mu + he_1$, h zero on the ellipse, the curl term survives and equals $-\int \varphi\varphi' \partial_2 h \, \mathrm{d}\rho \cdot (e_1 e_2^\top - e_2 e_1^\top)$	`NR::test_circulation_kernel_closed_form_and_vanishing_for_a_zero_current_orbit`, `NR::test_perfect_fit_n2_leaves_only_the_circulation_kernel`, `NR::test_perfect_fit_on_the_orbit_alone_keeps_the_curl_term`
J, averaged objective	∇J_m against finite differences (10^{-7}); the critical point is a non-degenerate minimum with two distinct kernels	`Pe::test_gradient_of_J_m_matches_finite_differences_and_critical_point_is_nondegenerate`

continued on the next page

Table 47, continued

statement	what is checked	where
J(a), (b), (d)	Newton shooting finds the periodic orbit (residual $< 10^{-10}$); $\|\bar{\Theta}_p - \Theta^*\|$ and $\max\|e\|$ halve from $k_e = 20$ to 40; the vector error of the first-order prediction of the offset halves as well (it is 0.55 at $k_e = 40$ in the fast-sweep test instance, and 0.041 at $k_e = 40$ in instance A of the experiment); the parameter Floquet exponents match $\frac{1}{k_e}\lambda(G_m^{1/2}\nabla^2 J_m G_m^{1/2})$ with a maximal relative error below 2% at $k_e = 40$ that halves with k_e; both instances at $k_e = 10$–160 in the experiment; the error-channel exponent recovered from Liouville's trace identity is $k_e - L$ with $\|L\| < 0.2$	`Pe::test_theorem_J_periodic_orbit_offset_and_floquet_exponents`, `exp_periodic_orbit.py` → `results/periodic_orbit/table.md`
K(i), (ii)	the Hessian of J_m at a collapsed configuration equals $I_m \otimes (\sigma_0 G^{-1} - R_2) + \frac{1}{m}(\mathbf{1}\mathbf{1}^\top) \otimes M_2$ (finite differences, 10^{-5} relative); σ_{crit} solves $\sigma = \lambda_{\max}(G^{1/2} R_2(\bar{\vartheta}(\sigma)) G^{1/2})$ (10^{-8}) and the separation block is singular there; the separation Floquet exponents of the collapsed orbit come in $(m-1)$-fold multiplets	`Pe::test_proposition_K_hessian_splitting_and_critical_leak`, `exp_orientation_stability.py`
K(iii), (iv)	the orientation difference of the softest separation exponent at $\sigma_0 = 0.5$: slope -2.18 against k_e, within 14% → 3% of the prediction at $\bar{\Theta}_2$ for $k_e = 10 \to 80$, while the prediction at Θ^* converges to a value 30% off; at σ_{crit} both orientations are unstable, predictions within 14% at $k_e = 40$; at the designed σ_0 the signs differ for $k_e = 10$ and 20	`exp_orientation_stability.py` → `results/orientation_stability/table.md`
J′ averaged data, solver	the single-kernel J, ∇J, b_{fr}, b_{osc} against the general H.7/H.7a code ($2 \cdot 10^{-5}$); the weighted Poisson solver against the Ornstein–Uhlenbeck closed forms (g, the mean shift $(G\lambda)^{-1}b$, the Green identity, zero current for a gradient drift); the stationary Fokker–Planck solver of the first-order SDE: current ratio → 1 linearly in ε, $\|m - \pi(1 + \varepsilon g)\|_1 \propto \varepsilon^2$	`tests/test_dither.py` (3 tests)
J′(b), (c), (d)	real loop, $2 \cdot 10^4$ replicas, $k_e = 10$–80, both orientations: the stationary means differ from the Gibbs values by $k_e^{-0.87}$ (α), $k_e^{-0.83}$ (γ), $k_e^{-1.09}$ ($\alpha\gamma$); the first-order prediction removes 93–97% of the discrepancy at $k_e = 10$ and 70–85% at $k_e = 20$–40, the residual at the Monte Carlo floor at $k_e = 80$ (second order not resolved); the orientation-odd shift matches its prediction to 7–8% at $k_e = 10$ and within the 10–30% resolution at $k_e = 20$–80; $k_e^2 \mathbb{E}e^2/(2\pi(\bar{L})) - 1 = 0.5$–$0.7/k_e$	`tests/test_dither.py::test_real_loop_means_are_closer_to_the_first_order_prediction_than_to_gibbs` ($k_e = 20$), `exp_dither_stationary.py` → `results/dither/table.md`
J′(e), J′.2	the circulation of the real loop is below the Monte Carlo resolution at first order ($k_e \geq 20$) and dominated by higher orders at $k_e = 10$; the current and density expansion of the first-order averaged SDE checked on the grid (ratios 1.47 → 1.03, ε^2 remainder 0.31); the antisymmetric part of $D(G^{-1}b)$ against the formula of J′.2(iii) to six digits (referee round 6)	`exp_dither_stationary.py` → `results/dither/table.md`; `review/REVIEW_round6_Jprime.md`

continued on the next page

Table 47, continued

statement	what is checked	where
Corollary J′.3	the Gibbs measures π_m (MALA, 4000 replicas, $m = 1$–32) converge to the mean-field minimiser μ_τ^* (fixed point to 10^{-11}) with a $1/m$ law: slopes -1.01 ($\mathbb{E}[\gamma_1]$) and -0.99 ($\mathbb{E}[\bar{L}(\mu_\Theta)]$), rescaled deviation 0.058 ± 0.002 at every m), $\mathbb{E}[\alpha_1^2]$ from 0.54 to 0.92 against 0.935 with rescaled deviation -0.41 ± 0.01; the $m = 1$ row agrees with the quadrature of π_1	`exp_gibbs_m.py` → `results/dither/gibbs_m.md`; `tests/test_dither.py::test_corollary_J3_mean_field_minimiser_and_gibbs_sampler_consistency`
H.7 + H.7a in the real closed loop	the orientation dependence of the learned kernels (same ρ) is predicted to within 3% in size and 11% as a vector, cosine 0.994 at $k_e = 40$; the measured difference scales like $k_e^{-1.01}$	`exp_orbit_orientation.py` → `results/orbit_orientation/table_sigma2_slow16.md`

Not checked in the repository:

- B.5 along (L), and B.6 and B.7;
- the Filippov form of C(i), the inequality of C(v), and Proposition C.5's $D^+V_{\mathrm{OT}} = \sigma_m(\delta - \delta^2)$;
- D(b), D's constant, and the one-sided constant of Remark D.1;
- the sharpness of the explicit rates of E beyond the tables of Remark E.0a and Lemma E.0′; Lemma E.0′ below its threshold k_0 (where it makes no claim) and for non-periodic PE; Corollary E.2 and Remark E.3;
- Theorem I(v) beyond the two instances (the analytic dichotomy is a proof only);
- Theorem J(c) (the basin; a single basin run is reported in `review/REVIEW_round5_J.md`);
- Proposition K(iii)–(iv) beyond the instance of `exp_orientation_stability.py`, and the second-order normal form it rests on;
- Theorem C′(iv) (its hypotheses are not met on any tested instance) and the convergence of the pure leak to a critical point beyond the tested instances; Theorem D′'s constant, its per-particle bound and its k-particle convergence (only the rate of (b) is checked);
- H.8, which is a proof only;
- Conjecture H.6.

References

[1] S. K. Ainsworth, J. Hayase, S. Srinivasa. *Git Re-Basin: merging models modulo permutation symmetries*. International Conference on Learning Representations (ICLR), 2023. arXiv:2209.04836.

[2] L. Ambrosio, N. Gigli, G. Savaré. *Gradient Flows in Metric Spaces and in the Space of Probability Measures*. 2nd edition, Lectures in Mathematics ETH Zürich. Birkhäuser, Basel, 2008. https://doi.org/10.1007/978-3-7643-8722-8.

[3] F. Aurenhammer, F. Hoffmann, B. Aronov. *Minkowski-type theorems and least-squares clustering*. Algorithmica, 20:61–76, 1998; J. Kitagawa, Q. Mérigot, B. Thibert, *Convergence of a Newton algorithm for semi-discrete optimal transport*, Journal of the European Mathematical Society, 21(9):2603–2651, 2019.

[4] D. Bakry, M. Émery. *Diffusions hypercontractives*. In J. Azéma, M. Yor (eds.), Séminaire de Probabilités XIX, 1983/84, Lecture Notes in Mathematics 1123, pp. 177–206. Springer, Berlin, 1985. https://doi.org/10.1007/BFb0075847.

[5] M. Balzer, T. Schlömer, O. Deussen. *Capacity-constrained point distributions*. ACM SIGGRAPH 2009; Q. Mérigot, *A multiscale approach to optimal transport*, Computer Graphics Forum, 30(5), 2011; F. de Goes, K. Breeden, V. Ostromoukhov, M. Desbrun, *Blue noise through optimal transport*, ACM Transactions on Graphics, 31(6), 2012.

[6] R. Barboni, G. Peyré, F.-X. Vialard. *Ultra-fast feature learning for the training of two-layer neural networks in the two-timescale regime*. arXiv:2504.18208, 2025.

[7] A. Blanchet, J. Bolte. *A family of functional inequalities: Łojasiewicz inequalities and displacement convex functions*. Journal of Functional Analysis, 275(7):1650–1673, 2018; S. Chewi, T. Maunu, P. Rigollet, A. J. Stromme, *Gradient descent algorithms for Bures–Wasserstein barycenters*, COLT 2020; Q. Mérigot, F. Santambrogio, C. Sarrazin, *Non-asymptotic convergence bounds for Wasserstein approximation using point clouds*, NeurIPS 2021; L. Ambrosio, N. Gigli, G. Savaré, *Gradient Flows in Metric Spaces and in the Space of Probability Measures*, Birkhäuser, 2008, §9.2.

[8] J. Bolte, A. Daniilidis, A. Lewis. *The Łojasiewicz inequality for nonsmooth subanalytic functions with applications to subgradient dynamical systems*. SIAM Journal on Optimization, 17(4):1205–1223, 2007; J. Bolte, A. Daniilidis, A. Lewis, M. Shiota, *Clarke subgradients of stratifiable functions*, SIAM Journal on Optimization, 18(2):556–572, 2007; D. Davis, D. Drusvyatskiy, S. Kakade, J. D. Lee, *Stochastic subgradient method converges on tame functions*, Foundations of Computational Mathematics, 20:119–154, 2020.

[9] E. A. Carlen, K. Craig. *Contraction of the proximal map and generalized convexity of the Moreau–Yosida regularization in the 2-Wasserstein metric*. Mathematics and Mechanics of Complex Systems, 1(1):33–65, 2013. https://doi.org/10.2140/memocs.2013.1.33. arXiv:1205.6565.

[10] P. Cattiaux, L. Mesnager. *Hypoelliptic non-homogeneous diffusions*. Probability Theory and Related Fields, 123:453–483, 2002.

[11] F. Chen, Z. Ren, S. Wang. *Uniform-in-time propagation of chaos for mean field Langevin dynamics*. Annales de l'Institut Henri Poincaré, Probabilités et Statistiques, 61(4):2357–2404, 2025. https://doi.org/10.1214/24-AIHP1499. arXiv:2212.03050.

[12] F. Chen, Z. Ren, S. Wang. *Uniform-in-time propagation of chaos for mean field Langevin dynamics*. arXiv:2212.03050, 2022; T. Suzuki, A. Nitanda, D. Wu, *Uniform-in-time propagation of chaos for the mean-field gradient Langevin dynamics*, ICLR 2023.

[13] J. Chevallier. *Uniform decomposition of probability measures: quantization, clustering and rate of convergence*. Journal of Applied Probability, 55(4):1037–1045, 2018. https://doi.org/10.1017/jpr.2018.69.

[14] L. Chizat, F. Bach. *On the global convergence of gradient descent for over-parameterized models using optimal transport*. Advances in Neural Information Processing Systems 31 (NeurIPS 2018), pp. 3036–3046, 2018. arXiv:1805.09545.

[15] L. Chizat. *Mean-field Langevin dynamics: exponential convergence and annealing*. Transactions on Machine Learning Research, 2022. arXiv:2202.01009.

[16] L. Chizat. *Sparse optimization on measures with over-parameterized gradient descent*. Mathematical Programming, 194(1–2):487–532, 2022. https://doi.org/10.1007/s10107-021-01636-z. arXiv:1907.10300.

[17] G. Chowdhary, E. Johnson. *Concurrent learning for convergence in adaptive control without persistency of excitation*. In Proc. 49th IEEE Conference on Decision and Control (CDC), pp. 3674–3679, 2010. https://doi.org/10.1109/CDC.2010.5717148.

[18] G. V. Chowdhary, E. N. Johnson. *Theory and flight-test validation of a concurrent-learning adaptive controller*. Journal of Guidance, Control, and Dynamics, 34(2):592–607, 2011. https://doi.org/10.2514/1.46866.

[19] J. Cortés, S. Martínez, T. Karatas, F. Bullo. *Coverage control for mobile sensing networks*. IEEE Transactions on Robotics and Automation, 20(2):243–255, 2004.

[20] K. Craig, K. Elamvazhuthi, M. Haberland, O. Turanova. *A blob method for inhomogeneous diffusion with applications to multi-agent control and sampling*. Mathematics of Computation, 92(344):2575–2654, 2023. https://doi.org/10.1090/mcom/3841. arXiv:2202.12927.

[21] M. Cuturi. *Sinkhorn distances: lightspeed computation of optimal transport*. NeurIPS 2013; G. Peyré, M. Cuturi, *Computational Optimal Transport*, Foundations and Trends in Machine Learning, 11(5–6), 2019; R. Cominetti, J. San Martín, *Asymptotic analysis of the exponential penalty trajectory in linear programming*, Mathematical Programming, 67:169–187, 1994; N. Courty, R. Flamary, D. Tuia, A. Rakotomamonjy, *Optimal transport for domain adaptation*, IEEE Transactions on Pattern Analysis and Machine Intelligence, 39(9):1853–1865, 2017; A. Ramdas, N. García Trillos, M. Cuturi, *On Wasserstein two-sample testing and related families of nonparametric tests*, Entropy, 19(2):47, 2017; A. Genevay, G. Peyré, M. Cuturi, *Learning generative models with Sinkhorn divergences*, AISTATS 2018; J. Feydy, T. Séjourné, F.-X. Vialard, S. Amari, A. Trouvé, G. Peyré, *Interpolating between optimal transport and MMD using Sinkhorn divergences*, AISTATS 2019; Y. M. Asano, C. Rupprecht, A. Vedaldi, *Self-labelling via simultaneous clustering and representation learning*, ICLR 2020.

[22] G. Da Prato, J. Zabczyk. *Ergodicity for Infinite Dimensional Systems*. London Mathematical Society Lecture Note Series 229. Cambridge University Press, Cambridge, 1996.

[23] Y. De Castro, S. Gadat, C. Marteau. *Fast Spawn&Prune (FS&P): global convergence of stochastic conic particle gradient descent via birth/death process*. arXiv:2605.19784, 2026.

[24] A. Dembo, O. Zeitouni. *Large Deviations Techniques and Applications*. 2nd edition. Springer, 1998.

[25] R. L. Dobrushin. *Vlasov equations*. Functional Analysis and Its Applications, 13:115–123, 1979.

[26] A. Durmus, A. Eberle, A. Guillin, R. Zimmer. *An elementary approach to uniform in time propagation of chaos*. Proceedings of the American Mathematical Society, 148:5387–5398, 2020.

[27] M. Emerick, B. Bamieh. *Continuum swarm tracking control: a geometric perspective in Wasserstein space*. In Proc. 62nd IEEE Conference on Decision and Control (CDC), pp. 1367–1374, 2023. https://doi.org/10.1109/CDC49753.2023.10384000. arXiv:2303.15638.

[28] S. Fabri, V. Kadirkamanathan. *Dynamic structure neural networks for stable adaptive control of nonlinear systems*. IEEE Transactions on Neural Networks, 7(5):1151–1167, 1996. https://doi.org/10.1109/72.536311.

[29] M. Fornasier, B. Piccoli, F. Rossi. *Mean-field sparse optimal control*. Philosophical Transactions of the Royal Society A, 372(2028):20130400, 2014. https://doi.org/10.1098/rsta.2013.0400. arXiv:1402.5657.

[30] N. Fournier, A. Guillin. *On the rate of convergence in Wasserstein distance of the empirical measure*. Probability Theory and Related Fields, 162(3–4):707–738, 2015. https://doi.org/10.1007/s00440-014-0583-7.

[31] A. Frieze, R. Kannan. *Log-Sobolev inequalities and sampling from log-concave distributions*. The Annals of Applied Probability, 9(1):14–26, 1999. https://doi.org/10.1214/aoap/1029962595.

[32] M. Fruchart, R. Hanai, P. B. Littlewood, V. Vitelli. *Non-reciprocal phase transitions*. Nature, 592(7854):363–369, 2021. https://doi.org/10.1038/s41586-021-03375-9.

[33] S. S. Ge, C. Wang. *Direct adaptive NN control of a class of nonlinear systems*. IEEE Transactions on Neural Networks, 13(1):214–221, 2002. https://doi.org/10.1109/72.977306.

[34] R. Giard, Y. De Castro, R. Denis, C. Marteau. *Riemannian gradient descent for Gaussian mixture models with unknown diagonal covariances*. arXiv:2609.30220, 2026.

[35] D. Gilbarg, N. S. Trudinger. *Elliptic Partial Differential Equations of Second Order*. 2nd edition. Springer, 1983 (reprinted 2001).

[36] M. Glasgow, J. Bruna. *Uniform-in-time weak propagation of chaos in shallow neural networks*. Advances in Neural Information Processing Systems 39 (NeurIPS 2026), to appear. arXiv:2605.22010, 2026.

[37] F. Golse. *On the dynamics of large particle systems in the mean field limit*. Lecture notes, arXiv:1301.5494, 2013. Published in A. Muntean, J. Rademacher, A. Zagaris (eds.), Macroscopic and Large Scale Phenomena: Coarse Graining, Mean Field Limits and Ergodicity, Lecture Notes in Applied Mathematics and Mechanics 3, pp. 1–144. Springer, 2016. https://doi.org/10.1007/978-3-319-26883-5_1.

[38] S. Graf, H. Luschgy. *Foundations of Quantization for Probability Distributions*. Lecture Notes in Mathematics 1730. Springer, Berlin, 2000. https://doi.org/10.1007/BFb0103945.

[39] J. Guo, S. T. Paruchuri, A. J. Kurdila. *Partial persistence of excitation in RKHS embedded adaptive estimation*. IEEE Transactions on Automatic Control, 68(10):5850–5861, 2023. https://doi.org/10.1109/TAC.2022.3226710. arXiv:2002.07963.

[40] M. Hairer. *Ergodic properties of Markov processes*. Lecture notes, University of Warwick, 2006.

[41] M. Hairer, J. C. Mattingly. *Yet another look at Harris' ergodic theorem for Markov chains*. In Seminar on Stochastic Analysis, Random Fields and Applications VI, Progress in Probability 63, pp. 109–117. Birkhäuser, 2011.

[42] A. J. Hoffman, J. B. Kruskal. *Integral boundary points of convex polyhedra*. In: H. W. Kuhn, A. W. Tucker (eds.), Linear Inequalities and Related Systems, Annals of Mathematics Studies 38, Princeton, 1956, pp. 223–246; V. Klee, C. Witzgall, *Facets and vertices of transportation polytopes*, in: Mathematics of the Decision Sciences, Part 1, Lectures in Applied Mathematics 11, AMS, 1968, pp. 257–282; A. Schrijver, *Theory of Linear and Integer Programming*, Wiley, 1986, Ch. 19.

[43] R. Holley, D. Stroock. *Logarithmic Sobolev inequalities and stochastic Ising models*. Journal of Statistical Physics, 46(5–6):1159–1194, 1987. https://doi.org/10.1007/BF01011161.

[44] L. Hörmander. *Hypoelliptic second order differential equations*. Acta Mathematica, 119:147–171, 1967; M. Hairer, *On Malliavin's proof of Hörmander's theorem*, Bulletin des Sciences Mathématiques, 135(6–7):650–666, 2011.

[45] D. Inoue, Y. Ito, H. Yoshida. *Optimal transport-based coverage control for swarm robot systems: generalization of the Voronoi tessellation-based method*. IEEE Control Systems Letters, 5(4):1483–1488, 2021. https://doi.org/10.1109/LCSYS.2020.3039008. arXiv:2011.08337.

[46] P. A. Ioannou, P. V. Kokotović. *Adaptive Systems with Reduced Models*. Lecture Notes in Control and Information Sciences 47. Springer, Berlin, 1983. https://doi.org/10.1007/BFb0006357.

[47] P. A. Ioannou, J. Sun. *Robust Adaptive Control*. Prentice Hall, Upper Saddle River, NJ, 1996.

[48] H. K. Khalil. *Nonlinear Systems*. 3rd edition. Prentice Hall, Upper Saddle River, NJ, 2002.

[49] P. V. Kokotović, H. K. Khalil, J. O'Reilly. *Singular Perturbation Methods in Control: Analysis and Design*. Academic Press, London, 1986. Reprinted as Classics in Applied Mathematics 25, SIAM, 1999, https://doi.org/10.1137/1.9781611971118.

[50] K. Kurdyka. *On gradients of functions definable in o-minimal structures*. Annales de l'Institut Fourier, 48(3):769–783, 1998; L. van den Dries, A. Macintyre, D. Marker, *The elementary theory of restricted analytic fields with exponentiation*, Annals of Mathematics, 140:183–205, 1994.

[51] N. Lanzetti, E. C. Balta, D. Liao-McPherson, F. Dörfler. *Stochastic Wasserstein gradient flows using streaming data with an application in predictive maintenance*. IFAC-PapersOnLine (IFAC World Congress 2023), 56(2):3954–3959, 2023. https://doi.org/10.1016/j.ifacol.2023.10.1334. arXiv:2301.12461.

[52] E. Lavretsky, T. E. Gibson. *Projection operator in adaptive systems*. arXiv:1112.4232, 2011.

[53] E. Lavretsky, K. A. Wise. *Robust and Adaptive Control: With Aerospace Applications*. Advanced Textbooks in Control and Signal Processing. Springer, London, 2013. https://doi.org/10.1007/978-1-4471-4396-3.

[54] F. L. Lewis, A. Yeşildirek, K. Liu. *Multilayer neural-net robot controller with guaranteed tracking performance*. IEEE Transactions on Neural Networks, 7(2):388–399, 1996. https://doi.org/10.1109/72.485674.

[55] F. L. Lewis, S. Jagannathan, A. Yeşildirek. *Neural Network Control of Robot Manipulators and Nonlinear Systems*. Taylor & Francis, London, 1999.

[56] C. Liu, L. Zhu, M. Belkin. *Loss landscapes and optimization in over-parameterized non-linear systems and neural networks*. Applied and Computational Harmonic Analysis, 59:85–116, 2022.

[57] S. P. Lloyd. *Least squares quantization in PCM*. IEEE Transactions on Information Theory, 28(2):129–137, 1982; Q. Du, V. Faber, M. Gunzburger, *Centroidal Voronoi tessellations: applications and algorithms*, SIAM Review, 41(4):637–676, 1999.

[58] F. Malrieu. *Logarithmic Sobolev inequalities for some nonlinear PDE's*. Stochastic Processes and their Applications, 95(1):109–132, 2001.

[59] P. Marion, R. Berthier. *Leveraging the two-timescale regime to demonstrate convergence of neural networks*. Advances in Neural Information Processing Systems 36 (NeurIPS 2023), pp. 64996–65029, 2023. arXiv:2304.09576.

[60] J. C. Mattingly, A. M. Stuart, M. V. Tretyakov. *Convergence of numerical time-averaging and stationary measures via Poisson equations*. SIAM Journal on Numerical Analysis, 48(2):552–577, 2010.

[61] C. McDiarmid. *On the method of bounded differences*. In Surveys in Combinatorics 1989, London Mathematical Society Lecture Note Series 141, pp. 148–188. Cambridge University Press, 1989.

[62] A. P. Morgan, K. S. Narendra. *On the stability of nonautonomous differential equations $\dot{x} = [A + B(t)]x$, with skew symmetric matrix $B(t)$*. SIAM Journal on Control and Optimization, 15(1):163–176, 1977. https://doi.org/10.1137/0315013.

[63] K. S. Narendra, A. M. Annaswamy. *A new adaptive law for robust adaptation without persistent excitation*. IEEE Transactions on Automatic Control, 32(2):134–145, 1987. https://doi.org/10.1109/TAC.1987.1104543.

[64] A. Nitanda, D. Wu, T. Suzuki. *Convex analysis of the mean field Langevin dynamics*. Proceedings of the 25th International Conference on Artificial Intelligence and Statistics (AISTATS), PMLR 151:9741–9757, 2022. arXiv:2201.10469.

[65] F. Otto, C. Villani. *Generalization of an inequality by Talagrand and links with the logarithmic Sobolev inequality*. Journal of Functional Analysis, 173(2):361–400, 2000. https://doi.org/10.1006/jfan.1999.3557.

[66] E. Pardoux, A. Yu. Veretennikov. *On the Poisson equation and diffusion approximation. I*. The Annals of Probability, 29(3):1061–1085, 2001 (and *II*, The Annals of Probability, 31(3):1166–1192, 2003).

[67] S. T. Paruchuri, J. Guo, A. J. Kurdila. *Kernel center adaptation in the reproducing kernel Hilbert space embedding method*. International Journal of Adaptive Control and Signal Processing, 36(7):1562–1583, 2022. https://doi.org/10.1002/acs.3407. arXiv:2009.02867.

[68] O. S. Patil, E. J. Griffis, W. A. Makumi, W. E. Dixon. *Simultaneous online system identification and control using composite adaptive Lyapunov-based deep neural networks*. arXiv:2311.13056, 2023 (revised 2026).

[69] O. S. Patil. *Adaptive control as an information gradient flow: excitation, constraints, energy, and stochastic learning*. arXiv:2610.04822, 2026.

[70] G. A. Pavliotis, A. M. Stuart. *Multiscale Methods: Averaging and Homogenization*. Texts in Applied Mathematics 53. Springer, New York, 2008.

[71] J. C. Perdomo, T. Zrnic, C. Mendler-Dünner, M. Hardt. *Performative prediction*. Proceedings of the 37th International Conference on Machine Learning (ICML), PMLR 119:7599–7609, 2020. arXiv:2002.06673.

[72] S. Plunder, B. Simeon. *Coupled systems of linear differential-algebraic and kinetic equations with application to the mathematical modelling of muscle tissue*. In T. Reis, S. Grundel, S. Schöps (eds.), Progress in Differential-Algebraic Equations II, Differential-Algebraic Equations Forum, pp. 357–395. Springer, 2020. https://doi.org/10.1007/978-3-030-53905-4_12. arXiv:1911.05468 (2019).

[73] S. Plunder, B. Simeon. *The mean-field limit for particle systems with uniform full-rank constraints*. Kinetic and Related Models, 16(6):884–912, 2023. https://doi.org/10.3934/krm.2023012. arXiv:2203.07249 (2022).

[74] B. T. Polyak. *Gradient methods for minimizing functionals*. Zhurnal Vychislitel'noi Matematiki i Matematicheskoi Fiziki, 3(4):643–653, 1963; H. Karimi, J. Nutini, M. Schmidt, *Linear convergence of gradient and proximal-gradient methods under the Polyak–Łojasiewicz condition*, ECML PKDD 2016.

[75] M. M. Polycarpou. *Stable adaptive neural control scheme for nonlinear systems*. IEEE Transactions on Automatic Control, 41(3):447–451, 1996. https://doi.org/10.1109/9.486648.

[76] J.-B. Pomet, L. Praly. *Adaptive nonlinear regulation: estimation from the Lyapunov equation*. IEEE Transactions on Automatic Control, 37(6):729–740, 1992. https://doi.org/10.1109/9.256328.

[77] K. Rose, E. Gurewitz, G. C. Fox. *Statistical mechanics and phase transitions in clustering*. Physical Review Letters, 65(8):945–948, 1990; K. Rose, *Deterministic annealing for clustering, compression, classification, regression, and related optimization problems*, Proceedings of the IEEE, 86(11):2210–2239, 1998.

[78] J. A. Sanders, F. Verhulst, J. Murdock. *Averaging Methods in Nonlinear Dynamical Systems*. 2nd edition, Applied Mathematical Sciences 59. Springer, New York, 2007. https://doi.org/10.1007/978-0-387-48918-6.

[79] R. M. Sanner, J.-J. E. Slotine. *Gaussian networks for direct adaptive control*. IEEE Transactions on Neural Networks, 3(6):837–863, 1992. https://doi.org/10.1109/72.165588.

[80] F. Santambrogio. {*Euclidean, metric, and Wasserstein*} *gradient flows: an overview*. Bulletin of Mathematical Sciences, 7(1):87–154, 2017. https://doi.org/10.1007/s13373-017-0101-1. arXiv:1609.03890.

[81] S. Sastry, M. Bodson. *Adaptive Control: Stability, Convergence, and Robustness*. Prentice Hall, Englewood Cliffs, NJ, 1989.

[82] U. Seifert. *Stochastic thermodynamics, fluctuation theorems and molecular machines*. Reports on Progress in Physics, 75:126001, 2012; C. Maes, K. Netočný, *Time-reversal and entropy*, Journal of Statistical Physics, 110:269–310, 2003.

[83] D. W. Stroock, S. R. S. Varadhan. *On the support of diffusion processes with applications to the strong maximum principle*. Proceedings of the Sixth Berkeley Symposium on Mathematical Statistics and Probability, III:333–359, 1972.

[84] N. J. Tatro, P.-Y. Chen, P. Das, I. Melnyk, P. Sattigeri, R. Lai. *Optimizing mode connectivity via neuron alignment*. Advances in Neural Information Processing Systems 33 (NeurIPS 2020), pp. 15300–15311, 2020. arXiv:2009.02439.

[85] J. N. Tsitsiklis, B. Van Roy. *An analysis of temporal-difference learning with function approximation*. IEEE Transactions on Automatic Control, 42(5):674–690, 1997. https://doi.org/10.1109/9.580874.

[86] C. Villani. *Optimal Transport: Old and New*. Grundlehren der mathematischen Wissenschaften 338, Springer, 2009, Theorem 5.10; F. Santambrogio, *Optimal Transport for Applied Mathematicians*, Birkhäuser, 2015, Ch. 1.

[87] F.-Y. Wang. *On estimation of the logarithmic Sobolev constant and gradient estimates of heat semigroups*. Probability Theory and Related Fields, 108(1):87–101, 1997. https://doi.org/10.1007/s004400050102.

[88] Y. Zhang, Q. Cai, Z. Yang, Y. Chen, Z. Wang. *Can temporal-difference and Q-learning learn representation? A mean-field theory*. Advances in Neural Information Processing Systems 33 (NeurIPS 2020), pp. 19680–19692, 2020. arXiv:2006.04761.

[89] G. M. Rotskoff, E. Vanden-Eijnden. *Trainability and accuracy of artificial neural networks: an interacting particle system approach*. Communications on Pure and Applied Mathematics, 2022; arXiv:1805.00915 (2018, as *Neural networks as interacting particle systems: asymptotic convexity of the loss landscape and universal scaling of the approximation error*).

[90] R. T. Rockafellar. *Monotone operators and the proximal point algorithm*. SIAM J. Control Optim. 14(5):877–898, 1976; M. Hanke, C. W. Groetsch, *Nonstationary iterated Tikhonov regularization*, J. Optim. Theory Appl. 98(1):37–53, 1998; S. Zhang, A. Choromanska, Y. LeCun, *Deep learning with elastic averaging SGD*, NeurIPS 2015; S. J. Nowlan, G. E. Hinton, *Simplifying neural networks by soft weight-sharing*, Neural Computation 4(4):473–493, 1992; K. Ullrich, E. Meeds, M. Welling, *Soft weight-sharing for neural network compression*, ICLR 2017.

[91] J. Sirignano, K. Spiliopoulos. *Mean field analysis of neural networks: a central limit theorem*. Stochastic Processes and their Applications 130(3):1820–1852, 2020; Z. Chen, G. M. Rotskoff, J. Bruna, E. Vanden-Eijnden, *A dynamical central limit theorem for shallow neural networks*, NeurIPS 2020.

[92] A. J. Kurdila, F. J. Narcowich, J. D. Ward. *Persistency of excitation in identification using radial basis function approximants*. SIAM J. Control Optim. 33(2):625–642, 1995.

[93] J. A. Farrell. *Stability and approximator convergence in nonparametric nonlinear adaptive control*. IEEE Trans. Neural Networks 9(5), 1998.

[94] T. Liu, C. Wang, D. J. Hill. *Learning from neural control of nonlinear systems in normal form*. Systems & Control Letters, 2009.

[95] K. S. Narendra, A. M. Annaswamy. *A new adaptive law for robust adaptation without persistent excitation*. IEEE TAC 32(2):134–145, 1987; P. A. Ioannou, *General robust adaptive law for tuning controllers*, Proc. 25th IEEE CDC, 474–476, 1986; P. A. Ioannou, J. Sun, *Theory and design of robust direct and indirect adaptive-control schemes*, Int. J. Control 47(3):775–813, 1988; W. Magnus, *On the exponential solution of differential equations for a linear operator*, Comm. Pure Appl. Math. 7(4):649–673, 1954; S. Blanes, F. Casas, J. A. Oteo, J. Ros, *The Magnus expansion and some of its applications*, Phys. Rep. 470(5–6):151–238, 2009; R. L. Kosut, B. D. O. Anderson, I. M. Y. Mareels, *Stability theory for adaptive systems: method of averaging and persistency of excitation*, IEEE TAC 32(1):26–34, 1987.

[96] R. Z. Khas'minskii. *A limit theorem for the solutions of differential equations with random right-hand sides*. Theory Probab. Appl. 11(3):390–406, 1966; N. Sadegh, R. Horowitz, *Stability and robustness analysis of a class of adaptive controllers for robotic manipulators*, Int. J. Robotics Research 9(3):74–92, 1990; T. Söderström, *Errors-in-Variables Methods in System Identification*, Springer, 2018.

[97] M. A. Hassouneh, H.-C. Lee, E. H. Abed. *Washout filters in feedback control: benefits, limitations and extensions*. Proc. 2004 American Control Conference, paper FrA02.2.

[98] L. Dinh, R. Pascanu, S. Bengio, Y. Bengio. *Sharp minima can generalize for deep nets.* ICML 2017; B. Neyshabur, R. Salakhutdinov, N. Srebro, *Path-SGD: path-normalized optimization in deep neural networks*, NeurIPS 2015; B. Şimşek, F. Ged, A. Jacot, F. Spadaro, C. Hongler, W. Gerstner, J. Brea, *Geometry of the loss landscape in overparameterized neural networks: symmetries and invariances*, ICML 2021.

[99] E. Hairer, C. Lubich, G. Wanner. *Geometric Numerical Integration: Structure-Preserving Algorithms for Ordinary Differential Equations.* 2nd ed., Springer, 2006.

[100] B. Widrow, J. M. McCool, M. G. Larimore, C. R. Johnson Jr. *Stationary and nonstationary learning characteristics of the LMS adaptive filter.* Proc. IEEE 64(8):1151–1162, 1976; K. Mayyas, T. Aboulnasr, *Leaky LMS algorithm: MSE analysis for Gaussian data*, IEEE Trans. Signal Process. 45(4):927–934, 1997; A. Benveniste, M. Métivier, P. Priouret, *Adaptive Algorithms and Stochastic Approximations*, Springer, 1990; R. Carmona, F. Delarue, D. Lacker, *Mean field games with common noise*, Ann. Probab. 44(6):3740–3803, 2016; B. D. O. Anderson, *Adaptive systems, lack of persistency of excitation and bursting phenomena*, Automatica 21(3):247–258, 1985; A. Dieuleveut, A. Durmus, F. Bach, *Bridging the gap between constant step size stochastic gradient descent and Markov chains*, Ann. Statist. 48(3):1348–1382, 2020, https://doi.org/10.1214/19-AOS1850; M. J. Box, *Bias in nonlinear estimation*, J. R. Stat. Soc. B 33(2):171–190, with discussion 190–201, 1971.

[101] T. Yucelen, W. M. Haddad. *Low-frequency learning and fast adaptation in model reference adaptive control.* IEEE Trans. Automat. Control 58(4):1080–1085, 2013.

[102] K. W. Chang. *Singular perturbations of a general boundary value problem.* SIAM J. Math. Anal. 3(3), 1972, https://doi.org/10.1137/0503050; P. C. Moan, J. Niesen, *Convergence of the Magnus series*, Found. Comput. Math. 8(3):291–301, 2008; T. Kato, *Perturbation Theory for Linear Operators*, Springer, 1966; P. V. Kokotović, H. K. Khalil, J. O'Reilly, [49], Ch. 4, Ch. 5 and App. A; A. H. Haddad, P. V. Kokotović, *Stochastic control of linear singularly perturbed systems*, IEEE TAC 22(5):815–821, 1977; X. Yang, J. J. Zhu, *A generalization of Chang transformation for linear time-varying systems*, 49th IEEE CDC, 2010, and *Chang transformation for decoupling of singularly-perturbed linear slowly time-varying systems*, 51st IEEE CDC, 2012; N. Berglund, B. Gentz, *Geometric singular perturbation theory for stochastic differential equations*, J. Differential Equations 191(1):1–54, 2003, https://doi.org/10.1016/S0022-0396(03)00020-2, and *Noise-Induced Phenomena in Slow-Fast Dynamical Systems*, Springer, 2006.